個股分析

ASML (ASML US)

獲利與展望雙升，擴產仍難滿足需求

永豐觀點

ASML 3Q26 獲利優於市場預期，主要受惠既有裝機服務與設備升級需求強勁。管理層將 2026 年營收展望上修至 440–450 億歐元、毛利率約 56%，全年成長主要由 EUV 及服務業務帶動。客戶預期至少至 2028 年供給仍難滿足需求，促使擴產加速，ASML 已承接大量 2028 年訂單，並見客戶提前下訂 2029–2030 年設備的強烈意願；原規劃 2027、2028 年產能各增 30%仍不足因應需求，凸顯設備供給持續偏緊。我們認為，新機種與既有設備升級將協助客戶提升有效產能，價值導向定價則自 2027 年起逐步貢獻獲利。

投資評價與建議

ASML 客戶需求能見度了拉長，以及 ASML 本身持續尋求更多的生產力提升，有助於帆宣設備業務的成長。我們看好帆宣受惠此一需求，進而在未來兩年高毛利業務上具有更強勁動能。現階段維持帆宣目標價 666 元(17.5x 2027EPSF)，待後續財報公告後出具報告更新。

營運現況與分析

服務業務超預期，帶動獲利改善：ASML F3Q26 營收 116.6 億歐元(以下同)(+25.0%QoQ，+55.2%YoY)，毛利率 57.5%，營業利益 51.7 億(+49.7%QoQ，+109.6%YoY)，稅後 EPS 12.95 歐元，高於市場預期的 10.7 歐元。本季獲利優於預期，主要受惠客戶設備升級需求強勁，既有裝機服務與升級業務(Installed Base Management，IBM)單季營收達 32.8 億，高於先前展望的 29 億，並帶動毛利率改善。此外，稅後淨利亦包涵約 5 億歐元的 Mistral AI 投資部位重新評價。

表一：3Q26 ASML 財報優於市場預期(US GAAP)

	實際值	Guidance	市場預估	與市場差異
營收(€M)	11,662	11,000~12,000	11,703	-0.4%
IBM(€M)	3,289	2,900		
毛利率	57.5%	55~57%	56.1%	+1.3pts
研發費用(€M)	1,174	1,200		
管銷費用(€M)	353	400		
營業利益(€M)	5,173	4,840	4,912	+5.3%
EPS(€)-basic	12.95		10.7	+21.0%

*IBM=Installed Base Management

資料來源：Bloomberg、ASML，永豐投顧研究部整理，Oct. 2026

繼續上修全年展望，業務規模與組合共同支撐毛利率：管理層預估 4Q26 營收 142-152 億歐元、毛利率約 57%，其中 IBM 營收達到 37 億，主要由設備升級需求帶動。同時 2026 年營收展望上修至 440-450 億(原估 430-450 億)，毛利率預估約 56%(原估 54-56%)，均朝原財測需求上緣收斂。全年營收成長 36.2%YoY，主要動能來自 EUV 業務(>45%YoY)及 IBM(~50%YoY)，在非 EUV 業務則年增約 20%。毛利率明顯優於年初展望的 51-53%，主要受惠營收規模擴大帶動固定成本攤提改善，以及 EUV、浸潤式曝光設備，以及設備升級業務帶來較佳的產品組合。此外，主要客戶的**價值導向定價協商已完成，預期 2027 年開始部分貢獻，2028 年起完整反映**。

客戶需求能見度延伸，設備供給仍須加速：管理層表示，客戶需求不僅短期增加，也延伸至更長期間，目前預期至少到 2028 年仍供不應求，因此正加速擴產及新廠建置。ASML 已承接大量 2028 年訂單，客戶亦對 2029-2030 年提前下單展現強烈意願。管理層認為，原規劃 2027 及 2028 年產能各增加 30%，仍不足以滿足客戶需求，正尋求於 2028 年及以後進一步擴充產能。我們認為，短中期設備供給仍偏緊，除增加出貨台數外，新機種如 NXE:3800F 及既有設備升級，也將成為協助客戶提高有效產能的重要途徑，並支撐 ASML 營收與獲利成長。

其他技術進展：(1) High NA 採用時程逐步明確： 台積電與 ASML 共同推動產業聯盟，目標於 2030 年代前半段導入 12 吋光罩；台積電並確認預計於 2030 年起將 High NA EUV 導入量產。三星規劃於 2028 年導入，Intel 則已於量產中使用，並持續增加採用層數。(2) 量測效率提升，支援先進製程需求：YieldStar 550B 為新一代光學疊對量測設備，用於檢確認不同製程層圖案的對齊精，並可將顯影完後的資料即時回傳給曝光機和製程控制軟體，調整後續晶圓/下一批的曝光參數，可支援 1.4nm 和 1D DRAM 節點。管理層表示，其量測生產效率提升 60%，有助於承接先進製程日益增加的量測需求，降低量測站形成產能瓶頸的風險。

王彥鈞

ASML (ASML US)：獲利與展望雙升，擴產仍難滿足需求

帆宣 (6196 TT，B，666；1.5%)

玉山投顧 E.SUN Investment Consulting

產業趨勢

台股各產業月營收暨科技類股掃描

上市櫃月營收總覽

*統計樣本依已公布營收的公司為主

分類	9月營收（億元）				年增率（%）	
	單月金額	累計金額	去年單月	去年累計	單月	累計
上市櫃加總	**65,986.9**	**485,549.9**	**44,937.8**	**352,383.8**	**46.8**	**37.8**
電子	53,243.8	380,432.0	34,467.5	258,770.9	54.5	47.0
金融	376.8	3,653.3	314.8	2,668.8	19.7	36.9
傳產	11,931.7	98,191.1	9,828.9	88,246.3	21.4	11.3
軟體	434.6	3,273.5	326.6	2,697.8	33.1	21.3

電子產業月營收總覽（月營收規模前30名）

次產業－電子類	9月營收（億元）				年增率（%）	
	單月金額	累計金額	去年單月	去年累計	單月	累計
電子中游-EMS	15,930.6	98,490.5	10,719.4	71,752.1	48.62	37.27
電子下游-筆記型電腦	10,375.6	83,225.5	6,354.5	48,223.5	63.28	72.58
電子上游-IC-通路	6,675.6	45,459.5	3,238.8	24,186.9	106.11	87.95
電子上游-IC-代工	5,548.8	42,299.7	3,632.0	30,345.9	52.78	39.39
電子上游-IC-封測	1,260.5	9,178.3	893.6	7,070.9	41.06	29.80
電子上游-IC-設計	1,135.8	9,675.6	1,036.4	8,825.9	9.59	9.63
電子中游-主機板	967.8	6,720.6	572.0	4,848.0	69.20	38.63
電子中游-電源供應器	949.8	7,830.8	830.5	6,049.9	14.37	29.44
電子上游-PCB-製造	926.4	6,490.6	685.4	5,166.6	35.15	25.62
電子中游-網通	837.8	6,211.0	567.8	4,484.8	47.56	38.49
電子上游-記憶體製造	817.1	4,995.4	174.9	1,204.2	367.24	314.82
電子中游-儀器設備工程	556.6	4,176.3	373.7	2,665.2	48.92	56.70
電子下游-電信服務	527.7	4,247.5	460.5	3,901.0	14.59	8.88
電子中游-LCD-TFT面板	489.4	4,423.0	495.2	4,291.9	-1.16	3.05
電子上游-連接元件	483.8	3,875.3	475.3	3,566.3	1.78	8.66
電子上游-PCB-材料設備	443.5	3,053.0	208.9	1,760.0	112.36	73.46
電子下游-工業電腦	427.8	3,222.0	284.4	2,402.2	50.44	34.12
電子上游-記憶體IC設計	423.0	2,659.6	94.5	713.0	347.43	273.03
電子上游-記憶體銷售	398.2	2,989.7	122.7	859.1	224.53	248.02
電子上游-被動元件	388.9	3,026.0	280.8	2,356.6	38.50	28.40
電子上游-ABF	294.2	2,122.6	184.5	1,541.0	59.43	37.74
電子中游-NB與手機零組件	268.0	2,088.6	204.4	1,414.7	31.13	47.64
電子下游-資訊通路	261.6	2,054.6	196.1	1,652.3	33.37	24.35
電子中游-機殼	253.7	2,002.1	228.8	1,952.7	10.89	2.53
電子下游-顯示器	239.2	1,891.0	215.4	1,799.7	11.06	5.07
電子下游-掃描器	234.2	1,344.5	192.4	1,191.2	21.71	12.87
電子上游-IP/ASIC	193.9	1,008.7	75.6	704.4	156.39	43.20
電子上游-IC-半導體設備	172.7	1,267.4	111.9	894.5	54.40	41.69
電子下游-電腦周邊	155.8	1,417.3	168.8	1,445.8	-7.65	-1.97
電子中游-LCD-STN面板	148.3	1,156.5	120.0	1,169.4	23.57	-1.10

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金融產業月營收總覽

次產業－金融	9月營收（億元）				年增率（%）	
	單月金額	累計金額	去年單月	去年累計	單月	累計
金融-銀行	180.8	1,735.6	169.1	1,573.4	6.92	10.31
金融-金控	96.4	949.0	74.7	615.1	29.07	54.29
金融-證券	85.7	842.4	58.2	365.4	47.16	130.58
金融-保險	14.0	126.2	12.8	115.0	9.06	9.74

傳統產業月營收總覽

次產業－傳產	9月營收（億元）				年增率（%）	
	單月金額	累計金額	去年單月	去年累計	單月	累計
傳產-塑膠	1,950.3	14,449.1	1,347.1	12,424.7	44.78	16.29
傳產-航運	1,718.0	13,050.2	1,144.9	11,073.6	50.05	17.85
傳產-其他	1,027.8	9,005.5	890.3	8,224.1	15.45	9.50
傳產-鋼鐵	962.4	8,123.9	812.8	7,562.5	18.40	7.42
傳產-食品	944.9	8,102.7	895.1	7,754.9	5.57	4.49
傳產-營建	718.6	5,219.3	472.7	4,075.5	52.03	28.07
傳產-百貨	662.8	6,708.4	752.4	6,182.3	-11.91	8.51
傳產-紡織纖維	550.4	4,806.8	527.5	4,615.7	4.33	4.14
傳產-汽車	494.3	4,404.8	426.5	4,163.9	15.90	5.79
傳產-電機	485.2	4,107.1	407.2	3,597.0	19.15	14.18
傳產-生技	365.6	2,946.3	300.7	2,560.0	21.57	15.09
傳產-化學工業	324.3	2,675.5	249.7	2,319.7	29.87	15.34
傳產-電線電纜	257.9	2,178.7	222.8	2,022.2	15.78	7.74
傳產-汽車零組件	245.0	2,063.6	219.0	1,927.4	11.88	7.07
傳產-水泥	239.2	1,885.3	229.5	1,894.0	4.22	-0.46
傳產-橡膠	209.5	1,790.3	170.1	1,564.0	23.16	14.47
傳產-紙業	184.7	1,545.6	164.8	1,435.8	12.11	7.65
傳產-觀光	168.3	1,563.0	153.3	1,461.8	9.74	6.92
傳產-綠能環保	116.2	988.1	161.8	922.5	-28.15	7.10
傳產-運動休閒	97.8	752.7	85.8	687.1	14.07	9.55
傳產-自行車	90.4	789.2	87.8	820.0	2.95	-3.76
傳產-玻璃陶瓷	60.3	456.7	47.3	405.2	27.51	12.73
傳產-高爾夫球	38.4	396.5	39.5	377.6	-2.95	5.03
傳產-文創娛樂	17.6	165.5	18.6	158.7	-5.36	4.27
傳產-照明	1.7	16.3	1.7	16.3	3.60	0.27

軟體產業月營收總覽

次產業－軟體	9月營收（億元）				年增率（%）	
	單月金額	累計金額	去年單月	去年累計	單月	累計
軟體-系統整合	300.1	2,157.7	215.6	1,747.7	39.16	23.46
軟體-其他	79.3	653.8	67.2	577.8	18.06	13.16
軟體-遊戲	55.2	462.0	43.7	372.3	26.19	24.11

資料來源：CMoney

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重點產業及公司月營收概況

代碼	公司	收盤價	股本（億元）	單月營收（億元）			月增率(%)	年增率(%)	2026累計營收（億元）	年增率(%)
				7月	8月	9月				
晶圓代工										
2330	台積電	2,540.00	2,593.2	4,675.8	5,148.1	5,118.6	-0.6	54.7	38,987.3	41.1
2303	聯電	145.50	1,257.0	238.4	250.4	253.5	1.2	27.2	2,040.1	16.1
5347	世界	181.00	187.7	47.1	50.9	59.0	15.9	15.5	424.8	18.0
6770	力積電	71.40	471.2	66.7	70.1	76.7	9.4	94.0	522.1	52.5
3707	漢磊	82.20	40.1	6.5	6.6	6.6	0.6	24.5	53.0	26.2
6789	采鈺	470.50	31.8	9.1	9.2	8.8	-4.5	18.0	75.0	12.8
	產業小計			5,043.6	5,535.3	5,523.2	-0.2	52.9	42,102.2	39.4
IC封測										
3711	日月光投控	762.00	447.1	737.8	822.5	881.3	7.2	45.5	6,088.9	30.3
6239	力成	304.50	75.9	82.7	85.6	85.1	-0.6	27.0	697.6	30.4
3374	精材	481.50	27.1	8.6	9.1	9.3	2.6	21.3	68.0	31.5
6451	訊芯-KY	414.00	11.8	7.8	6.9	7.2	3.6	39.9	57.5	5.3
3264	欣銓	272.50	49.0	17.7	17.9	17.8	-1.0	46.9	138.5	35.4
6257	矽格	287.00	48.4	20.1	20.7	20.9	1.0	26.3	173.5	21.2
3265	台星科	262.00	13.6	4.2	4.0	4.0	-0.5	11.5	39.4	15.9
6147	頎邦	229.00	74.5	25.2	26.2	24.4	-6.9	28.0	199.7	24.4
8150	南茂	123.00	70.5	28.2	27.9	27.9	0.2	33.8	227.2	30.5
8131	福懋科	67.60	44.2	10.9	11.4	12.4	9.0	35.5	95.6	37.2
8110	華東	49.55	51.8	6.9	8.0	7.8	-2.7	19.1	61.8	16.0
2449	京元電子	300.00	128.4	39.9	40.8	40.7	-0.2	24.5	334.8	34.1
2351	順德	212.50	19.6	13.3	13.3	13.7	3.5	57.4	103.9	35.3
	產業小計			1,003.3	1,094.2	1,152.4	5.3	41.5	8,286.3	29.9
IC耗材/零組件										
6223	旺矽	6,130.00	9.8	18.1	20.5	20.5	-0.0	82.2	150.7	58.0
6510	精測	3,420.00	3.3	6.1	6.4	6.5	2.1	55.9	49.0	35.8
6515	穎崴	6,025.00	3.6	16.0	17.1	12.5	-26.7	87.7	110.6	96.7
6683	雍智科技	1,440.00	2.8	2.3	2.3	2.5	5.9	33.6	20.3	30.3
6217	中探針	179.50	11.7	4.2	4.2	3.5	-16.1	0.7	34.6	29.7
7828	創新服務	1,830.00	4.1	1.3	1.6	1.6	-0.6	102.3	9.5	196.5
3413	京鼎	288.50	11.1	21.6	22.5	23.6	4.9	14.6	186.3	19.3
3680	家登	569.00	9.7	8.7	8.4	9.9	18.4	67.7	68.9	35.6
8028	昇陽半導體	281.50	18.5	5.1	5.1	5.1	1.3	33.2	42.6	29.5
1560	中砂	1,000.00	15.1	8.9	9.0	9.1	1.0	30.1	74.8	25.1
8091	翔名	219.50	5.8	3.3	3.0	2.5	-16.6	26.5	24.3	53.8
7768	頌勝科技	292.50	7.0	1.8	1.9	1.9	-0.7	34.4	16.9	11.6
7810	捷創科技	237.00	2.0	0.8	0.8	0.9	13.7	95.0	6.7	81.1
6488	環球晶	1,115.00	52.8	49.8	47.6	51.3	7.7	-13.1	440.7	-4.4
3532	台勝科	485.50	38.8	13.0	13.3	13.5	1.3	29.0	109.0	20.3
3016	嘉晶	179.00	29.6	4.8	5.1	5.1	1.2	52.7	38.7	36.9
3289	宜特	140.00	8.7	4.0	3.7	3.6	-3.1	-21.5	34.0	-5.9
3587	閎康	242.00	7.2	5.6	5.7	5.7	0.6	12.3	47.6	17.0
6830	汎銓	540.00	6.1	2.4	2.5	2.6	4.2	36.7	19.7	24.4
	產業小計			178.0	180.6	181.9	0.7	18.3	1,484.9	19.8
IC載板										
3037	欣興	1,255.00	164.4	162.5	177.5	186.5	5.1	64.7	1,329.9	37.7
8046	南電	1,440.00	64.6	54.4	53.5	55.9	4.4	57.8	411.3	41.8
3189	景碩	1,060.00	52.7	45.5	48.1	51.8	7.7	44.5	381.4	33.6
	產業小計			262.4	279.1	294.2	5.4	59.4	2,122.6	37.7

資料來源：CMoney

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				7月	8月	9月				
IC設計										
2454	聯發科	4,625.00	160.5	484.7	641.8	554.0	-13.7	2.0	4,693.9	5.3
5274	信驊	17,695.00	4.2	15.2	16.3	16.4	0.6	100.9	118.0	77.7
5269	祥碩	1,540.00	7.4	7.8	8.8	9.9	12.8	-30.0	88.8	-10.2
4966	譜瑞-KY	553.00	7.9	13.5	13.3	15.0	13.3	-8.4	124.2	-1.7
3443	創意	8,100.00	13.4	57.7	58.4	71.5	22.4	97.8	441.0	102.8
3035	智原	197.50	26.1	10.6	11.8	14.2	19.8	4.4	95.7	-37.0
3529	力旺	2,850.00	7.5	6.8	3.6	1.8	-50.3	13.5	34.0	21.4
3661	世芯-KY	4,395.00	8.6	74.3	87.8	99.8	13.7	496.0	378.8	44.6
6643	M31	476.50	4.3	1.7	1.8	1.5	-13.1	23.0	13.1	6.0
6533	晶心科	345.00	5.8	1.2	1.1	1.9	69.2	-10.1	13.2	29.7
6531	愛普*	947.00	8.2	11.0	10.3	11.9	15.3	90.2	77.9	106.0
2379	瑞昱	746.00	51.6	129.0	124.6	123.8	-0.7	23.4	1,117.1	15.9
4968	立積	95.20	9.2	3.2	3.2	3.3	2.4	0.0	28.8	2.5
3034	聯詠	529.00	60.9	103.0	103.3	103.5	0.2	25.0	827.8	6.3
3592	瑞鼎	236.00	7.6	21.0	22.1	22.5	1.8	26.3	188.2	10.1
8016	矽創	291.00	12.0	20.4	19.1	19.8	3.7	19.4	175.0	27.8
4961	天鈺	174.50	12.1	19.7	21.1	21.8	3.1	57.0	161.4	15.2
3227	原相	187.00	15.2	8.9	8.2	8.2	0.4	9.5	72.8	6.4
6415	矽力*-KY	440.00	9.7	23.6	24.2	25.1	3.9	50.7	181.4	35.2
8081	致新	262.50	8.6	8.4	8.0	8.3	3.0	14.2	68.2	3.0
6732	昇佳電子	150.50	4.9	3.0	3.0	3.1	3.1	-9.9	29.6	-13.8
6679	鈺太	192.50	5.5	1.7	1.6	1.5	-6.5	-2.8	15.7	5.5
2458	義隆	133.00	30.4	8.8	7.4	6.0	-18.7	-46.9	86.5	-8.6
3014	聯陽	144.50	16.6	6.4	6.3	6.1	-3.6	0.5	54.5	0.2
3006	晶豪科	260.50	29.8	67.8	79.0	74.6	-5.6	470.2	433.6	343.9
6202	盛群	58.00	23.0	3.7	3.6	3.4	-5.9	40.2	28.8	21.1
4919	新唐	154.00	45.2	25.7	25.3	27.6	9.4	4.7	237.9	1.4
6138	茂達	265.00	7.6	7.0	6.7	6.1	-8.4	-9.2	60.2	8.3
8261	富鼎	176.50	11.9	2.9	2.9	2.7	-6.9	-1.8	25.0	5.8
6435	大中	254.50	3.8	3.3	3.4	3.1	-9.9	8.8	28.8	13.7
	產業小計			1,152.1	1,327.9	1,268.3	-4.5	26.5	9,899.9	14.9
IC設備										
7769	鴻勁	6,490.00	18.0	52.5	52.2	57.3	9.8	100.2	407.1	93.9
7734	印能科技	2,915.00	2.8	4.1	2.6	4.5	75.7	122.3	29.0	60.4
7751	竑騰	1,640.00	2.7	2.4	2.4	2.8	14.0	57.0	21.1	60.8
7728	光焱科技	624.00	1.4	0.3	0.3	0.5	65.2	69.0	3.2	41.1
6937	天虹	291.50	6.8	2.1	2.7	3.9	47.1	29.1	21.6	42.1
6187	萬潤	1,675.00	9.8	10.5	11.1	11.6	4.6	109.2	70.9	58.3
3131	弘塑	2,525.00	3.0	7.9	6.8	7.5	9.7	61.9	55.4	27.0
5443	均豪	115.50	16.5	8.4	6.2	6.6	6.8	55.2	42.9	25.2
6640	均華	1,155.00	2.9	6.9	4.7	5.0	6.6	87.2	33.1	64.0
3551	世禾	197.50	6.2	3.0	3.0	3.1	2.3	17.7	25.8	20.2
3583	辛耘	755.00	8.4	9.7	10.7	10.4	-2.6	2.8	92.0	7.5
2360	致茂	2,365.00	42.5	54.3	56.3	68.6	21.8	206.5	433.2	119.5
	產業小計			162.2	158.9	181.7	14.3	106.7	1,235.3	75.0
IC特化										
4749	新應材	737.00	9.5	4.1	4.8	4.3	-11.2	11.8	38.1	19.9
4770	上品	270.00	8.0	5.3	4.9	5.8	17.7	69.0	39.6	7.8
4768	晶呈科技	363.00	4.8	1.1	1.3	1.4	9.4	23.9	12.0	106.1
4772	台特化	270.00	14.8	3.2	3.2	3.6	12.4	30.3	27.6	146.2
4755	三福化	124.00	10.1	4.1	4.4	4.8	7.9	16.2	37.0	3.3
5234	達興材料	319.50	10.3	4.5	4.5	4.8	6.3	19.5	38.6	12.2
1717	長興	74.70	118.4	40.9	40.7	44.3	8.9	29.4	347.6	13.8
	產業小計			63.2	64.0	69.1	8.0	28.8	540.6	17.2

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				7月	8月	9月				
IC工程										
2404	漢唐	1,235.00	19.1	99.5	93.9	81.7	-13.0	12.6	738.6	62.2
6196	帆宣	657.00	22.0	75.2	72.2	71.7	-0.6	88.5	574.2	55.5
5536	聖暉*	931.00	6.2	38.9	48.8	55.1	13.0	41.1	392.6	28.7
6613	朋億*	280.00	3.9	12.9	16.9	16.0	-5.5	139.3	107.1	71.7
6691	洋基工程	747.00	13.4	47.6	52.9	46.0	-13.1	114.3	327.6	113.0
6788	華景電	368.50	3.9	2.9	2.4	3.1	25.9	41.5	22.7	21.2
6139	亞翔	750.00	23.6	109.3	110.2	107.0	-2.9	6.0	786.0	52.2
6903	巨漢	332.00	6.7	6.0	6.2	8.1	30.4	109.0	60.4	206.9
6725	矽科宏晟	213.00	3.7	3.6	3.4	3.1	-7.3	3.0	28.9	3.5
	產業小計			395.8	406.9	391.8	-3.7	36.1	3,038.1	57.5
記憶體/模組										
2408	南亞科	527.00	345.0	438.7	446.9	450.9	0.9	576.6	2,652.8	627.0
2344	華邦電	174.00	450.0	267.7	273.1	282.5	3.4	256.7	1,804.3	187.4
2337	旺宏	115.50	198.6	77.2	81.5	83.7	2.7	188.2	538.3	154.5
8299	群聯	2,055.00	22.1	271.6	282.8	306.4	8.4	370.3	1,949.3	290.9
5289	宜鼎	1,260.00	9.8	84.9	86.8	87.1	0.4	535.7	615.1	550.6
3260	威剛	367.50	35.5	183.8	189.3	157.3	-16.9	200.0	1,173.1	215.6
2451	創見	286.50	43.1	52.0	48.4	60.4	24.8	290.4	485.1	370.0
8271	宇瞻	197.00	13.0	27.9	27.9	30.0	7.6	157.7	253.0	216.8
4967	十銓	269.50	9.9	25.4	35.1	40.8	16.2	109.7	267.7	90.2
	產業小計			1,429.3	1,471.6	1,499.1	1.9	325.0	9,738.8	290.6
IC通路										
3702	大聯大	116.00	175.2	1,461.0	1,422.4	2,195.9	54.4	146.5	12,827.1	72.5
5434	崇越	593.00	19.4	81.7	77.9	77.0	-1.2	26.4	630.3	25.3
3010	華立	132.50	25.9	85.4	79.9	90.1	12.9	41.3	712.2	22.5
3036	文曄	216.50	136.5	1,829.7	1,983.8	2,159.3	8.9	60.0	16,822.5	101.3
2459	敦吉	65.60	10.6	5.1	4.5	5.0	10.4	16.7	40.4	12.3
3388	崇越電	84.60	8.6	7.3	6.2	6.5	3.3	5.3	63.9	15.6
	產業小計			3,470.2	3,574.8	4,533.9	26.8	90.9	31,096.4	83.2
NB/OEM										
2317	鴻海	244.50	1,407.6	9,465.1	9,217.7	11,586.3	25.7	38.4	76,697.7	39.5
2382	廣達	322.00	410.8	3,662.7	4,239.7	3,253.0	-23.3	76.7	29,613.6	99.4
2324	仁寶	36.60	440.7	810.1	673.4	944.3	40.2	35.5	6,823.8	20.4
2356	英業達	58.10	358.8	901.9	832.7	986.4	18.5	62.9	7,422.6	42.8
4938	和碩	98.60	267.6	860.2	883.0	2,147.5	143.2	86.9	9,076.3	13.8
3231	緯創	189.50	343.0	3,082.2	4,600.7	3,808.1	-17.2	87.2	28,908.4	97.3
6669	緯穎	2,275.00	55.9	1,176.9	1,443.1	1,726.2	19.6	100.0	9,892.8	50.3
6933	AMAX-KY	382.50	4.2	14.4	13.6	14.3	5.2	228.9	87.1	66.3
2357	華碩	977.00	74.3	841.3	949.6	972.4	2.4	17.7	7,424.3	38.5
2353	宏碁	31.35	304.8	269.0	301.8	347.4	15.1	17.6	2,495.5	24.0
	產業小計			21,083.8	23,155.2	25,785.9	11.4	54.5	178,442.0	52.1
面板/零組件										
3481	群創	51.30	798.9	188.9	187.0	199.9	6.9	0.7	1,879.3	10.6
2409	友達	35.55	754.7	203.7	231.0	234.0	1.3	-4.3	2,068.0	-2.1
8069	元太	149.00	115.4	28.8	34.5	36.1	4.9	13.5	287.9	-1.1
6456	GIS-KY	102.00	33.8	40.8	42.1	61.6	46.4	30.8	410.4	-18.5
3673	TPK-KY	74.70	40.7	63.9	63.6	65.9	3.6	27.8	564.7	16.6
6176	瑞儀	90.50	34.9	37.6	38.2	37.1	-2.8	-6.9	344.1	-7.4
3718	中光電投控	62.10	39.1	0.0	34.3	33.8	-1.3	0.0	68.1	0.0
6706	惠特	153.50	7.9	0.7	0.8	0.9	13.2	131.7	5.7	-26.9
6278	台表科	272.50	29.2	46.7	46.3	51.6	11.5	15.9	402.4	6.5
	產業小計			611.1	677.7	721.1	6.4	4.2	6,030.6	1.9

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				7月	8月	9月				
手機零組件										
3008	大立光	7,325.00	13.1	48.8	50.1	57.8	15.3	-7.3	448.6	2.1
3406	玉晶光	1,070.00	11.3	24.2	27.2	26.7	-1.8	0.5	196.4	10.1
4915	致伸	63.40	47.2	58.6	61.3	62.5	2.0	15.0	513.6	13.5
2455	全新	564.00	18.9	4.2	4.7	4.5	-4.6	29.5	32.7	35.0
4971	IET-KY	590.00	4.7	1.1	1.1	1.1	2.0	11.4	9.1	19.3
3105	穩懋	580.00	42.4	18.9	19.4	20.3	4.8	30.0	157.1	32.6
8086	宏捷科	148.50	19.7	4.2	4.2	4.2	0.1	8.4	37.8	32.1
4991	環宇-KY	510.00	12.1	1.5	2.6	4.4	65.9	112.9	23.3	54.6
5222	全訊	100.50	10.0	0.7	0.7	0.3	-61.5	-69.8	5.7	-17.8
4958	臻鼎-KY	581.00	113.3	176.0	206.7	263.6	27.5	36.1	1,537.8	22.4
6269	台郡	63.50	32.4	20.5	20.0	21.0	5.1	-2.7	169.9	1.3
2354	鴻準	63.20	141.5	214.2	218.7	144.2	-34.1	11.2	1,133.3	-0.2
2474	可成	194.50	53.9	10.6	10.0	10.3	2.5	-34.2	95.4	-33.3
2439	美律	81.00	24.9	36.5	41.2	51.9	26.1	4.1	343.7	6.7
3042	晶技	204.50	34.3	13.7	14.1	14.2	0.0	14.6	112.3	12.1
	產業小計			633.4	681.9	686.8	0.7	15.8	4,816.6	9.6
NB零組件										
2308	台達電	1,900.00	259.8	670.7	645.9	654.0	1.3	14.6	5,396.7	37.2
2301	光寶科	288.00	231.6	190.1	195.9	188.5	-3.8	22.2	1,535.6	26.2
2385	群光	103.50	76.6	70.6	65.2	62.1	-4.7	-25.0	650.5	-10.4
6412	群電	73.30	40.1	25.3	24.7	24.1	-2.3	-21.0	236.4	-11.2
6409	旭隼	875.00	8.8	20.1	17.8	18.7	5.3	7.3	154.2	-2.9
3617	碩天	258.50	9.5	12.5	12.6	13.0	3.4	13.3	100.7	10.5
6121	新普	364.50	18.5	71.1	67.5	60.5	-10.4	-20.6	592.9	1.5
3211	順達	382.50	15.4	12.3	16.8	7.8	-53.7	-34.1	105.2	13.9
6781	AES-KY	1,015.00	8.5	17.2	17.5	17.6	0.5	25.5	145.7	26.3
3376	新日興	173.00	19.6	8.1	8.3	8.5	2.4	3.3	74.8	-11.0
3548	兆利	72.80	6.8	6.6	6.5	6.2	-4.2	-12.2	59.9	-18.7
6805	富世達	2,270.00	6.9	13.6	15.3	16.1	5.6	32.2	124.3	47.3
3324	雙鴻	1,540.00	9.4	37.7	31.4	28.8	-8.2	23.2	270.4	72.4
3653	健策	7,160.00	14.7	32.1	32.3	30.5	-5.3	80.0	220.7	47.1
3017	奇鋐	3,385.00	39.3	185.9	194.8	210.2	7.9	44.9	1,572.5	71.2
8996	高力	1,720.00	9.4	9.2	12.0	16.2	34.2	153.8	94.7	120.8
6230	尼得科超眾	113.50	8.6	7.9	7.6	7.9	3.5	3.4	70.1	6.1
3533	嘉澤	1,880.00	11.3	33.7	32.0	33.6	5.0	16.1	288.4	17.5
3665	貿聯-KY	2,355.00	20.0	96.3	88.9	93.3	5.0	39.2	719.6	39.9
3023	信邦	323.00	24.0	31.7	32.8	33.5	2.1	25.5	271.8	15.2
3526	凡甲	274.50	6.8	4.0	4.2	4.4	6.1	16.4	35.7	26.9
3217	優群	127.00	9.0	3.4	3.0	3.4	12.3	-16.8	28.7	-7.7
5243	乙盛-KY	100.50	16.9	12.7	13.7	14.9	8.6	46.9	105.0	14.3
8210	勤誠	861.00	12.6	24.6	24.6	24.8	0.8	13.8	223.2	46.3
3693	營邦	727.00	4.3	12.2	12.7	15.4	21.3	138.9	81.3	37.7
2376	技嘉	356.00	67.0	442.1	474.0	631.0	33.1	125.5	4,039.8	63.2
2377	微星	155.00	84.5	193.4	176.9	199.1	12.5	1.8	1,631.2	-5.5
3515	華擎	225.50	12.4	66.1	38.2	65.2	70.7	27.3	404.1	18.4
	產業小計			2,311.3	2,273.0	2,489.2	9.5	31.6	19,234.2	31.2
其他										
2059	川湖	10,580.00	9.5	64.1	64.7	44.6	-31.0	196.2	336.2	167.8
6584	南俊國際	522.00	7.1	2.8	2.8	2.7	-3.3	30.5	25.4	47.7
2049	上銀	338.00	35.4	26.8	27.1	27.2	0.5	30.1	220.0	23.9
1590	亞德客-KY	1,210.00	20.0	39.9	37.1	38.1	2.8	26.6	336.9	33.2
4551	智伸科	181.00	11.5	8.2	8.0	7.9	-1.4	18.1	69.7	20.7
2233	宇隆	163.50	6.0	2.8	2.8	2.9	3.4	6.5	25.6	1.9
	產業小計			144.7	142.5	123.5	-13.3	59.1	1,013.8	54.5

資料來源：CMoney

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代碼	公司	收盤價	股本（億元）	單月營收（億元）			月增率(%)	年增率(%)	2026累計營收（億元）	年增率(%)
				7月	8月	9月				
PCB相關										
2313	華通	252.00	123.4	69.6	70.1	86.2	22.9	15.0	621.2	13.0
3044	健鼎	579.00	52.6	98.2	102.2	106.5	4.1	53.5	765.2	40.6
2368	金像電	1,235.00	53.3	100.1	103.8	93.2	-10.3	55.5	733.8	69.1
3715	定穎投控	139.00	28.4	25.3	29.3	32.3	10.0	84.7	212.0	51.3
5439	高技	254.50	9.3	10.5	11.1	10.5	-5.0	-4.6	88.0	28.1
8155	博智	319.50	6.3	5.6	5.6	5.6	0.4	30.6	48.1	35.0
6191	精成科	93.50	49.9	39.2	38.0	42.5	11.7	36.1	312.6	28.2
5469	瀚宇博	83.50	48.6	65.9	65.4	71.3	8.9	36.3	528.0	26.2
2383	台光電	6,150.00	35.8	192.1	201.5	215.3	6.9	168.9	1,412.3	103.7
6274	台燿	1,735.00	29.9	59.1	59.1	61.3	3.6	114.8	423.0	99.4
6213	聯茂	812.00	36.4	47.5	50.4	58.2	15.5	110.7	362.9	50.1
8358	金居	522.00	25.3	10.1	10.6	11.0	4.5	64.6	85.1	48.9
8039	台虹	296.00	26.4	8.5	8.1	8.4	4.1	-19.0	76.1	-3.3
3563	牧德	730.00	6.4	3.6	3.6	3.7	3.2	63.2	30.0	18.8
6664	群翊	375.00	6.1	2.7	2.8	2.8	0.5	24.0	22.1	19.4
3455	由田	291.50	6.0	1.5	2.1	2.1	2.7	23.4	10.7	-14.0
2467	志聖	590.00	16.2	11.7	8.7	9.5	9.2	90.7	80.2	84.2
	產業小計			751.1	772.4	820.3	6.2	69.1	5,811.3	52.2
網通光纖										
2345	智邦	2,130.00	56.1	394.7	408.7	404.0	-1.2	66.1	2,864.0	62.5
5388	中磊	74.10	30.4	79.8	70.4	68.1	-3.4	47.4	590.9	54.2
6285	啟碁	263.50	48.4	169.1	157.0	160.8	2.4	64.1	1,172.9	40.9
3596	智易	157.00	22.0	48.5	47.3	50.6	7.0	8.5	413.1	3.3
3558	神準	134.00	5.9	12.9	16.0	17.1	7.2	45.9	112.3	-2.8
3491	昇達科	1,525.00	6.9	2.3	3.2	2.9	-9.2	52.5	27.7	70.9
6863	永道-KY	102.50	7.5	5.7	4.4	6.0	35.6	90.7	42.9	48.4
3704	合勤控	39.00	41.5	19.2	26.2	32.2	23.1	46.2	228.1	16.2
6143	振曜	77.90	8.6	7.2	6.7	6.1	-9.8	-26.9	63.2	6.1
6146	耕興	197.00	10.2	4.2	4.2	4.2	0.5	10.9	36.8	9.7
6840	東研信超	92.00	3.2	0.9	0.8	0.9	3.0	10.2	7.5	12.3
3081	聯亞	3,025.00	10.2	5.0	5.2	5.4	3.9	190.2	36.9	136.3
4979	華星光	582.00	14.3	5.1	5.5	5.7	5.3	97.8	41.1	27.3
4977	眾達-KY	184.00	9.1	1.8	2.1	2.3	11.4	125.8	10.5	27.7
6442	光聖	1,725.00	8.2	12.3	16.3	20.6	26.6	97.3	113.1	50.2
3450	聯鈞	519.00	14.6	12.2	12.5	12.9	3.3	99.8	88.2	36.5
3163	波若威	689.00	9.5	2.1	2.1	2.4	18.1	27.4	19.1	18.9
3363	上詮	602.00	11.4	1.7	2.0	2.2	9.7	28.4	14.1	-6.0
7717	萊德光電-KY	553.00	2.5	1.1	1.2	1.1	-13.1	80.3	9.3	36.8
	產業小計			785.7	791.9	805.6	1.7	57.1	5,891.6	44.8
被動元件										
2327	國巨*	636.00	51.5	161.3	163.3	170.7	4.5	46.1	1,321.5	36.3
2478	大毅	129.50	14.5	6.6	6.4	5.9	-7.2	29.8	50.1	23.6
2492	華新科	418.00	48.5	44.4	45.7	48.9	7.0	53.9	348.3	25.8
6173	信昌電	0.00	0.0	5.4	5.5	5.3	-4.5	54.8	40.2	29.3
6284	佳邦	78.60	14.9	6.4	6.5	7.1	9.0	4.0	55.8	-4.8
3357	臺慶科	222.50	11.9	8.8	8.2	8.6	4.9	47.8	64.6	34.1
6449	鈺邦	265.50	9.2	5.1	4.6	5.1	9.6	25.6	39.8	24.0
2428	興勤	281.50	12.8	9.8	9.4	10.1	7.0	41.6	76.9	29.3
5425	台半	100.50	26.4	17.3	17.1	18.0	5.0	15.3	147.2	8.4
2481	強茂	171.00	38.2	14.5	13.8	15.3	10.8	38.2	117.3	20.4
3090	日電貿	171.50	28.8	21.0	19.1	18.5	-3.1	31.3	155.1	30.2
	產業小計			300.5	299.7	313.4	4.6	41.7	2,416.6	29.3

資料來源：CMoney

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				7月	8月	9月				
IPC/POS/機器人										
2395	研華	721.00	86.9	112.7	104.2	107.5	3.2	73.1	789.5	49.1
6414	樺漢	404.50	15.7	164.0	154.9	177.6	14.7	45.3	1,349.4	31.3
6416	瑞祺電通	120.00	7.3	4.5	5.5	7.1	29.0	59.2	41.0	30.6
6206	飛捷	136.50	14.3	5.7	5.8	5.9	1.5	30.0	51.4	30.1
8114	振樺電	180.50	10.8	19.7	15.4	14.4	-6.6	44.6	136.7	-10.4
7750	新代	1,915.00	7.2	28.7	24.4	25.0	2.6	125.1	209.3	103.9
4585	達明	290.50	10.5	1.4	1.5	2.0	34.0	15.0	15.0	11.0
4576	大銀微系統	233.50	12.0	3.7	4.3	4.3	0.1	74.7	31.0	58.1
2359	所羅門	127.50	17.2	4.1	2.8	3.2	17.7	16.7	33.6	5.9
3167	大量	1,085.00	9.1	11.5	12.2	13.0	6.9	172.3	84.8	140.2
6579	研揚	151.00	17.0	9.1	9.8	9.7	-1.1	19.7	81.0	27.3
3479	安勤	132.00	7.3	9.6	9.6	9.5	-0.8	15.3	74.5	17.0
產業小計				374.7	350.4	379.3	8.3	56.4	2,897.0	37.2
電機										
1503	士電	204.00	52.1	29.7	28.5	31.4	10.3	8.2	300.8	9.4
1513	中興電	169.00	50.3	24.1	25.0	24.2	-3.4	11.3	213.5	5.0
1519	華城	724.00	34.8	20.5	19.2	20.3	5.9	7.2	162.1	-0.7
1514	亞力	105.00	27.9	8.9	7.7	9.8	26.8	10.4	80.2	8.2
7792	安葆	411.00	4.7	6.2	6.8	6.9	1.4	49.0	51.6	24.6
產業小計				89.4	87.2	92.5	6.2	11.3	808.2	6.7

資料來源：CMoney

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2026Q4 PCB產業-
需求推動漲價，傳輸靠PTFE解方

福邦投顧 研究部
2026/10

26Q4結論-上游材料加速更新，漲價仍為主旋律

1.PCB上游:供不應求推動漲價動力，新材料應用下帶來新機會

- 玻纖布:E-Glass(高階)/LowDK-2/T-Glass 在2026Q4~2027年呈現全面緊缺，成為推動當前上游材料漲價的主要因素，也是決定結構性缺貨的關鍵原料，**玻布廠迎來產品組合改善+報價上揚雙重催化。**
- 銅箔:HVLP4加速驗證日台以外的供應商，中系擴產積極+中系設備加速採用下，**維持2027年供需缺口放緩預期。**
- 銅箔基板:PTFE樹脂為美系客戶解決互聯的確定性解方，成為銅箔基板材料中新晉路線；傳統的PPO/CH樹脂仍有一片天，26Q4 VR300 設計將定案，PTFE+M8/M9設計呼聲較高。
- 鑽針:高階鍍膜針需求攀升，供應商在2026~2028年加快投入建置產能，新材料(PTFE)仍有一定的鑽針消耗，**產品組合的改變遠比漲價來的重要。**

2. IC載板: 產能全數售罄，缺料與缺產能並行，關注供應商漲價態度。

- 供需缺口:維持2026/2027年 20%+/40%+ 缺口預期。
- 主邏輯為面積放大+層數增加，大廠產能售罄，客戶需要付出更高的價格預定產能+料源。

3. PCB:關注AWS、Google新品放量，以及2027~2028年HDI設計的變化，**聚焦Tier1供應體系和伺服器級HDI稀缺產能。**

4. 建議首選關注個股:

- 高階稀缺HDI產能:2368金像電、3037欣興、4958臻鼎
- 產品組合優化顯著:1303南亞、1815富喬、8021尖點、8039台虹
- 缺貨27~28年不緩解+AI Agent受惠:8046南電、3189景碩、3037欣興、4958 臻鼎

僅供福邦投顧客戶使用

重點載板、CCL公司獲利預估

股票代號	股票名稱	股本(百萬)	25EPS	26EPS(F)	27EPS(F)	28EPS(F)	景氣上行期PE
PCB板-國際同業評價P/E評價20~30X							
2368	金像電	5,306	19.02	41.2	82.9	99.9	20~30
載板-國際同業評價P/E評價20~30X							
8046	南電	6,462	3.01	14.5	55	85	20~30
3037	欣興	16,435	4.4	16.8	33	44.5	20~30
3189	景碩	5,268	3.5	13.7	27.1	42	20~30
4958	臻鼎	10,773	7.1	14.3	26	34	20~30
材料-國際同業P/E評價20~30X							
1303	南亞	79,308	0.51	15	3~4(材料) + 8(南電) + 12~20(南亞科)	N/A	20~25
1815	富喬	5,855	1.5	4.8	7.01	10	15~23
8021	尖點	1,473	2.7	10.2	25	36	16~25
8358	金居	2,526	4.2	12	17.2	27	19~25
2383	台光電	3,583	41	117	260	380	20~25
6213	聯茂	3,636	4.2	9.7	22.3	40	15~25
6274	台燿	2,993	11.8	40.2	85	115	20~25
8039	台虹	2,640	2.04	2.8	3(原業務) +15(PTFE)	3(原業務) +22(PTFE)	20~25
資料來源:福邦投顧整理預估(2026.09)							

僅供福邦投顧客戶使用

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【僅供內部教育訓練使用，嚴禁外流】

全年核心:政策主導AI為基本方針

- AI 成為美國政策上的基本方針，OBBBA法案刺激企業加大投資AI算力，AAAP計畫掃除建置AI算力的瓶頸和主權AI方針的確立。

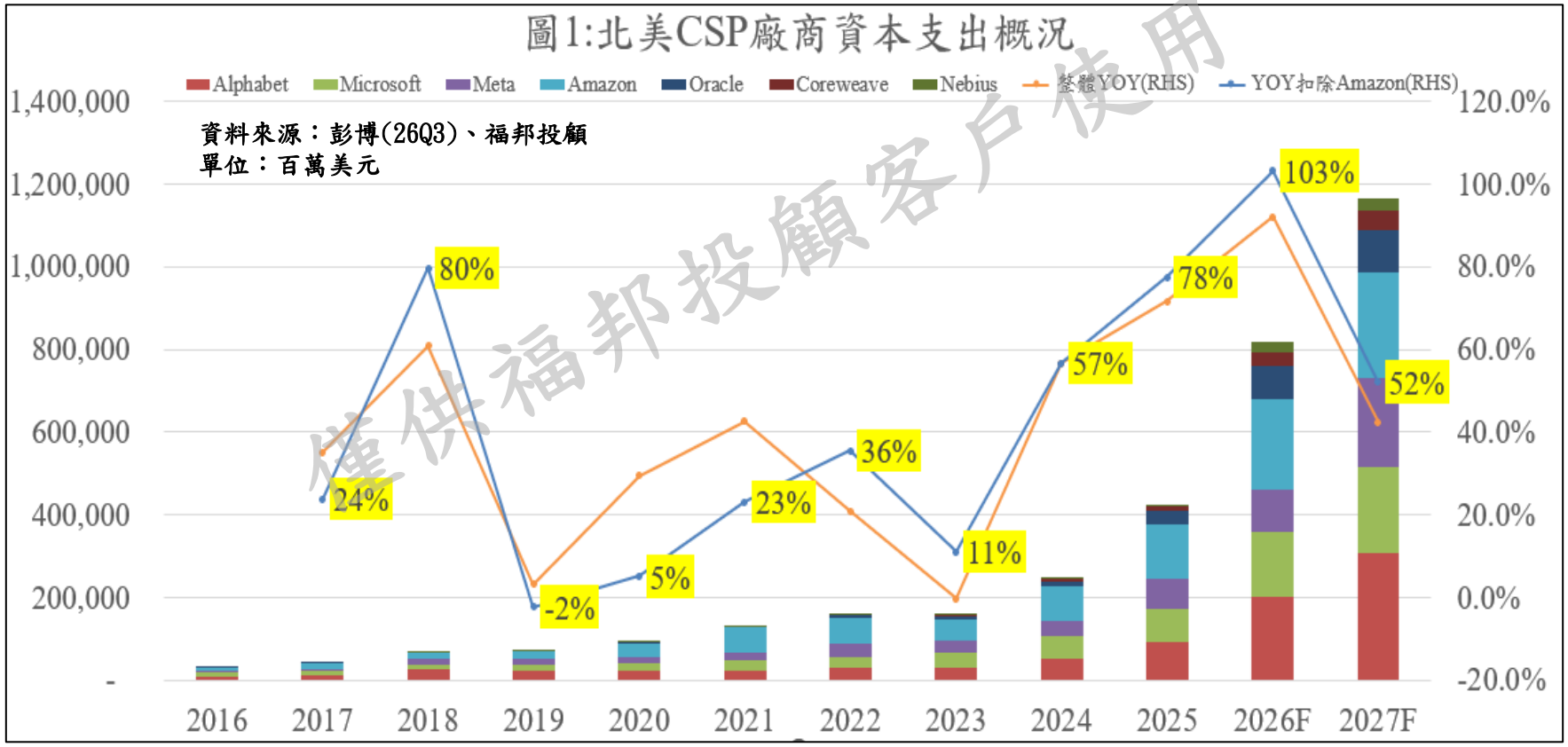
表1:OBBA法案和AI行動計畫　　資料來源:福邦投顧整理

法案	大而美法案(One Big Beautiful Bill Act)	AI行動計劃(America's AI Action Plan)
內容	1. 下調聯邦所得稅:從35%下調至21%，減輕企業稅賦負擔。 2. 恢復2017年減稅與就業法案項目中允許企業**當年資本支出可以全額抵減稅賦(折舊扣除)**，有利重資本支出廠商加快投資回收時間，受惠廠商如CSP大廠。 3. 國內R&D全額費用化:為了獎勵企業在美國進行生產研發，**符合條件的R&D費用可以於當年度全面費用化**，受惠廠商如Open AI、Anthropic等。	1. 加速AI創新:確保美國擁有全球最強大的AI系統，放鬆相關管制與障礙，並鼓勵開源等。 2. 增加AI基礎設施建置:提升美國AI算力基礎設施硬體基礎，以滿足AI推動經濟和安全的需求；加快關鍵項目的審核，**加快建設資料中心**、晶圓代工廠、電力基礎設施，掃除一切AI基礎設施擴充的障礙。 3. 維護國際外交與AI安全:在全球AI競爭中掌握規則制定權，輸出美國AI技術，同時阻止對手利用美國創新超越美國，嚴防高階算力流向競爭國家，向盟友出口美國AI算力技術等，有利**主權AI(Sovereign AI)**的發展。。

僅供福邦投顧客戶使用

雲端巨頭對資本支出態度維持積極

■2026年雲端CSP業者和算力租賃業者正加快擴建算力基礎設施，2026年預估資本支出成長上修至103%(含Amazon成長92%);2027年目前資本支出預估成長52%。



僅供福邦投顧客戶使用

全球伺服器出貨2026~2027年預期雙位數成長

■參考Digitimes最新數據預估，2026年預估出貨量YoY+17%，2027年目前預估11.2%。

圖2:全球伺服器出貨量成長預測

		2023	2024	2025	1Q26	2Q26	3Q26	4Q26	2026(F)	2027(F)
Gartner	出貨量	11,259	11,991	12,544					14,015	15,678
26Q2	YoY	-18.6%	6.5%	4.6%					11.7%	11.9%
	QoQ									
Digitimes	出貨量	14,432	14,888	16,597	4,699	4,877	5,015	4,896	19,487	21,670
26Q3	YoY	-20.4%	3.2%	11.5%	23.5%	26.0%	19.5%	9.0%	17.4%	11.2%
	QoQ				4.6%	3.8%	2.8%	-2.4%		
IDC	出貨量	12,357	14,461	16,779	3,773	3,975	4,261	4,421	16,430	18,083
26Q2	YoY	-17.4%	17.0%	16.0%	2.0%	1.9%	3.0%	0.2%	-2.1%	10.1%
	QoQ				-14.2%	5.4%	7.2%	3.8%		

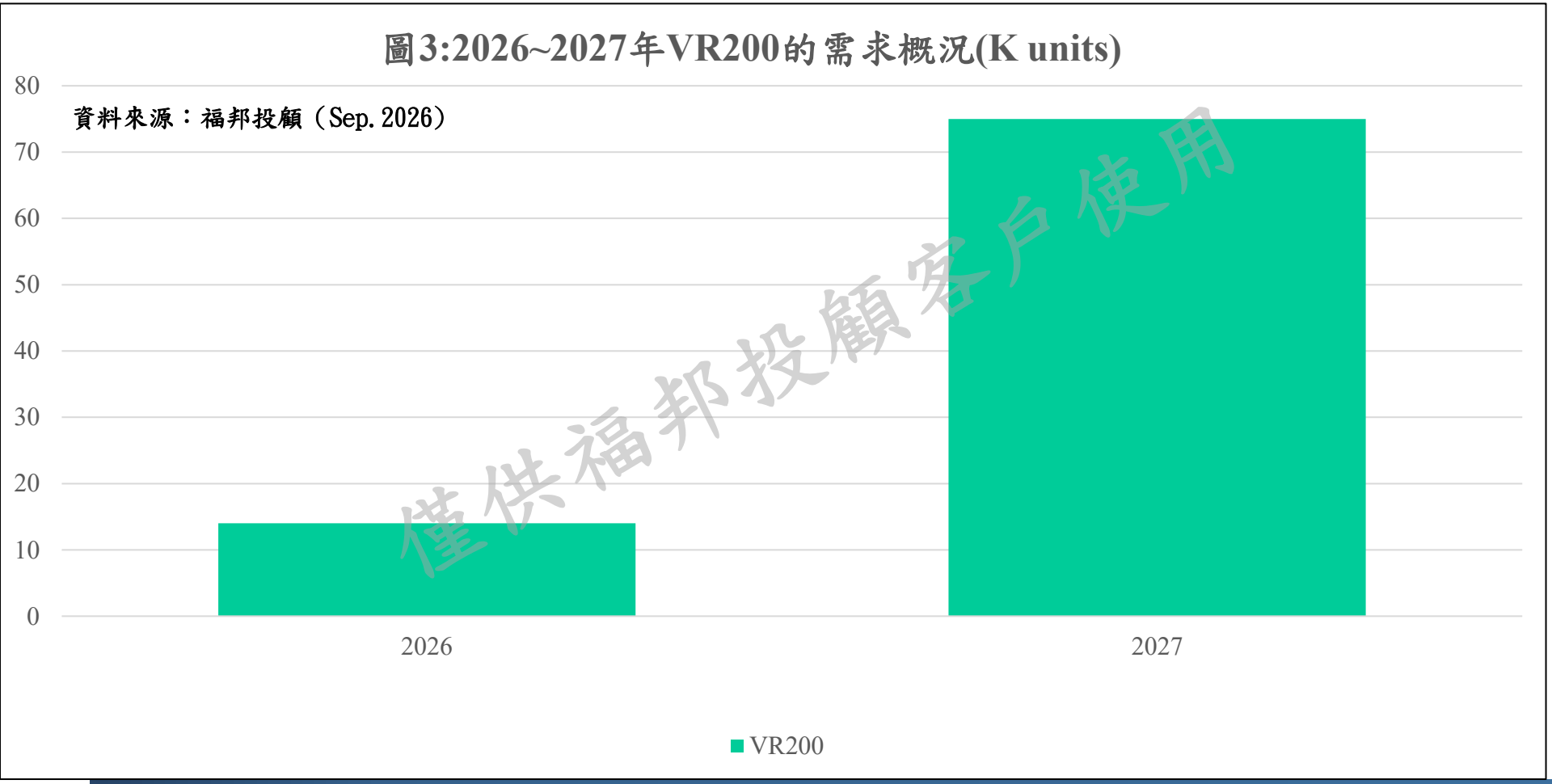
資料來源：福邦投顧整理（26Q2）

僅供福邦投顧客戶使用

GPU:26Q4~2027年VR200合計88K櫃

■根據供應鏈調查統計，2026年Rubin(VR200)預估組裝量約13~14K櫃，2027年預估組裝量約75K櫃。

■Rubin Ultra (VR300)於2027年預估需求量約5~6K組，2028潛在12K組(NVL576)。



僅供福邦投顧客戶使用

GB300/VR200/VR300規格:南亞、台虹為亮點

表2:GB200/300 VS. VR200　　資料來源:產業調研、福邦投顧整理

名稱	GB300 NVL72(72GPU)	VR200 NVL144(72GPU)	VR300 (RubinUltra)(9+9+9)
時間	2026放量	**2027放量**	**2028放量(26Q4 定案相關規格)**
晶圓製程	N4P	N3P	N3P
配合的PCIE版本	6	6	6
CoWoS製程	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝
單晶圓可切割數(顆)	14~15	8(包含I/O chiplet 面積增加)	8(包含I/O chiplet 面積增加)
TDP/散熱方式	1400W	2300W	1,800/2,600
載板供應商	Ibiden80~90% /欣興10~20%	Ibiden90~95% /欣興5~10%	Ibiden90~95% /欣興5~10%
CCL材料供應	計算板:斗山M8(LDK)+M4	計算板:斗山(M8+LDK+HVLP4)+M4	計算板:斗山(M8+LDK2+HVLP4)+M6
	交換板:M8(LDK2)+M2 生益科技(70%)台光電 (30%)	**交換板:M8(LDK2+HVLP4)** 生益科技/台光電/**南亞**	**交換板:M8(K2)/M9+PTFE** **生益科技+台虹/台虹**
		CPX:M8(K2)	CPX:M8(K2)
		中介板:M8(LDK2)	中介板:M9?
CCL上游材料供應	玻布:台玻、Asahi、泰山玻纖	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡
	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi
	HVLP3銅箔:三井/古河/盧森堡	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福
ComputeTray	5壓22L	5壓26L	**5壓30L**
PCB供應商	勝宏>欣興	勝宏>欣興>臻鼎 …定穎?(認證中)	勝宏/欣興/臻鼎/ 定穎(認證中)
NVSwitch Tray	22L HLC	32L(N+N HLC)	**32L(N+N HLC)**
PCB供應商	VGT50%/TTM40%/景旺10%	VGT/WUS/TTM	VGT/WUS /TTM/深南?/景旺?
CPX		Delay(需求不佳)	Delay(需求不佳)
PCB供應商			
小中介板(Midplane)		44L(N+N HLC)	44L(N+N HLC)
PCB供應商		VGT/景旺/滬電/欣興	VGT/景旺/滬電/欣興

僅供福邦證券客戶使用

雲端廠商ASIC相關規格-高階材料/製程加快導入

表3:雲端大廠&LLM廠商方案　　資料來源:產業調研、福邦投顧整理預估

客戶	Design Service/ Code Name	時間	主板 設計規格	L6	L10~L12	CCL Supply	Compute Board PCB Supply
Google	Broadcom - TPU v7 (Ironwood)	26H1	22/24L	Celestica80%/ Jabil 20%	Celestica /Flex/ 鴻海(5%)	M7-Panasonic (80~90%)/EMC	主:ISU/LCS/WUS/TTM/VGT
	Broadcom - TPU v8 Sunfish	**26Q4**	34L			**M8(LK2)-** EMC/松下	主:WUS/TTM/LCS/VGT
	MTK - TPU v8 Zebrafish	**27Q1**	34L			**M8(LK2)-** EMC/Doosan/松下	主:ISU/WUS/TTM/VGT/ **GCE**
	MTK- TPU v9 Humufish	**28Q2**	**6壓28L**			**M9+M8(LK1)** EMC	主:ISU/VGT/WUS/TTM/**欣興** 備用:**臻鼎/LCS**
Amazon	Marvell - AWS Trainium2e (Cayman)	26H1	26L	智邦(氣冷) / 緯穎(水冷)	緯穎 /Jabil/Flex	M8(LK)-EMC/TUC	GCE/SY/FHT
	Annapurna - Trainium3 (Mariana)	26Q2	主: 26L(氣) 40L(水) SW:22L			M8(LK)- EMC(氣+水)/ TUC(氣)	**ComTray:GCE+**SY **(氣+水)** /WUS(水)/**FHT(氣)** **Swtray:GCE/**SY/**ZDT/FHT**
	Annapurna - Trainium4 (Maverick)	27Q4	**主:3/4壓34L SW:30L**			**M9-** EMC/南亞?	**GCE/**SY/**ZDT**/Meiko
Meta	Broadcom - MTIA T V1.0 (Minerva)	25H2	40L	廣達/ Celestica	廣達	M8(LK)- EMC	WUS/TTM/ISU
	Broadcom - MTIA I V1.5 (Iris+ / Santa Barbara)	26H2	36L	廣達 or Celestica	廣達 or Celestica	M8(LK)- EMC	WUS/GCE/ TTM/ISU
	Broadcom - MTIA V2 (Olympus / Santa Cruz)	28E	?	廣達 or Celestica	廣達 or Celestica	**M9-** EMC	WUS/GCE/TTM/ ISU/DY?
MSFT	Marvell-MAIA 300 (Griffin)	27H1	40L+	廣達	廣達	**M9-** EMC/Doosan?	GCE/TTM/WUS/大德/ISU
AMD	MI4系列	27H2	48L	緯創	Celestica、緯穎	M8(LK2)	SCC(90%)/欣興/大德
	MI5系列	28H2	52L	緯創	Celestica、緯穎	M9	

僅供福邦投顧客戶使用

鑽針主邏輯—PCB廠積極擴產

- PCB設計日益複雜，耗用大量產能：過去伺服器板層數約16~2X層，目前主流已經來到20~30層，未來30~50層越顯常見。
- PCB用量增加，耗用大量產能:隨著各家設計陸續參考NVIDIA Rack設計，除了原有的主板(GPU/ASIC 使用)，亦新增更多副板，例如Switchboard、midplane等，終端客戶積極驗證新供應廠商。

表4:全球主要GPU/ASIC陣營PCB供應商擴產概況

億元	產值/產能	2025	2026(F)	YoY	2027(F)	YoY	2028(F)	YoY
勝宏科技	多層板(M^2)	28K	120K	+328%	350K	+190%	590K	+69%
	HDI(M^2)	130K	320K	+146%	640K	+100%	680K	+6%
金像電	sqf	6,800K	10,000K	+47%	11,000K	+10%	12,000K	+9%
ISU Petasy	(M^2)	240K	280K	+16%	320K	+14%	360K	+13%
滬電股份	(M^2)	160K	210K	+30%	340K	+62%	440K	+29%

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-CCL材料影響

■ PCB用鑽針受到板材影響，在同樣板厚下，每針可鑽孔次數大幅縮水:

- M6~M7時主針鑽孔壽命約600~800次
- M8材料下可鑽400~500次
- M9材料下僅剩下100~200次
- 材料越高階，鑽針使用量會是過往的6倍以上。

表5:鑽針基於不同板材的使用壽命&各應用對應的鑽針方案

	過去	目前	未來
對應材質	M6~M7	M8/(有布/無布+球型矽粉)	M9~M10
主針壽命	1000/600~800次	400~500次	100~200次

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-長徑比帶動報價上揚

- 當前主流GPU用PCB約5mm厚度，隨著AI用PCB板厚的提升，每增加1mm，製造上可能會採用增加一根預鑽針，採用預鑽針預鑽至特定深度，後續用主針完成加工。
- GPU伺服器:GB200採用「1主針配2白針」→GB300採用「1主針配3白針」→VR200採用「1主針配4白針」，

表6:鑽針各應用對應的鑽針方案　　資料來源:產業調研、福邦投顧整理預估

	GPU Server	ASIC Server	交換機
對應鑽針方案	1主針 +1~4根預鑽白針	2主針 +1根預鑽白針	3~4根主針 +1~2根預鑽白針

- AI用的鑽針，長度/直徑比超過30規格的鑽針，報價上會採用**一個方案一個價格**，因為加工精度要求提高，難度變大；目前AI用的鍍膜鑽針報價較傳統白針多20~50%，若是這種報價方式，特別是板厚超過7mm，產品售價將會多40~50%；未來約在2027~2028年時，7.5~8.5mm的鑽針將成為主流，報價將會是一般白針的6倍以上報價。

表7:鑽針不同長徑比的報價倍數　　資料來源:產業調研、福邦投顧整理預估

長徑比	一般(10~20)/鍍膜	30	40	50
售價倍數	1/1.2~1.5	3~4x	6~8x	12~16x

- **毛利率方向:AI用的鍍膜鑽針鑽針，根據產業訪查，產品毛利率約50%+**。

僅供福邦投顧客戶使用

全球主要鑽針廠亦積極擴充產能，供需維持吃緊

- 目前全球主要伺服器PCB廠採用的鑽針廠，高階鑽針以中鎢高新旗下的金洲科技和尖點為主、中階以下以鼎泰高科為主；日本佑能主要用於載板生產用針。
- 鑽針廠的外溢效果:由於前四大廠優先用於AI伺服器相關生產用針，使得中低階針有排擠效應，白針類產品外溢給凱崴等廠商。

表8:全球主要鑽針廠產能/客戶概況　　資料來源:產業調研、福邦投顧整理預估

公司	產能(萬支/月)	高階鑽針占比	主要客戶	規劃
鼎泰高科	1.2億支	>40%	深南、景旺、勝宏、健鼎、生益、方正、廣合、定穎	2026年:1.8~2億支/月
中鎢高新(金洲)	9,000	>15%~30%	勝宏、深南、高技、生益、景旺等((**中系高階主供**)	2026年底至1.2億支/年 2027年新增3600萬支+/年 2028年新增1.3億支/年
佑能	4000	>60%	專注日本、台灣市場、載板廠	2026年底至5,000萬支/月 2027年至6,000萬支/月
尖點	3,500(50%以上高階鍍膜針)	>50%	金像電、ISU、勝宏、TTM、滬電、臻鼎、欣興等	2026年:4,500萬支+/月， 2027年:˙7,000萬支+/月 2028年達到9,000萬支+/月

僅供福邦投顧客戶使用

HVLP4銅箔需求受良率影響，需求驚人

- 根據供應鏈調查，依照良率80~90%的假設下，Nvidia Rubin的需求約300~400噸/、LPU約200~250噸/月，Amazon AWS 800~850噸/月，Meta MTIA 200噸/月，AMD MI450 100~200噸/月，800G交換機 400噸/月，合計**1,200~1,300噸/月/2,700~2,800噸/月。**

- 若依照實際PCB生產狀況，良率約在60~70%不等，總需求可達**1,400噸/月/3,000~3,100/噸/月。**

表9:全球主要HVLP需求估算與供應鏈格局

	HVLP2	HVLP3	HVLP4	HVLP5
加工費	USD15/kg	USD20~25/kg	25~30美金/kg	35+
供應商	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田
客戶	AWS T2 (2024-2025年)	NVIDIA GB200 NVSW	AWS T2MAX(2026年)、Meta M9 896K2/K3、NVIDA Rubin、Google V8、AMD MI450、800G Switch	Panasonic M9
	Google V5/V6	Google V7		SY5300N
需求量	AWS 300-400噸/月	NV 38~40噸/月	AWS800~850噸/月(100萬張/月)	
		Google v7	Nvidia 300~1000噸/月	
			800G 交換機 400噸/月	
			Meta 20噸/月	
	資料來源:產業調研、福邦投顧整理預估		AMD 20噸/月	
			Google Zebra&Sun 80噸/月	

僅供福邦投顧客戶使用

全球主要供應商概況:加快導入新供應體系

表10:全球主要HVLP供應鏈格局

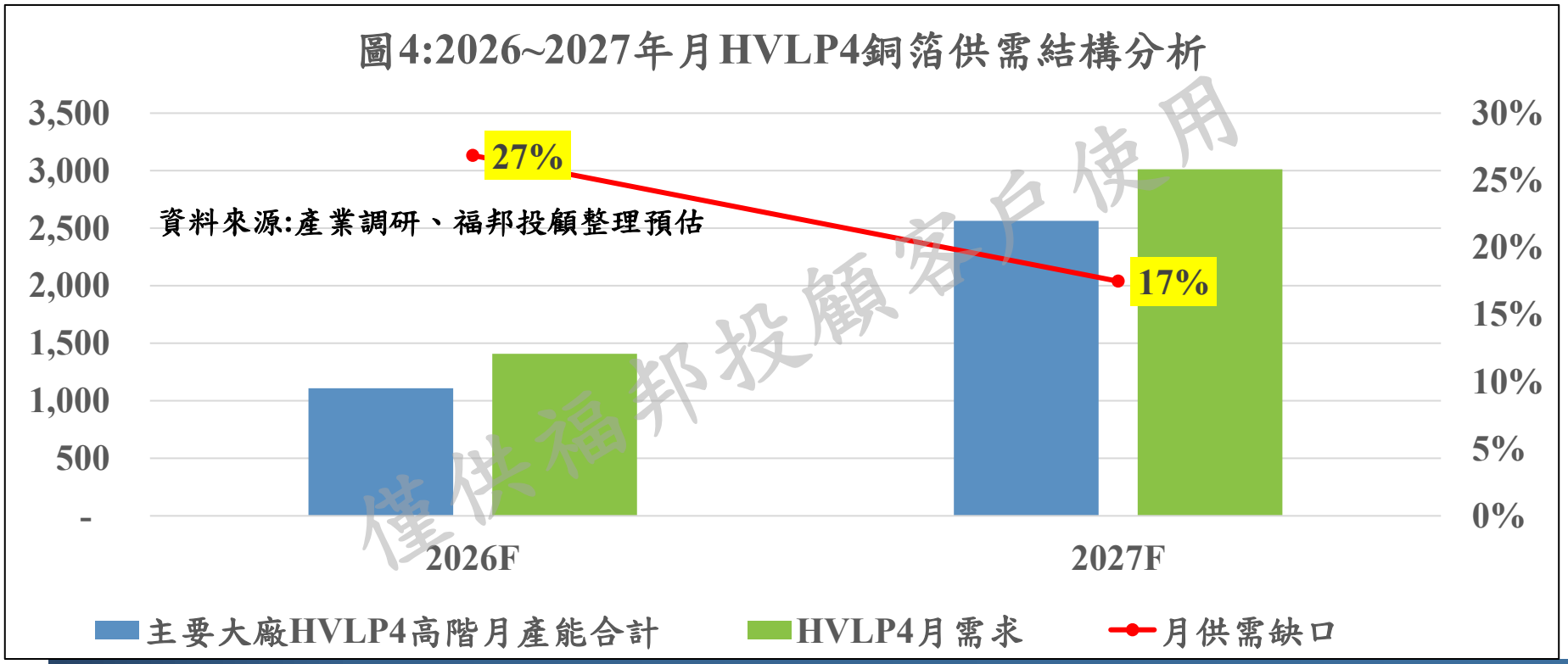
資料來源:產業調研、福邦投顧整理預估

	地區	產品分布	關鍵客戶	關鍵終端客戶	高階產能規劃
三井金屬	日	HVLP1~5	日:松下 台:台光電、台燿 中:生益	Nvidia/Google/AWS等	26Y:550噸/月 27Y:1,000噸/月 28Y:1,200噸/月 30Y:1400噸/月
古河電工	日	HVLP1~4	日:松下 台:台光電	Nvidia/Google/AWS等	26Y:100~200噸/月 27Y:200~300噸/月
福田金屬	日	HVLP1~5	日:松下 台:台光電、台燿、南亞 中:生益	Nvidia/Google	26Y:100~200噸/月
盧森堡	歐	HVLP1~4	韓:斗山 台:台光電、台燿 中:生益	Nvidia/AWS	26Y:100~200噸/月
金居	台	HVLP1~4	韓:斗山 台:台光電 中:南亞新材?	NVidia	26Y:330噸/月 27Y:700噸/月 28Y:800噸/月
德福科技	中	HVLP1~4	日:松下 中:生益 台:聯茂、台光電	NVidia/Google	26Y:200噸/月 27Y:500~600噸/月 (潛在1,200噸)
銅冠銅箔	中	HVLP1~4	中:生益、南亞新材 台:台光電、台燿	NVidia	26Y:50~100噸/月 27Y:200噸/月
長春	台	HVLP4	台:台光電	Nvidia/AWS	27Y:200噸/月
Lotte(樂天)	韓	HVLP2~4	韓:斗山	NVidia	26Y:300~350噸/月 27Y: 900~1,000噸/月

僅供福邦投顧客戶使用

銅箔供需缺口往2027年將縮小，供應廠商增加

- 根據供應鏈調查，由於供需缺口明顯，主要CCL廠商(台光電、生益科技等)積極導入新的供應體系，2026年主要生產廠商擴產後，可達1,200~1,300噸/月，2027年預估達2,300~2,400噸/月。(供應體系:台銅/古河/盧森堡/金居/福田/德福/銅冠/樂天)、台系長春後續有望驗證進入供應體系。
- **根據福邦投顧對於供需格局的假設預估，基於客戶加快認證相關可用產能，2026~2027年的供需缺口更新預估為27%(下修)/17%(上修)。**



僅供福邦投顧客戶使用

玻纖布使用種類

表11:玻纖布使用種類彙整

應用		元件	基板類型		材料等級要求	玻布類型	
						高階	中階
資通訊	AI伺服器/資料中心/交換機/基地台等	處理器	**封裝載板**	**CPU/GPU/ASIC**	**Low CTE**	**T**	**E**
				NAND	**Low CTE**	**T**	**E**
				DRAM	**Low Dk**	**NE/NER**	**E**
		主板	伺服器(AI/一般型)		Low Dk	NE/NER/Q	E
			交換機(核心/AI)		Low Dk	NE/NER/NEZ/Q	E
			網卡(NIC)		Low Dk	NE/NER	E
邊緣端	手機/平板等	處理器	封裝載板	AP/CPU	Low CTE	Ultra-Thin T, T	Ultra-ThinE
		非動態記憶體		NAND	Low CTE	Ultra-Thin T	Super Ultra-Thin E
		動態記憶體		DRAM	Low CTE	Ultra-Thin T(行動)	
				DRAM	Low Dk	NE(PC)	
		主板	主板		Low Dk	Ultra-Thin NE	Ultra-ThinE
		無線通訊	RF 封裝載板		Low Dk	Ultra-Thin NE	Ultra-ThinE
	DT/NB/AI PC	CPU/記憶體	封裝載板	CPU/GPU/NPU	Low CTE	T	E
		主板		DRAM	Low Dk	NE(PC)	E
	AR/VR	先進SoC	封裝載板		Low CTE	T	Super Ultra-Thin E
車用	EV/ADAS	先進SoC	封裝載板		Low CTE	T	E
		mmWave 雷達	模組載板		Low Dk	Ultra-Thin NE	E

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

高階電子布供應商概況:台光電開始取得日東紡料源

表12:電子玻纖布供應商與其主要客戶

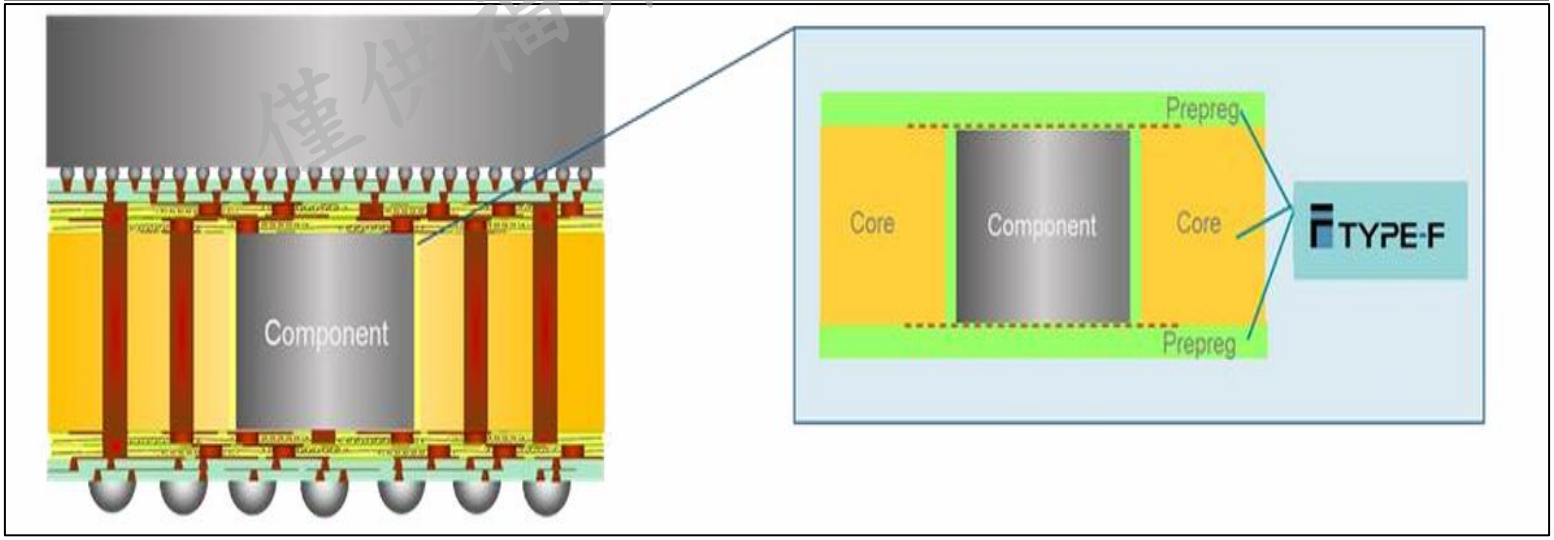
廠商	地區別	供應電子布產品	主要客戶
Nittobo	日	E/lowDK/LowCTE	• LowDK:松下、Doosan、TUC、南亞、EMC • LowCTE:Resonac、MGC、松下
Asahi	日	E/lowDK/LowCTE/Q	• LowDK:EMC、Doosan • Q:EMC、Doosan、松下
台玻	台	E/lowDK/LowCTE	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC
德宏	台	E/Q	• Q:EMC、南亞、聯茂
富喬	台	E/lowDK/LowCTE	• LowDK:EMC、Doosan、TUC、Asahi(紗) • LowCTE:Asahi(紗)
南亞塑膠	台	E/lowDK/LowCTE	• LowDK:南亞 • LowCTE:Resonac、MGC、南亞
泰山玻纖	中	E/lowDK/LowCTE/Q	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC • Q:EMC、生益科技、Asahi(紗)
光遠新材	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技
宏和科技	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技 • LowCTE:Resonac、MGC、EMC
建滔積層板	中	E/lowDK/T	• LowDK:EMC、生益科技
菲力華	中	Q	• Q:生益科技、EMC、松下、TUC
國際複材CPIC	中	E/lowDK/LowCTE	• LowDK:生益科技

僅供福邦投顧客戶使用

T-Glass:客戶積極導入抗翹曲設計

- 下個世代的半導體封裝需要更大的面積來放置更多元件、支持2D~3D封裝(CoWoS~SOIC)，因此材料的要求需要Low CTE、高彈性變量來達到低翹曲度 (Warpage)和連接可靠度。
- 根據Resonac白皮書，為了解決內埋元件載板對於核心層的厚度精度要求，較薄的核心層或不均勻的厚度，會導致內埋元件的損壞，因此之後許多載板核心的設計會採用厚Core+ low CTE 膠片 (採用T/Q Glass)。
- **2026年的TPU，2027年的Maia300都有見到這類核心Core CCL+PP的設計方案(Multilayer Core)，此也會多消耗T-Glass的供應量。**

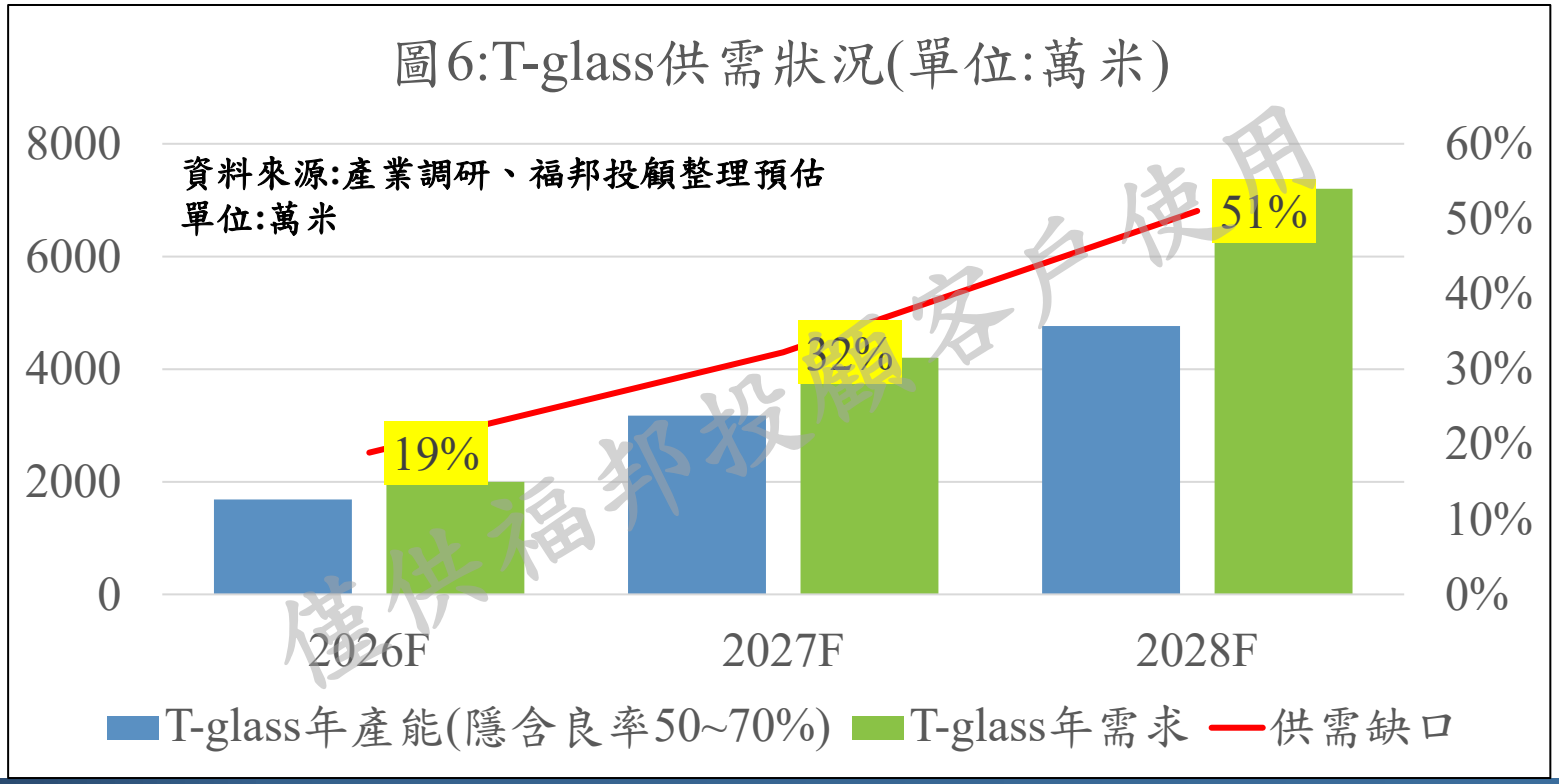
圖5:Resonac Type F產品



僅供福邦投顧客戶使用

T-glass至2027年仍將呈現供不應求格局，缺口30%+

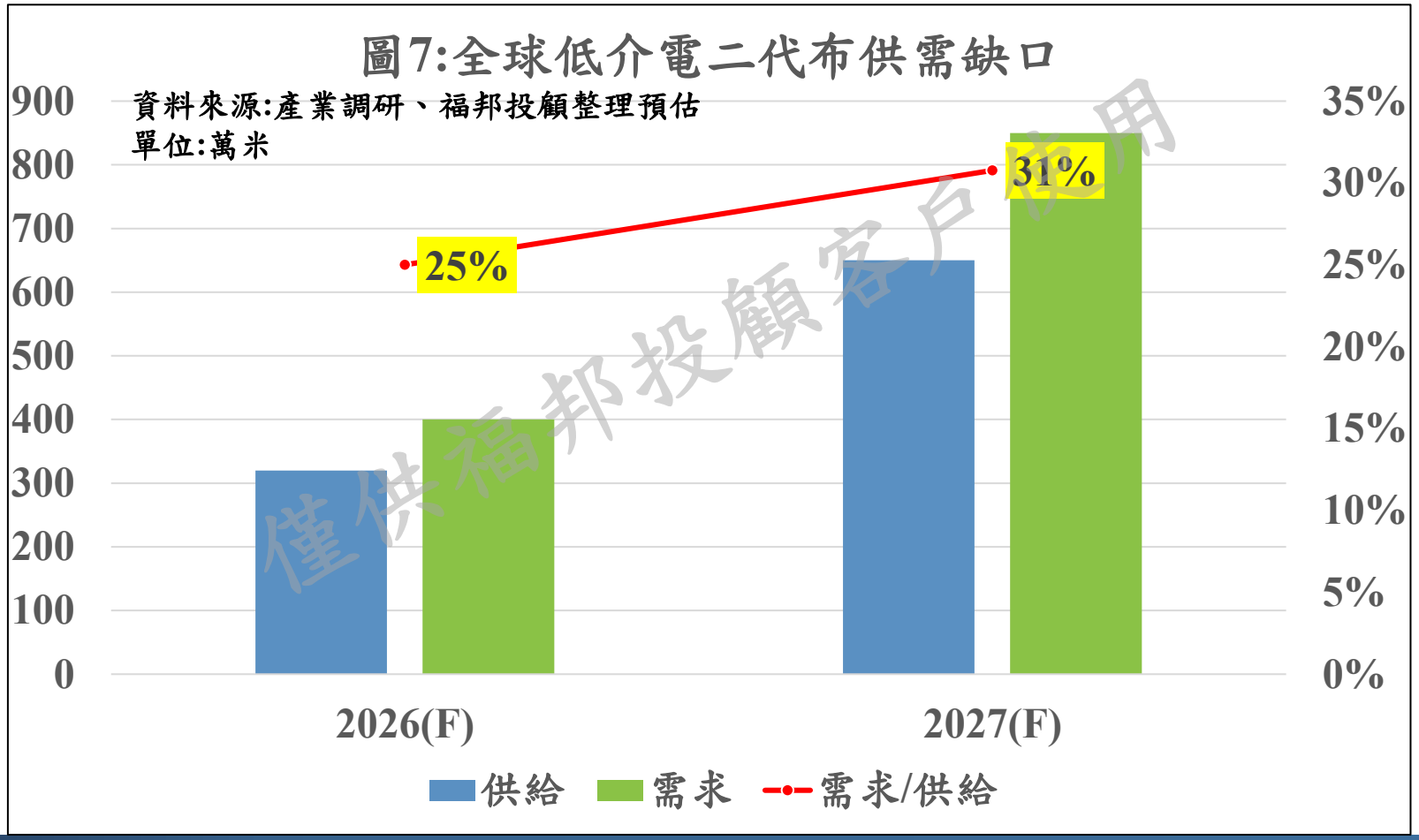
- 目前需求來看，T-glass需求中有70%用於FCBGA用的ABF載板，30%用於手機等FCCSP用的BT載板上。
- T-glass產能劃分來看，日東紡佔有全球主要產能的40%，台玻約15~20%，宏和約20~30%，其他(包含泰山玻纖等)15~20%。
- **福邦投顧預期儘管日東紡於2027年陸續開產能出來，但只緩解了Intel/AMD/Nvidia的用料，對於ASIC陣營的Broadcom/Marvell/MTK/Annapurna而言，仍難以取得足夠料源，維持T-Glass持續供不應求的判斷，缺口仍會逐年往上。**



僅供福邦投顧客戶使用

LowDK Gen2 2027年預期缺口31%

- 根據供應鏈訪查，2026H2開始，低介電二代布需求約400~500萬米/月，其中主要需求來源是Nvidia VR200、Google V8T/V8I。
- 以其銅箔基板廠商供應來看，主要以台玻、泰山玻纖、國際複材、Asahi為主，不足部分將由富喬、光遠、Nittobo補足，2027年供應量預估達650萬米/月以上。



僅供福邦投顧客戶使用

Q-Glass:LPU產品將率先採用

- 目前最新的NVIDIA LPX Rack從原先的全M9Q設計，改採用M8(K2)+M9(Q)的設計，配置為1:1，其中Q Glass主要採用10系列的極薄布，供應廠商少。
- 根據產業調查，預估NVIDIA LPU產品需求約在12~13K(2026~2027Y)，潛在Q glass需求約110萬米/月，隱含良率損耗則需要210~220萬米/月。

表13：LPU 主要供應資訊

	LPU Rack
PCB設計規格	52L(26+26)
Q-Glass供應商	菲利華、泰山玻纖、德宏
HVLP4供應商	三井、古河、盧森堡、金居
M9 CCL供應商	生益>台光電
PCB供應商	勝宏、Wus、TTM(30%)
組裝	鴻海

資料來源:福邦投顧預估

表14：LPU 供應試算

LPU 產品試算(單櫃)	
Tray 數量 per Rack	32
PCB/Tray	1
LPU/PCB	8
LPU總量(顆)	256
CCL 用量/PCB	8
總用量	256
M9Q/張 ASP(TWD)	10,000
CCL總產值(TWD) per Rack	2,560,000
CCL ASP per PCB (USD)	2,667
PCB ASP (USD)	4,444
Q布 ASP per PCB (USD)	1,440
Q布 ASP (USD)	45
Q布用量(米) per PCB	32

僅供福邦投顧客戶使用

Q-Glass：供應者寡，關注非日系供應

- 根據產業調查，統計台系(1家)、日系(2家)、中系(2家)產能規劃，2026年底預估產能超過300萬米/月，2027年預估產能可達600萬米/月以上。

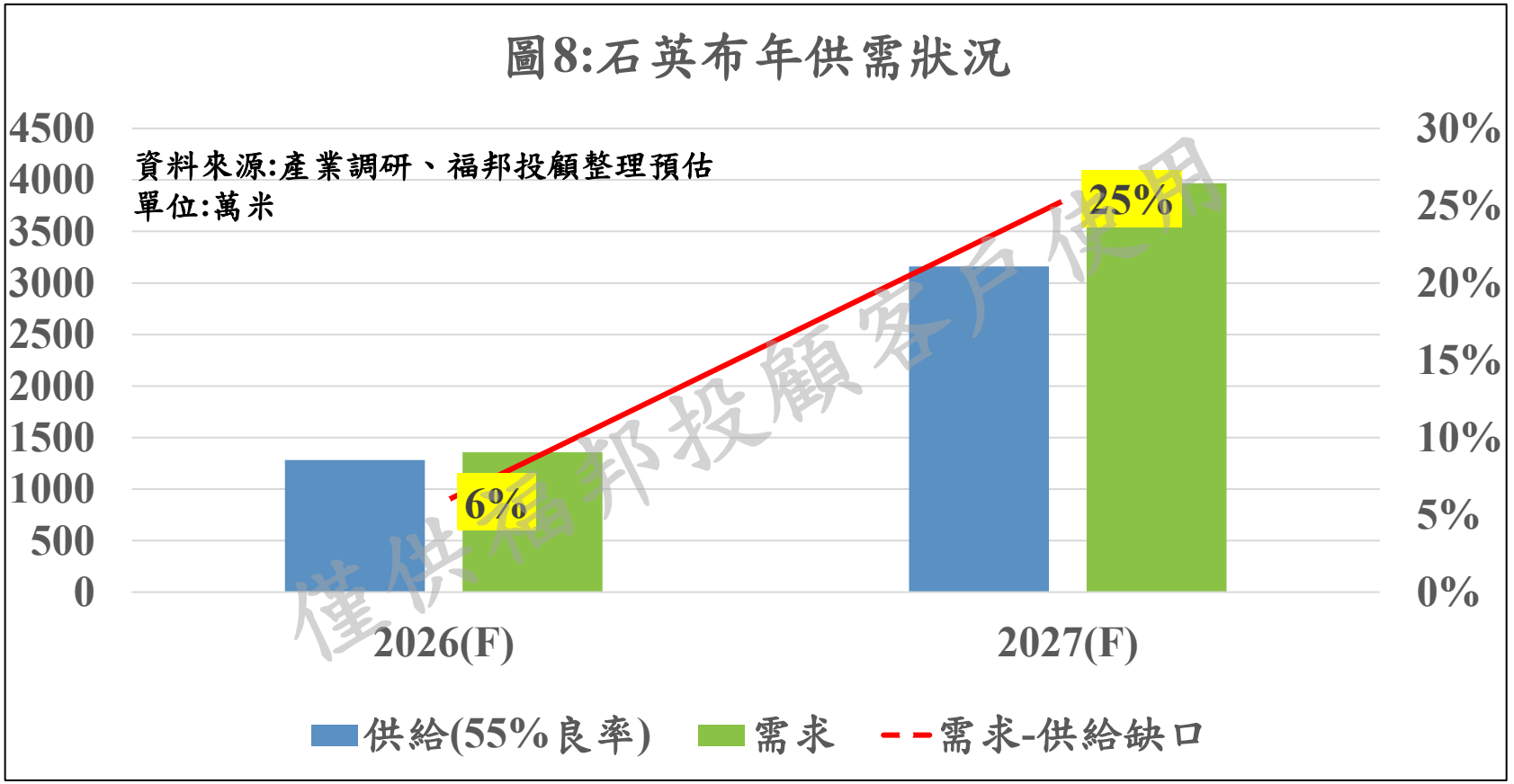
表15:Q-Glass供應商概況　　資料來源:福邦投顧整理

公司	地區	關鍵客戶	產能(K米/月)		說明
			2026F	2027F	
信越化學	日	日:松下、Resonac、MGC 台:台光電	75	240	外觀與同業不同，影響採用意願
Asahi	日	日:松下、Resonac、MGC 台:台光電	50	50	大客戶優先指定選用
泰山玻纖	中	台:台光電 中:生益科技	1,000	1,500~2,000	• 優先供應台光電
菲利華	中	日:松下 中:生益科技 台:台光電、台燿、南亞 韓:斗山	960	1,500~2,000	• 70%+供應生益科技 • 26H1 30W米/月 • →26H2 90~100W米/月
德宏	台	台:台光電、南亞、台燿(?)、聯茂 韓:斗山(?)	1,000	1,500~2,000	• 優先供應台光電 • 26H1 60W米/月→26年底100W米/月

僅供福邦投顧客戶使用

Q-Glass:26客戶進入庫存建立期，缺口下修至25%

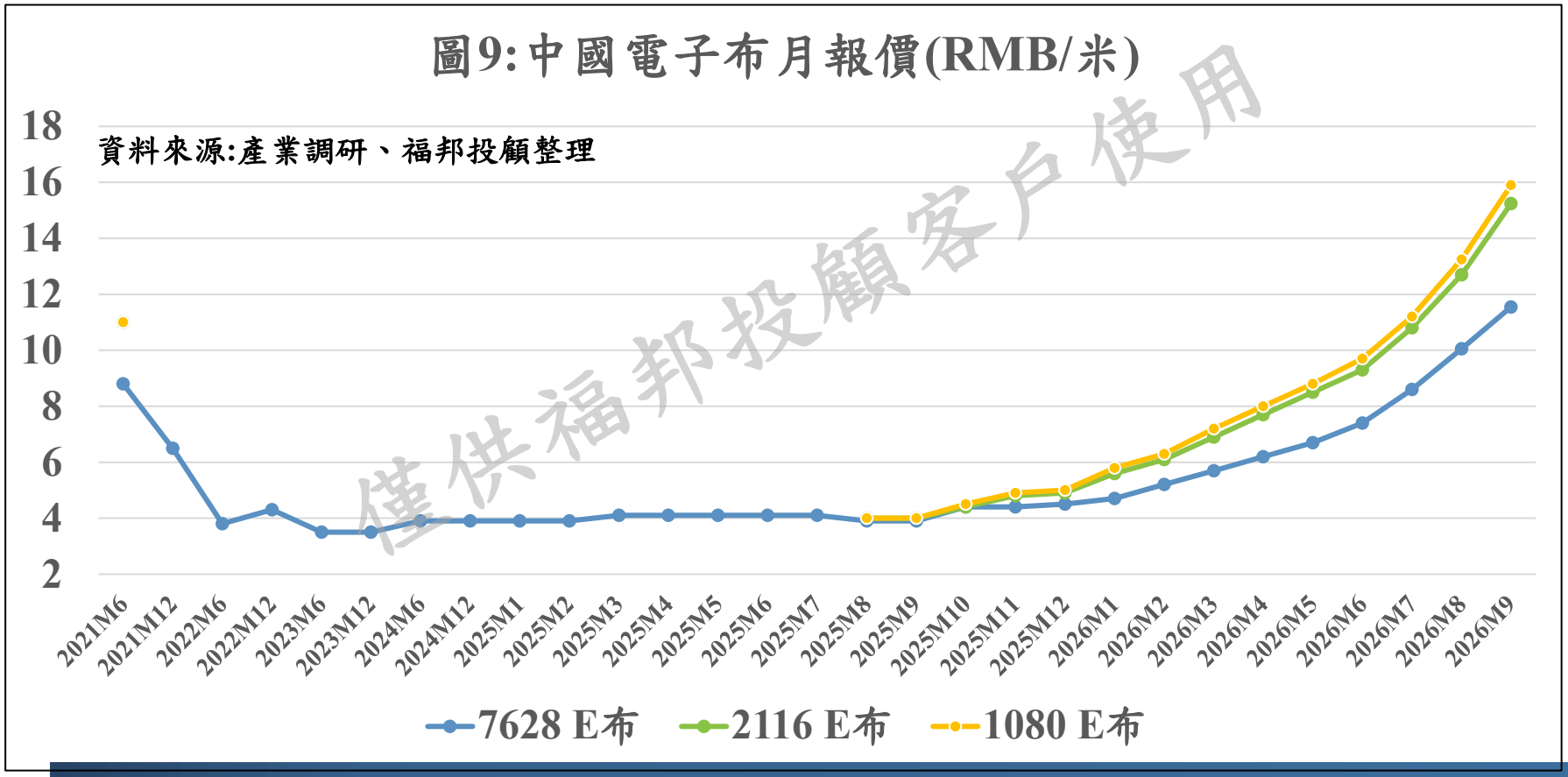
- 目前供應鏈的良率約在55~60%，26H2 進入Nvidia LPU 產品的備貨期，將帶動銅箔基板廠商的備貨需求，其中以**生益科技需求最為迫切，台光電供過於求。**
- ASIC陣營態度:目前Meta選用M9Q進行打樣，Amazon/MSFT選用M9-(K2)進行主板打樣，Google TPU V9至少選用M9以上材料。



僅供福邦投顧客戶使用

價格:E glass漲幅驚人，2H26將續漲至創新高價

- 玻纖布於2021~2022年週期係受到需求端推動帶動漲價，以7628系列為例，報價達到8.8RMB/米，本輪低階起漲，除受原物料之外，也受到主要供應商陸續轉做高階玻布(283，排擠低階布生產(打破供需平衡情況)，進而帶動漲價。
- 7628布累計至26Q3 YTD150%+，2116/1080 布YTD 210%+，均已創下歷史新高，2027年2116/1080的規格將繼續延續漲價趨勢。



僅供福邦投顧客戶使用

價格:高階電子布漲價，漲勢可望持續至2027H1

- 高階布種由於生產效率較傳統布下滑，以及良率因素影響，配合需求持續攀升下，每年均有約20~30%的漲幅，考量缺貨持續至2027年，漲價趨勢將延續至27H1。

表16:玻纖布漲價幅度

產品	目前ASP(USD/米)	2025	2026
LowDK	6~8	+20~30%	維持全年20~30%漲幅預期
LowDK2	18~22	+20~30%	• 26H1 YTD 漲幅+30~50% • **26Q3 漲幅+10%**。
T-Glass	23~30	+20~30%	• 日東紡26H1 YTD漲幅30% • 中系廠商26H1漲幅20% • **26H2預期會有10%+漲幅**。
Q-Glass	40~45	0%	26H1漲幅10~20%

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

銅箔基板漲價循環直至2027H2

- 目前AI相關CCL交期延長至4~6個月，上游料況吃緊（庫存1~2周），**福邦投顧預期考量上游材料漲價循環至少漲至27H1，將推動下游CCL客戶漲價續漲至2027年。**

表17:主要CCL廠商漲價幅度　　資料來源:產業調研、福邦投顧整理

公司名稱	地區	過去產品漲價幅度	26H2漲幅預期
Panasonic	日	• 26M5 高速CCL漲20%、FR4漲30%、 • PP漲15%	• 26M8 高速CCL漲10%、FR4漲15%、PP漲15%、載板材料漲5%
Resonac	日	• 26M3 CCL漲30%	
MGC	日	• 26M4 全品項漲40%	
台光電	台	• 26M3全品項漲15%+，M6以下漲20~25% • 26M4 M7以上漲5~10%	• 26Q3 M7以上漲10%。
台燿	台	• 26M4 漲幅20~40%。	• 26Q3 M7以上漲15%，M6以下上漲25%。
南亞	台	• 26H1 YTD+30%。	• 26Q3漲幅30~40%；26Q4會再繼續調漲
聯茂	台	• 26H1 YTD+40%+(4/6月)。	• 26H2每季預計漲20%+
建滔積層板	中	• 26H1累計五次漲價函，YTD 漲幅達50%+	• 26Q3漲幅30%+
生益科技	中	• 26M5 高速CCL /PP漲20%	• 26Q3漲幅30%+

僅供福邦投顧客戶使用

【僅供內部教育訓練使用，嚴禁外流】

Rubin Ultra帶起的材料變化-PTFE的方案發展

- 由於傳輸的需求(337G)，PPO或HC系列樹脂的M9無法滿足NVIDIA要求，NVIDIA 在Rubin Ultra 將導入混壓PTFE材料應用於SW Tray、Feynman 將導入純PTFE板材應用於正交背板。
- 目前銅板基板廠商正在積極送樣測試純無布的方案因應客戶需求。

表18:PTFE 方案組成　　資料來源:產業調研、福邦投顧整理

	PTFE CCL(無布) +CH PP(有布)	PTFE CCL(無布) +CH PP(無布)	PTFECCL +PTFE PP(無布)
樹脂配方	PTFE	PTFE	PTFE
銅箔搭配	HVLP4/5	HVLP4/5	HVLP4/5
供應商	三井金屬SI3-VSP		
電子布	LowDK 4(二代Q)	X	X
膠片配方	CH樹脂	CH樹脂	PTFE樹脂
供應商	生益科技、台光電、聯茂	生益科技、台光電	生益科技、台虹
電性	優於M10(CH樹脂系列)	次之	最佳
加工性	最佳	次之	最差
評論	備案，加工性較佳	供應鏈目前積極測試的方案	最終目標
PCB廠加工良率	30%	10~30%	10~30%
CCL供應商	走FCCL製程，生益科技SG5300N(塗佈+熱壓法)、台虹(熱壓法)		

僅供福邦投顧客戶使用

無布VS.有布方向選擇

- PTFE 由於Dk/Df的性能優於電子布，如果PTFE搭配電子布，會導致其Dk/Df轉差。

表19:PTFE與電子布的性能比較

	PTFE樹脂	LowDK 電子布	LowDK 3~4(1/2代Q-glass)
Dk(10Ghz)	~2.1	4.6~4.8	3.7~3.8
Df(10Ghz)	~0.0002	0.002~0.003	0.0002~0.0003

- 未來走向:

1. 有布方案(M9~M10搭配LowDK2~4電子布):若PCB上須放置CPU/GPU/ASIC等晶片，考量尺寸安定性因素，會需要搭配具有電子布的銅箔基板，例如UBB、ComputeBoard、SwitchBoard、MotherBoard。
2. 無布方案(搭配無布膠片):若僅考連傳輸性能(不放置晶片)，目前採取的無布方案，樹脂填矽粉/球型矽微粉的方案為主流，以改善膨脹性。

表20:PTFE的CTE比較

	PTFE樹脂	球型矽微粉	LowDK 3~4(1/2代Q-glass)
CTE(ppm/°C)	100~200+	0.4~0.6	0.5~0.55
說明		X/Y/Z軸的保護	X/Y軸保護

僅供福邦投顧客戶使用

不同填料對於銅箔基板的影響

- 在樹脂中加入填料能降低熱膨脹係數(CTE)、提高尺寸安定性和降低Dk/Df，近年的填料主要以球型粉料為主。

表21:各類填料的對應用途

填料	用途	相關填料供應商
二氧化矽粉(Si02)	降低CTE	日:電氣化學、Admatechs、龍森、新日鐵 中:聯瑞新材、錦藝新材、國瓷材料
陶瓷粉	調整Dk/Df	
氧化鋁	提高散熱性能	
碳酸鈣/滑石粉	通用材料	
氫氧化鈣/鎂	作為阻燃劑使用	
氧化鋁	改善導熱等	
碳酸鈣	降低複合材料成本	

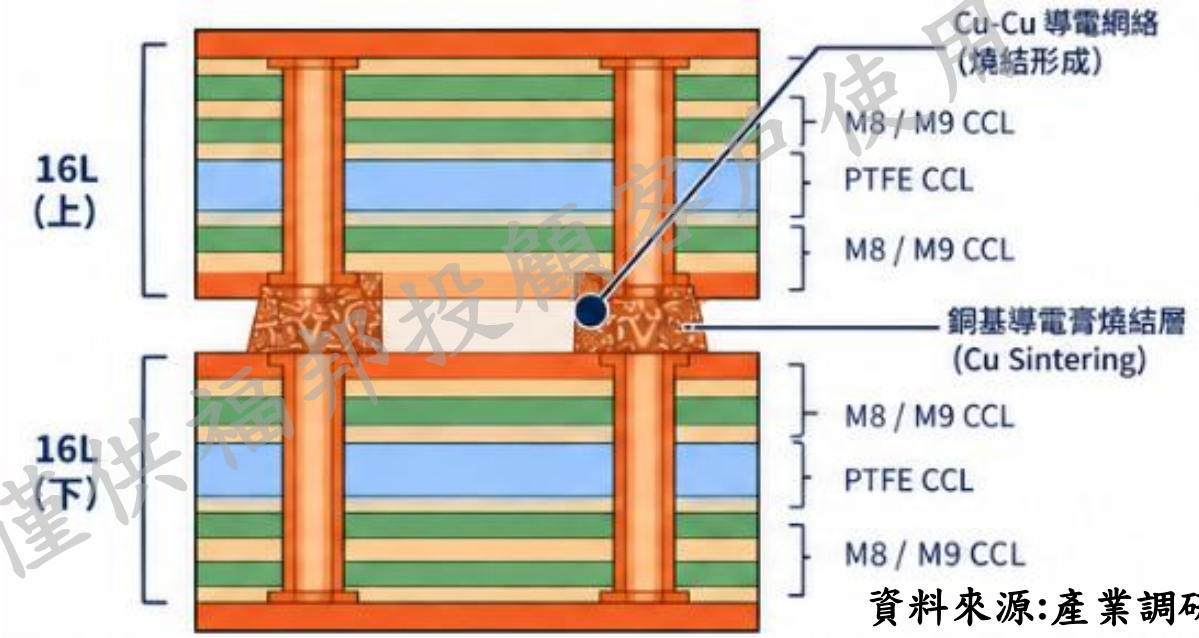
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PCB製造端的製程變化-銅漿燒結

- 銅漿燒結製程源於瞬態液相燒結(TLPS)技術，主要用於實現PCB盲/埋孔電路層間的Z軸互連。
- 過去傳統高多層多採用背鑽的製程，但在NV陣營，開始嘗試使用”銅漿燒結“的製程來處理高多層版的訊號連接，以解決背鑽精度控制的問題。
- 目前具有【燒結製程+大板製造能力】的主要為板廠，例如**深南電路、景旺電子、勝宏科技、滬電股份、生益電子；台系則有金像電具備此製程能力。**

圖10:PCB銅燒結概念示意圖



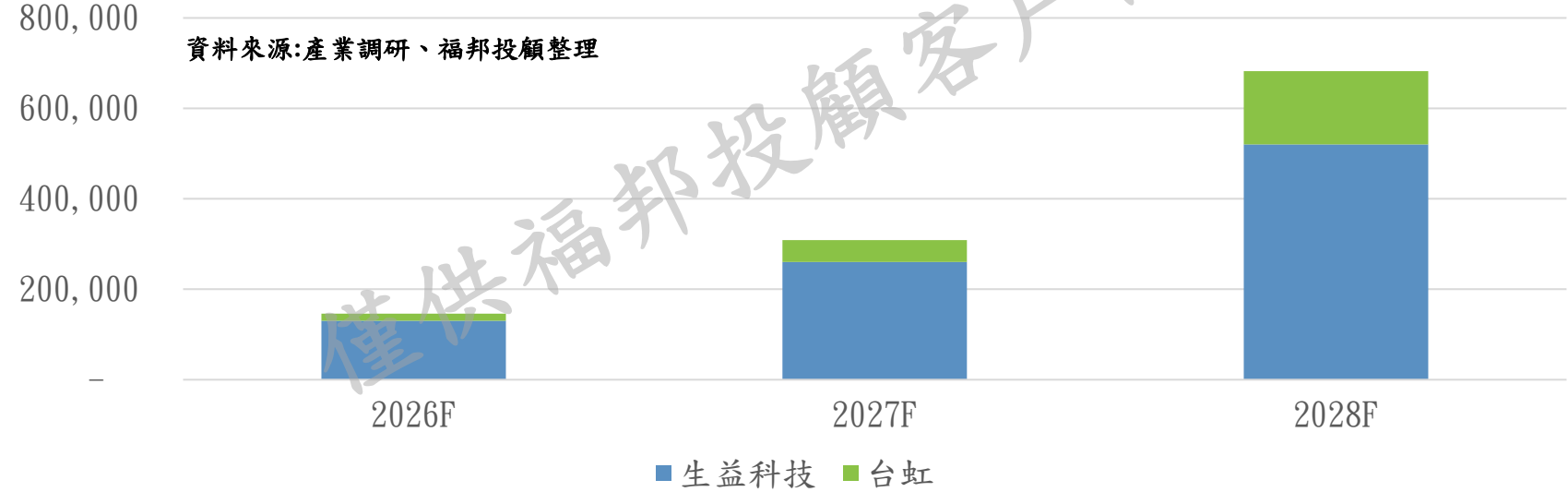
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PTFE供應商產能預期

- 目前Nvidia的供應來看，主要會以生益科技為第一供應商(合作最久)，第二供應商則從非中國供應商中選擇，目前與生益科技與台虹入列的優先供應商，**生益科技具有CCL+PP(PTFE/CH)的製造能力，台虹PTFE CCL (PFA)可搭配生益、台光電等CCL的CH PP方案，亦能搭配自產的PFA PP，可提供客戶彈性的產能需求。**
- 其餘具有PTFE樹脂CCL開發能力的供應商，開發進度較落後，不在考量範圍內。
- PCB段的生產: 以目前客戶的意願上，採用**無布方案**為優先目標，PCB生產良率約在10~30%，主要量產生產廠商預期為深南電路、景旺電子、勝宏科技、滬電股份；欣興僅參與配合RD階段(台郡負責燒結部分,但設備有限制)。

圖:PTFE主要供應商產能(米/月)



僅供福邦投顧客戶使用

PTFE CCL 潛在使用需求預估

- 假設: Rubin Ultra 於2027~2028年的晶片需求量預估400萬、900萬顆，所有機櫃均採用18層SWITCH TRAY設計，PCB板材採用混壓結構下，PTFE於2027~2028年的月均需求26、63萬張。

表22:PTFE CCL的需求試算

	SwitchTray	
產品	Rubin Ultra(VR300)	
層數	16+16=32L	
PTFE per PCB (米)	3	
Yield rate	30%	
實際用量	10	
	2027F	2028F
機櫃需求	35,000	84,000
單櫃SW Tray數	9	9
SW Board 量(張)	315,000	756,000
PTFE per PCB(米)	10	10
PTFE CCL 年總用量(米)	3,150,000	7,560,000
月均量(米)	262,500	630,000
產值預估(USD)	91M/月	220M/月

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

OpenClaw:加速從LLM往AI Agent 發展

- OpenClaw:開放式AI助理項目，可以根據使用者指令在應用程式和線上服務之間自行處理複雜任務，並被設計為可以代替使用者執行任務的AI軟體，並非過往的聊天機器人。
- OpenAI推出GPT5.4，此次更新提供Thinking和Pro兩種模型； Pro主要適用於需要進行複雜任務的使用者，Thinking模型可以在產出答案前提出思考計畫，使用者可以從中調整任務方向→強化Agent能力，可以支持智慧裝置執行複雜的工作流程。
- NVIDIA將推出類似OpenClaw的開放式平台-NemoClaw。

表23：LLM 與 Agent 推理比較

	傳統LLM推理	AI Agent 推理
流程	輸入→LLM(GPU)→輸出	感知→規劃→推理→工具與執行(CPU)
運作本質	單向生成	循環決策
互動對象	與使用者進行Q&A	與環境/工具/API互動
對硬體影響	主要消耗GPU算力	CPU需要介入成為主要調度，其中涉及沙盒隔離、儲存、記憶，複雜工具調用，

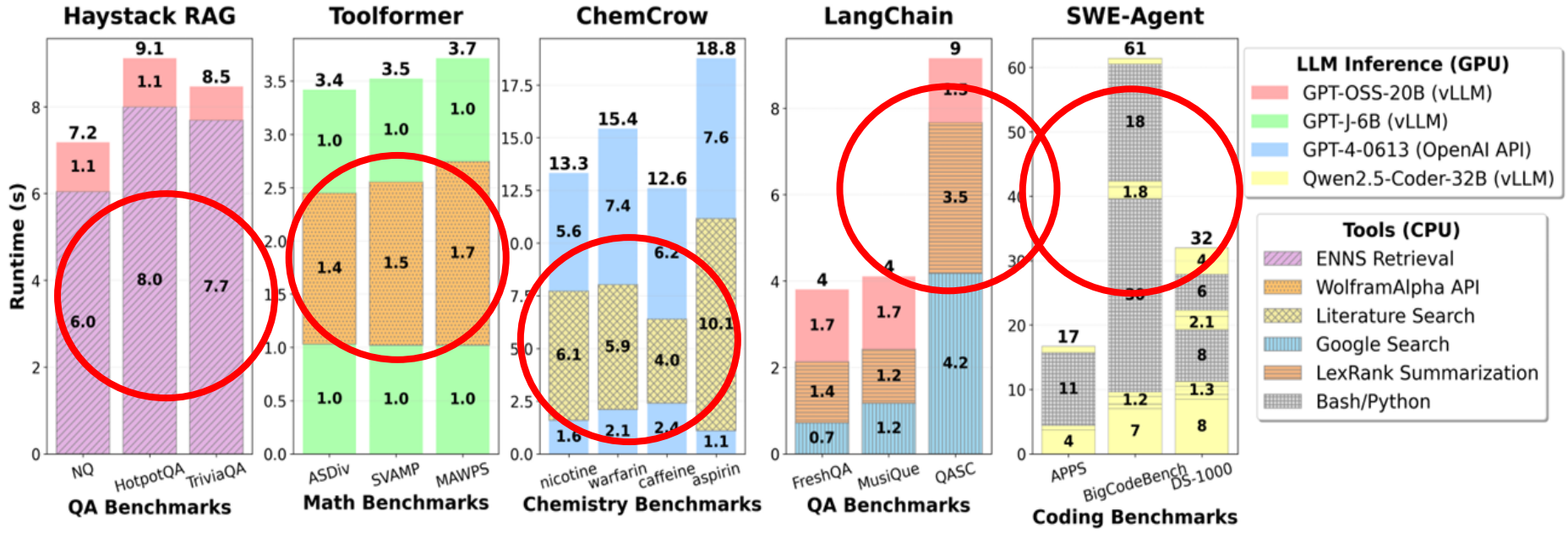
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

Agentic AI的瓶頸主要來自CPU

- 根據論文《A CPU-Centric Perspective on Agentic AI》一文指出，可以看出執行時間中CPU延遲占了大比例（紅圈），最高可達到90%以上；吞吐量瓶頸除了來自GPU，亦會來自CPU;能耗方面，大批次處理中(Batch size提升至128)，CPU平行處理效率低於GPU，佔到總能耗超過44%，延遲會從2.8秒提升至6秒以上，超過2倍。

圖11:Agent下，CPU佔據了大部分的延遲



資料來源:《A CPU-Centric Perspective on Agentic AI》 、福邦投顧整理

說明:Haystack RAG（問答檢索）**/Toolformer**（數學運算與 API 呼叫）**/ChemCrow**（化學文獻檢索）**Langchain**（網路搜尋與摘要）**/SWE-Agent**（程式碼與 Bash/Python 執行)

【僅供內部教育訓練使用，嚴禁外流】

CPU朝向多核發展→推升載板面積

- Agent執行:API處理、資料解析、腳本執行，90%由CPU完成，為了降低延遲，CPU往多核心發展是趨勢。
- 記憶體與上下文切換:CPU需要頻繁在不同Agent狀態做切換，多核心架構能讓每個Agent有獨立的處理單元，降低上下文切換帶來的效率損耗。
- 沙盒隔離/安全:Agent 需要大量獨立的一般伺服器建立Sandbox來執行，使用量與使用者和使用量正相關，需要更多核心數、高頻寬。

表24：伺服器CPU核心數

廠商	2023	2024	2025	2026	2027
Intel	Purley/Whitley	EagleStream	BirchStream	Oakstream	?
核心數	28/28	56	128	144~288	500?
AMD	Milan/Rome	Genoa/Bergamo	Turin	VENICE	VERANO
核心數	64/64	96/128	128~192	256	256~320?

資料來源:Intel、AMD、福邦投顧整理預估

【僅供內部教育訓練使用，嚴禁外流】

資料來源:各公司、福邦投顧整理

IC載板:主流晶片使用面積攀升

表25：主流晶片面積　　資料來源:各公司、福邦投顧整理

	產品名稱	推出時間	封裝方式	面積*層數	與前代的增幅
Intel	OAK Stream	2026	LGA	?	
	Birch Stream	2024H1	LGA	6,000*20	**+15%**
	Eagle Stream	2023	LGA	5,000*20	
AMD	**Venice**	**2026**	**FCLGA**	**6,100*26**	**+44%**
	Turin	2024Q4	FCLGA	5,500*20	-
	Genoa	2022Q4	FCLGA	5,500*20	
NV	**Rubin**	**2026~2027**	**CoWoS-L**	**9,000*18**	**+75%**
	Blackwell	2024~2025	CoWoS-L	5,900*14	**+116%**
	Hopper	2023~2024	CoWoS-L	3,200*12	
	Vera	2024~2025	CoWoS-R	5,700*18	**+46%**
	Grace	2023~2024		5,000*14	
Google	**HumuFish**	**2027**	**EMIB-T**	**14,000*24**	**+160%**
	Zebrafish	2026	CoWoS-S	5,300*24	**+22%**
	SunFish	2026	CoWoS-S	7,000*18	**+21%**
	GhostFish	2025	CoWoS-S	6,500*16	
Amazon	**Tranium4**	**2027**	**EMIB-T**	**14,000*24**	**+238%**
	Trainium3	2026	CoWoS-R	6,200*16	

僅供福邦投顧客戶使用

IC載板:產業能見度拉長

- 根據供應鏈訪查，目前主流載板對於供需的意見陸續在26H1看到供不應求，與之對應的生產相關輔材訂單能見度可至2028年。

表26：主流大廠對於供需的想法

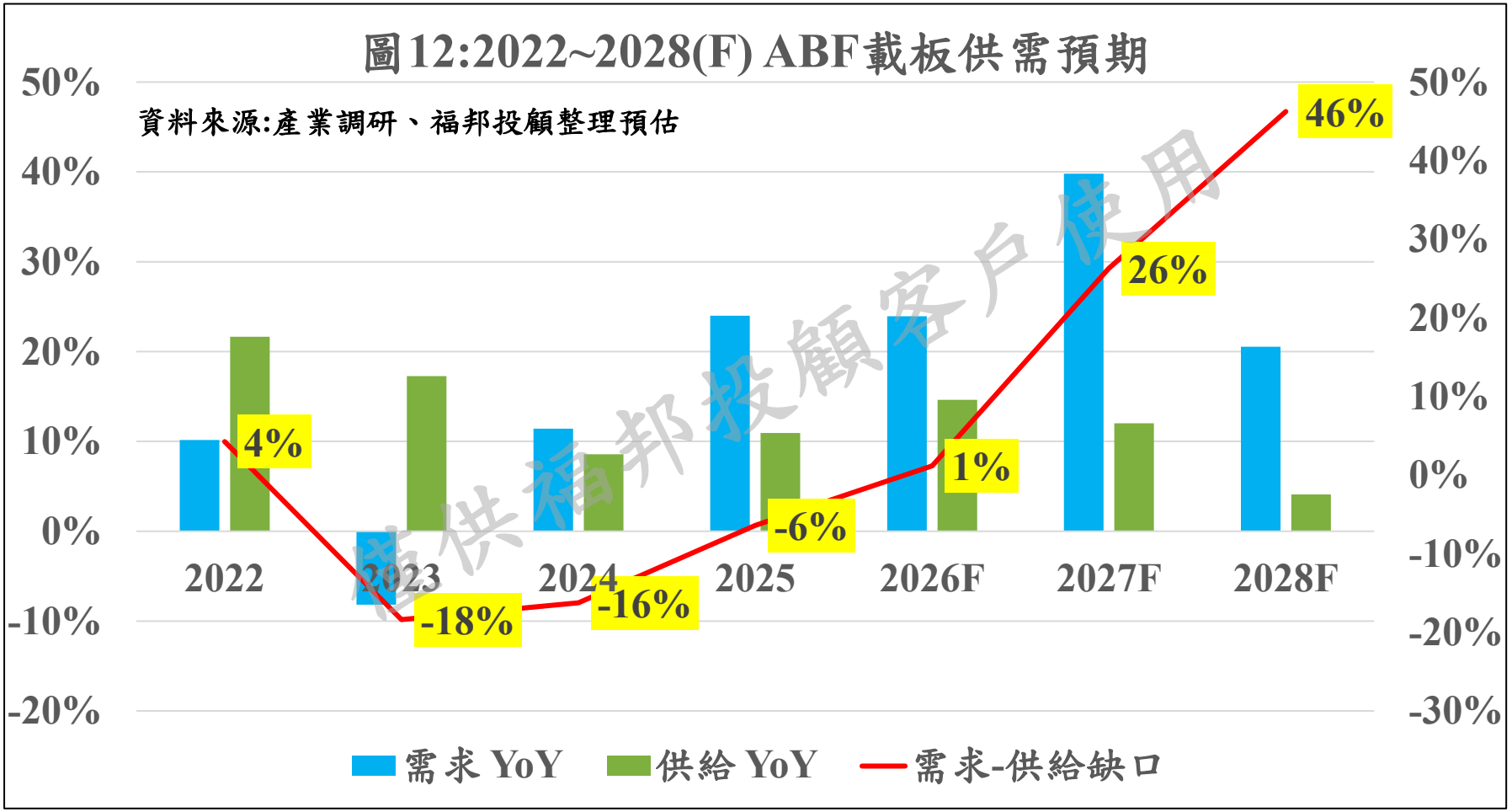
供應	公司	稼動率概況	對供需的想法	對應客戶
載板	Ibiden	AI相關滿載	2027年載板產能供不應求	NV/Intel/AMD
	Semco	26H2滿載	2027年載板產能供不應求	Intel/AMD/NV/AMZN
	AT&S	26Q1滿載	2027年載板產能供不應求	Intel/AMD/Marvell
	欣興	26Q1滿載	2026年玻纖布缺料，2027年供不應求	NV/Intel/AMD/Broadcom/AMZN
	南電	26Q1滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	Broadcom(主要)/AMD
	景碩	26H2滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	NV/AMD
生產輔材	鉅橡	26H2滿載	訂單能見度可至2028年	全球主要載板廠
上游材料	味之素	滿載	若需求持續暢旺，2027年產能嚴重不足：公司於2028年才有70%新產能開出	全球主要載板廠
	日東紡	滿載	2027年開始量產新產能。	全球主要載板廠
	Resonac	滿載	公司供給仍明顯低於需求，2027年有機會再次調價	全球主要載板廠

僅供福邦證券客戶使用

【僅供內部教育訓練使用，嚴禁外流】

IC載板:ABF載板供需將從2026年開始見到供不應求

- 根據福邦投顧預估，2025年ABF載板仍處於供過於求的格局，2026年開始在AI的需求強勢帶動影響下，ABF載板將開始見到供不應求格局，並於2027~28年逐步擴大。（**本圖不含缺T-glass影響**）



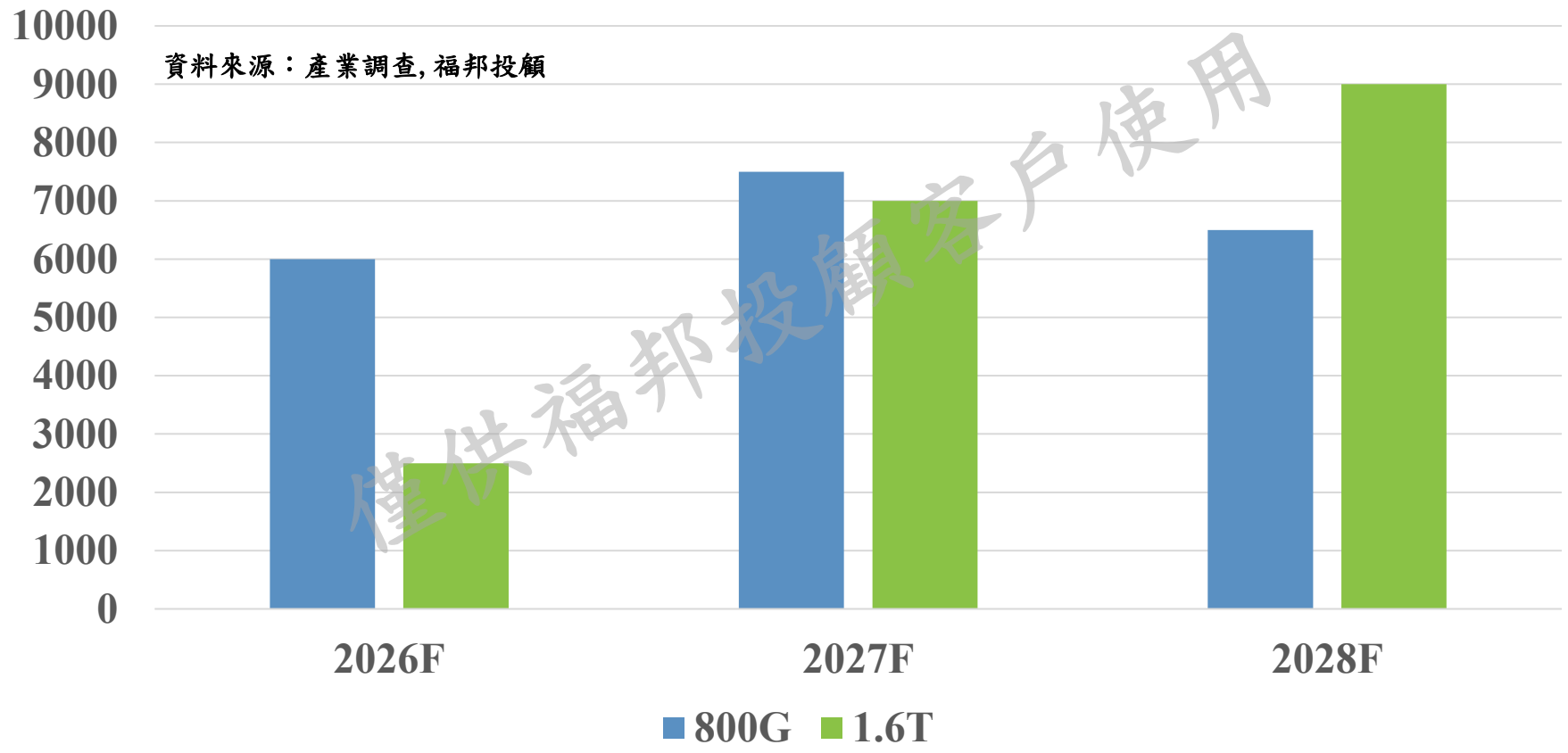
僅供福邦投顧客戶使用

可插拔光通訊模組:800G/1.6T潛在需求上升

■根據福邦投顧產業調查，2026/27/28年800G的採購”需求”可望達到6,000/7,500/6,500萬支，1.6T需求預估2,500/7,000/9,000萬支。

(隱含終端客戶對於良率的考量造成的Overbooking)

圖13: 2026~2028年光通訊模組需求預期(萬支)



僅供福邦投顧客戶使用

傳統光通訊模組板-HDI/mSAP供應體系的新機會

■因應高度整合和小型化需求，光模組開始大量採用MSAP來縮小產品體積和損耗；進入1.6T，L/S 要求微縮至**20μm-25μm**；因此1.6T開始，強制採用MSAP。

■材料規格上，亦往M7~M8等級以上升級，但在MSAP部分，會大量採用載體銅箔(三井金屬90%市佔率)、日東紡的Low CTE布，目前有面臨缺料的情況。

表27 :400G~3.2T PCB方案比較

速率	400G	800G	1.6T	3.2T
製程	HDI	HDI/mSAP	mSAP	mSAP
層數	10L	12~14L	14~18L	20-22L
L/S	40~50μm	30~40μm	20~30μm	10~15μm
CCL等級	M4/M6	M6/M7	M8 (斗山/台光電)	M9~M10+陶瓷基板?
ASP(USD)	~10	12→14~15	30→35	150+
毛利率	20~30%	40~50%	50~60%	70~80%
供應鏈	SCC、欣興、興森等	臻鼎/SCC/欣興/方正/景旺/華通/WUS等	臻鼎、SCC、欣興、華通等	臻鼎、SCC、欣興

資料來源：產業調查、福邦投顧

僅供福邦投顧客戶使用

傳統光通訊模組PCB供應商概況:27年可望緩解供應

表28 :主要供應商客戶與產能概況　　資料來源：產業調查、福邦投顧整

供應商	關鍵客戶	mSAP月產能規劃預期	客戶2027年月產能(支)
欣興	中:中際旭創 美:Coherent、Mellanox	26Y:7,500平米(=250~300萬支) **27Y: 10,000平米(=400萬支)**	■ 中際旭創: 800G/1.6T 125W/210W ■ 新易盛: 800G/1.6T 80W/35~40W ■ Coherent: 800G/1.6T 125W/125W ■ AAOI:90~100W ■ 華工科技: 800G/1.6T 30W/25W ■ 索爾斯:80W
深南電路	中:中際旭創、新易盛 美:Coherent	26H1:3,200平米(=120~130萬支) 26H2:6,000平米(=240萬支) **2027Y:8,000平米(=320萬支)**	
臻鼎	中:中際旭創	26Y: 2,800平米(=110萬支) **27Y:10,000平米(=400萬支)**	
華通	中:中際旭創 美:AAOI、Coherent	27Y:7,500平米(=300萬支) **28Y: 8,000~8,500平米(=330~340萬支)**	
方正科技	中:新易盛、 華工科技、光迅科技	26Y:1,500~2,000平米(=80萬支) **27Y:5,000平米(=200萬支)**	
景旺電子	中:中際旭創、新易盛 美: Coherent	26Y:1000平米(=40萬支) **27Y:10,000平米(=400萬支)**	
興森科技	中:華工科技、光迅科技	28Y:擴產10,000平米/月產能	
勝宏科技	潛在客戶:中際旭創	規劃20,000平米/月	
AT&S	中 ： 中際旭創 美:Coherent、Mellanox	26Y:2800片/月	

僅供福邦投顧客戶使用

雲端資料中心採用之交換器品牌

■雲端資料中心多採用品牌+白牌的混合方案

■Spine和DCI 因為結構複雜，所以多採用品牌廠，但Google和Amazon 由於架構和軟體上較強，所以選擇白牌來配合他們的結構。

■TOR和 Leaf 結構簡單，可以看到較多白牌(裸機、白盒)的方案。

■xAI的交換機品牌主要採用Juniper的方案，主權AI體系(ex:星際之門、中東等)多會採用Cisco等品牌交換機。

表29:各大雲廠商交換機廠商彙整

	TOR(機櫃)	Leaf（接入）	Spine（匯聚）	DCI（核心）
GOOGLE	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌/Juniper/CSCO
Amazon(AWS)	白牌(CLS/Accton/CSCO)	白牌(CLS/Accton)	白牌(CLS/Accton)	Juniper
微軟(Azure)	Arista/CSCO/Dell	Arista/CSCO/Dell	Arista	Arista
Meta	白牌(CLS/智邦/CSCO)	Arista/Edgecore（智邦）	Arista/Edgecore（智邦）	Arista/Juniper/白牌
百度	白牌/H3C	華為/H3C	華為/H3C	華為/H3C
阿里巴巴	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	華為/H3C
騰訊	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為	華為/H3C	華為/H3C
APPLE	Arista/CSCO/白牌	Arista/CSCO	Arista	CSCO/Juniper

資料來源：650 GROUP、福邦投顧整理

【僅供內部教育訓練使用，嚴禁外流】

交換機板:主流AI層數較低，傳統乙太網多層為主

■ 資料中心需要承載流量大+高速傳輸，在此需求下，高多層板成為此需求的解決方案，其優點在於電子元件間連線縮短下，訊號傳輸速度提高，而且也方遍佈線。

■乙太網交換機為各家CSP+主權主要Scale out的主流，800G報價是400G的2倍以上，1.6T是800G的4倍以上。

表30 :交換機供應鏈劃分

板層數劃分		產品	銅箔基板主要供應商	PCB主要供應商	出貨價格
Cisco	400G:24~36L PTH 800G:38~48L PTH (NPO:28L) 1.6T:46~52L PTH / 18~20L HDI	800G	台光電/松下/Doosan	金像電、TTM、滬士電、ISU	400G :1500 800G:4,000 1.6T:16,000+ (美金/平方米)
		1.6T	台光電/松下		
Arista		800G	台光電、Doosan	TTM、滬士電、金像電、ISU	
		1.6T	台燿、台光電		
Celestica		800G	台光電/Doosan/松下/台燿	金像電、ISU	
		1.6T	台光電/松下/台燿	金像電、ISU	
Accton		800G	Doosan、台光電	TTM、ISU、LCS	
		1.6T	台光電	TTM、ISU、LCS 金像電	
Mellanox		800G	台光電	TTM、滬士電	
		1.6T	台光電	TTM、滬士電	

福邦投顧產業調查

交換機的材料

■在交換機升級的過程中，由於高速傳輸需求，對於耗損也是嚴格規範，**800G材料價格則會較400G材料增加60%+，Tomahawk6 將會使用於2026年的1.6T交換機可使用材料約M8U~M9-(碳氫+LK2)等級，2027年後的規格可望往M9發展。**

表31:各速率材料概況

	100G	400G、**800G**	800G、1.6T	1.6T~2.4T	3.2T~
材料耗損等級	Very low loss	Ultra low loss	Extreme Ultra low loss	Super low loss	?
Df	0.005~0.006	~0.003	~0.002	~0.001	~0.0003
頻率	13Ghz(NRZ)	13Ghz(PAM4)	28Ghz(PAM4)	56Ghz(PAM4)	100+Ghz?
設計	PPO	PPO+Low K Glass	PPO+LowK/K2 +VLP2~4	PPO+CH樹脂+ VLP4 +Q/LK3/LK2	CH/PTFE樹脂+LK4/5 +VLP4/5
平均單價/張	2,000~3,000	**3,000~4,000**	**5.000~6,000**	6,000~7,000/ 10,000	12,000+
代表型號	松下M6	松下M7	松下M8	松下M9	松下M10
競爭同業產品儲備	EM-528/626	EM-528K/EM-890K	EM-892K/K2	EM-896K3	EM-898K4
	TU883	ThunderClad2/3	ThunderClad4	ThunderClad 5	?
	IT 968/968G	IT988G	IT998GSE/2	IT999GSE3	ITQ10G(K4)
	NP186/188U,H	NPG188UK/199	NPG199K/HK	NPG-998Q	NPG-999Q
	Synamic 6/6N	Synamic 8GN	Synamic 9GN	Synamic 10/11GQ	SG5300N
	DS7409DV（N)	DS7409DJG(N)	DS7409DQN/N2/T2	DS7409DYQ/L2/N	?
	NY6300S	NY-P1~3	NY-P4	NY-P5	NY-P6

資料來源：各公司，福邦投顧整理

謝謝指教
Q&A

【揭露事項與免責聲明】

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2026Q4 PCB產業-
需求推動漲價，傳輸靠PTFE解方

福邦投顧 研究部
2026/10

26Q4結論-上游材料加速更新，漲價仍為主旋律

1.PCB上游:供不應求推動漲價動力，新材料應用下帶來新機會

- 玻纖布:E-Glass(高階)/LowDK-2/T-Glass 在2026Q4~2027年呈現全面緊缺，成為推動當前上游材料漲價的主要因素，也是決定結構性缺貨的關鍵原料，**玻布廠迎來產品組合改善+報價上揚雙重催化。**
- 銅箔:HVLP4加速驗證日台以外的供應商，中系擴產積極+中系設備加速採用下，**維持2027年供需缺口放緩預期。**
- 銅箔基板:PTFE樹脂為美系客戶解決互聯的確定性解方，成為銅箔基板材料中新晉路線；傳統的PPO/CH樹脂仍有一片天，26Q4 VR300 設計將定案，PTFE+M8/M9設計呼聲較高。
- 鑽針:高階鍍膜針需求攀升，供應商在2026~2028年加快投入建置產能，新材料(PTFE)仍有一定的鑽針消耗，**產品組合的改變遠比漲價來的重要。**

2. IC載板: 產能全數售罄，缺料與缺產能並行，關注供應商漲價態度。

- 供需缺口:維持2026/2027年 20%+/40%+ 缺口預期。
- 主邏輯為面積放大+層數增加，大廠產能售罄，客戶需要付出更高的價格預定產能+料源。

3. PCB:關注AWS、Google新品放量，以及2027~2028年HDI設計的變化，**聚焦Tier1供應體系和伺服器級HDI稀缺產能。**

4. 建議首選關注個股:

- 高階稀缺HDI產能:2368金像電、3037欣興、4958臻鼎
- 產品組合優化顯著:1303南亞、1815富喬、8021尖點、8039台虹
- 缺貨27~28年不緩解+AI Agent受惠:8046南電、3189景碩、3037欣興、4958 臻鼎

僅供福邦投顧客戶使用

重點載板、CCL公司獲利預估

股票代號	股票名稱	股本(百萬)	25EPS	26EPS(F)	27EPS(F)	28EPS(F)	景氣上行期PE
PCB板-國際同業評價P/E評價20~30X							
2368	金像電	5,306	19.02	41.2	82.9	99.9	20~30
載板-國際同業評價P/E評價20~30X							
8046	南電	6,462	3.01	14.5	55	85	20~30
3037	欣興	16,435	4.4	16.8	33	44.5	20~30
3189	景碩	5,268	3.5	13.7	27.1	42	20~30
4958	臻鼎	10,773	7.1	14.3	26	34	20~30
材料-國際同業P/E評價20~30X							
1303	南亞	79,308	0.51	15	3~4(材料) + 8(南電) + 12~20(南亞科)	N/A	20~25
1815	富喬	5,855	1.5	4.8	7.01	10	15~23
8021	尖點	1,473	2.7	10.2	25	36	16~25
8358	金居	2,526	4.2	12	17.2	27	19~25
2383	台光電	3,583	41	117	260	380	20~25
6213	聯茂	3,636	4.2	9.7	22.3	40	15~25
6274	台燿	2,993	11.8	40.2	85	115	20~25
8039	台虹	2,640	2.04	2.8	3(原業務) +15(PTFE)	3(原業務) +22(PTFE)	20~25
資料來源:福邦投顧整理預估(2026.09)							

僅供福邦投顧客戶使用

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【僅供內部教育訓練使用，嚴禁外流】

全年核心:政策主導AI為基本方針

- AI 成為美國政策上的基本方針，OBBBA法案刺激企業加大投資AI算力，AAAP計畫掃除建置AI算力的瓶頸和主權AI方針的確立。

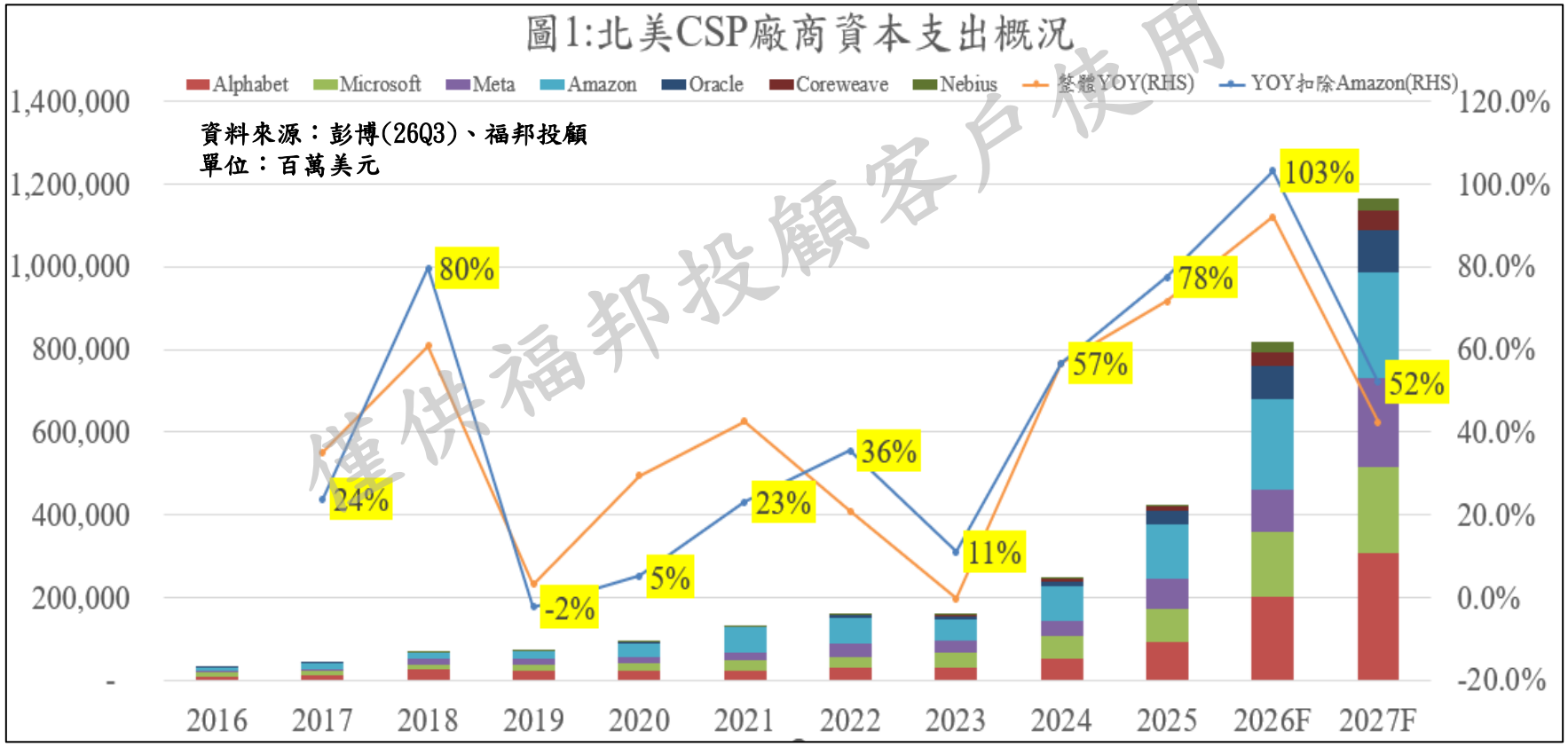
表1:OBBA法案和AI行動計畫　　資料來源:福邦投顧整理

法案	大而美法案(One Big Beautiful Bill Act)	AI行動計劃(America's AI Action Plan)
內容	1. 下調聯邦所得稅:從35%下調至21%，減輕企業稅賦負擔。 2. 恢復2017年減稅與就業法案項目中允許企業**當年資本支出可以全額抵減稅賦(折舊扣除)**，有利重資本支出廠商加快投資回收時間，受惠廠商如CSP大廠。 3. 國內R&D全額費用化:為了獎勵企業在美國進行生產研發，**符合條件的R&D費用可以於當年度全面費用化**，受惠廠商如Open AI、Anthropic等。	1. 加速AI創新:確保美國擁有全球最強大的AI系統，放鬆相關管制與障礙，並鼓勵開源等。 2. 增加AI基礎設施建置:提升美國AI算力基礎設施硬體基礎，以滿足AI推動經濟和安全的需求；加快關鍵項目的審核，**加快建設資料中心**、晶圓代工廠、電力基礎設施，掃除一切AI基礎設施擴充的障礙。 3. 維護國際外交與AI安全:在全球AI競爭中掌握規則制定權，輸出美國AI技術，同時阻止對手利用美國創新超越美國，嚴防高階算力流向競爭國家，向盟友出口美國AI算力技術等，有利**主權AI(Sovereign AI)**的發展。。

僅供福邦投顧客戶使用

雲端巨頭對資本支出態度維持積極

■2026年雲端CSP業者和算力租賃業者正加快擴建算力基礎設施，2026年預估資本支出成長上修至103%(含Amazon成長92%);2027年目前資本支出預估成長52%。



僅供福邦投顧客戶使用

全球伺服器出貨2026~2027年預期雙位數成長

■參考Digitimes最新數據預估，2026年預估出貨量YoY+17%，2027年目前預估11.2%。

圖2:全球伺服器出貨量成長預測

		2023	2024	2025	1Q26	2Q26	3Q26	4Q26	2026(F)	2027(F)
Gartner	出貨量	11,259	11,991	12,544					14,015	15,678
26Q2	YoY	-18.6%	6.5%	4.6%					11.7%	11.9%
	QoQ									
Digitimes	出貨量	14,432	14,888	16,597	4,699	4,877	5,015	4,896	19,487	21,670
26Q3	YoY	-20.4%	3.2%	11.5%	23.5%	26.0%	19.5%	9.0%	17.4%	11.2%
	QoQ				4.6%	3.8%	2.8%	-2.4%		
IDC	出貨量	12,357	14,461	16,779	3,773	3,975	4,261	4,421	16,430	18,083
26Q2	YoY	-17.4%	17.0%	16.0%	2.0%	1.9%	3.0%	0.2%	-2.1%	10.1%
	QoQ				-14.2%	5.4%	7.2%	3.8%		

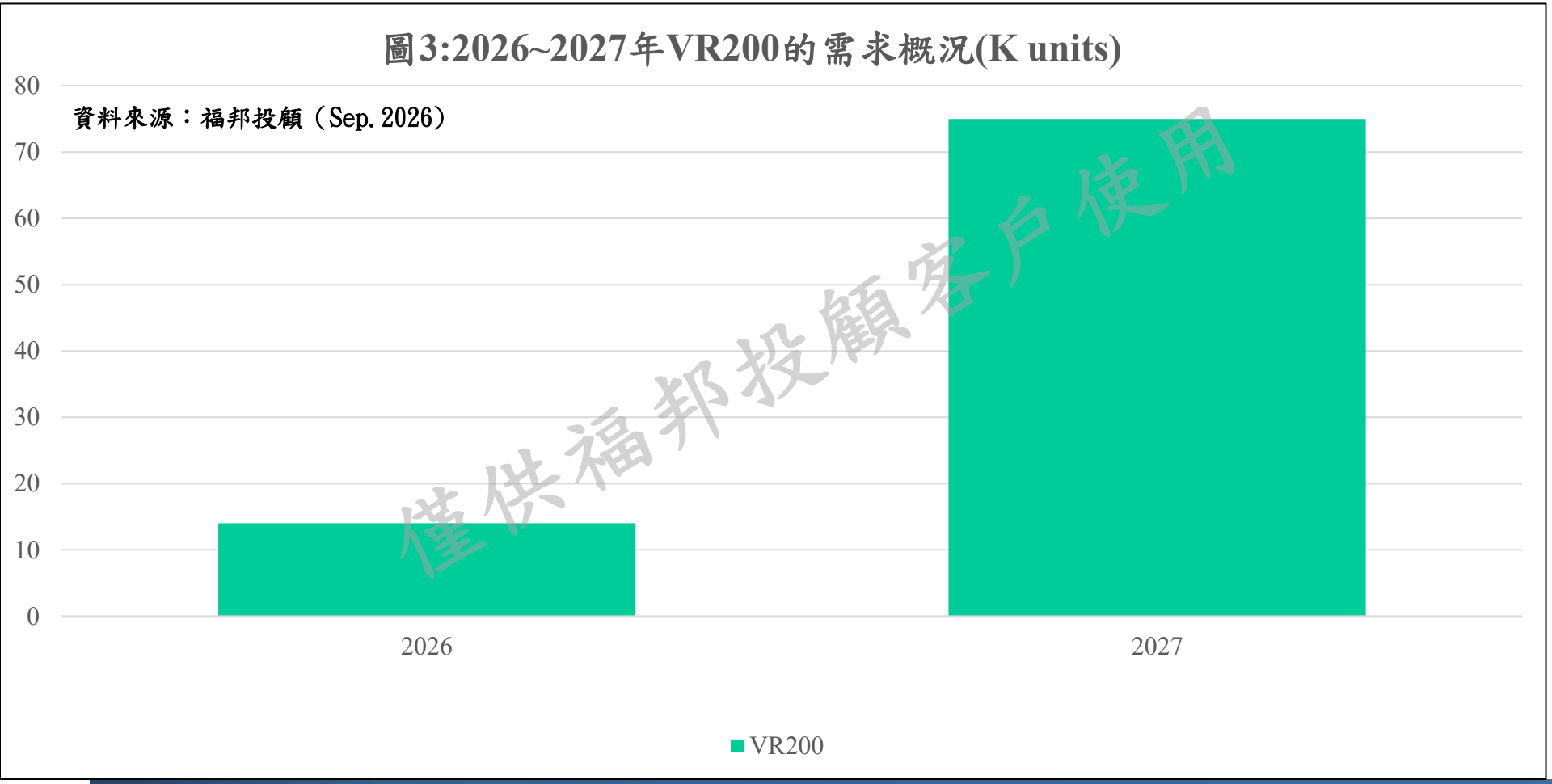
資料來源：福邦投顧整理（26Q2）

僅供福邦投顧客戶使用

GPU:26Q4~2027年VR200合計88K櫃

■根據供應鏈調查統計，2026年Rubin(VR200)預估組裝量約13~14K櫃，2027年預估組裝量約75K櫃。

■Rubin Ultra (VR300)於2027年預估需求量約5~6K組，2028潛在12K組(NVL576)。



僅供福邦投顧客戶使用

GB300/VR200/VR300規格:南亞、台虹為亮點

表2:GB200/300 VS. VR200　　資料來源:產業調研、福邦投顧整理

名稱	GB300 NVL72(72GPU)	VR200 NVL144(72GPU)	VR300 (RubinUltra)(9+9+9)
時間	2026放量	**2027放量**	**2028放量(26Q4 定案相關規格)**
晶圓製程	N4P	N3P	N3P
配合的PCIE版本	6	6	6
CoWoS製程	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝
單晶圓可切割數(顆)	14~15	8(包含I/O chiplet 面積增加)	8(包含I/O chiplet 面積增加)
TDP/散熱方式	1400W	2300W	1,800/2,600
載板供應商	Ibiden80~90% /欣興10~20%	Ibiden90~95% /欣興5~10%	Ibiden90~95% /欣興5~10%
CCL材料供應	計算板:斗山M8(LDK)+M4	計算板:斗山(M8+LDK+HVLP4)+M4	計算板:斗山(M8+LDK2+HVLP4)+M6
	交換板:M8(LDK2)+M2 生益科技(70%)台光電 (30%)	**交換板:M8(LDK2+HVLP4)** 生益科技/台光電/**南亞**	**交換板:M8(K2)/M9+PTFE** **生益科技+台虹/台虹**
		CPX:M8(K2)	CPX:M8(K2)
		中介板:M8(LDK2)	中介板:M9?
CCL上游材料供應	玻布:台玻、Asahi、泰山玻纖	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡
	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi
	HVLP3銅箔:三井/古河/盧森堡	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福
ComputeTray	5壓22L	5壓26L	**5壓30L**
PCB供應商	勝宏>欣興	勝宏>欣興>臻鼎 …定穎?(認證中)	勝宏/欣興/臻鼎/ 定穎(認證中)
NVSwitch Tray	22L HLC	32L(N+N HLC)	**32L(N+N HLC)**
PCB供應商	VGT50%/TTM40%/景旺10%	VGT/WUS/TTM	VGT/WUS /TTM/深南?/景旺?
CPX		Delay(需求不佳)	Delay(需求不佳)
PCB供應商			
小中介板(Midplane)		44L(N+N HLC)	44L(N+N HLC)
PCB供應商		VGT/景旺/滬電/欣興	VGT/景旺/滬電/欣興

僅供福邦證券客戶使用

雲端廠商ASIC相關規格-高階材料/製程加快導入

表3:雲端大廠&LLM廠商方案　　資料來源:產業調研、福邦投顧整理預估

客戶	Design Service/ Code Name	時間	主板 設計規格	L6	L10~L12	CCL Supply	Compute Board PCB Supply
Google	Broadcom - TPU v7 (Ironwood)	26H1	22/24L	Celestica80%/ Jabil 20%	Celestica /Flex/ 鴻海(5%)	M7-Panasonic (80~90%)/EMC	主:ISU/LCS/WUS/TTM/VGT
	Broadcom - TPU v8 Sunfish	**26Q4**	34L			**M8(LK2)-** EMC/松下	主:WUS/TTM/LCS/VGT
	MTK - TPU v8 Zebrafish	**27Q1**	34L			**M8(LK2)-** EMC/Doosan/松下	主:ISU/WUS/TTM/VGT/ **GCE**
	MTK- TPU v9 Humufish	**28Q2**	**6壓28L**			**M9+M8(LK1)** EMC	主:ISU/VGT/WUS/TTM/**欣興** 備用:**臻鼎/LCS**
Amazon	Marvell - AWS Trainium2e (Cayman)	26H1	26L	智邦(氣冷) / 緯穎(水冷)	緯穎 /Jabil/Flex	M8(LK)-EMC/TUC	GCE/SY/FHT
	Annapurna - Trainium3 (Mariana)	26Q2	主: 26L(氣) 40L(水) SW:22L			M8(LK)- EMC(氣+水)/ TUC(氣)	**ComTray:GCE+**SY **(氣+水)** /WUS(水)/**FHT(氣)** **Swtray:GCE/**SY/**ZDT/FHT**
	Annapurna - Trainium4 (Maverick)	27Q4	**主:3/4壓34L SW:30L**			**M9-** EMC/南亞?	**GCE/**SY/**ZDT**/Meiko
Meta	Broadcom - MTIA T V1.0 (Minerva)	25H2	40L	廣達/ Celestica	廣達	M8(LK)- EMC	WUS/TTM/ISU
	Broadcom - MTIA I V1.5 (Iris+ / Santa Barbara)	26H2	36L	廣達 or Celestica	廣達 or Celestica	M8(LK)- EMC	WUS/GCE/ TTM/ISU
	Broadcom - MTIA V2 (Olympus / Santa Cruz)	28E	?	廣達 or Celestica	廣達 or Celestica	**M9-** EMC	WUS/GCE/TTM/ ISU/DY?
MSFT	Marvell-MAIA 300 (Griffin)	27H1	40L+	廣達	廣達	**M9-** EMC/Doosan?	GCE/TTM/WUS/大德/ISU
AMD	MI4系列	27H2	48L	緯創	Celestica、緯穎	M8(LK2)	SCC(90%)/欣興/大德
	MI5系列	28H2	52L	緯創	Celestica、緯穎	M9	

僅供福邦投顧客戶使用

鑽針主邏輯—PCB廠積極擴產

- PCB設計日益複雜，耗用大量產能：過去伺服器板層數約16~2X層，目前主流已經來到20~30層，未來30~50層越顯常見。
- PCB用量增加，耗用大量產能:隨著各家設計陸續參考NVIDIA Rack設計，除了原有的主板(GPU/ASIC 使用)，亦新增更多副板，例如Switchboard、midplane等，終端客戶積極驗證新供應廠商。

表4:全球主要GPU/ASIC陣營PCB供應商擴產概況

億元	產值/產能	2025	2026(F)	YoY	2027(F)	YoY	2028(F)	YoY
勝宏科技	多層板(M^2)	28K	120K	+328%	350K	+190%	590K	+69%
	HDI(M^2)	130K	320K	+146%	640K	+100%	680K	+6%
金像電	sqf	6,800K	10,000K	+47%	11,000K	+10%	12,000K	+9%
ISU Petasy	(M^2)	240K	280K	+16%	320K	+14%	360K	+13%
滬電股份	(M^2)	160K	210K	+30%	340K	+62%	440K	+29%

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-CCL材料影響

■ PCB用鑽針受到板材影響，在同樣板厚下，每針可鑽孔次數大幅縮水:

- M6~M7時主針鑽孔壽命約600~800次
- M8材料下可鑽400~500次
- M9材料下僅剩下100~200次
- 材料越高階，鑽針使用量會是過往的6倍以上。

表5:鑽針基於不同板材的使用壽命&各應用對應的鑽針方案

	過去	目前	未來
對應材質	M6~M7	M8/(有布/無布+球型矽粉)	M9~M10
主針壽命	1000/600~800次	400~500次	100~200次

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-長徑比帶動報價上揚

- 當前主流GPU用PCB約5mm厚度，隨著AI用PCB板厚的提升，每增加1mm，製造上可能會採用增加一根預鑽針，採用預鑽針預鑽至特定深度，後續用主針完成加工。
- GPU伺服器:GB200採用「1主針配2白針」→GB300採用「1主針配3白針」→VR200採用「1主針配4白針」，

表6:鑽針各應用對應的鑽針方案　　資料來源:產業調研、福邦投顧整理預估

	GPU Server	ASIC Server	交換機
對應鑽針方案	1主針 +1~4根預鑽白針	2主針 +1根預鑽白針	3~4根主針 +1~2根預鑽白針

- AI用的鑽針，長度/直徑比超過30規格的鑽針，報價上會採用**一個方案一個價格**，因為加工精度要求提高，難度變大；目前AI用的鍍膜鑽針報價較傳統白針多20~50%，若是這種報價方式，特別是板厚超過7mm，產品售價將會多40~50%；未來約在2027~2028年時，7.5~8.5mm的鑽針將成為主流，報價將會是一般白針的6倍以上報價。

表7:鑽針不同長徑比的報價倍數　　資料來源:產業調研、福邦投顧整理預估

長徑比	一般(10~20)/鍍膜	30	40	50
售價倍數	1/1.2~1.5	3~4x	6~8x	12~16x

- **毛利率方向:AI用的鍍膜鑽針鑽針，根據產業訪查，產品毛利率約50%+**。

僅供福邦投顧客戶使用

全球主要鑽針廠亦積極擴充產能，供需維持吃緊

- 目前全球主要伺服器PCB廠採用的鑽針廠，高階鑽針以中鎢高新旗下的金洲科技和尖點為主、中階以下以鼎泰高科為主；日本佑能主要用於載板生產用針。
- 鑽針廠的外溢效果:由於前四大廠優先用於AI伺服器相關生產用針，使得中低階針有排擠效應，白針類產品外溢給凱崴等廠商。

表8:全球主要鑽針廠產能/客戶概況　　資料來源:產業調研、福邦投顧整理預估

公司	產能(萬支/月)	高階鑽針占比	主要客戶	規劃
鼎泰高科	1.2億支	>40%	深南、景旺、勝宏、健鼎、生益、方正、廣合、定穎	2026年:1.8~2億支/月
中鎢高新(金洲)	9,000	>15%~30%	勝宏、深南、高技、生益、景旺等((**中系高階主供**)	2026年底至1.2億支/年 2027年新增3600萬支+/年 2028年新增1.3億支/年
佑能	4000	>60%	專注日本、台灣市場、載板廠	2026年底至5,000萬支/月 2027年至6,000萬支/月
尖點	3,500(50%以上高階鍍膜針)	>50%	金像電、ISU、勝宏、TTM、滬電、臻鼎、欣興等	2026年:4,500萬支+/月， 2027年:˙7,000萬支+/月 2028年達到9,000萬支+/月

僅供福邦投顧客戶使用

HVLP4銅箔需求受良率影響，需求驚人

- 根據供應鏈調查，依照良率80~90%的假設下，Nvidia Rubin的需求約300~400噸/、LPU約200~250噸/月，Amazon AWS 800~850噸/月，Meta MTIA 200噸/月，AMD MI450 100~200噸/月，800G交換機 400噸/月，合計**1,200~1,300噸/月/2,700~2,800噸/月。**

- 若依照實際PCB生產狀況，良率約在60~70%不等，總需求可達**1,400噸/月/3,000~3,100/噸/月。**

表9:全球主要HVLP需求估算與供應鏈格局

	HVLP2	HVLP3	HVLP4	HVLP5
加工費	USD15/kg	USD20~25/kg	25~30美金/kg	35+
供應商	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田
客戶	AWS T2 (2024-2025年)	NVIDIA GB200 NVSW	AWS T2MAX(2026年)、Meta M9 896K2/K3、NVIDA Rubin、Google V8、AMD MI450、800G Switch	Panasonic M9
	Google V5/V6	Google V7		SY5300N
需求量	AWS 300-400噸/月	NV 38~40噸/月	AWS800~850噸/月(100萬張/月)	
		Google v7	Nvidia 300~1000噸/月	
			800G 交換機 400噸/月	
			Meta 20噸/月	
	資料來源:產業調研、福邦投顧整理預估		AMD 20噸/月	
			Google Zebra&Sun 80噸/月	

僅供福邦投顧客戶使用

全球主要供應商概況:加快導入新供應體系

表10:全球主要HVLP供應鏈格局

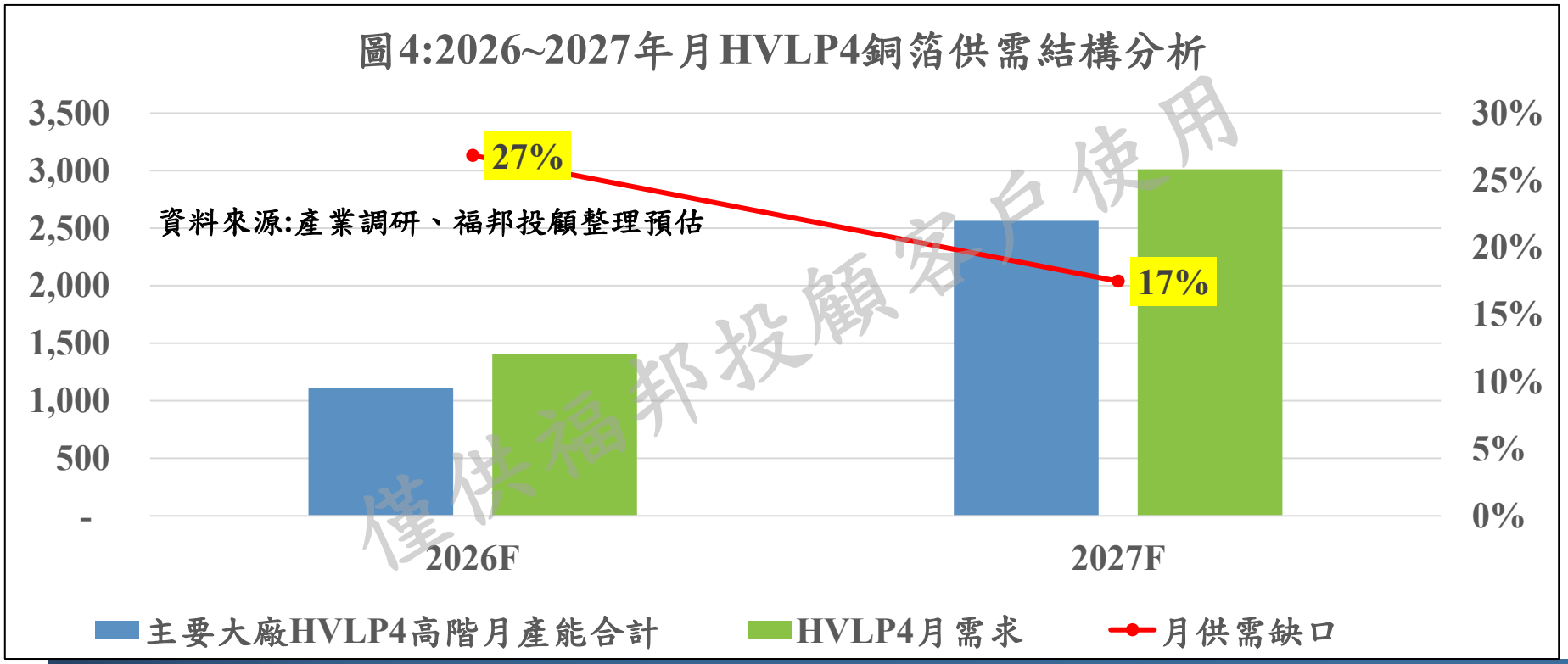
資料來源:產業調研、福邦投顧整理預估

	地區	產品分布	關鍵客戶	關鍵終端客戶	高階產能規劃
三井金屬	日	HVLP1~5	日:松下 台:台光電、台燿 中:生益	Nvidia/Google/AWS等	26Y:550噸/月 27Y:1,000噸/月 28Y:1,200噸/月 30Y:1400噸/月
古河電工	日	HVLP1~4	日:松下 台:台光電	Nvidia/Google/AWS等	26Y:100~200噸/月 27Y:200~300噸/月
福田金屬	日	HVLP1~5	日:松下 台:台光電、台燿、南亞 中:生益	Nvidia/Google	26Y:100~200噸/月
盧森堡	歐	HVLP1~4	韓:斗山 台:台光電、台燿 中:生益	Nvidia/AWS	26Y:100~200噸/月
金居	台	HVLP1~4	韓:斗山 台:台光電 中:南亞新材?	NVidia	26Y:330噸/月 27Y:700噸/月 28Y:800噸/月
德福科技	中	HVLP1~4	日:松下 中:生益 台:聯茂、台光電	NVidia/Google	26Y:200噸/月 27Y:500~600噸/月 (潛在1,200噸)
銅冠銅箔	中	HVLP1~4	中:生益、南亞新材 台:台光電、台燿	NVidia	26Y:50~100噸/月 27Y:200噸/月
長春	台	HVLP4	台:台光電	Nvidia/AWS	27Y:200噸/月
Lotte(樂天)	韓	HVLP2~4	韓:斗山	NVidia	26Y:300~350噸/月 27Y: 900~1,000噸/月

僅供福邦投顧客戶使用

銅箔供需缺口往2027年將縮小，供應廠商增加

- 根據供應鏈調查，由於供需缺口明顯，主要CCL廠商(台光電、生益科技等)積極導入新的供應體系，2026年主要生產廠商擴產後，可達1,200~1,300噸/月，2027年預估達2,300~2,400噸/月。(供應體系:台銅/古河/盧森堡/金居/福田/德福/銅冠/樂天)、台系長春後續有望驗證進入供應體系。
- **根據福邦投顧對於供需格局的假設預估，基於客戶加快認證相關可用產能，2026~2027年的供需缺口更新預估為27%(下修)/17%(上修)。**



僅供福邦投顧客戶使用

玻纖布使用種類

表11:玻纖布使用種類彙整

應用		元件	基板類型		材料等級要求	玻布類型	
						高階	中階
資通訊	AI伺服器/資料中心/交換機/基地台等	處理器	**封裝載板**	**CPU/GPU/ASIC**	**Low CTE**	**T**	**E**
				NAND	**Low CTE**	**T**	**E**
				DRAM	**Low Dk**	**NE/NER**	**E**
		主板	伺服器(AI/一般型)		Low Dk	NE/NER/Q	E
			交換機(核心/AI)		Low Dk	NE/NER/NEZ/Q	E
			網卡(NIC)		Low Dk	NE/NER	E
邊緣端	手機/平板等	處理器	封裝載板	AP/CPU	Low CTE	Ultra-Thin T, T	Ultra-ThinE
		非動態記憶體		NAND	Low CTE	Ultra-Thin T	Super Ultra-Thin E
		動態記憶體		DRAM	Low CTE	Ultra-Thin T(行動)	
				DRAM	Low Dk	NE(PC)	
		主板	主板		Low Dk	Ultra-Thin NE	Ultra-ThinE
		無線通訊	RF 封裝載板		Low Dk	Ultra-Thin NE	Ultra-ThinE
	DT/NB/AI PC	CPU/記憶體	封裝載板	CPU/GPU/NPU	Low CTE	T	E
		主板		DRAM	Low Dk	NE(PC)	E
	AR/VR	先進SoC	封裝載板		Low CTE	T	Super Ultra-Thin E
車用	EV/ADAS	先進SoC	封裝載板		Low CTE	T	E
		mmWave 雷達	模組載板		Low Dk	Ultra-Thin NE	E

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

高階電子布供應商概況:台光電開始取得日東紡料源

表12:電子玻纖布供應商與其主要客戶

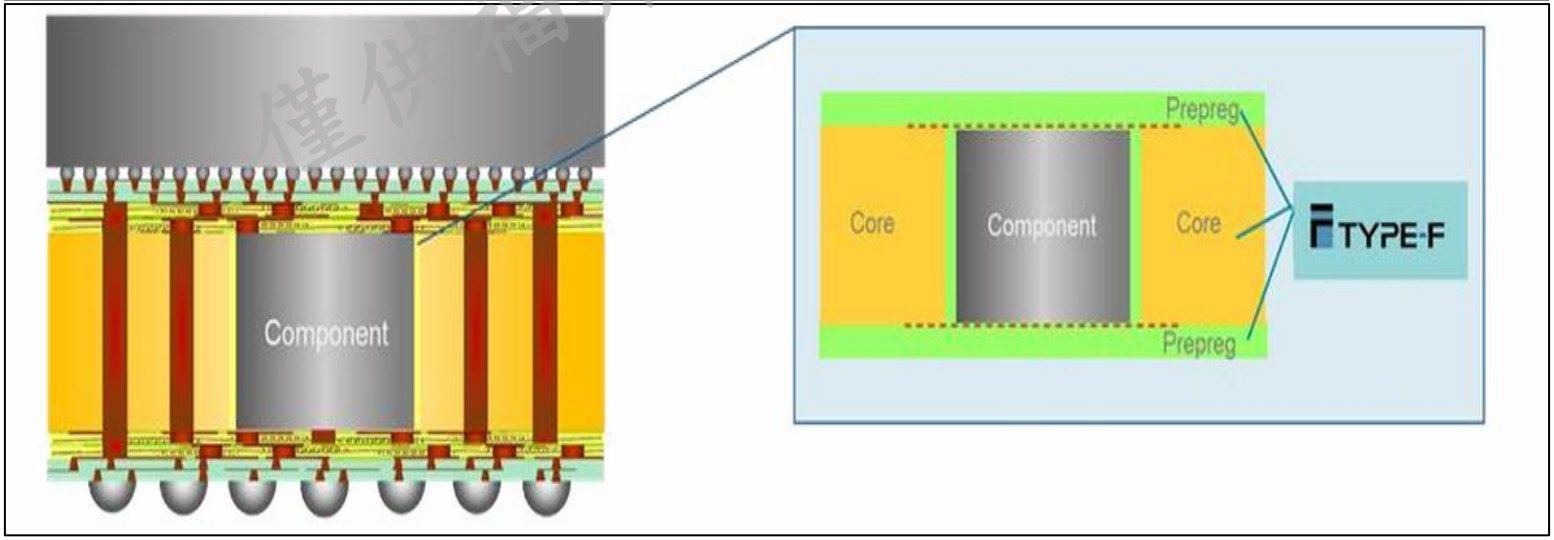
廠商	地區別	供應電子布產品	主要客戶
Nittobo	日	E/lowDK/LowCTE	• LowDK:松下、Doosan、TUC、南亞、EMC • LowCTE:Resonac、MGC、松下
Asahi	日	E/lowDK/LowCTE/Q	• LowDK:EMC、Doosan • Q:EMC、Doosan、松下
台玻	台	E/lowDK/LowCTE	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC
德宏	台	E/Q	• Q:EMC、南亞、聯茂
富喬	台	E/lowDK/LowCTE	• LowDK:EMC、Doosan、TUC、Asahi(紗) • LowCTE:Asahi(紗)
南亞塑膠	台	E/lowDK/LowCTE	• LowDK:南亞 • LowCTE:Resonac、MGC、南亞
泰山玻纖	中	E/lowDK/LowCTE/Q	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC • Q:EMC、生益科技、Asahi(紗)
光遠新材	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技
宏和科技	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技 • LowCTE:Resonac、MGC、EMC
建滔積層板	中	E/lowDK/T	• LowDK:EMC、生益科技
菲力華	中	Q	• Q:生益科技、EMC、松下、TUC
國際複材CPIC	中	E/lowDK/LowCTE	• LowDK:生益科技

僅供福邦投顧客戶使用

T-Glass:客戶積極導入抗翹曲設計

- 下個世代的半導體封裝需要更大的面積來放置更多元件、支持2D~3D封裝(CoWoS~SOIC)，因此材料的要求需要Low CTE、高彈性變量來達到低翹曲度 (Warpage)和連接可靠度。
- 根據Resonac白皮書，為了解決內埋元件載板對於核心層的厚度精度要求，較薄的核心層或不均勻的厚度，會導致內埋元件的損壞，因此之後許多載板核心的設計會採用厚Core+ low CTE 膠片 (採用T/Q Glass)。
- **2026年的TPU，2027年的Maia300都有見到這類核心Core CCL+PP的設計方案(Multilayer Core)，此也會多消耗T-Glass的供應量。**

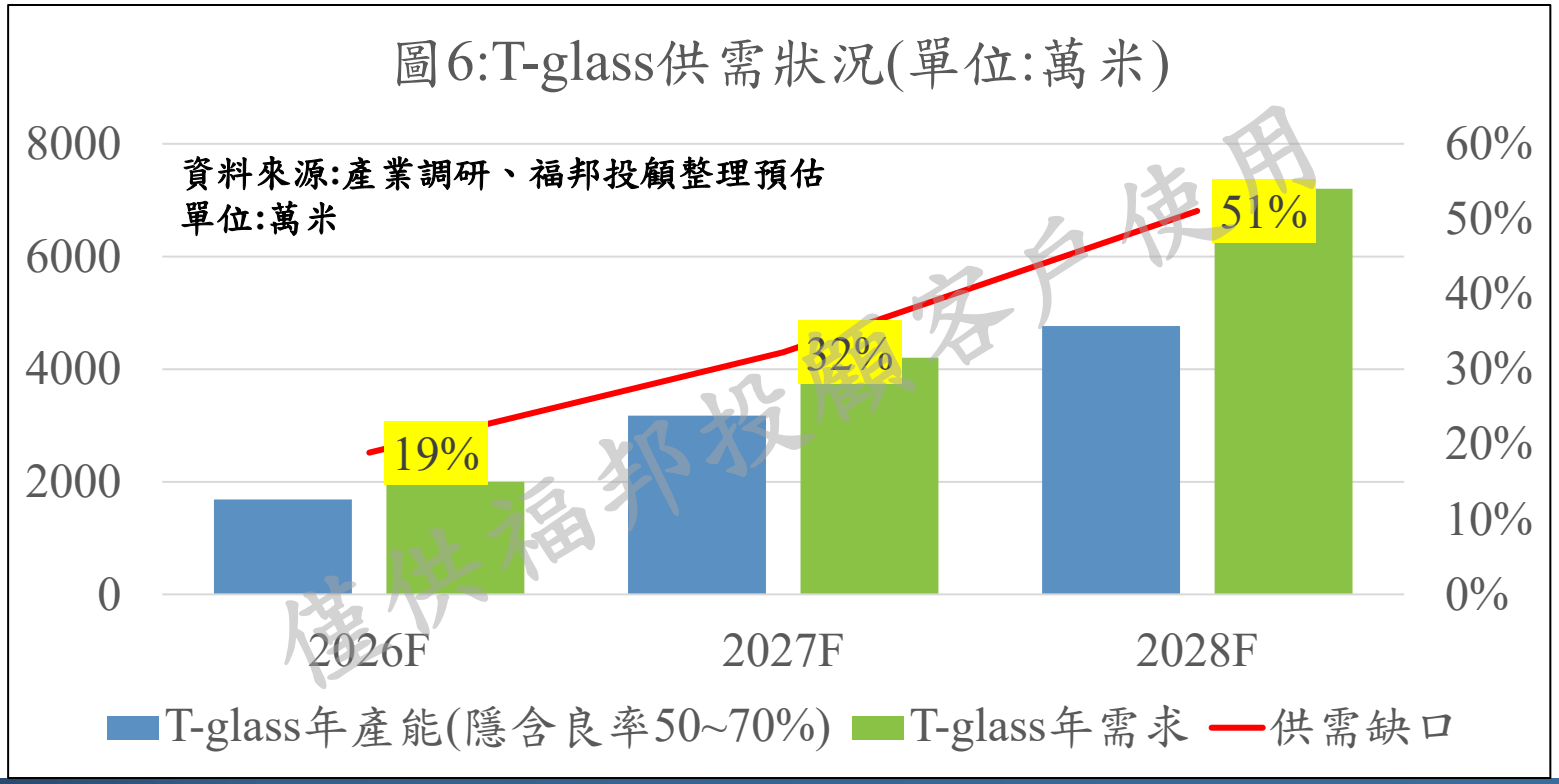
圖5:Resonac Type F產品



僅供福邦投顧客戶使用

T-glass至2027年仍將呈現供不應求格局，缺口30%+

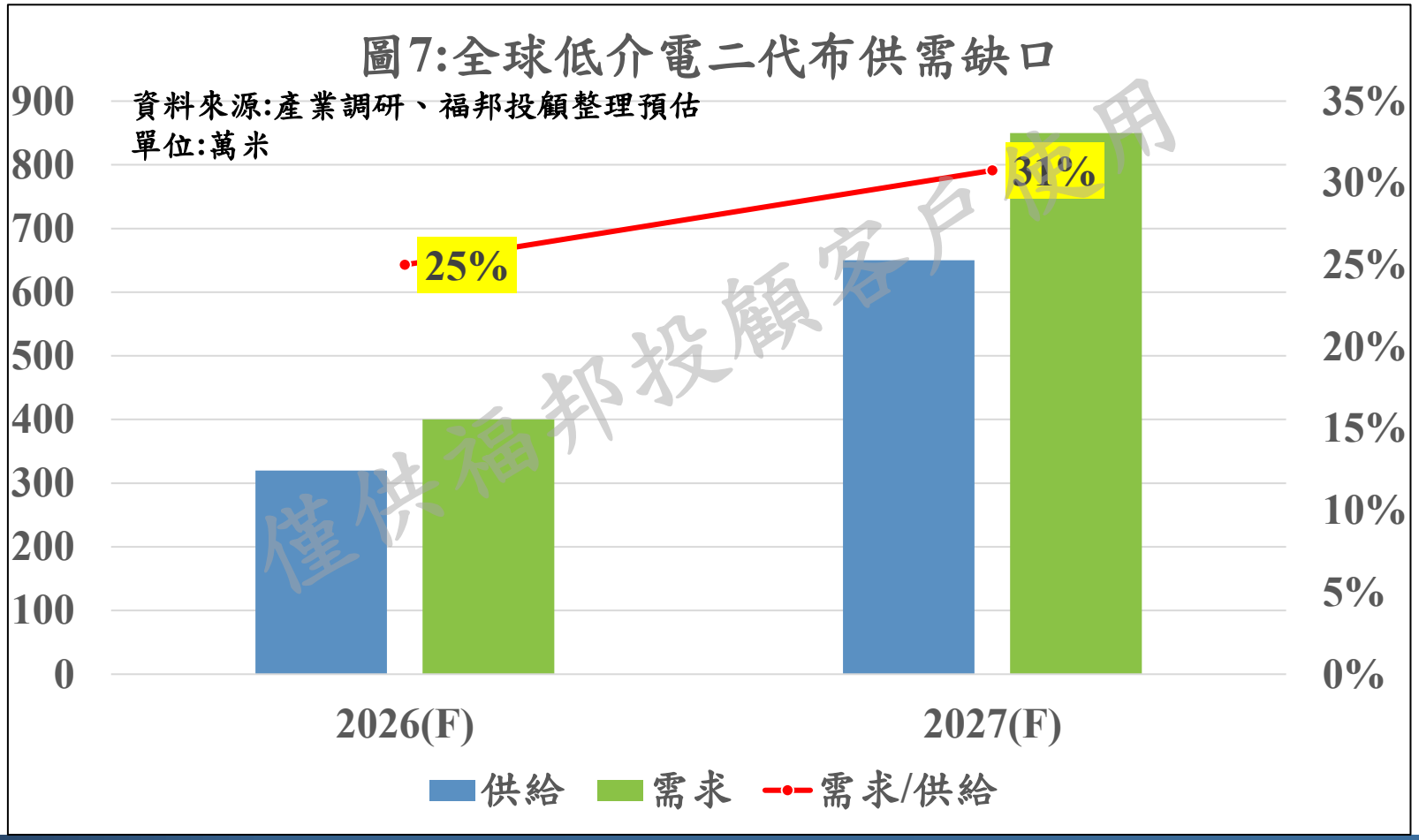
- 目前需求來看，T-glass需求中有70%用於FCBGA用的ABF載板，30%用於手機等FCCSP用的BT載板上。
- T-glass產能劃分來看，日東紡佔有全球主要產能的40%，台玻約15~20%，宏和約20~30%，其他(包含泰山玻纖等)15~20%。
- **福邦投顧預期儘管日東紡於2027年陸續開產能出來，但只緩解了Intel/AMD/Nvidia的用料，對於ASIC陣營的Broadcom/Marvell/MTK/Annapurna而言，仍難以取得足夠料源，維持T-Glass持續供不應求的判斷，缺口仍會逐年往上。**



僅供福邦投顧客戶使用

LowDK Gen2 2027年預期缺口31%

- 根據供應鏈訪查，2026H2開始，低介電二代布需求約400~500萬米/月，其中主要需求來源是Nvidia VR200、Google V8T/V8I。
- 以其銅箔基板廠商供應來看，主要以台玻、泰山玻纖、國際複材、Asahi為主，不足部分將由富喬、光遠、Nittobo補足，2027年供應量預估達650萬米/月以上。



僅供福邦投顧客戶使用

Q-Glass:LPU產品將率先採用

- 目前最新的NVIDIA LPX Rack從原先的全M9Q設計，改採用M8(K2)+M9(Q)的設計，配置為1:1，其中Q Glass主要採用10系列的極薄布，供應廠商少。
- 根據產業調查，預估NVIDIA LPU產品需求約在12~13K(2026~2027Y)，潛在Q glass需求約110萬米/月，隱含良率損耗則需要210~220萬米/月。

表13：LPU 主要供應資訊

	LPU Rack
PCB設計規格	52L(26+26)
Q-Glass供應商	菲利華、泰山玻纖、德宏
HVLP4供應商	三井、古河、盧森堡、金居
M9 CCL供應商	生益>台光電
PCB供應商	勝宏、Wus、TTM(30%)
組裝	鴻海

資料來源:福邦投顧預估

表14：LPU 供應試算

LPU 產品試算(單櫃)	
Tray 數量 per Rack	32
PCB/Tray	1
LPU/PCB	8
LPU總量(顆)	256
CCL 用量/PCB	8
總用量	256
M9Q/張 ASP(TWD)	10,000
CCL總產值(TWD) per Rack	2,560,000
CCL ASP per PCB (USD)	2,667
PCB ASP (USD)	4,444
Q布 ASP per PCB (USD)	1,440
Q布 ASP (USD)	45
Q布用量(米) per PCB	32

僅供福邦投顧客戶使用

Q-Glass：供應者寡，關注非日系供應

- 根據產業調查，統計台系(1家)、日系(2家)、中系(2家)產能規劃，2026年底預估產能超過300萬米/月，2027年預估產能可達600萬米/月以上。

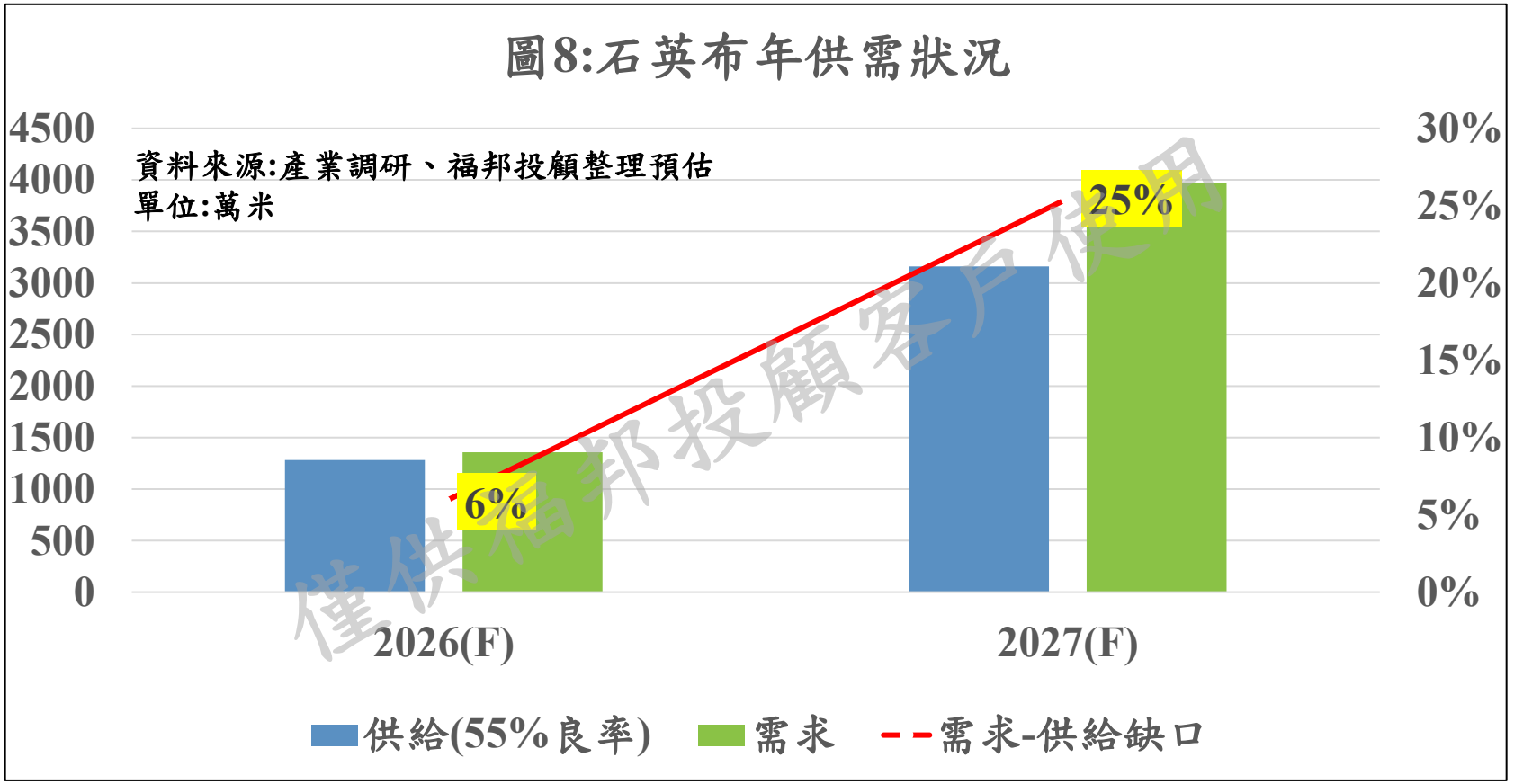
表15:Q-Glass供應商概況　　資料來源:福邦投顧整理

公司	地區	關鍵客戶	產能(K米/月)		說明
			2026F	2027F	
信越化學	日	日:松下、Resonac、MGC 台:台光電	75	240	外觀與同業不同，影響採用意願
Asahi	日	日:松下、Resonac、MGC 台:台光電	50	50	大客戶優先指定選用
泰山玻纖	中	台:台光電 中:生益科技	1,000	1,500~2,000	• 優先供應台光電
菲利華	中	日:松下 中:生益科技 台:台光電、台燿、南亞 韓:斗山	960	1,500~2,000	• 70%+供應生益科技 • 26H1 30W米/月 • →26H2 90~100W米/月
德宏	台	台:台光電、南亞、台燿(?)、聯茂 韓:斗山(?)	1,000	1,500~2,000	• 優先供應台光電 • 26H1 60W米/月→26年底100W米/月

僅供福邦投顧客戶使用

Q-Glass:26客戶進入庫存建立期，缺口下修至25%

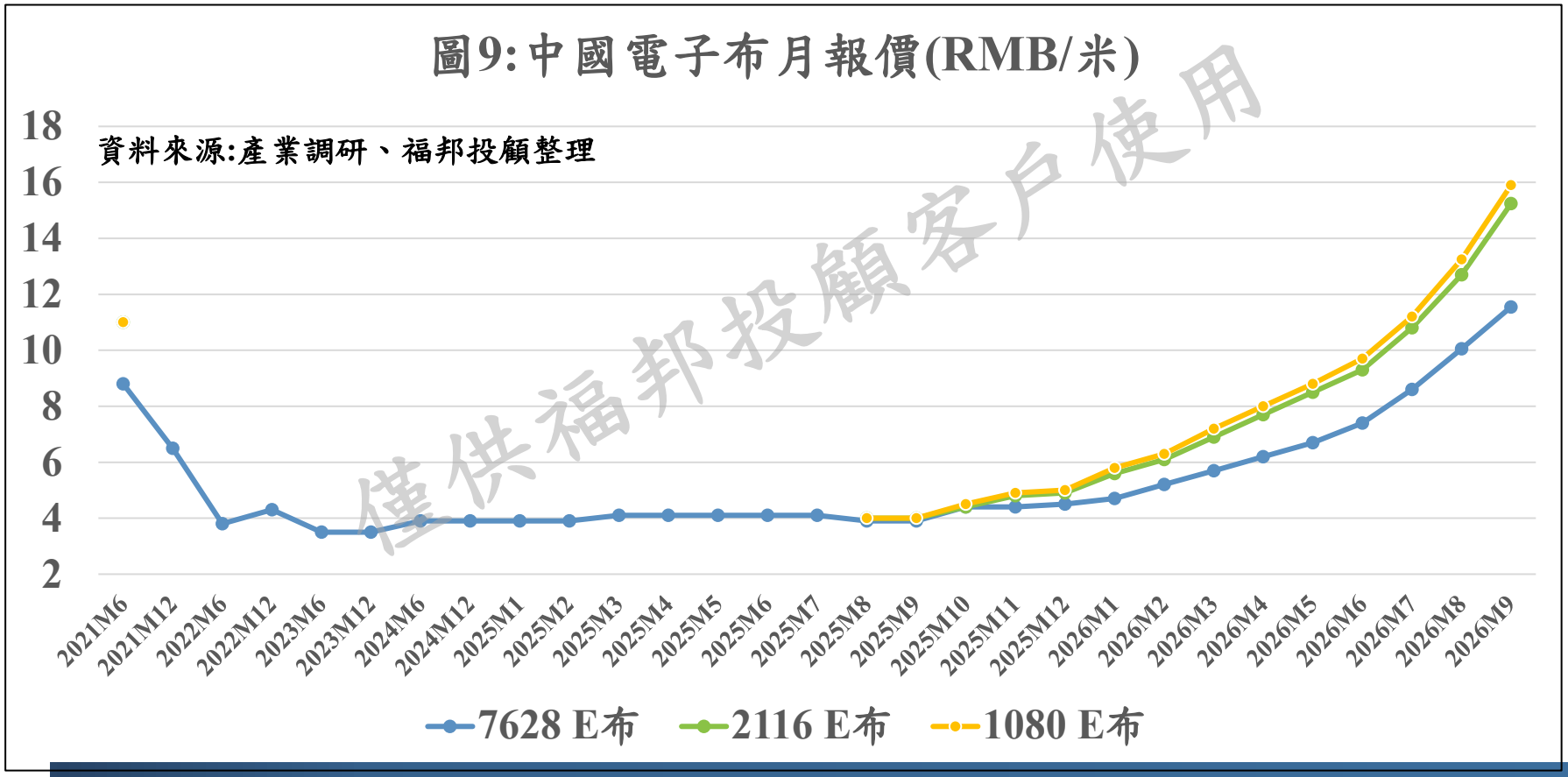
- 目前供應鏈的良率約在55~60%，26H2 進入Nvidia LPU 產品的備貨期，將帶動銅箔基板廠商的備貨需求，其中以**生益科技需求最為迫切，台光電供過於求。**
- ASIC陣營態度:目前Meta選用M9Q進行打樣，Amazon/MSFT選用M9-(K2)進行主板打樣，Google TPU V9至少選用M9以上材料。



僅供福邦投顧客戶使用

價格:E glass漲幅驚人，2H26將續漲至創新高價

- 玻纖布於2021~2022年週期係受到需求端推動帶動漲價，以7628系列為例，報價達到8.8RMB/米，本輪低階起漲，除受原物料之外，也受到主要供應商陸續轉做高階玻布(283，排擠低階布生產(打破供需平衡情況)，進而帶動漲價。
- 7628布累計至26Q3 YTD150%+，2116/1080 布YTD 210%+，均已創下歷史新高，2027年2116/1080的規格將繼續延續漲價趨勢。



僅供福邦投顧客戶使用

價格:高階電子布漲價，漲勢可望持續至2027H1

- 高階布種由於生產效率較傳統布下滑，以及良率因素影響，配合需求持續攀升下，每年均有約20~30%的漲幅，考量缺貨持續至2027年，漲價趨勢將延續至27H1。

表16:玻纖布漲價幅度

產品	目前ASP(USD/米)	2025	2026
LowDK	6~8	+20~30%	維持全年20~30%漲幅預期
LowDK2	18~22	+20~30%	• 26H1 YTD 漲幅+30~50% • **26Q3 漲幅+10%**。
T-Glass	23~30	+20~30%	• 日東紡26H1 YTD漲幅30% • 中系廠商26H1漲幅20% • **26H2預期會有10%+漲幅**。
Q-Glass	40~45	0%	26H1漲幅10~20%

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

銅箔基板漲價循環直至2027H2

- 目前AI相關CCL交期延長至4~6個月，上游料況吃緊（庫存1~2周），**福邦投顧預期考量上游材料漲價循環至少漲至27H1，將推動下游CCL客戶漲價續漲至2027年。**

表17:主要CCL廠商漲價幅度　　資料來源:產業調研、福邦投顧整理

公司名稱	地區	過去產品漲價幅度	26H2漲幅預期
Panasonic	日	• 26M5 高速CCL漲20%、FR4漲30%、 • PP漲15%	• 26M8 高速CCL漲10%、FR4漲15%、PP漲15%、載板材料漲5%
Resonac	日	• 26M3 CCL漲30%	
MGC	日	• 26M4 全品項漲40%	
台光電	台	• 26M3全品項漲15%+，M6以下漲20~25% • 26M4 M7以上漲5~10%	• 26Q3 M7以上漲10%。
台燿	台	• 26M4 漲幅20~40%。	• 26Q3 M7以上漲15%，M6以下上漲25%。
南亞	台	• 26H1 YTD+30%。	• 26Q3漲幅30~40%；26Q4會再繼續調漲
聯茂	台	• 26H1 YTD+40%+(4/6月)。	• 26H2每季預計漲20%+
建滔積層板	中	• 26H1累計五次漲價函，YTD 漲幅達50%+	• 26Q3漲幅30%+
生益科技	中	• 26M5 高速CCL /PP漲20%	• 26Q3漲幅30%+

僅供福邦投顧客戶使用

【僅供內部教育訓練使用，嚴禁外流】

Rubin Ultra帶起的材料變化-PTFE的方案發展

- 由於傳輸的需求(337G)，PPO或HC系列樹脂的M9無法滿足NVIDIA要求，NVIDIA 在Rubin Ultra 將導入混壓PTFE材料應用於SW Tray、Feynman 將導入純PTFE板材應用於正交背板。
- 目前銅板基板廠商正在積極送樣測試純無布的方案因應客戶需求。

表18:PTFE 方案組成　　資料來源:產業調研、福邦投顧整理

	PTFE CCL(無布) +CH PP(有布)	PTFE CCL(無布) +CH PP(無布)	PTFECCL +PTFE PP(無布)
樹脂配方	PTFE	PTFE	PTFE
銅箔搭配	HVLP4/5	HVLP4/5	HVLP4/5
供應商	三井金屬SI3-VSP		
電子布	LowDK 4(二代Q)	X	X
膠片配方	CH樹脂	CH樹脂	PTFE樹脂
供應商	生益科技、台光電、聯茂	生益科技、台光電	生益科技、台虹
電性	優於M10(CH樹脂系列)	次之	最佳
加工性	最佳	次之	最差
評論	備案，加工性較佳	供應鏈目前積極測試的方案	最終目標
PCB廠加工良率	30%	10~30%	10~30%
CCL供應商	走FCCL製程，生益科技SG5300N(塗佈+熱壓法)、台虹(熱壓法)		

僅供福邦投顧客戶使用

無布VS.有布方向選擇

- PTFE 由於Dk/Df的性能優於電子布，如果PTFE搭配電子布，會導致其Dk/Df轉差。

表19:PTFE與電子布的性能比較

	PTFE樹脂	LowDK 電子布	LowDK 3~4(1/2代Q-glass)
Dk(10Ghz)	~2.1	4.6~4.8	3.7~3.8
Df(10Ghz)	~0.0002	0.002~0.003	0.0002~0.0003

- 未來走向:

1. 有布方案(M9~M10搭配LowDK2~4電子布):若PCB上須放置CPU/GPU/ASIC等晶片，考量尺寸安定性因素，會需要搭配具有電子布的銅箔基板，例如UBB、ComputeBoard、SwitchBoard、MotherBoard。
2. 無布方案(搭配無布膠片):若僅考連傳輸性能(不放置晶片)，目前採取的無布方案，樹脂填矽粉/球型矽微粉的方案為主流，以改善膨脹性。

表20:PTFE的CTE比較

	PTFE樹脂	球型矽微粉	LowDK 3~4(1/2代Q-glass)
CTE(ppm/°C)	100~200+	0.4~0.6	0.5~0.55
說明		X/Y/Z軸的保護	X/Y軸保護

僅供福邦投顧客戶使用

不同填料對於銅箔基板的影響

- 在樹脂中加入填料能降低熱膨脹係數(CTE)、提高尺寸安定性和降低Dk/Df，近年的填料主要以球型粉料為主。

表21:各類填料的對應用途

填料	用途	相關填料供應商
二氧化矽粉(Si02)	降低CTE	日:電氣化學、Admatechs、龍森、新日鐵 中:聯瑞新材、錦藝新材、國瓷材料
陶瓷粉	調整Dk/Df	
氧化鋁	提高散熱性能	
碳酸鈣/滑石粉	通用材料	
氫氧化鈣/鎂	作為阻燃劑使用	
氧化鋁	改善導熱等	
碳酸鈣	降低複合材料成本	

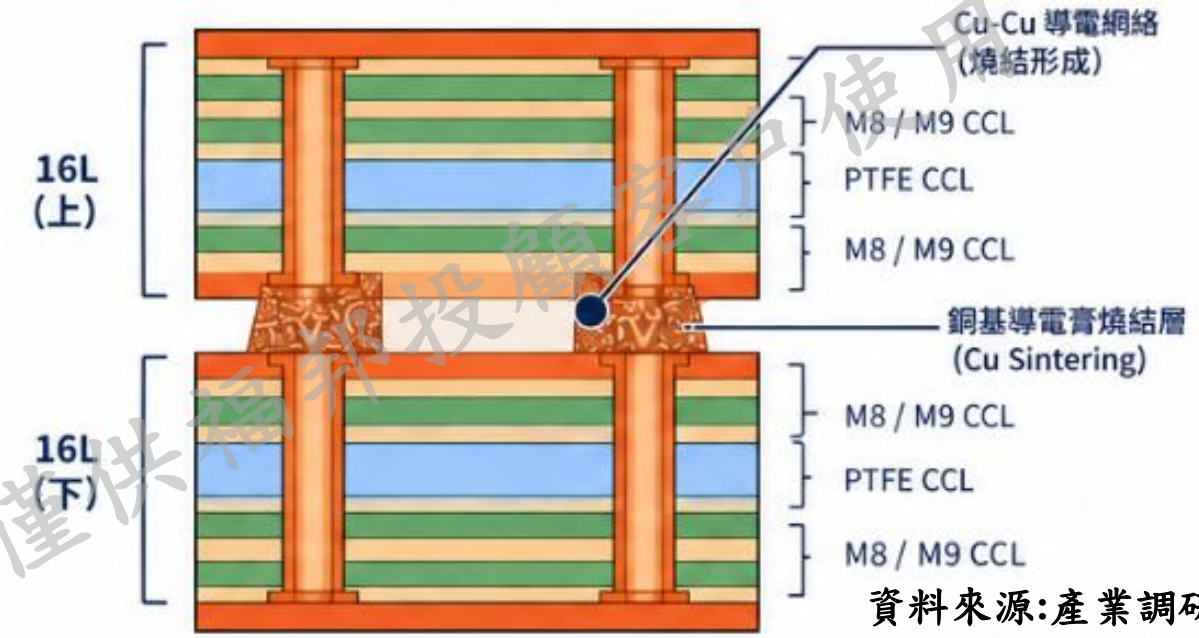
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PCB製造端的製程變化-銅漿燒結

- 銅漿燒結製程源於瞬態液相燒結(TLPS)技術，主要用於實現PCB盲/埋孔電路層間的Z軸互連。
- 過去傳統高多層多採用背鑽的製程，但在NV陣營，開始嘗試使用”銅漿燒結“的製程來處理高多層版的訊號連接，以解決背鑽精度控制的問題。
- 目前具有【燒結製程+大板製造能力】的主要為板廠，例如**深南電路、景旺電子、勝宏科技、滬電股份、生益電子；台系則有金像電具備此製程能力。**

圖10:PCB銅燒結概念示意圖



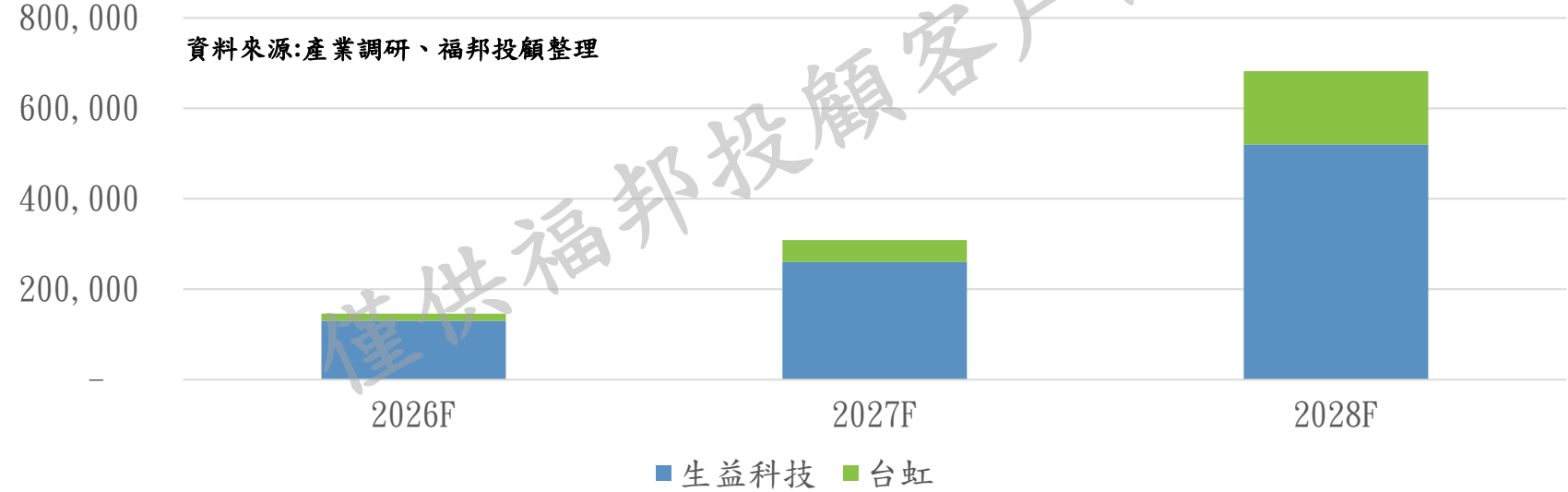
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PTFE供應商產能預期

- 目前Nvidia的供應來看，主要會以生益科技為第一供應商(合作最久)，第二供應商則從非中國供應商中選擇，目前與生益科技與台虹入列的優先供應商，**生益科技具有CCL+PP(PTFE/CH)的製造能力，台虹PTFE CCL (PFA)可搭配生益、台光電等CCL的CH PP方案，亦能搭配自產的PFA PP，可提供客戶彈性的產能需求。**
- 其餘具有PTFE樹脂CCL開發能力的供應商，開發進度較落後，不在考量範圍內。
- PCB段的生產: 以目前客戶的意願上，採用**無布方案**為優先目標，PCB生產良率約在10~30%，主要量產生產廠商預期為深南電路、景旺電子、勝宏科技、滬電股份；欣興僅參與配合RD階段(台郡負責燒結部分,但設備有限制)。

圖:PTFE主要供應商產能(米/月)



僅供福邦投顧客戶使用

PTFE CCL 潛在使用需求預估

- 假設: Rubin Ultra 於2027~2028年的晶片需求量預估400萬、900萬顆，所有機櫃均採用18層SWITCH TRAY設計，PCB板材採用混壓結構下，PTFE於2027~2028年的月均需求26、63萬張。

表22:PTFE CCL的需求試算

	SwitchTray	
產品	Rubin Ultra(VR300)	
層數	16+16=32L	
PTFE per PCB (米)	3	
Yield rate	30%	
實際用量	10	
	2027F	2028F
機櫃需求	35,000	84,000
單櫃SW Tray數	9	9
SW Board 量(張)	315,000	756,000
PTFE per PCB(米)	10	10
PTFE CCL 年總用量(米)	3,150,000	7,560,000
月均量(米)	262,500	630,000
產值預估(USD)	91M/月	220M/月

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

OpenClaw:加速從LLM往AI Agent 發展

- OpenClaw:開放式AI助理項目，可以根據使用者指令在應用程式和線上服務之間自行處理複雜任務，並被設計為可以代替使用者執行任務的AI軟體，並非過往的聊天機器人。
- OpenAI推出GPT5.4，此次更新提供Thinking和Pro兩種模型； Pro主要適用於需要進行複雜任務的使用者，Thinking模型可以在產出答案前提出思考計畫，使用者可以從中調整任務方向→強化Agent能力，可以支持智慧裝置執行複雜的工作流程。
- NVIDIA將推出類似OpenClaw的開放式平台-NemoClaw。

表23：LLM 與 Agent 推理比較

	傳統LLM推理	AI Agent 推理
流程	輸入→LLM(GPU)→輸出	感知→規劃→推理→工具與執行(CPU)
運作本質	單向生成	循環決策
互動對象	與使用者進行Q&A	與環境/工具/API互動
對硬體影響	主要消耗GPU算力	CPU需要介入成為主要調度，其中涉及沙盒隔離、儲存、記憶，複雜工具調用，

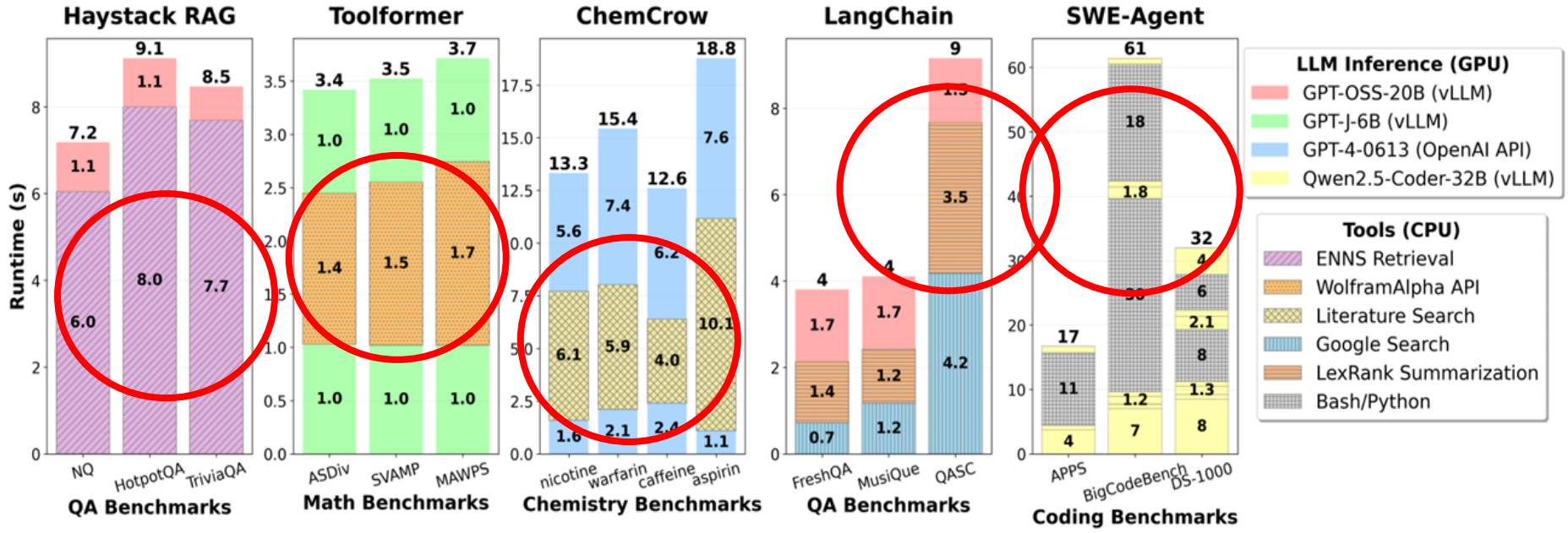
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

Agentic AI的瓶頸主要來自CPU

- 根據論文《A CPU-Centric Perspective on Agentic AI》一文指出，可以看出執行時間中CPU延遲占了大比例（紅圈），最高可達到90%以上；吞吐量瓶頸除了來自GPU，亦會來自CPU;能耗方面，大批次處理中(Batch size提升至128)，CPU平行處理效率低於GPU，佔到總能耗超過44%，延遲會從2.8秒提升至6秒以上，超過2倍。

圖11:Agent下，CPU佔據了大部分的延遲



資料來源:《A CPU-Centric Perspective on Agentic AI》 、福邦投顧整理

說明:Haystack RAG（問答檢索）**/Toolformer**（數學運算與 API 呼叫）**/ChemCrow**（化學文獻檢索）**Langchain**（網路搜尋與摘要）**/SWE-Agent**（程式碼與 Bash/Python 執行)

【僅供內部教育訓練使用，嚴禁外流】

CPU朝向多核發展→推升載板面積

- Agent執行:API處理、資料解析、腳本執行，90%由CPU完成，為了降低延遲，CPU往多核心發展是趨勢。
- 記憶體與上下文切換:CPU需要頻繁在不同Agent狀態做切換，多核心架構能讓每個Agent有獨立的處理單元，降低上下文切換帶來的效率損耗。
- 沙盒隔離/安全:Agent 需要大量獨立的一般伺服器建立Sandbox來執行，使用量與使用者和使用量正相關，需要更多核心數、高頻寬。

表24：伺服器CPU核心數

廠商	2023	2024	2025	2026	2027
Intel	Purley/Whitley	EagleStream	BirchStream	Oakstream	?
核心數	28/28	56	128	144~288	500?
AMD	Milan/Rome	Genoa/Bergamo	Turin	VENICE	VERANO
核心數	64/64	96/128	128~192	256	256~320?

資料來源:Intel、AMD、福邦投顧整理預估

【僅供內部教育訓練使用，嚴禁外流】

資料來源:各公司、福邦投顧整理

IC載板:主流晶片使用面積攀升

表25：主流晶片面積　　資料來源:各公司、福邦投顧整理

	產品名稱	推出時間	封裝方式	面積*層數	與前代的增幅
Intel	OAK Stream	2026	LGA	?	
	Birch Stream	2024H1	LGA	6,000*20	**+15%**
	Eagle Stream	2023	LGA	5,000*20	
AMD	**Venice**	**2026**	**FCLGA**	**6,100*26**	**+44%**
	Turin	2024Q4	FCLGA	5,500*20	-
	Genoa	2022Q4	FCLGA	5,500*20	
NV	**Rubin**	**2026~2027**	**CoWoS-L**	**9,000*18**	**+75%**
	Blackwell	2024~2025	CoWoS-L	5,900*14	**+116%**
	Hopper	2023~2024	CoWoS-L	3,200*12	
	Vera	2024~2025	CoWoS-R	5,700*18	**+46%**
	Grace	2023~2024		5,000*14	
Google	**HumuFish**	**2027**	**EMIB-T**	**14,000*24**	**+160%**
	Zebrafish	2026	CoWoS-S	5,300*24	**+22%**
	SunFish	2026	CoWoS-S	7,000*18	**+21%**
	GhostFish	2025	CoWoS-S	6,500*16	
Amazon	**Tranium4**	**2027**	**EMIB-T**	**14,000*24**	**+238%**
	Trainium3	2026	CoWoS-R	6,200*16	

僅供福邦投顧客戶使用

IC載板:產業能見度拉長

- 根據供應鏈訪查，目前主流載板對於供需的意見陸續在26H1看到供不應求，與之對應的生產相關輔材訂單能見度可至2028年。

表26：主流大廠對於供需的想法

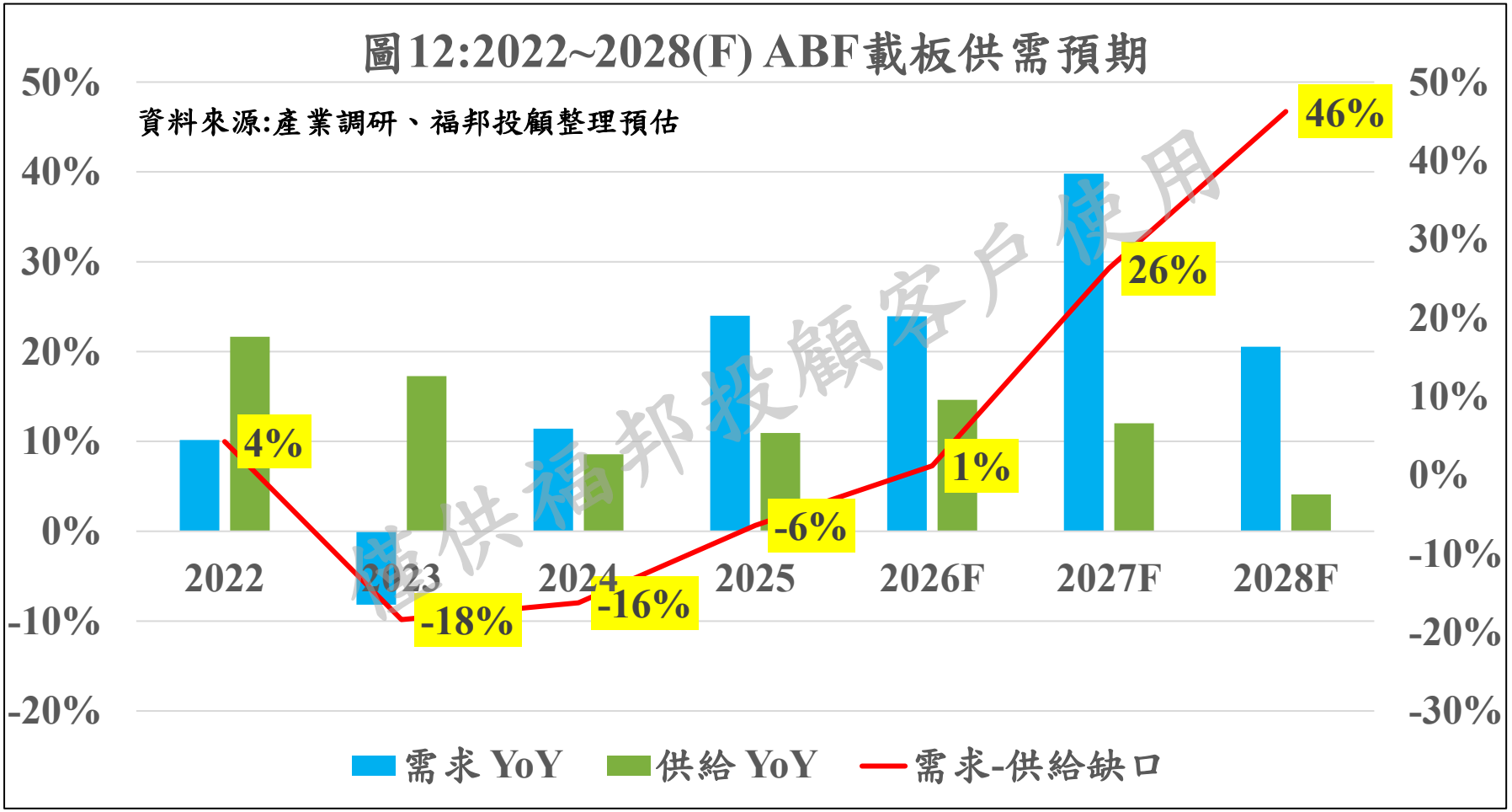
供應	公司	稼動率概況	對供需的想法	對應客戶
載板	Ibiden	AI相關滿載	2027年載板產能供不應求	NV/Intel/AMD
	Semco	26H2滿載	2027年載板產能供不應求	Intel/AMD/NV/AMZN
	AT&S	26Q1滿載	2027年載板產能供不應求	Intel/AMD/Marvell
	欣興	26Q1滿載	2026年玻纖布缺料，2027年供不應求	NV/Intel/AMD/Broadcom/AMZN
	南電	26Q1滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	Broadcom(主要)/AMD
	景碩	26H2滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	NV/AMD
生產輔材	鉅橡	26H2滿載	訂單能見度可至2028年	全球主要載板廠
上游材料	味之素	滿載	若需求持續暢旺，2027年產能嚴重不足：公司於2028年才有70%新產能開出	全球主要載板廠
	日東紡	滿載	2027年開始量產新產能。	全球主要載板廠
	Resonac	滿載	公司供給仍明顯低於需求，2027年有機會再次調價	全球主要載板廠

僅供福邦證券客戶使用

【僅供內部教育訓練使用，嚴禁外流】

IC載板:ABF載板供需將從2026年開始見到供不應求

- 根據福邦投顧預估，2025年ABF載板仍處於供過於求的格局，2026年開始在AI的需求強勢帶動影響下，ABF載板將開始見到供不應求格局，並於2027~28年逐步擴大。（**本圖不含缺T-glass影響**）



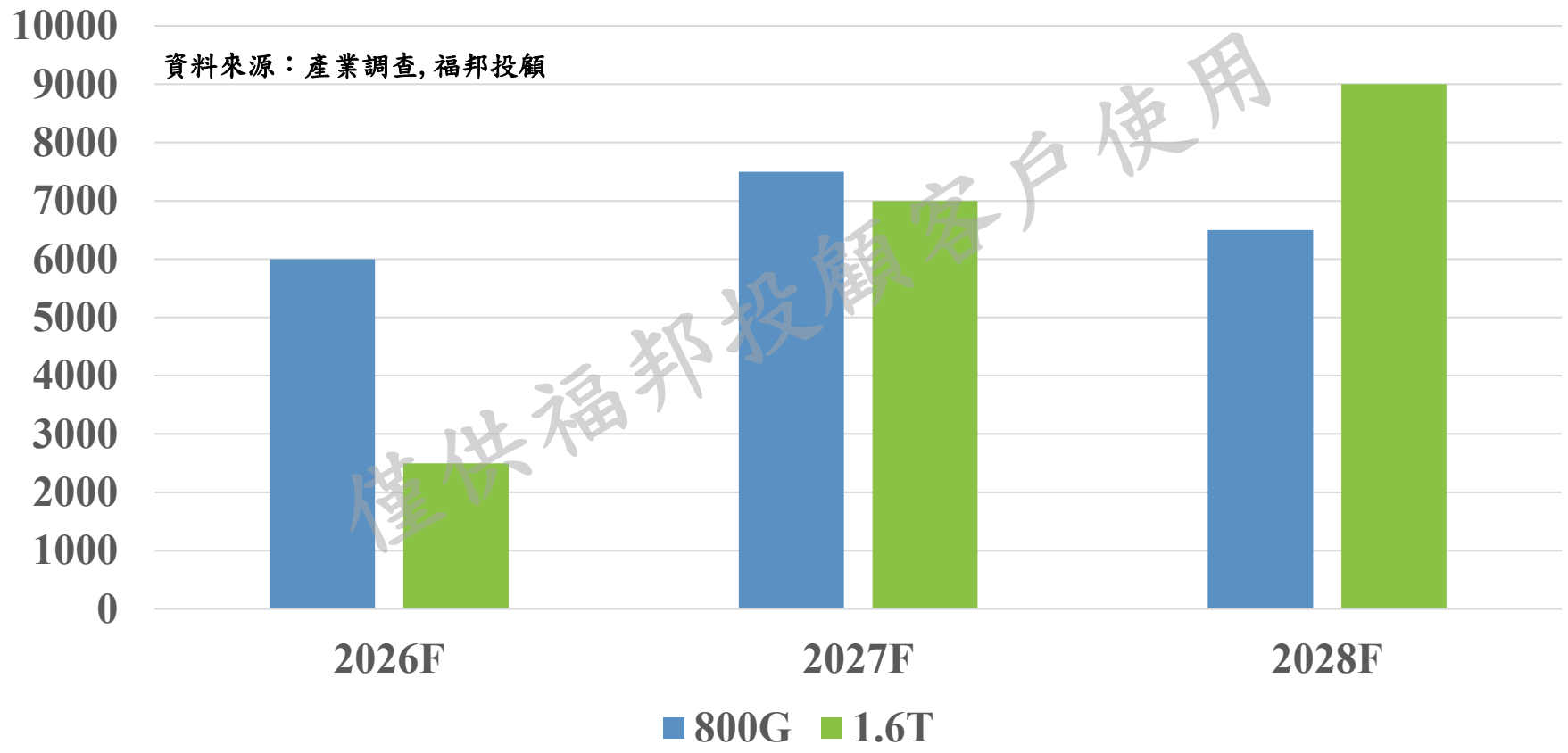
僅供福邦投顧客戶使用

可插拔光通訊模組:800G/1.6T潛在需求上升

■根據福邦投顧產業調查，2026/27/28年800G的採購”需求”可望達到6,000/7,500/6,500萬支，1.6T需求預估2,500/7,000/9,000萬支。

(隱含終端客戶對於良率的考量造成的Overbooking)

圖13: 2026~2028年光通訊模組需求預期(萬支)



僅供福邦投顧客戶使用

傳統光通訊模組板-HDI/mSAP供應體系的新機會

■因應高度整合和小型化需求，光模組開始大量採用MSAP來縮小產品體積和損耗；進入1.6T，L/S 要求微縮至**20μm-25μm**；因此1.6T開始，強制採用MSAP。

■材料規格上，亦往M7~M8等級以上升級，但在MSAP部分，會大量採用載體銅箔(三井金屬90%市佔率)、日東紡的Low CTE布，目前有面臨缺料的情況。

表27 :400G~3.2T PCB方案比較

速率	400G	800G	1.6T	3.2T
製程	HDI	HDI/mSAP	mSAP	mSAP
層數	10L	12~14L	14~18L	20-22L
L/S	40~50μm	30~40μm	20~30μm	10~15μm
CCL等級	M4/M6	M6/M7	M8 (斗山/台光電)	M9~M10+陶瓷基板?
ASP(USD)	~10	12→14~15	30→35	150+
毛利率	20~30%	40~50%	50~60%	70~80%
供應鏈	SCC、欣興、興森等	臻鼎/SCC/欣興/方正/景旺/華通/WUS等	臻鼎、SCC、欣興、華通等	臻鼎、SCC、欣興

資料來源：產業調查、福邦投顧

僅供福邦投顧客戶使用

傳統光通訊模組PCB供應商概況:27年可望緩解供應

表28 :主要供應商客戶與產能概況　　資料來源：產業調查、福邦投顧整

供應商	關鍵客戶	mSAP月產能規劃預期	客戶2027年月產能(支)
欣興	中:中際旭創 美:Coherent、Mellanox	26Y:7,500平米(=250~300萬支) **27Y: 10,000平米(=400萬支)**	■ 中際旭創: 800G/1.6T 125W/210W ■ 新易盛: 800G/1.6T 80W/35~40W ■ Coherent: 800G/1.6T 125W/125W ■ AAOI:90~100W ■ 華工科技: 800G/1.6T 30W/25W ■ 索爾斯:80W
深南電路	中:中際旭創、新易盛 美:Coherent	26H1:3,200平米(=120~130萬支) 26H2:6,000平米(=240萬支) **2027Y:8,000平米(=320萬支)**	
臻鼎	中:中際旭創	26Y: 2,800平米(=110萬支) **27Y:10,000平米(=400萬支)**	
華通	中:中際旭創 美:AAOI、Coherent	27Y:7,500平米(=300萬支) **28Y: 8,000~8,500平米(=330~340萬支)**	
方正科技	中:新易盛、 華工科技、光迅科技	26Y:1,500~2,000平米(=80萬支) **27Y:5,000平米(=200萬支)**	
景旺電子	中:中際旭創、新易盛 美: Coherent	26Y:1000平米(=40萬支) **27Y:10,000平米(=400萬支)**	
興森科技	中:華工科技、光迅科技	28Y:擴產10,000平米/月產能	
勝宏科技	潛在客戶:中際旭創	規劃20,000平米/月	
AT&S	中 ： 中際旭創 美:Coherent、Mellanox	26Y:2800片/月	

僅供福邦投顧客戶使用

雲端資料中心採用之交換器品牌

■雲端資料中心多採用品牌+白牌的混合方案

■Spine和DCI 因為結構複雜，所以多採用品牌廠，但Google和Amazon 由於架構和軟體上較強，所以選擇白牌來配合他們的結構。

■TOR和 Leaf 結構簡單，可以看到較多白牌(裸機、白盒)的方案。

■xAI的交換機品牌主要採用Juniper的方案，主權AI體系(ex:星際之門、中東等)多會採用Cisco等品牌交換機。

表29:各大雲廠商交換機廠商彙整

	TOR(機櫃)	Leaf（接入）	Spine（匯聚）	DCI（核心）
GOOGLE	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌/Juniper/CSCO
Amazon(AWS)	白牌(CLS/Accton/CSCO)	白牌(CLS/Accton)	白牌(CLS/Accton)	Juniper
微軟(Azure)	Arista/CSCO/Dell	Arista/CSCO/Dell	Arista	Arista
Meta	白牌(CLS/智邦/CSCO)	Arista/Edgecore（智邦）	Arista/Edgecore（智邦）	Arista/Juniper/白牌
百度	白牌/H3C	華為/H3C	華為/H3C	華為/H3C
阿里巴巴	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	華為/H3C
騰訊	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為	華為/H3C	華為/H3C
APPLE	Arista/CSCO/白牌	Arista/CSCO	Arista	CSCO/Juniper

資料來源：650 GROUP、福邦投顧整理

【僅供內部教育訓練使用，嚴禁外流】

交換機板:主流AI層數較低，傳統乙太網多層為主

■ 資料中心需要承載流量大+高速傳輸，在此需求下，高多層板成為此需求的解決方案，其優點在於電子元件間連線縮短下，訊號傳輸速度提高，而且也方遍佈線。

■乙太網交換機為各家CSP+主權主要Scale out的主流，800G報價是400G的2倍以上，1.6T是800G的4倍以上。

表30 :交換機供應鏈劃分

板層數劃分		產品	銅箔基板主要供應商	PCB主要供應商	出貨價格
Cisco	400G:24~36L PTH 800G:38~48L PTH (NPO:28L) 1.6T:46~52L PTH / 18~20L HDI	800G	台光電/松下/Doosan	金像電、TTM、滬士電、ISU	400G :1500 800G:4,000 1.6T:16,000+ (美金/平方米)
		1.6T	台光電/松下		
Arista		800G	台光電、Doosan	TTM、滬士電、金像電、ISU	
		1.6T	台燿、台光電		
Celestica		800G	台光電/Doosan/松下/台燿	金像電、ISU	
		1.6T	台光電/松下/台燿	金像電、ISU	
Accton		800G	Doosan、台光電	TTM、ISU、LCS	
		1.6T	台光電	TTM、ISU、LCS 金像電	
Mellanox		800G	台光電	TTM、滬士電	
		1.6T	台光電	TTM、滬士電	

福邦投顧產業調查

交換機的材料

■在交換機升級的過程中，由於高速傳輸需求，對於耗損也是嚴格規範，**800G材料價格則會較400G材料增加60%+，Tomahawk6 將會使用於2026年的1.6T交換機可使用材料約M8U~M9-(碳氫+LK2)等級，2027年後的規格可望往M9發展。**

表31:各速率材料概況

	100G	400G、**800G**	800G、1.6T	1.6T~2.4T	3.2T~
材料耗損等級	Very low loss	Ultra low loss	Extreme Ultra low loss	Super low loss	?
Df	0.005~0.006	~0.003	~0.002	~0.001	~0.0003
頻率	13Ghz(NRZ)	13Ghz(PAM4)	28Ghz(PAM4)	56Ghz(PAM4)	100+Ghz?
設計	PPO	PPO+Low K Glass	PPO+LowK/K2 +VLP2~4	PPO+CH樹脂+ VLP4 +Q/LK3/LK2	CH/PTFE樹脂+LK4/5 +VLP4/5
平均單價/張	2,000~3,000	**3,000~4,000**	**5.000~6,000**	6,000~7,000/ 10,000	12,000+
代表型號	松下M6	松下M7	松下M8	松下M9	松下M10
競爭同業產品儲備	EM-528/626	EM-528K/EM-890K	EM-892K/K2	EM-896K3	EM-898K4
	TU883	ThunderClad2/3	ThunderClad4	ThunderClad 5	?
	IT 968/968G	IT988G	IT998GSE/2	IT999GSE3	ITQ10G(K4)
	NP186/188U,H	NPG188UK/199	NPG199K/HK	NPG-998Q	NPG-999Q
	Synamic 6/6N	Synamic 8GN	Synamic 9GN	Synamic 10/11GQ	SG5300N
	DS7409DV（N)	DS7409DJG(N)	DS7409DQN/N2/T2	DS7409DYQ/L2/N	?
	NY6300S	NY-P1~3	NY-P4	NY-P5	NY-P6

資料來源：各公司，福邦投顧整理

謝謝指教
Q&A

【揭露事項與免責聲明】

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【富邦法人】台積電(2330)外資報告彙整

- **富邦&外資 preview 彙整**

券商	Rating	TP	2026F EPS	2027F EPS	2028F EPS	摘要
富邦 (10/12)	Buy	$3,500	$112.44 ↑ (from $108.80)	$161.21 ↑ (from $155.41)	$207.9	- 預期毛利率將超越先前指引，估 **3Q26 毛利率達 67.6%，EPS 估 29.97 元**。 - 預期管理層將指引 **4Q26 營收季增 10%至 12%**，毛利率則維持持平。受惠於下半年強勁表現，預估全年營收將年增 45%至 46%，有望再次超越原先指引。 - 資本支出方面，預期台積電維持全年 620 億美元的指引，估計 2027 年及 2028 年資本支出將大幅提升至 850 億及 1,000 億美元。 - 2027 年供應缺口將進一步擴大，因此認為台積電將於 2027 年再次調漲晶圓價格，平均漲幅超過 10%，其中先進製程節點漲價 10-15%，成熟製程節點漲價 5-10%。隨著 ASP 正向趨勢，認為 FY27 毛利率有望進一步攀升至 70%。
MS (10/5)	OW	$3,088 ↑ (from $2,988)	$115.32 ↑ (from $111.08)	$152.74↑ (from $144.93)	$189.91↑ (from $183.55)	- 預估 **3Q26 毛利率 67.5%，EPS 30.74 元**。 - 預估 **4Q26 營收季增 12.5%，毛利率 68.6%，EPS 估 35.25 元**。 - 預期法說會將全年營收成長指引上調至 low-to-mid-40%，較目前超過 40%的指引進一步提高，意味第 4 季營收可望季增 10%至 15%，主要受強勁的 3 奈米需求帶動。 - 預估 2026 年資本支出維持 540 億美元，2027 年則預估達 750~800 億美元。 - 2027 年晶圓價格方面，預期 7 奈米以下先進製程價格將上漲 5%至 10%，成熟製程則上漲 0%至 3%，帶動台積電 2027 年毛利率有機會接近 70%。
GS (10/5)	Buy	$3,300 ↑ (from $3,100)	$110.77 ↑ (from $109.21)	$150 ↑ (from $140.91)	$195.67 ↑ (from $180.83)	- 預估 **3Q26 毛利率 67.5%，EPS 29.07 元**。 - 預估 **4Q26 營收季增 11.0%**，AI 及 HPC 需求持續強勁，但 N2 製程持續擴產將帶來些微毛利率稀釋，**預估 67.3%，EPS 估 32.38 元**。 - 2027 年毛利率預估小幅提升至 67.5%，受惠於價格提升、有利產品組合、高稼動率及持續生產效率改善。 - 2026 年資本支出預估維持 640 億美元，但 2027 年及 2028 年分別上調至 850 億美元及 980 億美元，主要反映設備供應商成本上升，以及可能新增德州擴廠的初期支出。預估 2027 年底及 2028 年底 N3/N2 產能分別達到每月 20 萬片/14 萬片及 22 萬片/20 萬片。

Citi (10/7)	Buy	$4,000 ↑ (from $3,800)	$110.72 ↓ (from $110.73)	$158.30 ↑ (from $151.93)	$217.02 ↑ (from $207.24)	- 預估 **3Q26 毛利率 67.4%，EPS 33.16 元**。 - 預估 **4Q26 營收季增 16%，毛利率 66.7%，EPS 估 28.24 元**。 - 台積電在 AI 運算需求強勁、Agentic AI 潛在上行空間及 CPO 帶動新一輪網路設備週期的支持下，預期 2027 年營收可望年增超過 40%。隨著 N2 全年貢獻增加，以及 AI 運算工作負載自 2026 年下半年起逐步轉向 N3，並於 2028 年進一步轉向 N2 及 A16。 - 儘管近期市場傳出台積電可能與德州 Terafab 合作，但台積電目前已規劃亞利桑那州 6 期先進製程晶圓廠及 2 期先進封裝產能，管理層過去也曾表示，長期目標是讓亞利桑那州具備約 30%的先進製程產能，以滿足美國需求。第二及第三期預計於 2027 年底至 2028 年開始量產 N3 及 N2，第四至第六期則預計逐步建置至 2030 年，因此未來幾年暫不將德州新增產能及投資納入預估。
JPM (10/8)	OW	$3,300	$108.62 ↑ (from $108.15)	$147.80 ↑ (from $141.37)	$190.42 ↑ (from $180.01)	- 預估 **3Q26 毛利率 66.8%**，主要受惠於 N3/N5 產能利用率維持高檔及晶圓漲價，但部分受到 N2 初期量產、海外晶圓廠稀釋效應及下半年新台幣升值影響。**EPS 估 27.92 元**。 - 估 **4Q26 營收季增 11%，毛利率 66.8%，EPS 31.37 元**。 - 估 2026 年美元營收將年增 41%，新台幣營收年增 44%，全年毛利率約 66.9%。 - 預估 N3 產能將於 2026 至 2028 年底分別達到每月 17.8 萬、22.3 萬及 25 萬片。N2 方面，台積電為避免重演 N3 供不應求的情況，正加速擴產，預估 2027 及 2028 年底產能將分別達每月 11.8 萬及 21 萬片。此外，A14 進度優於原先預期，預估最快可於 2028 年下半年量產，早於市場普遍預期的 2029 年。 - 預估 2026 至 2028 年資本支出將分別達 640 億、860 億及 1,000 億美元，其中 2028 年仍有上修空間。 - 預期 2027 年漲價將涵蓋多數製程，先進製程漲幅約 10%，N7 及以上成熟製程則調漲低至中個位數百分比。預估 2027 年整體平均售價將年增 25%，高於 2026 年預估的 17%。

UBS (9/28)	Buy	$3,650	$109.06 ↑ (from $108.96)	$148.98 ↓ (from $149.14)	$202.52 ↑ (from $198.23)	- 預估 **3Q26 毛利率 67.1%，EPS 28.47 元**。 - 預估 **4Q26 營收季增 10%**，受惠於旗艦手機季節性備貨及 HPC 需求加速放量，但考量 N2 初期量產及海外擴廠的成本稀釋效應，**毛利率預估持平於 67.0%，EPS 估 31.26 元**。 - 雲端 AI 長期趨勢持續推動台積電營收成長，預估 2026 年營收將年增逾 40%，2027 年也可望維持約 40%的成長。近期產業調查顯示，雲端 AI 需求尚未出現降溫跡象，且 2027 至 2028 年 CPU、GPU 及 ASIC 需求展望進一步轉強。 - 資本支出方面，維持 2026 年預估 630 億美元，並將 2027 及 2028 年預估由原先的 800 億及 950 億美元，上調至 900 億及 1,050 億美元。主要考量台積電加速擴充 N3 及 N2 產能，並提前 A14 設備交付時程。將 2028 年底 N3 及 N2 月產能預估分別上調至 22 萬及 21 萬片。
廣發 (10/7)	Hold	$2,960 ↑ (from $2,700)	$110.9 ↑ (from $108.2)	$148.7 ↑ (from $138.3)	$184.6	- 預估 **3Q26 毛利率 67.8%，EPS 29.2 元**。 - 預估 **4Q26 營收季增 12%，毛利率小幅降至 67.2%，EPS 32.4 元**。 預計維持 2026 Capex 指引；考慮設備通脹，上調 2027E/2028E Capex 至 US$83bn/95bn。 - 台積電可能與 Terafab 合作的敘事出現變化，此前報道台積電負責運營晶圓廠或成立 JV，及為 Tesla/SpaceX 提供專屬產能；但 Elon Musk 10/7 表示 Tesla/SpaceX 將自行“build and run the fab”，認為市場預期將重新調整，而 Intel 仍參與 Terafab 項目。

- **上季法說重點(2026/7/16)**

2Q26 法說會重點摘要	
Revenue	**- 3Q26 指引：US$44.6-45.8bn(中值 QoQ+12%、YoY+37%)；匯率假設為 USD/TWD 32** **- 將 2026 年全年美元營收成長預估上調至略高於 40%**
GM	**- 3Q26 指引：65-67%，QoQ-1.7ppts** **- 2Q26 毛利率 67.7%，QoQ+1.5ppts 並略優於公司財測，成本改善措施與整體稼動率略高於預期，但部分被海外廠初期營運稀釋抵銷** **- 2nm 將於 2H26 快速 Ramp-Up，稀釋毛利率 3~4%** **- 海外廠未來幾年前期每年約稀釋公司毛利率 2~3%，後期擴大到 3~4%**
OP Margin	**- 3Q26 指引：56-58%**
2Q26 product mix	**- N2 3%；N3 30%；N5 33%；N7 11%；先進製程技術(N7 及以下)占晶圓營收 77%** **- HPC 66%(QoQ+20%)；Smartphone 22%(QoQ-4%)；IoT 5%(QoQ+4%)；Automotive 4%(QoQ+15%)；DCE 1%(QoQ+5%)**

CAPEX&折舊	**- 2026 年 CAPEX 上調至 US$60-64bn(原為 US$52-56bn，接近上緣)；70~80% 於先進製程，10% 特殊製程、10~20% 先進封裝、測試光罩** **- 資本支出通常對應未來數年更高的成長機會，目前看未來三年 CAPEX 會比過去三年再更為強勁** **- 已提前與設備供應商規畫產能，目前不預期設備供應會成為擴產計畫的重大瓶頸** **- 宣布將在亞利桑那追加 US$100bn 投資(總額提高至 US$265bn)，新增數座 N2 及更先進製程邏輯晶圓廠、先進封裝廠**
AI	**- 先前提出的 2024-29 年 AI 相關營收「中高 50% CAGR」展望，管理層表示需求已變得「更強、更強、再更強」，因為展望持續在調高所以不公布數字，實際趨勢高於先前說法**
先進封裝	**- 目前主流方案仍為 CoWoS，同時持續開發其他替代技術以降低封裝成本** **- 對 Intel EMIB-T 等先進封裝競爭，管理層認為後段和前段是完全不同等級，競爭者取得封裝訂單不代表可切入前段邏輯晶圓**

Sales Trader Commentary: TAIEX 49k Wobble - July De Ja Vu, or Just Pausing Before the Santa Rally? | GSTW

13 October 2026 | 4:11 PM Taiwan Standard Time

Robert Wang
Goldman Sachs (Asia) L.L.C. Taipei Branch
FICC & Equities

Index lost the 49k points threshold and saw the biggest decline on a point basis since the end of July. Market seemed to lose the steam to break through 50k points after the past choppy sessions. Is this July de ja vu or some healthy pullback? Foreigners and domestic investors seem to share different narrative. Check out below for our investor feedback.

Key Investor Feedback and Observations

- **Positioning**. we saw meaningful outflows today from all kinds of investors today, again focusing in ex-TSMC large caps. Today foreigners **net sold US$5.2bn** today, the second largest in the history (simultaneously drove TWD weakness over 32.23 today). Domestic investors, especially retail, on the other hand, anchored the market with the record high margin loan balance (as of 10/12/2026). Local funds remained buyer as well mainly thanks to ETFs inflow.
- **Macro**. CPI headline brought uncertainty and forced global investors to turn cautious, which was exacerbated by high yields and crude. For Taiwan, AI trade is relatively resilient to the Middle Eastern tension and high energy prices. However, at some point, it is going to have a real economic impact either through Fed or some other macro data.
- **Sentiment**. Surprisingly, people are still bullish in Taiwan, though they also acknowledge some choppy trading sessions ahead. Local investor feedback is not calling for capitulation; in fact, most believe this is just another "two-steps forward one-step backward" cadence towards another record high. Nonetheless, some retail start taking chips off the table. This echoes our observation among the foreign investors. Yes, the AI fundamental future is all sunshine, but as we head into the end of year, it is natural for them to protect their great fun thus far this year instead of adding on immediate risk. ***I'd imagine the next few weeks are going to be full of 1) rapid sector rotation; 2) weak or unsustainable momentum before Santa comes into town after the midterm election for our year-end rally Christmas gift.***
- **TSMC Earnings and Others**. We know the numbers are going to be good, and we also know people are going to sell-on-facts after TSMC earnings call; recall most recent one happened right after the meeting that scarred a lot of investors since. Perhaps this time people started trimming ahead (and maybe too early?), which provides some window for a quick rebound into small and mid cap. OCP summit is happening right now, and TPCA Show will start next week. Watch out for single stock movement mentioned in the conferences.

- **Sector**. **1) ABF/CCL** is still the crowded long sectors for both foreign and domestic investor, but echoing on above points, investors need more catalysts to add on risks or drive the re-ratings. **2) Power** is another big issues that data centers need to resolve, power solutions and fuel cells name are also key focus in Taiwan. 3) Let's not forget about **OSATs**. We need more equipment and testing capacity to suffice enlarging demand from both ASIC and GPU.
  - **Key Stocks I Follow**: Unimicron, Kinsus, NYPCB, ZDT, EMC, TUC, Delta, Lite-On, Kaori, ASE, KYEC, All Ring
- **Risk**. Upside looks thin in the short term. Good earnings are not enough - look at Nanya Tech and Memory sector today. Chasing momentum now may get hurt. I also smelled some short unwind today (i.e. funding short names outperformance such as Realtek, Novatek, Shipper, etc) - signs of de-leveraging. Directionless is how I would describe the past few days, but maybe it's time to find some value Buy?

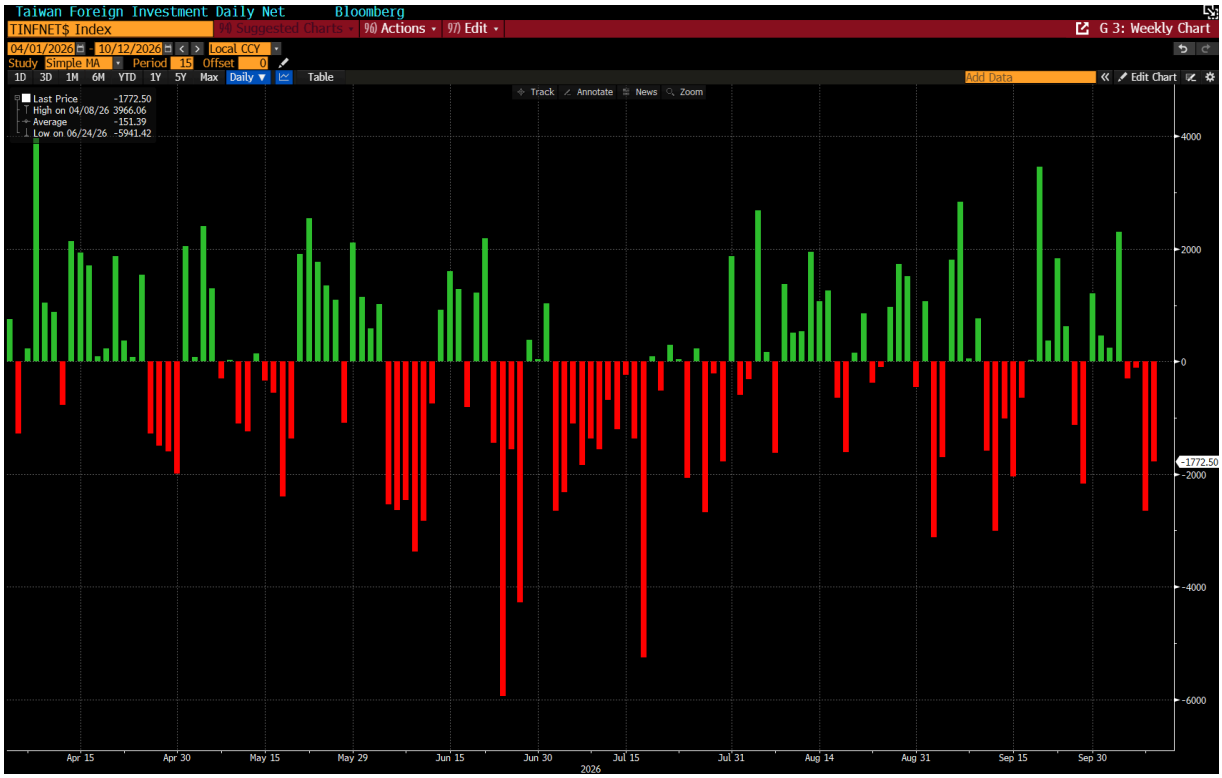

Exhibit 1. Foreign Investors Outflow since 4/1/2026

Please feel free to reach out to your GS representative to discuss any ideas.

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Past performance is not indicative of future results

Max loss amount invested is 100%

Source for all: Goldman Sachs FICC and Equities, Company Data, Bloomberg as of 10/13/2026

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臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎科技控股 (4958 TT)

投資人簡報

2026年10月

臻鼎科技控股
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(前身華虹電子 1978)

台灣桃園 總部

52,108
2025年底員工數

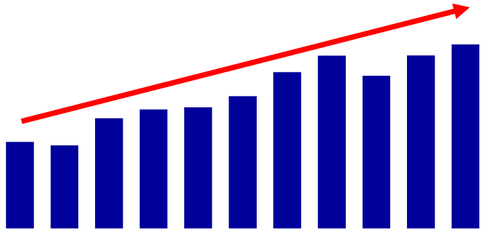
28 棟工廠
位於大陸、台灣、泰國，
其中 **19** 棟為智能工廠

NT$1,825.2億
2025年營收(6.3% YoY)

NT$5,666億
2026年9月30日市值

11.9%
10年平均股東權益報酬率
(2016-2025)

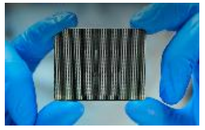
+8%
2015-2025年
營收年複合成長率



未來關鍵成長動能

AI伺服器

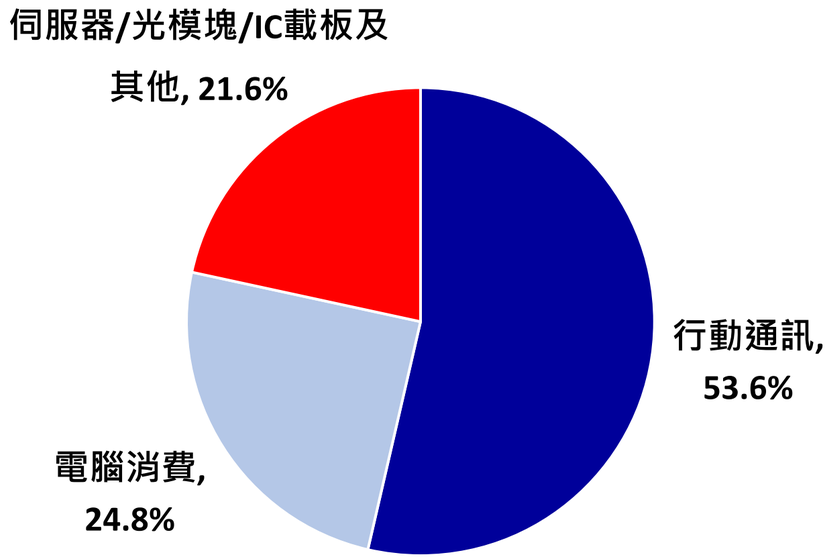
IC載板



光模塊

邊緣AI

銷售分析–產品應用別 (2026 H1)



臻鼎科技控股
Zhen Ding Tech. Holding

營運概況重點更新

營運成果回顧暨未來展望

1. **1H26營收再創同期新高，AI產品組合升級帶動獲利提升：**受惠於高階AI應用需求持續成長，1H26營收達新台幣892億元，年增13.9%，再創歷年同期新高。其中，伺服器 / 光模塊 / IC載板合計營收年增113%，營收占比達21.6%。隨高附加價值產品比重增加，1H26毛利率及營業利益率分別提升至22.4%及7.0%，顯示公司成長動能正由營收規模擴張，進一步轉化為獲利提升。

2. **AI算力建設開啟高階PCB長期成長週期，臻鼎憑藉完整技術布局邁入新一輪成長：**隨下一世代AI平台陸續量產，公司伺服器 / 光模塊業務自第二季起逐季增強，2026年相關營收目標維持成倍成長。中長期而言，AI需求將由伺服器與光模塊進一步擴展至Edge AI及人形機器人等終端應用，持續提升高階PCB的使用量、技術門檻與產品價值，公司對2026-2030 年的成長深具信心。

3. **以業界最大的擴產規模打造下一階段營運動能：**高階PCB產能建置與客戶認證均需較長前置期，唯有糧草先行，才能掌握AI帶來的長期成長機會。公司2026年資本支出為新台幣800億+，2027-2028年亦將維持較高投資水準，聚焦iHDI / mSAP、HLC、IC載板及高階軟板等高成長、高附加價值領域。公司目前共有22棟新廠建設中、9棟新廠規劃中，新增產能將依客戶導入及量產進度，於2026至2030年間分階段開出，逐步將資本投入轉化為營收與獲利貢獻。

4. **禮鼎於2026年7月2日正式向港交所遞交主板上市申請：**上市申請目前仍處於審查程序，相關資訊皆以港交所正式刊載的文件為準。

各產品應用營運概況

光模塊

- 光模塊方面，1.6T及後續世代產品須採用mSAP技術，全球高階光模塊供給缺口高，臻鼎正積極擴大相關產能。
- 公司訂單以 1.6T高速率板為主，預期2026年營收將成倍成長，同時客戶已開始預訂2027年產能。**目標2027年躍升為全球最大高階光模塊PCB供應商。**

AI伺服器

- AI伺服器方面，GPU與ASIC客戶之Intelligent HDI (iHDI) 與HLC產品已陸續轉量產，目標後續份額持續提升。
- 持續與客戶針對未來2-3個世代平台的產品進行開發。

邊緣AI相關應用

AI手機

AI眼鏡/VR/AR

- 2026年手機新機規格升級，帶動軟硬板單機價值同步提升，產品結構有利於毛利率表現。
- AI眼鏡朝更輕薄、高整合發展，不僅軟板使用數量提高，硬板亦導入更高線路等級與mSAP製程，PCB技術門檻顯著提升，預期AI眼鏡營收在2026/2027年皆保持快速成長。

參與SEMICON Taiwan 2026
1.6T光模塊、34層AI Server板齊發，對接次世代AI基建需求

參與SEMICON Taiwan 2026

光模塊

- 展出 800G 及最新 1.6T 高階光模塊展品，涵蓋 XPO 與 NPO 等次世代應用，鎖定新世代資料中心的大頻寬與低延遲需求。
- 產品結合 SLP（mSAP）精密製程與埋孔技術，為市場提供更具節能效益的傳輸解決方案。

AI Server 板

- 展示高達 34層之超高層伺服器板。
- 該系列產品採用 Ultra/Extreme Low Loss 材料與 M8 級技術標準，全面導入 VIPPO、Back drilling 與高厚徑比電鍍等極致工藝。

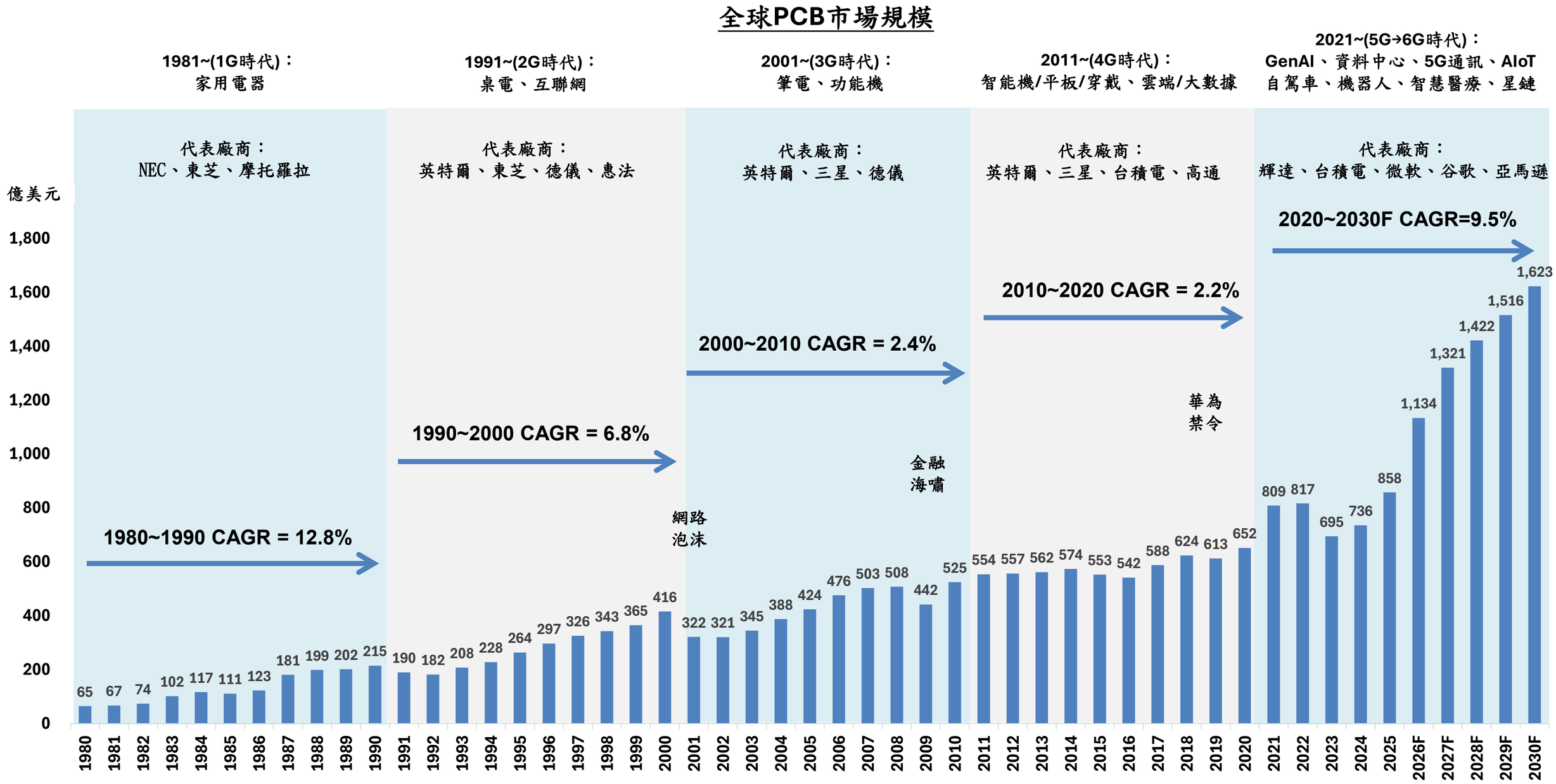
臻鼎科技控股
Zhen Ding Tech. Holding

強化領先優勢，

全面掌握HDI、HLC、與IC載板的黃金時代

臻鼎科技控股 Zhen Ding Tech. Holding

PCB產業隨電子產品迭代更新，朝高階高值化發展



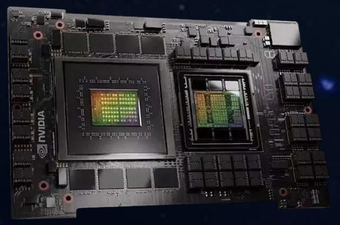
資料來源：Prismark (2026/9)

臻鼎科技集團
Zhen Ding Tech. Group

掌握AI潮流與趨勢

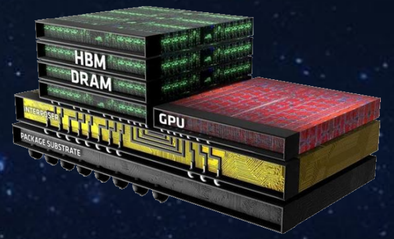
AI + PCB

雲

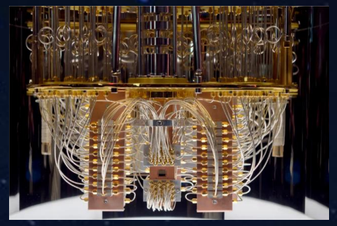


AI 伺服器

邊緣運算



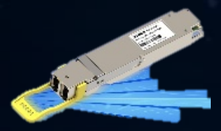
高速運算記憶體



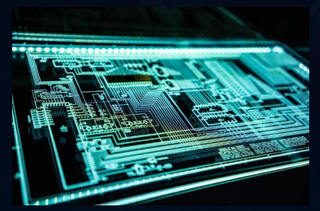
量子計算

管

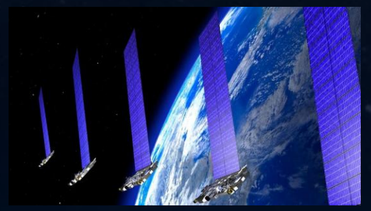
5G通訊設備



光模塊



衛星通信



光通訊

6G通訊感知一體

端

邊緣AI

AI 手機

AI PC/平板

智能穿戴

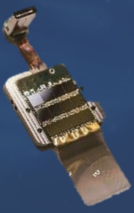
AR/VR

實體AI

智慧座艙

自駕車

人型機器人



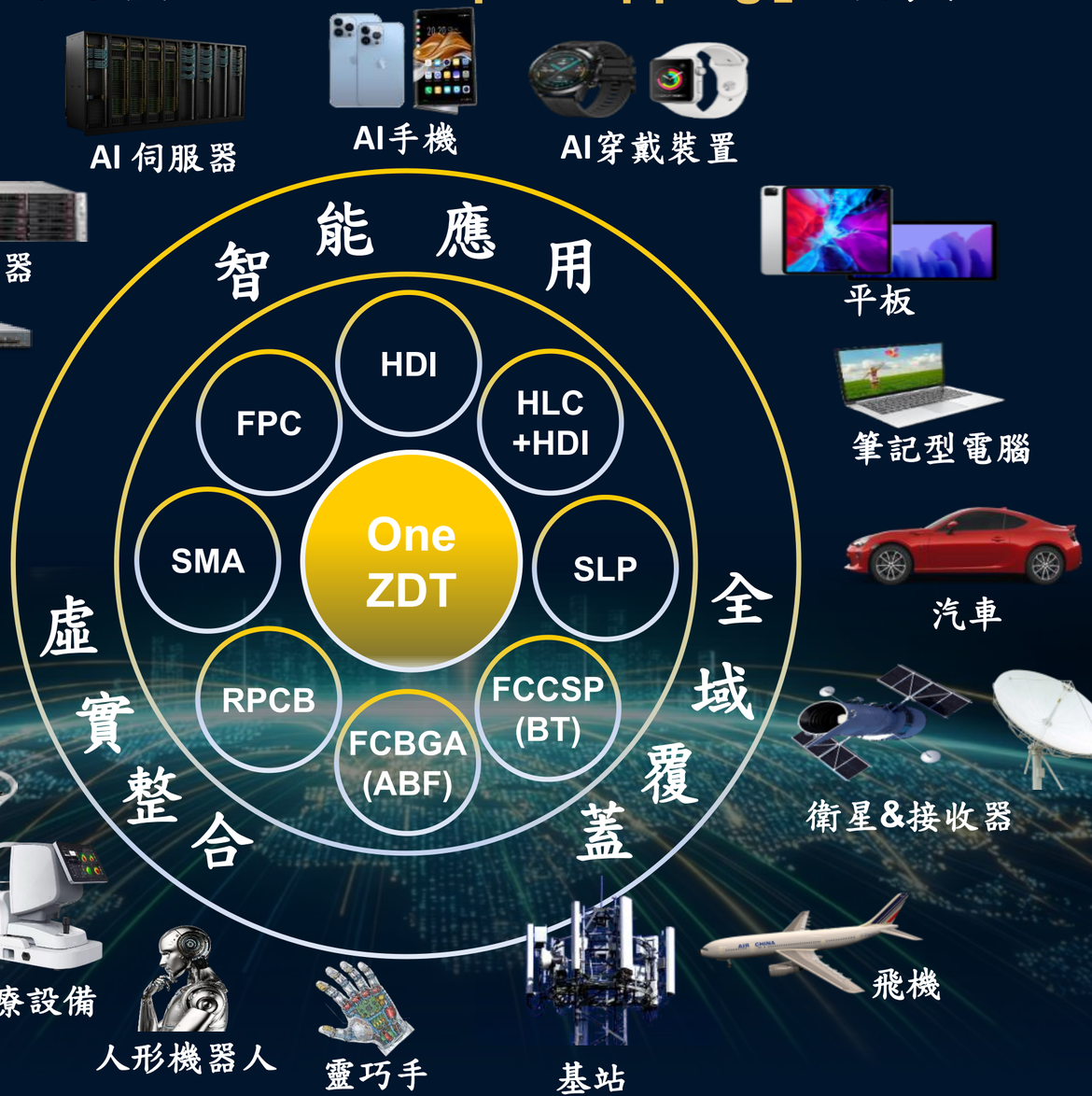
腦機介面

One ZDT, 多元化, 發展高端, 數位轉型, 擁抱AI

臻鼎科技集團
Zhen Ding Tech. Group

打造One ZDT平臺

臻鼎科技集團擁有一流產品的技術、品質、製造服務能力，並提供全方位電路板系列產品，滿足客戶「One Stop Shopping」的需求。

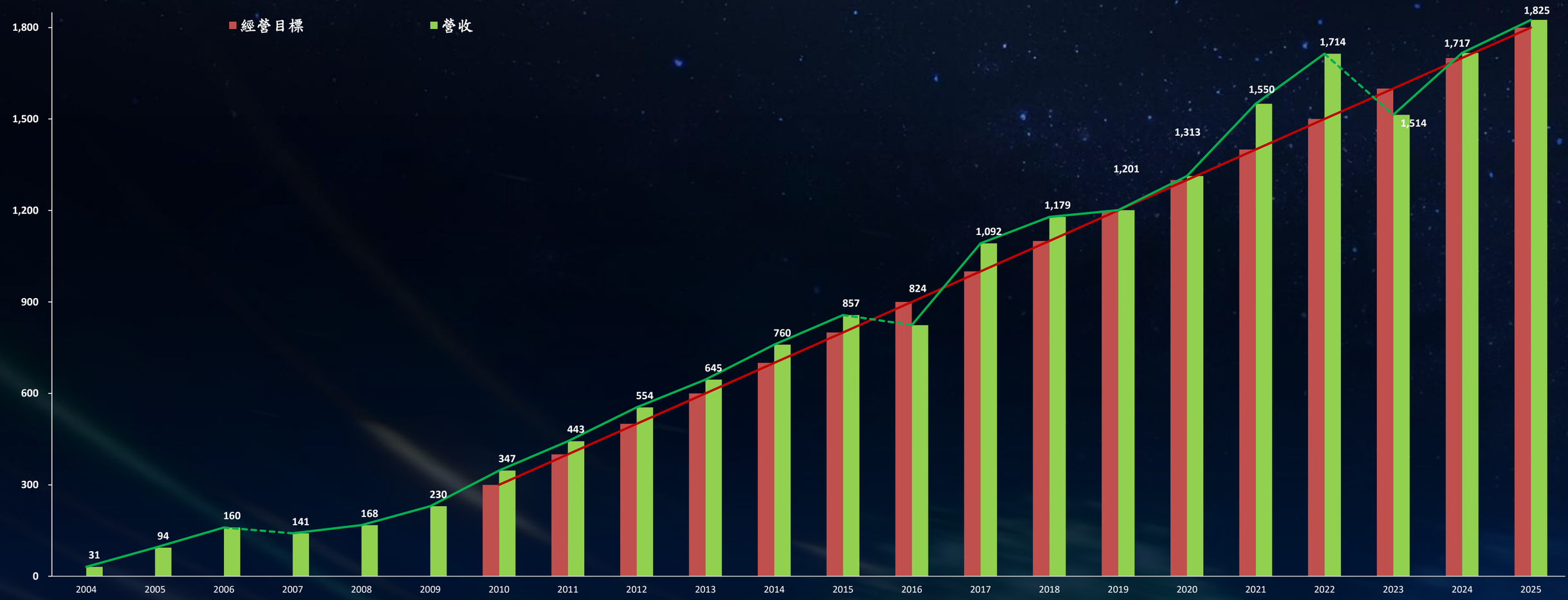


臻鼎科技集團
Zhen Ding Tech. Group

經營目標與成果

2010年4月製表,每年成長百億

單位：新台幣億元



年度	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
經營目標	-	-	-	-	-	-	300	400	500	600	700	800	900	1,000	1,100	1,200	1,300	1,400	1,500	1,600	1,700	1,800
營收	31	94	160	141	168	230	347	443	554	645	760	857	824	1,092	1,179	1,201	1,313	1,550	1,714	1,514	1,717	1,825
員工人數	3,396	4,605	8,805	13,006	17,009	18,142	22,983	23,034	25,392	27,820	33,212	37,801	39,943	40,622	38,588	35,953	40,521	44,330	42,425	41,478	49,149	50,694

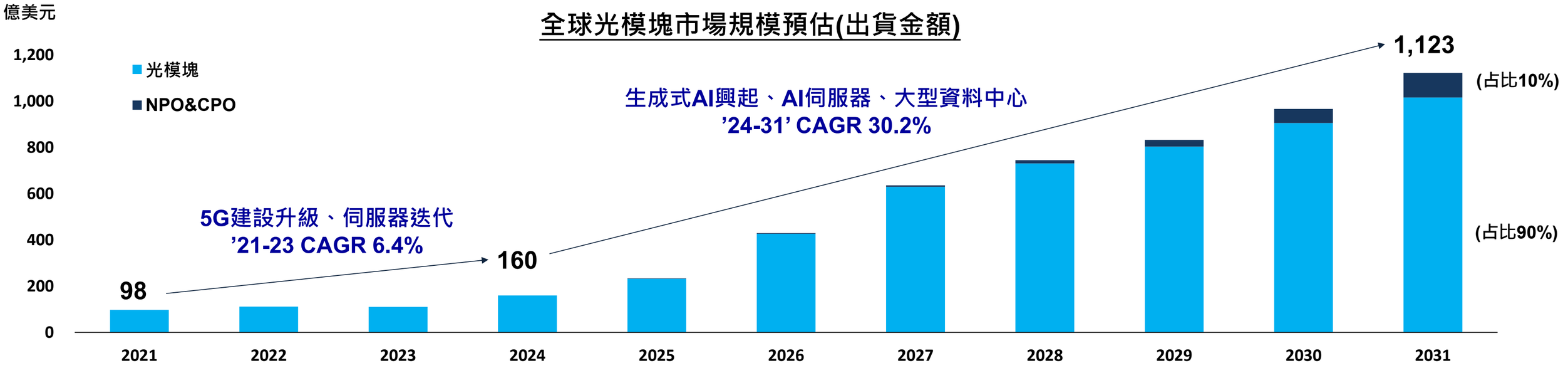
臻鼎科技集團
Zhen Ding Tech. Group

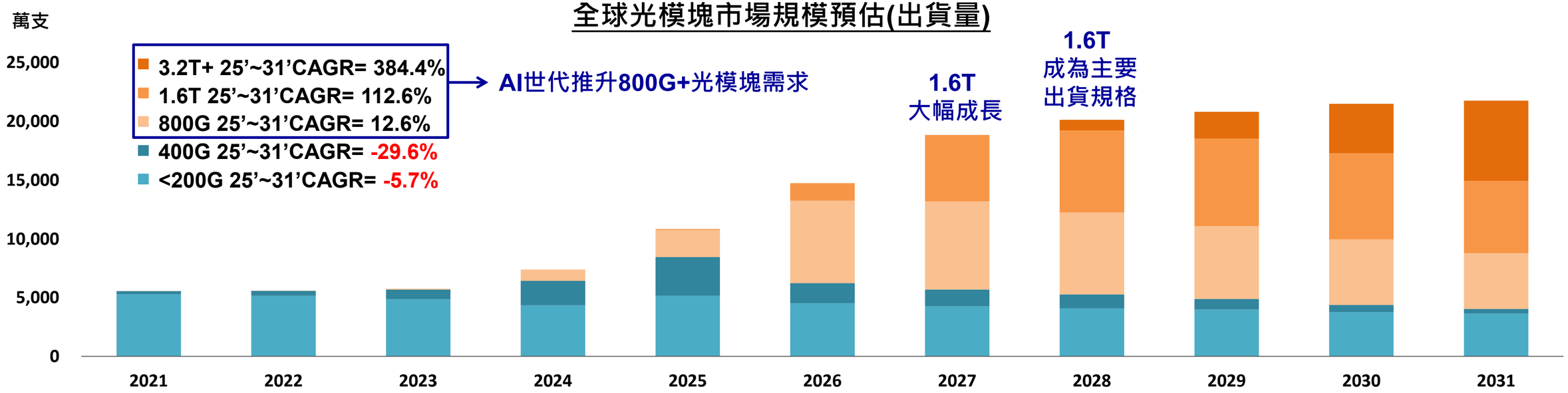
攜手共創，產業共生共榮

臻鼎科技集團於2026年7月24日在深圳舉辦全球供應商大會，匯聚247家供應商、400餘位代表共襄盛舉。

臻鼎科技控股 Zhen Ding Tech. Holding

光模塊市場需求快速攀升 1.6T時代提前來臨





資料來源：Yole Group(2026/5)

AI伺服器 PCB技術要求：臻鼎具備Tier 1技術能力

• GPU Servers

	2022	2023	2024		2025		2026	2027
PCB	NVIDIA HGX A100	NVIDIA HGX H200	NVIDIA HGX B200	NVIDIA NVL72 (GB200)	NVIDIA HGX B300	NVIDIA NVL72 (GB300)	NVIDIA NVL72 (VR200)	NVIDIA NVL576 (Rubin Ultra)
OAM	4+N+4 iHDI	5+N+5 iHDI	5+N+5 iHDI	5+N+5 iHDI (Compute Board)	5+N+5 iHDI	5+N+5 iHDI (Compute Board)	7+N+7 iHDI (Compute Board)	8+N+8 iHDI (Compute Board)
UBB	24-26L	26-28L	28-30L		28-30 L			
Mainboard	16L +	16-18L	16-20L		16-20 L			
Midplane							44L RPCB	44L RPCB
Backplane								
Switch board	20L +	20-22L	22-24L	22L RPCB	22-24L	22L RPCB	5+N+5 iHDI	6+N+6 iHDI
Others				4+N+4 iHDI		4+N+4 iHDI RPCB-網路卡	4+N+4 iHDI RPCB-網路卡 MSAP-光模塊	4+N+4 iHDI

• ASIC Servers

	2026	2027	2025	2026	2027	2025	2026	2027	2026-2027
PCB	Microsoft Maia200	Microsoft Maia300	Google TPU v7	Google TPU v8	Google TPU v9	Amazon Trainium 2+2.5	Amazon Trainium 3	Amazon Trainium 4	China ASIC Servers
OAM					iHDI (TPU board)	3+N+3 iHDI	4+N+4 iHDI	iHDI	5+N+5 iHDI
UBB	26L+ RPCB (ASIC Board)	RPCB (ASIC Board)	28 L RPCB (TPU board)	30/34 L RPCB (TPU board)		24L RPCB	28/30L RPCB		26-28L RPCB
Others	CPU-RPCB 配版-RPCB	CPU-RPCB 配版-RPCB		配版-RPCB			CPU-RPCB Switch-RPCB 網路卡-RPCB	CPU-RPCB Switch-RPCB 網路卡-RPCB	Huawei 光模塊-MSAP NPU-iHDI

資料來源： Goldman Sachs Research(2026/8)

臻鼎科技控股 Zhen Ding Tech. Holding

臻鼎將全面掌握HDI與高多層板(HLC)的黃金時代

Global AI Server PCB Market Opportunity

單位：百萬美元

應 用	2025	2026E	2027E	2028E
GPU Server	2,247	4,728	18,803	42,568
OAM(加速板模組)	966	2,132	9,343	24,168
UBB(通用基板)	555	628	1,183	784
Mainboard(主板)	208	221	223	116
Midplane(中介板)		185	2,451	5,787
Backplane(背板)			225	900
Switch board(交換器板)	439	1,094	3,964	9,010
Others	79	468	1,415	1,803
ASIC Server	2,734	8,832	18,726	41,077
合計	4,981	13,560	37,529	83,645

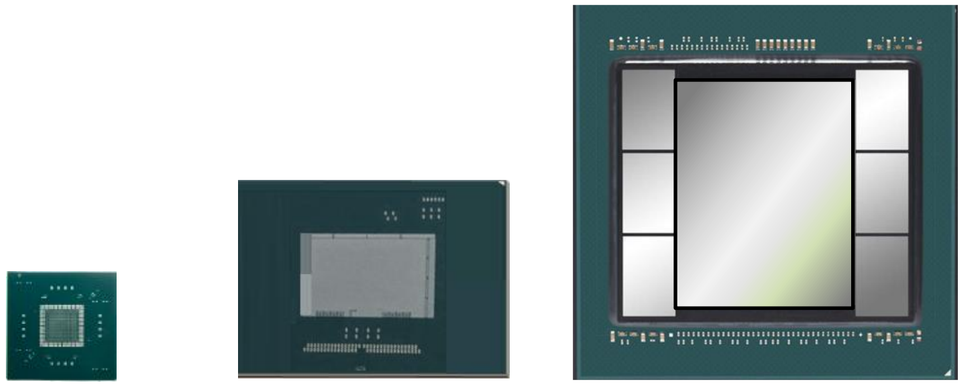
產 品 技 術	2025	2026E	2027E	2028E
RPCB	2,960	7,864	19,609	40,673
<20 layers	509	1,277	2,190	4,980
20-30 layers	2,087	5,171	8,474	16,451
30 layers+	364	1,417	8,946	19,241
HDI	2,021	5,695	17,920	42,973
4+N+4	79	468	1,415	1,803
5+N+5	1,730	3,508	10,195	12,783
6+N+6 or above	212	1,719	6,310	28,386
合計	4,981	13,560	37,529	83,645

資料來源： Goldman Sachs Research(2026/8)

臻鼎ABF載板技術能力與Tier 1競爭者相當，致力爭取全球領先半導體客戶

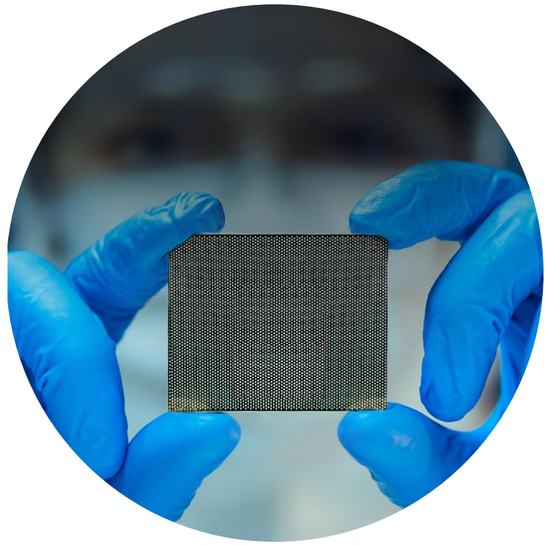
ABF載板發展趨勢

層數高、尺寸大、表面平、產品精度準

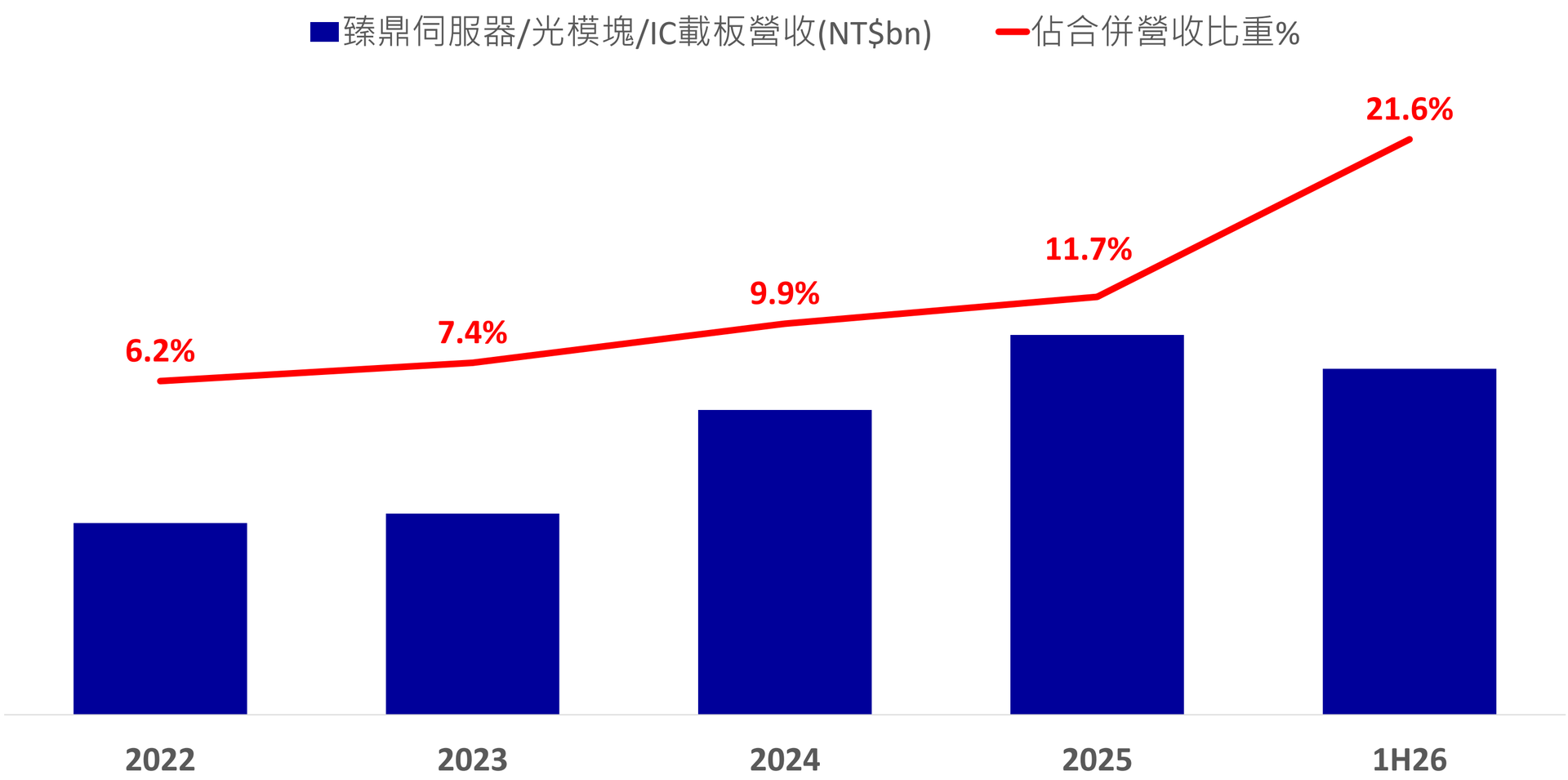


	2005	2020	2026
Body Size (mm)	31 x 31	75 x 60	>150 x 150
Layer count (L)	6	20	28
Bump Count	1K	100K	300~500k

臻鼎專注高階需求，
致力爭取全球領先半導體客戶

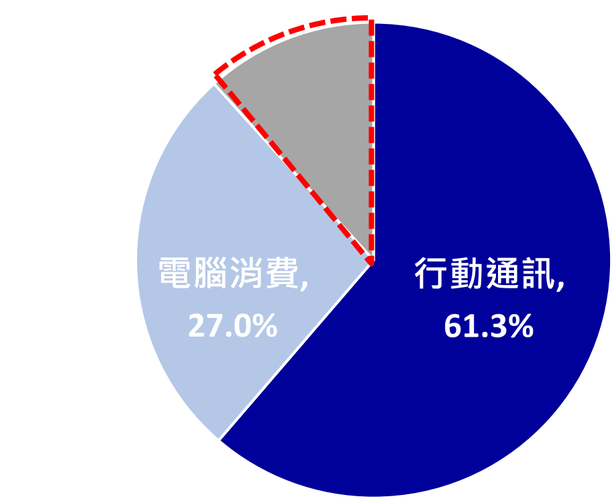


One ZDT 驅動結構升級，
1H26 伺服器 / 光模塊 / IC載板營收佔比突破20%

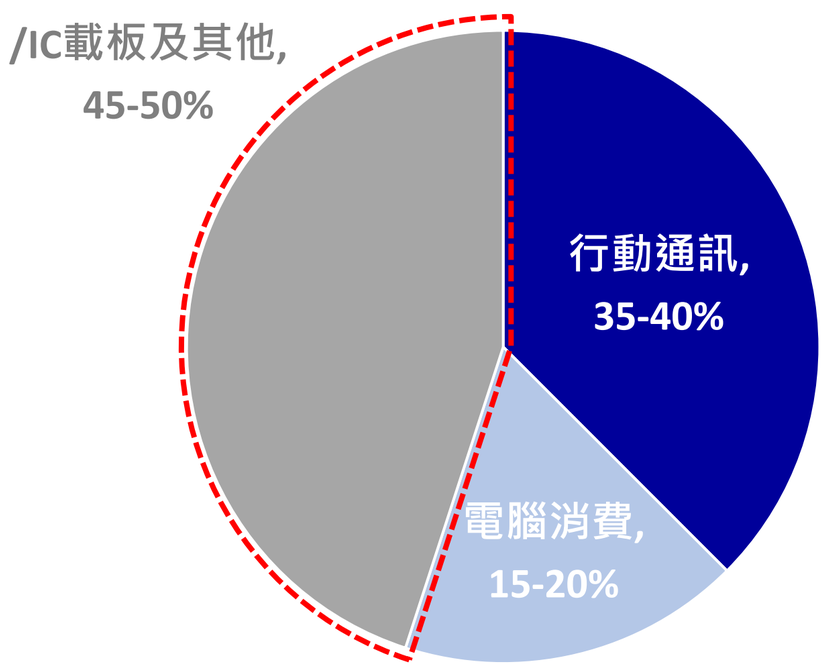


穩步推進伺服器/光模塊與IC載板佈局，持續擴大營收貢獻

臻鼎：2025年產品應用營收佔比



臻鼎：2030年目標產品應用營收佔比



臻鼎科技控股
Zhen Ding Tech. Holding

擴產有依，穩健前行，
搶占市場先機

臻鼎科技集團
Zhen Ding Tech. Group

臻鼎集團戰略佈局：產能藍圖

2026年
274公頃(4,110畝)

深圳第一園區

深圳第二園區

深圳第三園區

深圳禮鼎園區

淮安HA園區

淮安HB園區

淮安HC園區

淮安HD園區

秦皇島園區

秦皇島禮鼎園區

高雄園區

泰國巴真園區

2026年

深圳第一園區

深圳第二園區

深圳第三園區

深圳禮鼎園區

淮安HA園區

淮安園區

淮安園區

淮安HD園區

28座

秦皇島園區

秦皇島禮鼎園區

高雄園區

泰國巴真園區

2030年

59座

(建設中22座、規劃中9座)

全球生產基地加速擴產，效益將於2026年起逐步顯現

擴產規劃

大陸廠區

- 淮安園區：
  - 規劃至2026年底，淮安HA、HB、HC園區將有6棟高階PCB新廠完成建設。
  - 另HD園區已於今年四月動土，規劃新建HDI/MSAP與HLC廠，8月10日封頂，預計明年三月加入生產。
- 深圳園區：
  - 第三園區智慧製造基地於今年七月動土，預計興建三棟智慧製造廠房，重點布局AI伺服器用及光模塊用的高階類載板，及AI終端用高階軟板等高端產品。
  - 禮鼎深圳第二園區於今年九月正式動土，總投資金額不低於人民幣120億元，規劃建設數座工廠，聚焦FCBGA等高階IC載板。新產能主要面向AI、高效能運算及先進封裝需求，進一步強化公司高階FCBGA產品的研發、製造與規模化交付能力。

泰國巴真府廠區

- 泰國一廠：
  - 主要生產高階iHDI、HLC及光模塊產品，於3Q25進入量產。
  - 量產爬坡順利，已通過多家客戶認證及量產，尚有多家伺服器與光模塊全球領先客戶正積極進行認證中。
  - 7月已達損益兩平。
- 目前另有三棟新廠房及一棟機械鑽孔中心同步建置中，規劃將於4Q26起逐步進入試產階段。

臻鼎科技集團
Zhen Ding Tech. Group

260922

臻鼎科技集團淮安科技城

(總用地面積：2,151畝)

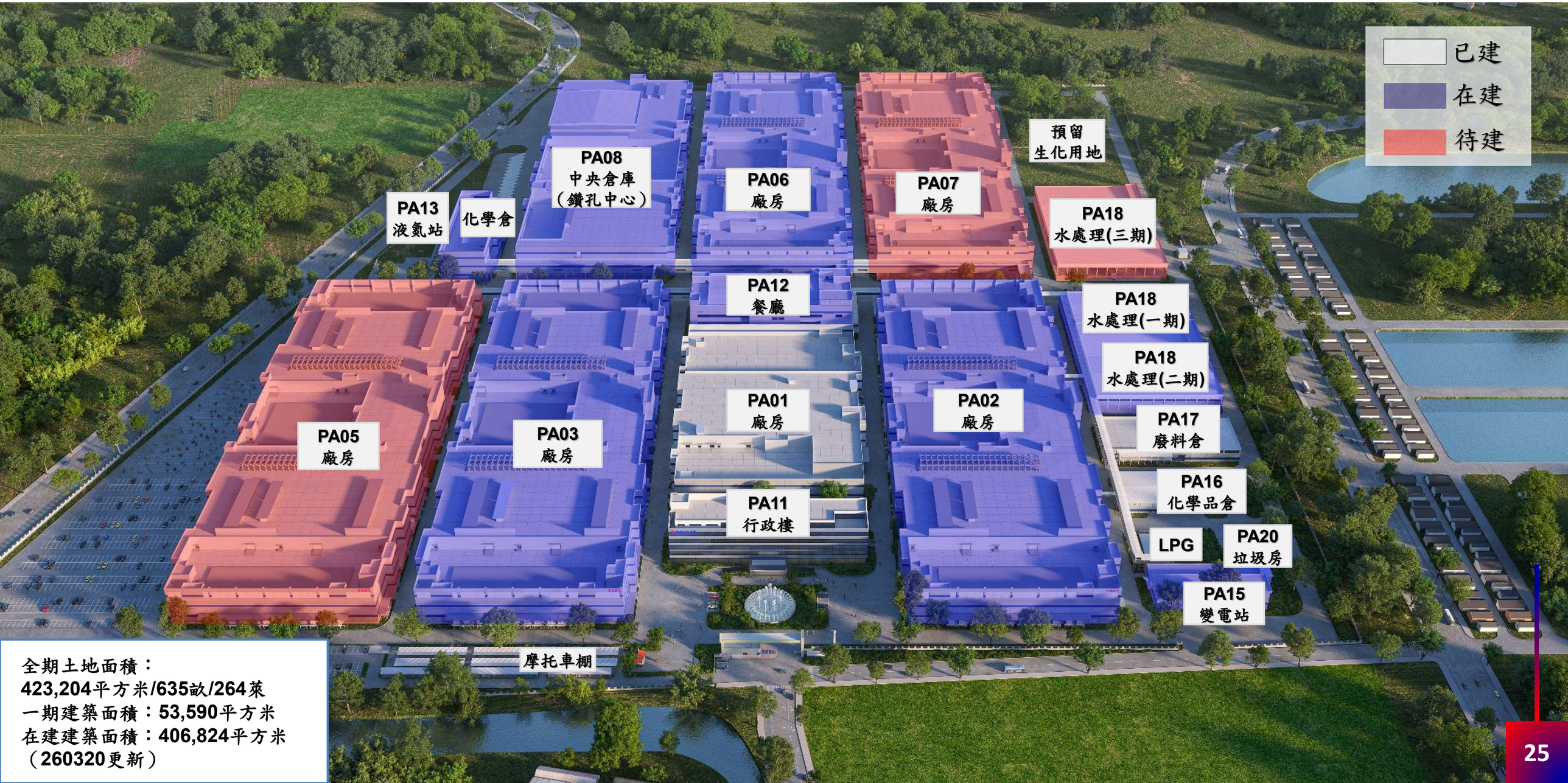
項目總用地面積：1,434,322平方米/2,151畝
項目總建築面積：2,038,683平方米
（260924更新）

臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎深圳廠區規劃

臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎泰國巴真府廠區規劃



全期土地面積：
423,204平方米/635畝/264萊
一期建築面積：53,590平方米
在建建築面積：406,824平方米
（260320更新）

臻鼎集團產品線分佈概覽

V: Will be ready

臻鼎科技控股 Zhen Ding Tech. Holding

禮鼎科技 LEADING TECH　佰鼎科技 Brai Technology　鵬鼎控股 AVARY HOLDING　先豐通訊 BoardTek Elec.

	大陸 深圳廠	大陸 秦皇島廠	台灣 高雄廠	大陸 秦皇島廠	大陸 深圳廠	大陸 淮安廠	泰國 巴真廠	印度 清奈廠	台灣 高雄廠	台灣 高雄廠
FPC				V	V	V	V		V	
SMA				V	V	V	V	V	V	
HDI (含SLP/MSAP)				V	V	V	V			V
HLC					V	V	V			V
Rigid-Flex						V				
IC Substrate	V	V	V							

臻鼎科技控股 Zhen Ding Tech. Holding

透過智慧製造，提高人均產值，創造更佳獲利

智慧製造
Smart Manufacturing for Industry 3.5/4.0

PLM | SCM | ERP | HRM | i-Foundry

非結構化數據系統 | 結構化數據系統 | 資料倉儲系統 | AI 算力平台

MES 製造執行系統與聰明生產
Manufacturing Execution System
APS/Data-driven DSS

QMS 品質管理系統與良率提升
Quality Management System
SPC/APC/AQC/AEC

EAP (Equipment Automation System) / IIOT

半導體設備通訊標準介面SECS / 通用設備模型GEM/ IPC-CFX /WEBAPI/SebService/

壓合

雷射

曝光 LDI

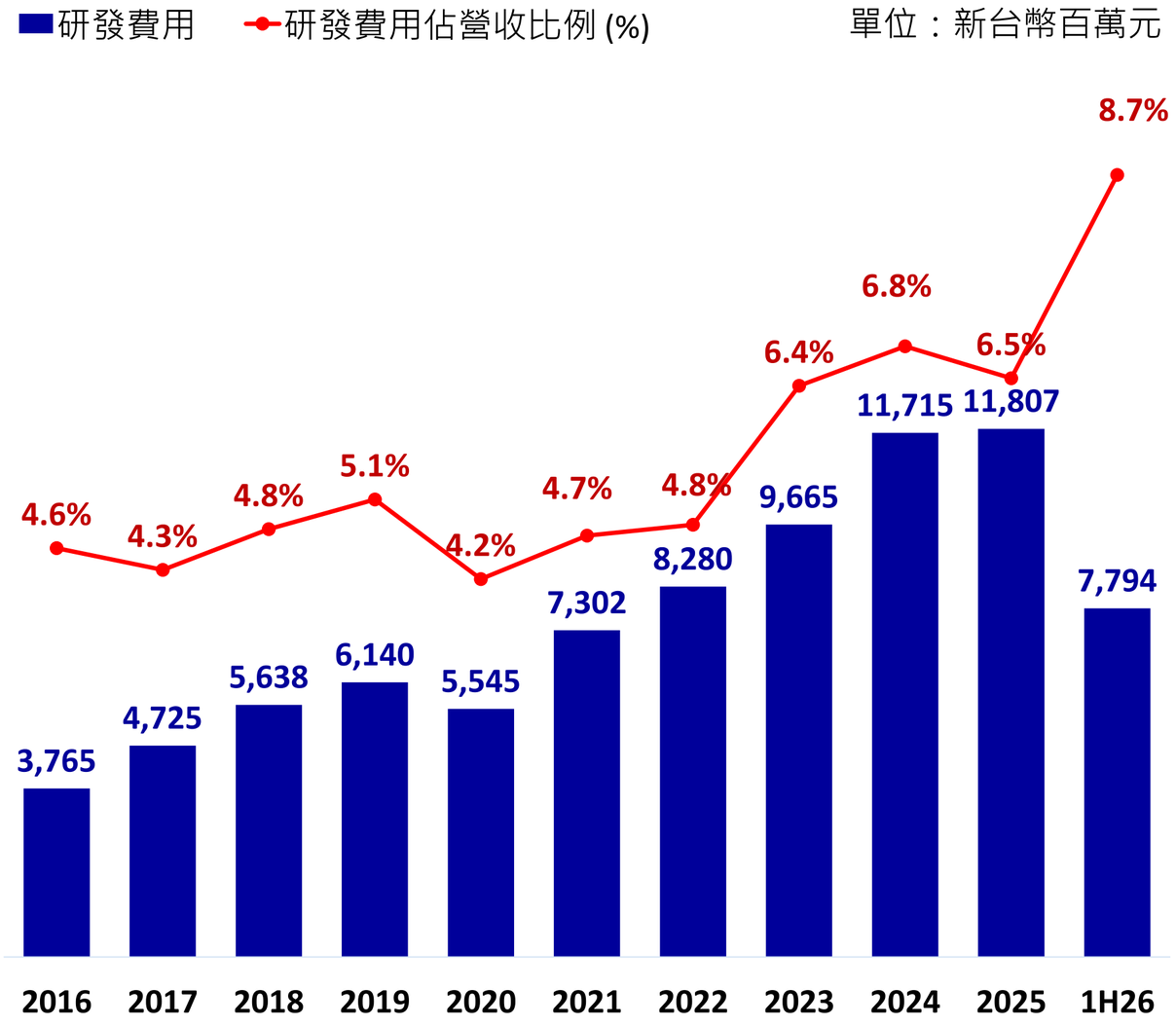
垂直顯影

防焊印刷

電測

作業數位化、數位平台化、平台智能化、智能實戰化

持續投入高階產品研發，鞏固產業領導地位



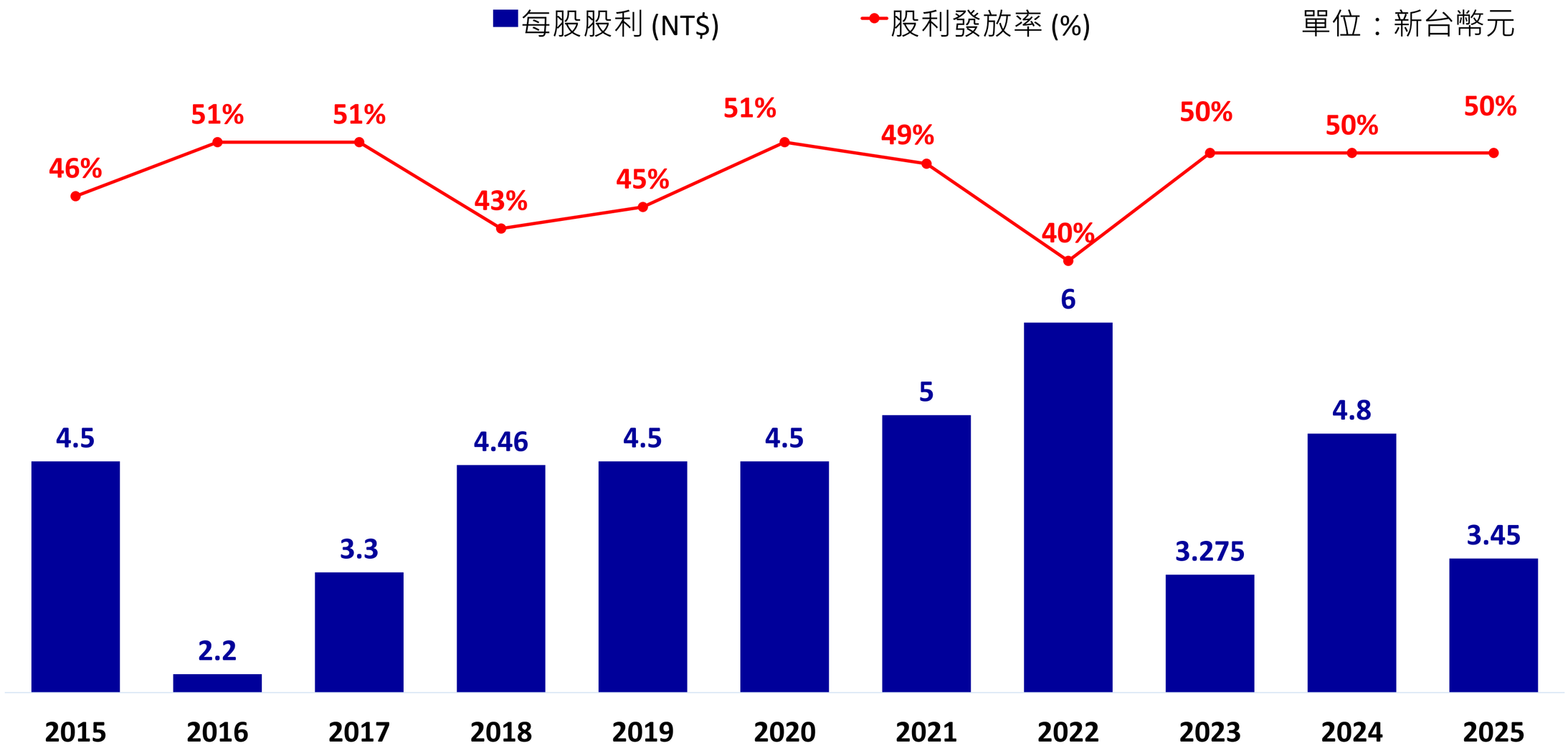
在印刷電路板和半導體相關開發計畫上，臻鼎遵循12 字方針，研發高階技術並引入到各項具有前瞻性之領域應用中。

12 字方針：

輕 輕型化	薄 薄型化	短 流程短	小 小型化
高 高頻、高速、高散熱	低 低污染、低成本、低功耗	多 多功能、多層	快 快速研發、快速製造
精 公差精進	美 美觀	細 細線路	智 智能化

實施台灣及大陸雙研發中心，持續提升新技術開發能力，以保持高端產品及技術之領先優勢。

維持高股息發放率，與股東共享經營成果



致力推動EPS + ESG，國際ESG評比持續提升

▶ 標普全球企業永續評鑑 S&P Global ESG

2025年S&P Global ESG評級進步至88分，並入選**道瓊領先世界指數 (DJBIC World) 及道瓊領先新興市場指數 (DJBIC Emerging Markets)**，亦於2022年至2025年連續4年入選標普全球永續年鑑 (Sustainability Yearbook)。

▶ ISS ESG 評級

最新年度的ISS ESG評級為「C+」，**獲得「優秀 (Prime)」等級**。

▶ 臺灣公司治理評鑑

2025年公司治理評鑑排名上市公司6%-20%，再度**入選臺灣證券交易所公司治理100指數**。

▶ CDP 碳揭露專案

2025年在氣候變遷與水安全管理皆獲得**「A」最高等級殊榮**，其中氣候變遷為首度獲得A級，水安全則已連續兩年獲得A級。

▶ Sustainalytics ESG 風險評級

最新Sustainalytics ESG風險評級為16.1，為**低等風險等級**。

▶ 富時羅素 FTSE Russell ESG

最新年度的FTSE Russell ESG評級獲得4.5分（滿分為5分），成為**台灣PCB企業第一名**。

臻鼎科技控股
Zhen Ding Tech. Holding

營運成果摘要

臻鼎科技控股 Zhen Ding Tech. Holding

季度營運成果

單位: 新台幣佰萬元，除另予註明者外

	2Q26	2Q25	年變化 (%)
營業收入	48,431	38,203	+26.8%
營業毛利	11,186	7,011	+59.6%
營業毛利率	**23.1%**	**18.4%**	**+4.7ppts**
營業費用	7,425	4,586	+61.9%
營業利益	3,761	2,425	+55.1%
營業利益率	**7.8%**	**6.3%**	**+1.5ppts**
營業外收入及支出	1,170	(153)	
匯兌損益	(668)	(742)	
本期淨利	3,727	1,387	+168.6%
淨利率	**7.7%**	**3.6%**	**+4.1ppts**
歸屬於母公司之淨利	2,392	605	+295.3%
每股盈餘 (新台幣元) (1)	**2.19**	**0.63**	
研發費用	4,218	2,611	+61.6%
折舊及攤銷	5,171	4,516	+14.5%
資本支出	17,009	8,751	+94.4%
現金及約當現金(2)	97,098	74,194	+30.9%
股東權益報酬率 (%) (3)	**7.8%**	**3.9%**	

附註：(1) 2026第二季加權平均流通在外股數為1,090,870仟(實際發行股本1,108,887仟股-庫藏股424仟股) 。(2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 股東權益報酬率為以母公司股東平均股權計算的年化數據。

上半年營運成果

單位: 新台幣佰萬元，除另予註明者外

	1H26	1H25	年變化 (%)
營業收入	89,159	78,285	+13.9%
營業毛利	19,998	12,896	+55.1%
營業毛利率	**22.4%**	**16.5%**	**+5.9ppts**
營業費用	13,733	9,415	+45.9%
營業利益	6,265	3,481	+80.0%
營業利益率	**7.0%**	**4.4%**	**+2.6ppts**
營業外收入及支出	1,067	248	+330.4%
匯兌損益	(1,598)	(639)	
本期淨利	5,774	2,413	+139.3%
淨利率	**6.5%**	**3.1%**	**+3.4ppts**
歸屬於母公司之淨利	3,817	1,237	+208.5%
每股盈餘 (新台幣元) (1)	**3.55**	**1.30**	
研發費用	7,794	5,173	+50.7%
折舊及攤銷	10,122	9,255	+9.4%
資本支出	29,688	14,388	+106.3%
現金及約當現金(2)	97,098	74,194	+30.9%
股東權益報酬率 (%) (3)	**6.1%**	**3.4%**	

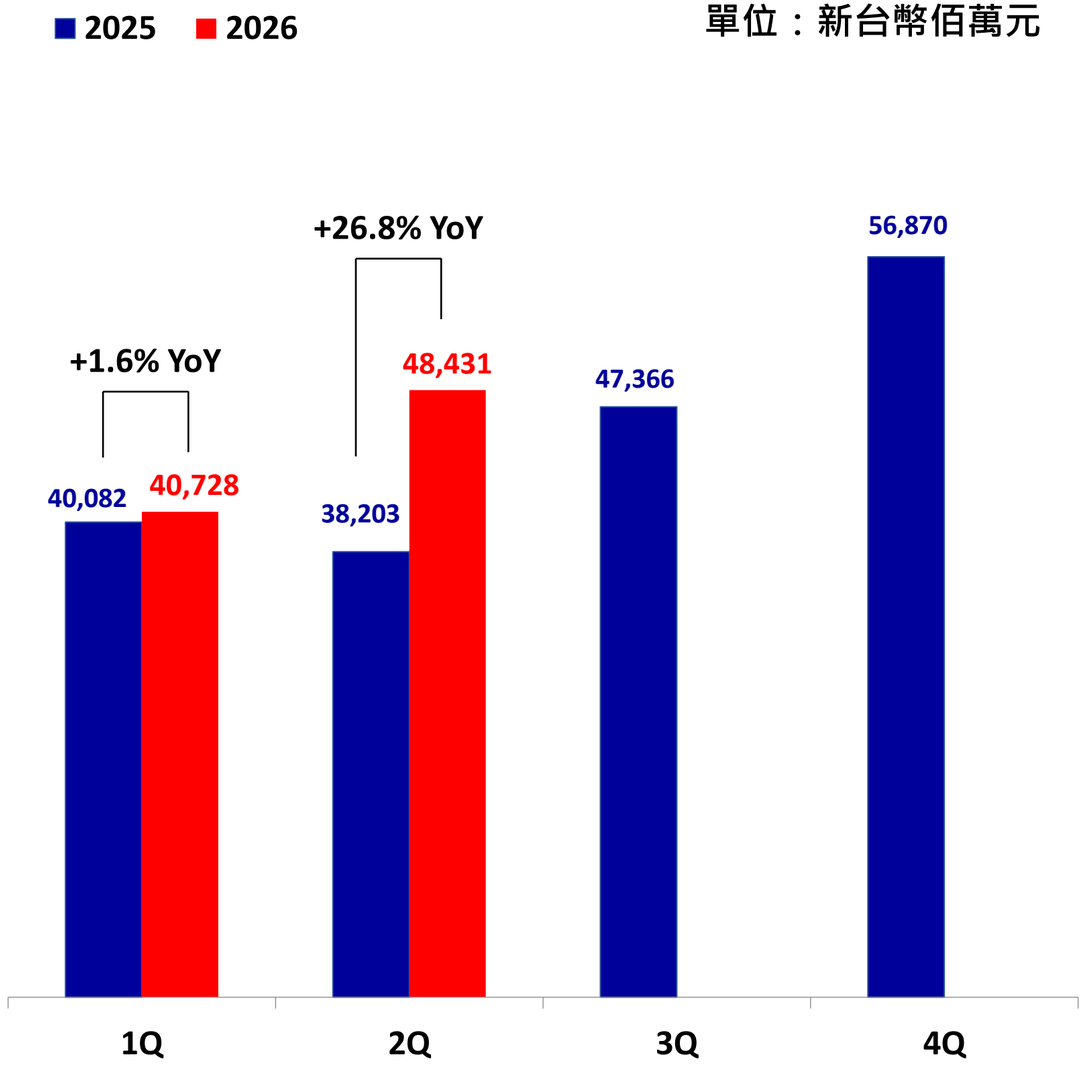
附註：(1) 2026上半年加權平均流通在外股數為1,074,541仟股 (實際發行股本1,108,887仟股-庫藏股424仟股)。(2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 股東權益報酬率為以母公司股東平均股權計算的年化數據。

臻鼎科技控股 Zhen Ding Tech. Holding

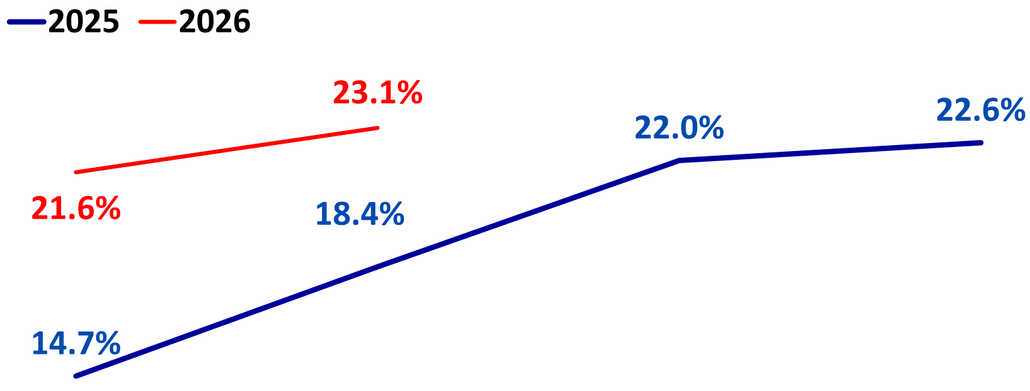
季度營運成果

營業收入

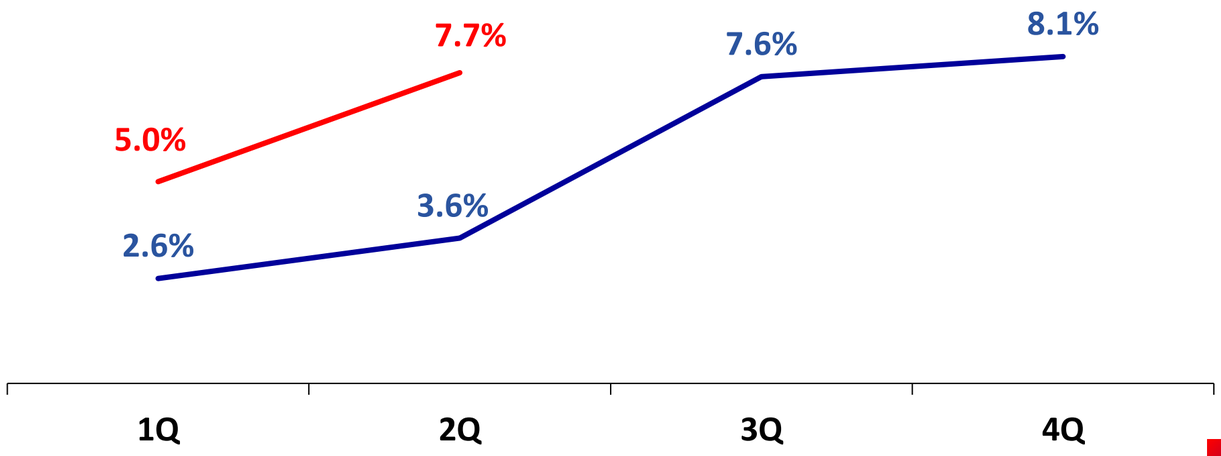
單位：新台幣佰萬元



毛利率 (%)

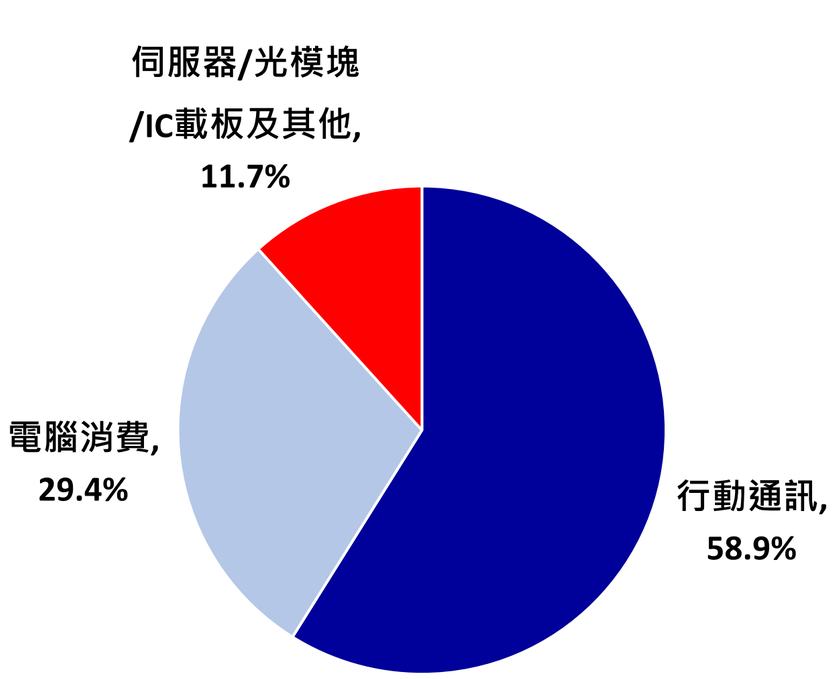


淨利率 (%)

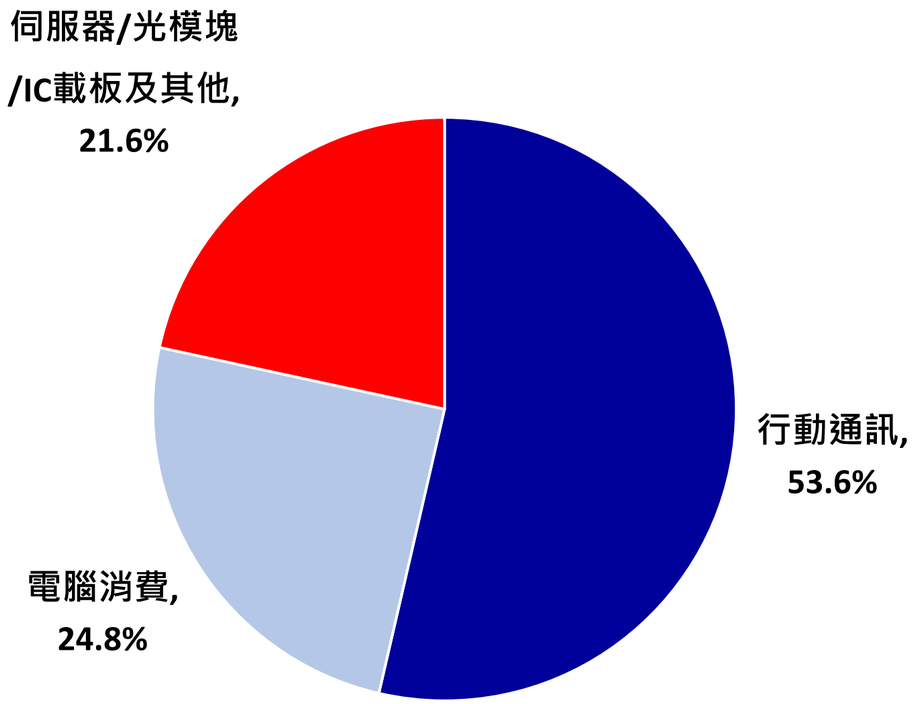


銷售分析–產品應用別

1H25營收：新台幣783億



1H26營收：新台幣892億



資產負債表及重要財務比率

單位：新台幣佰萬元

	2026-6-30		2025-6-30		差異	
	金額	%	金額	%	金額	%
現金及約當現金[2]	97,098	27.8%	74,194	30.6%	22,904	+30.9%
應收款項	30,359	8.7%	18,409	7.6%	11,950	+64.9%
存貨	28,590	8.2%	16,178	6.7%	12,412	+76.7%
不動產、廠房及設備[3]	152,794	43.7%	109,642	45.3%	43,152	+39.4%
資產總額	**349,706**	**100.0%**	**242,172**	**100.0%**	**107,534**	
借款	70,950	51.1%	55,438	22.9%	15,512	+28.0%
應付款項	53,780	15.4%	41,265	17.0%	12,515	+30.3%
負債總額	**138,872**	**39.7%**	**109,026**	**45.0%**	**29,846**	
股東權益總額	**210,834**	**60.3%**	**133,145**	**55.0%**	**77,689**	
重要財務比率						
平均收現日數(天)	62		57		5	
平均銷貨日數(天)	67		51		16	
流動比率(倍)	1.61		1.54		0.07	
資產生產力(倍)[4]	1.28		1.40		(0.13)	

附註：(1) 2026上半年加權平均流通在外股數為1,090,870仟(實際發行股本1,108,887仟股-庫藏股424仟股) 。 (2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 不動產、廠房 及設備 包含 投資性不動產。(4) 資產生產力 =年化營業收入淨額 /平均不動產、廠房及設備淨額。

臻鼎科技控股 Zhen Ding Tech. Holding

歷年營運成果

單位：新台幣佰萬元

年度	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
營業收入	82,393	109,238	117,913	120,068	131,279	155,022	171,356	151,398	171,664	182,522
營業毛利	12,542	17,833	26,061	27,222	26,584	30,537	39,888	27,459	32,461	36,136
本期淨利	3,456	6,772	11,536	12,402	11,508	13,694	20,535	9,432	13,096	10,605
母公司淨利	3,456	5,172	8,448	8,685	8,095	9,651	14,197	6,189	9,180	6,791
折舊及攤銷	5,295	5,679	6,820	7,955	8,405	11,875	14,638	16,323	17,749	18,552
每股盈餘 (元)	4.29	6.43	10.50	9.93	8.90	10.21	15.02	6.55	9.67	6.91
每股股利 (元)	2.20	3.30	4.46	4.50	4.50	5.00	6.00	3.275	4.80	3.45
分派比率 (%)	51%	51%	43%	45%	51%	49%	40%	50%	50%	50%
現金及約當現金 *	30,241	33,296	49,154	43,071	46,775	35,179	57,599	65,970	79,830	71,152
不動產、廠房及設備	32,262	36,681	41,913	46,243	68,177	86,073	104,814	109,965	113,462	126,808
資本額	8,047	8,047	8,047	9,022	9,470	9,470	9,470	9,470	9,567	10,706
資本支出	9,682	12,747	16,224	16,211	21,645	31,149	29,032	26,066	16,431	33,257
股東權益報酬率 (%)	8.59%	14.49%	17.30%	14.72%	11.84%	12.59%	16.67%	7.10%	9.15%	6.57%
負債比率 (%)	59.72%	55.33%	44.25%	35.41%	42.56%	42.01%	42.87%	44.67%	42.85%	39.04%

* 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等

臻鼎科技控股
Zhen Ding Tech. Holding

Thank You

Largan Precision

Core weakness outweighs progress in AI optics

Equities

Taiwan

Electric Components & Equipment

12-month rating	**Sell**
12m price target	**NT$5,000.00**
Price (08 Oct 2026)	**NT$6,660**

RIC: 3008.TW **BBG:** 3008 TT

Trading data and key metrics

52-wk range	NT$7,810.00-2,025.00
Market cap.	NT$875b/US$27.4b
Shares o/s	131m (ORD)
Free float	70%
Avg. daily volume ('000)	2,379
Avg. daily value (m)	NT$13,134.2
Common s/h equity (12/26E)	NT$199b
P/BV (12/26E)	4.4x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	188.21	160.04	-15	184.91
12/27E	214.00	180.00	-16	204.02
12/28E	250.66	220.00	-12	238.78

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CPO progress is more tangible but substantial success is already priced in

We maintain Sell and our NT$5,000 target price as erosion in the core smartphone business outweighs CPO progress. The Q326 call confirmed that Largan's fiber array has passed customer qualification although its customer qualified another vendor's prism micro-lens assembly (PMLA) that Largan would now buy in. If Largan qualifies the full FAU plus carrier assembly for 2H27, content rises but with more pass-through cost and less margin upside. We roll our valuation to 2028E, the first full year of CPO production, though 25% below the NT$6,660 close on October 8.

Full FAU assembly would lift content, but with more pass-through

We believe Largan's FA customer is one of two module suppliers on NVIDIA's 2H27 program, which we estimate could reach 40% share. The program is still subject to successful yield ramp and could move to metalens in 2028, which would reopen supplier qualification. Assembling the full FAU could lift Largan's ASP to US$125-150 leveraging its precision alignment and automation but the PMLA, V-groove and carrier parts would be pass through that cuts into margin leverage. We expand our FAU TAM analysis (Figure 9) with FAU market in 2028/30 expanding to 12.3mn/35.6mn units, with Largan base case set at 25% of the TAM.

Core smartphone business stays soft into 2027

Q326 sales of NT$15.7bn, +15% QoQ/-11% YoY missedUBSe/Street +36%/+33% on weaker flagship smartphone demand. GMs fell 7.2ppts QoQ to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (+1% YoY vs. -11% YoY). We cut 2027 from NT$77bn to NT $67bn, before a recovery to NT$79bn in 2028 with the next periscope upgrade. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass through in 2028 pricing.

Maintain Sell and our NT$5,000 target

We cut 2026E/27E EPS by 14%/16% to NT$160/NT$180 and introduce 2028E EPS of NT$220 for the core smartphone business. We value the lens business at NT$3,300 (15x 2028E ex-CPO EPS of NT$220, the middle of Largan's 13–18x historical range) and CPO option at NT$1,700 or US$7bn (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US $1.85bn FAU market at US$150 per unit and a 50% gross margin. That credits it with yields and margins it still needs to prove in a quarter when yields and energy costs cut 7.2ppts from its lens margin. At ~NT$6,660, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
EBIT (UBS)	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Net earnings (UBS)	18,047	22,480	23,613	20,953	23,567	28,805	28,397	29,473
EPS (UBS, diluted) (NT$)	135.22	168.43	176.92	160.04	180.00	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	81.97	90.34	110.41	108.85
Net (debt) / cash	111,655	117,781	122,309	109,131	120,267	127,842	144,348	160,697
Profitability/valuation	**12/23**	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
ROIC (EBIT) %	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
EV/EBITDA (UBS core) x	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
P/E (UBS, diluted) x	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,660.00 on 12-Oct-2026 06:29:19 CST

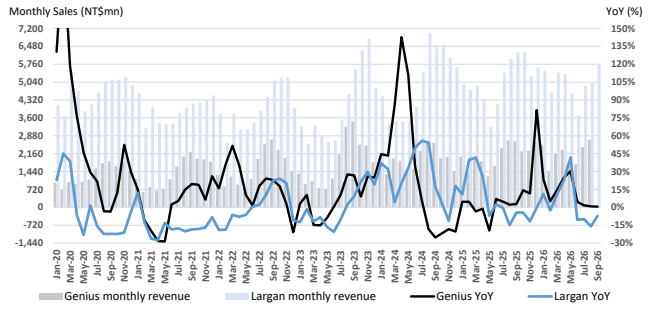
This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 13.**

Core smartphone lens seeing weakness with declining margins

Q326 sales was reported light at NT$15.7bn, +15% QoQ missing UBSe/Street +36%/ +33% amid weaker flagship smartphone demand. GMs fell 720bp to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. Renewable-energy costs are structural under Largan's long-term power commitments and rise with Apple's share of the mix; with 2027 pricing set, pass-through is more likely in 2028.

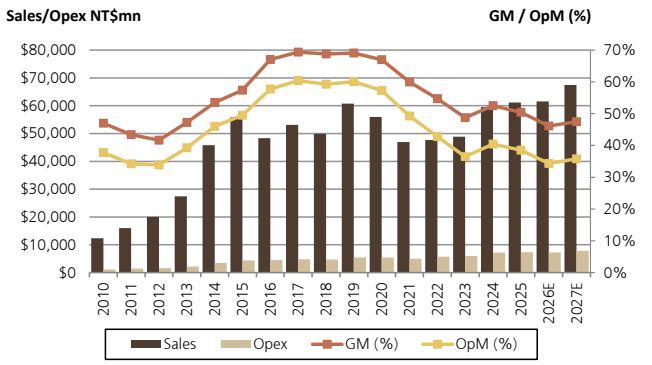
We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (flat vs. -11% YoY). October revenue should be flat MoM and November lower as the key customer has shortened its forecast after cutting flagship orders in September. The spring-2027 model starts shipping in October but will not offset the flagship decline.

Figure 1: Genius outpacing Largan in recent months



Source: Company data

Figure 2: Largan margins dropping in recent years



Source: Company data, UBS estimates

We cut 2026E/27E revenue 9%/13% to NT$61.6bn/NT$67.4bn and EPS 14%/16% to NT$163/NT$180, and introduce 2028E revenue of NT$79.5bn, +18% YoY for the new smartphone lens upgrade, taking EPS to NT$220. Our estimates exclude CPO, which we value separately under base, bull and bear scenarios. Including our base case, 2008E EPS would be NT$255, 6% above the NT$240 consensus which likely includes some CPO. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass-through in 2028 pricing. With a still a range of outcomes on 2028 CPO, we keep it in our valuation as a scenario for our upside/downside/base cases.

Figure 3: UBS vs. VA Consensus Q126A-Q426E and 2025A-28E

	Q126	Q226	Q326E			Q426E			2025	2026E			2027E			2028E		
(NT$m)	Actual	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
Revenue	**15,544**	**13,665**	**15,651**	**18,566**	**18,147**	**16,690**	**20,094**	**16,173**	**61,148**	**61,550**	**67,869**	**61,033**	**67,438**	**77,189**	**68,359**	**79,464**	**83,044**	**115,261**
- QoQ / YoY	-9.7%	-12.1%	14.5%	35.9%	32.8%	6.6%	28.4%	3.3%	2.8%	0.7%	11.0%	-0.2%	9.6%	13.7%	12.0%	17.8%	7.6%	68.6%
Gross profit	**7,680**	**6,753**	**6,607**	**9,242**	**8,968**	**7,346**	**10,044**	**6,980**	**30,837**	**28,386**	**33,718**	**28,019**	**32,008**	**38,405**	**30,319**	**38,823**	**42,004**	**53,520**
- Gross margin	49.4%	49.4%	42.2%	49.8%	49.4%	44.0%	50.0%	43.2%	50.4%	46.1%	49.7%	45.9%	47.5%	49.8%	44.4%	48.9%	50.6%	46.4%
Operating profit	**5,812**	**5,148**	**4,735**	**7,144**	**6,819**	**5,443**	**7,833**	**5,105**	**23,558**	**21,139**	**25,937**	**20,800**	**24,150**	**29,796**	**21,482**	**30,110**	**34,945**	**39,917**
- OP margin	37.4%	37.7%	30.3%	38.5%	37.6%	32.6%	39.0%	31.6%	38.5%	34.3%	38.2%	34.1%	35.8%	38.6%	31.4%	37.9%	42.1%	34.6%
Non-op profit	**1,476**	**951**	**1,357**	**1,027**	**1,071**	**1,064**	**1,064**	**1,477**	**2,342**	**4,847**	**4,518**	**5,261**	**4,467**	**4,467**	**4,734**	**4,595**	**4,595**	**4,937**
Pre-tax profit	**7,288**	**6,099**	**6,092**	**8,171**	**7,889**	**6,507**	**8,897**	**6,582**	**25,900**	**25,986**	**30,455**	**26,061**	**28,618**	**34,263**	**26,543**	**34,705**	**39,540**	**44,854**
Tax	**-1,112**	**-1,373**	**-853**	**-1,385**	**-1,250**	**-1,041**	**-1,499**	**-1,306**	**-4,340**	**-4,379**	**-5,369**	**-4,763**	**-5,050**	**-6,244**	**-5,150**	**-5,900**	**-6,722**	**-8,572**
- Tax rate	15.3%	22.5%	14.0%	17.0%	15.8%	16.0%	16.9%	19.8%	16.8%	16.9%	17.6%	18.3%	17.6%	18.2%	19.4%	17%	17%	19%
Net profit	**6,123**	**4,670**	**5,182**	**6,736**	**6,614**	**5,409**	**7,348**	**5,275**	**21,275**	**21,385**	**24,877**	**21,298**	**23,567**	**28,019**	**21,414**	**28,805**	**32,818**	**36,282**
EPS (NT$)	**46.63**	**35.70**	**39.62**	**51.50**	**49.69**	**41.36**	**56.18**	**40.04**	**159.41**	**163.33**	**190.00**	**161.25**	**180.00**	**214.00**	**161.64**	**220.00**	**250.66**	**275.28**

Source: Company data, Visible Alpha, UBS estimates. Note: EPS based on reported and may differ with cover page diluted EPS.

CPO fiber array qualified, but rising pass-through as it upgrades to FAU production

Largan's fiber array (FA) has passed qualification with a volume customer, and management expects mass-production capacity by mid-2027 in a newly completed fab dedicated to FA/FAU, using in-house equipment. The company's next step is to produce the full detachable FAU mounted on a carrier, a board with three alignment beams that sits on the optical engine's receptor. Largan can build FAUs now and says the customer

would prefer it to supply the full assembly, but qualification is pending with the customer preferring FAU and carrier sampled separately.

Even as its potential to supply the full FAU improve, Largan will buy in most of the content. The customer has qualified another PMLA vendor and prefers semiconductor-made optics; V-grooves and carrier beams and boards are outsourced and the customer does not want Largan to make MT ferrules. Management described its role as closer to a contractor and said margins hinge on yields, as the FA carries a high ASP but also a large bill of materials. Supplying the complete FAU could raise billed content, but a larger purchased bill of materials also dilutes the relationship between revenue and value added. Our full-FAU ASP assumption requires Largan to capture that invoice scope.

Next year's product is grating-coupled and works for both CPO and NPO while edge-coupled designs are being explored with the customer. The current product does not go directly to end customers, but management expects to work directly with end customers on next-generation designs (likely 2028) as it progresses toward multi-row FAs.

To ramp production, Largan has bought several land parcels and will build new fabs from scratch, with construction finishing around 2030. It may outsource some capacity in the interim and said it cannot yet be sure capacity will meet demand.

Annual FAU TAM reaches US$1.85bn in 2028 and US$4.5bn in 2030 in our base case

We update our FAU (fiber array unit) TAM factoring CPO/NPO penetration into AI networking switches, NVIDIA multi-rack systems, scale-up on compute and switch boards, and non-NVIDIA accelerators. We assume one passive FAU assembly connecting a fiber array to a photonic engine with one unit per optical engine.

We assume one FAU per optical engine and 32 per switch ASIC in our base case, matching NVIDIA's 32x3.2T engines on its 102.4T Spectrum-X package. Broadcom's 102.4Tb/s Davisson uses 16 engines at 6.4Tb/s, so our bear case assumes and even mix of these two designs (24 per ASIC). We note that higher switch bandwidth does not automatically increase FAU count if engine bandwidth rises at the same rate. Our 2030 assumption maintains the 32 FAUs per switch based on bandwidth rising to 204.8Tb/s ASICs using 6.4Tb/s optical engines.

Figure 4: UBS scenario based on 32 FAUs per switch although switches to date carry a range of specifications

	TH5 Bailly	TH6 Davisson	Quantum-X Q3450	Spectrum-X SN6810	Spectrum-X SN6800
Year of Introduction	**2023**	**H225**	**H225**	**H226**	**H226**
Switch ASIC	Tomahawk 5	Tomahawk 6	Quantum-3	Spectrum 6	Spectrum 6
Bandwidth per Switch box	51.2T	102.4T	115.2T	102.4T	409.6T
Switch ASICs per box	1	1	4	1	4
OE bandwidth (Tb/s each way)	6.4T	6.4T	1.6T	3.2T	3.2T
Optical engines per ASIC	8	16	18	32	32
FAU per Engine	1	1	1	1	1
FAU per Switch	**8**	**16**	**72**	**32**	**128**

Source: Company Data, UBS

Our base case for the FAU market sums up 4 opportunities for CPO/NPO attach:

1. Scale-out and scale-across switches

We estimate annual switch ASIC shipments x CPO + NPO penetration x optical engines per ASIC x FAUs per engine. In 2028, 25% CPO/NPO penetration or 1.0m 102.4T switch ASICs at 32 FAUs each produces 8.0m FAUs. In 2030, 55% of 1.1mn 204.8T switch ASICs at 32 FAUs drives a market size of 19.4mn FAUs.

Figure 5: Scale-out/scale-across could drive 8.0/19.4mn FAUs in 2028/2030

	2028			2030			Assumption
Scale out/Scale across	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
Switch ASIC shipments (k units)	800	1,000	1,050	880	1,100	1,155	
NPO penetration	3%	5%	7%	8%	10%	12%	Optics on the board beside the ASIC
CPO penetration	10%	20%	25%	35%	45%	55%	Optics inside the switch package.
NPO/CPO switch ASICs (k units)	**104**	**250**	**336**	**378**	**605**	**774**	
Bandwidth per Switch (Tbps)	102.4	102.4	102.4	204.8	204.8	204.8	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs per switch ASIC	24	32	32	24	32	32	NVIDIA: 32 OE per ASIC, Broadcom 16
Scale out/Scale across # of FAUs (mn)	**2.5**	**8.0**	**10.8**	**9.1**	**19.4**	**24.8**	

Source: UBS

2. NVIDIA's multi-rack connections for Rubin Ultra and Feynman

NVIDIA at its 2026 GTC San Jose proposed a multi-rack option to network together 8 Oberon NVL72 racks for the Rubin Ultra generation and 8 Kyber NVL144 racks. Feedback at the time showed possible transceiver or NPO connections for Rubin Ultra and CPO connections for Feynman. Our 2028 base case uses 110,000 racks, 15% multi-rack adoption and 75% CPO/NPO penetration, giving approximately 12,400 connected racks.

For 2030, our rack count falls to 35,000 racks as we assume Kyber would have 4x the number of GPUs, equivalent to 140k Oberon equivalents. We assume the 35,000 racks at 55% multi-rack adoption x 80% CPO/NPO attach gives 15,400 connected racks. We assume about 216 and 576 inter-rack FAUs per rack, respectively, driving a market opportunity of 2.7mn FAUs in 2028 and 8.9mn in 2030.

Figure 6: NVIDIA rack connected pods could drive 2.7mn/8.9mn FAUs in 2028/2030

	2028			2030			Assumption
NVIDIA Pod Connected GPUs	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV platform	Oberon / Rubin Ultra			Kyber / Feynman			
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
8-rack pods penetration (%)	10%	15%	20%	40%	55%	70%	
NPO/CPO penetration within pods (%)	50%	75%	80%	65%	80%	95%	Pluggable FAUs excluded from this model
NPO/CPO connected racks (k)	**4.8**	**12.4**	**20.0**	**7.8**	**15.4**	**26.6**	
GPUs per rack	72	72	72	144	144	144	
Switches per rack	72	72	72	144	144	144	72 ASICs in Oberon, 144 per Kyber
Optical engine (Tb) bandwidth	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs attached per switch	2.5	3.0	4.0	2.5	4.0	4.5	3.2T OE in 2028; 6.4T in 2030 (each way)
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
FAUs per connected racks	**180**	**216**	**288**	**360**	**576**	**648**	
NV scale up # of FAUs (mn)	**0.9**	**2.7**	**5.8**	**2.8**	**8.9**	**17.2**	

Source: UBS. Switch count increases to connect to other racks

3. Initial CPO/NPO optical connections on compute and switch boards by 2030

We include by 2030 CPO native optical connections on compute and switch boards in 5% of racks each adding 1,152 FAUs across the GPUs and switches or 2.0mn FAUs by 2030.

Figure 7: NVIDIA compute board optics could drive 2.0mn FAUs in 2030

	2028			2030			Assumption
NVIDIA Compute Board Optics	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
Optical GPU penetration	0%	0%	0%	3%	5%	15%	CPU-native GPUs
Compute tray FAUs #	0	0	0	144	144	144	36 trays x 16
GPUs per rack	72	72	72	144	144	144	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	0	0	0	720	1,152	1,296	
NV optical GPU # of FAUs (mn)	**0.0**	**0.0**	**0.0**	**0.6**	**2.0**	**7.8**	

Source: UBS.

4. Non NVIDIA scale-up opportunity

Non-NVIDIA scale-up uses accelerator rack shipments x optical attach rate x FAUs per rack, producing about 0.5m units in 2028 and 3.7m in 2030.

Figure 8: ASIC market could drive additional compute board optics could drive 3.7mn FAUs in 2030

	2028			2030			Assumption
Non-NV scale up	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
ASIC rack shipments (k)	43.6	48.5	50.9	76.6	85.1	89.3	*Trainium 4/5/6, MI500, MI600, Jalapeno
Addressable NPO/CPO attach rate (%)	0%	5%	10%	5%	15%	25%	
NPO/CPO connected racks (k)	**0.0**	**2.4**	**5.1**	**3.8**	**12.8**	**22.3**	
Accelerators per rack	72	72	72	72	72	72	Average accelerators per rack
FAUs per accelerator	2.5	3.0	4.0	2.5	4.0	4.5	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
Scale-up per non-NVIDIA accelerator (Tb)	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	180	216	288	180	288	324	
Non-NV # of FAUs (mn)	**0.0**	**0.5**	**1.5**	**0.7**	**3.7**	**7.2**	

Source: UBS

Overall market opportunity: 12.3mn/35.6mn generates a US$1.85bn/US $4.5bn TAM in 2028/2030

We sum four pools: scale-out/scale-across switches, NVIDIA multi-rack pods, CPO native NVIDIA GPUs and non-NVIDIA scale-up. We add 10%/5% for FAUs lost at downstream attach and test and for field spares, driving a market size of in our base case of 12.3mn FAUs in 2028 and 35.6mn in 2030. At US$150/US$125 per FAU, that is a US$1.85bn market in 2028 and US$4.5bn in 2030 (bear/bull US$0.5bn/US$3.5bn in 2028; US $1.4bn/US$9.0bn in 2030). The expansion to 2030 comes from wider optical adoption and greater content in scale-up systems, despite an assumed FAU ASP decline from US $150 to US$125.

Figure 9: Summary of FAU drivers: 12.3mn/35.6mn base case in 2028-2030

	2028			2030		
Summary of FAU drivers	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
Scale out/Scale across # of FAUs (mn)	2.5	8.0	10.8	9.1	19.4	24.8
NV scale up # of FAUs (mn)	0.9	2.7	5.8	2.8	8.9	17.2
NV optical GPU # of FAUs (mn)	0.0	0.0	0.0	0.6	2.0	7.8
Non-NV # of FAUs (mn)	0.0	0.5	1.5	0.7	3.7	7.2
Downstream yield loss and spares	10%	10%	10%	5%	5%	5%
FAUs (mn units)	**3.7**	**12.3**	**19.8**	**13.9**	**35.6**	**59.9**
FAU ASP	$125	$150	$175	$100	$125	$150
Total FAU TAM (US$bn)	**$0.5**	**$1.8**	**$3.5**	**$1.4**	**$4.5**	**$9.0**

Source: UBS

Scenario analysis on Largan's CPO opportunity

Our base case gives Largan the benefit of the doubt for successfully ramping up a full FAU production and capturing 25% of the overall FAU TAM, yielding NT$35 of 2028E EPS, about 16% of our 2028E lens EPS. A 25% share means 3.1mn FAUs in its first full year of production: about 29% of NVIDIA-linked units and roughly three-quarters of the allocation our checks suggest its module partner holds. We book these at the full US $150 FAU price and 50% gross margin, allowing for an early-supply premium even though the optics, V-grooves and carrier parts are bought in. That gives 2028E CPO sales of US$462mn, rising to US$1.1bn and NT$84 of EPS in 2030 on a larger market.

Our bear case (15% share, 45% GM) gives NT$5 of CPO EPS in 2028E and NT$14 in 2030E; our bull case (35% share, 55% GM) gives NT$103 and NT$267. Each 5ppt of share moves 2028E CPO EPS by ~NT$6.9 (~NT$340/share at 49x), and each 5ppt of gross margin by ~NT$4.6 (~NT$225/share). If gross margin were 35% rather than 50%, holding operating expenses at 12% of sales, EPS would fall to approximately NT$21. A next-generation shift in lens or coupling design could also change content and alignment requirements including a potential metalens design which could invite in different suppliers.

Our 2030 base case implies ~8.9mn Largan FAUs a year (21mn in the upside case), though with risks around competition in a crowded field, yields and technology. A shift to semiconductor-made optics or metalens would loosen alignment tolerances and erode the precision premium behind Largan's pitch. The upside lies in direct work with cloud providers on multi-row FAs from 2028, where Largan's multi-layer alignment is most differentiated.

Figure 10: Largan downside/base/upside cases for 2028 and 2030

Largan CPO	2028			2030		
Scenario analysis	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
FAUs (mn units)	3.7	12.3	19.8	13.9	35.6	59.9
FAU ASP	125	150	175	100	125	150
Total FAU TAM (US$mn)	**$461**	**$1,847**	**$3,461**	**$1,389**	**$4,452**	**$8,980**
Largan Sales (US$mn)	**$69.1**	**$461.9**	**$1,211.3**	**$208.3**	**$1,113.1**	**$3,142.8**
Largan share of the TAM (%)	15%	25%	35%	15%	25%	35%
GM %	45%	50%	55%	45%	50%	55%
Pretax %	33%	38%	43%	33%	38%	43%
Pretax income (US$mn)	**$22.8**	**$175.5**	**$520.8**	**$68.7**	**$423.0**	**$1,351.4**
Tax	18%	18%	18%	18%	18%	18%
Net income (US$mn)	**$18.7**	**$143.9**	**$427.1**	**$56.4**	**$346.8**	**$1,108.2**
Net income (NT$mn)	**$589**	**$4,533**	**$13,453**	**$1,776**	**$10,925**	**$34,907**
EPS (NT$)	**$5**	**$35**	**$103**	**$14**	**$84**	**$267**
Shares	130.8	130.8	130.8	130.8	130.8	130.8

Source: UBS

Valuation and investment implications

We roll our valuation forward to 2028E and value Largan on a sum of the parts. We value the lens business at NT$3,300 (15x 2028E EPS, the middle of Largan's 13–18x historical range) and CPO at NT$1,700 (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US$1.85bn FAU market at US$150 per unit and a 50% gross margin. The 49x multiple also assigns a substantial value to a business whose complete assembly still requires qualification and whose margins still need to demonstrate can reach similar levels to its mature smartphone business. At ~NT$6,660 recent share price level, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Upside risks include faster full-FAU qualification, firm volume commitments from additional customers, stronger lens demand and better yields. Near-term catalysts are the customer's capacity forecast and its visit to the new fab in the coming month. A large forecast or faster FAU plus carrier qualification would be upside risks to our sell.

Figure 11: UBS SOTP valuation for Largan

(NT$)		UBSe			Prior UBS		Upside Case			Downside Case		
	2025	**2026**	**2027**	**2028**	**2026**	**2027**	**2026**	**2027**	**2028**	**2026**	**2027**	**2028**
Lens Business ex-CPO												
Sales	$61,148	$61,550	$67,438	$79,464	$67,869	$77,189	$62,166	$70,135	$82,642	$59,704	$64,066	$75,491
YoY Growth	2.8%	0.7%	9.6%	17.8%	11.0%	13.7%	1.7%	12.8%	17.8%	-2.4%	7.3%	17.8%
Gross Profit	**$30,837**	**$28,386**	**$32,008**	**$38,823**	**$33,718**	**$38,405**	**$28,981**	**$34,410**	**$41,699**	**$26,997**	**$29,446**	**$35,750**
GM%	**50.4%**	**46.1%**	**47.5%**	**48.9%**	**49.7%**	**49.8%**	**46.6%**	**49.1%**	**50.5%**	**45.2%**	**46.0%**	**47.4%**
Opex	$7,279	$7,247	$7,857	$8,714	$7,781	$8,609	$7,283	$8,015	$8,888	$7,138	$7,661	$8,496
Op Profit	**$23,558**	**$21,139**	**$24,150**	**$30,110**	**25,937**	**29,796**	**21,697**	**$26,396**	**$32,811**	**19,858**	**$21,785**	**$27,254**
OpM%	**38.5%**	**34.3%**	**35.8%**	**37.9%**	**38.2%**	**38.6%**	**34.9%**	**37.6%**	**39.7%**	**33.3%**	**34.0%**	**36.1%**
Non-Op	$2,342	$4,847	$4,467	$4,595	$4,518	$4,467	$4,847	$4,467	$4,595	$4,847	$4,467	$4,595
Pretax	$25,900	$25,986	$28,618	$34,705	$30,455	$34,263	$26,545	$30,863	$37,405	$24,706	$26,253	$31,849
Tax	($4,340)	($4,379)	($5,050)	($5,900)	($5,369)	($6,244)	($4,473)	($5,447)	($6,601)	($4,163)	($4,633)	($5,621)
Net income	**$21,275**	**$21,385**	**$23,567**	**$28,805**	**$24,877**	**$28,019**	**$21,849**	**$25,416**	**$30,804**	**$20,321**	**$21,620**	**$26,228**
EPS	**$159.41**	**$163.33**	**$180.00**	**$220.00**	**$190.00**	**$214.00**	**$166.88**	**$194.12**	**$235.28**	**$155.20**	**$165.13**	**$200.32**
Shares	133	131	131	131	131	131	131	131	131	131	131	131
EPS vs. base case							**-94.1%**	**7.8%**	**6.9%**	**-5.0%**	**-8.3%**	**-8.9%**
Current Price		**$6,660**										
P/E				15		15			16			14
Implied Valuation				**$3,300**		**$3,270**			**$3,670**			**$2,810**
CPO Scenario												
EPS				**$35**		**$35**			**$103**			**$5**
P/E				49		50			52			47
Implied Valuation				**$1,700**		**$1,730**			**$5,300**			**$210**
Total Largan												
Implied Share Price				**$5,000**		**$5,000**			**$8,970**			**$3,020**
Upside/Downside				**-25%**					**35%**			**-55%**

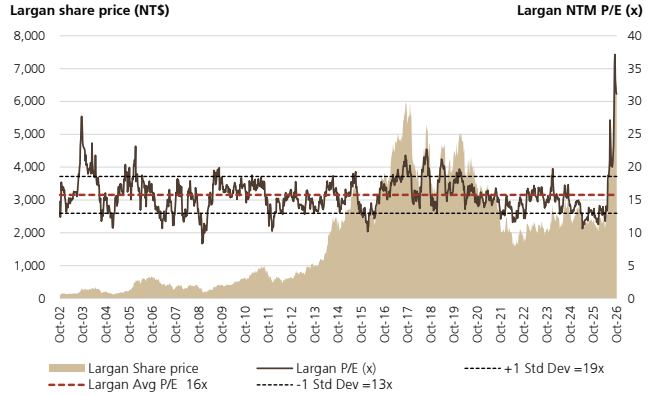
Source: Company data, UBS estimates

Figure 12: Optical component peers average 27x 2028 with a broad range of 12-75x 2028 P/E

Optical Company	Ticker	Market cap (US$m)	Price 11-Oct (LC$)	Price target (LC$)	P/E target (x)	Rating	EPS (LC$) 2025	EPS (LC$) 2026E	EPS (LC$) 2027E	EPS (LC$) 2028E	Earnings growth (%) 2026E	Earnings growth (%) 2027E	Earnings growth (%) 2028E	PE (x) 2026E	PE (x) 2027E	PE (x) 2028E	ROE (%) 2026E	ROE (%) 2027E	ROE (%) 2028E	Historical P/E (x) -1 Std Dev	Historical P/E (x) Avg.	Historical P/E (x) +1 Std Dev
Lens																						
Largan	3008.TW	27,378	6,660.0	5,000	23.4	Sell	176.92	188.21	214.00	250.66	4.4	13.7	17.1	35.4	31.1	26.6	12.7	13.7	14.8	13.1	15.9	18.8
Genius	3406.TW	3,741	1,060.0	1,250	16.9	Buy	36.28	38.72	41.47	47.75	6.7	7.1	15.1	27.4	25.6	22.2	15.8	15.6	16.4	1.1	2.3	3.5
Average											**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**
CPO Supply Chain																						
Lumentum	LITE.O	93,344	1,103.4	960	40.4	Neutral	2.12	10.52	23.78	35.81	434.4	160.4	51.7	52.0	56.0	37.9	27.1	36.1	37.9	8.5	19.4	30.3
Corning	GLW.N	137,095	156.7	201	42.1	Buy	2.57	3.34	4.78	7.12	30.6	43.5	47.3	47.7	33.4	22.4	22.9	31.8	41.1	-168.2	32.9	234.0
Coherent	COHR.N	123,129	312.6	-	-	-	4.49	7.42	11.74	16.82	86.0	64.0	42.0	42.1	26.6	18.6	13.5	19.7	23.8	14.1	19.6	25.1
Senko	9069.T	416,040	2,368.0	-	-	-	119.98	140.89	159.70	171.34	19.5	12.8	7.4	16.8	14.8	13.8	10.4	11.3	11.6	9.2	11.6	14.0
TFC	300394.SZ	256,388	235.0	-	-	-	2.02	3.32	5.36	6.88	65.5	60.7	28.5	70.8	43.9	34.2	47.0	47.4	42.1	21.2	33.9	46.5
FOCI	3363.TWO	75,798	667.0	-	-	-	0.85	0.22	13.89	43.01	(66.3)	6684.0	210.2	3,102.3	48.0	15.5	-1.6	39.2	74.6	51.6	115.2	178.8
EZconn	6442.TW	146,044	1,785.0	-	-	-	20.95	62.01	130.00	NULL	206.9	109.6	n.a.	28.8	13.7	n.a.	56.1	63.9	NULL	12.9	23.7	34.5
Himax	HIMX.N	2,320	13.3	-	-	-	0.32	0.52	1.01	1.96	62.3	88.2	93.5	25.5	13.2	6.8	10.5	14.5	26.1	0.6	21.5	42.3
Browave	3163.TWO	65,204	689.0	-	-	-	4.36	9.33	21.17	NULL	n.a.	n.a.	n.a.	73.8	32.5	n.a.	22.2	38.6	NULL	10.1	32.2	54.2
VisEra	6789.TW	153,102	481.0	-	-	-	3.60	6.13	7.45	8.95	70.9	21.5	20.2	78.4	64.6	53.7	10.7	13.3	12.2	31.3	50.4	69.6
Average											**101.1**	**805.0**	**62.6**	**353.8**	**34.7**	**25.4**	**21.9**	**31.6**	**33.7**	**-0.9**	**36.0**	**72.9**
Optical Average											**76.4**	**606.4**	**50.7**	**325.1**	**35.5**	**27.2**	**19.9**	**27.5**	**28.6**	**1.2**	**30.5**	**59.8**

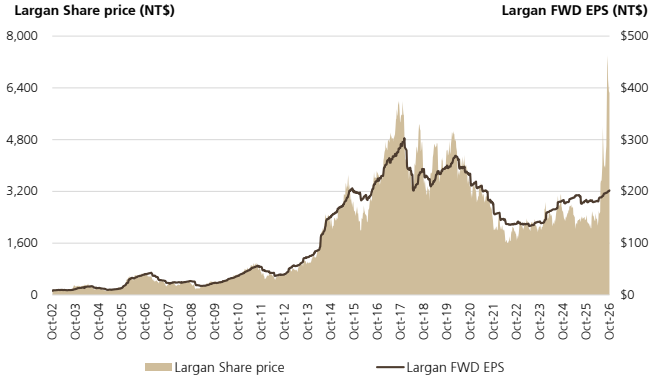
Source: UBSe, Eikon

Figure 13: Share price rally has re-rated Largan well above its historical range



Source: LSEG Workspace

Figure 14: Largan's share price has run well ahead of forward EPS as investors price in CPO



Source: LSEG Workspace

Figure 15: UBS P&L for Largan - mild growth at lower margins in recent years

Ticker		3008.TW	Current Price (NT$)			$6,660
Shares outstanding (mn)		130.8	Mkt cap (NT$ mn)			$871,115
Year	**2022**	**2023**	**2024**	**2025**	**2026E**	**2027E**
Revenues (NT$ mn)	**$47,675**	**$48,842**	**$59,458**	**$61,148**	**$61,550**	**$67,438**
YoY Growth (%)		**2.4%**	**21.7%**	**2.8%**	**0.7%**	**9.6%**
Gross profit (NT$mn)	$26,083	$23,794	$31,209	$30,837	$28,386	$32,008
GM (%)	**54.7%**	**48.7%**	**52.5%**	**50.4%**	**46.1%**	**47.5%**
Operating profit (NT$mn)	$20,384	$17,807	$24,033	$23,558	$21,139	$24,150
OPM (%)	**42.8%**	**36.5%**	**40.4%**	**38.5%**	**34.3%**	**35.8%**
Net profit (NT$ mn)	$22,625	$17,902	$25,915	$21,275	$21,385	$23,567
EPS (NT$)	**$169.52**	**$134.13**	**$194.17**	**$159.41**	**$163.33**	**$180.00**
EPS growth (%)		-21%	45%	-18%	2%	10%
P/E (x)	**39.3**	**49.7**	**34.3**	**41.8**	**40.8**	**37.0**
Dividend yield (%)	2%	1.1%	1.2%	1.3%	2.0%	1.2%
P/B (x)	**5.7**	**5.4**	**4.8**	**4.7**	**4.4**	**4.1**
ROE (%)	15%	11%	14%	11%	11%	12%

Quarters	**Q325**	**Q425**	**Q126**	**Q226**	**Q326E**	**Q426E**
Revenues (NT$ mn)	**$17,677**	**$17,219**	**$15,544**	**$13,665**	**$15,651**	**$16,690**
QoQ Growth (%)		**-2.6%**	**-9.7%**	**-12.1%**	**14.5%**	**6.6%**
Gross profit (NT$mn)	$8,352	$8,260	$7,680	$6,753	$6,607	$7,346
GM (%)	**47.2%**	**48.0%**	**49.4%**	**49.4%**	**42.2%**	**44.0%**
Operating profit (NT$mn)	$6,265	$6,329	$5,812	$5,148	$4,735	$5,443
OPM (%)	**35.4%**	**36.8%**	**37.4%**	**37.7%**	**30.3%**	**32.6%**
Net profit (NT$ mn)	$7,080	$6,720	$6,123	$4,670	$5,182	$5,409
EPS (NT$)	**$53.05**	**$50.35**	**$45.88**	**$34.99**	**$39.58**	**$41.32**

Source: Company data, UBS estimates. Note: 2026E EPS based on reported.

Figure 16: UBS Cash Flow Summary - staying positive to invest in CPO

Annual (NT$mn)	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
Revenue	$46,962	$47,675	$48,842	$59,458	$61,148	$61,550	$67,438	$52,817
Capital spending	$5,850	$8,256	$8,080	$11,126	$12,338	$11,721	$11,135	$9,130
Capex/revenue (%)	**12%**	**17%**	**17%**	**19%**	**20%**	**19%**	**17%**	**17%**
Dep and amort	$4,650	$5,031	$5,299	$6,038	$7,511	$9,221	$10,460	$5,706
Depr/revenue (%)	**10%**	**11%**	**11%**	**10%**	**12%**	**15%**	**16%**	**11%**
Operating cash flow	$20,878	$44,207	$18,199	$31,579	$26,080	$20,614	$33,003	$28,188
Free cash flow	**$15,028**	**$35,951**	**$10,118**	**$20,453**	**$13,743**	**$8,893**	**$21,868**	**$19,058**

FCF and Dividends	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
FCF / share (NT$)	$112	$269	$76	$153	$103	$68	$167	$143
FCF / EV Yield (%)	**2%**	**5%**	**1%**	**3%**	**2%**	**1%**	**3%**	**3%**
FCF Yield (%)	**2%**	**4%**	**1%**	**2%**	**2%**	**1%**	**3%**	**2%**
Dividend/share (NT$)	$91.51	$109.66	$72.50	$81.00	$85.50	$131.95	$81.89	$88.03
Payout rate (%)	**50%**	**79%**	**43%**	**60%**	**44%**	**83%**	**50%**	**55%**
Dividend amount (NT$)	$12,274	$14,636	$9,676	$10,811	$11,412	$17,276	$10,722	$11,762
Pre-Dividend close (NT$)	$3,040.00	$1,990.00	$2,310.00	$3,120.00	$2,260.00	$6,660.00	$6,660.00	
Dividend Yield (%)	**3.0%**	**5.5%**	**3.1%**	**2.6%**	**3.8%**	**2.0%**	**1.2%**	**3.6%**

Source: Company data, UBS estimates

Largan Precision (3008.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**48,842**	**59,458**	**61,148**	**61,550**	*0.7*	**67,438**	*9.6*	**79,464**	**78,546**	**81,569**
Gross profit	23,794	31,209	30,837	28,386	*-7.9*	32,008	*12.8*	38,823	38,398	39,903
EBITDA (UBS)	**23,106**	**30,070**	**31,069**	**30,360**	*-2.3*	**34,610**	*14.0*	**41,028**	**41,108**	**42,938**
Depreciation & amortisation	(5,299)	(6,038)	(7,511)	(9,221)	*-22.8*	(10,460)	*-13.4*	(10,918)	(11,621)	(12,290)
EBIT (UBS)	**17,807**	**24,033**	**23,558**	**21,139**	*-10.3*	**24,150**	*14.2*	**30,110**	**29,487**	**30,648**
Associates & investment income	212	0	3	12	*352.1*	11	*-7.1*	11	11	11
Other non-operating income	294	303	419	458	*9.5*	200	*-56.4*	200	200	200
Net interest	3,934	4,404	4,259	3,946	*-7.3*	4,256	*7.9*	4,384	4,516	4,651
Exceptionals (incl goodwill)	(145)	3,435	(2,338)	432	*-*	0	*-*	0	0	0
Pre-tax profit	**22,102**	**32,174**	**25,900**	**25,986**	*0.3*	**28,618**	*10.1*	**34,705**	**34,213**	**35,510**
Tax	(4,200)	(5,963)	(4,340)	(4,379)	*-0.9*	(5,050)	*-15.3*	(5,900)	(5,816)	(6,037)
Profit after tax	**17,902**	**26,211**	**21,560**	**21,607**	*0.2*	**23,567**	*9.1*	**28,805**	**28,397**	**29,473**
Preference dividends	0	0	0	0	*-*	0	*-*	0	0	0
Minorities	0	(296)	(285)	(222)	*22.1*	0	*-*	0	0	0
Extraordinary items	0	0	0	0	*-*	0	*-*	0	0	0
Net earnings (local GAAP)	**17,902**	**25,915**	**21,275**	**21,385**	*0.5*	**23,567**	*10.2*	**28,805**	**28,397**	**29,473**
Net earnings (UBS)	**18,047**	**22,480**	**23,613**	**20,953**	*-11.3*	**23,567**	*12.5*	**28,805**	**28,397**	**29,473**
Tax rate (%)	19.0	18.5	16.8	16.9	*0.6*	17.6	*4.7*	17.0	17.0	17.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
EPS (local GAAP, diluted)	134.13	194.17	159.41	163.33	*2.5*	180.00	*10.2*	220.00	216.89	225.11
EPS (UBS, basic)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	*-17.9*	81.97	*2.5*	90.34	110.41	108.85
Cash EPS (UBS, diluted) [1]	174.92	213.67	233.20	230.47	*-1.2*	259.89	*12.8*	303.39	305.65	318.97
Book value per share	1,234.64	1,382.91	1,430.07	1,518.08	*6.2*	1,616.11	*6.5*	1,745.87	1,849.64	1,963.23
Average shares (diluted)	133	133	133	131	*-1.9*	131	*0.0*	131	131	131

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	111,655	117,985	122,309	109,131	*-10.8*	120,267	*10.2*	127,842	144,348	160,697
Other current assets	22,666	27,779	31,977	42,218	*32.0*	45,034	*6.7*	56,368	55,712	57,851
Total current assets	**134,321**	**145,764**	**154,285**	**151,350**	*-1.9*	**165,301**	*9.2*	**184,210**	**200,060**	**218,547**
Net tangible fixed assets	41,135	46,936	51,472	54,428	*5.7*	55,103	*1.2*	54,763	53,191	50,448
Net intangible fixed assets	0	0	0	0	*-*	0	*-*	0	0	0
Investments / other assets	19,682	23,827	15,029	23,254	*54.7*	24,367	*4.8*	24,215	24,225	24,236
Total assets	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**
Trade payables & other ST liabilities	29,517	30,375	29,749	30,078	*1.1*	32,982	*9.7*	34,573	35,276	36,158
Short term debt	0	203	0	0	*-*	0	*-*	0	0	0
Total current liabilities	**29,517**	**30,578**	**29,749**	**30,078**	*1.1*	**32,982**	*9.7*	**34,573**	**35,276**	**36,158**
Long term debt	0	0	0	0	*-*	0	*-*	0	0	0
Other long term liabilities	110	561	171	194	*13.7*	194	*0.0*	31	31	31
Preferred shares	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities (incl pref shares)	**29,627**	**31,139**	**29,919**	**30,272**	*1.2*	**33,176**	*9.6*	**34,604**	**35,307**	**36,189**
Common s/h equity	**165,510**	**185,388**	**190,868**	**198,760**	*4.1*	**211,594**	*6.5*	**228,584**	**242,170**	**257,042**
Minority interests	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities & equity	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	17,902	25,915	21,275	21,385	*0.5*	23,567	*10.2*	28,805	28,397	29,473
Depreciation & amortisation	5,299	6,038	7,511	9,221	*22.8*	10,460	*13.4*	10,918	11,621	12,290
Net change in working capital	(8,990)	(6,633)	(7,328)	(1,082)	*85.2*	(6,064)	*NM*	(9,743)	1,359	(1,256)
Other operating	(212)	295	282	210	*-25.5*	5,040	*NM*	(11)	(11)	(11)
Operating cash flow	**13,999**	**25,615**	**21,740**	**29,735**	*36.8*	**33,003**	*11.0*	**29,968**	**41,366**	**40,496**
Tangible capital expenditure	(8,080)	(11,126)	(12,338)	(11,721)	*5.0*	(11,135)	*5.0*	(10,578)	(10,049)	(9,547)
Intangible capital expenditure	0	0	0	0	*-*	0	*-*	0	0	0
Net (acquisitions) & disposals	(2,111)	(13)	(1,200)	962	*-*	0	*-*	0	0	0
Other investing	(959)	(4,927)	4,302	(965)	*-*	0	*-*	0	0	0
Investing cash flow	**(11,151)**	**(16,066)**	**(9,236)**	**(11,724)**	*-26.9*	**(11,135)**	*5.0*	**(10,578)**	**(10,049)**	**(9,547)**
Equity dividends paid	(9,676)	(10,811)	(11,412)	(17,276)	*-51.4*	(10,732)	*37.9*	(11,816)	(14,811)	(14,601)
Share issues / (buybacks)	0	0	0	0	*-*	0	*-*	0	0	0
Other financing	(34)	1,255	(2,937)	(4,133)	*-40.7*	0	*-*	0	0	0
Change in debt & pref shares	(19)	212	(212)	0	*-*	0	*-*	0	0	0
Financing cash flow	**(9,729)**	**(9,344)**	**(14,561)**	**(21,408)**	*-47.0*	**(10,732)**	*49.9*	**(11,816)**	**(14,811)**	**(14,601)**
Cash flow inc/(dec) in cash	**(6,881)**	**205**	**(2,057)**	**(3,398)**	*-65.2*	**11,136**	*-*	**7,575**	**16,506**	**16,349**
FX / non cash items	6,823	6,125	6,380	(9,780)	*-*	0	*-*	0	0	0
Balance sheet inc/(dec) in cash	**(59)**	**6,330**	**4,324**	**(13,177)**	*-*	**11,136**	*-*	**7,575**	**16,506**	**16,349**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Largan Precision (3008.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	16.4	13.1	14.8	40.8	37.0	30.3	30.7	29.6
P/E (UBS, diluted)	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
P/CEPS	12.6	11.9	10.1	28.9	25.6	22.0	21.8	20.9
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6
P/BV	1.8	1.8	1.6	4.4	4.1	3.8	3.6	3.4
EV/revenues (core)	3.6	3.6	3.1	NM	NM	9.3	9.3	8.7
EV/EBITDA (UBS core)	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
EV/EBIT (core)	10.0	8.9	8.0	35.4	31.1	24.6	24.7	23.2
EV/OpFCF (core)	8.1	7.4	6.3	25.1	22.0	18.0	17.7	16.5
EV/op. invested capital	3.9	4.1	3.3	10.7	9.3	8.6	8.2	8.2

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	294,465	340,820	315,906	874,578	874,578	874,578	874,578	874,578
Net debt (cash)	(111,675)	(114,718)	(120,045)	(115,720)	(114,699)	(124,054)	(136,095)	(152,522)
Buy out of minorities	0	0	0	0	0	0	0	0
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**182,790**	**226,102**	**195,861**	**758,858**	**759,879**	**750,524**	**738,483**	**722,055**
Non core assets	(4,916)	(12,460)	(8,517)	(9,934)	(9,945)	(9,955)	(9,966)	(9,976)
Core enterprise value	**177,874**	**213,642**	**187,344**	**748,924**	**749,934**	**740,568**	**728,517**	**712,079**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	2.4	21.7	2.8	0.7	9.6	17.8	(1.2)	3.8
EBITDA (UBS)	(9.1)	30.1	3.3	(2.3)	14.0	18.5	0.2	4.5
EBIT (UBS)	(12.6)	35.0	(2.0)	(10.3)	14.2	24.7	(2.1)	3.9
EPS (UBS, diluted)	3.8	24.6	5.0	(9.5)	12.5	22.2	(1.4)	3.8
Net DPS	21.9	(21.1)	44.4	(17.9)	2.5	10.2	22.2	(1.4)

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	48.7	52.5	50.4	46.1	47.5	48.9	48.9	48.9
EBITDA margin	47.3	50.6	50.8	49.3	51.3	51.6	52.3	52.6
EBIT (UBS) margin	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
Net earnings (UBS) margin	37.0	37.8	38.6	34.0	34.9	36.2	36.2	36.1
ROIC (EBIT)	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
ROIC post tax	31.8	36.6	34.6	25.1	24.8	29.1	27.5	29.3
ROE (UBS)	11.3	12.8	12.6	10.8	11.5	13.1	12.1	11.8

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(4.8)	(3.9)	(3.9)	(3.6)	(3.5)	(3.1)	(3.5)	(3.7)
Net debt / total equity %	(67.5)	(63.5)	(64.1)	(54.9)	(56.8)	(55.9)	(59.6)	(62.5)
Net debt / (net debt + total equity) %	NM	NM	NM	NM	NM	NM	NM	NM
Net debt/EV %	(61.1)	(50.7)	(61.3)	(15.2)	(15.1)	(16.5)	(18.4)	(21.1)
Capex / depreciation %	152.5	184.3	164.3	127.1	106.5	96.9	86.5	77.7
Capex / revenue %	16.5	18.7	20.2	19.0	16.5	13.3	12.8	11.7
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.6	2.5	1.8	2.0	2.2	2.4	2.0	2.1
Div. payout ratio (UBS) %	63.2	40.1	55.1	50.0	45.5	41.1	50.9	48.4

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
Total	**48,842**	**59,458**	**61,148**	**61,550**	**67,438**	**79,464**	**78,546**	**81,569**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Total	**17,807**	**24,033**	**23,558**	**21,139**	**24,150**	**30,110**	**29,487**	**30,648**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	-24.9%
Forecast dividend yield	1.2%
Forecast stock return	-23.7%
Market return assumption	6.3%
Forecast excess return	-30.0%

Company Description

Founded in 1987, Largan Precision (Largan) is one of the world's leading manufacturers of lenses used in cameras and IT products. It specialises in plastic/hybrid lenses and, to a lesser extent, glass lenses and modules. In 2022, 20MP+ camera lenses were 10-20% of Largan's shipment mix, 10MP lenses were 50-60%, 8MP lenses were 0-10% and other lenses were 30-40%.

Valuation Method and Risk Statement

Our price target is based on SOTP valuation on the CPO opportunity and traditional lens business.
We think Largan's downside risks include: 1) ongoing pricing pressure from smartphone OEM clients and increasingly intense competition among lens suppliers; and 2) a slower-than-expected smartphone demand recovery in China, and 3) CPO commercial success below the market's elevated expectations.
We believe the key upside risk is a faster-than-expected CPO qualification process at Largan. In addition, if the company demonstrates exceptionally efficient and highly accurate automation capabilities, it could support greater market share gains within the industry than we currently anticipate.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Largan Precision

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	3
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	2
5. What's driving the difference?	We expect downside to consensus in 2026 due to slower smartphone sales through August. We indicate things getting better for the stock in the past 3-6 months on the smartphone traction.
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	2
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Negative Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Irene Chang, Randy Abrams.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Largan Precision[28]	3008.TW	Sell	NT$6,660	08 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

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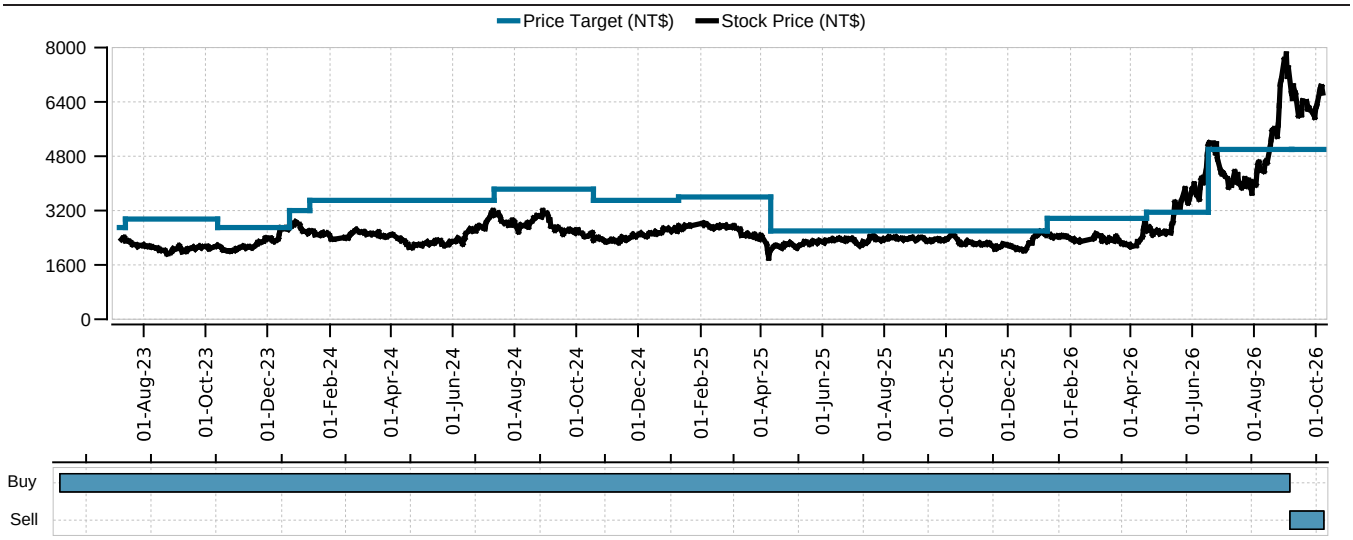
Largan Precision (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	2340.00	2700.00	Buy
2023-07-13	2405.00	2950.00	Buy
2023-10-12	2150.00	2700.00	Buy
2023-12-22	2650.00	3200.00	Buy
2024-01-11	2550.00	3500.00	Buy
2024-07-11	3200.00	3830.00	Buy
2024-10-17	2545.00	3500.00	Buy
2025-01-09	2595.00	3600.00	Buy
2025-04-10	1980.00	2600.00	Buy
2026-01-08	2475.00	2970.00	Buy
2026-04-16	2820.00	3150.00	Buy
2026-06-16	4860.00	5000.00	Buy
2026-09-06	7400.00	5000.00	Sell

Source: UBS Global Research; LSEG Eikon as of 08-Oct-2026. All prices as of local market close. Ratings as of date shown.

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Largan (3008.TW)

FAU to drive strong earnings growth and re-rating; TP up to NT$11,000, Buy

3008.TW	12m Price Target: **NT$11,000.00**	Price: **NT$6,660.00**	Upside: **65.2%**

BUY

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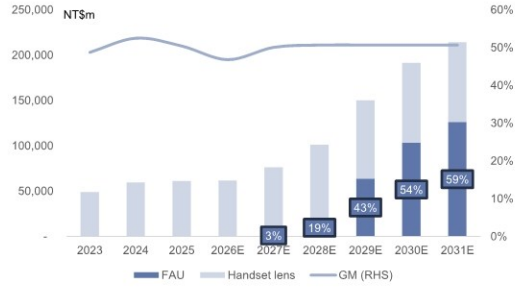
We are positive on Largan and expect its strong alignment technology to continue supporting its expansion from handset lens into FA and FAU for CPO switches. The company's FA has passed customer qualification with strong positive feedback, while customers prefer to purchase its integrated product (FA+FAU), reaffirming our positive view on Largan's business expansion. Management highlights that its strong alignment and automated production provide strong precision, fast production and testing, supporting customer penetration. We model FAU revenue contribution at 3% in 2027E and above 50% in 2030E, driving Largan's earnings growth, margin expansion, and valuation re-rating. Maintain Buy with TP up to NT$11,000 (implies 39.8x 2028E PE).

Exhibit 1: Largan revenues by products and GM trend



Source: Company data, Goldman Sachs Global Investment Research

3-month revenue preview: Largan delivered weak revenue growth in 3Q26 with revenue -11% YoY, which we attribute to a slow smartphone market amid high memory costs, fewer model launches of its major smartphone brand customer, and foldable phones released in 4Q26 (report link). The miss also reflects a slower handset lens pixel mix upgrade than we had previously estimated.

Key Data

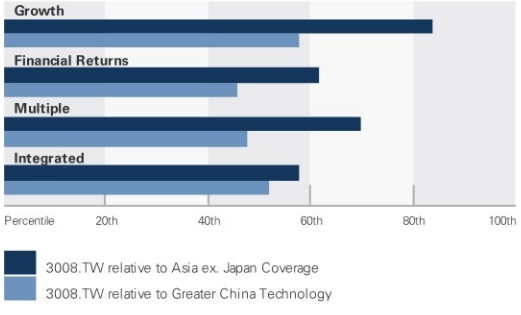
Market cap: NT$896.1bn / $28.1bn
Enterprise value: NT$782.7bn / $24.5bn
3m ADTV: NT$13.2bn / $413.8mn
Taiwan
Greater China Technology
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**61,147.9**	**61,653.5**	**76,334.7**	**101,306.6**
Revenue (NT$ mn) Old	61,147.9	65,503.4	76,561.8	85,020.5
EBITDA (NT$ mn)	31,290.3	30,251.2	42,329.3	53,616.0
EPS (NT$) New	**159.41**	**162.60**	**207.41**	**278.57**
EPS (NT$) Old	159.41	187.34	217.11	239.29
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
Dividend yield (%)	3.3	1.2	1.5	2.1
CROCI (%)	11.2	11.5	14.4	16.2

	9/26	12/26E	3/27E	6/27E
EPS (NT$)	39.62	40.65	49.77	36.05

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates. See disclosures for details.

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BUY

Largan (3008.TW)

Rating since Nov 12, 2024

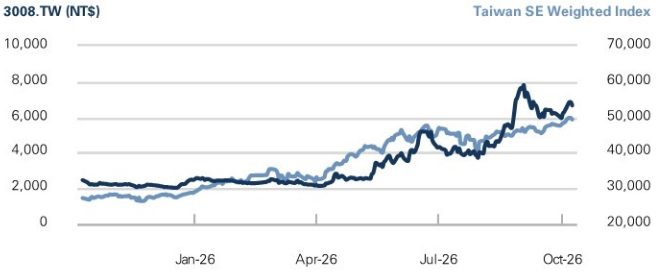
Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	14.8	41.0	32.1	23.9
P/B (X)	1.7	4.5	4.2	3.9
FCF yield (%)	4.3	1.1	3.7	3.3
EV/EBITDAR (X)	6.4	25.6	17.8	13.8
EV/EBITDA (excl. leases) (X)	6.4	25.6	17.8	13.8
CROCI (%)	11.2	11.5	14.4	16.2
ROE (%)	11.4	11.1	13.4	16.7
Net debt/equity (%)	(64.1)	(60.3)	(66.6)	(68.0)
Net debt/equity (excl. leases) (%)	(64.1)	(60.3)	(66.6)	(68.0)
Interest cover (X)	15,611.8	7,608.9	9,940.7	13,702.5
Days inventory outst, sales	37.1	47.9	40.9	40.4
Receivable days	62.1	62.0	50.0	50.0
Days payable outstanding	21.6	22.0	22.0	22.0
DuPont ROE (%)	11.1	10.7	12.9	15.9
Turnover (X)	0.3	0.3	0.3	0.4
Leverage (X)	1.2	1.2	1.2	1.2
Gross cash invested (ex cash) (NT$)	222,420.9	241,679.6	267,828.7	298,740.5
Average capital employed (NT$)	73,429.5	80,655.1	82,379.7	79,780.5
BVPS (NT$)	1,395.59	1,484.98	1,579.62	1,727.59

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	2.8	0.8	23.8	32.7
EBITDA growth	3.4	(3.3)	39.9	26.7
EPS growth	(19.7)	2.0	27.6	34.3
DPS growth	(19.0)	2.4	27.0	35.0
EBIT margin	38.5	35.2	39.2	40.7
EBITDA margin	51.2	49.1	55.5	52.9
Net income margin	34.8	35.1	36.3	36.7

Price Performance



	3m	6m	12m
Absolute	68.6%	192.1%	168.0%
Rel. to the Taiwan SE Weighted Index	56.4%	105.9%	47.1%

Source: FactSet. Price as of 8 Oct 2026 close.

Income Statement (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	61,147.9	61,653.5	76,334.7	101,306.6
Cost of goods sold	(30,310.8)	(32,786.7)	(38,112.7)	(49,958.8)
SG&A	(1,985.5)	(1,885.5)	(1,832.0)	(1,620.9)
R&D	(5,293.3)	(5,295.8)	(6,488.4)	(8,509.8)
Other operating inc./(exp.)	–	–	–	–
EBITDA	**31,290.3**	**30,251.2**	**42,329.3**	**53,616.0**
Depreciation & amortization	(7,732.0)	(8,565.7)	(12,427.8)	(12,398.9)
EBIT	**23,558.3**	**21,685.5**	**29,901.5**	**41,217.1**
Net interest inc./(exp.)	4,258.6	3,896.3	3,913.0	4,104.2
Income/(loss) from associates	87.0	83.3	–	–
Pre-tax profit	**25,900.2**	**26,446.2**	**33,814.6**	**45,321.3**
Provision for taxes	(4,340.0)	(4,539.3)	(5,883.7)	(7,891.5)
Minority interest	(284.9)	(241.0)	(248.7)	(250.2)
Preferred dividends	–	–	–	–
Net inc. (pre-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
Post-tax exceptionals	–	–	–	–
Net inc. (post-exceptionals)	**21,275.4**	**21,665.9**	**27,682.2**	**37,179.6**
EPS (basic, pre-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, pre-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
EPS (basic, post-except) (NT$)	**159.41**	**162.60**	**207.41**	**278.57**
EPS (diluted, post-except) (NT$)	**157.21**	**161.03**	**204.55**	**276.22**
DPS (NT$)	77.87	79.76	101.32	136.83
Div. payout ratio (%)	48.9	49.1	48.9	49.1

Balance Sheet (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	115,941.7	115,177.0	136,906.6	152,609.6
Accounts receivable	10,461.7	10,483.6	10,430.0	17,325.2
Inventory	6,713.2	9,455.6	7,669.0	14,778.3
Other current assets	21,168.8	21,168.8	21,168.8	21,168.8
Total current assets	**154,285.5**	**156,285.0**	**176,174.5**	**205,881.9**
Net PP&E	51,472.4	60,978.8	57,943.2	52,866.6
Net intangibles	474.6	351.6	344.5	213.7
Total investments	26.5	26.5	26.5	26.5
Other long-term assets	14,528.1	14,528.1	14,528.1	14,528.1
Total assets	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Accounts payable	1,727.9	2,224.5	2,369.9	3,652.5
Short-term debt	0.0	0.0	0.0	0.0
Short-term lease liabilities	–	–	–	–
Other current liabilities	28,020.6	28,214.1	31,194.1	35,898.7
Total current liabilities	**29,748.5**	**30,438.5**	**33,564.1**	**39,551.2**
Long-term debt	–	–	–	–
Long-term lease liabilities	–	–	–	–
Other long-term liabilities	170.6	170.6	170.6	170.6
Total long-term liabilities	**170.6**	**170.6**	**170.6**	**170.6**
Total liabilities	**29,919.1**	**30,609.1**	**33,734.6**	**39,721.7**
Preferred shares	–	–	–	–
Total common equity	188,867.2	199,801.1	213,771.1	232,534.0
Minority interest	**2,000.8**	**1,759.8**	**1,511.1**	**1,261.0**
Total liabilities & equity	**220,787.0**	**232,169.9**	**249,016.8**	**273,516.7**
Net debt, adjusted	(122,308.6)	(121,543.8)	(143,273.5)	(158,976.5)

Cash Flow (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Net income	25,900.2	21,665.9	27,682.2	37,179.6
D&A add-back	7,732.0	8,565.7	12,427.8	12,398.9
Minority interest add-back	(284.9)	(241.0)	(248.7)	(250.2)
Net (inc)/dec working capital	(1,208.2)	(2,267.7)	1,985.6	(12,721.8)
Other operating cash flow	(6,058.9)	–	–	–
Cash flow from operations	**26,080.2**	**27,723.0**	**41,846.9**	**36,606.6**
Capital expenditures	(12,340.0)	(17,849.1)	(9,160.2)	(7,091.5)
Acquisitions	(2,040.4)	–	–	–
Divestitures	(2.6)	(100.0)	(225.0)	(100.0)
Others	5,147.2	–	–	–
Cash flow from investing	**(9,235.7)**	**(17,949.1)**	**(9,385.2)**	**(7,191.5)**
Repayment of lease liabilities	–	–	–	–
Dividends paid (common & pref)	(13,013.1)	(10,538.6)	(10,732.1)	(13,712.2)
Inc/(dec) in debt	(203.4)	–	–	–
Other financing cash flows	(1,344.6)	0.0	0.0	0.0
Cash flow from financing	**(14,561.2)**	**(10,538.6)**	**(10,732.1)**	**(13,712.2)**
Total cash flow	**2,283.4**	**(764.7)**	**21,729.7**	**15,703.0**
Free cash flow	13,740.3	9,873.9	32,686.7	29,515.1

Source: Company data, Goldman Sachs Research estimates.

We expect Oct revenues to be at a similar level to Sep, and to sequentially decline in Nov and Dec, resulting in 4Q26 revenue growth of +7% QoQ.

Exhibit 2: Largan monthly revenues preview

	Jul 2026	Aug 2026	Sep 2026	Oct 2026E	Nov 2026E	Dec 2026E	2Q26	3Q26	4Q26E
Rev (NT$m)	4,877	5,012	5,779	5,895	5,600	5,299	13,665	15,651	16,794
YoY	-11%	-16%	-7%	-6%	6%	-6%	17%	-11%	-2%
MoM/QoQ	31%	3%	15%	2%	-5%	-5%	-12%	15%	7%
GS estimates (NT$m)	4,826	6,756	7,285				13,750	18,866	
Act. Vs. GS	1%	-26%	-21%				-1%	-17%	

Source: Company data, Goldman Sachs Global Investment Research

Earnings revision

Our net income estimates are revised by -13% / -4% / +16% in 2026-28E mainly on changes in revenues, GM, and Opex ratio, reflecting: **(1) handset lens:** we continue to expect Largan's lens pixel mix upgrade, but at a slower pace, reflecting the 3Q26 miss and slow smartphone market amid high memory costs, **(2) FAU:** we estimate revenue contribution will begin in 2H27, with 0.6m / 5m in 2027 / 28E, along with product mix upgrade toward higher bandwidth FAU that should support ASP and GM. We expect FAU revenue contribution of 3% in 2027E and exceed 50% in 2030E, **(3) GM:** we lower our GM estimates to reflect the slower pace of handset lens pixel mix upgrade and the company's higher costs on green energy. We expect GM to sustain at ~50% in 2026-28E, supported by continuous handset lens pixel mix upgrade and strong demand of FAU, **(4) Opex ratio:** we continue to expect rising opex, while the opex ratio would improve as strong revenue growth driven by FAU and FAU's higher ASP relative to handset lens enhance operating efficiency.

Exhibit 3: Largan earnings revision

	2026E			2027E			2028E		
NT m	Old	New	Chg	Old	New	Chg	Old	New	Chg
Revenue	65,503	61,654	-6%	76,562	76,335	0%	85,021	101,307	19%
GP	33,681	28,867	-14%	42,012	38,222	-9%	47,968	51,348	7%
OP	25,995	21,685	-17%	32,893	29,902	-9%	37,757	41,217	9%
Net income	24,904	21,666	-13%	28,976	27,682	-4%	31,937	37,180	16%
EPS (Diluted)	185.12	161.03	-13%	214.48	204.55	-5%	237.27	276.22	16%
Margins									
GM	51.4%	46.8%		54.9%	50.1%		56.4%	50.7%	
OPM	39.7%	35.2%		43.0%	39.2%		44.4%	40.7%	
NM	38.0%	35.1%		37.8%	36.3%		37.6%	36.7%	

Source: Goldman Sachs Global Investment Research

Exhibit 4: Largan revenues and GM revision

Revenues (NT$m)	2026E	2027E	2028E	GM	2026E	2027E	2028E
NEW	61,654	76,335	101,307	NEW	46.8%	50.1%	50.7%
FAU	-	2,074	19,191	FAU		46.0%	47.4%
Periscope lens	19,653	23,178	26,702	Periscope lens	38.4%	38.3%	38.2%
20MPx+ handset lens	11,911	26,562	34,714	20MPx+ handset lens	70.5%	70.4%	70.4%
Other handset lens	17,833	13,078	9,657	Other handset lens	57.1%	56.4%	55.6%
Others	12,256	11,443	11,042	Others	22.5%	20.3%	20.3%
OLD	65,503	76,562	85,021	OLD	51.4%	54.9%	56.4%
FAU	63	1,067	4,518	FAU	50.3%	50.4%	50.5%
Periscope lens	18,610	18,570	20,741	Periscope lens	38.9%	38.5%	38.5%
20MPx+ handset lens	16,705	31,766	38,447	20MPx+ handset lens	72.9%	73.0%	75.0%
Other handset lens	17,796	13,029	9,611	Other handset lens	62.7%	62.3%	62.0%
Others	12,330	12,130	11,705	Others	24.8%	24.8%	24.9%
CHG	-6%	0%	19%	CHG (NEW - OLD)	-5%	-5%	-6%
FAU	-100%	94%	325%	FAU	-50%	-4%	-3%
Periscope lens	6%	25%	29%	Periscope lens	-1%	0%	0%
20MPx+ handset lens	-29%	-16%	-10%	20MPx+ handset lens	-2%	-3%	-5%
Other handset lens	0%	0%	0%	Other handset lens	-6%	-6%	-6%
Others	-1%	-6%	-6%	Others	-2%	-5%	-5%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: Largan 3Q26 results snapshot

NT m	3Q25	2Q26	3Q26	QoQ	YoY	GS	Act / GS	Cons.	Act/Cons.
Revenue	17,677	13,665	15,651	15%	-11%	18,866	-17%	18,196	-14%
GP	8,352	6,753	6,607	-2%	-21%	9,339	-29%	9,120	-28%
OP	6,265	5,148	4,735	-8%	-24%	7,094	-33%	6,946	-32%
Net income	7,080	4,670	5,182	11%	-27%	6,699	-23%	6,723	-23%
Margins									
GM	47.2%	49.4%	42.2%			49.5%		50.1%	
OPM	35.4%	37.7%	30.3%			37.6%		38.2%	
NM	40.1%	34.2%	33.1%			35.5%		36.9%	

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Compared to Bloomberg consensus, we are 4% / 16% ahead in 2027 / 28E net income, mainly on higher revenues and lower opex ratio, reflecting our positive view on Largan's FAU, which is in strong demand with bandwidth upgrade.

Exhibit 6: GS vs. Bloomberg consensus

	2027E			2028E		
NT m	GS	BB	Diff %	GS	BB	Diff %
Revenue	76,335	72,074	6%	101,307	90,479	12%
GP	38,222	36,929	4%	51,348	45,410	13%
OP	29,902	28,454	5%	41,217	34,680	19%
Net income	27,682	26,732	4%	37,180	31,942	16%
Margins						
GM	50.1%	51.2%		50.7%	50.2%	
OPM	39.2%	39.5%		40.7%	38.3%	
NM	36.3%	37.1%		36.7%	35.3%	

Source: Goldman Sachs Global Investment Research, Bloomberg

Valuation

We change our valuation methodology from 2027E PE to discounted PE to better capture the company's long term opportunities in FAU. Our target PE multiple (29.8x) is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at

19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our new target price is NT$11,000 (vs. NT$6,263 previously). Our target price implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Exhibit 7: Largan discounted PE

NT$m	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Key assumptions									
Revenues (NT$m)	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
FAU	-	-	-	-	2,074	19,191	63,908	103,530	126,348
Others	48,842	59,458	61,148	61,654	74,261	82,115	86,221	87,945	87,945
YoY	**2%**	**22%**	**3%**	**1%**	**24%**	**33%**	**48%**	**28%**	**12%**
FAU						825%	233%	62%	22%
Others	2%	22%	3%	1%	20%	11%	5%	2%	0%
FAU revenues contribution					*3%*	*19%*	*43%*	*54%*	*59%*
FAU shipment (k units)					592	4,789	14,368	21,551	25,862
YoY						*710%*	*200%*	*50%*	*20%*
FAU ASP (US$)					117	134	148	160	163
YoY						*14%*	*11%*	*8%*	*2%*
P&L									
Revenue	**48,842**	**59,458**	**61,148**	**61,654**	**76,335**	**101,307**	**150,128**	**191,476**	**214,294**
Gross profit	**23,808**	**31,209**	**30,837**	**28,867**	**38,222**	**51,348**	**76,093**	**97,050**	**108,616**
Gross margin	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%	50.7%	50.7%	50.7%
OPEX	5,986	7,177	7,279	7,181	8,320	10,131	15,013	19,148	21,429
Opex ratio	12.3%	12.1%	11.9%	11.6%	10.9%	10.0%	10.0%	10.0%	10.0%
Operating profit	**17,821**	**24,033**	**23,558**	**21,685**	**29,902**	**41,217**	**61,081**	**77,903**	**87,186**
OPM	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%	40.7%	40.7%	40.7%
Non-op	4,280	8,142	2,342	4,761	3,913	4,104	4,104	4,104	4,104
Pre-tax profit	**22,102**	**32,174**	**25,900**	**26,446**	**33,815**	**45,321**	**65,185**	**82,007**	**91,291**
Tax	4,200	5,963	4,340	4,539	5,884	7,891	11,350	14,279	15,896
Tax rate	19.0%	18.5%	16.8%	17.2%	17.4%	17.4%	17.4%	17.4%	17.4%
Minority	-	296	285	241	249	250	250	250	250
Net profit	**17,902**	**25,915**	**21,275**	**21,666**	**27,682**	**37,180**	**53,584**	**67,478**	**75,145**
EPS (NTD, diluted)	**132.62**	**191.44**	**157.21**	**161.03**	**204.55**	**276.22**	**398.10**	**501.32**	**558.28**
YoY	*-21%*	*44%*	*-18%*	*2%*	*27%*	*35%*	*44%*	*26%*	*11%*
TP implied P/E					53.8	39.8	27.6	21.9	19.7

2030E target P/E	29.8
Target multiple x EPS	14,939
Discounted back to 2027; TP (NTD)	**11,000**
Implied 2027 P/E	53.8

COE assumption	
Beta	1.1
Risk free	3.5%
Market risk premium	6.5%
COE	**10.7%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 8: Largan peer comparison

Company	Ticker	Rating	PE	NI YoY		OPM		Ratio
			2030E	2030E	2031E	2030E	2031E	
Largan	3008.TW	Buy	13.3	26%	11%	40.7%	40.7%	0.2
Largan (TP implied)	3008.TW	Buy	29.8	26%	11%	40.7%	40.7%	0.5
Company	**Ticker**	**Rating**	**PE**	**NI YoY**		**OPM**		**Ratio**
			2027E	**2027E**	**2028E**	**2027E**	**2028E**	
Peers								
AAC	2018.HK	Neutral	12.4	17%	12%	9.7%	10.0%	0.5
T&S	300570.SZ	NC	47.1	96%	62%	24.2%	25.4%	0.5
VPEC	2455.TW	Buy	48.9	67%	55%	35.2%	37.3%	0.5
Sumitomo	5802.JT	Buy	17.2	18%	25%	9.8%	11.2%	0.5
Average			**31.4**	**50%**	**38%**	**19.7%**	**21.0%**	**0.5**

NC: Not Covered; data from Bloomberg consensus. Price as of Oct 9, 2026.

Source: Goldman Sachs Global Investment Research, Bloomberg

Exhibit 9: Taiwan AI server supply chain enjoys strong re-rating when expanding from consumer electronics to AI data center

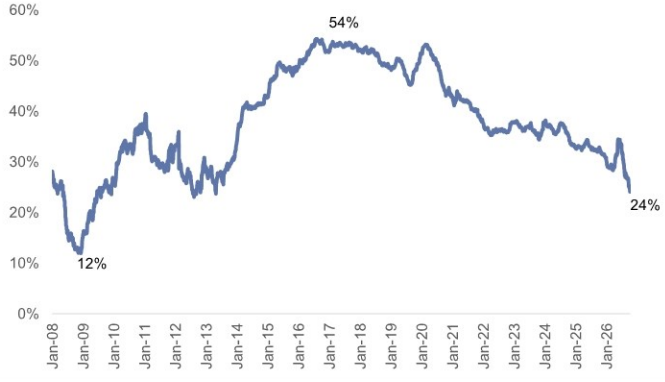
Company	Ticker	2020-2025 PE		
		Trough	Peak	Diff
Largan	3008.TW	10.6	28.0	163%
Taiwan AI server supply chain				
Landmark	3081.TWO	28.9	144.1	398%
VPEC	2455.TW	18.1	74.1	310%
Auras	3324.TWO	8.8	54.6	523%
Fositek	6805.TW	11.3	49.9	343%
AVC	3017.TW	4.7	32.8	604%
Inventec	2356.TW	9.9	31.0	213%
King Slide	2059.TW	9.0	30.2	236%
Gigabyte	2376.TW	3.8	29.6	685%
Mitac	3706.TW	2.5	28.7	1044%
Chenbro	8210.TW	6.7	25.6	282%
Wiwynn	6669.TW	6.1	23.4	287%
Wistron	3231.TW	6.0	21.8	262%
Quanta	2382.TW	6.8	20.4	201%
Hon Hai	2317.TW	7.7	17.5	129%
Avg.		**9.3**	**41.7**	**394%**

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 10: Largan 12M forward PE ratio

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 11: Largan QFII holdings



Source: TEJ

Financial tables

Exhibit 12: Largan P&L

NT$m	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Revenues	15,544	13,665	15,651	16,794	18,095	13,322	19,655	25,263	48,842	59,458	61,148	61,654	76,335	101,307
GP	7,680	6,753	6,607	7,827	9,110	6,375	9,995	12,743	23,808	31,209	30,837	28,867	38,222	51,348
OP income	5,812	5,148	4,735	5,990	7,137	4,923	7,852	9,989	17,821	24,033	23,558	21,685	29,902	41,217
Net income	6,123	4,670	5,182	5,690	6,643	4,812	7,230	8,996	17,902	25,915	21,275	21,666	27,682	37,180
EPS, diluted (NT$)	46.11	35.61	39.52	39.79	49.09	35.56	53.43	66.48	132.62	191.44	157.21	161.03	204.55	276.22
YoY														
Revenues	7%	17%	-11%	-2%	16%	-3%	26%	50%	2%	22%	3%	1%	24%	33%
GP	-4%	8%	-21%	-5%	19%	-6%	51%	63%	-9%	31%	-1%	-6%	32%	34%
OP income	-5%	6%	-24%	-5%	23%	-4%	66%	67%	-13%	35%	-2%	-8%	38%	38%
Net income	-5%	352%	-27%	-15%	8%	3%	40%	58%	-21%	45%	-18%	2%	28%	34%
EPS, diluted (NT$)	-3%	362%	-25%	-19%	6%	0%	35%	67%	-21%	44%	-18%	2%	27%	35%
QoQ														
Revenues	-10%	-12%	15%	7%	8%	-26%	48%	29%						
GP	-7%	-12%	-2%	18%	16%	-30%	57%	27%						
OP income	-8%	-11%	-8%	27%	19%	-31%	60%	27%						
Net income	-9%	-24%	11%	10%	17%	-28%	50%	24%						
EPS, diluted (NT$)	-6%	-23%	11%	1%	23%	-28%	50%	24%						
Margins														
GM	49.4%	49.4%	42.2%	46.6%	50.3%	47.9%	50.9%	50.4%	48.7%	52.5%	50.4%	46.8%	50.1%	50.7%
OPM	37.4%	37.7%	30.3%	35.7%	39.4%	37.0%	40.0%	39.5%	36.5%	40.4%	38.5%	35.2%	39.2%	40.7%
NM	39.4%	34.2%	33.1%	33.9%	36.7%	36.1%	36.8%	35.6%	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
Ratios														
Opex ratio	12%	12%	12%	11%	11%	11%	11%	11%	12%	12%	12%	12%	11%	10%
Tax rate	15%	23%	14%	17%	17%	17%	17%	17%	19%	19%	17%	17%	17%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 13: Largan balance sheet

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Balance Sheet						
Cash and equivalents	107,490	113,658	115,942	115,177	136,907	152,610
Accounts receivable	10,091	10,360	10,462	10,484	10,430	17,325
Inventory	4,591	5,733	6,713	9,456	7,669	14,778
Other current assets	12,149	16,012	21,169	21,169	21,169	21,169
Current assets	**134,321**	**145,764**	**154,285**	**156,285**	**176,174**	**205,882**
Net PP&E/Fixed assets	41,135	46,936	51,472	60,979	57,943	52,867
Net intangibles	239	505	475	352	345	214
Other long-term assets	19,443	23,322	14,555	14,555	14,555	14,555
Non-current assets	**60,817**	**70,763**	**66,502**	**75,885**	**72,842**	**67,635**
Total assets	**195,138**	**216,527**	**220,787**	**232,170**	**249,017**	**273,517**
Accounts payable	1,731	1,855	1,728	2,224	2,370	3,653
Short-term debt	-	203	-	-	-	-
Other current liabilities	27,787	28,520	28,021	28,214	31,194	35,899
Current liabilities	**29,517**	**30,578**	**29,749**	**30,439**	**33,564**	**39,551**
Long-term debt	-	-	-	-	-	-
Other long-term liabilities	110	561	171	171	171	171
Non-current liabilities	**110**	**561**	**171**	**171**	**171**	**171**
Total liabilities	**29,627**	**31,139**	**29,919**	**30,609**	**33,735**	**39,722**
Common stock	1,335	1,335	1,335	1,335	1,335	1,335
Retained earnings	160,871	175,969	185,818	196,752	210,722	229,485
Other common equity	3,305	6,215	1,714	1,714	1,714	1,714
Total common equity	**165,510**	**183,519**	**188,867**	**199,801**	**213,771**	**232,534**
Minority interest	-	1,869	2,001	1,760	1,511	1,261
Total equity	**165,510**	**185,388**	**190,868**	**201,561**	**215,282**	**233,795**
BVPS	1,226.07	1,369.51	1,410.38	1,498.06	1,590.78	1,736.96
Cash conversion cycle						
Account receivable days	68	63	62	62	50	50
Inventory days	71	67	75	90	82	82
Net payable days	25	23	22	22	22	22
Cash conversion cycle	114	106	116	130	110	110
Ratios						
ROE (common equity)	11%	15%	11%	11%	13%	17%
ROA	9%	13%	10%	10%	12%	14%
Net debt to total equity	-67%	-64%	-64%	-60%	-67%	-68%
Total liabilities to total assets	15%	14%	14%	13%	14%	15%
Dupont analysis						
Asset turnover	0.3	0.3	0.3	0.3	0.3	0.4
Leverage (assets to equity)	1.2	1.2	1.2	1.2	1.2	1.2
Net margin	36.7%	43.6%	34.8%	35.1%	36.3%	36.7%
ROE (total equity)	11%	15%	11%	11%	13%	17%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: Largan cash flow

NT$ mn	2023	2024	2025	2026E	2027E	2028E
Cash flow statement						
Net income	22,102	32,174	25,900	21,666	27,682	37,180
Minority interest add-back	-	(296)	(285)	(241)	(249)	(250)
Depreciation and amortization add-back	5,422	6,230	7,732	8,566	12,428	12,399
(Increase)/decrease in working capital	(1,400)	(1,287)	(1,208)	(2,268)	1,986	(12,722)
Other operating cash flow items	(7,925)	(5,243)	(6,059)	-	-	-
Cash flow from operating	**18,199**	**31,579**	**26,080**	**27,723**	**41,847**	**36,607**
Capital expenditure	(8,237)	(11,133)	(12,340)	(17,849)	(9,160)	(7,091)
Other investment cash flow items	(2,915)	(4,933)	3,104	(100)	(225)	(100)
Cash flow from investing	**(11,151)**	**(16,066)**	**(9,236)**	**(17,949)**	**(9,385)**	**(7,191)**
Dividends paid	(11,412)	(9,009)	(13,013)	(10,539)	(10,732)	(13,712)
Change in common stock	-	-	-	-	-	-
Increase/(decrease) in short-term debt	(19)	203	(203)	-	-	-
Increase/(decrease) in long-term debt	-	-	-	-	-	-
Other financing cash flow items	1,680	(1,958)	(241)	-	-	-
Cash flow from financing	**(9,750)**	**(10,764)**	**(13,458)**	**(10,539)**	**(10,732)**	**(13,712)**
Net change in cash	(2,682)	6,168	2,283	(765)	21,730	15,703
FCF	9,962	20,446	13,740	9,874	32,687	29,515
Ratio						
Capex to revenue	17%	19%	20%	29%	12%	7%

Source: Company data, Goldman Sachs Global Investment Research

Price Target Risks and Methodology - Largan

Valuation: We are Buy rated on Largan and have a 12-month target price of NT$11,000, which is based on discounted PE. Our target P/E multiple of 29.8x is derived from (1) peers' avg. ratio of trading PE to forward year fundamentals (NI YoY and OPM), which is at 0.5x, and (2) Largan's forward year fundamentals: 2030-31E NI YoY at 19% on avg. and OPM at 41% on avg. We apply the target PE multiple to Largan's 2030E EPS and discount it back to 2027E via a COE of 10.7%, which is based on beta at 1.1x, risk free at 3.5%, and market risk premium at 6.5%. Our target price of NT$11,000 implies 39.8x 2028E PE, the first full year of FAU contribution, which is within Largan's historical trading range of 8.0x to 45.1x, reflecting our positive view on Largan's expansion from handset lens to FAU, which broadens its exposure from consumer electronics to AI data center that enjoys strong market growth, and could support a valuation re-rating. We note that Taiwan AI supply chain also enjoys re-rating when expanding from consumer electronics to AI data center. Maintain Buy.

Key risks: (1) slower-than-expected smartphone market growth, (2) fiercer competition in handset lenses, (3) slower-than-expected smartphone camera lens specification upgrades or Largan's handset lens pixel mix upgrade, (4) slower-than-expected FAU ramp up, and (5) fiercer competition in FAU.

Disclosure Appendix

Reg AC

We, Verena Jeng and Allen Chang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

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Disclosures

Rating and pricing information

AAC (Neutral, HK$37.36), Sumitomo Electric Industries (Buy, ¥2,361) and VPEC (Buy, NT$558.00)

The rating(s) for Largan is/are relative to the other companies in its/their coverage universe: AAC, ACM Research, AMEC, ASMPT, AVC, AccoTest, Anji Micro, Asus, Auras, BOE, BYDE, Biren, CR Micro, CXMT, Cambricon, Chenbro, China Mobile (HK), China Telecom, China Tower Corp., China Unicom, Chinasoft Intl, Compal, Desay SV, E Ink, E-Town Semis, EHang, Empyrean, Eoptolink, FOCI, Fositek, Foxconn Industrial Internet, Gigabyte, Gigadevice, Glodon Co., HTC Corp., Hikvision, Hon Hai, Horizon Robotics, Hua Hong, Huace Navigation, Huaqin Co.(A), Huaqin Co.(H), Hwatsing, Iluvatar, InnoScience, Innolight (A), Innolight (H), Inspur, Insta360, Inventec, JCET, Kematek, King Slide, Kingdee, Kingsoft Office, LandMark, Largan, Lenovo, Lingyi, Luxshare (A), Luxshare (H), Maxscend, Meitu, MetaX, Mitac, Montage (A), Montage (H), NAURA, NSIG, Nexchip, OmniVision, Pegatron, Pony AI Inc. (ADR), Pony AI Inc. (H), Quanta, RoboTechnik, Ruijie Networks, SG Micro, SICC, SMIC (A), SMIC (H), SZS, Sangfor, SenseTime, Shengyi Tech, Shennan Circuits, StarPower, Sunny Optical, TFC Optical, Thundersoft, Tongyu Communication, Transsion, UMT, UNIS, VPEC, Vanchip, VeriSilicon, Victory Giant, WNC, WUS, WeRide, Wistron, Wiwynn, YJ Semitech, YOFC, Yonyou, ZTE (A), ZTE (H), iFlytek

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There are no company-specific disclosures for: Largan (NT$6,660.00)

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
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Global	50%	34%	16%	65%	59%	43%

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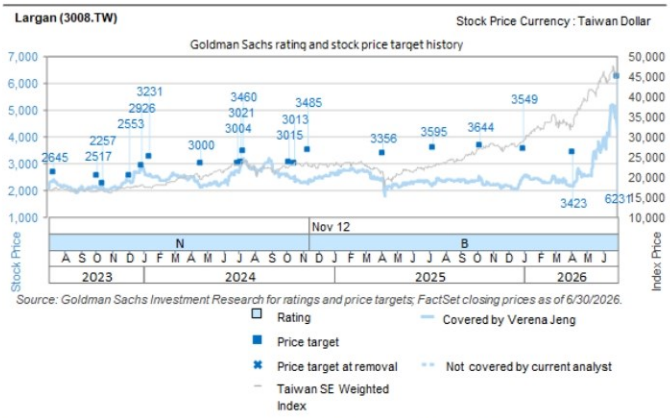
Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Largan (3008.TW)

Date of report	Target price (NT$)	Closing price (NT$)
13-Jul-26	6,263.00	4,345.00
30-Jun-26	6,231.00	4,290.00
07-Apr-26	3,423.00	2,170.00
01-Jan-26	3,549.00	2,495.00
09-Oct-25	3,644.00	2,470.00
11-Jul-25	3,595.00	2,285.00
06-Apr-25	3,356.00	2,450.00
12-Nov-24	3,485.00	2,255.00
18-Oct-24	3,013.00	2,325.00
07-Oct-24	3,015.00	2,525.00
11-Jul-24	3,460.00	3,200.00
08-Jul-24	3,021.00	3,030.00
01-Jul-24	3,004.00	2,700.00
21-Apr-24	3,000.00	2,120.00
12-Jan-24	3,231.00	2,600.00
29-Dec-23	2,926.00	2,870.00
06-Dec-23	2,553.00	2,320.00
15-Oct-23	2,257.00	2,175.00

Price targets shown in table(s) are unadjusted for corporate actions.

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UPDATE

October 8, 2026 11:00 AM GMT

Largan Precision | Asia Pacific

3Q26 Results Miss on Lower GPM, but FAU/CPO Development Well on Track

AlphaSignals Earnings Reaction

Unchanged	**Meaningful shortfall**	**Modest revision lower**
Impact to our thesis	Financial results versus consensus	Direction of next 12-month consensus EPS

Source: Company data, Morgan Stanley Research

Key Takeaways

- 3Q26 gross margin is 42.2%, which is lower-than-expected and triggers a 31-36% GP and OP miss vs. our estimates.
- Revenue is likely to be flat MoM in October and decline sequentially in November.
- CPO-related (FAU) has seen good progress with new product verification passed.

Largan reported 3Q26 net income of NT$5.182bn, down 27% YoY.

The company reported a 11% YoY revenue decline, to NT$15.651bn; gross margin declined YoY and QoQ to 42.2% in 3Q26, lower than expected due to lower new-project yields and structural green electricity cost hikes.

Management expects shipments to be flat MoM in October and decline sequentially in November as order momentum is weaker-than-expected.

FAU/CPO development well on track: First FA project passed customer verification; expanding from FA into FAU, including coupling and potential FAU-to-carrier assembly, with a dedicated new facility expected to support mass production from around mid-2027. We maintain our long-term positive view on Largan's FAU/CPO opportunities.

Exhibit 1: 3Q26 results summary

NT$mn	3Q26	2Q26	QoQ	3Q25	YoY
Net sales	15,651	13,665	15%	17,677	-11%
Gross profit	6,607	6,753	-2%	8,352	-21%
% margin	42.2%	49.4%	-7%	47.2%	-5%
OPEX	-1,872	-1,605	17%	-2,088	-10%
% OPEX	12.0%	11.7%	0%	11.8%	0%
Operating income	4,735	5,148	-8%	6,265	-24%
% margin	30.3%	37.7%	-7%	35.4%	-5%
Net income	5,182	4,670	11%	7,080	-27%
% margin	33.1%	34.2%	-1%	40.1%	-7%
EPS (NT$)	39.62	35.73	11%	53.05	-25%

Source: Company data, Morgan Stanley Research

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Largan Precision (3008.TW, 3008 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Equal-weight**
Industry View	**In-Line**
Price target	NT$7,100.00
Up/downside to price target (%)	7
Shr price, close (Oct 8, 2026)	NT$6,660.00
52-Week Range	NT$7,995.00-2,020.00
Sh out, dil, curr (mn)	134
Mkt cap, curr (mn)	NT$893,374
EV, curr (mn)	NT$762,165
Avg daily trading value (mn)	NT$5,658

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	159.40	184.85	204.18	215.63
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	61,148	66,920	74,757	77,025
EBITDA (NT$ mn)	31,260	33,813	37,386	39,037
ModelWare net inc (NT $ mn)	21,275	24,179	26,707	28,204
P/E	15.7	36.0	32.6	30.9
P/BV	1.7	4.1	3.6	3.2
ROE (%)	11.5	12.7	12.4	11.7

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
e = Morgan Stanley Research estimates

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Exhibit 2: 3Q26 results vs. our estimates and consensus

NT$ mn	3Q26	MSe	Diff	Cons.	Diff
Net sales	15,651	19,255	-19%	18,403	-15%
Gross profit	6,607	9,627	-31%	9,131	-28%
% margin	42.2%	50.0%	-8%	49.6%	-7%
OPEX	-1,872	-2,274	-18%	-2,169	-14%
% OPEX	12.0%	11.8%	0%	11.8%	0%
Operating income	4,735	7,353	-36%	6,962	-32%
% margin	30.3%	38.2%	-8%	37.8%	-8%
Net income	5,182	6,988	-26%	6,743	-23%
% margin	33.1%	36.3%	-3%	36.6%	-4%
EPS (NT$)	39.62	53.43	-26%	51.35	-23%

Source: Company data, FactSet, Morgan Stanley Research estimates

Valuation Methodology and Risks

Largan Precision (3008.TW)

Base case: Our price target is based on an SOTP valuation, comprising the traditional business and potential FAU opportunity. We value the traditional business using a residual income model, consistent with our Greater China Hardware Tech coverage, assuming an 8.5% cost of equity, a 7% net profit CAGR over 2026-36E and a 3% terminal growth rate. We separately value the FAU business, assuming a 40% market share in 2028.

Risks to Upside

- Better-than-expected high-end smartphone demand
- Faster-than-expected penetration of periscopes
- Stronger-than-expected CPO volume growth

Risks to Downside

- Greater-than-expected competition, which would erode Largan's ASP and gross margin
- Weaker-than-expected smartphone demand, especially for high-end models
- Market share gains and more pricing pressure from Sunny
- Weaker-than-expected CPO volume growth

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Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

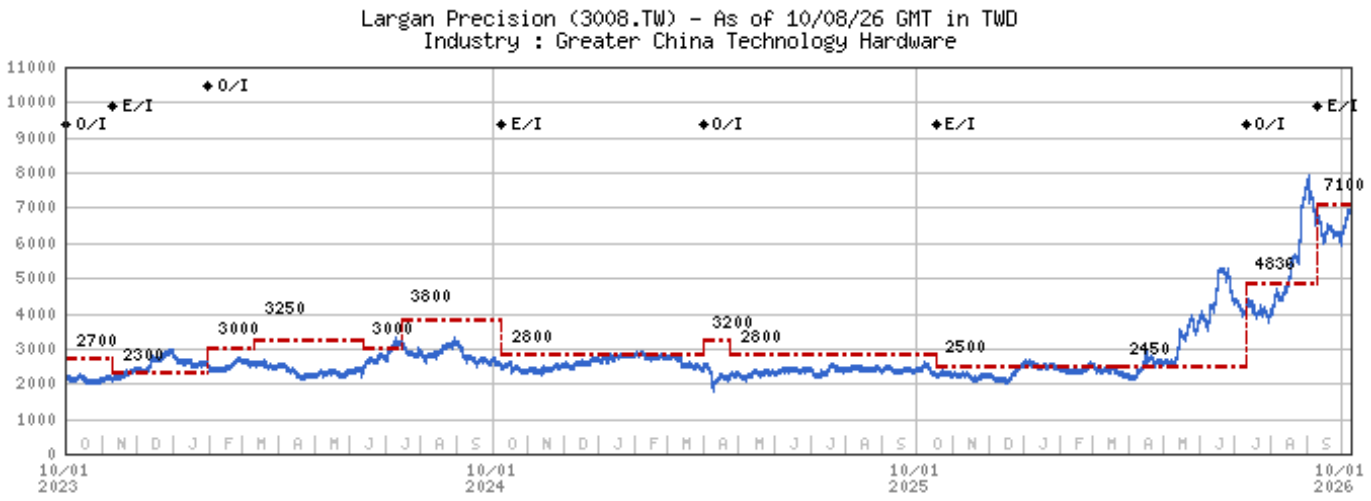
Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : E/I; 12/17/21 : O/I; 9/13/22 : E/I; 1/9/23 : U/I; 5/16/23 : O/I; 11/10/23 : E/I; 1/31/24 : O/I; 10/9/24 : E/I; 3/31/25 : O/I; 10/17/25 : E/I; 7/10/26 : O/I; 9/10/26 : E/I

Price Target History: 9/7/21 : 3050; 10/7/21 : 2300; 12/17/21 : 2600; 4/5/22 : 2300; 7/25/22 : 2400; 9/13/22 : 2200; 10/14/22 : 1900; 1/9/23 : 1700; 5/16/23 : 2700; 11/10/23 : 2300; 1/31/24 : 3000; 3/11/24 : 3250; 6/12/24 : 3000; 7/15/24 : 3800; 10/9/24 : 2800; 3/31/25 : 3200; 4/23/25 : 2800; 10/17/25 : 2500; 3/23/26 : 2450; 7/10/26 : 4836; 9/10/26 : 7100

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/08/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$37.24
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb152.01
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.14
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.18
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.29
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb378.50
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,060.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb21.70
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.10
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.00
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,660.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.83
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.42
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.18
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$50.80
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb48.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.39
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.84
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb68.49
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$35.06
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb46.10
Sunny Optical (2382.HK)	E (03/23/2026)	HK$60.05
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb235.07

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.00
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.66
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb393.99
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$167.60
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb101.20
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb23.15
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb783.00
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$999.50
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.70
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb29.32
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,070.00
Advantech (2395.TW)	O (01/20/2021)	NT$727.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,205.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,440.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$36.60
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,520.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,360.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.50
Chenbro (8210.TW)	O (07/23/2025)	NT$874.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,330.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,965.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$144.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,370.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$332.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.00
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,615.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb30.04
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$297.50
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.63
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.18
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb175.19
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.30
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$996.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$195.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.50
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,000.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.67
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb55.69
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$363.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,285.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$249.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb64.56
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$50.15
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,960.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.32
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,915.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,475.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$89.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$325.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb131.86
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb368.02
Unimicron (3037.TW)	O (02/23/2026)	NT$1,270.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$558.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$188.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,240.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$635.00
Zhen Ding (4958.TW)	++	NT$592.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$535.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,740.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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Initiation of Coverage

Hermes Testing Solutions

Shifting-left, scaling-up: dual catalysts in chip probing and CPO

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,880.00**
	Prior :
Price (06 Oct 2026)	**NT$4,190**

RIC: 7856.TWO **BBG:** 7856 TT

Trading data and key metrics

52-wk range	NT$5,295.00-516.55
Market cap.	NT$114b/US$3.58b
Shares o/s	27.2m (ORD)
Free float	94%
Avg. daily volume ('000)	240
Avg. daily value (m)	NT$1,068.4
Common s/h equity (12/26E)	NT$15.2b
P/BV (12/26E)	9.1x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From		To	% ch	Cons.
12/26E	-		29.11	-	32.19
12/27E	-		153.28	-	89.82
12/28E	-		286.48	-	71.44

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Jack-of-all-trades for bespoke testing solutions in the semiconductor industry

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 PT. HTSI is uniquely positioned to benefit from two key semiconductor testing trends: shift-left testing and the emerging CPO testing market. Leveraging its long-standing relationship with TSMC, HTSI has established capabilities across electrical test engineering, customised test equipment, thermal control modules, probe cards, and CPO testing solutions. We forecast revenue growth of 127%/152%/78% in 2026E/27E/28E, driven by increasing adoption of thermal control modules for wafer-level test and initial deployment of CPO insertion 2 and insertion 3 test equipment through ficonTEC. Additional 2028E growth drivers include reflow tools, laser clean solutions, and MEMS probe cards. The stock trades at 27/15x 2027/28E PE, versus ~35x NTM PE for handler-equipment peers, offering re-rating potential as sales scale, new products are qualified, and its CPO presence expands. See our companion testing deep dive sector note.

Shift-left beneficiary with exceptional thermal control modules

Thermal control modules should be HTSI's largest growth driver over the next several years. We expect increasing chip power consumption, rising package complexity, and tighter probe pitches to drive adoption of advanced thermal control solutions used alongside leading-edge probers, in additional to more wafer level insertions and longer test time. We estimate HTSI will ship approximately 3,000 thermal-control modules in 2027E, up from around 1,200 units in 2026E, sales contribution rise from ~30% in Q226 to >50% of sales by 2027E. The launch of its second-generation platform in 2027E, which carries 25-50% higher ASPs, should further improve profitability. We forecast GM expansion from 55.5% in 2026E to 58.3%/60.0% in 2027E/28E.

Unique position to capture CPO upside

Through its partnership with German optical-automation leader ficonTEC, HTSI is positioned to participate in the initial ramp of CPO testing equipment. HTSI supports local manufacturing, installation, automation and after-sales service, making it a key enabler of volume production. We estimate HTSI captures approximately 30% of equipment value across the 40 insertion 2 and 70 insertion 3 systems we forecast for delivery in 2027E, with further potential upside in 2028E as adoption expands.

Valuation: initiate coverage with a Buy rating and price target NT$8,880

We forecast EPS of NT$29.1 in 2026E (including a one-off employee share purchase programme impact), rising to NT$153.3/286.5 in 2027E/28E. Our NT$8,880 price target is based on 31x average 2028E PE, versus handler-equipment peers forward PE at roughly 35x, reflecting HTSI's unique exposure to both shift-left testing, CPO, and our estimate of 32% LT earnings CAGR. We initiate with a Buy rating.

Highlights (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	1,598	2,425	5,505	13,879	24,713	29,218	32,810
EBIT (UBS)	54	297	1,076	6,154	11,604	14,294	16,170
Net earnings (UBS)	57	268	862	5,059	9,454	11,664	13,227
EPS (UBS, diluted) (NT$)	2.71	10.65	29.11	153.28	286.48	353.44	400.81
DPS (net) (NT$)	0.00	1.10	11.64	61.32	114.60	141.38	160.33
Net (debt) / cash	(469)	618	14,360	17,474	22,786	29,613	36,992
Profitability/valuation	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	3.4	12.2	19.5	44.3	47.0	48.9	49.3
ROIC (EBIT) %	-	41.7	147.5	381.2	334.1	282.5	261.7
EV/EBITDA (UBS core) x	-	35.1	80.1	15.1	7.7	5.8	4.7
P/E (UBS, diluted) x	-	68.6	NM	27.3	14.6	11.9	10.5
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 4,190.00 on 06-Oct-2026

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 19.**

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will HTSI be able to keep its moat within chip probing thermal control modules?
Yes. We expect HTSI's competitive position to strengthen as chip power consumption, package complexity, and probe density continue to increase. The company is developing next-generation thermal control solutions, including liquid cooling technologies, while also expanding into specialized chuck solutions designed to address wafer warpage. In addition, HTSI's close collaboration with TSMC, TEL, Advantest, and other OSATs should reinforce its position within advanced wafer-level testing.

Q: Will insertion 2 of CPO testing process be skipped?
No. We believe concerns surrounding insertion 2 adoption are overdone. While some industry participants favor bypassing insertion 2 due to throughput and ROI considerations, doing so increases the risk of defective optical engines being identified only after the components are delivered to OSATs, creating yield accountability challenges and higher packaging losses. Based on our industry studies, we expect insertion 2 testing to remain an important part of the CPO workflow over at least the next two years as the industry focuses on improving PIC yields and overall testing efficiency.

UBSVIEW

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 price target. HTSI offers unique exposure to two of the semiconductor testing industry's most attractive growth drivers: 1) shift-left testing and 2) CPO testing. The industry's shift-left testing strategy will have more test insertions move to the wafer level before packaging, in addition to longer CP time. Leveraging its longstanding relationship with TSMC, HTSI has established capabilities across thermal control modules, electrical-test engineering services, customized equipment, and emerging CPO testing solutions. We believe there could be re-rating potential as sales scale, new products are qualified, and its CPO presence expands.

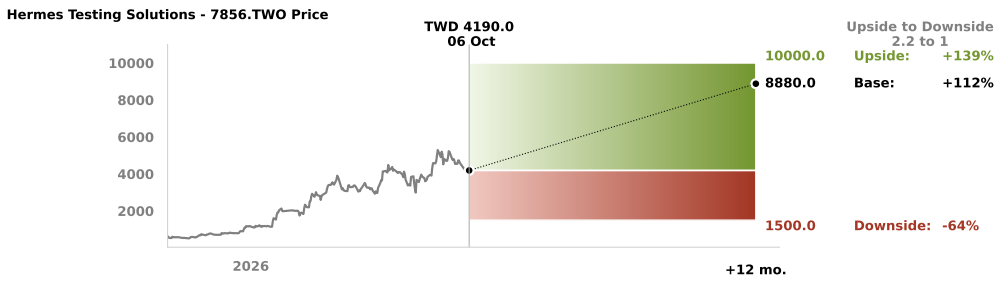
EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

WHAT´S PRICED IN?

Despite a 264% YTD share-price increase, HTSI currently trades at just 15x of our 2028E EPS versus handler-equipment peers forward PE at roughly 35x. We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM



Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&180	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

COMPANY DESCRIPTION

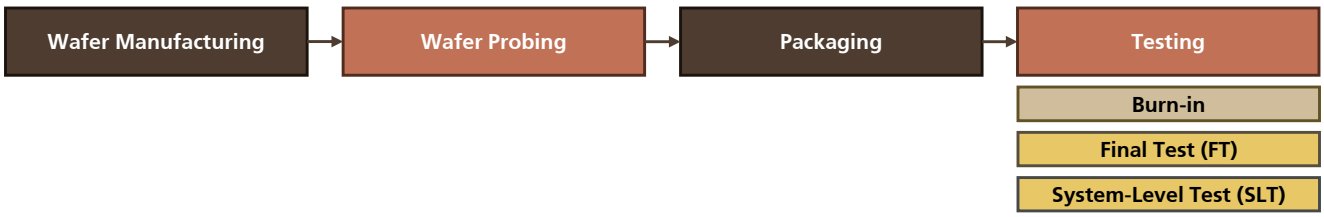
Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

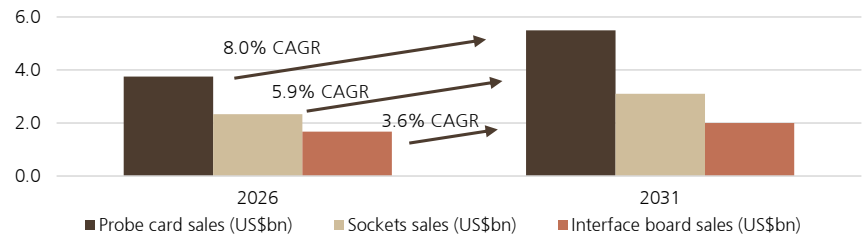
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now got substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damages.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.

- **Thermal control module ASP expansion.** Gen 2 thermal control modules will start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027 with scale-out CPO adoption, we see HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not happening anytime soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 3: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

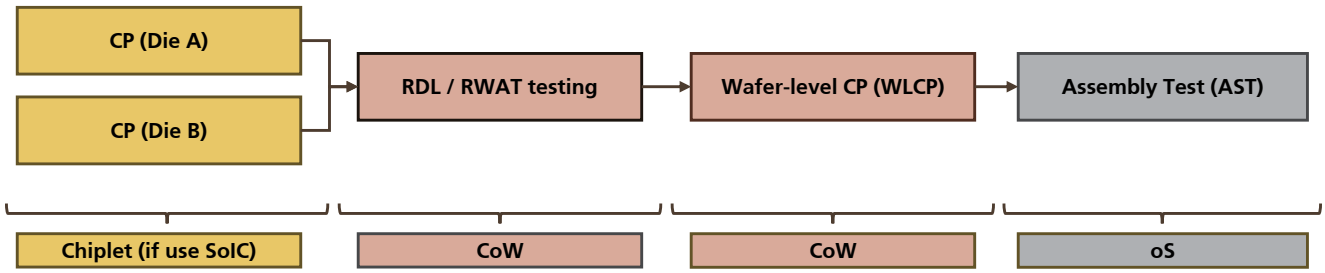
Robust growth drivers across semiconductor testing

We believe HTSI is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see HTSI is uniquely positioned as the main beneficiary of two of the strongest growth drivers within semiconductor testing: testing shift-left and CPO.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging.

Figure 4: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

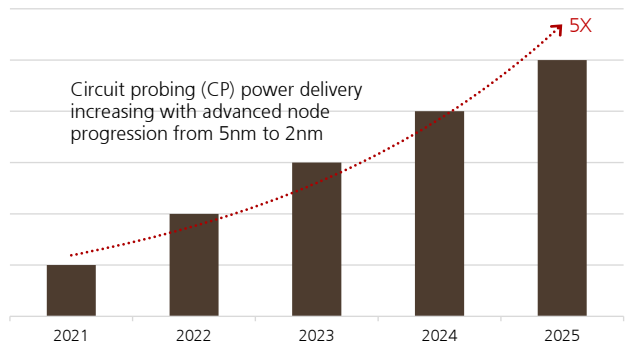


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

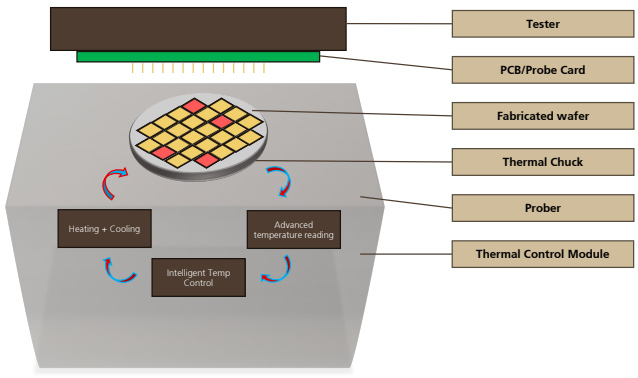
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 5: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 6: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are required for high TDP semiconductor wafer probe in order to maintain the temperature

of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

CPO testing fuelling future growth for semiconductor testing

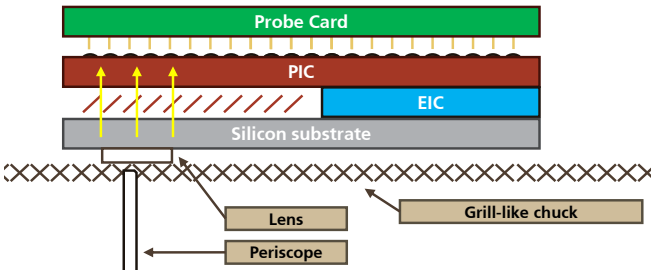
As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

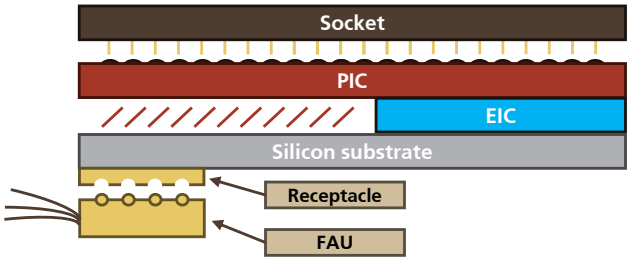
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma has noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 7: An illustration for insertion 2 test method



Source: UBS

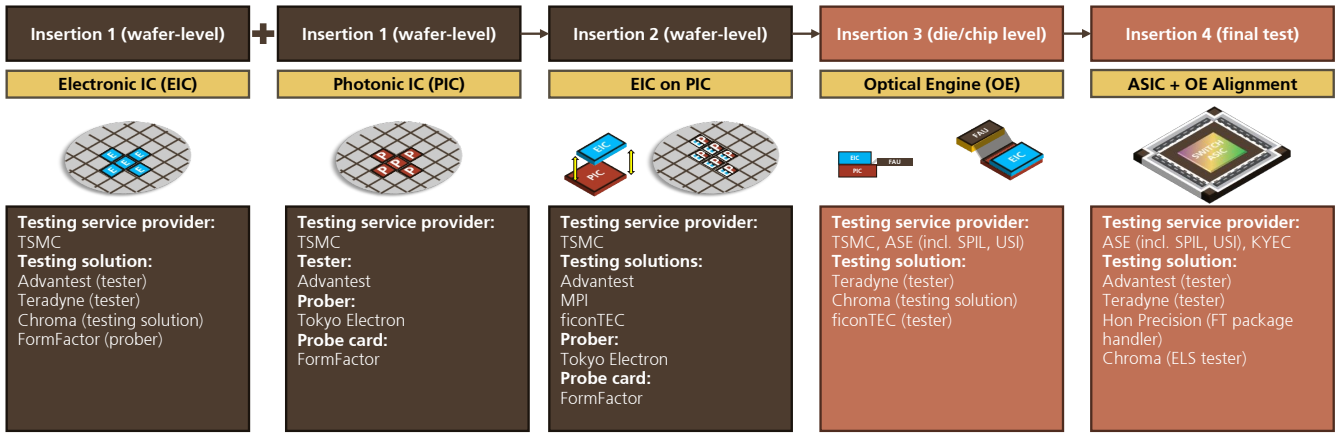
Figure 8: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main target and area of competition for the major players in the field. Notable ones for each insertions are listed.

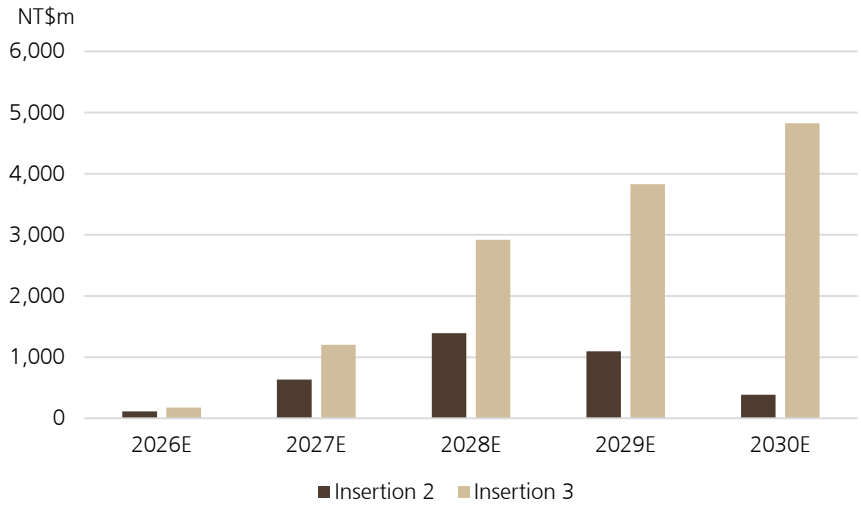
Figure 9: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **HTSI**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs are still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with potential upside to the numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

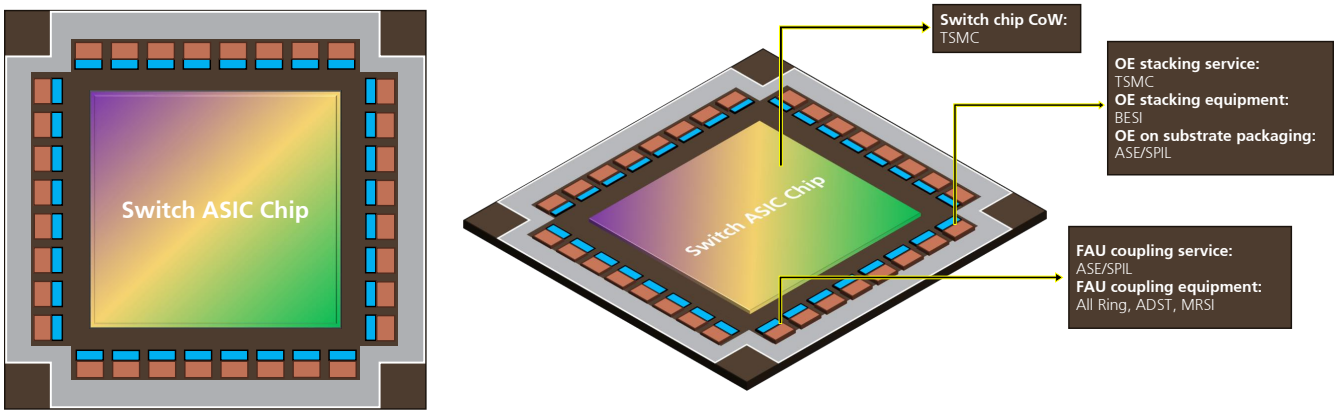
Figure 10: We believe HTSI CPO equipment sales will pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Figure 11: A switch chip with 32 OEs and the CPO packaging supply chain



Source: UBS

HTSI's unique position for new TAM of thermal control modules and CPO test equipment

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan, as a spin-off from Hermes-Epitek. The company provides semiconductor testing and inspection solutions, engineering services, customized equipment, and test consumables. HTSI maintains a global service footprint across Taiwan (Hsinchu, Tainan), Mainland China (Suzhou), Japan (Tokyo, Kumamoto), Singapore, Malaysia, and the United States (Arizona), enabling close support for leading semiconductor customers.

Its main business segments are structured around three pillars:

- **Semiconductor equipment and customized products:** HTSI utilises a dual-track strategy of independent R&D and also equipment agency sales. They offers

highly customized solutions designed to address specific client pain points and bottlenecks. Key products include: thermal control modules that pair with a 1:1 ratio with wafer-level test probers, electron beam inspection equipment, white light interference optical inspection systems, laser probe cleaning machines, and the SYMPHONY fluxless reflow system.

- **Testing equipment engineering services:** The company operates the largest engineering service team in Taiwan, consisting of over 340 technical professionals. The engineering service team provides extensive equipment life-cycle support, including installation, relocations, modification, testing, routine maintenance, and repair services. CPO related equipment sales/services will be recognized within this segment, as CPO are not direct HTSI products, but rather sales recognition via its partnership with ficonTEC.
- **Probe card design, manufacturing, and cleaning materials:** HTSI delivers a comprehensive product portfolio that supports the entire probe card life-cycle. For probe cards, they supply membrane, cantilever, and vertical probe cards. As they are small in scale, HTSI targets niche probe card applications such as RF, 5G/6G, millimeter wave, LCD driver, low end logics, TIA (Transimpedance Amplifier), etc. In addition to the three existing probe technologies, HTSI also collaborates with MJC in Japan for MEMS probe development, targeting sales in 2028E for high-end, AI/HPC related applications. Cleaning materials is another important business within this segment, as HTSI source and process sand paper or other materials to offer cleaning sheets to clean the probe cards, removing oxidations, unwanted particles and substances that accumulated on the probe cards during the testing process.

The sales resiliency in the coming years mainly stems from its robust relationship with key industry players, especially as an engineering and service provider with TSMC for testing solutions. This sets up a solid foundation for its customized equipment business, as they know the specific testing bottleneck to tackle. For instance, as Hermes-Epitek manufacturers the probers for Tokyo Electron (TEL) and HTSI helps with installation, services, and maintenance of the probers, they understand the thermal bottlenecks that the wafer-level test industry has been facing and can effectively come up with module solutions to solve this.

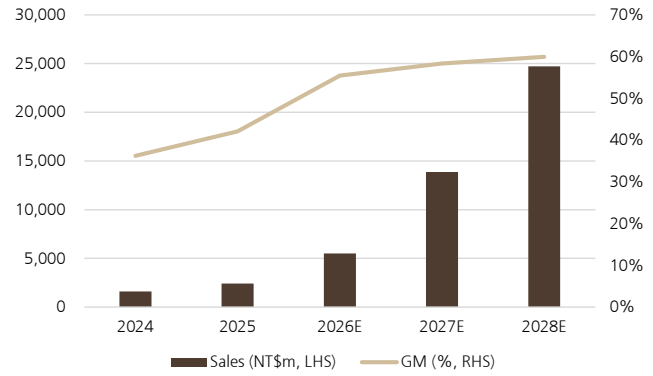
Margins set to expand with value-added products and services

The company has demonstrated strong execution in advanced wafer-level testing, with H126 revenue of NT$2.2bn already exceeding 90% of FY25 revenue (NT$2.4bn). Profitability has also improved significantly, with GM increasing from 42.1% in 2025 to 54.9% in H126.

With share of value-added products and services set to rise in 2027E in sales mix, HTSI is set to see GM and operating leverage expansion. As a niche engineering service provider that has robust ties to leading foundries and OSATs, we believe HTSI has substantial pricing power and estimate HTSI' GM for thermal control modules to be around 65%, with GM for CPO equipment of around 60% as HTSI provides an engineering-service-like business for it. We believe these factors will help the company to reach a 60% long-term GM range.

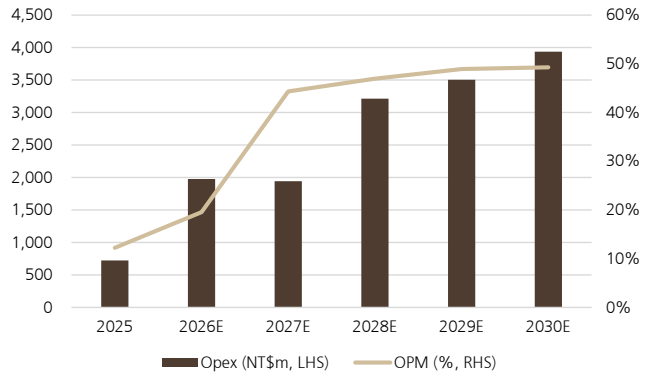
OPM is also set to expand with customized equipment sales rise in total sales mix. If we exclude the one-time expense to be recognized during Q326 for employee stock benefits (we estimate around NT$1bn), YoY opex growth will be around 99% for 2027E, with downside risk if the company conducts tight cost control measures. We forecast the company's OPM will expand from 19.5% in 2026E, to 44.3%/47.0% in 2027/28E.

Figure 12: HTSI sales set to continue to climb with 152% growth and GM at 58.3% in 2027E



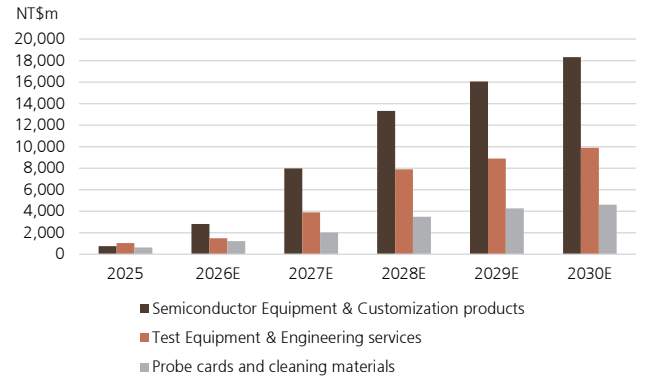
Source: UBS estimates

Figure 13: OPM set to grow to 44.3%/47.0% in 2027/28E



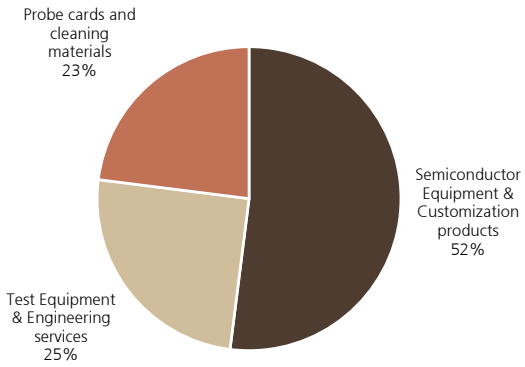
Source: UBS estimates. Note 2026E opex includes a one-time NT$1bn employee stock benefit.

Figure 14: We see customized equipment to continue to drive company growth



Source: Company data, UBS estimates

Figure 15: HTSI H126 sales mix breakdown



Source: Company data, UBS

Figure 16: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**2,425**	**985**	**1,200**	**1,513**	**1,807**	**5,505**	**2,634**	**3,197**	**3,760**	**4,289**	**13,879**	**24,713**	**29,218**	**32,810**
- YoY chg (%)	52%	123%	108%		28%	127%	167%	166%	149%	137%	152%	78%	18%	12%
- QoQ chg (%)		-30%	-51%	54%	51%		46%	-42%	43%	34%				
Gross profit	**1,021**	**545**	**655**	**843**	**1,012**	**3,054**	**1,534**	**1,856**	**2,184**	**2,523**	**8,097**	**14,817**	**17,801**	**20,107**
- Gross margin	42.1%	55.3%	54.5%	55.7%	56.0%	55.5%	58.3%	58.1%	58.1%	58.8%	58.3%	60.0%	60.9%	61.3%
Operating profit	**297**	**323**	**411**	**-399**	**741**	**1,076**	**1,166**	**1,408**	**1,657**	**1,923**	**6,154**	**11,604**	**14,294**	**16,170**
- Operating margin	12.2%	32.8%	34.3%	-26.4%	41.0%	19.5%	44.3%	44.1%	44.1%	44.8%	44.3%	47.0%	48.9%	49.3%
Pre-tax profit	**322**	**326**	**412**	**-396**	**743**	**1,085**	**1,205**	**1,449**	**1,700**	**1,968**	**6,323**	**11,818**	**14,580**	**16,534**
Net profit	**268**	**254**	**330**	**-317**	**594**	**862**	**964**	**1,160**	**1,360**	**1,574**	**5,059**	**9,454**	**11,664**	**13,227**
- YoY chg (%)	373%	586%	301%		299%	221%	280%	251%	-529%	165%	487%	87%	23%	13%
- QoQ chg (%)		70%	23%	-225%	80%		62%	35%	41%	36%				
Basic EPS (NT$)	**10.83**	**9.35**	**12.15**	**-10.20**	**18.01**	**29.11**	**29.22**	**35.14**	**41.22**	**47.70**	**153.29**	**286.49**	**353.45**	**400.82**
- YoY chg (%)	281%	492%	279%		200%	169%	213%	189%	-504%	165%	427%	87%	23%	13%
- QoQ chg (%)		55%	12%	-209%	48%		62%	21%	41%	36%				

Source: UBS estimates

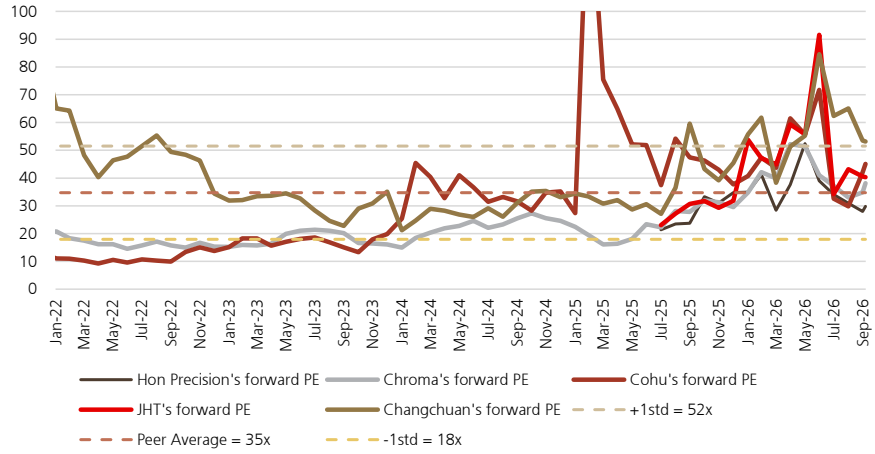
Figure 17: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

WHAT´S PRICED IN?

HTSI is trading at NT$4,190 as of 6 Oct 2026



Source: LSEG, UBS

Despite a 264% YTD share-price increase, HTSI still trades at just 15x 2028E P/E. We believe market expectations for 2027E CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

We believe the market is not fully reflecting three key growth drivers:

1) Shift-left testing is still underappreciated.

As advanced packaging costs rise and AI accelerators become increasingly complex, more thorough testing is done at the wafer-level. This should drive sustained demand for HTSI's thermal control modules, probe card consumables, and other wafer-level testing solutions.

2) CPO revenue potential remains underestimated.

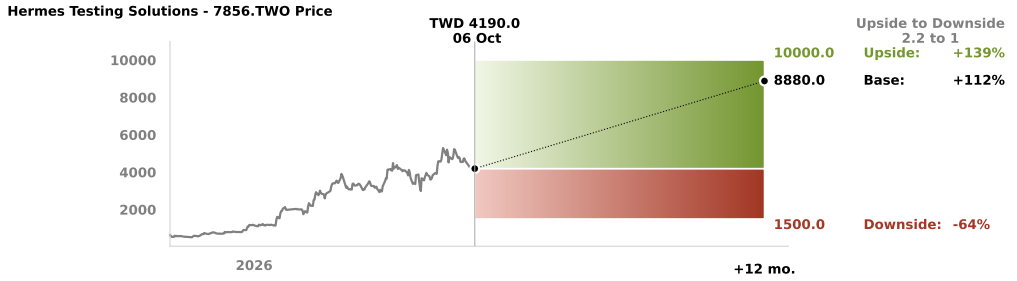
Investor expectations for 2027 CPO equipment shipments, particularly insertion 2 testing systems, remain conservative. Based on our industry studies, we forecast HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, with further potential upside in 2028E as CPO adoption broadens.

3) New products provide additional optionality.

Beyond its core businesses, HTSI is expanding into higher-value solutions including mass-reflow tools, MEMS probe cards and laser cleaning systems, creating incremental growth opportunities that we believe are largely absent from current expectations.

We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028E PE as our view is that 2028 is the year for thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics as active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment. For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE (see What's Priced In chart).

UPSIDE/DOWNSIDE SPECTRUM



HTSI is trading at NT $4,190 as of 6 Oct 2026

Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&175	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

Risk to the current share price is heavily skewed (2.2:1) to the upside.

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected thermal module and CPO equipment shipment in 2027-28E, with additional upside from better traction of probe cards and mass reflow tools, forming 157%/91% YoY sales growth in 2027/28E and even higher operating leverage with OPM of 46.3%/50.0% in 2027/28E. This scenario implies a 34x 2028E PE, a upside driving by more equipment demand stemming for robust shift-left test induced equipment bottleneck, alongside CPO proliferation with scale-up CPO equipment pull-in in 2028E.

BASE (NT$8,880): Our price target of NT$8,880 is based on 31x 2028E PE. We expect an earnings CAGR of 32% in 2027-31E. With shift-left trend for more thermal module requirement during testing, alongside 40/70 CPO insertion 2/3 shipment. We see these growth drivers driving YoY sales growth from a high base of 127% in 2026E to 152%/78% YoY in 2027/2028E, with OPM of 44.3%/47.0% in 2027/28E.

DOWNSIDE (NT$1,500): Our downside scenario reflects lower shipment of thermal control modules due to less advanced prober demand with less insertion 2 and 3 equipment shipment due to bottlenecks with CPO production, alongside limited additional growth drivers to be picked up in 2028 such as MEMS probe card and mass reflow tools. This leads to weaker YoY sales growth of 67%/50% in 2027/28E and operating margin of 29.3%/32.0% in 2027/28E as equipment shipment fails to scale. This scenario implies a 14x 2028E PE, as valuation de-rate with dimmed outlook without CPO and limited upside from thermal control module shipments.

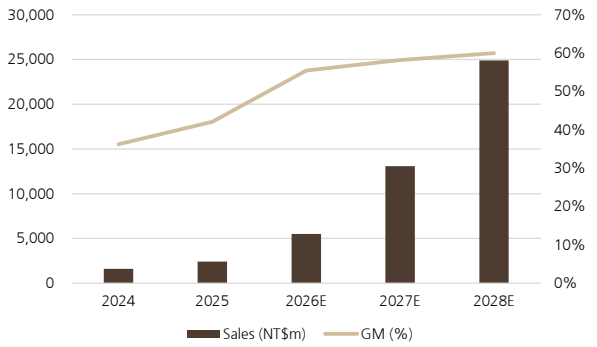
COMPANY DESCRIPTION

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Industry Outlook

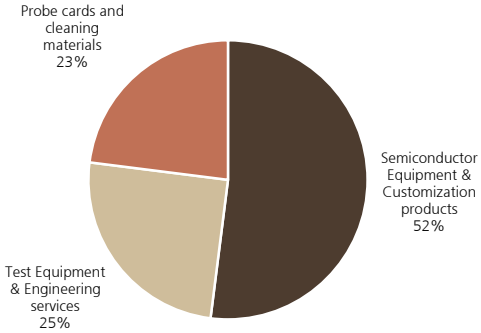
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

HTSI sales and GM trend



Source: Company data, UBS estimates

HTSI H126 sales mix breakdown



Source: Company data, UBS

Hermes Testing Solutions (7856.TWO)

Income Statement (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**1,598**	**2,425**	**5,505**	*127.0*	**13,879**	*152.1*	**24,713**	**29,218**	**32,810**
Gross profit	579	1,021	3,054	*199.2*	8,097	*165.1*	14,817	17,801	20,107
EBITDA (UBS)	**280**	**556**	**1,329**	*139.0*	**6,512**	*389.8*	**12,138**	**15,053**	**17,182**
Depreciation & amortisation	(226)	(260)	(254)	*2.3*	(358)	*-41.0*	(534)	(759)	(1,012)
EBIT (UBS)	**54**	**297**	**1,076**	*262.7*	**6,154**	*NM*	**11,604**	**14,294**	**16,170**
Associates & investment income	0	0	0	-	0	-	0	0	0
Other non-operating income	13	35	1	*-96.8*	0	-	0	0	0
Net interest	(10)	(9)	8	-	169	*NM*	214	286	364
Exceptionals (incl goodwill)	0	0	0	-	0	-	0	0	0
Pre-tax profit	**57**	**322**	**1,085**	*236.7*	**6,323**	*NM*	**11,818**	**14,580**	**16,534**
Tax	0	(54)	(223)	*-314.2*	(1,265)	*NM*	(2,364)	(2,916)	(3,307)
Profit after tax	**57**	**268**	**862**	*221.2*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Preference dividends	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	*-83.3*	0	-	0	0	0
Extraordinary items	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Net earnings (UBS)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Tax rate (%)	0.0	16.7	20.6	*23.0*	20.0	*-2.7*	20.0	20.0	20.0

Per Share (NT$)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (local GAAP, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (UBS, basic)	2.84	10.83	29.11	*168.8*	153.29	*NM*	286.49	353.45	400.82
DPS (net) (NT$)	0.00	1.10	11.64	*NM*	61.32	*NM*	114.60	141.38	160.33
Cash EPS (UBS, diluted) [1]	13.49	20.94	37.67	*79.9*	164.12	*335.6*	302.66	376.44	431.48
Book value per share	12.85	46.45	459.82	*NM*	602.66	*31.1*	827.84	1,066.69	1,326.13
Average shares (diluted)	21	25	30	*17.5*	33	*11.5*	33	33	33

Balance Sheet (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	182	709	14,360	*NM*	17,474	*21.7*	22,786	29,613	36,992
Other current assets	589	806	1,732	*114.7*	3,913	*125.9*	6,302	6,856	7,786
Total current assets	**771**	**1,515**	**16,092**	*NM*	**21,386**	*32.9*	**29,088**	**36,469**	**44,777**
Net tangible fixed assets	509	441	429	*-2.7*	765	*78.5*	1,467	2,169	2,797
Net intangible fixed assets	0	0	0	-	0	-	0	0	0
Investments / other assets	122	111	224	*101.6*	224	*0.0*	224	224	224
Total assets	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**
Trade payables & other ST liabilities	384	694	1,485	*113.8*	2,401	*61.8*	3,374	3,575	3,950
Short term debt	525	0	0	-	0	-	0	0	0
Total current liabilities	**909**	**694**	**1,485**	*113.8*	**2,401**	*61.8*	**3,374**	**3,575**	**3,950**
Long term debt	125	90	0	-	0	-	0	0	0
Other long term liabilities	28	19	86	*349.5*	86	*0.0*	86	86	86
Preferred shares	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,062**	**804**	**1,571**	*95.4*	**2,488**	*58.4*	**3,461**	**3,662**	**4,037**
Common s/h equity	**339**	**1,263**	**15,174**	*NM*	**19,888**	*31.1*	**27,319**	**35,201**	**43,762**
Minority interests	0	0	0	*-1.1*	0	*0.0*	0	0	0
Total liabilities & equity	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**

Cash Flow (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	57	268	862	*221.1*	5,059	*NM*	9,454	11,664	13,227
Depreciation & amortisation	226	260	254	*-2.3*	358	*41.0*	534	759	1,012
Net change in working capital	(254)	(65)	(374)	*NM*	(1,264)	*-238.1*	(1,416)	(354)	(554)
Other operating	88	139	256	*84.3*	0	-	0	0	0
Operating cash flow	**117**	**602**	**998**	*65.9*	**4,152**	*316.1*	**8,572**	**12,069**	**13,685**
Tangible capital expenditure	(157)	(122)	(275)	*-126.4*	(694)	*-152.1*	(1,236)	(1,461)	(1,641)
Intangible capital expenditure	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(29)	17	13	*-24.0*	0	-	0	0	0
Other investing	(19)	(20)	(2)	*89.5*	0	-	0	0	0
Investing cash flow	**(206)**	**(125)**	**(265)**	*-111.9*	**(694)**	*-162.2*	**(1,236)**	**(1,461)**	**(1,641)**
Equity dividends paid	0	0	(27)	-	(345)	*NM*	(2,023)	(3,782)	(4,666)
Share issues / (buybacks)	23	626	13,102	*NM*	0	-	0	0	0
Other financing	(46)	(54)	(33)	*38.9*	0	-	0	0	0
Change in debt & pref shares	141	(525)	(125)	*76.2*	0	-	0	0	0
Financing cash flow	**117**	**46**	**12,916**	*NM*	**(345)**	-	**(2,023)**	**(3,782)**	**(4,666)**
Cash flow inc/(dec) in cash	**29**	**523**	**13,649**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**
FX / non cash items	-	4	2	*-42.4*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**-**	**527**	**13,652**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Hermes Testing Solutions (7856.TWO)

Valuation (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/E (UBS, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/CEPS	-	34.3	NM	25.5	13.8	11.1	9.7
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8
P/BV	-	15.7	9.1	7.0	5.1	3.9	3.2
EV/revenues (core)	-	8.1	NM	7.1	3.8	3.0	2.5
EV/EBITDA (UBS core)	-	35.1	80.1	15.1	7.7	5.8	4.7
EV/EBIT (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/OpFCF (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/op. invested capital	-	27.4	>100	60.7	27.0	17.3	13.1

Enterprise value (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	-	19,581	113,884	113,884	113,884	113,884	113,884
Net debt (cash)	-	(75)	(7,489)	(15,917)	(20,130)	(26,200)	(33,302)
Buy out of minorities	28	19	86	86	86	86	86
Pension provisions/other	0	0	0	0	0	0	0
Total enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**
Non core assets	(29)	0	0	0	0	0	0
Core enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**

Growth (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	-	51.8	127.0	152.1	78.1	18.2	12.3
EBITDA (UBS)	-	98.9	139.0	NM	86.4	24.0	14.1
EBIT (UBS)	-	NM	NM	NM	88.6	23.2	13.1
EPS (UBS, diluted)	-	NM	173.4	NM	86.9	23.4	13.4
Net DPS	-	-	NM	NM	86.9	23.4	13.4

Margins & Profitability (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	36.2	42.1	55.5	58.3	60.0	60.9	61.3
EBITDA margin	17.5	22.9	24.1	46.9	49.1	51.5	52.4
EBIT (UBS) margin	3.4	12.2	19.5	44.3	47.0	48.9	49.3
Net earnings (UBS) margin	3.6	11.1	15.7	36.4	38.3	39.9	40.3
ROIC (EBIT)	-	41.7	NM	NM	NM	NM	NM
ROIC post tax	-	34.7	NM	NM	NM	NM	NM
ROE (UBS)	-	33.5	10.5	28.9	40.1	37.3	33.5

Capital structure & Coverage (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	1.7	(1.1)	(10.8)	(2.7)	(1.9)	(2.0)	(2.2)
Net debt / total equity %	138.1	(49.0)	(94.6)	(87.9)	(83.4)	(84.1)	(84.5)
Net debt / (net debt + total equity) %	58.0	(95.9)	NM	NM	NM	NM	NM
Net debt/EV %	-	(0.4)	(7.0)	(16.2)	(21.5)	(29.8)	(41.3)
Capex / depreciation %	69.5	46.8	108.5	194.1	NM	192.5	162.1
Capex / revenue %	9.8	5.0	5.0	5.0	5.0	5.0	5.0
EBIT / net interest	5.3	31.3	-	-	-	-	-
Dividend cover (UBS)	-	9.9	2.5	2.5	2.5	2.5	2.5
Div. payout ratio (UBS) %	-	10.1	40.0	40.0	40.0	40.0	40.0

Revenues by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	1,598	2,425	5,505	13,879	24,713	29,218	32,810
Total	**1,598**	**2,425**	**5,505**	**13,879**	**24,713**	**29,218**	**32,810**

EBIT (UBS) by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	54	297	1,076	6,154	11,604	14,294	16,170
Total	**54**	**297**	**1,076**	**6,154**	**11,604**	**14,294**	**16,170**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	111.9%
Forecast dividend yield	0.3%
Forecast stock return	112.2%
Market return assumption	6.3%
Forecast excess return	105.9%

Company Description

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Valuation Method and Risk Statement

We value Hermes Testing based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected front-end/CoWoS capacity expansion: This could result in Hermes Testing gaining/losing equipment orders from major foundries and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in Hermes Testing losing market share from major foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to Hermes Testing, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future equipment orders.
- Increase of raw material costs: Rising raw material costs may directly impact Hermes Testing's margins for equipment shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Hermes Testing Solutions

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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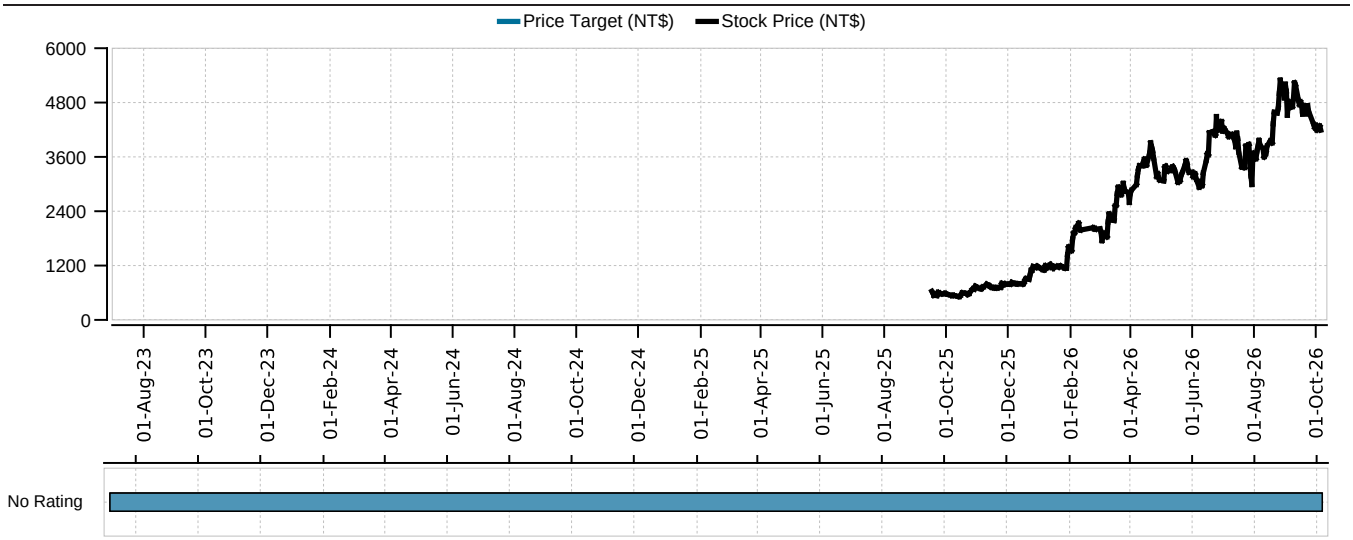
Hermes Testing Solutions (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-06	NaN	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 06-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Chroma ATE, NT$2350.00 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Hon.Precision, Inc., NT$5905.00 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Initiation of Coverage

Hermes Testing Solutions

Shifting-left, scaling-up: dual catalysts in chip probing and CPO

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,880.00**
	Prior :
Price (06 Oct 2026)	**NT$4,190**

RIC: 7856.TWO **BBG:** 7856 TT

Trading data and key metrics

52-wk range	NT$5,295.00-516.55
Market cap.	NT$114b/US$3.58b
Shares o/s	27.2m (ORD)
Free float	94%
Avg. daily volume ('000)	240
Avg. daily value (m)	NT$1,068.4
Common s/h equity (12/26E)	NT$15.2b
P/BV (12/26E)	9.1x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	-	29.11	-	32.19
12/27E	-	153.28	-	89.82
12/28E	-	286.48	-	71.44

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Jack-of-all-trades for bespoke testing solutions in the semiconductor industry

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 PT. HTSI is uniquely positioned to benefit from two key semiconductor testing trends: shift-left testing and the emerging CPO testing market. Leveraging its long-standing relationship with TSMC, HTSI has established capabilities across electrical test engineering, customised test equipment, thermal control modules, probe cards, and CPO testing solutions. We forecast revenue growth of 127%/152%/78% in 2026E/27E/28E, driven by increasing adoption of thermal control modules for wafer-level test and initial deployment of CPO insertion 2 and insertion 3 test equipment through ficonTEC. Additional 2028E growth drivers include reflow tools, laser clean solutions, and MEMS probe cards. The stock trades at 27/15x 2027/28E PE, versus ~35x NTM PE for handler-equipment peers, offering re-rating potential as sales scale, new products are qualified, and its CPO presence expands. See our companion testing deep dive sector note.

Shift-left beneficiary with exceptional thermal control modules

Thermal control modules should be HTSI's largest growth driver over the next several years. We expect increasing chip power consumption, rising package complexity, and tighter probe pitches to drive adoption of advanced thermal control solutions used alongside leading-edge probers, in additional to more wafer level insertions and longer test time. We estimate HTSI will ship approximately 3,000 thermal-control modules in 2027E, up from around 1,200 units in 2026E, sales contribution rise from ~30% in Q226 to >50% of sales by 2027E. The launch of its second-generation platform in 2027E, which carries 25-50% higher ASPs, should further improve profitability. We forecast GM expansion from 55.5% in 2026E to 58.3%/60.0% in 2027E/28E.

Unique position to capture CPO upside

Through its partnership with German optical-automation leader ficonTEC, HTSI is positioned to participate in the initial ramp of CPO testing equipment. HTSI supports local manufacturing, installation, automation and after-sales service, making it a key enabler of volume production. We estimate HTSI captures approximately 30% of equipment value across the 40 insertion 2 and 70 insertion 3 systems we forecast for delivery in 2027E, with further potential upside in 2028E as adoption expands.

Valuation: initiate coverage with a Buy rating and price target NT$8,880

We forecast EPS of NT$29.1 in 2026E (including a one-off employee share purchase programme impact), rising to NT$153.3/286.5 in 2027E/28E. Our NT$8,880 price target is based on 31x average 2028E PE, versus handler-equipment peers forward PE at roughly 35x, reflecting HTSI's unique exposure to both shift-left testing, CPO, and our estimate of 32% LT earnings CAGR. We initiate with a Buy rating.

Highlights (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	1,598	2,425	5,505	13,879	24,713	29,218	32,810
EBIT (UBS)	54	297	1,076	6,154	11,604	14,294	16,170
Net earnings (UBS)	57	268	862	5,059	9,454	11,664	13,227
EPS (UBS, diluted) (NT$)	2.71	10.65	29.11	153.28	286.48	353.44	400.81
DPS (net) (NT$)	0.00	1.10	11.64	61.32	114.60	141.38	160.33
Net (debt) / cash	(469)	618	14,360	17,474	22,786	29,613	36,992
Profitability/valuation	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	3.4	12.2	19.5	44.3	47.0	48.9	49.3
ROIC (EBIT) %	-	41.7	147.5	381.2	334.1	282.5	261.7
EV/EBITDA (UBS core) x	-	35.1	80.1	15.1	7.7	5.8	4.7
P/E (UBS, diluted) x	-	68.6	NM	27.3	14.6	11.9	10.5
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 4,190.00 on 06-Oct-2026

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will HTSI be able to keep its moat within chip probing thermal control modules?
Yes. We expect HTSI's competitive position to strengthen as chip power consumption, package complexity, and probe density continue to increase. The company is developing next-generation thermal control solutions, including liquid cooling technologies, while also expanding into specialized chuck solutions designed to address wafer warpage. In addition, HTSI's close collaboration with TSMC, TEL, Advantest, and other OSATs should reinforce its position within advanced wafer-level testing.

Q: Will insertion 2 of CPO testing process be skipped?
No. We believe concerns surrounding insertion 2 adoption are overdone. While some industry participants favor bypassing insertion 2 due to throughput and ROI considerations, doing so increases the risk of defective optical engines being identified only after the components are delivered to OSATs, creating yield accountability challenges and higher packaging losses. Based on our industry studies, we expect insertion 2 testing to remain an important part of the CPO workflow over at least the next two years as the industry focuses on improving PIC yields and overall testing efficiency.

UBSVIEW

We initiate coverage of Hermes Testing (HTSI) with a Buy rating and NT$8,880 price target. HTSI offers unique exposure to two of the semiconductor testing industry's most attractive growth drivers: 1) shift-left testing and 2) CPO testing. The industry's shift-left testing strategy will have more test insertions move to the wafer level before packaging, in addition to longer CP time. Leveraging its longstanding relationship with TSMC, HTSI has established capabilities across thermal control modules, electrical-test engineering services, customized equipment, and emerging CPO testing solutions. We believe there could be re-rating potential as sales scale, new products are qualified, and its CPO presence expands.

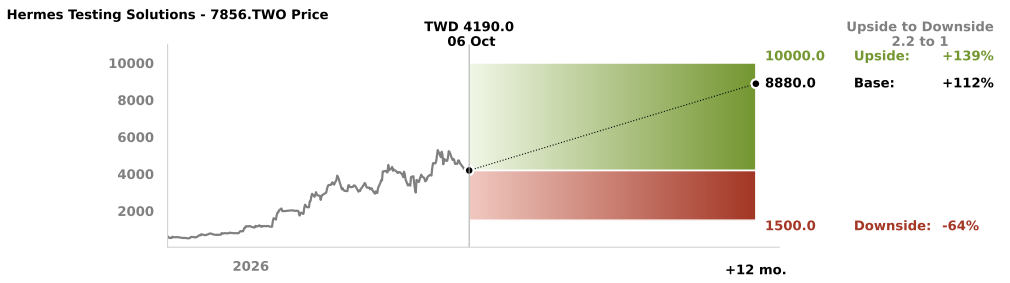
EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

WHAT´S PRICED IN?

Despite a 264% YTD share-price increase, HTSI currently trades at just 15x of our 2028E EPS versus handler-equipment peers forward PE at roughly 35x. We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM



Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&180	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

COMPANY DESCRIPTION

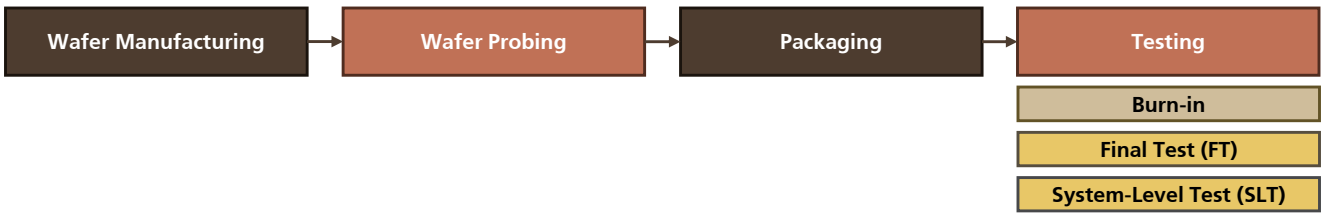
Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

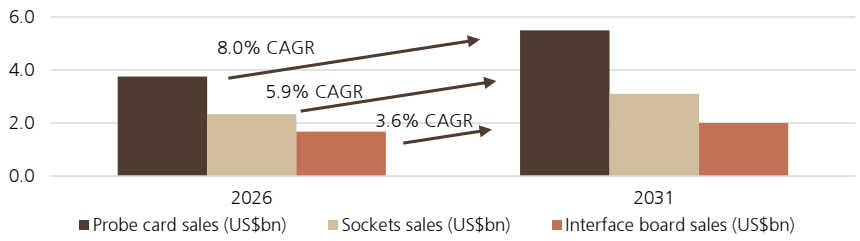
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now got substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damages.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.

- **Thermal control module ASP expansion.** Gen 2 thermal control modules will start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027 with scale-out CPO adoption, we see HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not happening anytime soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 3: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

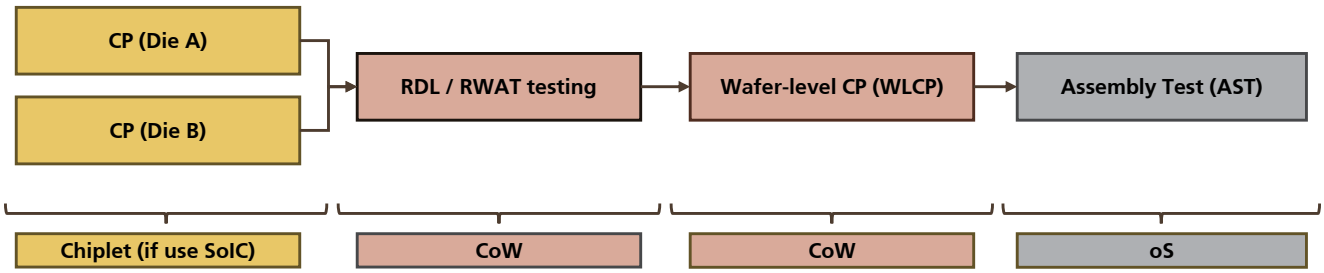
Robust growth drivers across semiconductor testing

We believe HTSI is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see HTSI is uniquely positioned as the main beneficiary of two of the strongest growth drivers within semiconductor testing: testing shift-left and CPO.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging.

Figure 4: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

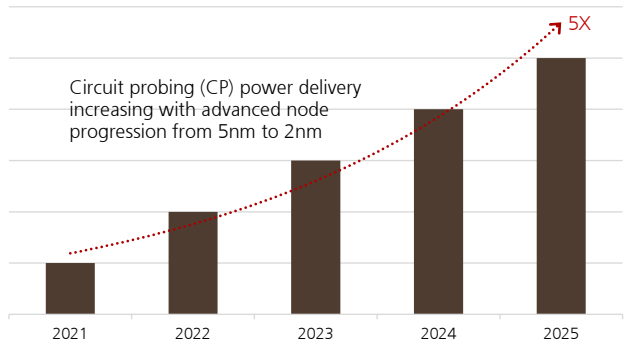


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

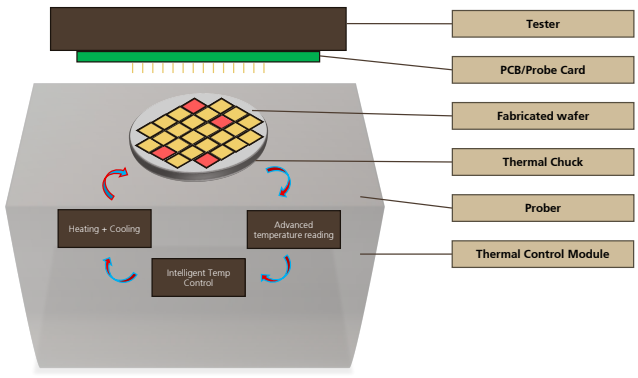
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 5: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 6: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are required for high TDP semiconductor wafer probe in order to maintain the temperature

of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

CPO testing fuelling future growth for semiconductor testing

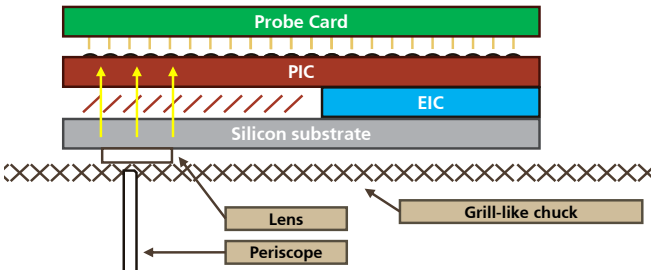
As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

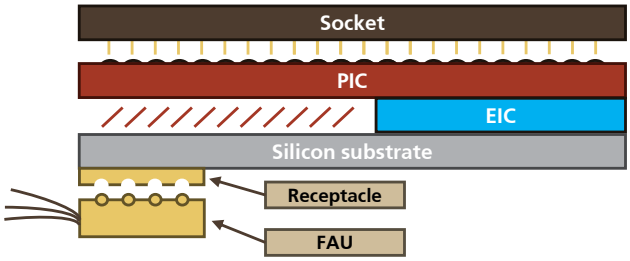
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma has noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 7: An illustration for insertion 2 test method



Source: UBS

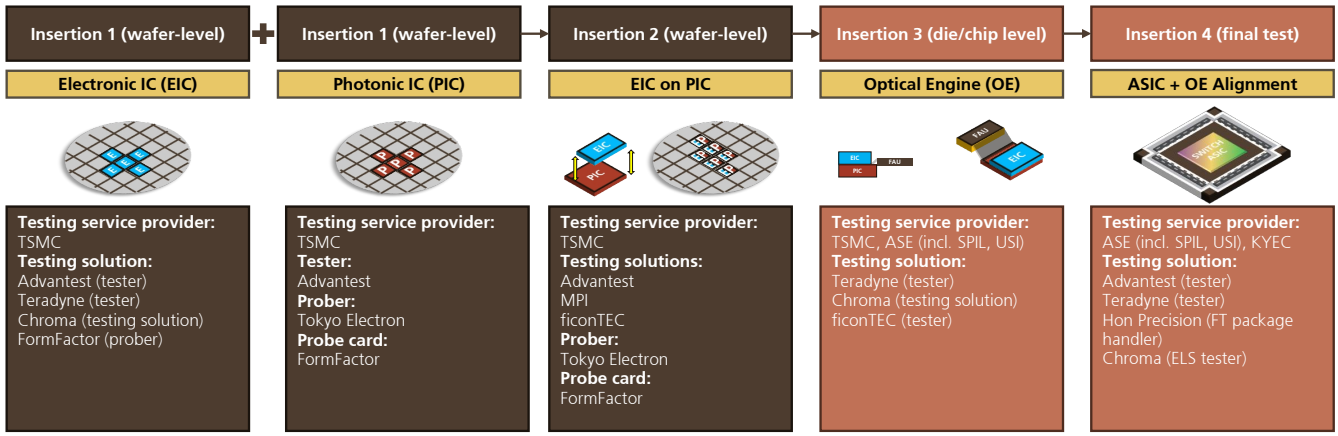
Figure 8: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main target and area of competition for the major players in the field. Notable ones for each insertions are listed.

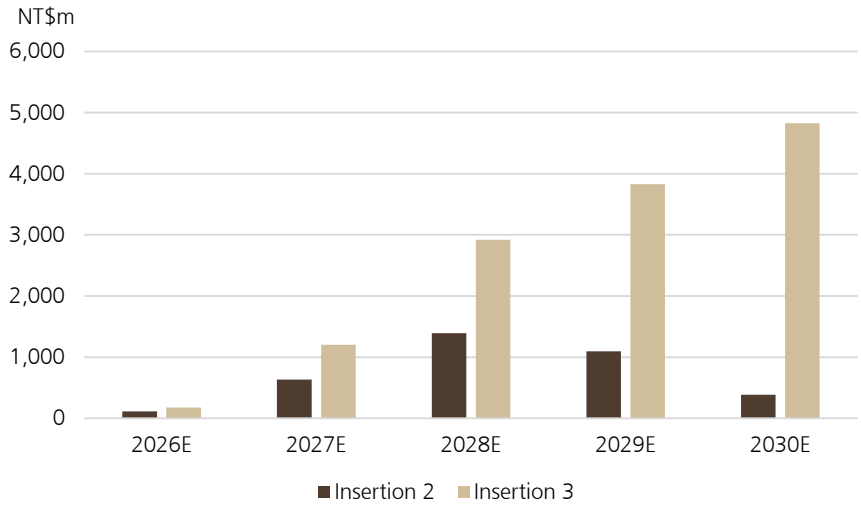
Figure 9: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **HTSI**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs are still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with potential upside to the numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

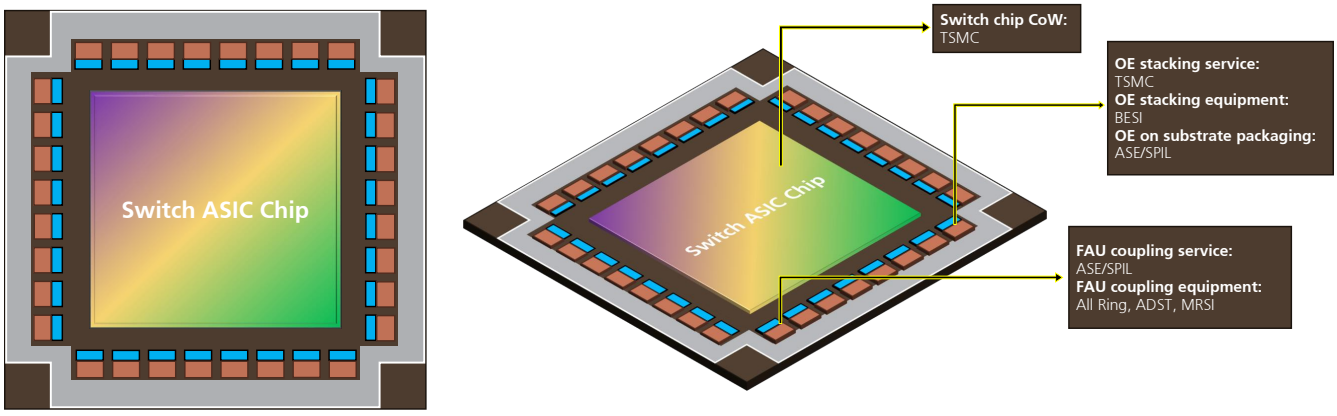
Figure 10: We believe HTSI CPO equipment sales will pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Figure 11: A switch chip with 32 OEs and the CPO packaging supply chain



Source: UBS

HTSI's unique position for new TAM of thermal control modules and CPO test equipment

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan, as a spin-off from Hermes-Epitek. The company provides semiconductor testing and inspection solutions, engineering services, customized equipment, and test consumables. HTSI maintains a global service footprint across Taiwan (Hsinchu, Tainan), Mainland China (Suzhou), Japan (Tokyo, Kumamoto), Singapore, Malaysia, and the United States (Arizona), enabling close support for leading semiconductor customers.

Its main business segments are structured around three pillars:

- **Semiconductor equipment and customized products:** HTSI utilises a dual-track strategy of independent R&D and also equipment agency sales. They offers

highly customized solutions designed to address specific client pain points and bottlenecks. Key products include: thermal control modules that pair with a 1:1 ratio with wafer-level test probers, electron beam inspection equipment, white light interference optical inspection systems, laser probe cleaning machines, and the SYMPHONY fluxless reflow system.

- **Testing equipment engineering services:** The company operates the largest engineering service team in Taiwan, consisting of over 340 technical professionals. The engineering service team provides extensive equipment life-cycle support, including installation, relocations, modification, testing, routine maintenance, and repair services. CPO related equipment sales/services will be recognized within this segment, as CPO are not direct HTSI products, but rather sales recognition via its partnership with ficonTEC.
- **Probe card design, manufacturing, and cleaning materials:** HTSI delivers a comprehensive product portfolio that supports the entire probe card life-cycle. For probe cards, they supply membrane, cantilever, and vertical probe cards. As they are small in scale, HTSI targets niche probe card applications such as RF, 5G/6G, millimeter wave, LCD driver, low end logics, TIA (Transimpedance Amplifier), etc. In addition to the three existing probe technologies, HTSI also collaborates with MJC in Japan for MEMS probe development, targeting sales in 2028E for high-end, AI/HPC related applications. Cleaning materials is another important business within this segment, as HTSI source and process sand paper or other materials to offer cleaning sheets to clean the probe cards, removing oxidations, unwanted particles and substances that accumulated on the probe cards during the testing process.

The sales resiliency in the coming years mainly stems from its robust relationship with key industry players, especially as an engineering and service provider with TSMC for testing solutions. This sets up a solid foundation for its customized equipment business, as they know the specific testing bottleneck to tackle. For instance, as Hermes-Epitek manufacturers the probers for Tokyo Electron (TEL) and HTSI helps with installation, services, and maintenance of the probers, they understand the thermal bottlenecks that the wafer-level test industry has been facing and can effectively come up with module solutions to solve this.

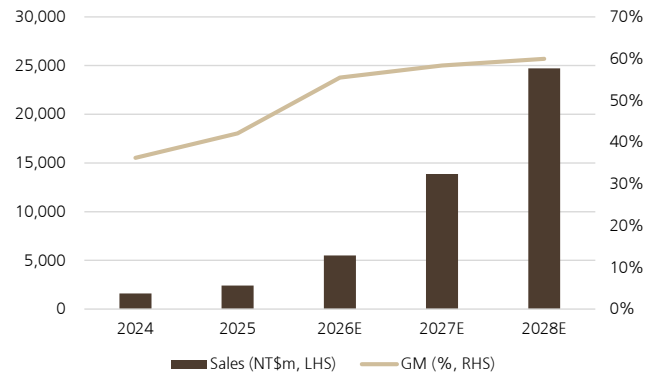
Margins set to expand with value-added products and services

The company has demonstrated strong execution in advanced wafer-level testing, with H126 revenue of NT$2.2bn already exceeding 90% of FY25 revenue (NT$2.4bn). Profitability has also improved significantly, with GM increasing from 42.1% in 2025 to 54.9% in H126.

With share of value-added products and services set to rise in 2027E in sales mix, HTSI is set to see GM and operating leverage expansion. As a niche engineering service provider that has robust ties to leading foundries and OSATs, we believe HTSI has substantial pricing power and estimate HTSI' GM for thermal control modules to be around 65%, with GM for CPO equipment of around 60% as HTSI provides an engineering-service-like business for it. We believe these factors will help the company to reach a 60% long-term GM range.

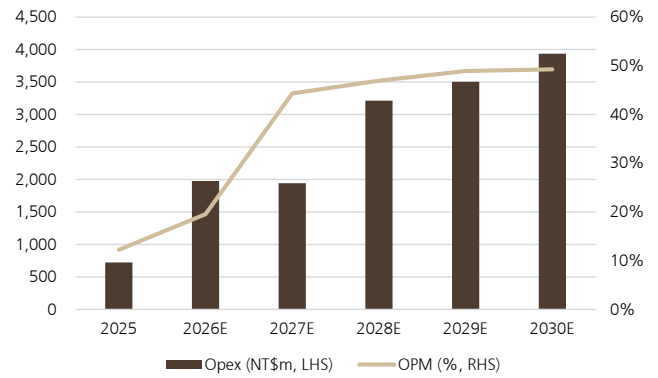
OPM is also set to expand with customized equipment sales rise in total sales mix. If we exclude the one-time expense to be recognized during Q326 for employee stock benefits (we estimate around NT$1bn), YoY opex growth will be around 99% for 2027E, with downside risk if the company conducts tight cost control measures. We forecast the company's OPM will expand from 19.5% in 2026E, to 44.3%/47.0% in 2027/28E.

Figure 12: HTSI sales set to continue to climb with 152% growth and GM at 58.3% in 2027E



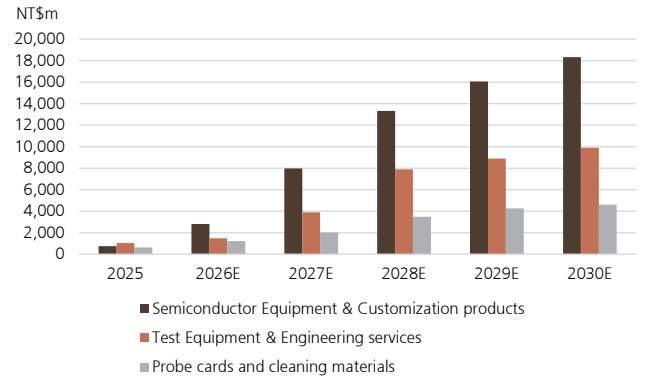
Source: UBS estimates

Figure 13: OPM set to grow to 44.3%/47.0% in 2027/28E



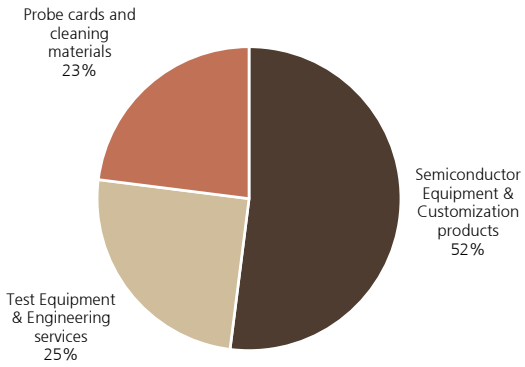
Source: UBS estimates. Note 2026E opex includes a one-time NT$1bn employee stock benefit.

Figure 14: We see customized equipment to continue to drive company growth



Source: Company data, UBS estimates

Figure 15: HTSI H126 sales mix breakdown



Source: Company data, UBS

Figure 16: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**2,425**	**985**	**1,200**	**1,513**	**1,807**	**5,505**	**2,634**	**3,197**	**3,760**	**4,289**	**13,879**	**24,713**	**29,218**	**32,810**
- YoY chg (%)	52%	123%	108%		28%	127%	167%	166%	149%	137%	152%	78%	18%	12%
- QoQ chg (%)		-30%	-51%	54%	51%		46%	-42%	43%	34%				
Gross profit	**1,021**	**545**	**655**	**843**	**1,012**	**3,054**	**1,534**	**1,856**	**2,184**	**2,523**	**8,097**	**14,817**	**17,801**	**20,107**
- Gross margin	42.1%	55.3%	54.5%	55.7%	56.0%	55.5%	58.3%	58.1%	58.1%	58.8%	58.3%	60.0%	60.9%	61.3%
Operating profit	**297**	**323**	**411**	**-399**	**741**	**1,076**	**1,166**	**1,408**	**1,657**	**1,923**	**6,154**	**11,604**	**14,294**	**16,170**
- Operating margin	12.2%	32.8%	34.3%	-26.4%	41.0%	19.5%	44.3%	44.1%	44.1%	44.8%	44.3%	47.0%	48.9%	49.3%
Pre-tax profit	**322**	**326**	**412**	**-396**	**743**	**1,085**	**1,205**	**1,449**	**1,700**	**1,968**	**6,323**	**11,818**	**14,580**	**16,534**
Net profit	**268**	**254**	**330**	**-317**	**594**	**862**	**964**	**1,160**	**1,360**	**1,574**	**5,059**	**9,454**	**11,664**	**13,227**
- YoY chg (%)	373%	586%	301%		299%	221%	280%	251%	-529%	165%	487%	87%	23%	13%
- QoQ chg (%)		70%	23%	-225%	80%		62%	35%	41%	36%				
Basic EPS (NT$)	**10.83**	**9.35**	**12.15**	**-10.20**	**18.01**	**29.11**	**29.22**	**35.14**	**41.22**	**47.70**	**153.29**	**286.49**	**353.45**	**400.82**
- YoY chg (%)	281%	492%	279%		200%	169%	213%	189%	-504%	165%	427%	87%	23%	13%
- QoQ chg (%)		55%	12%	-209%	48%		62%	21%	41%	36%				

Source: UBS estimates

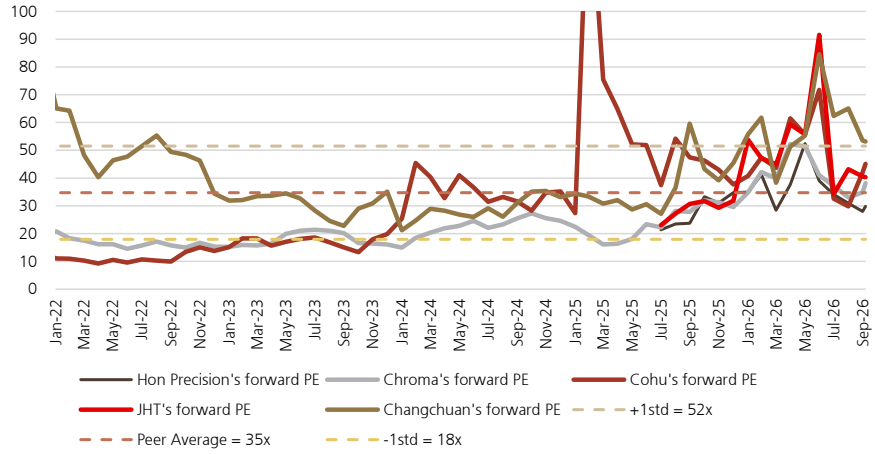
Figure 17: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

WHAT´S PRICED IN?

HTSI is trading at NT$4,190 as of 6 Oct 2026



Source: LSEG, UBS

Despite a 264% YTD share-price increase, HTSI still trades at just 15x 2028E P/E. We believe market expectations for 2027E CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands. Our reverse DCF analysis indicates that the current market price implies only a 17% long-term sales CAGR vs our estimate of 27% and long-term GM level of 33% vs our estimate of 61%. We see scope for re-rating.

We believe the market is not fully reflecting three key growth drivers:

1) Shift-left testing is still underappreciated.

As advanced packaging costs rise and AI accelerators become increasingly complex, more thorough testing is done at the wafer-level. This should drive sustained demand for HTSI's thermal control modules, probe card consumables, and other wafer-level testing solutions.

2) CPO revenue potential remains underestimated.

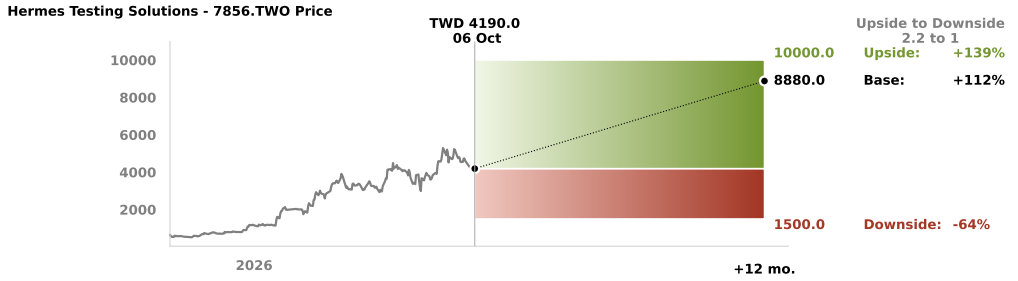
Investor expectations for 2027 CPO equipment shipments, particularly insertion 2 testing systems, remain conservative. Based on our industry studies, we forecast HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, with further potential upside in 2028E as CPO adoption broadens.

3) New products provide additional optionality.

Beyond its core businesses, HTSI is expanding into higher-value solutions including mass-reflow tools, MEMS probe cards and laser cleaning systems, creating incremental growth opportunities that we believe are largely absent from current expectations.

We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028E PE as our view is that 2028 is the year for thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics as active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment. For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE (see What's Priced In chart).

UPSIDE/DOWNSIDE SPECTRUM



HTSI is trading at NT $4,190 as of 6 Oct 2026

Value drivers (2027/28E)	Thermal control module shipments	CPO insertion 2 & 3 shipments	GM
NT$10,000 upside	3,300/5,000	60&100 / 100&175	61.2%/63.6%
NT$8,880 base	3,000/4,100	40&70 / 80&160	58.3%/60.0%
NT$1,500 downside	1,300/2,000	5&20 / 10&50	53.3%/54.0%

Source: UBS estimates

Risk to the current share price is heavily skewed (2.2:1) to the upside.

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected thermal module and CPO equipment shipment in 2027-28E, with additional upside from better traction of probe cards and mass reflow tools, forming 157%/91% YoY sales growth in 2027/28E and even higher operating leverage with OPM of 46.3%/50.0% in 2027/28E. This scenario implies a 34x 2028E PE, a upside driving by more equipment demand stemming for robust shift-left test induced equipment bottleneck, alongside CPO proliferation with scale-up CPO equipment pull-in in 2028E.

BASE (NT$8,880): Our price target of NT$8,880 is based on 31x 2028E PE. We expect an earnings CAGR of 32% in 2027-31E. With shift-left trend for more thermal module requirement during testing, alongside 40/70 CPO insertion 2/3 shipment. We see these growth drivers driving YoY sales growth from a high base of 127% in 2026E to 152%/78% YoY in 2027/2028E, with OPM of 44.3%/47.0% in 2027/28E.

DOWNSIDE (NT$1,500): Our downside scenario reflects lower shipment of thermal control modules due to less advanced prober demand with less insertion 2 and 3 equipment shipment due to bottlenecks with CPO production, alongside limited additional growth drivers to be picked up in 2028 such as MEMS probe card and mass reflow tools. This leads to weaker YoY sales growth of 67%/50% in 2027/28E and operating margin of 29.3%/32.0% in 2027/28E as equipment shipment fails to scale. This scenario implies a 14x 2028E PE, as valuation de-rate with dimmed outlook without CPO and limited upside from thermal control module shipments.

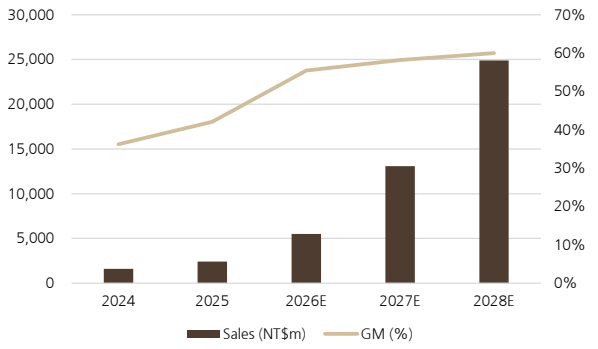
COMPANY DESCRIPTION

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Industry Outlook

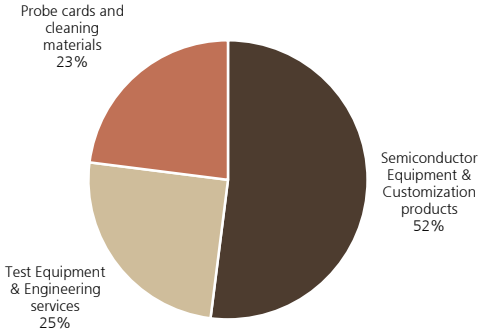
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

HTSI sales and GM trend



Source: Company data, UBS estimates

HTSI H126 sales mix breakdown



Source: Company data, UBS

Hermes Testing Solutions (7856.TWO)

Income Statement (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**1,598**	**2,425**	**5,505**	*127.0*	**13,879**	*152.1*	**24,713**	**29,218**	**32,810**
Gross profit	579	1,021	3,054	*199.2*	8,097	*165.1*	14,817	17,801	20,107
EBITDA (UBS)	**280**	**556**	**1,329**	*139.0*	**6,512**	*389.8*	**12,138**	**15,053**	**17,182**
Depreciation & amortisation	(226)	(260)	(254)	*2.3*	(358)	*-41.0*	(534)	(759)	(1,012)
EBIT (UBS)	**54**	**297**	**1,076**	*262.7*	**6,154**	*NM*	**11,604**	**14,294**	**16,170**
Associates & investment income	0	0	0	-	0	-	0	0	0
Other non-operating income	13	35	1	*-96.8*	0	-	0	0	0
Net interest	(10)	(9)	8	-	169	*NM*	214	286	364
Exceptionals (incl goodwill)	0	0	0	-	0	-	0	0	0
Pre-tax profit	**57**	**322**	**1,085**	*236.7*	**6,323**	*NM*	**11,818**	**14,580**	**16,534**
Tax	0	(54)	(223)	*-314.2*	(1,265)	*NM*	(2,364)	(2,916)	(3,307)
Profit after tax	**57**	**268**	**862**	*221.2*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Preference dividends	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	*-83.3*	0	-	0	0	0
Extraordinary items	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Net earnings (UBS)	**57**	**268**	**862**	*221.1*	**5,059**	*NM*	**9,454**	**11,664**	**13,227**
Tax rate (%)	0.0	16.7	20.6	*23.0*	20.0	*-2.7*	20.0	20.0	20.0

Per Share (NT$)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (local GAAP, diluted)	2.71	10.65	29.11	*173.4*	153.28	*NM*	286.48	353.44	400.81
EPS (UBS, basic)	2.84	10.83	29.11	*168.8*	153.29	*NM*	286.49	353.45	400.82
DPS (net) (NT$)	0.00	1.10	11.64	*NM*	61.32	*NM*	114.60	141.38	160.33
Cash EPS (UBS, diluted) [1]	13.49	20.94	37.67	*79.9*	164.12	*335.6*	302.66	376.44	431.48
Book value per share	12.85	46.45	459.82	*NM*	602.66	*31.1*	827.84	1,066.69	1,326.13
Average shares (diluted)	21	25	30	*17.5*	33	*11.5*	33	33	33

Balance Sheet (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	182	709	14,360	*NM*	17,474	*21.7*	22,786	29,613	36,992
Other current assets	589	806	1,732	*114.7*	3,913	*125.9*	6,302	6,856	7,786
Total current assets	**771**	**1,515**	**16,092**	*NM*	**21,386**	*32.9*	**29,088**	**36,469**	**44,777**
Net tangible fixed assets	509	441	429	*-2.7*	765	*78.5*	1,467	2,169	2,797
Net intangible fixed assets	0	0	0	-	0	-	0	0	0
Investments / other assets	122	111	224	*101.6*	224	*0.0*	224	224	224
Total assets	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**
Trade payables & other ST liabilities	384	694	1,485	*113.8*	2,401	*61.8*	3,374	3,575	3,950
Short term debt	525	0	0	-	0	-	0	0	0
Total current liabilities	**909**	**694**	**1,485**	*113.8*	**2,401**	*61.8*	**3,374**	**3,575**	**3,950**
Long term debt	125	90	0	-	0	-	0	0	0
Other long term liabilities	28	19	86	*349.5*	86	*0.0*	86	86	86
Preferred shares	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,062**	**804**	**1,571**	*95.4*	**2,488**	*58.4*	**3,461**	**3,662**	**4,037**
Common s/h equity	**339**	**1,263**	**15,174**	*NM*	**19,888**	*31.1*	**27,319**	**35,201**	**43,762**
Minority interests	0	0	0	*-1.1*	0	*0.0*	0	0	0
Total liabilities & equity	**1,402**	**2,067**	**16,745**	*NM*	**22,376**	*33.6*	**30,780**	**38,863**	**47,799**

Cash Flow (NT$m)	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	57	268	862	*221.1*	5,059	*NM*	9,454	11,664	13,227
Depreciation & amortisation	226	260	254	*-2.3*	358	*41.0*	534	759	1,012
Net change in working capital	(254)	(65)	(374)	*NM*	(1,264)	*-238.1*	(1,416)	(354)	(554)
Other operating	88	139	256	*84.3*	0	-	0	0	0
Operating cash flow	**117**	**602**	**998**	*65.9*	**4,152**	*316.1*	**8,572**	**12,069**	**13,685**
Tangible capital expenditure	(157)	(122)	(275)	*-126.4*	(694)	*-152.1*	(1,236)	(1,461)	(1,641)
Intangible capital expenditure	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(29)	17	13	*-24.0*	0	-	0	0	0
Other investing	(19)	(20)	(2)	*89.5*	0	-	0	0	0
Investing cash flow	**(206)**	**(125)**	**(265)**	*-111.9*	**(694)**	*-162.2*	**(1,236)**	**(1,461)**	**(1,641)**
Equity dividends paid	0	0	(27)	-	(345)	*NM*	(2,023)	(3,782)	(4,666)
Share issues / (buybacks)	23	626	13,102	*NM*	0	-	0	0	0
Other financing	(46)	(54)	(33)	*38.9*	0	-	0	0	0
Change in debt & pref shares	141	(525)	(125)	*76.2*	0	-	0	0	0
Financing cash flow	**117**	**46**	**12,916**	*NM*	**(345)**	-	**(2,023)**	**(3,782)**	**(4,666)**
Cash flow inc/(dec) in cash	**29**	**523**	**13,649**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**
FX / non cash items	-	4	2	*-42.4*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**-**	**527**	**13,652**	*NM*	**3,113**	*-77.2*	**5,313**	**6,827**	**7,379**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Hermes Testing Solutions (7856.TWO)

Valuation (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/E (UBS, diluted)	-	68.6	NM	27.3	14.6	11.9	10.5
P/CEPS	-	34.3	NM	25.5	13.8	11.1	9.7
Equity FCF (UBS) yield %	-	2.5	0.6	3.0	6.4	9.3	10.6
Dividend yield (net) %	-	0.2	0.3	1.5	2.7	3.4	3.8
P/BV	-	15.7	9.1	7.0	5.1	3.9	3.2
EV/revenues (core)	-	8.1	NM	7.1	3.8	3.0	2.5
EV/EBITDA (UBS core)	-	35.1	80.1	15.1	7.7	5.8	4.7
EV/EBIT (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/OpFCF (core)	-	65.8	99.0	15.9	8.1	6.1	5.0
EV/op. invested capital	-	27.4	>100	60.7	27.0	17.3	13.1

Enterprise value (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	-	19,581	113,884	113,884	113,884	113,884	113,884
Net debt (cash)	-	(75)	(7,489)	(15,917)	(20,130)	(26,200)	(33,302)
Buy out of minorities	28	19	86	86	86	86	86
Pension provisions/other	0	0	0	0	0	0	0
Total enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**
Non core assets	(29)	0	0	0	0	0	0
Core enterprise value	**-**	**19,526**	**106,481**	**98,054**	**93,841**	**87,771**	**80,668**

Growth (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	-	51.8	127.0	152.1	78.1	18.2	12.3
EBITDA (UBS)	-	98.9	139.0	NM	86.4	24.0	14.1
EBIT (UBS)	-	NM	NM	NM	88.6	23.2	13.1
EPS (UBS, diluted)	-	NM	173.4	NM	86.9	23.4	13.4
Net DPS	-	-	NM	NM	86.9	23.4	13.4

Margins & Profitability (%)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	36.2	42.1	55.5	58.3	60.0	60.9	61.3
EBITDA margin	17.5	22.9	24.1	46.9	49.1	51.5	52.4
EBIT (UBS) margin	3.4	12.2	19.5	44.3	47.0	48.9	49.3
Net earnings (UBS) margin	3.6	11.1	15.7	36.4	38.3	39.9	40.3
ROIC (EBIT)	-	41.7	NM	NM	NM	NM	NM
ROIC post tax	-	34.7	NM	NM	NM	NM	NM
ROE (UBS)	-	33.5	10.5	28.9	40.1	37.3	33.5

Capital structure & Coverage (x)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	1.7	(1.1)	(10.8)	(2.7)	(1.9)	(2.0)	(2.2)
Net debt / total equity %	138.1	(49.0)	(94.6)	(87.9)	(83.4)	(84.1)	(84.5)
Net debt / (net debt + total equity) %	58.0	(95.9)	NM	NM	NM	NM	NM
Net debt/EV %	-	(0.4)	(7.0)	(16.2)	(21.5)	(29.8)	(41.3)
Capex / depreciation %	69.5	46.8	108.5	194.1	NM	192.5	162.1
Capex / revenue %	9.8	5.0	5.0	5.0	5.0	5.0	5.0
EBIT / net interest	5.3	31.3	-	-	-	-	-
Dividend cover (UBS)	-	9.9	2.5	2.5	2.5	2.5	2.5
Div. payout ratio (UBS) %	-	10.1	40.0	40.0	40.0	40.0	40.0

Revenues by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	1,598	2,425	5,505	13,879	24,713	29,218	32,810
Total	**1,598**	**2,425**	**5,505**	**13,879**	**24,713**	**29,218**	**32,810**

EBIT (UBS) by division (NT$m)	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	54	297	1,076	6,154	11,604	14,294	16,170
Total	**54**	**297**	**1,076**	**6,154**	**11,604**	**14,294**	**16,170**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	111.9%
Forecast dividend yield	0.3%
Forecast stock return	112.2%
Market return assumption	6.3%
Forecast excess return	105.9%

Company Description

Hermes Testing Solutions (HTSI) was founded in 2004 in Hsinchu, Taiwan. It focuses on providing semiconductor testing and inspecting solutions, equipment installation, technical support, and maintenance services. Its main business segments are customized semiconductor equipment, engineering services, and probe cards and other materials.

Valuation Method and Risk Statement

We value Hermes Testing based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected front-end/CoWoS capacity expansion: This could result in Hermes Testing gaining/losing equipment orders from major foundries and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in Hermes Testing losing market share from major foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to Hermes Testing, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future equipment orders.
- Increase of raw material costs: Rising raw material costs may directly impact Hermes Testing's margins for equipment shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Hermes Testing Solutions

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

Required Disclosures

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Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

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Company Name	Reuters	12-month rating	Price	Price date
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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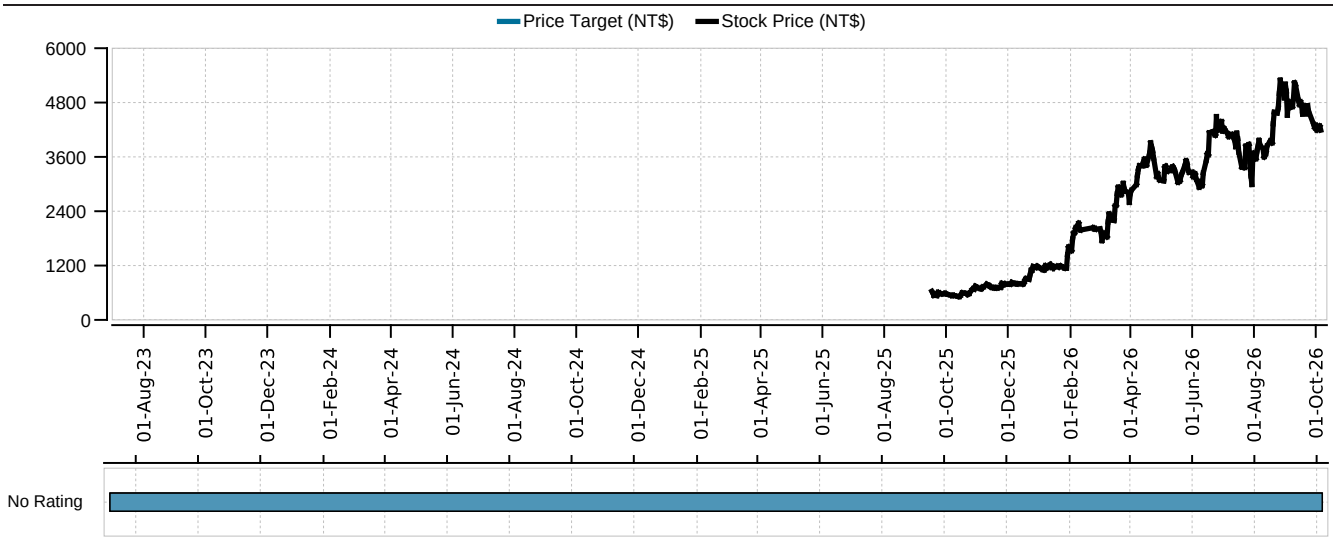
Hermes Testing Solutions (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-06	NaN	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 06-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Chroma ATE, NT$2350.00 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Hon.Precision, Inc., NT$5905.00 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Asia Semiconductor Insight

AI put to the test: the new era of chip testing

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Time to revisit semiconductor testing as content growth accelerates

Our core thesis is that AI-driven chip complexity is drastically increasing testing content across wafer, package, and system levels, creating a multi-year growth opportunity that the market has yet to fully price in. Testing companies in Taiwan have underperformed TAIEX in the past three months (indexing at -9% vs TAIEX +7%), yet to recover from the July market deleveraging. Investor hesitation currently stems from 1) fear of test content growth expectations already priced-in, 2) lingering concerns over shorter test times, and 3) near-term capacity constraints among major suppliers of probe cards, load boards, socket pins, testers, and handlers, caused by heightened testing demand. Ultimately, these concerns overshadow the structural increase in test insertions required by AI accelerators, chiplets, and advanced packaging. As advanced packaging scales, testing intensity must rise in tandem, driving growth through higher content per device rather than semiconductor unit growth alone.

Expanding testing opportunities with shift-left for CP and increasing FT content

Chip probing (CP) supply chain should be the key beneficiary of the industry's shift-left testing trend, as more test insertions move to the wafer level before packaging, in addition to longer CP time. We believe investors underestimate this trend, which should accelerate with broader adoption of 2.5D/3D packaging, chiplets and larger AI packages, driving demand for high pin count probe cards, probers, and thermal-control equipment. We expect strong growth for **Hermes Testing (HTSI)** as thermal-management requirements rise with increasing chip power and complexity, with prober thermal-control module shipments reaching 3,000/4,100 units in 2027E/28E, up from 1,200 in 2026E. We forecast **Chunghwa Precision Test (CHPT)**'s sales to exceed consensus by 7%/15% in 2027E/28E, supported by structural AI/HPC testing demand and rising value-added content. Importantly, shift-left testing does not diminish the role of final test (FT) and mission-mode testing. Test content and test time should continue to increase as validation becomes increasingly critical for complex, large-form-factor AI accelerators. We expect larger package sizes and rising system-level testing requirements to drive demand for **WinWay**'s high-pin-count sockets, supporting 74%/71% YoY sales growth in 2027E/28E.

Future of testing: co-packaged optics (CPO) will be a major opportunity

CPO testing represents a significant new market opportunity, though we expect the ecosystem to remain fragmented rather than consolidating around a single winner. As AI scaling shifts the industry bottleneck from memory to interconnect, CPO adoption is becoming increasingly important, creating new testing challenges that will require collaboration across equipment, optics, and test solution providers. Leveraging ficonTEC's optical alignment expertise and **HTSI**'s probe, testing, and manufacturing capabilities, we estimate the HTSI will deliver 40 units of insertion 2 equipment and 70 units of insertion 3 testing equipment in 2027, with further upside in 2028 as CPO deployment expands from scale-out towards scale-up.

Initiate Hermes Testing, Chunghwa Precision Test, WinWay with Buy ratings

We initiate coverage on semiconductor testing solution provider **Hermes Testing with a Buy rating, NT$8,880 price target** based on 31x 2028E PE, for thermal control/CPO with the most attractive risk/reward in our testing coverage; test interface PCB supplier **CHPT with a Buy rating, NT$5,000 price target** based on 31x avg 2027-28E PE, for CPU/TPU programs and test board content gain; and semiconductor test socket leader **WinWay with a Buy rating, NT$8,900 price target** based on 30x avg 2027-28E PE, for socket content growth and margin recovery. Among our testing coverage, we also like Hon Precision and Chroma ATE.

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 22.**

UBS Research THESIS MAP	MOST FAVORED	LEAST FAVORED
	HTSI, CHPT, WinWay, Hon Precision, Chroma ATE	N/A

PIVOTAL QUESTIONS

Q: Is chip probing becoming increasingly important in the testing supply chain?

Yes. As package costs rise, foundries must improve known-good-die (KGD) screening before packaging, increasing demand for wafer-level testing. This should drive demand for probe cards, probers and advanced thermal control solutions capable of managing higher power consumption, wafer warpage and tighter test tolerances. We forecast HTSI/CHPT/WinWay' probe card business to see a 31%/35%/33% sales CAGR for 2026-2031E, as they are positioned to grow strongly given their smaller scale and robust growth drivers from shift left, CPU, AI accelerators. We forecast HTSI to ship 1,200/3,000/4,100 thermal control modules in 2026E/27E/28E, as advanced wafer-level test for AI accelerators will require thermal control solutions in order to prevent probe card needle burnt.

Q: Will FT and SLT content continue to grow in the coming years?

Yes. Despite concerns that final-test (FT) time could flatten or decline with Nvidia's Rubin GPUs, we expect overall FT content to continue growing as advanced packaging complexity and chip sizes increase. We also expect broader adoption of system-level testing (SLT)/mission-mode testing, as AI accelerators require increasingly stringent validation to ensure optimal system and rack-level performance. We forecast WinWay's socket business to see a 33% sales CAGR for 2026-2031E.

Q: Will CPO testing be a revenue generator in 2027-28 to supply chain players?

Yes. Testing remains a key bottleneck, creating a new equipment opportunity beginning in 2027E. We expect initial CPO volumes in 2027, followed by broader adoption in 2028. Testing remains one of the key bottlenecks to commercialization, driving demand for new equipment and test solutions. We believe early leaders such as ficonTEC (partnership with HTSI) are well positioned to benefit from the initial ramp, alongside suppliers including Chroma ATE and Hon Precision.

WHAT´S PRICED IN?

1) We believe upside from agentic AI CPU demand is not reflected in CHPT and WinWay shares. We estimate CPU-related demand could contribute 20-30% of 2027E sales for CHPT and WinWay, respectively, with further upside in 2028E. **2)** We believe market expectations for 2027 CPO testing equipment shipments remain conservative, particularly for insertion 2 equipment. Based on our industry analysis, we estimate HTSI will deliver 40 insertion 2 and 70 insertion 3 systems in 2027E, this should be higher than the market's expectation of no insertion 2 and limited insertion 3 shipment due to engineering bottleneck. We also see further upside in 2028E as CPO adoption expands.

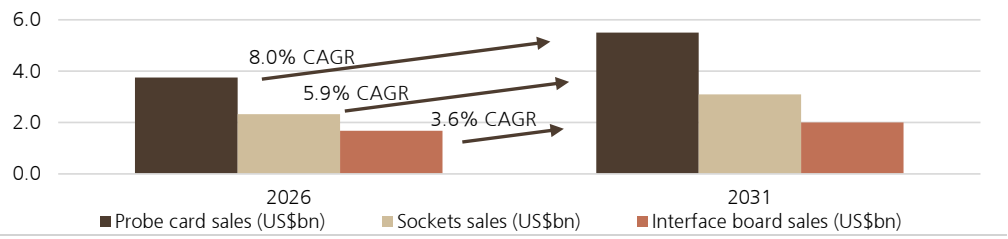
UBSVIEW

Testing is becoming one of the largest incremental beneficiaries of AI-driven semiconductor complexity. As more test insertions are required across wafer, package and system levels, testing content per device should increase materially, expanding the industry's addressable market and supporting multi-year growth across the supply chain.

EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand.

Figure 1: Testing consumables TAM to continue to climb in the next 5 years



Source: Yole, UBS

Executive Summary

Figure 2: Summary of our investment thesis for HTSI, CHPT, and WinWay

Company	Value Chain Positioning	Primary Product Focus	Investment Thesis	Price Target	Upside
Hermes Testing Solutions (HTSI)	Wafer-level Test	Semiconductor testing related installation and services, customized equipment (e.g. thermal control modules for probers), engineering services (CPO test equipment), probe cards.	A key semiconductor testing solution provider and a shift-left beneficiary, with thermal management capabilities that positions the company in the center of AI/HPC chip probing. Uniquely positioned with CPO exposure through its engineering expertise and partnership with ficonTEC.	**NT$8,880**	
Chunghwa Precision Test (CHPT)	Wafer-level Test, Final Test, System-level Test, Burn-in Test	Semiconductor test PCB, final test load board (interface board), probe cards, burn-in board.	A key semiconductor test PCB maker with unique vertical integration capabilities for fast product cadence to enable design-wins. Probe card business is benefiting from the shift-left trend, driving robust growth with design-wins from projects such as CPU and AI ASICs.	**NT$5,000**	
WinWay Technology	Wafer-level Test, Final Test, System-level Test, Burn-in Test	Final test sockets, system-level test sockets, probe cards.	Leading semiconductor test socket provider, beneficiary of final test and mission-level test content expansion with advanced packaging and chiplet proliferation. With its partnership with Technoprobe, WinWay's probe card strength is adding another leg of growth.	**NT$8,900**	

Source: UBS estimates

Shift-left igniting secular growth for chip probing

With the industry increasingly adopting shift-left testing, chip probing (CP) and its supply chain should be key beneficiaries of a multi-year structural growth trend. This underpins our above-consensus forecasts for **Chunghwa Precision Test (CHPT)**, with 2027E/28E sales 7%/15% above consensus driven by stronger AI/HPC demand, capacity expansion, and rising value-added content. **WinWay** should also benefit through its partnership with Technoprobe, with probe card revenue growth remaining robust at 94%/72% in 2026E/27E following 161%/260% growth in 2024/25.

Hermes Testing (HTSI) should be another major beneficiary as increasing chip power and complexity drive demand for advanced thermal-management modules used alongside high-end probing systems. With TEL's leading AI-oriented probers typically paired with advanced thermal-control solutions, we expect HTSI's thermal-module shipments to reach 3,000 units in 2027E from 1,200 in 2026E. Combined with testing, installation and service revenue, this should drive HTSI sales growth of 127%/152% in 2026E/27E.

Accelerators still in focus with TPU as another significant growth driver

Nvidia remains a key growth driver for both CHPT and WinWay. We expect both companies to gain share with Rubin, with WinWay's socket share improving from roughly 50% historically to 60% with Rubin, while CHPT is positioned to supply around 70% of probe card and final test PCB content. Nvidia currently accounts for roughly 50% of WinWay's revenue and 30% of CHPT's.

Beyond Nvidia, we view Google's TPU roadmap as a meaningful incremental opportunity. Following the successful ramp of TPU v8t with MediaTek as design partner, **CHPT** is positioned to supply probe card PCBs, final test load boards, system-level test boards and burn-in boards. **WinWay** will supply sockets for both TPU v8t and v9, with the larger v9 package providing another growth leg. We estimate production of 4mn TPU v8t units in 2027E and 2.3mn TPU v9 packages (9.2mn dies) alongside 2m v8t units in 2028E.

CPU business as a supplemental catalyst for test interface in an agentic world

With agentic AI proliferation, CPU demand is increasing significantly and this directly correlates to increasing demand for testing consumables. WinWay and CHPT should both benefit from the strong ramp up of Nvidia's ARM based Vera CPU and Google's ARM based Axion CPU in 2026E and 2027E. WinWay will be supplying sockets and probe cards for Vera and Axion, and CHPT will be the probe card and PCB supplier for both Vera and Axion. WinWay will have the additional upside from AMD's x86 Venice CPU, and CHPT could capture some share through AMD's equipment vendor procurement. We estimate **CHPT**'s CPU related sales to be low to mid-teens % of total sales in 2026E and growing to 20-30% in 2027E with CHPT's share expansions with turnkey probe card solutions for Google's Axion. For **WinWay**, we believe CPU should account for around 30% of sales in 2026E/27E.

CPO testing fuelling future growth for semiconductor testing

ficonTEC's 25 years of precision optical-test expertise makes it an early leader in CPO test equipment, particularly in optical alignment, a critical capability for high-volume manufacturing. Through its partnership with ficonTEC, **HTSI** is well positioned to benefit from the initial CPO ramp in 2027E-28E by supporting manufacturing scale-up.

We expect both insertion 2 and insertion 3 testing to be required over the next two years. Despite the technical challenges of double-sided wafer probing, insertion 2 testing remains important given current optical-engine yields and the need to identify known-good optical engines before fibre-array-unit (FAU) coupling. This becomes increasingly critical as switch designs integrate larger numbers of optical engines.

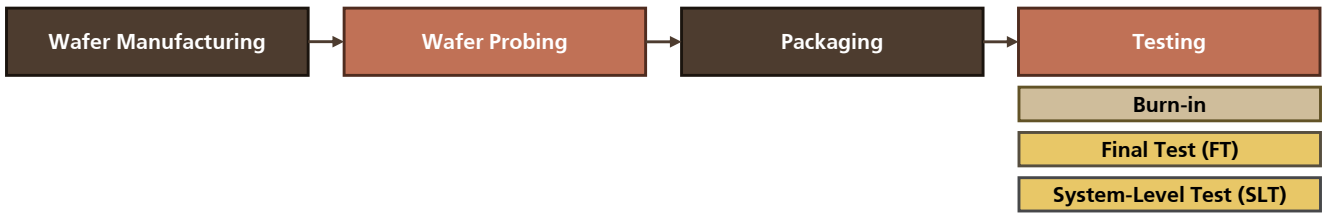
We estimate HTSI will ship 40 insertion 2 and 70 insertion 3 systems in 2027E, with further upside in 2028E as CPO adoption expands and next-generation platforms move into volume production.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 3 illustrates the semiconductor manufacturing flow.

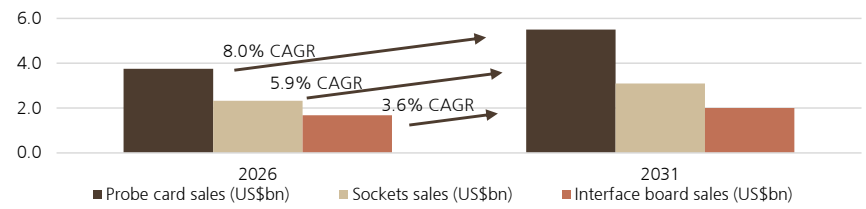
Figure 3: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 4: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector are likely to stand out among competition with greater sales CAGR comparing to the industry. Below we highlight **Hermes Testing (HTSI)**, **Chunghwa Precision Test (CHPT)**, and **WinWay**'s impressive growth drivers and how this compares with broader industry sales CAGR.

We see **HTSI**'s sales for the next few years to mainly be driven by 1) thermal control module shipment and 2) CPO equipment shipment. In the past, thermal control modules weren't necessary for chip probing, due to less TDP for the chips. However, as new iterations of AI accelerators now have substantial power draw, this unlocks a brand new TAM for HTSI to enter with their thermal control modules to pair with probers, in order to prevent probe card damage.

- **Robust thermal control module shipments.** We estimate HTSI to ship 3,000/4,100 thermal control modules in 2027E/28E, with continuous demand in future years, bringing their 2026-31E semiconductor equipment sales CAGR to 48%.
- **Thermal control module ASP expansion.** Gen 2 thermal control modules should start to ship in 2027E which will drive tool ASP up around 25-50%.
- **Demand supported by Advantest expansions.** We believe the 3,000 thermal control modules shipment forecast for 2027E is not far-fetched, as we read-across from Advantest's announced capacity expansion plan (testers will be paired with probers with a 1:1 ratio) with increasing capacity up to 5,000 testers within FY2027/3 and they also announced a 10,000-unit target within CY2028. Although the testers capacity expansion plan includes testers for final test, we do see a good portion to be allocated to chip probing given the shift-left testing trend increasing both testing insertions and testing time before chip packaging.
- **CPO is another brand new TAM for semiconductor testing.** As insertion 2 and 3 testing equipment demand starts to rise in 2027E with scale-out CPO adoption, we expect HTSI to ship 40/70 insertion 2 and 3 equipment in 2027E, with this number growing in 2028E to 80/160 and higher in future years as CPO proliferates, and driving 52% 2026-31E test equipment & engineering services sales CAGR.
- **CPO to see content increase.** Gen 2 insertion 3 equipment may start to ship in 2027E, with higher throughput and tool ASP.
- **Insertion 2 might scale down, but not likely any time soon.** Insertion 2 equipment shipment might come down in 2029E and beyond, as foundries scale-up yield for photonic IC (PIC) in order to lessen the time consuming insertion 2 testing burden.

Figure 5: Hermes Testing is set to benefit with a 46% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Semiconductor Equipment & Customization products	**$2,815**	**$7,980**	**$13,324**	**$16,061**	**$18,315**	**$20,104**	**48%**
YoY		*183%*	*67%*	*21%*	*14%*	*10%*	
Test Equipment & Engineering services	**$1,480**	**$3,887**	**$7,901**	**$8,887**	**$9,898**	**$11,853**	**52%**
YoY		*163%*	*103%*	*12%*	*11%*	*20%*	
Probe cards and cleaning materials	**$1,210**	**$2,012**	**$3,488**	**$4,270**	**$4,597**	**$4,681**	**31%**
YoY		*66%*	*73%*	*22%*	*8%*	*2%*	
Key products shipment units, ASP, sales	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Thermal control module shipment	**1,200**	**3,000**	**4,100**	**4,500**	**4,700**	**4,900**	
Thermal control product blended ASP (NT$m)	$1.9	$2.4	$2.8	$3.0	$3.3	$3.5	
Total sales from thermal control module shipment	**$2,316**	**$7,168**	**$11,417**	**$13,505**	**$15,591**	**$17,067**	**49%**
% of total sales	*42%*	*52%*	*46%*	*46%*	*48%*	*47%*	
CPO Insertion 2 shipment	**8**	**40**	**80**	**60**	**20**	**20**	
Ins2 blended ASP (NT$m)	$13.8	$15.8	$17.4	$18.3	$19.2	$20.1	
Total sales from insertion 2	**$111**	**$633**	**$1,391**	**$1,095**	**$383**	**$402**	**29%**
% of total sales	*2%*	*5%*	*6%*	*4%*	*1%*	*1%*	
CPO insertion 3 shipment	**11**	**70**	**160**	**200**	**240**	**280**	
Ins3 blended ASP (NT$m)	$16.0	$17.2	$18.2	$19.1	$20.1	$21.1	
Total sales from insertion 3	**$176**	**$1,202**	**$2,917**	**$3,829**	**$4,824**	**$5,910**	**102%**
% of total sales	*3%*	*9%*	*12%*	*13%*	*15%*	*16%*	
Key product sales - total	**$2,602**	**$9,002**	**$15,725**	**$18,429**	**$20,798**	**$23,379**	**55%**
% of total sales	*47%*	*65%*	*64%*	*63%*	*63%*	*64%*	
Others - total	**$2,903**	**$4,877**	**$8,988**	**$10,789**	**$12,012**	**$13,260**	**36%**
% of total sales	*53%*	*35%*	*36%*	*37%*	*37%*	*36%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	**$5,505**	**$13,879**	**$24,713**	**$29,218**	**$32,810**	**$36,639**	**46%**
YoY		*152.1%*	*78.1%*	*18.2%*	*12.3%*	*11.7%*	
GM %	**55.5%**	**58.3%**	**60.0%**	**60.9%**	**61.3%**	**61.1%**	
OpM %	**19.5%**	**44.3%**	**47.0%**	**48.9%**	**49.3%**	**51.1%**	

Source: UBS estimates

Uniquely positioned as a company with vertical integration capabilities with MEMS probes, multi-layer PCB capabilities, and automation equipment all-in-house, **CHPT** is set to capture the growing test shift-left trend in the coming years, increasing both testing insertions and testing time before chip packaging.

- **CP to scale from a lower base.** As a smaller probe card vendor, CHPT can benefit from the growing demand for AI probe cards, with 2026-31E sales CAGR for turnkey probe card robust at 41%.
- **Also a FT/SLT/burn-in beneficiary.** The company will also see final test, system-level test, burn-in content increase to play a role in their sales growth from their IC Test Board sales CAGR of 39% in 2026-31E.
- **Capacity expansion supporting growth.** As CHPT is currently capacity constrained, the robust projected sales growth is enabled by CHPT's rapid capacity expansion strategy with strong demand for its products, with one production expansion scheduled for August 2026 and a second in March 2027, doubling and tripling year-end 2025 capacity, respectively. Its Fab 3 will come online in Q428, doubling existing capacity when fully ramped.
- **MEMS capability to support share gain.** CHPT has MEMS probe capabilities stemming from its history as a smartphone application processor (AP) probe card supplier. It is in a good position to capture the growing demand for MEMS probe cards as AI/HPC are shifting to MEMS probes.
- **Probe card content expect to rise.** We also expect probe card content per device to continue rising as AI accelerators grow in die size, pin count, power consumption and packaging complexity. This should drive secular growth for both probe card suppliers and upstream component providers, making chip probing one of the most attractive areas within the semiconductor testing value chain.

Figure 6: CHPT is set to benefit with a 35% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Probe Card	$1,753	$2,903	$4,535	$7,194	$8,361	$9,711	41%
YoY		*66%*	*56%*	*59%*	*16%*	*16%*	
Board Only	$3,574	$6,420	$8,836	$12,650	$13,899	$14,528	32%
YoY		*80%*	*38%*	*43%*	*10%*	*5%*	
Chip Probing	$5,327	$9,324	$13,371	$19,844	$22,261	$24,239	35%
YoY		*75%*	*43%*	*48%*	*12%*	*9%*	
IC Test Board	$1,315	$2,520	$3,579	$5,005	$5,996	$6,825	39%
YoY		*92%*	*42%*	*40%*	*20%*	*14%*	
Service and Others	$502	$640	$918	$1,137	$1,308	$1,504	25%
YoY		*28%*	*43%*	*24%*	*15%*	*15%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$7,144	$12,484	$17,868	$25,985	$29,565	$32,567	35%
YoY		*74.7%*	*43.1%*	*45.4%*	*13.8%*	*10.2%*	
GM %	56.4%	56.1%	56.2%	56.4%	56.3%	56.3%	
OpM %	35.3%	41.5%	44.0%	45.2%	47.1%	47.3%	

Source: UBS estimates

As the leader in AI/HPC sockets, **WinWay** is set to deliver 2026-31E socket sales CAGR of 33%, underpinned by robust demand for final test and system-level test; with test content and test times rising which each iteration of new AI accelerators/CPU, more sockets and equipment will be needed in order to increase the throughput.

- **Strong accelerator/CPU roadmap to support growth.** Robust testing demand can also be inferred from AI accelerator/CPU roadmaps of Nvidia, AMD, Google, alongside tester capacity expansions by Advantest.
- **Capacity expansion underpinning future strength.** WinWay's socket pin capacity expansion will support this exceptional sales growth. from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.
- **Probe card business to rise from smaller base.** Its probe card business is also set to benefit from the shift-left trend for high-end AI/HPC probe cards. WinWay leverages its existing customers' demand in addition to their MEMS partnership with Technoprobe, as we forecast WinWay's probe card sales to rise from a low base, with a 33% 2026-31E sales CAGR.

Figure 7: WinWay is set to benefit with a 33% 2026-31E sales CAGR

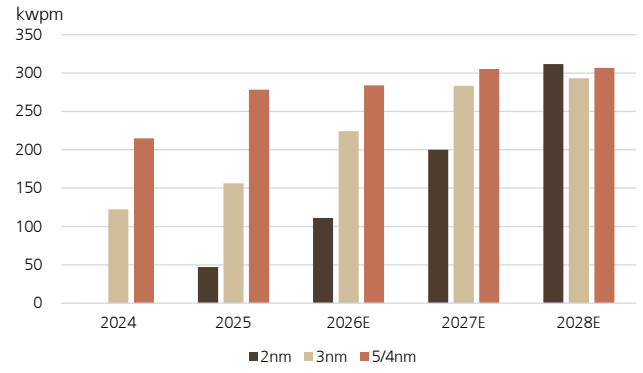
Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Coaxial Socket (HyperSocket)	$7,661	$13,705	$23,103	$25,872	$30,603	$34,726	35%
YoY		*79%*	*69%*	*12%*	*18%*	*13%*	
RF & Plastic Socket	$2,311	$3,835	$6,465	$6,960	$7,259	$7,252	26%
YoY		*66%*	*69%*	*8%*	*4%*	*0%*	
Burn-in Socket	$253	$306	$516	$715	$903	$1,075	34%
YoY		*21%*	*69%*	*39%*	*26%*	*19%*	
Test Socket	$10,224	$17,845	$30,083	$33,547	$38,764	$43,053	33%
YoY		*75%*	*69%*	*12%*	*16%*	*11%*	
Probe Card	$4,478	$7,700	$13,690	$15,507	$16,095	$18,403	33%
YoY		*72%*	*78%*	*13%*	*4%*	*14%*	
Contact Element	$937	$1,596	$2,691	$3,014	$3,465	$3,899	33%
YoY		*70%*	*69%*	*12%*	*15%*	*13%*	
Others	$444	$811	$1,367	$1,671	$2,133	$2,654	43%
YoY		*83%*	*69%*	*22%*	*28%*	*24%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$16,084	$27,952	$47,831	$53,739	$60,457	$68,010	33%
YoY		*73.8%*	*71.1%*	*12.4%*	*12.5%*	*12.5%*	
GM %	41.1%	43.1%	44.7%	44.7%	44.9%	45.1%	
OpM %	27.5%	32.1%	36.7%	36.7%	36.9%	37.1%	

Source: UBS estimates

The AI era has materially increased the value of semiconductor testing. Compared with traditional mobile and automotive chips, AI accelerators incorporate larger compute dies, more advanced nodes, chiplet architectures, advanced packaging technologies, multiple I/O dies, and increasing HBM content. These factors drive higher-value test consumables, including advanced MEMS probe cards, multilayer interface boards, and next-generation sockets designed to support higher pin counts, thermal loading, and power delivery requirements.

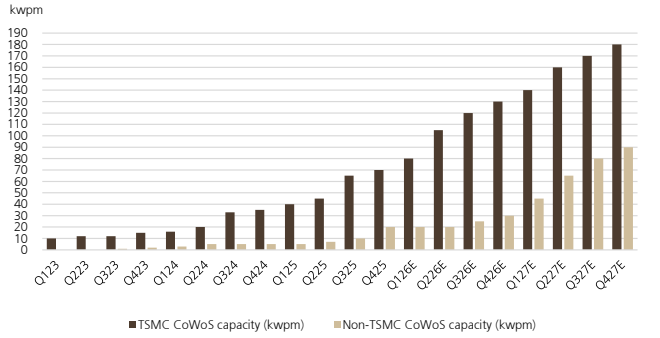
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations (i.e. Nvidia's roadmap from Blackwell to Rubin to Feynman or Google's TPU roadmap from v7 to v8 to v9). This trend is evident with TSMC's front-end and TSMC, Intel, and OSAT's back-end advanced packaging capacity expansions. For front-end, we see industry N3 capacity expanding from 156kwpm in year end 2025 to 224/283/293kpwm in year end 2026E/27E/28E, and N2 capacity from 47kwpm in year-end 2025 to 111/200/312kpwm in year end 2026E/27E/28E. For back-end, we are seeing industry CoWoS capacity rising from 90kwpm in 2025 to 160kwpm in 2026E and further expanding to 270kwpm in 2027E.

Figure 8: Industry's leading edge capacity



Source: UBS estimates

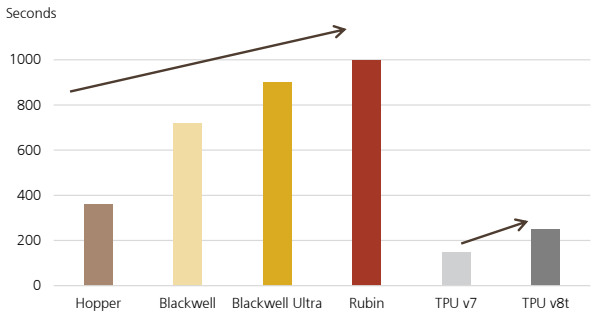
Figure 9: Industry CoWoS capacity



Source: UBS estimates

Final-test, system-level test, and burn-in content should continue to increase as AI accelerators become larger, more power-hungry and more complex. Rising thermal design power and packaging complexity increase validation requirements, driving longer test times and higher demand for test equipment to prevent testing from becoming a production bottleneck.

Figure 10: Our FT test time estimates



Source: UBS estimates

While architectural changes can affect test requirements for individual product generations, we expect the long-term trend of rising testing content to remain intact. For example, Nvidia's Rubin Ultra adopts a simpler package structure than previously expected, from a 4-die structure to a 2-die structure and Rubin Ultra HBM de-spec from HBM4E 12-Hi to initially use HBM4 8-Hi and upgrade to HBM4E 8-Hi at the latter stages, which may moderate near-term test growth. However, future products such as Nvidia's Feynman are expected to reintroduce greater package complexity, with a four-GPU per package structure. More importantly, any reduction in test time can be offset by higher production volumes, making throughput requirements equally important for testing demand.

A rising test trend that lifts all test interface vendors

Figure 11: Key vendors for each stage of the semiconductor testing industry

	Chip Probing	Final Test	System-Level Test
Testing service providers	TSMC, ASE, KYEC, Amkor, Intel	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu
Testers	Advantest, Teradyne	Advantest, Teradyne	N / A
Handlers (Probers)	TEL, Tokyo Seimitsu, TSK	Advantest, Chroma, Cohu, Hangzhou Changchuan Technology, Hon Precision, JHT, Teradyne	Hon Precision, Chroma
Probe Cards	FormFactor, Technoprobe, MPI, MJC, JEM, WinWay, CHPT	N / A	N / A
Socket	N / A	WinWay, Cohu, Leeno, Yokowo, Yamaichi, Smiths	WinWay
Load board	N / A	Yamaichi, Advantest, CHPT	Yamaichi, Advantest, CHPT

Source: UBS

Below we summarize key technical items of probe cards, sockets, and load boards/PCB used for semiconductor testing:

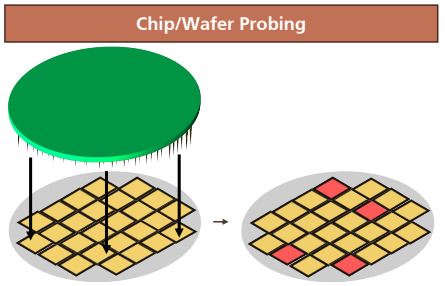
Probe cards and chip probing (CP)

Probe cards are used during chip probing (CP), the wafer-level stage of semiconductor testing that identifies known-good dies (KGD) before packaging. As advanced packaging technologies such as CoWoS, 3D stacking and hybrid bonding become more prevalent, CP is becoming increasingly important because a single defective die or HBM stack can compromise the entire package. This is driving the industry's shift-left testing strategy, with more testing conducted earlier in the manufacturing process to identify defects before costly packaging steps.

Probe cards consist of contact elements (probes) mounted on a PCB and are used together with probers and automated test equipment (ATE). During testing, wafers are placed on the prober chuck while the probe card makes electrical contact with individual dies. The tester then performs electrical, parametric, and functional tests to determine whether each die meets performance requirements. Dies that fail testing are removed from the production flow prior to dicing and packaging.

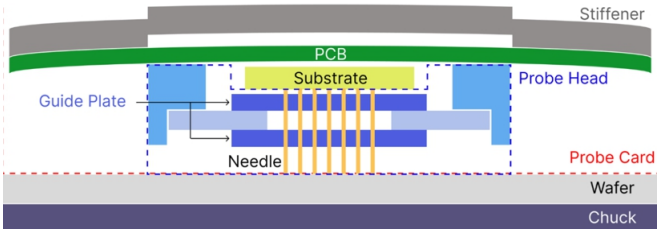
Furthermore, probe cards are highly customised products designed for specific chip architectures and testing platforms. As AI accelerators adopt larger dies, higher pin counts and more advanced packaging technologies, probe card complexity and content value continue to increase.

Figure 12: Chip probing utilise probe cards to find known-good dies (KGD) and eliminate faulty dies



Source: UBS

Figure 13: CP test illustration with probe card structure

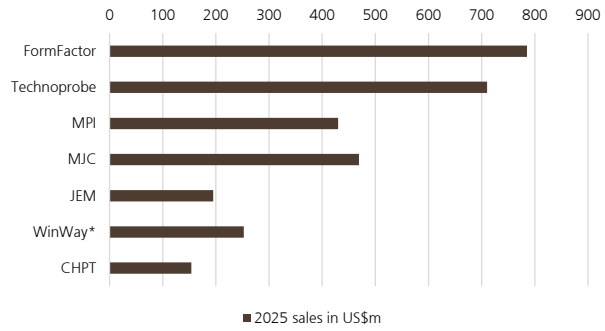


Source: Company data (CHPT)

The global probe card market remains concentrated among a handful of suppliers:

- **FormFactor (FORM.O)**: World's largest probe card company (including both logic and memory probe cards) by revenue as of year end 2025. Key supplier for HBM probe cards.
- **Technoprobe (TPRO.MI)**: World's largest logic probe card company by revenue as of year end 2025. Key supplier for AI accelerators MEMS probe cards, especially for GPU.
- **MJC (6871.T)**: Japanese company, mainly specializing in memory probe cards, 3rd largest probe card company by revenue as of year end 2025.
- **MPI (6223.TWO)**: Traditionally specializes in cantilever probe cards, now expanding its MEMS probe capacity. 4th largest probe card company by revenue as of year end 2025. A probe card supplier of AI ASICs.
- **JEM (6855.T)**: Mainly specialize in memory probe cards, 5th largest probe card company by revenue as of year end 2025.
- **WinWay (6515.TWO)**: One of the fastest-growing logic probe card suppliers. Following its collaboration with Technoprobe, WinWay rose from the 15th-largest probe card supplier in 2024 to the fifth-largest logic probe card supplier in 2025. Probe card revenue increased from 11% of sales in 2024 to 29% in 2025.
- **Chunghwa Precision Test Tech (6510.TW)**: A key supplier for smartphone application processor (AP) probe cards, and has been migrating to HPC related applications in recent years. Although its share within the probe card market is minor, it is a major supplier for probe card PCB, especially for high end probe cards with advanced PCB that has high layer counts (>40 layers, and over 80 layers for AI applications). Its key competitive advantage is its vertical integration as it manufacturers the MEMS probe and PCB in-house.

Figure 14: 2025 sales of major probe card suppliers



Source: Company data, UBS. Note: Only 29% of WinWay's 2025 sales are from probe cards.

Probe cards generally employ either **vertical** or **cantilever** architectures. **Vertical** probe cards use probes arranged perpendicular to the substrate, enabling higher pin density

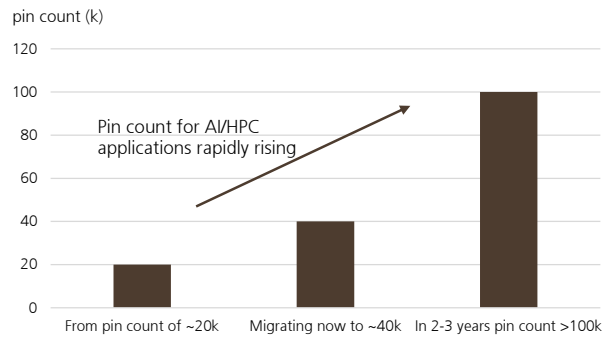
and finer pitch capability. They have become the standard for AI, HPC and other advanced semiconductor applications. **Cantilever** probe cards use angled probes extending from the PCB and are typically used in analog, driver IC and selected memory applications where pin-count and pitch requirements are less demanding. The latest generation of AI accelerators increasingly relies on **MEMS** (Micro-Electro-Mechanical Systems) based vertical probe cards, which support ultra-fine pitches, higher pin counts and improved probing precision. Technoprobe currently leads this segment with pitch capability reaching approximately 40 micrometres.

For leading-edge AI applications, probe cards must increasingly support:

1. Higher current-carrying capability (CCC).
2. Stable contact resistance.
3. Lower parasitic inductance for signal integrity.
4. Greater durability under varying thermal conditions.

For **CHPT**, with their MEMS probe capabilities stemming from their smartphone application processor probe card supply, they are in a good position to capture the growing demand for MEMS probe cards as AI/HPC are shifting to MEMS probes. We also expect probe card content per device to continue rising as AI accelerators grow in die size, pin count, power consumption and packaging complexity. This should drive secular growth for both probe card suppliers and upstream component providers, making chip probing one of the most attractive areas within the semiconductor testing value chain.

Figure 15: Probe card pin count growth trajectory

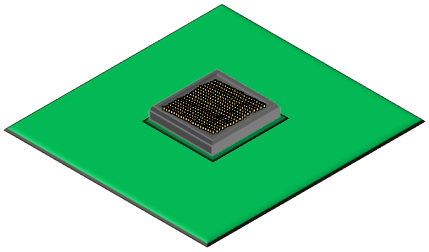


Source: UBS estimates

Sockets

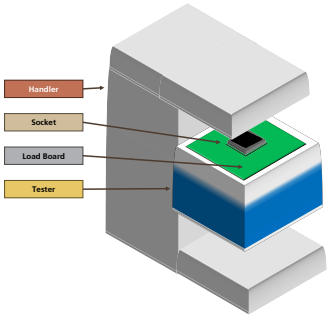
Semiconductor test sockets are electro-mechanical interfaces that are used post chip probing and chip packaging process to test the functionality of the chips. Sockets are placed on to load boards on top of testers during the final test/system level test processes. Handlers place the chips onto the sockets, and pins inside the sockets enable the testers to read the signals from the device under test (DUT, i.e. the packaged chips), with tester signals fanning out from beneath the PCB load board.

Figure 16: Socket illustration



Source: UBS

Figure 17: FT equipment/ consumables



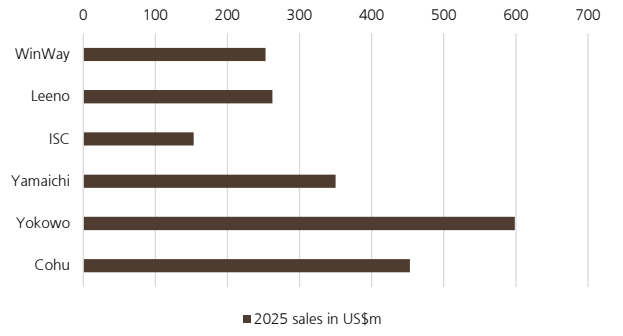
Source: UBS

As AI accelerators become larger and more complex, socket requirements continue to increase. High-end sockets typically comprise a socket housing, alignment plate, integrated thermal-control components and contact elements that connect the DUT to the tester. Spring probes (pogo pins) are the most common choice for HPC sockets, with high durability, individual probe replacement flexibility, and consistent contact force. FT and SLT sockets account for more than 70% of the total socket market, with burn-in sockets representing the remainder.

Compared with the probe card market, the socket industry is more fragmented. Key suppliers include:

- **WinWay (6515.TW)**: A major share gainer in AI/HPC sockets, with strong exposure to Nvidia and AMD. The company specializes in high-performance coaxial sockets and is developing its next-generation HyperSocket platform. It is also expanding into probe cards through its partnership with Technoprobe.
- **Leeno (058470.KQ)**: Vertically integrated supplier focused on smartphone AP and memory sockets.
- **ISC (095340.KQ)**: Leading memory socket supplier with elastomer expertise.
- **Smiths Interconnect (Acquired by Molex)**: A key player in AI/HPC socket industry, with its DaVinci coaxial sockets able to support high frequency connectivity testing.
- **Yamaichi (6941.T)**: Leading Japanese socket supplier with strong automotive exposure.
- **Yokowo (6800.T)**: Major provider of contact probes and socket pins.
- **Cohu (COHU.O)**: Broad test-solutions provider spanning ATE, handlers and sockets, with greater auto/industrial/consumer exposure.

Figure 18: 2025 sales of major socket suppliers



Source: Company data, UBS. Note: Most socket suppliers are companies with multiple business segments, i.e. Yokowo and Cohu.

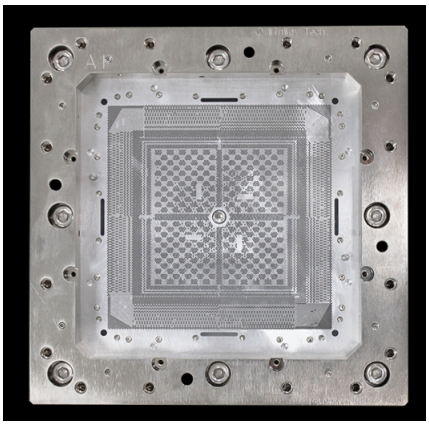
AI accelerators are pushing socket requirements beyond 10,000 pins, making **coaxial sockets** the industry standard for leading-edge AI/HPC devices. These sockets must support transmission speeds of up to 224G while handling significantly higher thermal design power (TDP) during testing.

Elastomer technology is increasingly being adopted alongside spring-probe (pogo-pin) architectures to reduce chip damage and improve contact reliability. WinWay's next-generation **HyperSocket** integrates elastomer materials between the DUT and spring probes, improving probe lifetime, reducing cleaning requirements and enhancing production efficiency. We believe HyperSocket could become an important differentiator as next-generation AI accelerators demand higher pin counts, bandwidth and thermal performance. We estimate HyperSocket could contribute approximately 5% of WinWay's revenue in 2027E while carrying a 20-30% ASP premium versus conventional coaxial sockets.

The transition toward larger AI packages, higher pin counts and faster data-transfer speeds should continue to increase socket content per device. We expect this trend to support long-term growth in the socket market, with technology leaders such as WinWay well positioned to gain share through next-generation solutions including

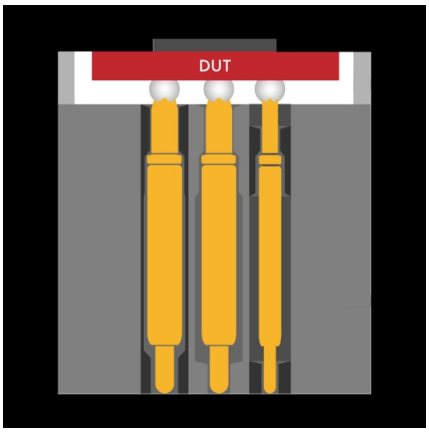
HyperSocket.

Figure 19: Coaxial socket



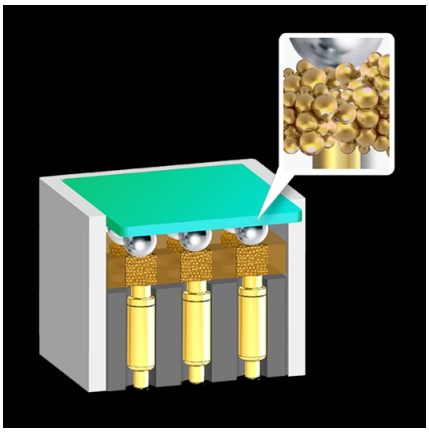
Source: Company data (WinWay)

Figure 20: DUT socket contact



Source: Company data (WinWay)

Figure 21: HyperSocket



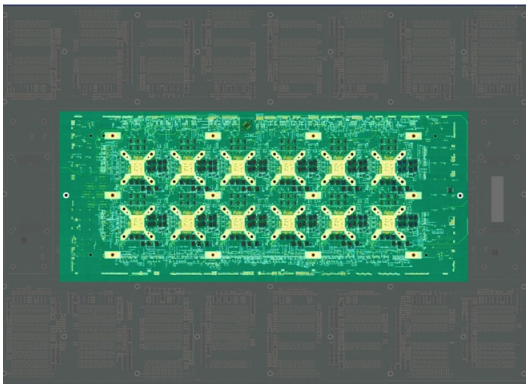
Source: Company data (WinWay)

Load Board/Interface board

Load boards are specialised printed circuit boards (PCBs) that connect the tester to the device under test (DUT) during final test (FT) and system-level test (SLT). Acting as the electrical interface between the packaged chip and automated test equipment (ATE), load boards are responsible for power delivery and high-speed signal transmission throughout the testing process. As AI and HPC devices become increasingly complex, load board requirements continue to rise. **Multilayer construction** is crucial as modern load boards often require more than 50 layers for smartphone application processors and over 80 layers for leading-edge AI accelerators to support higher pin counts, dense signal routing and greater power delivery demands. At the same time, **impedance control** is an important factor as maintaining signal integrity is becoming increasingly important as interconnect bandwidth rises and high-speed SerDes blocks become more prevalent. These trends are driving greater technology content and value per load board.

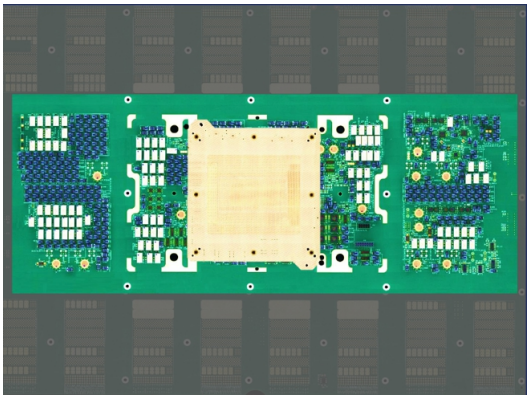
We expect load board content to increase alongside AI chip complexity, larger package sizes and higher interface speeds, benefiting suppliers with advanced multilayer PCB capabilities such as CHPT.

Figure 22: CHPT's AP load board



Source: Company data (CHPT)

Figure 23: CHPT's AI load board



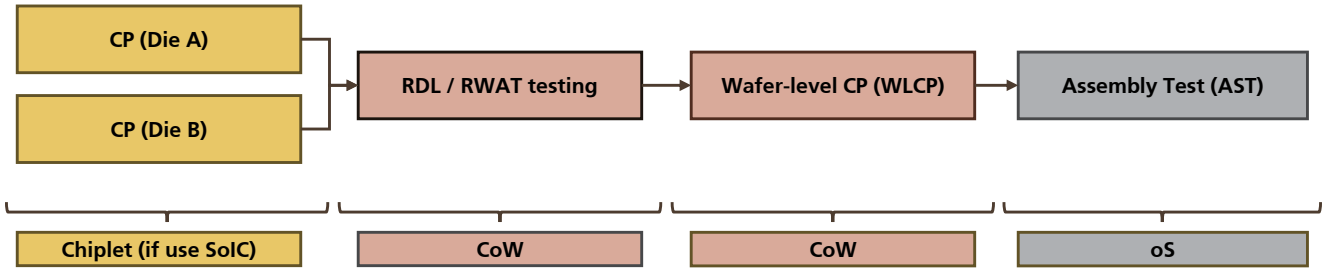
Source: Company data (CHPT)

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging. This

underpins our above-consensus forecasts for **Chunghwa Precision Test (CHPT)**, with 2027E/28E sales 7%/15% above consensus driven by structural AI/HPC probe card and test consumables demand, capacity expansion, and increasing test interface content as test insertions and test time increase. **WinWay** should also benefit through its growing probe card business alongside partner Technoprobe, with probe card sales expected to grow 94%/72% YoY in 2026E/27E following 161%/260% growth in 2024/25.

Figure 24: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

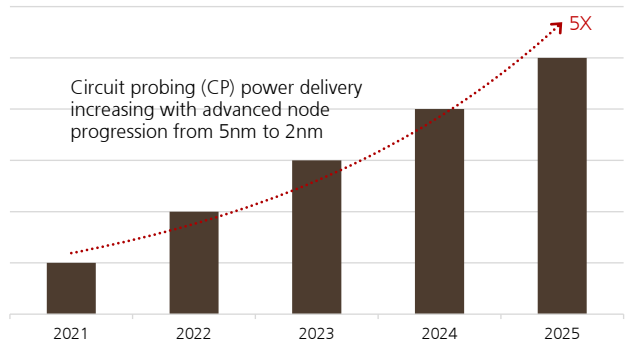


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

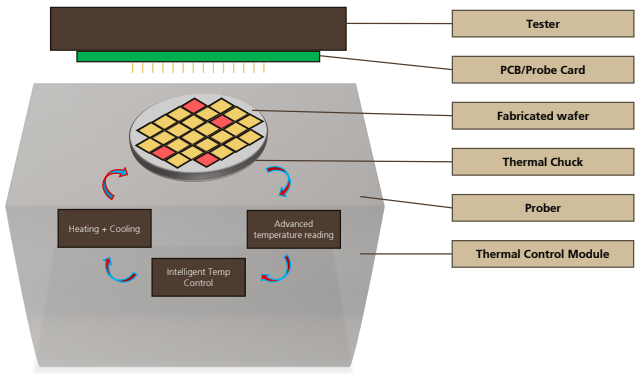
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal control solutions.

Figure 25: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 26: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

Hermes Testing (HTSI) is a key beneficiary of this trend. Leveraging its long-standing relationship with TSMC and expertise in semiconductor test solutions, HTSI supplies advanced thermal control modules used alongside leading-edge probers (made by Tokyo Electron), particularly for AI applications. These thermal control modules are

required for high TDP semiconductor wafer probe in order to maintain the temperature of the chuck to avoid needle burns or probe card damages. We estimate HTSI will ship approximately 3,000 thermal-management modules in 2027E, up from 1,200 in 2026E, supporting sales growth of 127%/152% in 2026E/27E. HTSI is also the partner for prober installation, prober maintenance and services. In addition, HTSI provides high-end chuck processing services, an area that should benefit from increasing package sizes and higher wafer warpage associated with advanced packaging.

Beyond thermal control modules, HTSI also supplies consumables and customised solutions for wafer-level testing, including probe card cleaning sheets and laser-based probe head cleaning systems. As probing intensity increases under the shift-left testing model, demand for cleaning and maintenance solutions should rise alongside probe card utilisation. Currently, cleaning sheets is around 5% of sales, but we believe it could gain traction in 2027/28E and beyond. We see additional upside from future modular laser cleaning systems that could be integrated directly with probers, further expanding HTSI's content opportunity within advanced wafer-level testing.

Rapid capacity expansions accelerating growth

Industry test interface suppliers are expanding capacity aggressively to meet rising AI and HPC demand. As advanced semiconductors require higher-performance probe cards, sockets and interface boards, capacity additions are becoming increasingly important to support growth across both wafer-level and package-level testing.

We believe **CHPT** and **WinWay** are well positioned to gain share within the high-end AI/HPC test interface market. Taiwanese suppliers benefit from their proximity to leading fabless customers and the growing preference for localised supply chains, particularly for advanced testing applications where responsiveness and engineering support are critical. Against a backdrop of tight industry supply, both companies have announced significant capacity expansion plans over the next two to three years.

CHPT: capacity expansion supporting AI/HPC share gains

CHPT's existing capacity is effectively fully utilised and the company is pursuing both near-term and long-term expansion. Near-term growth will come from leased facilities, with one production expansion scheduled for August 2026 and a second in March 2027, doubling and tripling year-end 2025 capacity, respectively. Longer term, CHPT's new Fab 3 facility in Pingzhen will significantly expand in-house PCB manufacturing and wet-processing capacity, with mass production expected around Q428, expected to double existing capacity when fully ramped.

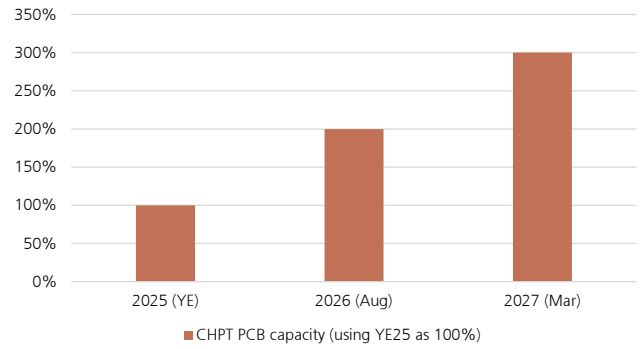
Combined with increasing exposure to higher-value AI/HPC products, including turnkey CPU probe cards and TPU FT/SLT/burn-in boards, we expect CHPT to deliver revenue growth of 49%/75%/43% in 2026E/27E/28E.

WinWay: scaling for the next phase of AI demand

WinWay's capacity is expected to increase from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

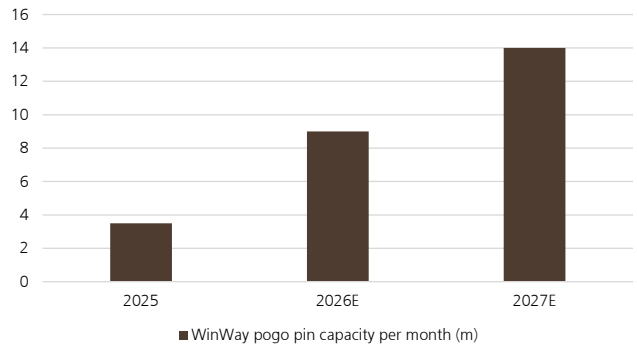
We estimate these capacity additions will support revenue growth of 105%/73%/71% in 2026E/27E/28E.

Figure 27: Announced capacity expansions for CHPT



Source: Company data, UBS

Figure 28: Announced capacity expansions for WinWay



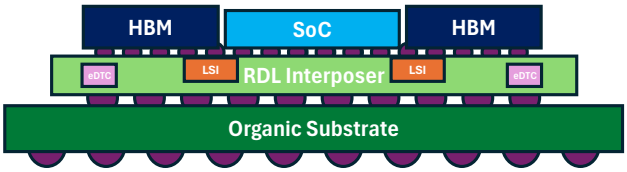
Source: Company data, UBS

Advanced packaging growth remains the key demand driver

Capacity expansion by test interface vendors is ultimately supported by continued investment throughout the advanced packaging ecosystem, from providers such as TSMC, ASE, Intel, and Amkor. During the Q226 earnings calls, TSMC raised its recent 2026 capex from US$52-56bn to US$60-64bn and ASE raised its 2026 capex from US $8.5bn to US$10.5bn, with both companies citing robust demand supporting the capex raise. We believe capex for the next two years will be even higher than 2026, implying continuous growth in future years. With advanced packaging technologies getting significantly more complex and with larger form factor, illustrated by the upcoming ramp of both TSMC's CoPoS (chip-on-panel-on-substrate) and Intel's EMIB-T (embedded multi-die interconnect bridge) packaging technologies, test interface solution providers will benefit with more value added consumables (see our CoPoS vs EMIB-T deep dive).

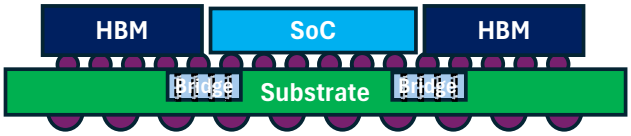
We expect test interface content growth to outpace unit growth. For instance, test socket's ASP directly correlates with its pin counts - with the addition of 20-30% more pins inside the sockets, socket ASP can grow 20-30% or more as the socket might require significant cooling capabilities as the TDP of the DUT rises with its larger form factor. Unlike prior semiconductor cycles, much of the current demand is driven by rising test interface content per device rather than unit growth alone.

Figure 29: TSMC's CoPoS



Source: UBS

Figure 30: Intel's EMIB-T



Source: UBS

Accelerators still in focus with TPU as another significant growth driver

Nvidia has been the primary driver of testing content growth over the past several years, benefiting the entire semiconductor testing supply chain. We expect this trend to continue as GPU architectures evolve from Blackwell to Rubin and then to Feynman, with increasing process node complexity, larger package sizes and more advanced packaging technologies driving higher testing value per device.

Nvidia should remain a major contributor to both CHPT and WinWay, with GPUs continuing to serve as their core growth engine. Both companies are positioned to gain share with Rubin. We estimate WinWay's socket share could improve from a 50/50 split with Smiths Interconnect in prior generations to 60/40 with Rubin, while CHPT is gaining

share in test interface PCBs and is positioned to supply around 70% of Rubin probe card and final test PCB content. Nvidia currently accounts for approximately 50% of WinWay's sales and 30% of CHPT's sales. Both Nvidia GPUs and Google TPUs increase testing demand through higher content per device rather than unit growth alone.

Beyond Nvidia, Google's TPU roadmap represents a significant incremental growth opportunity. Following the successful ramp of TPU v8t with MediaTek as design partner, **CHPT** is positioned to supply probe card PCBs, final test load boards, system-level test boards and burn-in boards. For v9, although the final test interface vendors are yet to be decided, we believe as the current supplier, CHPT will still be supplying PCB for final test, system level test, and burn-in, with potential to continue to supply PCB for probe cards. **WinWay** should also benefit as the socket supplier for both TPU v8t and the next-generation TPU v9 platform. We believe MediaTek's increasing share gain for Google TPUs will also benefit the local Taiwanese test interface supply chain.

We view TPU v9 as a particularly attractive opportunity given its large four-die package architecture based on Intel's EMIB-T packaging technology, which should drive higher testing content. We estimate production of 4mn TPU v8t units in 2027E, followed by 2.3mn TPU v9 packages (9.2mn dies) and an additional 2mn TPU v8t units in 2028E. Importantly, while EMIB-T packaging is expected to be performed in Malaysia, we believe the majority of advanced testing value should remain within the Taiwanese testing ecosystem. Leading testing providers, including ASE and KYEC, maintain substantial testing operations across Taiwan and Southeast Asia, supporting continued participation by Taiwanese test interface suppliers. See our latest MediaTek report.

CPU business as a supplemental catalyst for test interface in an agentic world

We believe semiconductor test interfaces are an underappreciated beneficiary of the emerging agentic AI ecosystem. UBS estimates the CPU market will expand to US $170bn by 2030E from US$31bn in 2025 and US$71bn in 2027E, driven by rising deployment of AI infrastructure. As CPU-to-GPU ratios increasingly approach 1:1 in AI clusters, demand for testing interfaces should expand alongside CPU volumes. TSMC is the main CP test provider in the industry, with probe card suppliers aiding its massive demand by supplying probes, such as TSMC's GPU probe cards supported by Technoprobe's probes. However, with capacity now becoming scarce with strong front and back-end demand, TSMC is outsourcing its CP business to ASE and KYEC, further broadening the testing supply chain. This is where new probe card vendors come in—smaller probe card players in the past, such as **WinWay** and **CHPT**, now have the opportunity to gain significant, non-GPU share from TSMC, as they have MEMS probe capabilities (WinWay's MEMS probes are from its strategic partnership with Technoprobe) and has a long established relationship with the fabless companies such as Nvidia via supplying non-probe card consumables such as sockets and PCB. Of which, CPU projects have become the main area for CHPT and WinWay to expand their footprint.

We expect both companies to benefit from the ramp of Nvidia's ARM-based Vera CPU and Google's ARM-based Axion CPU in 2026E-27E. WinWay is positioned to supply sockets and probe cards for Vera and sockets for Axion, while CHPT should provide probe cards and test interface PCBs for both Vera and Axion, with particularly strong exposure to Axion. Additional upside could come from AMD's Venice CPU, where WinWay is well positioned and CHPT could gain indirect exposure through testing-equipment procurement channels. We estimate CPU-related revenue will represent low to mid-teens % of CHPT sales in 2026E, increasing to 20-30% in 2027E as the company expands its turnkey probe card solutions for Axion. For WinWay, CPU-related demand could account for approximately 30% of sales in 2026E-27E. Importantly, CPU programmes diversify exposure beyond AI GPUs and provide a second structural growth driver for test interface suppliers.

CPO testing fuelling future growth for semiconductor testing

As interconnect bottlenecks increasingly shift from the memory wall to the copper wall, we believe co-packaged optics (CPO) adoption is inevitable over the longer term. Testing remains one of the key bottlenecks to CPO deployment, as optical validation differs

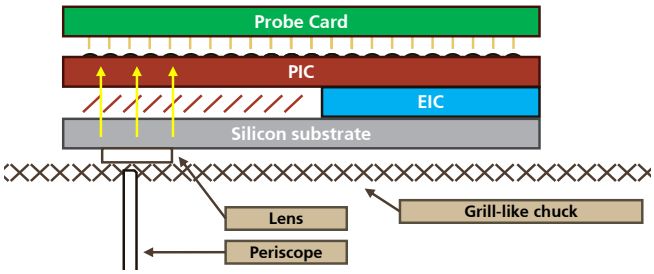
fundamentally from the electrical testing processes that have dominated semiconductor manufacturing for decades. As a result, CPO is creating a new testing opportunity across the semiconductor equipment and consumables ecosystem. We have written extensively on CPO and the opportunities that it brings to the semiconductor supply chain, see our CPO deep dive, our thoughts on CPO timing, 2026 SiPhIA conference takeaways, and SEMICON 2026 takeaways.

Various test equipment suppliers are scrambling to come up with ideal testing equipment across the four insertions of CPO testing, with solutions to ensure both optical and electrical signal integrity of the CPO solutions. The four insertions are:

- **Insertion 1: Electronic IC (EIC)/Photonic IC (PIC) test.** Single-side wafer test that has dedicated probe cards to test the fabricated EIC/PIC wafers.
- **Insertion 2: Optical/Electrical test.** Double sided wafer test, testing the hybrid bonded (e.g. via TSMC's COUPE) EIC and PIC wafers by an opto-electrical test that verifies the electrical signals on the top and optical signal on the bottom of the hybrid bonded wafers.
- **Insertion 3: Optical engine (OE) test.** The OE tests are done to ensure the quality of the OEs and with FAU (Fiber Array Unit) attached, to ensure the signal integrity of the OEs. The testing equipment needs to be able to handle 224G per channel data rates, requiring high unit per hour (UPH) throughput, meticulous dirt mitigation, and maintaining a near 100% contact rating on high-speed sockets.
- **Insertion 4: ASIC final test.** This final test step is established post the OEs are packaged together with the ASIC. The complete ASIC package (now with CPO) is now tested for opto-electrical functionalities.

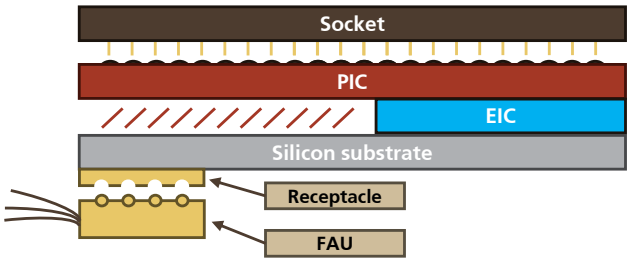
As it stands, insertion 2 and insertion 3 are the two areas where there are the most debates in the industry. Insertion 3 suppliers such as Chroma have noted the trend of "shift-to-middle" paradigm centered on insertion 3. In high-density scale-out switches deploying up to 32 optical engines (e.g. Nvidia's Spectrum-X architectures), any OE defect results in package scrap at the CoWoS level, making 100% known-good die (KGD) validation at the singulated OE stage an absolute commercial requirement.

Figure 31: An illustration for insertion 2 test method



Source: UBS

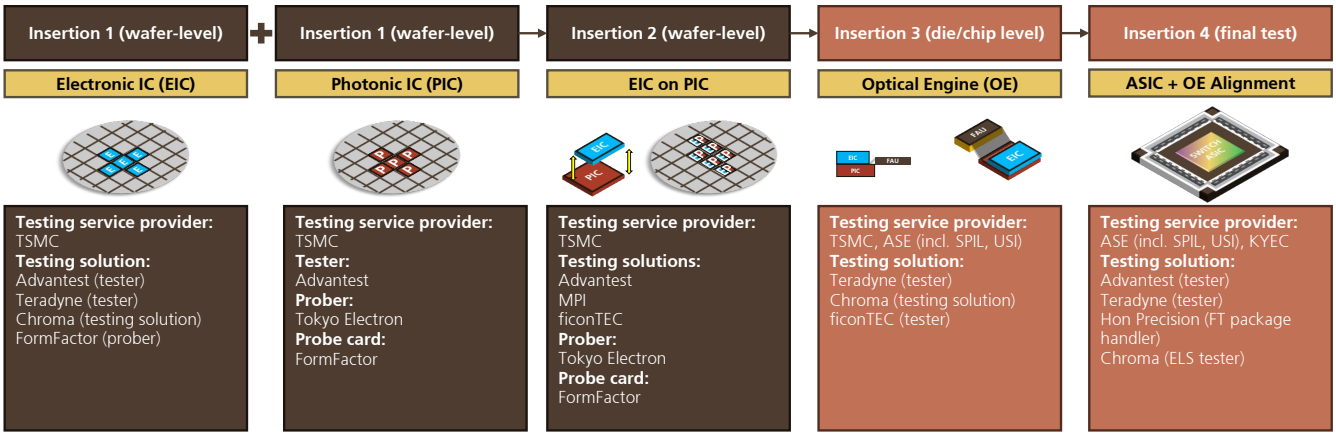
Figure 32: An illustration for insertion 3 test method



Source: UBS

As shown in the below figure, as there are no set standards for the testing equipment, throughput and stability are now the main targets and areas of competition for the major players in the field. Notable ones for each insertions are listed.

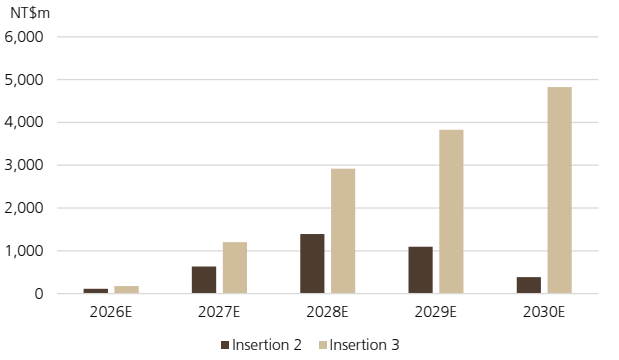
Figure 33: CPO testing flow chart with key suppliers



Source: UBS

With its 25 years of precision test equipment expertise, ficonTEC is an early mover for CPO test equipment and has key know-how for optical alignment, which is quintessential for optical test. **Hermes Testing**, through its partnership with ficonTEC, is set to be a major beneficiary once CPO proliferates in 2027E-28E. As the partner of ficonTEC, Hermes Testing is the key enabler that can help ficonTEC achieve HVM for its CPO testing equipment. As of now, our view is that both insertion 2 and insertion 3 should be required within the next two years, despite the difficulties surrounding double-sided wafer probing (insertion 2), as the yield for OEs is still not high enough to avoid defective dies. Finding out known-good OEs will be crucial before the FAU coupling process during the latter stages of CPO packaging, to prevent impeding the final package with multiple OEs attached to the switch ASIC (e.g. 32 OEs on Nvidia's Spectrum-X). We estimate Hermes Testing will ship 40/70 units of insertion 2/3 testing equipment in 2027E, with upside to these numbers as we view Hermes Testing will likely assist with the ramp up for Gen 2 insertion 3 equipment in 2027E. Gen 2 insertion equipment will come with better throughput and ASP.

Figure 34: HTSI CPO equipment sales to pick up in 2027/28E with 541%/135% YoY growth



Source: UBS estimates

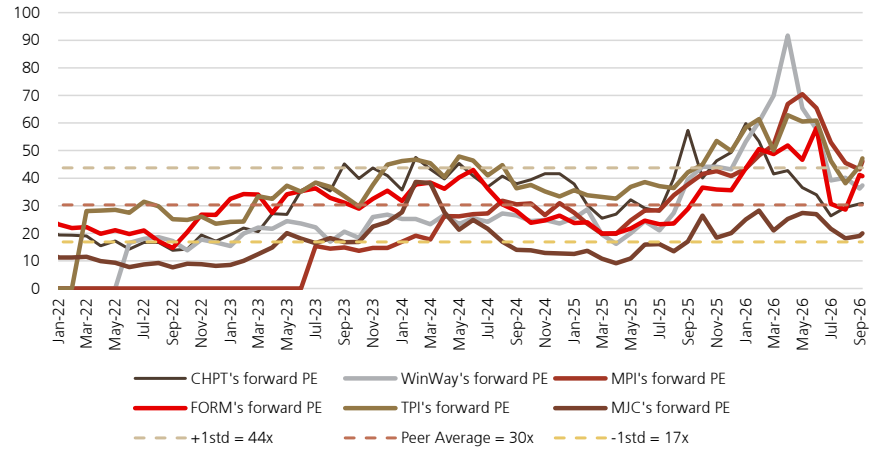
Beyond insertions 2 and 3, HTSI is also positioned to participate in insertion 1 testing through customised prober solutions for photonic integrated circuit (PIC) testing. We believe these solutions could eventually be packaged as licensable intellectual property, creating an additional long-term growth opportunity as CPO adoption broadens. While near-term testing growth is driven by AI accelerators, CPO represents the next major expansion cycle for semiconductor testing. We believe HTSI is uniquely positioned to participate across multiple test insertions, providing exposure to a potentially significant source of incremental revenue beginning in 2027E.

Valuation comparison

We set up three peer groups for valuation comparison: probe card, sockets, and equipment vendors.

For probe card vendors, the peer group's average NTM PE post 2022 is around 30x, and the group is currently trading at around 39x NTM PE. We value CHPT using 31x average 2027-28E PE, in-line with its historical average of 31x and underpinned by a 31% 2027-31E earnings CAGR.

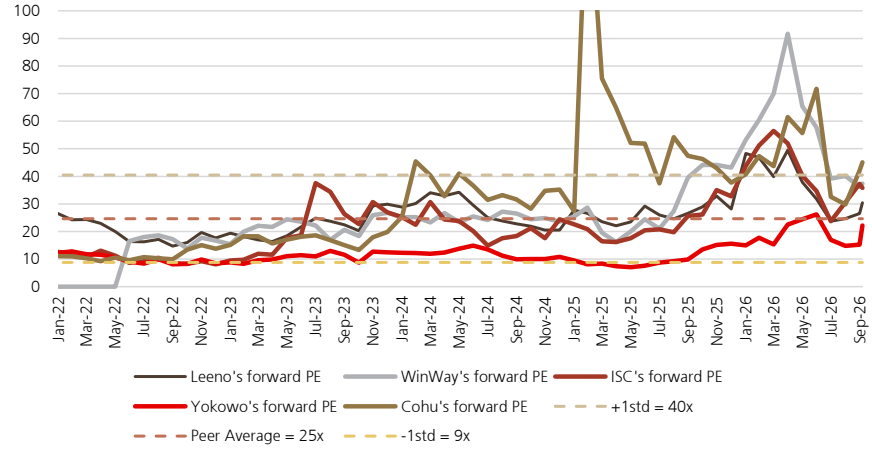
Figure 35: Key probe card vendors' 12-month forward PE band (x)



Source: LSEG, UBS

For socket vendors, the peer group's average NTM PE post 2022 is around 25x, and the group is currently trading at around 39x NTM PE. We value WinWay using 30x average 2027-28E PE, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR.

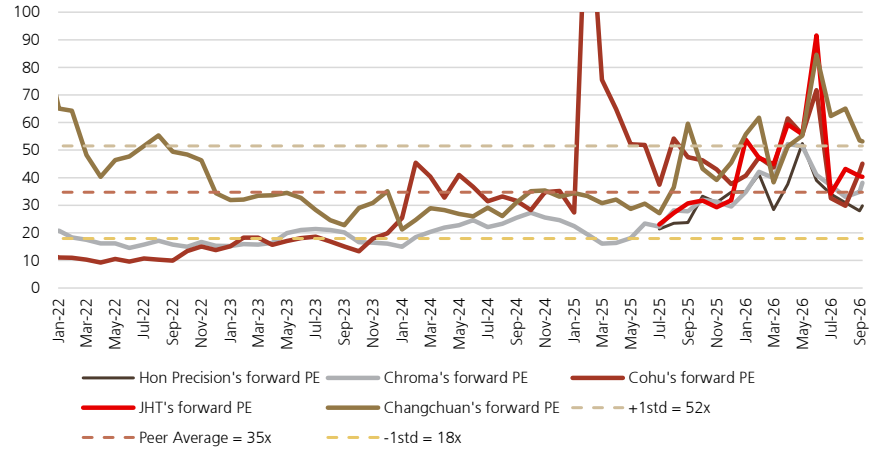
Figure 36: Key socket vendors' 12-month forward PE band (x)



Source: LSEG, UBS

For equipment vendors, the peer group's average NTM PE post 2022 is around 35x, and the group is currently trading at around 41x NTM PE. We value HTSI using 31x average 2028E PE, underpinned by a 32% 2027-31E earnings CAGR. We use 2028 PE as our view is that 2028 is the year thermal control modules and CPO adoption grow from the initial ramping up stage to mass adoption stage. We believe that using handlers as a peer group is justifiable for a unique company like HTSI, with thermal control modules for chip probing sharing similar characteristics to active thermal control used for final test, in addition to Hon Precision and Chroma having exposure to CPO test equipment.

Figure 37: Key equipment vendors' 12-month forward PE band (x)



Source: LSEG, UBS

Figure 38: Testing supply chain valuation table

Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

Valuation Method and Risk Statement

Tech investing involves risk. It is difficult for the investment community, UBS included, to project the financial results of tech companies, as their operating models are highly volatile and unpredictable, and they compete in a highly dynamic marketplace. In addition to their low predictability, valuing tech stocks can be challenging because neither traditional nor non-traditional valuation measures have provided much insight into how tech stocks trade.

We value Chroma ATE using a PE methodology.
Downside risks include weaker-than-expected demand for high-performance computing dampening its semiconductor testing business and slower-than-expected recovery in EV investments into battery capacity installations.

We value Chunghwa Precision Test, Hermes Testing and WinWay based on PE methodology.
Risks: Faster- or slower-than-expected front-end/CoWoS capacity expansion could result in CHPT, HTSI and WinWay gaining/losing probe card/board orders from major foundries and OSATs. Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in these companies losing market share from end customers, major foundries, and OSATs. Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of product orders from customers, major foundries, and OSATs to CHPT, HTSI and WinWay, affecting long-term revenue growth potential. A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and even CPO, would limit future growth potential. Rising raw material costs may directly impact their margins for equipment shipments.

We value Hon Precision based on PE multiple.
Faster- or slower-than-expected CoWoS capacity expansion could indirectly result in Hon Precision gaining or losing equipment orders from major customers; the risk of competition from either local or foreign companies achieving technological advancements could result in Hon Precision losing market share from its major customers; better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from customers to Hon Precision, affecting long-term revenue growth potential; a potential slowdown or delay in the rollout of technological advancements for technologies, such as MCL and CPO, would limit future equipment orders; rising raw material costs may directly impact Hon Precision's margins for equipment shipments.

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Chroma ATE[28]	2360.TW	Buy	NT$2,400	07 Oct 2026
Chunghwa Precision Test Tech	6510.TWO	No Rating	NT$3,055	06 Oct 2026
Hermes Testing Solutions	7856.TWO	No Rating	NT$4,190	06 Oct 2026
Hon.Precision, Inc.	7769.TW	Buy	NT$5,885	07 Oct 2026
WinWay Technology	6515.TW	No Rating	NT$5,520	07 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
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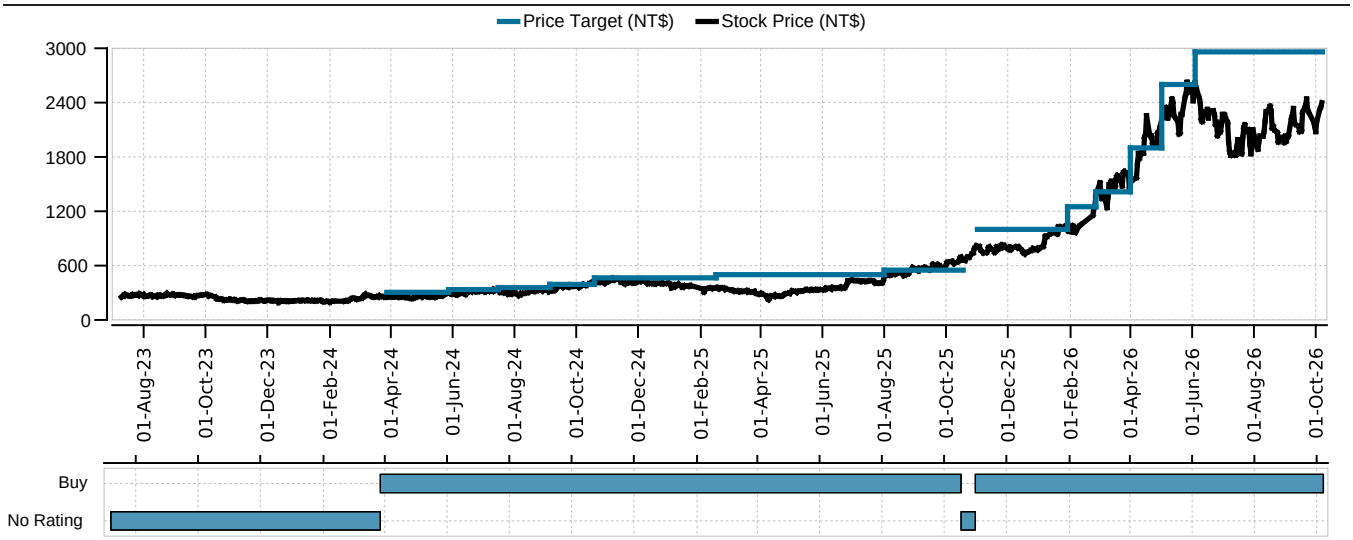
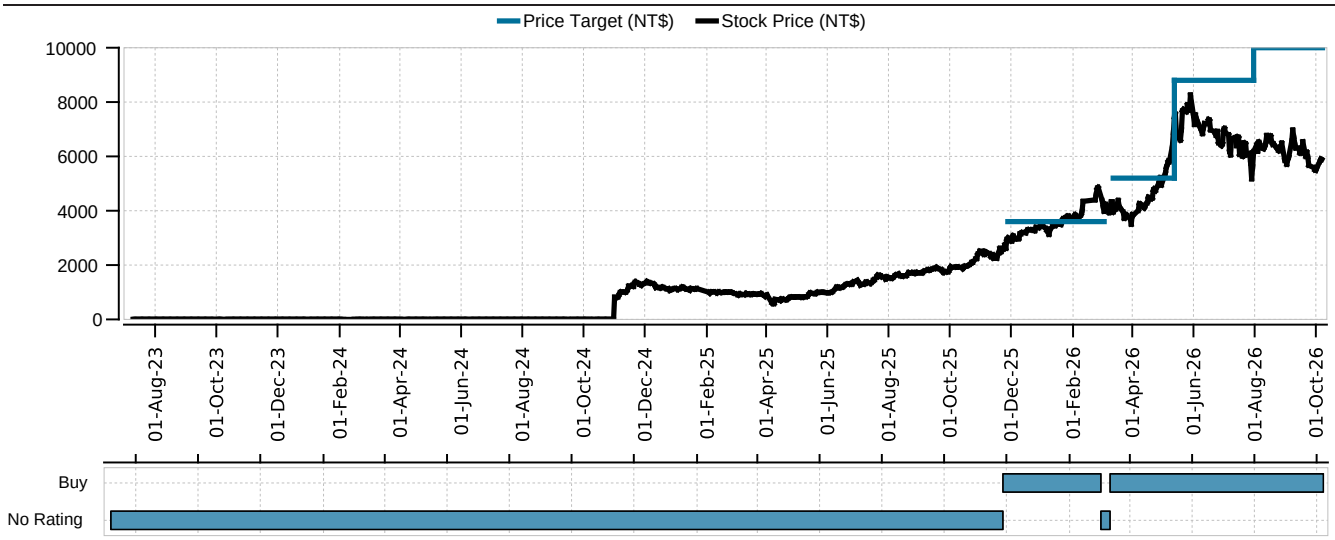
Chroma ATE (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	256.50	-	No Rating
2024-03-27	256.00	305.00	Buy
2024-05-27	293.50	335.00	Buy
2024-07-15	335.00	358.00	Buy
2024-09-04	311.00	393.00	Buy
2024-10-18	416.00	465.00	Buy
2025-02-15	350.00	500.00	Buy
2025-07-31	430.50	550.00	Buy
2025-10-17	662.00	-	No Rating
2025-10-31	819.00	1000.00	Buy
2026-01-28	1035.00	1250.00	Buy
2026-02-25	1255.00	1415.00	Buy
2026-03-31	1465.00	1900.00	Buy
2026-05-01	2120.00	2600.00	Buy
2026-06-03	2545.00	2960.00	Buy

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Hon.Precision, Inc. (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	0.00	-	No Rating
2025-11-27	2940.00	3600.00	Buy
2026-03-03	4150.00	-	No Rating
2026-03-12	4325.00	5200.00	Buy
2026-05-12	6870.00	8800.00	Buy
2026-07-30	5620.00	10000.00	Buy

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: Hangzhou Changchuan Technology, Rmb239.79 (30 Sep 2026); ASE Industrial, NT$744.00 (06 Oct 2026); Technoprobe SpA, €36.84 (06 Oct 2026); Alphabet Inc. - Class A, US$347.68 (06 Oct 2026); King Yuan Electronics, NT$298.00 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (06 Oct 2026); Tokyo Electron, ¥12805 (06 Oct 2026); MediaTek Inc., NT $4920.00 (06 Oct 2026); NVIDIA Corp, US$239.24 (06 Oct 2026); Teradyne Inc, US$430.32 (06 Oct 2026); Advantest, ¥41850 (06 Oct 2026); Advanced Micro Devices Inc, US$649.42 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Initiation of Coverage

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Final point of contact: securing AI yield through advanced testing socket

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
	Prior : No Rating
12m price target	**NT$8,900.00**
	Prior :
Price (06 Oct 2026)	**NT$6,130**

RIC: 6515.TW **BBG:** 6515 TT

Trading data and key metrics

52-wk range	NT$10,795.00-2,125.00
Market cap.	NT$221b/US$6.95b
Shares o/s	36.0m (ORD)
Free float	69%
Avg. daily volume ('000)	496
Avg. daily value (m)	NT$3,249.8
Common s/h equity (12/26E)	NT$10.0b
P/BV (12/26E)	22.0x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	-	99.30	-	97.71
12/27E	-	196.30	-	187.07
12/28E	-	383.04	-	348.32

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Solid growth ahead with robust capacity expansions

We initiate coverage of AI/HPC test socket leader WinWay with a Buy rating and NT $8,900 price target. WinWay's share price has corrected 43% from its April 2026 peak due to broader market de-rating and margin pressure associated with its Technoprobe partnership and capacity expansion. We believe the recent correction presents an opportunity to re-engage with WinWay's long-term growth trajectory, supported by three key drivers: 1) increasing content from AI GPUs, CPUs, and TPUs, 2) substantial capacity expansion, and 3) broader adoption of system-level testing (SLT). Together, these drivers should support revenue growth of 105%/74%/71% in 2026E/27E/28E. Importantly, WinWay's growth is increasingly being driven by higher socket content per device rather than semiconductor unit growth alone and the stock currently trades at 16x of our 2028E P/E vs historical average of 30x, potential to re-rate with GM recovery in the coming quarters. See our companion testing deep dive sector note.

Rising yields and utilisation to offset probe card dilution

Gross margin declined to 38.3% in Q226 from 45.3% in FY25, pressured by outsourced MEMS probes and expansion-related costs. While probe-card margins are unlikely to return to H125 levels of 48.9% given the Technoprobe partnership structure, we expect margin pressure to ease as utilization and yields improve, supporting GM recovery to 43.1% in 2027E and 44.7% in 2028E.

CPU and TPU as fundamental growth drivers

Beyond its core GPU socket franchise, WinWay is well positioned to benefit from the expansion of AI-oriented CPU and TPU platforms. We expect the company to supply sockets for AMD's Venice CPU, Nvidia's Vera CPU, and Google's TPU v8t and v9, and also Vera CPU probe cards, providing an additional growth engine beyond GPUs. WinWay is well positioned in this segment due to its historical collaboration with end customers and its proximity to the local Taiwanese test supply chain. We view AI CPUs and TPUs as an increasingly important contributor to test interface demand as compute architectures diversify in the agentic AI era, driving WinWay's socket and probe card sales growth of 132%/75%/69% and 94%/72%/78% in 2026E/27E/28E, respectively. We estimate CPU/TPU to contribute around 30%/20% of WinWay's 2027E sales.

Valuation: Initiate with a Buy rating, with price target of NT$8,900

We forecast EPS of NT$101.8 in 2026E, rising to NT$201.4/393.0 in 2027E/28E. Our NT $8,900 price target is based on 30x average 2027E-28E P/E, in line with the company's historical average. We believe this valuation is supported by WinWay's exposure to structural growth in AI/HPC testing, expanding socket content, and our long-term earnings CAGR estimate of 30%. We initiate with a Buy rating.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	3,682	5,798	7,857	16,084	27,952	47,831	53,739	60,457
EBIT (UBS)	557	1,368	2,070	4,428	8,986	17,560	19,699	22,336
Net earnings (UBS)	464	1,186	1,673	3,656	7,241	14,130	15,870	18,029
EPS (UBS, diluted) (NT$)	13.34	33.02	46.71	99.30	196.30	383.04	430.19	488.73
DPS (net) (NT$)	10.87	25.75	25.07	61.10	120.85	235.82	264.85	300.89
Net (debt) / cash	302	977	2,594	1,155	2,200	7,340	10,417	14,723
Profitability/valuation	**12/23**	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	15.1	23.6	26.3	27.5	32.1	36.7	36.7	36.9
ROIC (EBIT) %	20.7	35.3	49.2	70.5	83.8	116.7	100.9	93.9
EV/EBITDA (UBS core) x	33.4	21.5	22.9	46.1	23.0	11.7	10.1	8.6
P/E (UBS, diluted) x	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
Equity FCF (UBS) yield %	(1.1)	2.1	2.8	0.4	1.5	4.3	5.2	6.3
Dividend yield (net) %	1.6	2.6	1.6	1.0	2.0	3.8	4.3	4.9

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,130.00 on 06-Oct-2026

This report has been prepared by UBS Securities Pte. Ltd., Taipei Branch. **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 24.**

UBS Research THESIS MAP a guide to our thinking and what´s where in this report

PIVOTAL QUESTIONS

Q: Will capacity constraints limit WinWay's growth in the next two to three years?
Near term, yes. Beyond 2026E, no. WinWay's current capacity remains constrained, but ongoing expansions should significantly increase output over the next two years. With new capacity coming online and the company shifting toward higher-value AI/HPC applications, including CPU probe cards, >10k-pin coaxial sockets and next-generation HyperSockets, we forecast revenue growth of 105%/74%/71% in 2026E/27E/28E, supported by capacity additions, rising AI/HPC content and expanding socket demand. We also expect operating leverage to improve as capacity scales and utilization increases, with OPM rising from 27.5% in 2026E to 32.1%/36.7% in 2027E/28E.

Q: Will WinWay's GM recover from Q226's 38.3%?
Yes. We view recent margin pressure as largely temporary. GM has been impacted by MEMS probe sourcing through Technoprobe and start-up costs associated with capacity expansion. Although margin dilution from probe card business is unlikely to unravel due to the MEMS partnership terms with Technoprobe, as new facilities ramp and yields improve, we expect profitability to recover gradually from the Q226 trough, with gross margin improving to 43.1% in 2027E and 44.7% in 2028E.

UBSVIEW

We initiate coverage of WinWay with a Buy rating and NT$8,900 price target. We believe the market is overly focused on near-term margin pressure and underappreciating WinWay's long-term growth outlook. Key drivers include 1) expanding content from AI GPUs, CPUs, and TPUs, 2) substantial capacity expansion, and 3) growing adoption of system-level testing (SLT). Together, these drivers should support revenue growth of 105%/74%/71% in 2026E/27E/28E.

EVIDENCE

TSMC and OSATs have been expanding their CoWoS capacities aggressively, from 40kwpm in 2024 to around 90 kwpm in 2025, and is tracking to expand CoWoS capacity toward 160kwpm in 2026E. TSMC also ramped up expanding N3/N2 capacity, with multiple product launches from customers' pipelines—Nvidia's Rubin, AMD's MI400 series/Venice CPU and other cloud ASICs—translating into foreseeable, strong testing demand as the majority are WinWay's customers. AI accelerator test times have increased meaningfully for new platforms. We expect continued front-end and advanced packaging investments to drive higher testing intensity and support long-term demand for advanced sockets and test interfaces.

WHAT´S PRICED IN?

WinWay's share price has corrected 43% from its April 2026 peak and currently trades at 16x of our 2028E P/E vs historical average of 30x. Our reverse DCF analysis indicates that the current market price implies only a 19% long-term sales CAGR vs our estimate of 25%, alongside long-term GM level of 32% vs our estimate of 45%. We see scope for re-rating.

UPSIDE/DOWNSIDE SPECTRUM

Value drivers (2027/28E)	Test socket sales growth	Probe card sales growth	GM %
NT$10,000 upside	90%/87%	87%/93%	45.1%/49.7%
NT$8,900 base	75%/69%	72%/78%	43.1%/44.7%
NT$4,000 downside	35%/34%	32%/38%	38.1%/39.7%

Source: UBS estimates

COMPANY DESCRIPTION

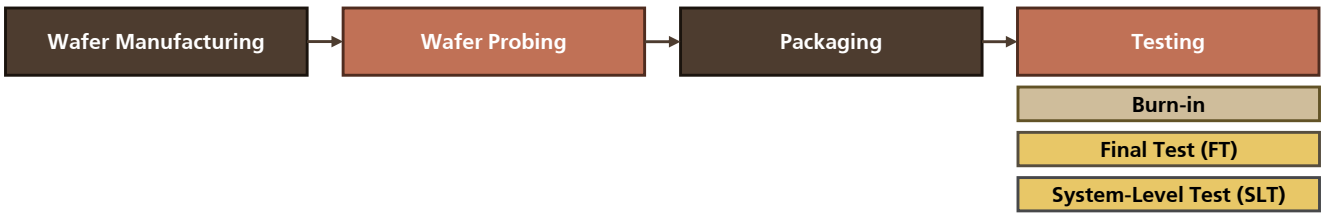
WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solutions provider, with final test socket as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own made probes and partnership with Technoprobe.

Testing TAM expanding with IC complexity: leaders likely to outperform industry growth

Our central view is that testing content per AI accelerator is growing faster than semiconductor unit volumes, creating a multi-year expansion in testing TAM. We believe semiconductor testing is entering a period of structural growth as AI-driven chip complexity increases testing intensity across wafer, package and system levels. The industry's shift-left strategy, which moves more testing upstream in the semiconductor manufacturing process, is resulting in longer and more frequent test insertions. As cloud AI GPUs and ASICs become increasingly complex, additional testing is required to ensure functionality, performance and yield, while reducing defective parts per million (DPPM). More robust testing should improve deployment efficiency and lower the cost of operating AI infrastructure.

Semiconductor testing generally comprises four stages: wafer/chip probing (CP), final test (FT), system-level test (SLT), and burn-in. CP is conducted at the wafer-level before packaging, while FT, SLT, and burn-in occur after packaging. The below Figure 1 illustrates the semiconductor manufacturing flow.

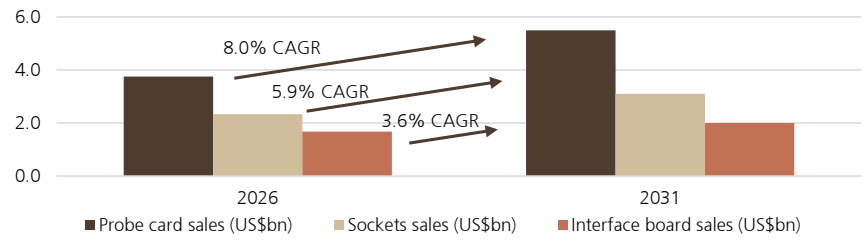
Figure 1: Semiconductor testing process: from wafer level to package level



Source: UBS

We estimate semiconductor testing historically accounts for ~1% of total semiconductor revenue. Within testing, consumables such as probe cards, sockets and interface boards represent ~10-15% of industry revenue. According to Yole, the probe card, socket and interface board markets should grow from US$3.8bn, US$2.3bn and US$1.7bn in 2026 to US$5.5bn, US$3.1bn and US$2.0bn by 2031, respectively.

Figure 2: Yole estimating testing consumables TAM to climb in the next 5 years



Source: Yole, UBS

Nonetheless, leaders within the testing sector will standout among competition, with greater sales CAGR comparing to the industry.

As the leader of AI/HPC sockets, **WinWay** is set to deliver 2026-31E socket sales CAGR of 33%, underpinned by robust demand for final test and system-level test, with test content and test times rising which each iteration of new AI accelerators/CPU, more sockets and equipment will be needed in order to increase the throughput.

- **Strong accelerator/CPU roadmap to support growth.** Robust testing demand can also be inferred from AI accelerator/CPU roadmaps of Nvidia, AMD, Google, alongside tester capacity expansions by Advantest.
- **Capacity expansion underpinning future strength.** WinWay's socket pin capacity expansion is expected by the company to support this exceptional sales growth from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching

9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

- **Probe card business to rise from smaller base.** Its probe card business is also set to benefit from the shift-left trend for high-end AI/HPC probe cards. WinWay leverages its existing customers' demand in addition to their MEMS partnership with Technoprobe, as we forecast WinWay's probe card sales to rise from a low base, with a 33% 2026-31E sales CAGR.

Figure 3: WinWay is set to benefit with a 33% 2026-31E sales CAGR

Sales by Segments NT$m	2026E	2027E	2028E	2029E	2030E	2031E	2026-31E CAGR
Coaxial Socket (HyperSocket)	$7,661	$13,705	$23,103	$25,872	$30,603	$34,726	35%
YoY		*79%*	*69%*	*12%*	*18%*	*13%*	
RF & Plastic Socket	$2,311	$3,835	$6,465	$6,960	$7,259	$7,252	26%
YoY		*66%*	*69%*	*8%*	*4%*	*0%*	
Burn-in Socket	$253	$306	$516	$715	$903	$1,075	34%
YoY		*21%*	*69%*	*39%*	*26%*	*19%*	
Test Socket	$10,224	$17,845	$30,083	$33,547	$38,764	$43,053	33%
YoY		*75%*	*69%*	*12%*	*16%*	*11%*	
Probe Card	$4,478	$7,700	$13,690	$15,507	$16,095	$18,403	33%
YoY		*72%*	*78%*	*13%*	*4%*	*14%*	
Contact Element	$937	$1,596	$2,691	$3,014	$3,465	$3,899	33%
YoY		*70%*	*69%*	*12%*	*15%*	*13%*	
Others	$444	$811	$1,367	$1,671	$2,133	$2,654	43%
YoY		*83%*	*69%*	*22%*	*28%*	*24%*	
Operating Metrics	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**2031E**	**2026-31E CAGR**
Revenue NT$m	$16,084	$27,952	$47,831	$53,739	$60,457	$68,010	33%
YoY		*73.8%*	*71.1%*	*12.4%*	*12.5%*	*12.5%*	
GM %	41.1%	43.1%	44.7%	44.7%	44.9%	45.1%	
OpM %	27.5%	32.1%	36.7%	36.7%	36.9%	37.1%	

Source: UBS estimates

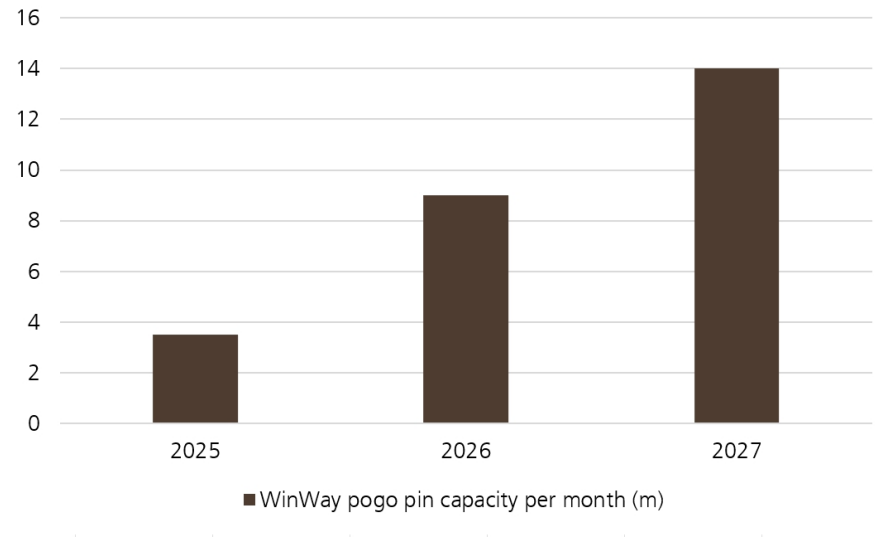
WinWay is expanding capacity rapidly to capture this growing TAM. Industry test interface suppliers are expanding capacity aggressively to meet rising AI and HPC demand. As advanced semiconductors require higher-performance probe cards, sockets and interface boards, capacity additions are becoming increasingly important to support growth across both wafer-level and package-level testing.

We believe WinWay is well positioned to gain share within the high-end AI/HPC test interface market. Taiwanese suppliers benefit from their proximity to leading fabless customers and the growing preference for localised supply chains, particularly for advanced testing applications where responsiveness and engineering support are critical. Against a backdrop of tight industry supply, WinWay announced significant capacity expansion plans over the next two to three years.

WinWay's capacity is expected by the company to increase from 3.5mn pins per month at year-end 2025 to 6mn currently, reaching 9mn by year-end 2026 and 14mn by year-end 2027 through expansions in Renwu, Kaohsiung. By 2027, approximately 60% of pogo-pin requirements should be internally sourced, strengthening both supply security and margins.

We estimate these capacity additions will support revenue growth of 105%/74%/71% in 2026E/27E/28E.

Figure 4: Announced capacity expansions for WinWay



Source: Company data

For WinWay, this growth opportunity is substantial, as the capacity expansions for its socket pins will satisfy around 60% of its own pogo pin demand in 2027E, we estimate socket sales will likely contribute for around NT$163 of EPS in 2027E and to further expand to NT$307 in 2028E. This paves the way for substantial share gaining opportunity for WinWay for AI/HPC sockets as we believe there are advantages to have footprint within Taiwan, where there's already an established testing ecosystem with testing providers such as ASE/SPIL/KYEC, and handler providers such as Hon Precision, all positioned locally in Taiwan to provide around the clock service and support for AI/HPC customers.

We see industry capacity tightness to support socket pin ASP growth, as we forecast pin ASP to grow from US$2.0 in 2026E to US$2.5 in 2027E and US$3.0 in 2028E. We believe WinWay's capacity expansion creates a significant growth opportunity with a better margin profile. By 2027E, approximately 60% of pogo pin demand should be internally sourced, improving both supply security and profitability. We estimate sockets could contribute around NT$123 of EPS in 2027E, rising to NT$261 in 2028E. Combined with its expanding capacity, strong customer relationships and proximity to Taiwan's semiconductor testing ecosystem, including ASE/SPIL, KYEC and major handler suppliers, WinWay is well positioned to gain share in the AI/HPC socket market.

Figure 5: We see socket pin expansions to drive significant growth in the next 2-3 years

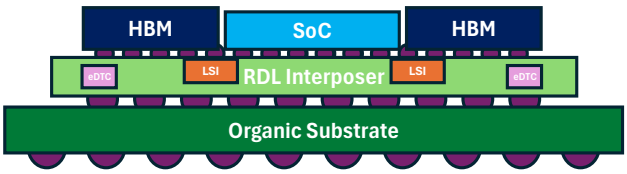
Socket Pin Analysis	2025	2026E	2027E	2028E
Pin Capacity (m/month)	3.2	6.7	11.2	18.3
Self sufficiency	*56%*	*50%*	*60%*	*70%*
Pin demand (m/month)	5.63	13.40	18.59	26.11
ASP (US$)	$2.0	$2.0	$2.5	$3.0
Sales contribution (US$m/month)	**$11**	**$27**	**$46**	**$78**
Sales contribution (US$m/year)	**$135**	**$322**	**$558**	**$940**
Socket sales (NT$m)	**$4,214**	**$10,224**	**$17,845**	**$30,083**
GM (%)	*50%*	*42%*	*47%*	*50%*
Gross Profit (NT$m)	**$2,107**	**$4,294**	**$8,387**	**$15,041**
OPEX (%)	*19%*	*17%*	*16%*	*11%*
Operating Profit (NT$m)	**$1,306**	**$2,556**	**$5,532**	**$11,732**
OPM (%)	*31%*	*25%*	*31%*	*39%*
Tax rate (%)	*19%*	*20%*	*20%*	*20%*
Net profit (NT$m)	**$1,060**	**$2,041**	**$4,426**	**$9,386**
EPS (NT$m)	**$29.7**	**$56.9**	**$123.1**	**$261.1**

Source: UBS estimates

The AI era has materially increased the value of semiconductor testing. Compared with traditional mobile and automotive chips, AI accelerators incorporate larger compute dies, more advanced nodes, chiplet architectures, advanced packaging technologies, multiple I/O dies, and increasing HBM content. These factors drive higher-value test consumables, including advanced MEMS probe cards, multilayer interface boards, and next-generation sockets designed to support higher pin counts, thermal loading, and power delivery requirements.

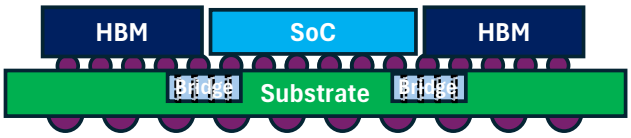
Capacity expansion by test interface vendors is ultimately supported by continued investment throughout the advanced packaging ecosystem, from providers such as TSMC, ASE, Intel, and Amkor. During the Q226 earnings calls, TSMC raised its recent 2026 capex from US$52-56bn to US$60-64bn and ASE raised its 2026 capex from US $8.5bn to US$10.5bn, with both companies citing robust demand supporting capex raise, we believe capex for the next two years will be even higher than 2026, indicating continuous growth in future years. With advanced packaging technologies getting significantly more complex and with larger form factor, illustrated by the upcoming ramp of both TSMC's CoPoS (chip-on-panel-on-substrate) and Intel's EMIB-T (embedded multi-die interconnect bridge) packaging technologies, test interface solution providers will benefit with more value added consumables (see our CoPoS vs EMIB-T deep dive).

Figure 6: TSMC's CoPoS



Source: UBS

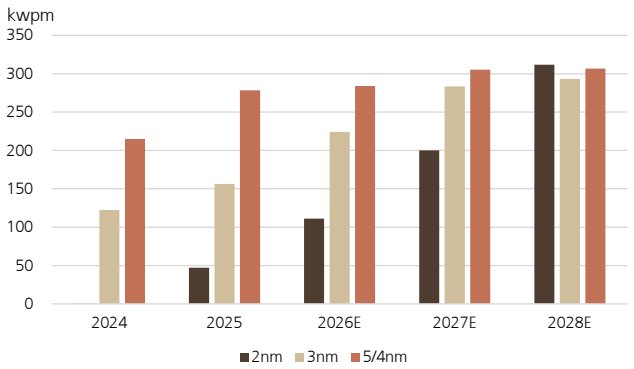
Figure 7: Intel's EMIB-T



Source: UBS

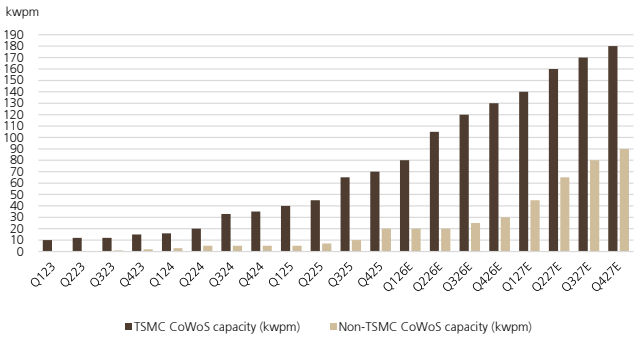
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations (i.e. Nvidia's roadmap from Blackwell to Rubin to Feynman or Google's TPU roadmap from v7 to v8 to v9). This trend is evident with TSMC's front-end and TSMC, Intel, and OSAT's back-end advanced packaging capacity expansions. For front-end, we see industry N3 capacity expanding from 156kwpm in year end 2025 to 224/283/293kpwm in year end 2026E/27E/28E, and N2 capacity from 47kwpm in year-end 2025 to 111/200/312kpwm in year end 2026E/27E/28E. For back-end, we are seeing industry CoWoS capacity to rise from 90kwpm in 2025 to 160kwpm in 2026E and further expand to 270kwpm in 2027E.

Figure 8: Industry's leading edge capacity



Source: UBS estimates

Figure 9: Industry CoWoS capacity

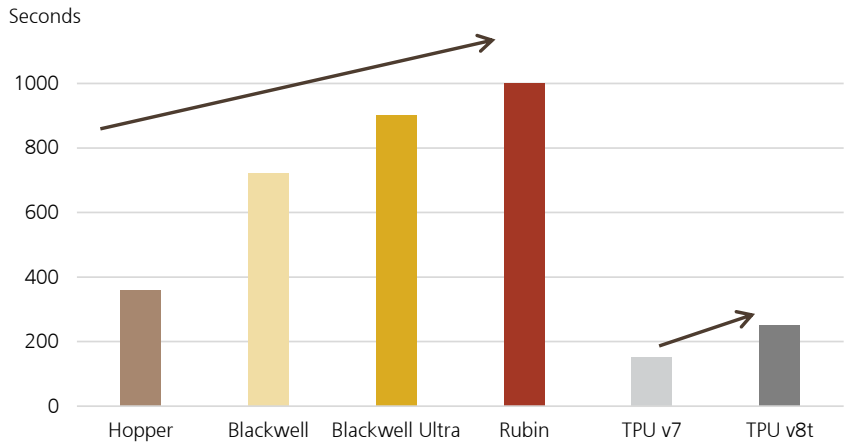


Source: UBS estimates

Final-test, system-level test, and burn-in content should continue to increase as AI accelerators become larger, more power-hungry and more complex. Rising thermal design power and packaging complexity increase validation requirements, driving

longer test times and higher demand for test equipment to prevent testing from becoming a production bottleneck. This is where **WinWay**'s sockets comes into play—as high-end chips with 2.5D/3D/3.5D packaging requires intricate FT sockets to handle the high thermal design power (TDP) and high data rates of up to 224G during the FT process for reliable test results.

Figure 10: Our FT test time estimates



Source: UBS estimates

While architectural changes can affect test requirements for individual product generations, we expect the long-term trend of rising testing content to remain intact. For example, Nvidia's Rubin Ultra adopts a simpler package structure than previously expected, from a 4-die structure to a 2-die structure and Rubin Ultra HBM de-spec from HBM4E 12-Hi to initially use HBM4 8-Hi and upgrade to HBM4E 8-Hi at the latter stages, which may moderate near-term test growth. However, future products such as Nvidia's Feynman are expected to reintroduce greater package complexity, with a four-GPU per package structure. More importantly, any reduction in test time can be offset by higher production volumes, making throughput requirements equally important for testing demand.

We expect test interface content growth to outpace unit growth, For instance, test socket's ASP directly correlates with its pin counts, with the addition of 20-30% more pins inside the sockets, socket ASP can grow 20-30% or more as the socket might require significant cooling capabilities as the TDP of the DUT rises with its larger form factor. Unlike prior semiconductor cycles, much of the current demand is driven by rising test interface content per device rather than unit growth alone.

Robust growth drivers across semiconductor testing

We believe WinWay is set to benefit from several semiconductor testing growth drivers and capture the AI induced, expanding testing TAM. We see three main growth drivers for WinWay: 1) AI accelerators (GPU + TPU), 2) agentic AI driven CPU demand, and 3) shift-left testing trend. Below we list the three main growth drivers in detail.

Accelerators still in focus with TPU as another significant growth driver

Nvidia has been the primary driver of testing content growth over the past several years, benefiting the entire semiconductor testing supply chain. We expect this trend to continue as GPU architectures evolve from Blackwell to Rubin and then to Feynman, with increasing process node complexity, larger package sizes and more advanced packaging technologies driving higher testing value per device.

We believe Nvidia should remain a major contributor to WinWay, with GPUs continuing to serve as their core growth engine. We estimate WinWay's socket share could improve from a roughly 50/50 split with Smiths Interconnect in prior generations to 60/40 with Rubin. Nvidia currently accounts for approximately 50% of WinWay's revenue. Both Nvidia GPUs and Google TPUs increase testing demand through higher content per device rather than unit growth alone.

Beyond Nvidia, Google's TPU roadmap represents a significant incremental growth opportunity. In our view, WinWay should benefit as the socket supplier for both TPU v8t and the next-generation TPU v9 platform. We believe MediaTek's increasing share gain for Google TPUs will also benefit the local Taiwanese test interface supply chain. We estimate TPU to contribute to around 20% of WinWay 2027E sales.

We view TPU v9 as a particularly attractive opportunity given its large four-die package architecture based on Intel's EMIB-T packaging technology, which should drive higher testing content. We estimate production of 4mn TPU v8t units in 2027E, followed by 2.3mn TPU v9 packages (9.2mn dies) and an additional 2mn TPU v8t units in 2028E. Importantly, while EMIB-T packaging is expected to be performed in Malaysia, we believe the majority of advanced testing value should remain within the Taiwanese testing ecosystem. Leading testing providers, including ASE and KYEC, maintain substantial testing operations across Taiwan and Southeast Asia, supporting continued participation by Taiwanese test interface suppliers. See our latest MediaTek report and EMIB-T deep dive.

CPU business as a supplemental catalyst for test interface companies in the agentic world

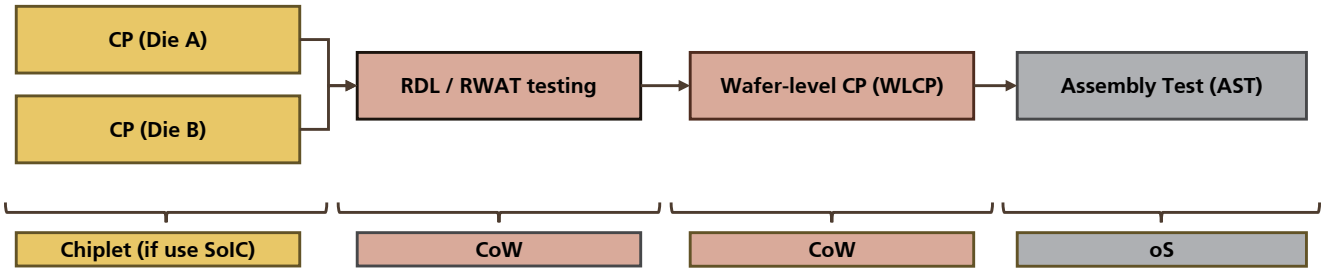
We believe semiconductor test interface are an underappreciated beneficiary of the emerging agentic AI ecosystem. UBS estimates the CPU market will expand to US $170bn by 2030E from US$31bn in 2025 and US$71bn in 2027E, driven by rising deployment of AI infrastructure. As CPU-to-GPU ratios increasingly approach 1:1 in AI clusters, demand for testing interfaces should expand alongside CPU volumes. TSMC is the main CP test provider in the industry, with probe card suppliers aiding its massive demand by supplying probes, such as TSMC's GPU probe cards supported by Technoprobe's probes. However, with capacity now becoming scarce with strong front and back-end demand, TSMC is outsourcing its CP business to ASE and KYEC, further broadening the testing supply chain. This is where new probe card vendors come in—smaller probe card players in the past, such as WinWay, now has the opportunity to gain significant, non-GPU share from TSMC, as they have MEMS probe capabilities (WinWay's MEMS probe are from its strategic partnership with Technoprobe) and has a long established relationship with the fabless companies such as Nvidia via supplying non-probe card consumables such as sockets. Of which, CPU projects has become the main area for WinWay to expand their footprint.

We expect WinWay to benefit from the ramp of Nvidia's ARM-based Vera CPU and Google's ARM-based Axion CPU in 2026E-27E. WinWay is positioned to supply sockets and probe cards for Vera and sockets for Axion. Additional upside could come from AMD's Venice CPU, where WinWay is well positioned. For WinWay, CPU-related demand could account for approximately 30% of sales in 2026E-27E. Importantly, CPU programmes diversify exposure beyond AI GPUs and provide a second structural growth driver for test interface suppliers.

Shift-left igniting secular growth for chip probing

The industry's adoption of shift-left testing should drive a multi-year structural growth opportunity for chip probing (CP) and its supply chain. As advanced packaging costs rise and AI accelerators become increasingly complex, semiconductor manufacturers are placing greater emphasis on identifying known-good dies (KGD) before packaging. The test shift-left trend will drive structural AI/HPC probe card and test consumables demand, with increasing test interface content as test insertions and test time increase. WinWay should also benefit through its partnership with Technoprobe, with probe card revenue growth remaining robust at 94%/72% in 2026E/27E following 161%/260% growth in 2024/25.

Figure 11: CP is set to grow substantially with shift-left, each packaging step set to be MORE costly than the prior step

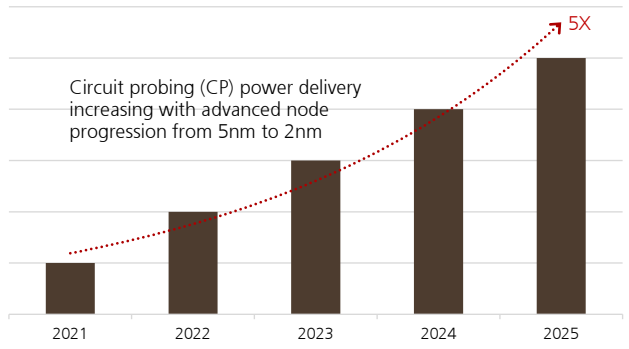


Source: UBS

The economics of advanced packaging reinforce this trend. Heterogeneous integration, chiplet architectures and 2.5D/3D packaging significantly increase package value, while defects identified late in the manufacturing process can lead to substantial scrap costs. To mitigate this risk, foundries and OSATs such as TSMC and ASE employ distributed testing flows incorporating KGD, known-good-stack (KGS) and assembly test screening, to catch defects before the components enter a multi-week packaging cycle (>2 weeks for CoW and 2 weeks for oS). The importance of these tests rises as AI clusters become increasingly dependent on component reliability and low defective-parts-per-million (DPPM) rates. Given the multiplicative yield degradation, a 300bps yield decline at just one manufacturing stage (e.g. dropping from 99.5% to 96.5%) reduces overall integrated four-stage yield from 98.0% to 95.0%; on a 500,000 unit ramp at US $10,000 per unit, this 3% scrap delta translates into around US$150m in unrecoverable losses given >6-month manufacturing lead times.

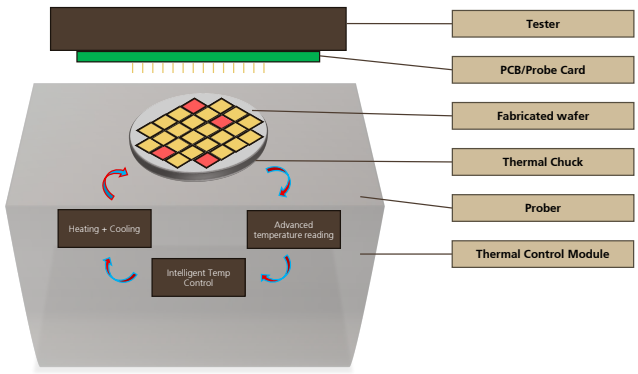
At the same time, rising chip power consumption is increasing the technical difficulty of wafer-level testing. As chip thermal design power scales beyond 1,000W and power density is expected by the industry to surge another 8x by 2030, traditional wafer-level chip probe (WLCP) encounters severe physical bottlenecks. Finer probe pitches and higher power densities increase the risk of overheating, probe needle burnout and yield degradation, creating demand for more advanced probing and thermal-control solutions.

Figure 12: CP power delivery trend rises with process tech evolution



Source: UBS

Figure 13: Thermal control module inside probers are essential to control wafer temperature



Source: UBS

WinWay set to benefit from the AI trend with its leading position within advanced test socket market

Figure 14: Key vendors for each stage of the semiconductor testing industry

	Chip Probing	Final Test	System-Level Test
Testing service providers	TSMC, ASE, KYEC, Amkor, Intel	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu	ASE, Amkor, KYEC, JCET, Tianshui Huatian, Tongfu
Testers	Advantest, Teradyne	Advantest, Teradyne	N / A
Handlers (Probers)	TEL, Tokyo Seimitsu, TSK	Advantest, Chroma, Cohu, Hangzhou Changchuan Technology, Hon Precision, JHT, Teradyne	Hon Precision, Chroma
Probe Cards	FormFactor, Technoprobe, MPI, MJC, JEM, WinWay, CHPT	N / A	N / A
Socket	N / A	WinWay, Cohu, Leeno, Yokowo, Yamaichi, Smiths	WinWay
Load board	N / A	Yamaichi, Advantest, CHPT	Yamaichi, Advantest, CHPT

Source: UBS

WinWay was founded in 2001 in Kaohsiung, Taiwan, and is a leading provider of semiconductor testing interface solutions, with test sockets representing its core business. The company focuses on advanced AI/HPC applications, supplying sockets for final test (FT), system-level test (SLT) and burn-in, as well as cobra probe cards and partnership with Technoprobe for MEMS probe cards.

WinWay's business is primarily divided into two segments:

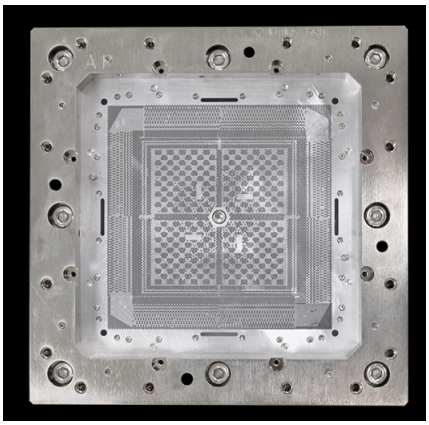
- **Test socket:** Sockets are WinWay's largest revenue contributor. The company supplies a broad range of testing sockets, including coaxial, PoP/plastic and burn-in sockets, serving applications ranging from consumer electronics to high-performance computing.
    - **Coaxial socket:** utilised as the mainstream option for nearly all AI/HPC chips, as traditional sockets are near its physical limits. Coaxial sockets need to handle data rates of up to 224G for high speed transmission and high chip TDP during the FT process.
    - **HyperSocket:** a rubber-silicone like layer called **elastomer** are placed on top of the pogo pins, these elastomer pins causes zero damage to the chips being tested under high contact forces. WinWay's next generation HyperSocket technology utilises elastomers to provide a softer contact between the DUT and the pogo pins, avoiding damage to solder ball. This structure significantly enhances spring probe's lifetime, reduces cleaning frequency, and achieves maximum production efficiency. We estimate it will contribute around 5% of sales in 2027E with a 20-30% ASP premium as compared to standard coaxial socket.
- **Probe cards:** WinWay historically supplies cobra probe cards (vertical probe cards), but with the company announcing their five-year probe supply deal with Technoprobe for MEMS probe in December 2025, they are now one of Nvida Vera CPU's main probe card suppliers.

AI accelerators are pushing socket requirements beyond 10,000 pins, making **coaxial sockets** the industry standard for leading-edge AI/HPC devices. These sockets must support transmission speeds of up to 224G while handling significantly higher thermal design power (TDP) during testing.

Elastomer technology is increasingly being adopted alongside spring-probe (pogo-pin) architectures to reduce chip damage and improve contact reliability. WinWay's next-generation **HyperSocket** integrates elastomer materials between the DUT and spring probes, improving probe lifetime, reducing cleaning requirements and enhancing production efficiency. We believe HyperSocket could become an important differentiator as next-generation AI accelerators demand higher pin counts, bandwidth and thermal performance. We estimate HyperSocket could contribute approximately 5% of WinWay's revenue in 2027E while carrying a 20-30% ASP premium versus conventional coaxial sockets. The transition toward larger AI packages, higher pin counts and faster data-transfer speeds should continue to increase socket content per device. We expect this trend to support long-term growth in the socket market, with technology leaders such as WinWay well positioned to gain share through next-generation solutions including HyperSocket.

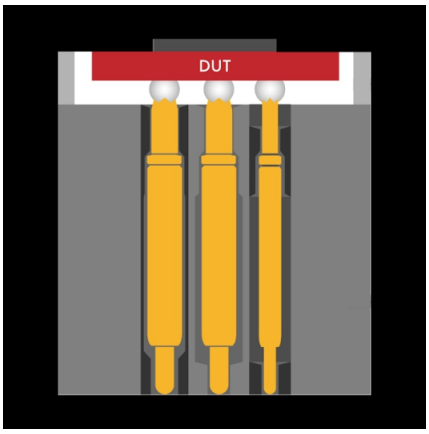
Beyond its existing product portfolio, WinWay is developing next-generation technologies including CPO electrical test sockets and liquid-cooled sockets for future high TDP AI accelerators, reinforcing its position within the AI/HPC testing ecosystem.

Figure 15: Coaxial socket



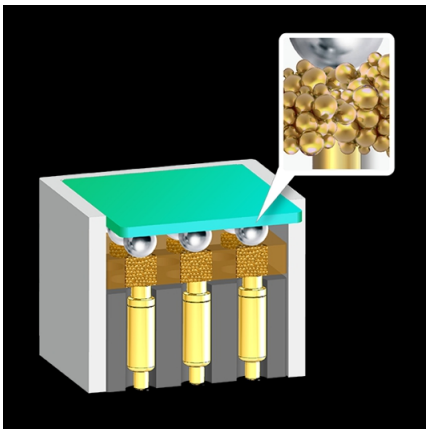
Source: Company data (WinWay)

Figure 16: DUT socket contact



Source: Company data (WinWay)

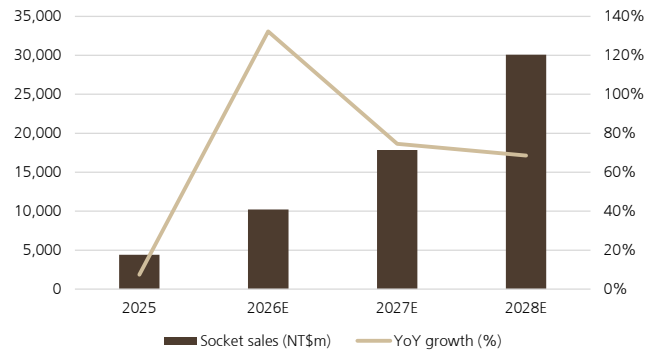
Figure 17: HyperSocket



Source: Company data (WinWay)

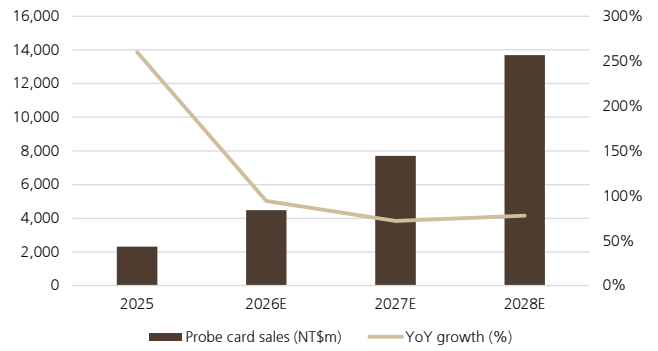
WinWay's competitive advantages are its 1) technology leadership with coaxial socket and HyperSocket, 2) proximity to the semiconductor testing value chain in Taiwan, and 3) historical relationship with key AI companies such as Nvidia and AMD. We believe these advantages may allow WinWay to capture significant upside in the growing semiconductor testing TAM driven by cloud AI proliferation, and gain share across its competition such as Smiths Interconnect for AI and HPC applications.

Figure 18: WinWay's socket sales growth



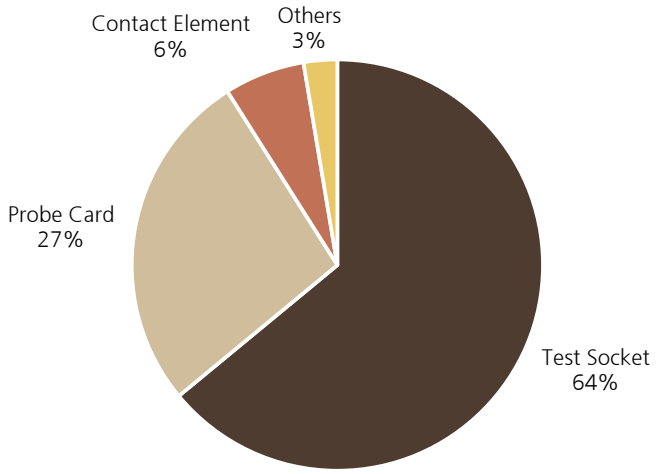
Source: Company data, UBS estimates

Figure 19: WinWay' probe card sales growth



Source: Company data, UBS estimates

Figure 20: WinWay's H126 product mix

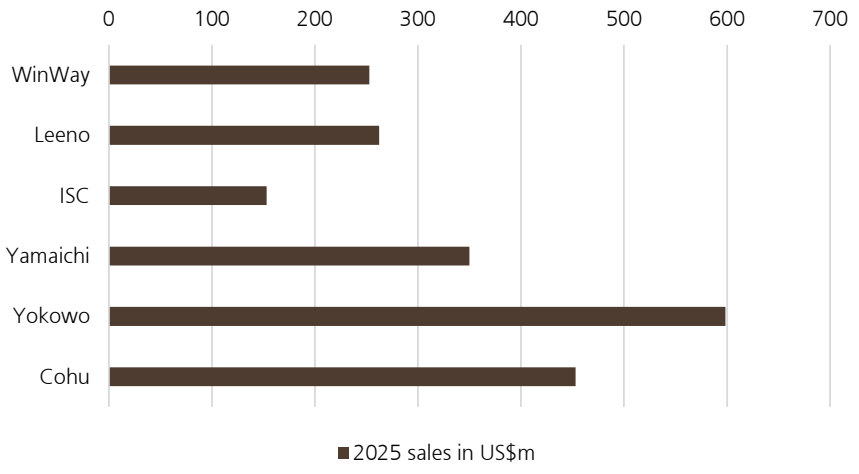


Source: Company data, UBS

Compared with the probe card market, the socket industry is more fragmented. Competition include:

- **WinWay (6515.TW)**: A major share gainer in AI/HPC sockets, with strong exposure to Nvidia and AMD. The company specializes in high-performance coaxial sockets and is developing its next-generation HyperSocket platform. It is also expanding into probe cards through its partnership with Technoprobe.
- **Leeno (058470.KQ)**: Vertically integrated supplier focused on smartphone AP and memory sockets.
- **ISC (095340.KQ)**: Memory socket supplier with elastomer expertise.
- **Smiths Interconnect (Acquired by Molex)**: A key player in AI/HPC socket industry, with its DaVinci coaxial sockets able to support high frequency connectivity testing.
- **Yamaichi (6941.T)**: Japanese socket supplier with substantial automotive exposure.
- **Yokowo (6800.T)**: Major provider of contact probes and socket pins.
- **Cohu (COHU.O)**: Broad test-solutions provider spanning ATE, handlers and sockets, with high exposure to automotive, industrial and consumer applications.

Figure 21: 2025 sales of major socket suppliers

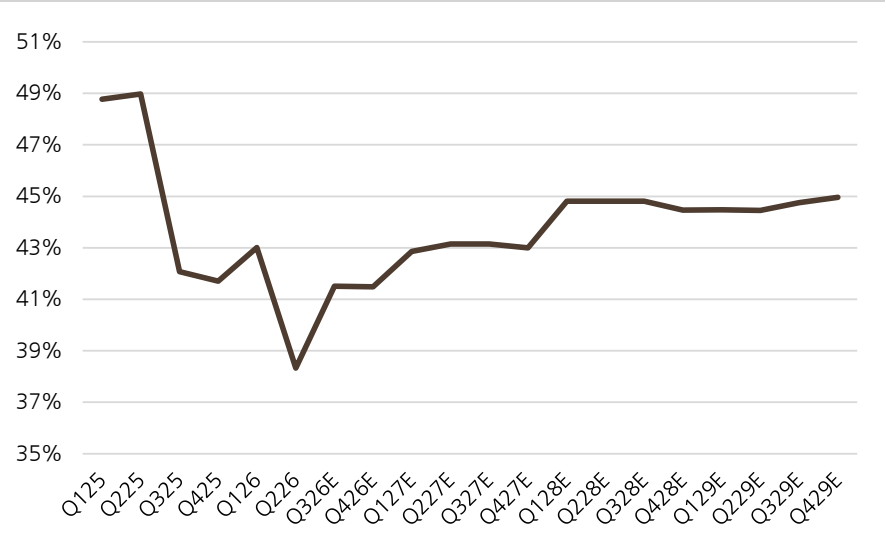


Source: Company data, UBS. Note: Most socket suppliers are companies with multiple business segments, i.e. Yokowo and Cohu.

WinWay's margin is set to recover in 2027E/28E

Company GM hit a trough in Q226, at 38.3%. We view recent margin pressure as largely temporary. GM has been impacted by MEMS probe sourcing through Technoprobe (probe card GM now below 40% and dilutive to >50% coaxial socket GM) and start-up costs associated with WinWay's rapid capacity expansion. During the early part of 2025, WinWay's GM was high at 48.9% in H125, with more cobra probe cards (60% GM) in mix with solid pogo pin and socket yield. With MEMS probe card now accounting for more than 70% of probe card revenue and GM at around low to mid 30%, margin dilution from probe card business is unlikely to unravel for GM to return to above 45% in the near term due to the MEMS partnership terms with Technoprobe. However, we expect that as new facilities ramp and yields improve, profitability to recover gradually from the Q226 trough. We see GM improving to 43.1% in 2027E and 44.7% in 2028E.

Figure 22: We believe WinWay' GM will gradually recover in the next two-three years



Source: UBS estimates

Figure 23: UBS earnings forecast

(NT$m)	2025	Q126	Q226	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E	2028E	2029E	2030E
Revenue	**7,857**	**2,980**	**3,523**	**4,556**	**5,024**	**16,084**	**5,488**	**6,978**	**7,327**	**8,159**	**27,952**	**47,831**	**53,739**	**60,457**
- YoY chg (%)	36%	30%	131%	153%	125%	105%	84%	98%	61%	62%	74%	71%	12%	13%
- QoQ chg (%)		33%	-55%	53%	43%		9%	-57%	33%	17%				
Gross profit	**3,556**	**1,282**	**1,350**	**1,891**	**2,084**	**6,608**	**2,353**	**3,011**	**3,162**	**3,508**	**12,034**	**21,387**	**23,998**	**27,173**
- Gross margin	45.3%	43.0%	38.3%	41.5%	41.5%	41.1%	42.9%	43.2%	43.2%	43.0%	43.1%	44.7%	44.7%	44.9%
Operating expense	1,487	505	525	547	603	2,180	659	768	806	816	3,048	3,827	4,299	4,837
Operating profit	**2,070**	**777**	**825**	**1,344**	**1,482**	**4,428**	**1,694**	**2,244**	**2,356**	**2,692**	**8,986**	**17,560**	**19,699**	**22,336**
- Operating margin	26.3%	26.1%	23.4%	29.5%	29.5%	27.5%	30.9%	32.2%	32.2%	33.0%	32.1%	36.7%	36.7%	36.9%
Non-op profit	-8	97	22	17	14	150	15	17	18	15	66	103	138	200
Pre-tax profit	**2,061**	**874**	**848**	**1,361**	**1,495**	**4,578**	**1,710**	**2,261**	**2,374**	**2,707**	**9,052**	**17,663**	**19,837**	**22,537**
Tax	388	175	175	272	299	922	342	452	475	541	1,810	3,533	3,967	4,507
- Effective tax rate	18.8%	20.1%	20.7%	20.0%	20.0%	20.1%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%
Outstanding shares (m)	36	36	36	36	36	36	36	36	36	36	36	36	36	36
Net profit	**1,673**	**699**	**672**	**1,089**	**1,196**	**3,656**	**1,368**	**1,809**	**1,899**	**2,166**	**7,241**	**14,130**	**15,870**	**18,029**
- YoY chg (%)	41%	14%	228%	193%	147%	119%	96%	169%	74%	81%	98%	95%	12%	14%
- QoQ chg (%)		45%	-60%	56%	78%		14%	-51%	39%	20%				
Basic EPS (NT$)	**46.93**	**19.54**	**18.70**	**30.28**	**33.27**	**101.83**	**38.04**	**50.31**	**52.83**	**60.24**	**201.42**	**393.03**	**441.42**	**501.48**
- YoY chg (%)	37%	14%	225%	190%	145%	117%	95%	169%	74%	81%	98%	95%	12%	14%
- QoQ chg (%)		44%	-60%	55%	78%		14%	-51%	39%	20%				

Source: UBS estimates

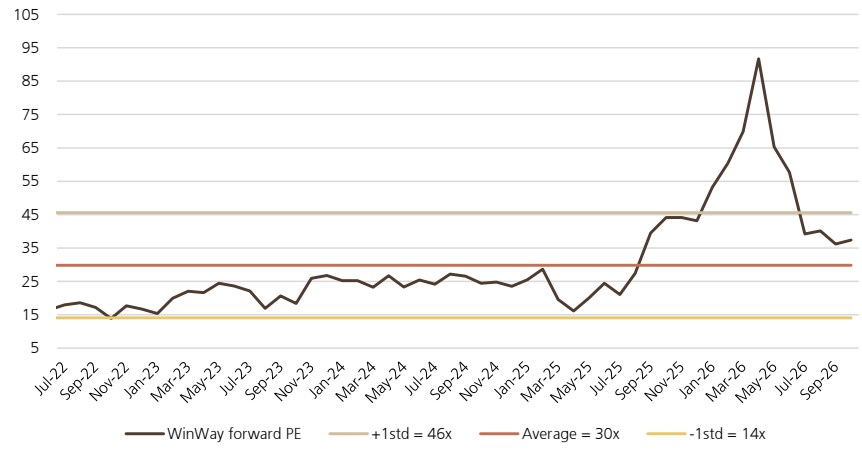
Figure 24: UBS estimates vs consensus

	UBS			Consensus			Difference		
(NT$m)	**2026E**	**2027E**	**2028E**	**2026F**	**2027F**	**2028F**	**2026E**	**2027E**	**2028E**
Revenue	**16,084**	**27,952**	**47,831**	**16,322**	**26,839**	**45,477**	**-1%**	**4%**	**5%**
- YoY chg (%)	105%	74%	71%	108%	64%	69%			
Gross profit	**6,608**	**12,034**	**21,387**	**6,825**	**12,136**	**20,622**	**-3%**	**-1%**	**4%**
- Gross margin	41.1%	43.1%	44.7%	41.8%	45.2%	45.3%			
Operating profit	**4,428**	**8,986**	**17,560**	**4,438**	**8,983**	**16,451**	**0%**	**0%**	**7%**
- Operating margin	27.5%	32.1%	36.7%	27.2%	33.5%	36.2%			
Pretax profit	**4,578**	**9,052**	**17,663**	**4,305**	**8,652**	**15,545**	**6%**	**5%**	**14%**
Net profit	**3,656**	**7,241**	**14,130**	**3,477**	**6,992**	**12,642**	**5%**	**4%**	**12%**
- Net margin	22.7%	25.9%	29.5%	21.3%	26.1%	27.8%			
Basic EPS (NT$)	**101.83**	**201.42**	**393.03**	**97.71**	**187.07**	**348.32**	**4%**	**8%**	**13%**
- YoY chg (%)	117%	98%	95%	108%	91%	86%			

Source: UBS estimates, LSEG

Our NT$8,900 price target is based on our valuation for WinWay at 30x average 2027E-28E P/E, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR.

Figure 25: WinWay's 12-month forward PE (x)



Source: LSEG, UBS

Figure 26: Testing supply chain valuation table

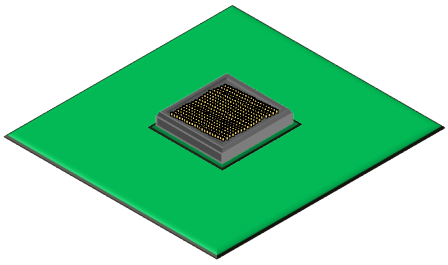
Company name	Ticker	Analyst	Market cap (US$m)	Price Target (LC)	Share price (LC)	Rating	EPS growth (%)		P/E (x)		P/BV (x)		ROE (%)	
							2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Testing service provider														
ASE Industrial	3711.TW	Sunny Lin	104,043	835	744	Buy	104	64	39.3	24.0	7.5	6.3	21	28
King Yuan Electronics	2449.TW	Sunny Lin	11,456	325	298	Buy	40	62	32.6	20.1	5.9	5.0	20	27
Amkor Technology	AMKR.O	Randy Abrams	13,114	80	53	Buy	73	8	20.6	19.1	2.6	2.4	14	13
JCET Group	600584.SS	Jimmy Yu	17,156	135	64	Buy	72	53	42.7	28.0	3.7	3.3	9	12
Tianshui Huatian Technology	002185.SZ	Jimmy Yu	7,928	13	16	Neutral	33	27	57.0	44.8	2.8	2.7	5	6
TongFu Microelectronics Co Ltd	002156.SZ	N/A	13,526	N/A	57	N/A	50	13	45.3	40.1	5.0	4.5	10	11
Shanghai V-Test Semiconductor Tech Co Ltd	688372.SS	N/A	2,517	N/A	100	N/A	36	106	40.6	19.7	3.8	3.2	11	18
Sector Average							**58**	**47**	**39.7**	**28.0**	**4.5**	**3.9**	**13**	**17**
Tester														
Advantest	6857.T	Atsuhiro Kinoshita	191,547	43,600	41,850	Buy	80	96	38.8	53.1	16.8	31.8	70	58
Teradyne Inc	TER.O	Timothy Arcuri	67,666	500	430	Buy	134	51	45.7	30.8	16.7	12.8	43	47
Sector Average							**107**	**74**	**42.2**	**42.0**	**16.7**	**22.3**	**57**	**52**
Prober/Prober solutions														
Hermes Testing Solutions	7856.TWO	Ryan Sun	3,580	8,880	4,190	Buy	221	487	143.9	27.3	9.1	7.0	10	29
Tokyo Electron	8035.T	Atsuhiro Kinoshita	184,031	18,720	12,805	Buy	-16	71	29.7	37.1	6.5	12.8	36	44
Sector Average							**102**	**279**	**86.8**	**32.2**	**7.8**	**9.9**	**23**	**36**
Handler														
Hon.Precision, Inc.	7769.TW	Sunny Lin	33,404	10,000	5,905	Buy	90	86	45.2	24.4	14.5	10.6	36	50
Chroma ATE	2360.TW	Jerry Su	31,434	2,960	2,350	Buy	64	38	51.8	37.7	28.6	20.8	58	64
Cohu, Inc.	COHU.O	N/A	3,460	N/A	73	N/A	n.a.	94	77.0	39.6	-	-	7	16
JHT Design Co Ltd	603061.SS	N/A	4,045	N/A	312	N/A	155	47	54.8	37.3	13.0	9.9	22	25
Hangzhou Changchuan Technology	300604.SZ	Yongwei Lai	22,690	310	240	Buy	60	53	76.1	49.8	23.4	16.5	36	39
Sector Average							**92**	**63**	**61.0**	**37.8**	**19.9**	**14.4**	**32**	**39**
Test Socket														
WinWay Technology	6515.TW	Ryan Sun	6,948	8,900	6,130	Buy	119	98	60.2	30.4	22.0	14.6	44	34
Leeno Industrial Inc	058470.KQ	N/A	4,936	N/A	86,900	N/A	33	17	35.4	30.3	8.0	6.8	24	24
ISC Co Ltd	095340.KQ	N/A	3,247	N/A	205,500	N/A	70	41	47.0	33.2	7.0	5.9	16	19
Yokowo Co Ltd	6800.T	N/A	744	N/A	4,925	N/A	28	93	38.3	19.8	-	1.8	-	-
Sector Average							**62**	**62**	**45.2**	**28.4**	**12.3**	**7.3**	**28**	**26**
Load Board														
Chunghwa Precision Test Tech	6510.TWO	Ryan Sun	3,149	5,000	3,055	Buy	112	99	47.5	23.8	9.2	7.3	21	34
Probe Card														
Technoprobe SpA	TPRO.MI	Harry Blaiklock, CFA	27,100	41	37	Buy	259	18	66.9	56.5	14.6	11.6	25	23
MPI Corp	6223.TWO	N/A	17,518	N/A	5,670	N/A	97	110	87.2	41.5	30.3	19.1	38	54
FormFactor Inc	FORM.O	N/A	11,038	N/A	141	N/A	159	24	46.2	37.1	9.3	8.9	19	23
MICRONICS JAPAN CO.,LTD.	6871.T	N/A	3,906	N/A	15,410	N/A	155	28	24.8	19.4	7.1	5.6	32	36
Japan Electronic Materials Corp	6855.T	N/A	804	N/A	8,650	N/A	46	43	24.9	17.4	-	-	-	-
Sector Average							**138**	**54**	**49.6**	**32.6**	**14.1**	**10.5**	**27**	**34**

Source: LSEG, UBS estimates. Priced as of October 6 (HKT).

Appendix: What are semiconductor test sockets?

Semiconductor test socket are electro-mechanical interface that are used post chip probing and chip packaging process to test the functionality of the chips. Sockets are placed onto load boards on top of testers during the final test/system level test processes. Handlers will place the chips onto the sockets, and pins inside the sockets will enable the testers to read the signals from the device under test (DUT, i.e. the packaged chips), with tester signals fan out from beneath the PCB load board.

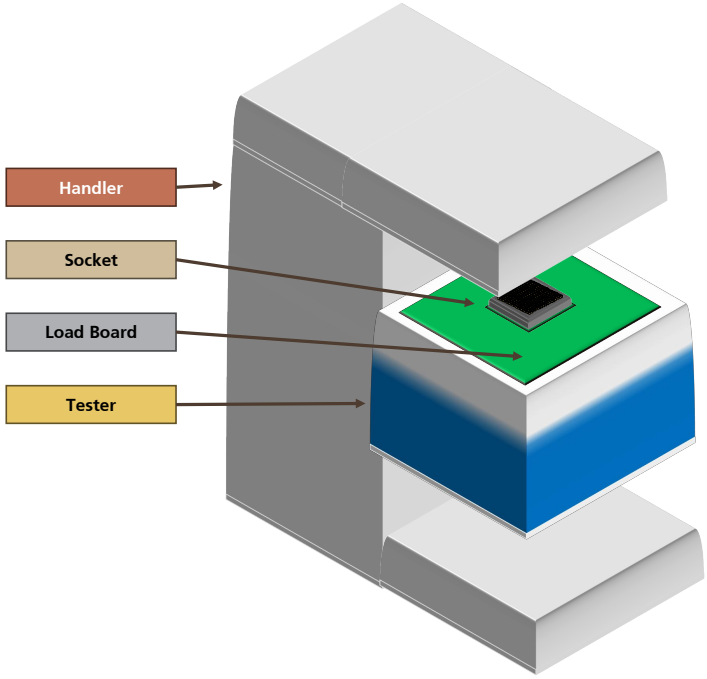
Figure 27: Socket illustration (with load board underneath)



Source: UBS

As AI accelerators become larger and more complex, socket requirements continue to increase. High-end sockets typically comprise a socket housing, alignment plate, integrated thermal-control components and contact elements that connect the DUT to the tester. Spring probes (pogo pins) are the most common choice for HPC sockets, with high durability, individual probe replacement flexibility, and consistent contact force. FT and SLT sockets account for more than 70% of the total socket market, with burn-in sockets representing the remainder.

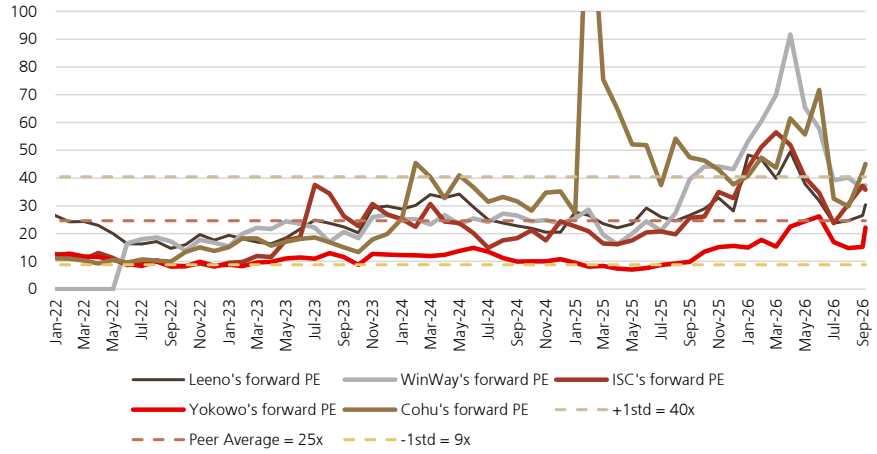
Figure 28: Final test equipment and consumables (simplified for illustration purposes, handlers should come with ATC)



Source: UBS

WHAT´S PRICED IN?

WinWay is trading at NT$6,130 as of 6 Oct 2026



Source: LSEG, UBS estimates

Despite a 120% YTD share-price increase, WinWay remains 43% below its April 2026 peak and trades at only 16x our 2028E P/E. We believe the market remains overly focused on near-term GM pressure while underestimating the company's multi-year growth opportunity. Our reverse DCF analysis indicates that the current market price implies only a 19% long-term sales CAGR vs our estimate of 25%, alongside long-term GM level of 32% vs our estimate of 45%.

We believe investors are not fully reflecting three key drivers:

1) CPU and TPU opportunities are becoming increasingly meaningful.

Beyond its core GPU socket franchise, WinWay is positioned to benefit from Nvidia's Vera CPU, AMD's Venice CPU, and Google's TPU. These programmes should broaden revenue sources and create an additional demand driver for high-end sockets and probe cards.

2) Larger AI packages are driving higher socket content per device.

As AI accelerators adopt larger form factors, higher pin counts and greater power consumption, socket complexity and ASPs continue to rise. We expect socket content growth to outpace semiconductor unit growth, creating a structural tailwind for WinWay.

3) Capacity ramp and margin recovery are not fully reflected.

Following substantial capacity expansion, we expect revenue growth of 105%/74%/71% in 2026E/27E/28E. At the same time, we forecast gross margin recovery from 38.3% in Q226 to 43.1%/44.7% in 2027E/28E as utilization and yields improve.

Our NT$8,900 price target is based on our valuation for WinWay at 30x average 2027E-28E P/E, in-line with its historical average of 30x and underpinned by a 30% 2027-31E earnings CAGR. Comparing with historical NTM peer average PE of roughly 25x (currently trading at 39x NTM PE), we believe WinWay deserves to trade at a premium given its leadership in AI/HPC sockets, expanding CPU and TPU exposure, and our long-term earnings CAGR estimate.

UPSIDE/DOWNSIDE SPECTRUM

WinWay is trading at NT $6,130 as of 6 Oct 2026

Value drivers (2027/28E)	Test socket sales growth	Probe card sales growth	GM %
NT$10,000 upside	90%/87%	87%/93%	45.1%/49.7%
NT$8,900 base	75%/69%	72%/78%	43.1%/44.7%
NT$4,000 downside	35%/34%	32%/38%	38.1%/39.7%

Source: UBS estimates

Risk to the current share price is skewed (1.8:1) to the upside

UPSIDE (NT$10,000): Our upside scenario reflects stronger than expected market share gain among Nvidia's Rubin sockets and Vera CPU probe card strength, with strong TPU v9 shipment in 2028E, forming 90%/87% YoY test socket sales growth and 87%/93% probe card sales growth in 2027/28E, translating to higher sales growth of 82%/75% in 2027/28E and steeper GM recovery in 2027/28E.

BASE (NT$8,900): Our price target of NT$8,900 is based on 30x average 2027-28E PE. We expect an earnings CAGR of 30% in 2027-31E. With extending FT time and larger form factor coaxial sockets, we see these growth drivers driving YoY sales growth from a high base of 105% in 2026E to 74%/71% YoY in 2027/2028E, with OPM of 32.1%/36.7% in 2027/28E.

DOWNSIDE (NT$4,000): Our downside scenario reflects slow GM recovery of 38.1%/39.7% in 2027/28E, with weaker growth for both socket and probe card businesses for slower capacity ramp up, leading to YoY sales growth of 22.3%/34.8% in 2027/28E, respectively.

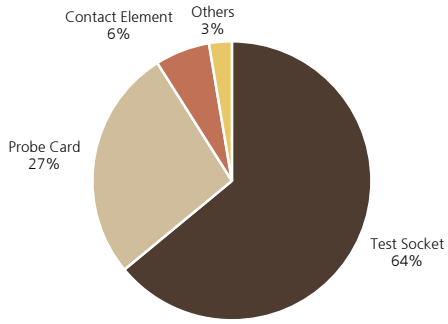
COMPANY DESCRIPTION

WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solution provider, with final test sockets as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own-made probes and a partnership with Technoprobe.

Industry Outlook

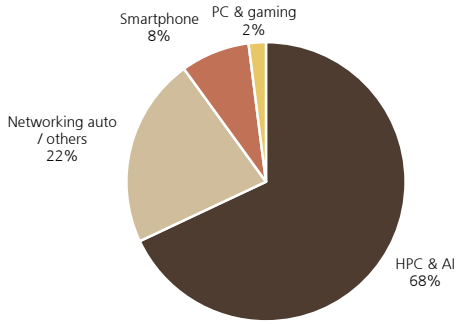
New iterations of semiconductors, especially for AI / HPC applications, are set to increase in both front-end wafer fabrication advancements and back-end packaging complexities with each iterations. With the industry now moves to utilize "shift left", chip probing and its relevant supply chain will be key beneficiaries of this trend which we believe will unlock multi-year, secular growth trends.

WinWay's H126 product mix



Source: Company data, UBS

WinWay's 4M26 revenue by end market



Source: Company data, UBS

WinWay Technology (6515.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**3,682**	**5,798**	**7,857**	**16,084**	*104.7*	**27,952**	*73.8*	**47,831**	**53,739**	**60,457**
Gross profit	1,365	2,534	3,556	6,608	*85.8*	12,034	*82.1*	21,387	23,998	27,173
EBITDA (UBS)	**684**	**1,605**	**2,328**	**4,753**	*104.2*	**9,531**	*100.5*	**18,478**	**21,047**	**24,147**
Depreciation & amortisation	(127)	(237)	(258)	(325)	*-25.9*	(546)	*-68.0*	(918)	(1,348)	(1,811)
EBIT (UBS)	**557**	**1,368**	**2,070**	**4,428**	*113.9*	**8,986**	*102.9*	**17,560**	**19,699**	**22,336**
Associates & investment income	0	0	0	0	-	0	-	0	0	0
Other non-operating income	(7)	57	(53)	98	-	0	-	0	0	0
Net interest	11	10	45	52	*14.6*	66	*27.6*	103	138	200
Exceptionals (incl goodwill)	0	0	0	0	-	0	-	0	0	0
Pre-tax profit	**561**	**1,435**	**2,061**	**4,578**	*122.1*	**9,052**	*97.7*	**17,663**	**19,837**	**22,537**
Tax	(97)	(249)	(388)	(922)	*-137.5*	(1,810)	*-96.3*	(3,533)	(3,967)	(4,507)
Profit after tax	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Preference dividends	0	0	0	0	-	0	-	0	0	0
Minorities	0	0	0	0	-	0	-	0	0	0
Extraordinary items	0	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Net earnings (UBS)	**464**	**1,186**	**1,673**	**3,656**	*118.5*	**7,241**	*98.1*	**14,130**	**15,870**	**18,029**
Tax rate (%)	17.3	17.3	18.8	20.1	*6.9*	20.0	*-0.7*	20.0	20.0	20.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	13.34	33.02	46.71	99.30	*112.6*	196.30	*97.7*	383.04	430.19	488.73
EPS (local GAAP, diluted)	13.34	33.02	46.71	99.30	*112.6*	196.30	*97.7*	383.04	430.19	488.73
EPS (UBS, basic)	13.52	34.31	46.93	101.83	*117.0*	201.42	*97.8*	393.03	441.42	501.48
DPS (net) (NT$)	10.87	25.75	25.07	61.10	*143.7*	120.85	*97.8*	235.82	264.85	300.89
Cash EPS (UBS, diluted) [1]	17.00	39.61	53.91	108.13	*100.6*	211.09	*95.2*	407.93	466.73	537.81
Book value per share	100.33	158.19	178.79	278.86	*56.0*	418.88	*50.2*	691.05	896.65	1,133.28
Average shares (diluted)	35	36	36	37	*2.8*	37	*0.2*	37	37	37

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	639	1,124	2,594	2,632	*1.5*	3,677	*39.7*	8,818	11,894	16,201
Other current assets	1,737	3,921	3,832	11,091	*189.4*	14,824	*33.7*	18,720	20,653	23,217
Total current assets	**2,377**	**5,045**	**6,426**	**13,724**	*113.6*	**18,501**	*34.8*	**27,537**	**32,547**	**39,418**
Net tangible fixed assets	2,109	2,256	2,206	2,991	*35.6*	4,401	*47.2*	6,831	9,245	11,667
Net intangible fixed assets	0	0	0	0	-	0	-	0	0	0
Investments / other assets	390	278	412	837	*103.2*	837	*0.0*	837	837	837
Total assets	**4,876**	**7,580**	**9,044**	**17,551**	*94.1*	**23,740**	*35.3*	**35,205**	**42,629**	**51,921**
Trade payables & other ST liabilities	979	1,812	2,524	5,968	*136.4*	7,109	*19.1*	8,790	8,822	9,606
Short term debt	0	0	0	0	-	0	-	0	0	0
Total current liabilities	**979**	**1,812**	**2,524**	**5,968**	*136.4*	**7,109**	*19.1*	**8,790**	**8,822**	**9,606**
Long term debt	337	147	0	1,477	-	1,477	*0.0*	1,477	1,477	1,477
Other long term liabilities	69	81	75	93	*25.2*	93	*0.0*	93	93	93
Preferred shares	0	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**1,386**	**2,040**	**2,599**	**7,539**	*190.1*	**8,680**	*15.1*	**10,361**	**10,393**	**11,177**
Common s/h equity	**3,489**	**5,540**	**6,445**	**10,011**	*55.3*	**15,059**	*50.4*	**24,845**	**32,236**	**40,744**
Minority interests	0	0	0	0	-	0	-	0	0	0
Total liabilities & equity	**4,876**	**7,580**	**9,044**	**17,551**	*94.1*	**23,740**	*35.3*	**35,205**	**42,629**	**51,921**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	464	1,186	1,673	3,656	*118.5*	7,241	*98.1*	14,130	15,870	18,029
Depreciation & amortisation	127	237	258	325	*25.9*	546	*68.0*	918	1,348	1,811
Net change in working capital	327	(784)	(44)	(2,212)	*NM*	(2,592)	*-17.2*	(2,215)	(1,901)	(1,780)
Other operating	(318)	411	(84)	443	-	0	-	0	0	0
Operating cash flow	**600**	**1,049**	**1,803**	**2,212**	*22.7*	**5,195**	*134.9*	**12,834**	**15,316**	**18,060**
Tangible capital expenditure	(856)	(298)	(277)	(1,287)	*-364.4*	(1,957)	*-52.1*	(3,348)	(3,762)	(4,232)
Intangible capital expenditure	0	0	0	0	-	0	-	0	0	0
Net (acquisitions) & disposals	(230)	(834)	1,009	(3,887)	-	0	-	0	0	0
Other investing	(9)	48	(10)	(5)	*53.3*	0	-	0	0	0
Investing cash flow	**(1,094)**	**(1,084)**	**722**	**(5,178)**	-	**(1,957)**	*62.2*	**(3,348)**	**(3,762)**	**(4,232)**
Equity dividends paid	(754)	(378)	(890)	(894)	*-0.4*	(2,193)	*-145.4*	(4,345)	(8,478)	(9,522)
Share issues / (buybacks)	0	0	0	0	-	0	-	0	0	0
Other financing	(14)	(15)	(17)	(11)	*33.4*	0	-	0	0	0
Change in debt & pref shares	400	909	(151)	3,896	-	0	-	0	0	0
Financing cash flow	**(369)**	**516**	**(1,057)**	**2,991**	-	**(2,193)**	-	**(4,345)**	**(8,478)**	**(9,522)**
Cash flow inc/(dec) in cash	**(863)**	**481**	**1,468**	**25**	*-98.3*	**1,045**	*NM*	**5,141**	**3,076**	**4,306**
FX / non cash items	(1)	4	2	14	*NM*	0	*-100.0*	0	0	0
Balance sheet inc/(dec) in cash	**(865)**	**485**	**1,470**	**39**	*-97.4*	**1,045**	*NM*	**5,141**	**3,076**	**4,306**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

WinWay Technology (6515.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
P/E (UBS, diluted)	51.2	30.5	32.9	61.7	31.2	16.0	14.2	12.5
P/CEPS	39.6	24.4	28.4	55.3	28.3	14.6	12.8	11.1
Equity FCF (UBS) yield %	(1.1)	2.1	2.8	0.4	1.5	4.3	5.2	6.3
Dividend yield (net) %	1.6	2.6	1.6	1.0	2.0	3.8	4.3	4.9
P/BV	6.8	6.4	8.6	22.0	14.6	8.9	6.8	5.4
EV/revenues (core)	6.2	5.9	6.8	NM	7.8	4.5	3.9	3.4
EV/EBITDA (UBS core)	33.4	21.5	22.9	46.1	23.0	11.7	10.1	8.6
EV/EBIT (core)	41.0	25.2	25.8	49.5	24.4	12.3	10.8	9.3
EV/OpFCF (core)	41.0	25.2	25.8	49.5	24.4	12.3	10.8	9.3
EV/op. invested capital	8.5	8.9	12.7	34.9	20.5	14.4	10.9	8.8

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	23,660	35,016	55,053	220,986	220,986	220,986	220,986	220,986
Net debt (cash)	(903)	(640)	(1,785)	(1,874)	(1,677)	(4,770)	(8,879)	(12,570)
Buy out of minorities	69	81	75	93	93	93	93	93
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**22,827**	**34,457**	**53,343**	**219,206**	**219,403**	**216,310**	**212,201**	**208,510**
Non core assets	0	0	(5)	(138)	(138)	(138)	(138)	(138)
Core enterprise value	**22,827**	**34,457**	**53,338**	**219,068**	**219,265**	**216,172**	**212,063**	**208,372**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	(28.1)	57.5	35.5	104.7	73.8	71.1	12.4	12.5
EBITDA (UBS)	(53.3)	134.6	45.0	104.2	100.5	93.9	13.9	14.7
EBIT (UBS)	(58.5)	145.7	51.3	113.9	102.9	95.4	12.2	13.4
EPS (UBS, diluted)	(58.1)	147.5	41.5	112.6	97.7	95.1	12.3	13.6
Net DPS	(50.2)	136.9	(2.6)	143.7	97.8	95.1	12.3	13.6

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	37.1	43.7	45.3	41.1	43.1	44.7	44.7	44.9
EBITDA margin	18.6	27.7	29.6	29.5	34.1	38.6	39.2	39.9
EBIT (UBS) margin	15.1	23.6	26.3	27.5	32.1	36.7	36.7	36.9
Net earnings (UBS) margin	12.6	20.5	21.3	22.7	25.9	29.5	29.5	29.8
ROIC (EBIT)	20.7	35.3	49.2	70.5	NM	NM	NM	NM
ROIC post tax	17.1	29.2	39.9	56.3	67.1	NM	NM	NM
ROE (UBS)	12.9	26.3	27.9	44.4	57.8	70.8	55.6	49.4

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(0.4)	(0.6)	(1.1)	(0.2)	(0.2)	(0.4)	(0.5)	(0.6)
Net debt / total equity %	(8.7)	(17.6)	(40.2)	(11.5)	(14.6)	(29.5)	(32.3)	(36.1)
Net debt / (net debt + total equity) %	(9.5)	(21.4)	(67.3)	(13.0)	(17.1)	(41.9)	(47.7)	(56.6)
Net debt/EV %	(4.0)	(1.9)	(3.3)	(0.9)	(0.8)	(2.2)	(4.2)	(6.0)
Capex / depreciation %	NM	126.1	107.4	NM	NM	NM	NM	NM
Capex / revenue %	23.2	5.1	3.5	8.0	7.0	7.0	7.0	7.0
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.2	1.3	1.9	1.7	1.7	1.7	1.7	1.7
Div. payout ratio (UBS) %	80.4	75.1	53.4	60.0	60.0	60.0	60.0	60.0

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	3,682	5,798	7,857	16,084	27,952	47,831	53,739	60,457
Total	**3,682**	**5,798**	**7,857**	**16,084**	**27,952**	**47,831**	**53,739**	**60,457**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	557	1,368	2,070	4,428	8,986	17,560	19,699	22,336
Total	**557**	**1,368**	**2,070**	**4,428**	**8,986**	**17,560**	**19,699**	**22,336**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	45.2%
Forecast dividend yield	1.0%
Forecast stock return	46.2%
Market return assumption	6.3%
Forecast excess return	39.9%

Company Description

WinWay was founded in 2001 in Kaoshiung, Taiwan. It is a semiconductor testing interface solution provider, with final test sockets as its main business segment. The company has high-end solutions for AI/HPC testing, providing final test/system-level test and burn-in test sockets and probe cards through its own-made probes and a partnership with Technoprobe.

Valuation Method and Risk Statement

We value WinWay based on PE methodology.
Risks for the semis industry:
- Geopolitical uncertainties: Geopolitical uncertainties including tariffs and more export control could negatively impact semis cycle, and a downturn would lead to slower demand in the overall market and moderating capacity expansion.
- Meaningful deceleration of new technology development: Slower demand and less investment are expected If there is meaningful deceleration of new technology development such as Cloud AI, autonomous driving, etc.
- Rising utility cost: Larger-than-expected utility cost increases would further pressure profitability. Rising utility costs in Taiwan have negatively impacted margins for Taiwan tech companies.
Company risks:
- Faster- or slower-than-expected CoWoS capacity expansion: This could result in WinWay gaining/losing socket/probe card orders from major end customers, foundries, and OSATs.
- Competitors achieving technological advancement: Potential risk of competition (from either local or foreign companies) achieving technological advancements could result in WinWay losing market share from major end customers, foundries and OSATs.
- Better- or worse-than-expected AI growth: Better- or worse-than-expected AI growth could lead to a ramp-up or a slowdown of equipment orders from major foundries and OSATs to WinWay, affecting long-term revenue growth potential.
- Slowdown or delay of new technological advancement: A potential slowdown or delay of the rollout for technological advancements for technologies such as SoIC, CoPoS, and CPO, would limit future socket/probe card orders.
- Increase of raw material costs: Rising raw material costs may directly impact WinWay's margins for socket/probe card shipments.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

WinWay Technology

Question	Response
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	3
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	No Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

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12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities Pte. Ltd., Taipei Branch: Ryan Sun, Sunny Lin.

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Company Name	Reuters	12-month rating	Price	Price date
WinWay Technology	6515.TW	No Rating	NT$5,520	07 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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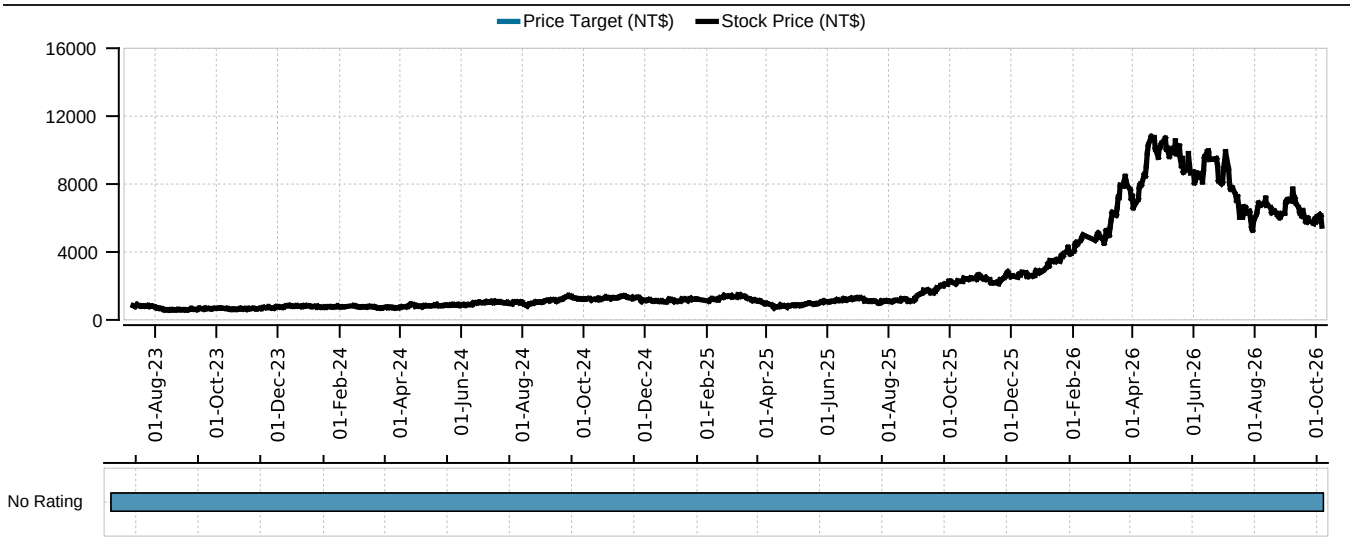
WinWay Technology (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	824.00	-	No Rating

Source: UBS Global Research; LSEG Eikon as of 07-Oct-2026. All prices as of local market close. Ratings as of date shown.

Additional Prices: ASE Industrial, NT$732.00 (07 Oct 2026); Alphabet Inc. - Class A, US$347.68 (06 Oct 2026); Taiwan Semiconductor Manufacturing, NT$2585.00 (07 Oct 2026); NVIDIA Corp, US$239.24 (06 Oct 2026); Advanced Micro Devices Inc, US$649.42 (06 Oct 2026); Source: UBS. All prices as of local market close.

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Daiwa Capital Markets Hong Kong Limited

Acter (5536 TT)

Target price: **TWD1,415.00**

Share price (7 Oct): **TWD1,045.00** | Up/downside: **+35.4%**

5 4 3 2 **1**

★ **Buy** (initiation)

Initiation: acting on the demand for cleanrooms

- Key beneficiary of the cleanroom shortage in the high-tech industry
- Fast-growing order backlog likely to be realised over 2026-28E
- Initiating with a Buy (1) rating and a 12-month TP of TWD1,415

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Investment case: We initiate coverage of Acter, an established multinational cleanroom engineering, procurement, construction and management (EPCM) turnkey service provider, with a Buy (1) rating. Its core business includes cleanroom electromechanical engineering, water/gas/chemical delivery systems, and other cleanroom services. We believe now is a good time to revisit the stock as it underperformed the TAIEX Index by 32.7% in 3Q26, mainly on weaker-than-expected 2Q26 results and soft sector sentiment. Given its stronger revenue momentum over recent months and upcoming TSMC results on 15 October, which may drive sentiment, we suggest investors to buy the stock. We forecast a 24.6% revenue CAGR and 36.9% net profit CAGR over 2026-28E.

Intact global demand for cleanroom technology, the edge AI market, and rapid cleanroom expansion in Asia further support Acter's order backlog. Its backlog reached TWD66bn in July 2026 vs. TWD46bn in 2025 and TWD30bn in 2022 (15.3% 2022-25 CAGR), which largely outpaced the global cleanroom technology market's forecast CAGR over 2023-2032E, according to Precedence Research.

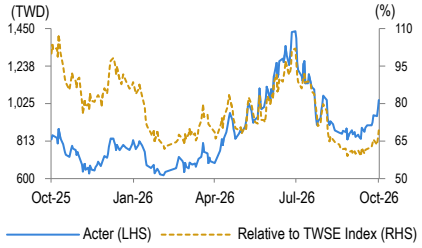
Strong competitive edge supports order share gains across multiple segments. Acter's competitive edge lies in 1) intelligent management enabled by digital twins and AI, using 3D building information modelling (BIM) and AI-driven digital twin management; 2) intellectualisation of green engineering, which comprises green design, green procurement, and green techniques; 3) experience across multiple industries (ie, data centres and electromechanical engineering for the consumer industry) which supports better integration; and 4) a complete network, enabling on-the-spot support. Acter believes it has the highest market share among the operators of three major US data centre players due largely to its capabilities in liquid cooling and its expertise in electromechanical and cleanroom engineering.

Catalysts: Near-term: project wins from key clients, supporting stronger monthly revenue momentum; mid- to long-term: an improving operating-margin trend and a resilient order backlog.

Valuation: We are the first non-domestic broker to cover Acter. We initiate coverage with a 12-month TP of TWD1,415, based on a target PER of 20x (equivalent to a 0.54x PEG over 2026-28E). We believe our valuation is fair, given Acter's strong earnings visibility, which we believe warrants a rerating. Our TP is supported by a DCF valuation of TWD1,454.

Risks: 1) weaker-than-expected expansion in the semiconductor industry; and 2) project delays or cancellations.

Share price performance



12-month range	618.00-1,435.00
Market cap (USDbn)	4.08
3m avg daily turnover (USDm)	31.93
Shares outstanding (m)	124
Major shareholder	Xiang-Hui Development (4.4%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	53,844	66,983	83,593
Operating profit (m)	9,853	13,598	18,014
Net profit (m)	6,325	8,797	11,851
Core EPS (fully-diluted)	50.788	70.635	95.161
EPS change (%)	79.3	39.1	34.7
Daiwa vs Cons. EPS (%)	(6.4)	(5.4)	n.a.
PER (x)	20.6	14.8	11.0
Dividend yield (%)	3.4	4.7	6.4
DPS	35.7	49.6	66.9
PBR (x)	9.3	7.2	5.5
EV/EBITDA (x)	11.6	8.3	6.1
ROE (%)	46.8	54.9	56.8

Source: FactSet, Daiwa forecasts

See important disclosures, including any required research certifications, beginning on page 25

Daiwa Capital Markets Hong Kong Limited

5 4 3 2 **1**

Buy (initiation)

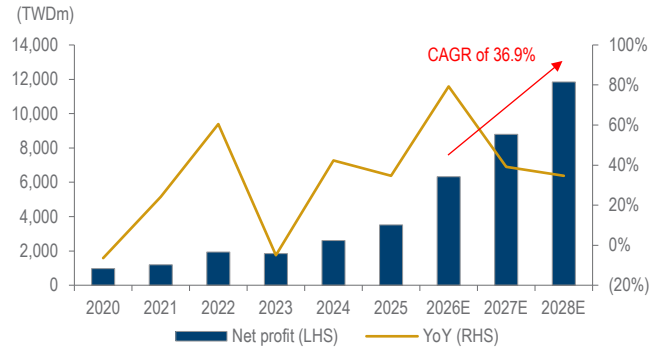
How do we justify our view?

Growth outlook	Valuation	Earnings revisions
✔ ✔ ✔ ✔ ✔	✔ ✔ ✔ ✔ ✔	✔ ✔ ✔ ✔ ✔

Growth outlook

We forecast Acter to record a 36.9% net profit CAGR over 2026-28E, on the back of a 24.6% revenue CAGR over the same period, driven by 1) larger project scale, as strong demand for cleanroom EPCM services from the high-tech sector is likely to support continued growth in Acter's order backlog and 2) improving gross margins and operating efficiency. We expect operating margin to expand by 2.0pp in 2027E and 1.3pp in 2028E, driven by favourable gross margins on new projects and operating leverage.

Acter: net profit and net-profit growth



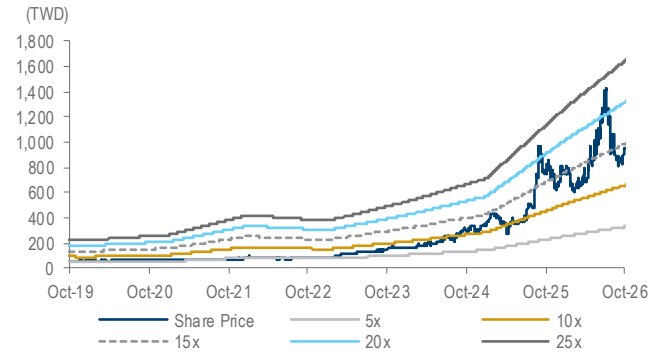
Source: Company, Daiwa forecasts

Valuation

Our 12-month TP of TWD1,415 is based on a target PER of 20x, applied to our 1-year-forward EPS forecast. Our target multiple of 20x is higher than the stock's past-3-year 1-year-forward PER (12.2x) and past-5-year 1-year-forward PER (9.6x) and implies a PEG of 0.54x on our 36.9% net profit CAGR forecast over 2026-28E. We believe the stock deserves a rerating in the next 12 months, largely supported by resilient cleanroom demand globally.

Our TP is also supported by an alternative DCF-derived valuation of TWD1,454/share.

Acter: 1-year-forward PER bands

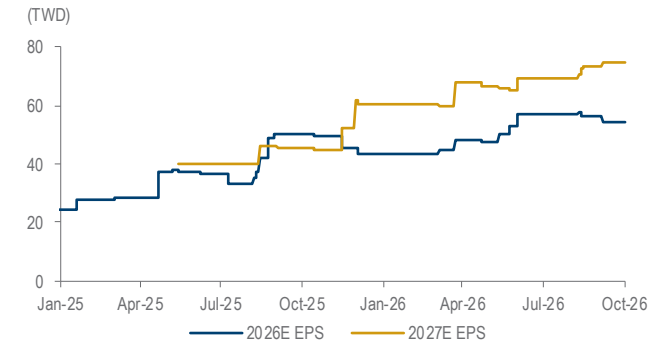


Source: Bloomberg, Daiwa forecasts

Earnings revisions

Bloomberg consensus EPS estimates for 2026E and 2027E have risen 24.8% and 24.3% YTD, respectively. This is likely due to upward revisions to order backlog and earnings forecasts following better-than-expected industry outlook. We are the first foreign broker to cover Acter.

Acter: Bloomberg consensus EPS forecast revisions



Source: Bloomberg

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cleanroom eletromechanical revenue growth (%)	36.9	44.1	(19.0)	30.6	70.7	18.0	15.1	15.8
Water, gas and chemical delivery systems revenue growth (%)	58.5	35.0	11.0	14.8	(16.8)	67.7	46.7	40.4
Consumer industry electromechanical revenue growth (%)	55.2	32.4	(7.3)	5.7	(4.7)	78.7	38.8	33.1
Biomedical revenue growth (%)	(10.8)	83.7	(27.6)	(38.5)	68.0	52.5	8.3	33.1
High-tech equipment and materials sales and services revenue growth (%)	143.9	2.5	(54.0)	14.7	98.0	(44.9)	(5.5)	0.0

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cleanroom electromechanical	11,279	16,256	13,173	17,200	29,359	34,641	39,859	46,141
Water, gas and chemical delivery systems	5,915	7,985	8,861	10,174	8,467	14,198	20,829	29,253
Other Revenue	3,024	4,021	3,026	2,879	3,656	5,005	6,296	8,198
Total Revenue	**20,217**	**28,262**	**25,061**	**30,254**	**41,482**	**53,844**	**66,983**	**83,593**
Other income	0	0	0	0	0	0	0	0
COGS	(16,934)	(23,484)	(20,143)	(23,676)	(33,634)	(41,190)	(50,238)	(61,859)
SG&A	(892)	(1,121)	(1,292)	(1,616)	(1,718)	(2,369)	(2,646)	(3,093)
Other op.expenses	(252)	(335)	(362)	(555)	(427)	(431)	(502)	(627)
Operating profit	**2,139**	**3,323**	**3,264**	**4,407**	**5,703**	**9,853**	**13,598**	**18,014**
Net-interest inc./(exp.)	9	7	95	160	164	227	50	50
Assoc/forex/extraord./others	30	90	157	290	195	92	42	42
Pre-tax profit	**2,179**	**3,420**	**3,516**	**4,857**	**6,062**	**10,172**	**13,690**	**18,106**
Tax	(625)	(921)	(916)	(1,269)	(1,458)	(2,447)	(3,293)	(4,355)
Min. int./pref. div./others	(349)	(566)	(761)	(971)	(1,077)	(1,400)	(1,600)	(1,900)
Net profit (reported)	**1,204**	**1,933**	**1,838**	**2,617**	**3,527**	**6,325**	**8,797**	**11,851**
Net profit (adjusted)	**1,204**	**1,933**	**1,838**	**2,617**	**3,527**	**6,325**	**8,797**	**11,851**
EPS (reported)(TWD)	**21.079**	**16.838**	**15.027**	**21.093**	**28.424**	**50.976**	**70.897**	**95.513**
EPS (adjusted)(TWD)	**21.079**	**16.838**	**15.027**	**21.093**	**28.424**	**50.976**	**70.897**	**95.513**
EPS (adjusted fully-diluted)(TWD)	**19.548**	**15.477**	**14.585**	**20.982**	**28.325**	**50.788**	**70.635**	**95.161**
DPS (TWD)	**15.034**	**8.823**	**6.087**	**10.000**	**14.500**	**35.683**	**49.628**	**66.859**
EBIT	**2,139**	**3,323**	**3,264**	**4,407**	**5,703**	**9,853**	**13,598**	**18,014**
EBITDA	**2,292**	**3,556**	**3,595**	**4,893**	**6,097**	**10,482**	**14,182**	**18,610**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	2,179	3,420	3,516	4,857	6,062	10,172	13,690	18,106
Depreciation and amortisation	123	144	174	196	199	537	542	554
Tax paid	(490)	(638)	(943)	(804)	(1,249)	(2,447)	(3,293)	(4,355)
Change in working capital	(1,809)	(1,544)	2,386	(2,536)	421	(2,122)	(1,006)	(1,946)
Other operational CF items	(82)	304	(775)	530	368	(227)	(50)	(50)
Cash flow from operations	**(80)**	**1,686**	**4,358**	**2,244**	**5,801**	**5,913**	**9,883**	**12,309**
Capex	(24)	(147)	(137)	(151)	(128)	(150)	(180)	(180)
Net (acquisitions)/disposals	(116)	(413)	(285)	(194)	(120)	(122)	(122)	(122)
Other investing CF items	(36)	22	4	21	35	0	0	0
Cash flow from investing	**(176)**	**(537)**	**(418)**	**(324)**	**(213)**	**(272)**	**(302)**	**(302)**
Change in debt	(231)	470	197	(80)	115	0	0	0
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(686)	(859)	(1,416)	(1,303)	(1,861)	(1,799)	(4,428)	(6,158)
Other financing CF items	(211)	1,883	(55)	127	(721)	0	0	0
Cash flow from financing	**(1,128)**	**1,494**	**(1,274)**	**(1,256)**	**(2,467)**	**(1,799)**	**(4,428)**	**(6,158)**
Forex effect/others	0	0	0	0	0	0	0	0
Change in cash	**(1,384)**	**2,643**	**2,665**	**664**	**3,120**	**3,841**	**5,153**	**5,849**
Free cash flow	(31)	1,228	4,900	1,403	5,141	5,763	9,703	12,129

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	3,699	6,411	8,951	9,899	12,996	16,838	21,991	27,839
Inventory	1,560	2,904	2,615	2,302	1,862	2,907	2,848	3,612
Accounts receivable	6,027	8,416	6,619	9,150	11,517	15,574	18,958	23,919
Other current assets	5,971	7,405	6,419	8,614	9,972	1,833	1,833	1,833
Total current assets	**17,258**	**25,136**	**24,604**	**29,964**	**36,346**	**37,153**	**45,630**	**57,204**
Fixed assets	573	690	768	833	890	4,464	4,101	3,728
Goodwill & intangibles	108	92	75	81	71	71	71	71
Other non-current assets	765	1,199	2,139	3,199	4,012	4,012	4,012	4,012
Total assets	**18,704**	**27,117**	**27,587**	**34,078**	**41,320**	**45,700**	**53,814**	**65,015**
Short-term debt	651	896	541	524	591	591	591	591
Accounts payable	6,568	8,708	7,727	9,454	11,676	14,657	16,975	20,754
Other current liabilities	2,554	4,893	4,977	6,160	8,199	7,403	7,403	7,403
Total current liabilities	**9,773**	**14,497**	**13,245**	**16,138**	**20,467**	**22,652**	**24,970**	**28,749**
Long-term debt	0	0	0	3	3	3	3	3
Other non-current liabilities	1,635	1,820	1,257	1,211	1,433	1,433	1,433	1,433
Total liabilities	**11,408**	**16,318**	**14,502**	**17,351**	**21,902**	**24,088**	**26,405**	**30,184**
Share capital	572	586	620	620	620	1,241	1,241	1,241
Reserves/R.E./others	5,063	6,998	8,191	10,350	12,499	12,673	16,870	22,391
Shareholders' equity	**5,635**	**7,585**	**8,812**	**10,970**	**13,119**	**13,913**	**18,110**	**23,632**
Minority interests	1,661	3,214	4,273	5,756	6,299	7,699	9,299	11,199
Total equity & liabilities	**18,704**	**27,117**	**27,587**	**34,078**	**41,320**	**45,700**	**53,814**	**65,015**
EV	128,278	127,364	125,527	126,048	123,561	121,120	117,566	113,618
Net debt/(cash)	**(3,048)**	**(5,514)**	**(8,410)**	**(9,373)**	**(12,402)**	**(16,244)**	**(21,397)**	**(27,245)**
BVPS (TWD)	**98.613**	**66.063**	**72.041**	**88.411**	**105.729**	**112.132**	**145.957**	**190.453**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	44.6	39.8	(11.3)	20.7	37.1	29.8	24.4	24.8
EBITDA (YoY)	30.5	55.2	1.1	36.1	24.6	71.9	35.3	31.2
Operating profit (YoY)	25.8	55.3	(1.8)	35.0	29.4	72.8	38.0	32.5
Net profit (YoY)	24.2	60.5	(4.9)	42.4	34.8	79.3	39.1	34.7
Core EPS (fully-diluted) (YoY)	18.1	(20.8)	(5.8)	43.9	35.0	79.3	39.1	34.7
Gross-profit margin	16.2	16.9	19.6	21.7	18.9	23.5	25.0	26.0
EBITDA margin	11.3	12.6	14.3	16.2	14.7	19.5	21.2	22.3
Operating-profit margin	10.6	11.8	13.0	14.6	13.7	18.3	20.3	21.6
Net profit margin	6.0	6.8	7.3	8.7	8.5	11.7	13.1	14.2
ROAE	22.3	29.2	22.4	26.5	29.3	46.8	54.9	56.8
ROAA	7.1	8.4	6.7	8.5	9.4	14.5	17.7	19.9
ROCE	28.7	33.8	25.8	28.5	30.6	46.7	54.2	56.8
ROIC	50.5	50.9	48.5	54.1	60.3	n.a	n.a	n.a
Net debt to equity	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Effective tax rate	28.7	26.9	26.1	26.1	24.1	24.1	24.1	24.1
Accounts receivable (days)	91.7	93.3	109.5	95.1	90.9	91.8	94.1	93.6
Current ratio (x)	1.8	1.7	1.9	1.9	1.8	1.6	1.8	2.0
Net interest cover (x)	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Net dividend payout	71.3	52.4	40.5	47.4	51.0	70.0	70.0	70.0
Free cash flow yield	n.a.	0.9	3.8	1.1	4.0	4.4	7.5	9.4

Source: FactSet, Daiwa forecasts

Company profile

Established in 1979, Acter Group (Acter) is a leading engineering services provider specialising in cleanroom, electromechanical, and high-tech facility integration solutions for semiconductor, optoelectronics, biotech, green energy, and advanced manufacturing clients across Taiwan, China and the rest of the world. As of 2025, its revenue comprised cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).

Acting on the demand for cleanrooms

Why are cleanrooms a bottleneck in the high-tech industry?

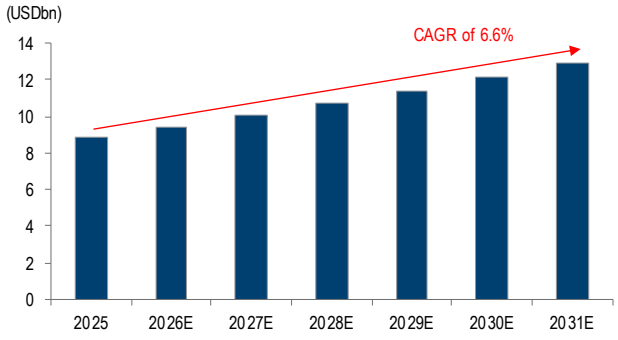
In recent years, cleanrooms have become a critical bottleneck for high-tech industries, especially semiconductor manufacturing, OSAT, AI hardware, and PCB, mainly because of its indispensable nature, severe supply-demand imbalance, labour shortages, and high technical and regulatory barriers.

Cleanrooms are a bottleneck in the high-tech industry, including semiconductor foundry, memory, and electronic assembling

Cleanrooms are an indispensable core infrastructure as they determine product yield and quality stability. Advanced manufacturing nodes in semiconductors, optoelectronics, and biopharmaceuticals require highly particle-free, constant-temperature, and constant-humidity environments to ensure reliable production. As process technologies continue to advance, such as 2nm node (or even newer nodes) and advanced packaging, existing cleanrooms must be rebuilt, and the complexity of building new ones thus increases. This keeps overall demand for cleanrooms at high levels.

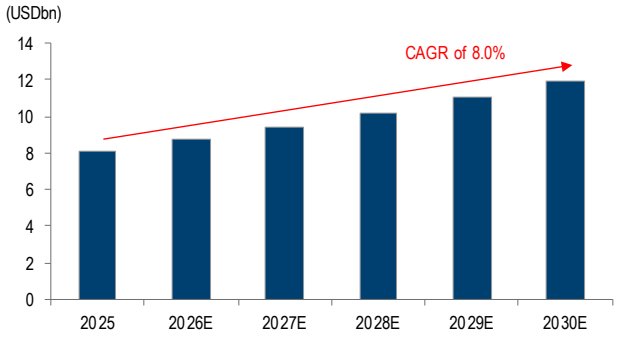
A severe global supply-demand imbalance with demand for cleanroom construction outpacing supply. The surge in AI servers, high-performance computing (HPC), and high-bandwidth memory (HBM) has triggered a global expansion for front-end wafer fabs, packaging and testing facilities, and PCB manufacturing. Order visibility remains extremely strong across the industry. According to Markets and Markets, the global cleanroom market size is set to expand at a CAGR of 6.6% over 2025-31E. At the same time, the global semiconductor cleanroom market size is set to grow at a CAGR of 8.0% over 2025-30E. For instance, Acter's order backlog exceeded TWD66bn at the end of July 2026, up from TWD46bn at the end of 2025, showing the urgency of demand for cleanroom construction.

Global: cleanroom market size and growth



Source: Markets and Markets

Global: semiconductor cleanroom market size and growth



Source: Markets and Markets

Human resources are an additional limitation to capacity growth. Cleanroom engineering is highly technical and labour-intensive, relying on specialised on-site installation and supervision that cannot be replaced by robots. At the same time, the industry faces severe labour shortages and an aging workforce, and younger generations are reluctant to join field operations. Cross-border staff deployment is also less cost-efficient than local hiring.

Cleanrooms involve complex system integration, creating high barriers to entry. A cleanroom construction includes 4 sequential phases, where each phase usually requires totally different areas of expertise. Therefore, the competitive landscape in each area differs. The process begins with **civil & structural (C&S) works** to establish the physical plant, followed by **mechanical, electrical, and plumbing (MEP) works** for essential utility generation, where Acter is one of the leading global experts. After MEP is finished, chemicals/gas/water are then distributed via **main systems** through primary piping network and finally delivered to production equipment through **hook-up systems**. As

global supply chains restructure under N+1 (domestic market + 1) strategies, companies building overseas facilities in places like the US face strict and complicated licensing processes, particularly on fire safety and hazardous chemicals. Without experience dealing with such engineering and safety issues, construction timelines can be severely delayed.

As a result, even with advanced equipment and chip designs, a high-tech company cannot begin production without adequate cleanroom space, skilled engineers, and system-integration capabilities. Therefore, cleanroom engineering contractors hold significant bargaining power and capacity allocation leverage in today's tech supply chain.

IC foundry

In the foundry sector, where the cleanroom supply-demand shortage is the tightest, project activity remains centred on advanced process capacity expansion in Taiwan and Arizona. Simultaneously, domestic advanced node development remains intense, with multi-phase fab builds advancing across Hsinchu, Taichung, and Kaohsiung. Local cleanroom and MEP integration demand is further boosted by advanced packaging facilities in Taiwan (such as Chiayi and Tainan) and upcoming projects in the US are set to contribute revenue for the cleanroom providers in the following years.

Memory

The memory segment is exhibiting robust momentum as well, mainly catalysed by HBM capacity additions. Projects such as the plant in Japan, transitioned to contribution for the cleanroom EPCM providers starting from 2026. In Taiwan, turnkey engineering contracts in Miaoli are also actively underway. This trend is reinforced by stronger-than-expected expansion activity from memory producers in China.

OSAT, PCB and ODM/OEM

For OSAT companies, demand is also surging due to the spill-over effect of advanced packaging protocols and N+1 supply chain restructuring. The OSAT names are expanding their footprint both in Southern Taiwan and in overseas locations including in the US and Southeast Asia. Major engineering contracts signed between OSAT clients and Acter show stable long-term workload for strong cleanroom demand over our forecast horizon. Similarly, overseas expansion for PCB manufacturing is peaking in Southeast Asia, especially in Thailand. We expect Chinese PCB players to also increase their capacity, just like memory companies in the past year. Except for front-end and back-end chip manufacturing, ODM/OEM and data centre infrastructure is also a key growth engine for the EPCM industry. Major Taiwanese ODMs are building production facilities in Texas and Mexico to meet local manufacturing needs. Cleanroom contractors, such as Acter, providing MEP and system integration in Texas are directly benefiting from these projects. The shift toward liquid cooling systems has also increased engineering complexity, favouring few cleanroom companies with the expertise to deliver complex integration for CSPs in both North America and Southeast Asia.

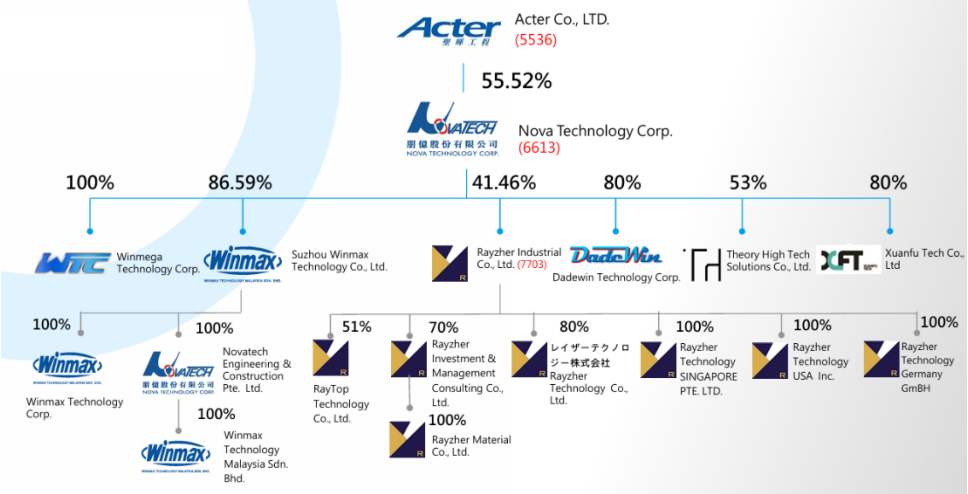
Acter: an integrated cleanroom solutions provider

Acter provides integrated cleanroom solutions via various subsidiaries

Acter Group (Acter), an integrated cleanroom solutions provider, maintains its competitive positioning within the high-tech cleanroom and MEP engineering sector, driven by its group synergy and global footprint. Acter operates as a comprehensive vendor through a group business model. The parent company, Acter, supervises overall cleanroom and MEP services, while subsidiary Nova Technology (Novatech, 6613 TT, not rated) mainly provides chemical primary system and hook-up, and Novatech's subsidiary Rayzher (7703 TT, not rated) provides gas primary system, hook-up and equipment. In China, Acter operates through Acter Technology Integration Group (603163 CH, not rated) and Suzhou Winmax (not listed) for Chinese EPCM (engineering, procurement, construction, and management) and equipment business. Except for C&S (civil and structural) works, where the margin is low (7-8%) and the contract duration is long, Acter's services span nearly the entire cleanroom construction process. Its one-stop solutions model supports customer retention

and improves bid-win rates, particularly for overseas expansion, where customers have few reliable providers to choose from.

Acter: organisational chart



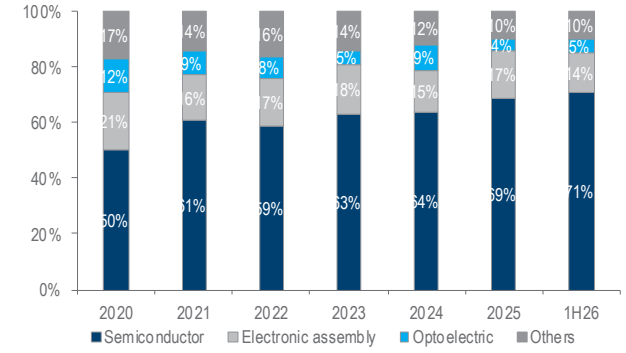
Source: Company, Daiwa

To avoid the risk of over-concentration, Acter diversifies its client portfolio across semiconductor foundries, memory, PCB, OSAT, ODM hardware assembly, CSP data centres and the consumer industry. By sector, semiconductor projects represent the largest share of Acter's portfolio, accounting for 66% of total revenue in 1H26, vs. 69% in 2025. Foundry manufacturing is the largest source of demand for cleanrooms in Taiwan, especially as advanced-node capacity expands at sites in Hsinchu, Taichung and Kaohsiung. In the memory segment, the company executes turnkey engineering builds to expand its capacity aggressively, creating cleanroom opportunities across Taiwan, Japan and Singapore. This multi-sector coverage decreases the risk of cyclical downturns in any individual segment, in our view.

Acter provides cleanroom services for numerous industries, across multiple regions

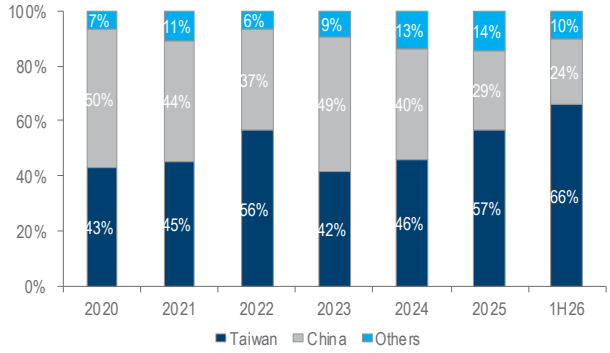
By establishing 33 operating entities across 11 international regions, the company effectively captures capex migration driven by global N+1 supply chain diversification. This allows Acter to support Taiwanese and international clients as they expand manufacturing capacity from Taiwan into Southeast Asia, Japan and the US. Due to the complexity of advanced-node IC manufacturing, packaging and testing, it is especially important for these Taiwanese semiconductor companies to work with familiar cleanroom builders when expanding in the US. This is one of the reasons why Taiwanese cleanroom companies are receiving overseas orders and starting to set up global bases, in our view. Despite higher management, material, and manpower costs in the US, most cleanroom companies are benefitting from the trend by improving margins, given the higher complexity and risks to operate overseas. Across overseas hubs in Texas and Phoenix (expected to begin from 2027), Acter prioritises local hiring and partners with regional civil contractors to shorten the learning curve and to avoid the high overhead. To further address the potential labour shortages in the future, the company has established cross-border academic and vocational partnerships to bring Southeast Asian talent to Taiwan for technical training before deploying them back to their home markets, securing future human capacity to support global expansion projects.

Acter: revenue by clients' industry



Source: Company, Daiwa

Acter: revenue by geography

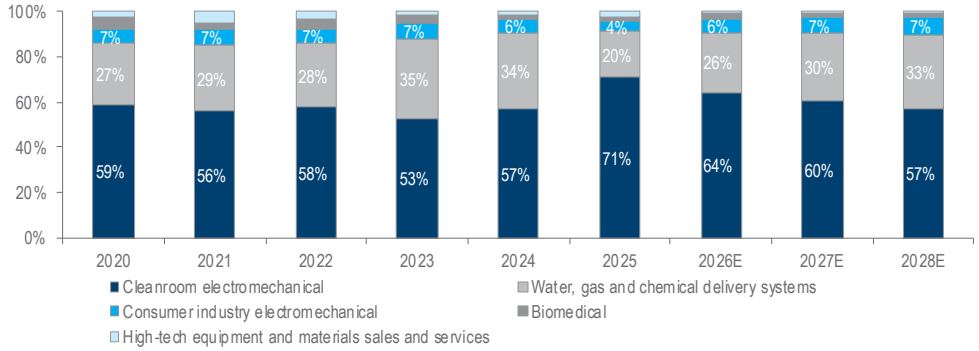


Source: Company, Daiwa

Semiconductor-related business in Taiwan is the major revenue stream for Acter

By product, cleanroom electromechanical services remain Acter's largest source of revenue (accounting for 66% of 1H26 revenue, per the company). These services include cleanroom architectural and MEP engineering, which powers airflow, filtration, environmental regulation and utility controls. Water, gas and chemical delivery systems are its second-largest source of revenue (accounting for 24% of 1H26 revenue, per the company). Contributed mainly by Novatech (6613 TT, not rated) and Rayzher (7703 TT, not rated), these services cover the main systems and hook-up systems, for ultra-pure water (UPW), specialty gases, chemical piping, etc., all of which require highly precise build-up work. Except for cleanroom electromechanical and water, gas and chemical delivery systems, the rest of Acter's revenue comes from the consumer industry services electromechanical (5%), biomedical (3%), and high-tech equipment and materials sales and services (1%).

Acter: revenue by product



Source: Company, Daiwa

Technically, Acter integrates digital transformation and green engineering into its business model, helping it differentiate from peers. Acter deploys digital twin technology and 1:1 3D BIM modelling to detect piping interferences and optimise spatial layout before installation, minimising schedule delays. This makes Acter a more reliable option, in our view, and thus being a favourable supplier, especially for the semiconductor clients that have extremely high quality and swiftness criterion.

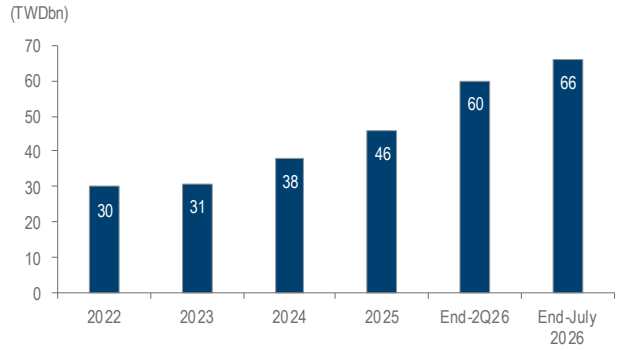
Bright outlook: strong order visibility

Rising orders on hand due to supply-demand imbalance for cleanrooms

Acter's order backlog has increased from TWD55bn in 1Q26 to TWD60bn in 2Q26 and TWD66bn in July 2026. We expect this figure to further increase in the following quarters, given the supply-demand imbalance in the cleanroom sector. This is composed of semiconductor (63%), electronic assembly (11%), consumer (9%) and others (17%), per the company. In 2H25, the company accelerated the digestion of small-scale projects below TWD1bn, and, as a result, starting from 1H26, it successfully shifted its project

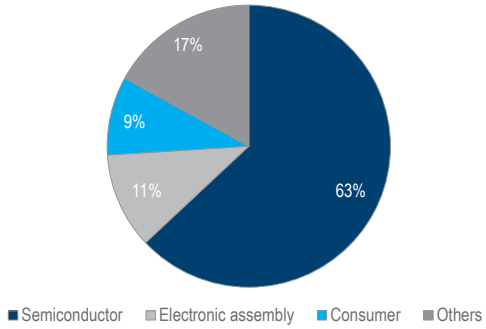
selection strategy to focus on large-scale projects valued at TWD1bn or above while reducing smaller contracts. This drives the growth in top-line.

Acter: order backlog



Source: Company, Daiwa

Acter: order backlog by clients' industry



Source: Company, Daiwa

OSAT customers have large order backlogs with Acter, which should drive future growth for the company

Back-end packaging, cloud infrastructure and hardware assembly are additional growth drivers. Acter serves as a contractor for both Taiwanese and overseas OSAT providers, including TWD21.7bn contracts YTD, awarded by a major OSAT client vs. TWD16.6bn contract obtained from the same client in 2025. In cloud infrastructure, Acter maintains qualified supplier status with US CSPs, providing mechanical, electrical and plumbing (MEP) integration, especially for those data centres that implement liquid-cooling systems. The company also wins factory construction projects for Taiwanese ODMs across the US and Southeast Asia, as well as PCB facility projects in Thailand and China. Non-tech sectors, including biotechnology, commercial buildings, smart logistics hubs and medical facilities, comprise 9% of the total backlog, per management.

Acter: contracts signed with one major OSAT client

Date of announcement	Type	Constructor	Amount (TWDm)
2026/1/7	EPCM	Acter	596
2026/1/16	EPCM	Acter	969
2026/1/22	EPCM	Acter	880
2026/2/23	EPCM	Acter	659
2026/4/15	EPCM	Acter	1,166
2026/4/27	EPCM	Acter	1,075
2026/5/6	EPCM & equipment	Acter	365
2026/5/19	EPCM	Acter	535
2026/5/27	EPCM	Novatech	947
2026/6/1	EPCM	Acter	550
2026/6/22	EPCM & equipment	Acter	349
2026/6/25	EPCM	Novatech	508
2026/7/2	EPCM	Acter	662
2026/7/8	EPCM	Acter	679
2026/7/13	EPCM	Acter	1,990
2026/7/20	EPCM	Acter	1,074
2026/8/5	EPCM	Acter	2,046
2026/8/5	EPCM	Novatech	714
2026/8/10	EPCM	Acter	841
2026/8/12	EPCM	Acter	582
2026/8/17	EPCM & equipment	Acter	389
2026/9/2	EPC	Acter	542
2026/9/17	EPC & equipment	Acter	462
2026/9/30	EPC	Acter	2,037
2026/10/1	EPC	Acter	1,055
Total for 2026 YTD			**21,675**
Total for 2025			**16,554**

Source: Companies, Daiwa

Competitive advantages

High entry barriers to the segment

Comprehensive product portfolio, empowered by subsidiaries

The supply of cleanroom services is limited due to high entry barriers in the segment

Acter provides a one-stop EPCM solution, leveraging its subsidiaries to deliver complete solutions. Within the group, the parent company Acter focuses on overall cleanroom and MEP system integration; subsidiary Novatech provides primary chemical systems and hook-up systems; and sub-subsidiary Rayzer specialises in primary specialty-gas systems and hook-up systems. In China, Suzhou Winmax, also a subsidiary of Novatech, focuses on the supply of wet processing and chemical handling equipment. This integrated model covers cleanroom engineering integration, water, power, chemical and gas-utility systems, as well as post-completion validation and maintenance, ensuring cross-system compatibility while mitigating risks associated with multi-contractor hiring.

Under global supply chain diversification trends, this unified group strategy strengthens order-winning capabilities in overseas markets. Clients expanding internationally face regulatory, linguistic and supply-chain complexities, favouring partners that offer one-stop services with a proven cross-border track record.

A diverse client base helps Acter weather economic cycles

Acter's main competitive advantage is its diversification of clients by industry and region. This guides the company through the peaks and troughs of the economy, which is especially crucial given the strong cyclicality of the semiconductor/high-tech industry. Acter serves a broad client base across semiconductors, OSAT, PCB, electronics assembly (ODM), CSP data centres, biopharmaceuticals and even the consumer industry (including commercial real estate and logistics), whereas many peers rely on a few large foundry clients. When semiconductor demand slows, capacity expansion in the PCB or data centre sectors offset the decline. Additionally, Acter avoids concentration risk and margin compression from major foundry accounts, as projects from leading foundries yield lower margins; by serving niche markets like US-based CSPs, the company achieves higher gross margins.

Geographically, Acter operates across Taiwan, China, Southeast Asia, the US and other countries. The company has OSAT and foundry customers in Taiwan and China, PCB clients expanding into Thailand, and ODM and data centre clients establishing operations in the US. Across different countries, local hiring and local supply chain implementation can further reduce costs and increase efficiency relative to peers reliant on a cross-border labour force.

Labour shortages

Amid global labour shortages, Acter minimises the operational impact and turns workforce constraints into a moat through localised talent development. Since engineering labour cannot easily be automated, Acter has built a cross-border talent pipeline via industry-academia partnerships. The company recruits Southeast Asian students for specialised M&E engineering training in Taiwan, deploying them to global project sites after their graduation to solve labour bottlenecks. This structure maintains an engineering team with an average tenure of more than 7 years, creating an operational barrier in an industry with a high employee turnover rate.

Compared with peers relying on high-cost personnel sent from Taiwan, we note that Acter's local hiring strategy creates cost and efficiency advantages. Prioritising compliant local hiring across overseas markets reduces travel expenses while accelerating integration with local supply chains and regulations. Localised teams also handle post-completion services and technical support directly on-site, building long-term client trust in overseas markets.

Numerous competitors, each with different specialisation

Some competitors, unlike Acter, lack a diversified client base

Acter faces different competitors across its main product lines. In the EPCM segment, including cleanroom and MEP works, key competitors include UIS (2404 TT, not rated), L&K (6139 TT, not rated), Yankey (6691 TT, not rated) and Jiuhan (6903 TT, not rated). Despite having business models similar to Acter, UIS, Yankey and Jiuhan focus more on major semiconductor players in both Taiwan and the US, and L&K concentrates on ASEAN semiconductor projects. In the equipment/material sales and agency space, Acter mainly competes with Marketech (6196 TT, not rated). In the water, gas and chemical delivery systems area, through Novatech and Rayzher, Acter group competes with Marketech, Mega Union (6944 TT, not rated), Taiwan Puritic (6826 TT, not rated), Trusval (6667 TT, not rated) and Cica-Huntek (6725 TT, not rated).

In our view, despite the competitive environment, Acter maintains a formidable competitive moat built on several key advantages. We attribute this to its more than 45 year experience in the industry, enabling it to develop specialised expertise that is hard for competitors to replicate.

Acter: peer comparison (cleanroom related)

Company	Ticker	Market Cap (USDm)	Market	Main thermal product	2025 Product mix	2025 Segment breakdown	2025 Market breakdown
Acter	**5536 TT**	**4,070**	**Taiwan**	**EPCM turnkey solutions**	**Cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).**	**Semiconductor (69%), electronics assembly (17%), optoelectronics (10%), others (4%)**	**Taiwan (57%), China (29%), others (14%)**
Novatech	6613 TT	708	Taiwan	Gas and chemical delivery systems and wet processing equipment	Equipment (46%), engineering (47%), eco-integration (7%)	Semiconductor (85%), optoelectronics (7%), others (8%)	China (50%), Taiwan (45%), others (5%)
Rayzher	7703 TT	225	Taiwan	Gas delivery systems	Hook-up systems (52%), primary system (28%), small-scale works (18%), equipment & material sales (3%)	Semiconductor, equipment and material providers, gas system providers, optoelectronics	Taiwan (99%), China & Japan (1%)
Acter Technology Integration	603163 CH	1,102	China	EPCM turnkey solutions	Cleanroom electromechanical (65%), other electromechanical (32%), equipment & material sales (2%)	Semiconductor (60%), precision manufacturing (34%), optoelectronics (2%), others (4%)	China (58%), others (42%)
UIS	2404 TT	7,957	Taiwan	EPCM turnkey solutions	System integration (100%)	Foundries, memory	Taiwan (67%), U.S. (25%), China (3%), Singapore (3%), Japan (1%)
L&K	6139 TT	5,820	Taiwan	EPCM turnkey solutions	Turnkey solutions (63%), electromechanical (25%), cleanroom builds (8%), delivery systems (4%)	Semiconductor (74%), railway/highway/airport (10%), MRT (7%), commercial (4%), energy (2%), expropriation (2%), others (1%)	ASEAN (59%), Taiwan (34%), China (7%)
Marketech	6196 TT	4,634	Taiwan	Gas and chemical delivery systems, Cleanroom equipment manufacturing/representative, EPCM turnkey solutions	Gas and chemical delivery systems (39%), representative for equipment (25%), equipment manufacturing (18%), turnkey solutions (18%)	Semiconductor (81%), optoelectronics (6%), others (13%)	Taiwan (45%), China (18%), U.S. (17%), others (20%)
Yankey	6691 TT	3,253	Taiwan	EPCM turnkey solutions	CSA (civil, structural & architectural), MEP (mechanical, electrical & plumbing), CR (cleanroom)	Semiconductor (58%), PCB (15%), optoelectronics (11%), data centres (6%), office (6%), electronic components (2%), others (1%)	Taiwan (89%), China (6%), others (5%)
Jiuhan	6903 TT	684	Taiwan	EPCM turnkey solutions	EPCM turnkey solutions (100%)	AI-related (83%), public works (17%)	Taiwan (100%)
Mega Union	6944 TT	2,842	Taiwan	UWS (ultrapure water systems), wastewater recycling, water treatment engineering	Water treatment engineering (80%), O&M/chemicals/consumables (20%)	Semiconductor and other high-tech industry	Taiwan (69%), U.S. (19%), Singapore (10%), China (2%)
Taiwan Puritic	6826 TT	1,223	Taiwan	Gas delivery systems	Gas primary system (83%), gas equipment (17%)	High-tech industry (100%)	Taiwan (73%), Americas (18%), Asia (9%)
Trusval	6667 TT	468	Taiwan	Water, gas and chemical delivery systems	Primary and hook-up systems (92%), eco-integration (4%)		Taiwan (96%), others (6%)
Cica-Huntek	6725 TT	253	Taiwan	Chemical delivery systems	Primary and hook-up systems (73%), system integration engineering (20%), system maintenance (7%)		Taiwan (87%), Asia (13%)

Source: Companies, Bloomberg, Daiwa

Financial analysis

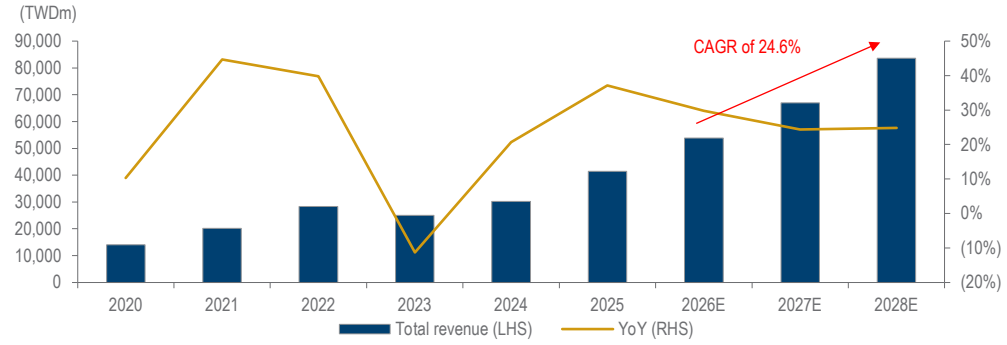
We forecast the company's net profit to rise at a CAGR of 36.9% over 2026-28E, driven by a revenue CAGR of 24.6% due to surging demand from the high-tech industry (an increase in the order backlog to TWD66bn at the end of July 2026 vs. TWD46bn at end-2025) and a 3.3pp expansion in operating margin over 2026-28E, driven by the gross margin improvement from more favourable project wins and operating leverage.

Strong order visibility, with revenue expected to grow across all sectors over 2026-28E

We forecast Acter's revenue to record a CAGR of 24.6% over 2026-28E

We forecast Acter's revenue to increase at a CAGR of 24.6% over 2026-28E, driven by scale expansion and a growing demand from various industries, including semiconductor, electronic assembly, optoelectronics and others.

Acter: revenue growth trend



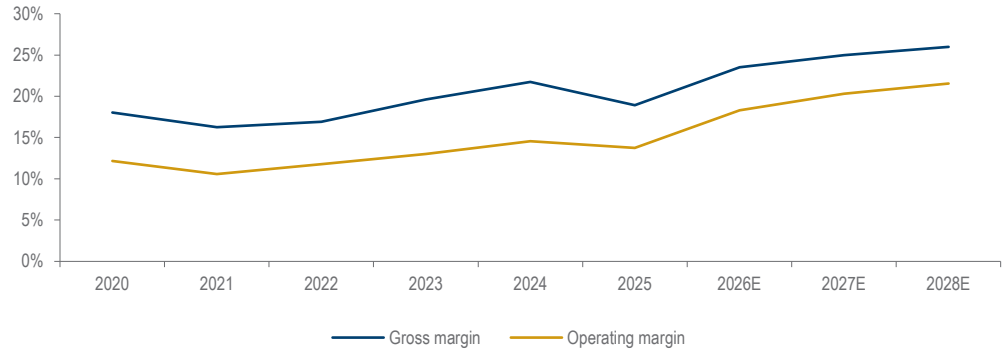
Source: Company, Daiwa forecasts

Gross- and operating-margin expansion trend

We forecast an upward operating-margin trend over 2026-28E

Over 2026-28E, we forecast Acter's gross and operating margins to expand on rising revenue scale and a higher complexity of cleanrooms for advanced manufacturing. We note that the gross margin from high-tech cleanrooms is higher than that from consumer industries, such as commercial buildings and biopharmaceuticals. Thus, we estimate the company's gross margin to reach 26.0% in 2028E from 18.9% in 2025 and its operating margin to reach 21.6% in 2028E from 13.7% in 2025.

Acter: gross- and operating-margin trend



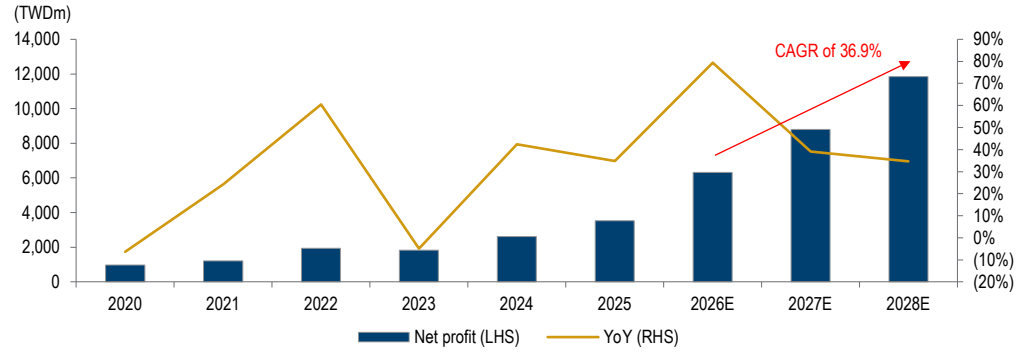
Source: Company, Daiwa forecasts

Net margin likely to rise over 2026-28E

We forecast a net profit CAGR of 36.9% for Acter over 2026-28E

We forecast Acter's net profit to rise to TWD8,797m in 2027E and TWD11,851m in 2028E (from TWD6,325m in 2026E). We expect its net margin to come in at 13.1% in 2027E and 14.2% in 2028E (vs. 11.7% in 2026E) and for its net profit to grow at a CAGR of 36.9% over 2026-28E.

Acter: net profit trend



Source: Company, Daiwa forecasts

Quarterly P&L

Acter: quarterly P&L

(TWDm)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Revenue	11,603	13,381	13,959	14,900	14,900	16,128	17,242	18,713
Gross profit	2,945	2,762	3,211	3,736	3,427	4,032	4,310	4,976
Operating profit	2300	2126	2499	2930	2697	3258	3517	4125
Profit before tax	2371	2232	2569	3000	2720	3281	3540	4148
Net profit	1583	1361	1547	1835	1786	2092	2269	2651
Basic EPS (TWD)	12.76	10.97	12.46	14.79	14.39	16.86	18.28	21.36
Margin								
Gross margin	25.4%	20.6%	23.0%	25.1%	23.0%	25.0%	25.0%	26.6%
Operating margin	19.8%	15.9%	17.9%	19.7%	18.1%	20.2%	20.4%	22.0%
Pre-tax margin	20.4%	16.7%	18.4%	20.1%	18.3%	20.3%	20.5%	22.2%
Net margin	13.6%	10.2%	11.1%	12.3%	12.0%	13.0%	13.2%	14.2%
YoY								
Revenue	35.6%	22.7%	26.4%	35.8%	28.4%	20.5%	23.5%	25.6%
Gross profit	91.7%	31.5%	63.1%	66.6%	16.4%	46.0%	34.3%	33.2%
Operating profit	124.6%	35.5%	58.3%	91.1%	17.3%	53.3%	40.8%	40.8%
Profit before tax	109.3%	45.8%	46.2%	82.8%	14.7%	47.0%	37.8%	38.3%
Net profit	146.3%	57.9%	51.6%	83.1%	12.8%	53.7%	46.7%	44.5%
QoQ								
Revenue	5.7%	15.3%	4.3%	6.7%	0.0%	8.2%	6.9%	8.5%
Gross profit	31.3%	-6.2%	16.3%	16.4%	-8.3%	17.7%	6.9%	15.4%
Operating profit	50.0%	-7.6%	17.5%	17.2%	-7.9%	20.8%	8.0%	17.3%
Profit before tax	44.5%	-5.9%	15.1%	16.8%	-9.3%	20.6%	7.9%	17.2%
Net profit	58.0%	-14.1%	13.7%	18.6%	-2.7%	17.1%	8.5%	16.8%

Source: Company, Daiwa forecasts

Intact demand to support revenue growth over our forecast horizon; ROE expansion underway

We expect stronger cash balances and FCF over 2026-28E due to better profitability

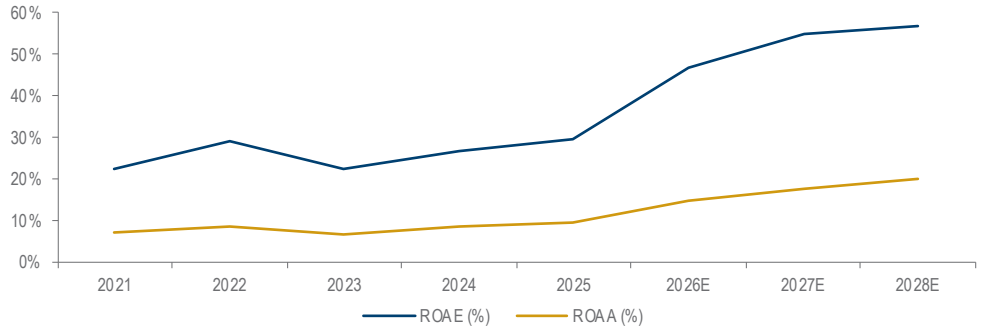
We forecast Acter's total cash and equivalents to reach TWD16,838m (36.8% of its total assets) in 2026E and increase to TWD27,839m in 2028E (42.8% of total assets) with net cash likely to come in at TWD16,244m in 2026E and TWD27,245m in 2028E.

After factoring in our 2027 and 2028 capex assumptions of TWD180m, we expect Acter's free cash flow (FCF) to improve to TWD12,129m in 2028E, from TWD5,763m in 2026E. We believe the capacity expansion cost will initially be low and that the company will continue to benefit from the better profitability.

Strong ROE outlook over 2026-28E

We expect Acter's ROE to improve to a record-high of 56.8% in 2028E from 29.3% in 2025 (following steady annual increases to 46.8% in 2026E and 54.9% in 2027E), which we attribute to margin expansion from the revenue size increase and gross margin expansion led by increases in cleanroom complexity.

Acter: ROAE and ROAA trend



Source: Company, Daiwa forecasts

Valuation and risks

Undemanding valuation, alongside its leading global positions in the OSAT and memory segments and an optimistic industry outlook

12-month TP of TWD1,415, based on a target PER of 20x

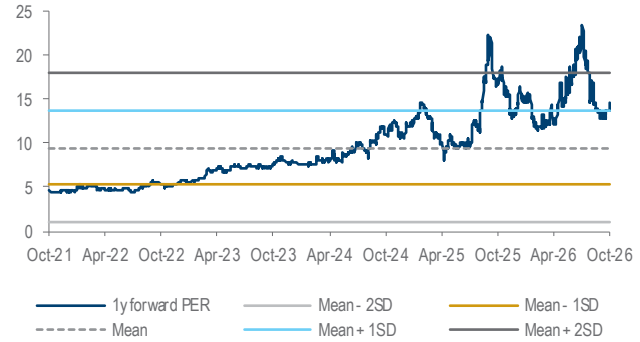
We initiate coverage of Acter with a Buy (1) rating and 12-month TP of TWD1,415, based on a target PER of 20x applied to our one-year-forward EPS forecast. This target multiple is higher than the average of its past-3-year 1-year-forward PER of 12.2x (ranged from 7.4x to 23.5x) and the average of its past-5-year 1-year-forward PER of 9.6x (ranged from 4.4x to 23.5x), which we view as undemanding as we forecast a 36.9% net profit CAGR over 2026-28E, implying a 0.54x PEG. This is supported by its business expansion in the next 2-3 years. We see potential for a rerating of the stock, given Acter's global leadership in cleanroom EPCM, particularly in the OSAT and memory industries.

The stock is trading currently at a 15.8x 1-year-forward PER vs. its earnings CAGR of 36.9% over 2026-28E, translating to a 0.43x PEG ratio. Its share price has underperformed TAIEX Index by 32.7% in 3Q26, mainly due to its weaker-than-expected 2Q26 results. However, given the potential improvement in sector sentiment following TSMC's results on 15 October 2026, along with the likelihood that Acter's revenue momentum will strengthen in the coming months, we recommend that investors buy the stock.

Acter: past-3-year-forward PER (average: 12.0x)

Source: Bloomberg, Daiwa estimates

Acter: past-5-year-forward PER (average: 9.5x)



Source: Bloomberg, Daiwa estimates

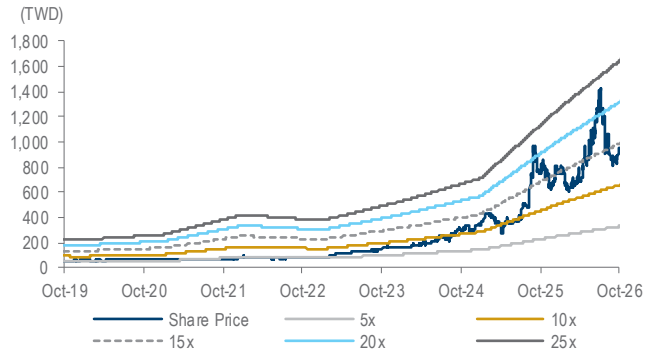
We compare Acter with other leading cleanroom EPCM companies such as UIS (2404 TT, TWD1,330.0, not rated), L&K (6139 TT, TWD787.0, not rated) and Yankey (6691 TT, TWD776.0, not rated). UIS, L&K and Yankey are trading at a PER of 14.7x (earnings CAGR of 40.7% over 2025-27E), 9.5x (earnings CAGR of 36.9% over 2026-28E) and 15.1x (earnings CAGR of 53.6% over 2025-27E), on their 1-year-forward EPS, respectively.

Stronger earnings growth outlook and more room to grow vs. peers

Acter's peers are trading at 18.7x for 2026E and 14.2x for 2027E. We forecast Acter to deliver an EPS CAGR of 36.9% over 2026-28E vs. 31.5% CAGR for the segment over the same period.

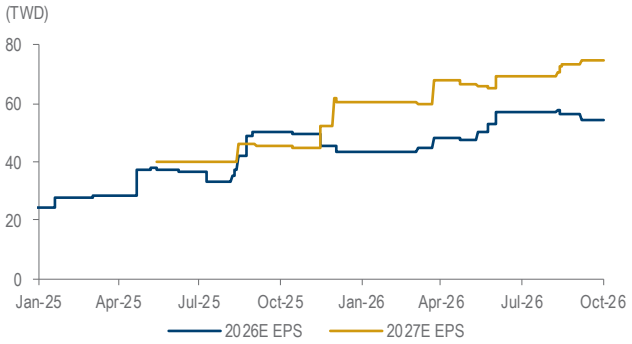
We believe our target one-year-forward PER of 20x (equivalent to a 0.54x PEG using 2026-28E EPS CAGR) is undemanding as: **1) Acter is ramping up capacity to meet growing demand from high-tech customers and is in an expansion cycle;** and **2) its leading position among OSAT and memory clients warrants a valuation premium, compared to its global peers**.

Acter: 1-year-forward PER bands



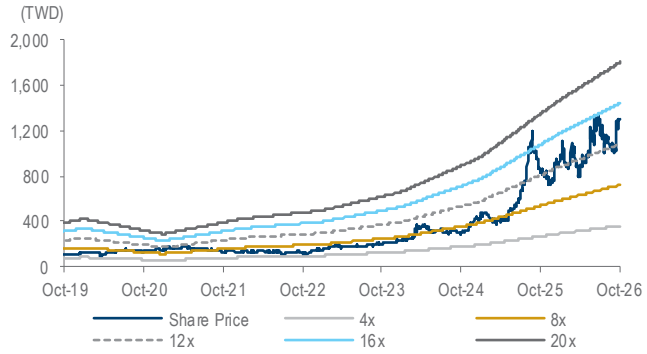
Source: Bloomberg, Daiwa forecasts

Acter: earnings revision trend



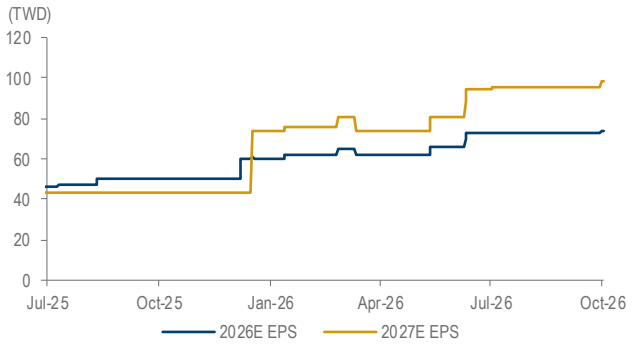
Source: Bloomberg, Daiwa

UIS: 1-year-forward PER bands



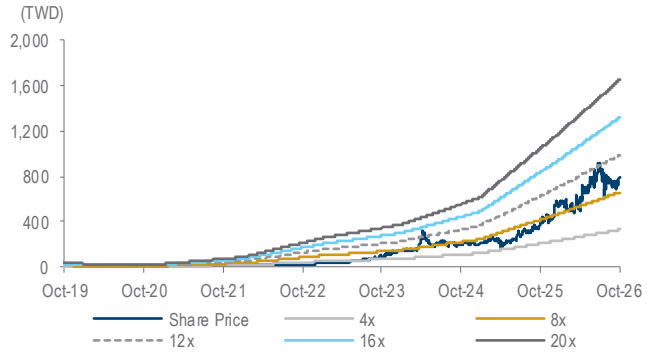
Source: Bloomberg, Daiwa forecasts

UIS: earnings revision trend



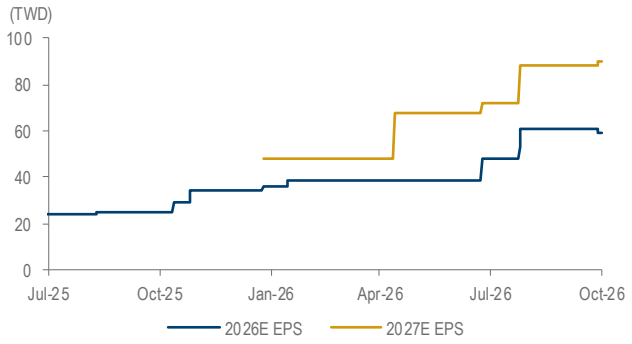
Source: Bloomberg, Daiwa

L&K: 1-year-forward PER bands



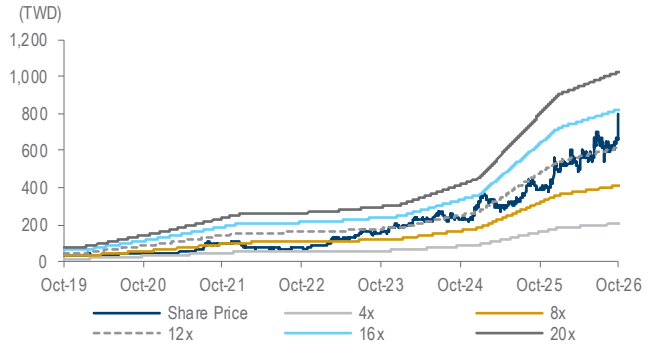
Source: Bloomberg, Daiwa forecasts

L&K: earnings revision trend



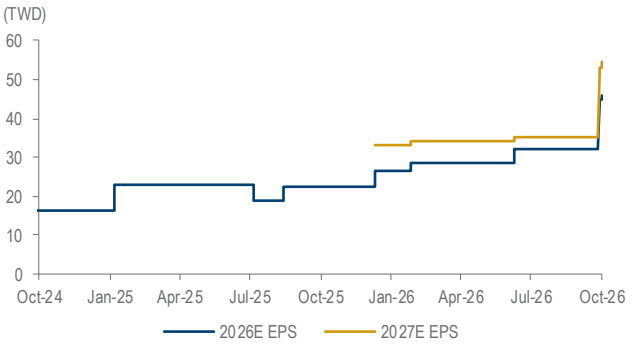
Source: Bloomberg, Daiwa

Yankey: 1-year-forward PER bands



Source: Bloomberg, Daiwa forecasts

Yankey: earnings revision trend



Source: Bloomberg, Daiwa

Acter: peers valuation comparison

Companies	Bloomberg code	Price (lc) 7/10/26	Daiwa rating	Market cap (USDm)	D. turnover (USDm)	PER (x) 2026E	PER (x) 2027E	PER (x) 2028E	EPS CAGR 2026-28E	ROE (%) 2026E	ROE (%) 2027E	ROE (%) 2028E	PBR (x) 2026E	PBR (x) 2027E	PBR (x) 2028E	Dividend yield (%) 2026E	Dividend yield (%) 2027E	Dividend yield (%) 2028E
Acter*	***5536 TT***	***1,045.0***	***Buy***	***4,070***	***33.4***	***20.6***	***14.8***	***11.0***	***36.9***	***46.8***	***54.9***	***56.8***	***9.3***	***7.2***	***5.5***	***3.4***	***4.7***	***6.4***
Novatech	6613 TT	290.0	Non rated	708	3.8	15.5	11.6	8.1	38.4	27.9	34.3	n.a.	n.a.	n.a.	n.a.	5.1	6.8	n.a.
Rayzher	7703 TT	206.0	Non rated	225	0.6	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Acter Technology Integration	603163 CH	73.9	Non rated	1,102	9.6	33.8	22.3	17.6	38.5	17.0	22.4	24.9	5.7	4.9	4.2	1.6	2.4	3.7
UIS	2404 TT	1,330.0	Non rated	7,957	68.0	17.9	13.5	11.0	27.7	56.0	48.0	47.3	n.a.	n.a.	n.a.	4.6	5.5	n.a.
L&K	6139 TT	787.0	Non rated	5,820	81.6	13.3	8.8	7.1	36.9	64.0	64.1	n.a.	n.a.	n.a.	n.a.	5.2	8.3	n.a.
Marketech	6196 TT	670.0	Non rated	4,634	27.8	23.5	20.5	19.3	10.4	34.5	32.3	30.2	7.2	6.1	5.4	2.3	2.6	2.6
Yankey	6691 TT	776.0	Non rated	3,253	16.8	16.9	14.2	n.a.	n.a.	67.2	62.1	n.a.	8.6	5.5	n.a.	4.9	5.8	n.a.
Jiuhan	6903 TT	327.0	Non rated	684	4.7	9.8	7.4	n.a.	n.a.	46.9	42.7	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Mega Union	6944 TT	908.0	Non rated	2,842	12.2	22.9	18.2	n.a.	n.a.	44.1	46.0	n.a.	6.6	8.2	n.a.	2.7	3.2	n.a.
Taiwan Puritic	6826 TT	455.5	Non rated	1,223	1.6	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Trusval	6667 TT	278.5	Non rated	468	2.4	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Cica-Huntek	6725 TT	217.0	Non rated	253	0.8	12.5	10.5	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Industry average					***18.7***	***14.2***	***12.4***	***31.5***	***44.9***	***45.2***	***39.8***	***7.5***	***6.4***	***5.0***	***3.7***	***4.9***	***4.2***

*Source: Bloomberg; *Daiwa forecasts*

Our 12-month TP of TWD1,415 is also supported by our DCF valuation

We also reference a DCF methodology (see charts below), which indicates a DCF-derived fair value estimate of TWD1,454/share, which supports our 12-month TP of TWD1,415.

In our DCF calculation, we apply a WACC of 19.3% to Acter's FCF over 2026-35E and a terminal growth rate of 7.0% thereafter. We expect the company to continue to record steadily increasing FCF over 2026-35E.

Acter: DCF calculation

Target gearing (debt/capital)	**0.7%**
Market risk premium (%)	**18.0%**
Risk-free rate (%)	**5.0%**
Cost of debt (%)	**5.0%**
Cost of equity (%)	**19.4%**
WACC (%)	**19.3%**
Terminal growth rate	7.0%
Terminal WACC	19.3%
Enterprise value (TWDm)	167,994
Less: net debt (as of end-2025) (TWDm)	-12,402
Equity value (TWDm)	180,396
No. shares (m)	124
Per share equity value (TWD)	**TWD1,454**

Source: Company, Daiwa estimates

Acter: DCF sensitivity

		WACC (%) 18.9	19.1	19.3	19.5	19.7
Terminal growth rate (%)	8.0	1,571	1,545	1,520	1,496	1,473
	7.5	1,533	1,509	1,486	1,463	1,442
	7.0	1,498	1,476	1,454	1,433	1,413
	6.5	1,466	1,445	1,425	1,405	1,386
	6.0	1,437	1,417	1,398	1,379	1,361

Source: Daiwa estimates

Acter: free cash flow forecasts

(TWDm)	2026E	2027E	2028E	2029E	...	2035E
EBITDA	**10,482**	**14,182**	**18,610**	**22,332**	...	**66,683**
Less: taxation (EBIT*tax rate)	-2,447	-3,293	-4,355	-5,226	...	-15,605
Plus: decrease in working capital	-2,122	-1,006	-1,946	-1,946	...	-1,946
Less: capital expenditure	-150	-180	-180	-180	...	-180
Free cash flow (TWDm)	**5,763**	**9,703**	**12,129**	**14,980**	...	**48,952**
Discount rate	19.3%	19.3%	19.3%	19.3%	...	19.3%
Discount factor	0.96	0.80	0.67	0.56	...	0.20
NPV of free cash flow	**5,510**	**7,776**	**8,145**	**8,432**	...	**9,558**

Source: Daiwa estimates and forecasts

Risks to our call

Primary/key risks:

Weaker-than-expected expansion in semiconductor industry

The biggest downside risk to our call for Acter is weaker-than-expected expansion from semiconductor clients, especially in advanced-node foundries, advanced packaging and OSAT. If global demand for AI servers, HPC or HBM slows, or if clients delay fab build-ups and overseas expansion plans, it will directly pressure order inflows and backlog conversion for Acter.

Project execution delays or cancellations

Acter's revenue recognition is tied to large, complex EPCM and integrated engineering projects, which are often subject to customer-driven delays. Any postponement in project

schedule, construction bottlenecks or cancellation of large projects could defer revenue recognition and hurt overall profit. This risk is more obvious in overseas projects, where licensing, fire-safety approvals and hazardous-chemical regulations are more complicated.

Key risks to our call: weaker-than-expected expansion in semiconductor industry, project delays or cancellations, and higher-than-expected raw-material price hikes

Higher-than-expected raw-material price hikes

Raw-material costs (especially copper and oil-related costs) account for 50-60% of COGS and thus the price volatility makes costs difficult to manage. Acter needs to maintain supplier relations, negotiate appropriate price-adjustment mechanisms with clients, and invest in R&D to identify composite material substitutes, thereby reducing reliance on any single component.

Secondary risks:

Labour issues or larger-than-expected increases in labour costs

Cleanroom engineering is highly labour-intensive and relies on on-site installation and supervision. Although Acter has built a local hiring and talent-development model, a larger-than-expected shortage of engineers or rising wage pressure could still raise costs and constrain capacity expansion. Any inability to secure enough skilled labour may also limit the company's ability to convert orders into revenue in time.

Margin pressure from lower-margin projects or mix shift

Acter's business mix includes a few low-margin C&S services as well as larger EPCM and system integration contracts. If the company takes on a higher proportion of lower-margin projects, or if competition intensifies in certain geographies or customer segments, gross margin could come under pressure. In addition, an unfavourable product mix shift towards less profitable work could offset benefits from backlog growth.

Overseas expansion and regulatory risks

Acter is expanding across markets in the US, Southeast Asia, Japan and China, which exposes it to FX movements, local compliance requirements, tax complexity and regulatory uncertainty. Delays in licensing, changes in safety standards or local-content requirements could increase project costs and reduce execution efficiency. Any misstep in managing overseas operations could damage profitability and client retention.

Appendix

Company background

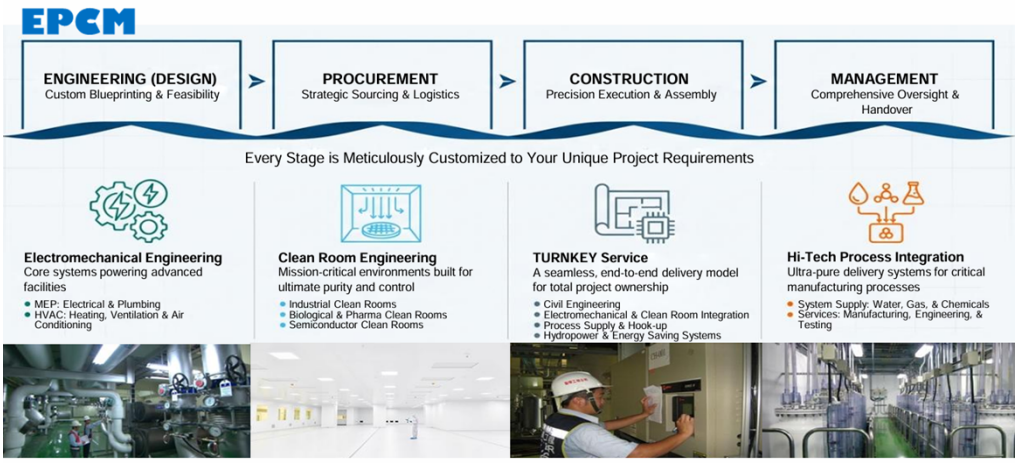
History

Comprehensive cleanroom EPCM provider in Taiwan; dominates both high-tech market and consumer industry

Acter (5536 TT), founded in 1979 and headquartered in Taichung, Taiwan, is a cleanroom EPCM provider with over four decades of experience. Starting with early mechanical and electrical (M&E) engineering through vertical and horizontal integration, Acter has now developed into an engineering technology group with cross-border execution capabilities. A cleanroom building process begins with **civil & structural (C&S) works** to establish the physical plant, followed by **mechanical, electrical, and plumbing (MEP) works** for essential utility generation, where Acter is one of the experts. After MEP is finished, chemicals/gas/water are then distributed via main systems **through primary piping networks**, and finally delivered to production equipment through **hook-up systems**. The company's core competency is providing one-stop integrated total solutions, spanning services, primary piping networks and hook-up systems, leaving C&S services to other engineering peers, given the nature of the latter segment's low margin and long timespan.

In terms of business model, Acter delivers EPCM (Engineering, Procurement, Construction, Management) capabilities. From initial customised planning and feasibility assessments to strategic procurement, installation, and post-completion maintenance, Acter builds production environments tailored to client process requirements.

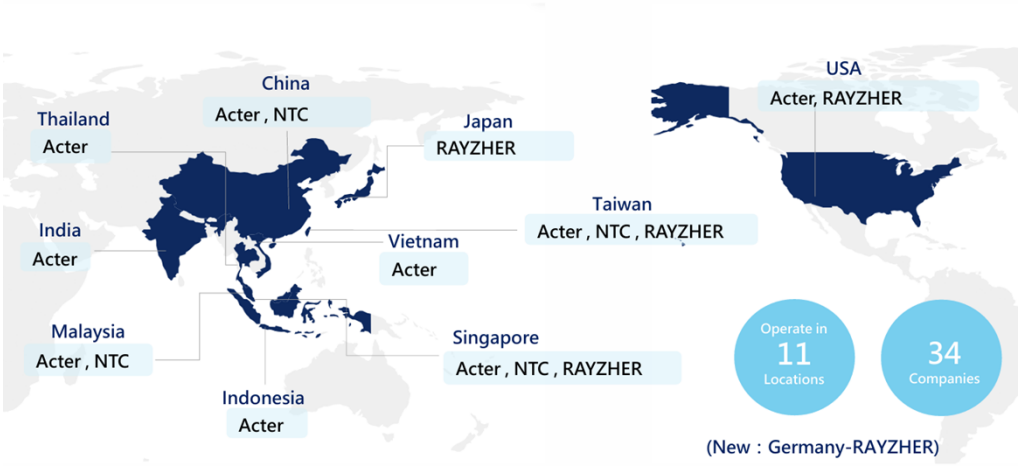
Acter: EPCM solutions offered



Source: Company, Daiwa

Regarding global expansion, Acter actively responds to the global supply chain's "N+1" strategy, operating across 11 regions with 33 companies globally. Its service network spans Taiwan, China, Southeast Asia (including Thailand, Malaysia, Vietnam, Indonesia, and Singapore), Japan, India, and the US. The company leverages a group synergy strategy by integrating resources from key subsidiaries – such as Novatech (6613 TT, not rated; provides chemical primary and hook-up systems), Rayzher (7703 TT, not rated; provides special gas primary and hook-up supply systems), and Acter Group (603163 CH, not rated; provides cleanroom engineering in China) – to form a complete technical ecosystem, ensuring cross-border capability and quality.

Acter: global footprint



Source: Company, Daiwa

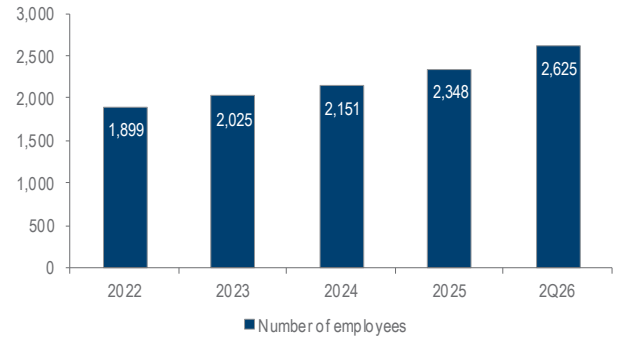
In 2Q26, Acter classified revenue as: cleanroom electromechanical (63%), water, gas and chemical delivery systems (28%), consumer industry electromechanical (6%), biomedical (2%) and high-tech equipment & materials sales & services (1%). In 2025, the product mix was: cleanroom electromechanical (71%), water, gas and chemical delivery systems (20%), consumer industry electromechanical (4%), biomedical (2%) and high-tech equipment & materials sales & services (3%).

Young, capable human resource is a crucial asset for Acter

Acter serves a diversified client base divided into high-tech and consumer divisions. The high-tech division focuses on advanced sectors including semiconductors, OSAT, PCB, AI data centres (for CSPs), and electronics assembly. The consumer division extends into healthcare, premium commercial properties, logistics facilities, and green engineering. This multi-industry strategy allows the company to capitalise on tech sector upgrades while maintaining resilient operations across technology (especially semiconductor) cycles.

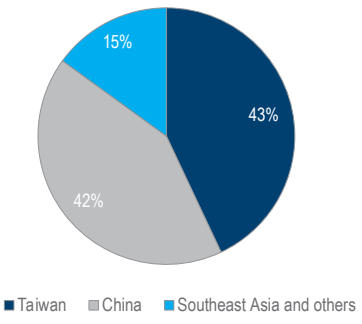
Acter adopts a management strategy involving local hiring in overseas markets, prioritising collaboration with local supply chains and compliance with local regulations to reduce cross-border deployment costs and thus build strong local teams. Acter has more than 2,600 employees worldwide (average age: 42 years; average length of service: 7 years). With more than 45 years of experience in EPCM, the company plays a reliable role for both high-tech and consumer industry.

Acter: number of employees



Source: Company, Daiwa

Acter: location of employees (as of June 2026)



Source: Company, Daiwa

Furthermore, Acter integrates ESG sustainability into its core corporate identity and technological innovation. As a green engineering provider, the company develops energy-saving construction methods, digital twins, ice-storage energy conservation, wastewater recycling, and AI smart facility management systems to help clients achieve carbon neutrality while expanding capacity. Backed by solid corporate governance, Acter

Acter integrates ESG capabilities into its core corporate value

consistently ranks among the top performers in Taiwan's corporate governance evaluations, maintaining a stable dividend payout policy and strong ROE as one of the leading facility engineering vendors.

Acter: management team (according to 2025 annual report)

Management	Position	Background and experience
Liang, Chin-Li	Chairman/CEO	Director since founding the company in 2008. Liang possesses working experience related to refrigeration and air-conditioning; graduated from EMBA, National Chiao Tung University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Hu, Tai-Tsen	Director	Director since founding the company in 2009. Hu possesses working experience related to refrigeration and air-conditioning; graduated from EMBA, Tunghai University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Lai, Ming-Kun	President	Graduated from EMBA, National Taiwan University of Science and Technology, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech
Wang, Chun-Sheng	President	Graduated from EMBA, Feng Chia University, and Department of Electrical Engineering – Refrigerating and Air-conditioning, Taipei Tech

Source: Company

Acter: major shareholders (as of August 2026)

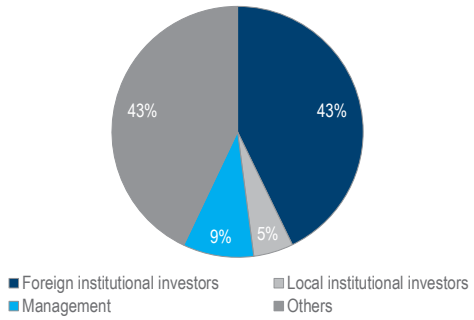
Holder	Share (%)
Xiang-Hui Development Co., Ltd.	4.35%
Chiu-Chang Investment Co., Ltd.	3.96%
Liang, Chin-Li	3.72%
JFE Engineering Co., Ltd.	2.23%

Source: TEJ

Acter: QFII holdings

Source: TEJ

Acter: shareholding structure (as of September 2026)



Source: TEJ

Glossary of terms

EPCM (engineering, procurement, construction, and management): Cleanroom construction services that cover the whole process including project design, purchasing, construction and management.

BIM (building information modelling): A digital modelling system of a building and its key information used in cleanroom construction.

C&S (civil & structural): Site infrastructure and the design and construction of load-bearing elements.

MEP (mechanical, electrical, and plumbing): Building essential systems for environment control, power, lighting, water, and drainage.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	1	Acter maintains a structured governance framework that supports disciplined management. The chairman is Chin-Li Liang. The company operates with defined responsibility across the board, management, cyber security oversight, operations, and audit functions. It has established an organisation function led by the President, a Chief Cyber Security Officer, a Response Team, an Operations and Policy Enforcement Team, and an Audit Team to supervise policy execution and review security implementation.
	Capital management	1	Acter manages capital in a stable and shareholder-friendly manner. Over the past several years, Acter has typically maintained a dividend payout ratio of 65-80%. It also maintains a strong financing network across multiple banks, supporting flexibility for project execution and expansion while preserving financial stability.
	Related party & transaction	1	Acter's business is supported by a transparent network of suppliers, project partners, and financial counterparties, including companies with equity ties and reciprocal business relationships that are common in its operating ecosystem. These relationships are managed through formal internal procedures for fund lending, endorsements and guarantees, and asset acquisition and disposal, all of which are approved through shareholder mechanisms and company rules. Such a structure helps keep inter-company and supply-chain transactions organised and traceable, reducing the risk of unclear dealings.
E	Product quality & safety	1	Acter executes its projects through a complete, disciplined process covering bidding assessment, project planning, design, procurement, construction management, verification and testing, completion and acceptance, closing review, and warranty service. Its cleanroom and electromechanical engineering business is built around maintaining temperature, humidity, and dust-free environments that support high-precision manufacturing. The company strengthens quality control through BIM, Revit, simulation software, ISO drawing output tools, and automated modeling, which improve design accuracy, construction efficiency, and project consistency.
E	Ecological impacts	1	Acter has built a concrete sustainability programme with measurable environmental outcomes. The company completed Scope 1, 2, and 3 greenhouse gas emissions calculation, obtained third-party verification under ISO 14064-1:2018, and publicly disclosed energy consumption and emissions data. It also implemented a strong waste reduction programme, achieving a 95% wastepaper recycling rate at headquarters in 2025 and 100% paper recycling at the 19F and 20F floors after establishing paper segregation and reuse procedures. The company also promoted sustainability measures through emissions inventory management, environmental disclosures, and resource-efficiency practices. These actions show active environmental management and a clear commitment to lowering operational impact.
	Health & safety		

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 7 Oct 2026

Source: Daiwa, Company

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Rating	Percentage of total
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Source: Daiwa

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- Before engaging in any trading, please thoroughly confirm accounting and tax treatments regarding your trading in financial instruments with such experts as certified public accountants.

* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

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國泰期貨 國泰證券子公司

產業更新 – 國防軍工產業

無人機、無人艇兵分二路，無人載具供應鏈蓄勢

What's New – 國防預算持續提高，六年編列 2,400 億元

2026 年國防總支出規劃達 9,495 億元，2027 年行政院總預算案進一步提高至 1 兆 1,225 億元，年增約 18%，後續待立法院審議。《強化國防自主暨無人載具產業發展條例》規劃六年編列 2,400 億元，每年以 400 億元為原則，支持可信任供應鏈、研發及測試能力建置。此外，行政院提出 1,457 億元國防追加預算，其中濱海監偵型、濱海攻擊型無人機及小型自殺無人艇合計 559 億元，盼於預算通過後加快採購，於 2027 年形成初始戰力。隨政策需求逐步落實，國內軍工供應鏈將由樣機開發與測試，進一步朝產線建置、零組件穩定供貨及批量交付發展。

Trends/Outlook – 無人載具供應鏈蓄勢，持續性發展可期

台灣無人機需求將沿低成本大量部署與高階情監偵兩條路線發展。前者重視採購成本、生產速度與快速補充能力，帶動模組化設計及共用零組件需求；後者強調長航時、感測品質與穩定資料鏈，商機涵蓋機體、光電酬載、導控系統及後勤服務。反無人機系統則由單一干擾設備走向多感測器、指揮控制與反制手段整合，跨品牌相容性、場域驗證及持續更新能力將成為競爭重點。無人艇方面，產業價值由船體製造延伸至自主航行、通信及任務酬載，台海風浪環境亦提高海試與可靠度驗證的重要性。研究部認為，隨部署規模擴大，具備系統整合、穩定交付及長期維保能力的業者，有機會建立持續性收入來源。

Investment Summary

1. 長榮航太(2645 TT)以軍用商規艦載監偵無人機切入市場，延伸航太製造與品質管理優勢；漢翔(2634 TT)與 Shield AI 合作，以 V-BAT 垂直起降無人機取得海巡案，後續可望拓展在地整合、維保及新增採購需求。創泓科技(7714 TT)布局 C2 無人機防禦系統與 Sentrycs 反無人機方案，商機涵蓋設備供應、偵測反制及指揮控制整合，後續關注標案導入與服務收入成長。
2. 台船(2208 TT)的「奮進魔鬼魚」採三體船與模組化酬載設計，並透過與 Havoc 合作布局自主水面載具；龍德造船(6753 TT)的「黑潮」以高速操縱與匿蹤設計為特色，結合既有軍用小型船建造經驗，爭取量產與後勤需求。研究部對台灣軍工產業中期發展持正面看法，後續隨預算、採購及產品驗證逐步落實，相關業者有望將技術與製造能力轉化為實質營運成長。

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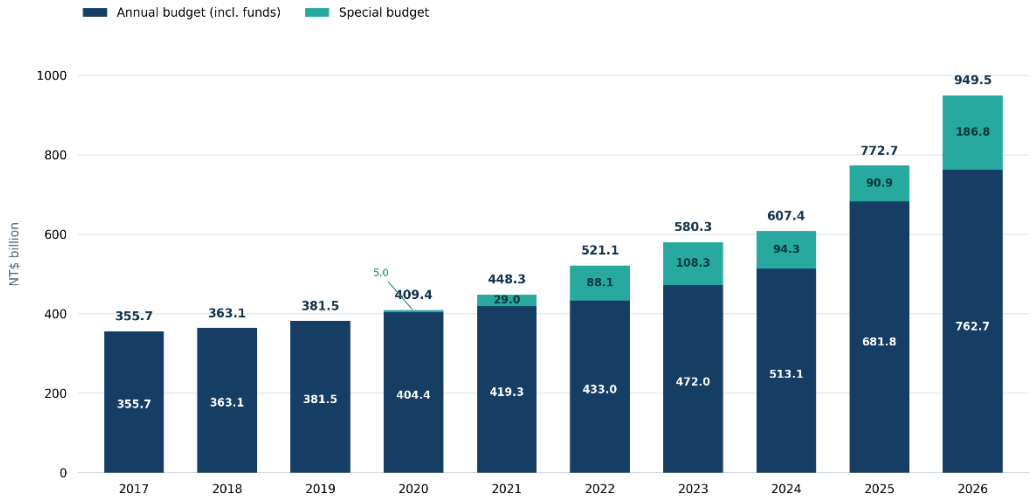
個股	代碼	預估每股盈餘(當地幣別)			每股盈餘成長率(%)			預估本益比/本淨比			目標價(當地幣別)	最新收盤價(當地幣別)	評等
		2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E			
長榮航太	2645 TT	5.46	8.63	9.68	11.4	58.0	12.2	31.6	20	17.8	220	172.5	買進
漢翔	2634 TT	0.78	2.58	2.45	-66.2	230.8	-6.2	82.2	24.8	26.2	73.5	64.1	中立
創泓科技	7714 TT	6.33	-	-	24.9	-	-	26.5	-	-	-	168.0	未評等
台船	2208 TT	-1.7	-	-	-	-	-	N/A	-	-	-	18.45	未評等
龍德造船	6753 TT	5.98	7.73	5.28	-4.3	29.3	-31.7	21.9	16.9	24.8	-	131.0	未評等

國泰期貨 國泰證券子公司

從預算增加到產線擴張，台灣軍工迎來新一輪成長

過去十年，台灣國防支出的成長，逐步為軍工產業累積更大的發展空間。如今隨著無人機、無人艇及反制系統成為國防重點，這股需求正從大型軍備採購，延伸至電子、航太與造船供應鏈。以廣義國防總支出觀察，2026 年原規劃規模達 9,495 億元，若將其分為年度預算與特別預算兩類，年度預算含基金約 7,627 億元，特別預算約 1,868 億元。年度預算支撐人員、裝備維持與持續性軍事投資，特別預算則加快重點裝備籌獲，兩者共同擴大產業需求。行政院先前也通過的 2027 年總預算案，進一步將廣義國防總支出推升至 1 兆 1,225 億元，較 2026 年原規劃增加約 18%，目前送至立法院審議，仍待後續觀察。

表一：Taiwan defense budget from 2017 to 2026



資料來源：行政院主計處、國防部、國泰證期研究部整理

六年 2,400 億元，行政院再提無人載具追加預算

如果說國防預算擴張為產業打開市場，無人載具專法則讓企業看見更長期的投入方向。9 月行政院副署的《強化國防自主暨無人載具產業發展條例》，要求循年度預算程序，連續六年編列 2,400 億元，每年以 400 億元為原則，必要時可依需求增編。政策範圍涵蓋可信任供應鏈、研發與測試場域，也將資源引導至具實質技術及製造能力的業者，讓國防採購同時成為產業升級的推力。此外，行政院也提出的 1,457 億元國防追加預算，其中「濱海監偵型」、「濱海攻擊型無人機」及「小型自殺無人艇」三項合計 559 億元。國防部已完成招商說明，希望在預算通過後加快採購，於 2027 年形成初始戰力。換句話說，產業正走到從樣機、測試走向批量交付的關鍵階段：無人機廠需要準備產線，零組件供應商需要建立穩定供貨能力，系統整合商則必須完成通信、控制與任務酬載的驗證。後續追加案等待立法院三讀。

國泰期貨 國泰證券子公司

國內無人機需求轉向大量部署及偵察視野延伸

烏克蘭讓產業看見小型無人機也能改變軍事裝備的需求結構。當無人機從少數專業單位的裝備，成為前線頻繁使用、需要持續補充的工具，產業競爭也開始從單架性能，延伸至生產速度、成本控制與快速改版。台灣正沿著這條路徑前進：政府已透過前幾年軍用商規無人機採購累積量產經驗，9 月經濟部亦公布中光電相關平台已完成軍用商規量產交付，顯示國內供應鏈逐步跨過展示與驗證階段。未來市場將形成兩條並行的成長路線。

1. **低成本、大量部署型無人機**，重點在可負擔的採購成本、穩定補充能力與持續更新；烏克蘭常見的 FPV(第一人稱視角)機型，即展現這類需求的規模潛力。台灣的濱海任務還需要兼顧飛行距離、海島環境與可信任供應鏈，因此，共用零組件、模組化設計及多家供應，將是逐步降低成本的重要方向。

2. **高階情監偵無人機**，透過更長滯空時間、更高品質的感測器與穩定資料鏈，擴大海域觀測範圍。如漢翔已確認與 Shield AI 合作取得海巡署 V-BAT 艦載無人機案，該平台具垂直起降與超過 12 小時續航能力，有利在有限甲板空間下執行長時間監偵。海軍公開規劃亦包括 2026~2029 年以約 167.9 億元籌購 32 架銳鳶二及 8 套導控站，另以 360 億元籌購 280 架艦載智慧垂直起降無人機及 140 套導控系統。這類跨年度計畫帶來的不僅是機體製造，還涵蓋光電酬載、地面站、訓練與後勤服務；後續採購與在地分工逐步落實後，具備整合及長期維保能力的廠商，有機會取得更高附加價值。

圖一：Ukrainian FPV Drone - General Cherry

資料來源：烏克蘭國防部、國泰證期研究部整理

當前線需要更多「看得見」的能力，後方的基地、機場與重要設施，也需要更完整的低空防護能力。烏克蘭國防部先前就曾指出，Mavic、Autel、Matrice 等多旋翼系統仍持續大量採購，用於前線觀測，並同步測試替代產品，也說明小型監偵與大型長航時平台各有用途。對台灣而言，這種任務分層將帶動不同等級的飛控、射頻、光學與電池管理需求，也讓資安、韌體更新及檢測認證成為供應鏈的重要環節。

另一方面，反無人機系統正從單一干擾器，走向涵蓋偵測、辨識、指揮與處置的完整架構：雷達與射頻感測發現目標，光電及熱影像協助確認，指揮控制平台整合資訊，再搭配適合的反制手段。北約 2026 年公布的反無人機框架採購，同樣採取多感測器與統一指管架構，顯示市場對整合能力的重視。國內創泓科技已推出 C2 無人機防禦系統，整合訊號偵測、光學辨識及反制模組，9 月進一步介紹 Sentrycs 代理及在地合作布局，商業機會可由設備供應延伸至系統整合、技術支援與持續更新。隨光纖操控與自主飛行等技術發展，單靠切斷無線通信的反制方式，也需要更多感測器與處

置方案配合。研究部認為，反無人機市場的競爭核心，將逐步轉向跨品牌整合、現場驗證及軟體更新能力；能在複雜環境中降低誤警、穩定辨識目標並持續維護系統的業者，更有機會建立長期客戶關係。後續觀察國防及關鍵基礎設施標案的導入進度。

圖二：Ukrainian Small ISR Drone - Mavic 3 Pro

資料來源：烏克蘭國防部、國泰證期研究部整理

無人艇從船體製造延伸至自主航行與海洋感測

當船艇能夠在無人乘載的情況下完成巡邏、觀測與運補，造船產業的競爭範圍，也從船體與推進系統延伸至自主航行、通信與感測器整合。水面無人艇(USV)可執行海域監偵、通信中繼、搜救、設施巡檢及軍事任務；水下無人載具(UUV)則將觀測範圍帶到海面之下，應用於海床測繪、海纜與管線檢查、反水雷及水下環境偵測。台灣既有的軍艦、商船與高速小型船經驗，也為這個新市場提供製造基礎。台船「奮進魔鬼魚」船長 8.6 公尺、寬 3.7 公尺，最高航速超過 35 節，採三體船與模組化酬載設計，並於 2025 年取得 ABS 原則性設計認可(AIP)，為後續產品驗證建立基礎。9 月台船進一步與 Havoc 宣布合作，擬推動自主水面載具授權共同生產，展現國內造船能力與國際自主控制技術結合的方向。龍德造船則以「黑潮」切入市場，船長 9 公尺、寬 2 公尺、最高航速超過 40 節，在 25 節巡航條件下航程約 250 公里，並具自主航行及目標識別功能。隨產品逐步成形，後續量產採購與任務驗證將成為主要觀察重點。水面平台有望先帶動船體、航控、資料鏈與任務模組需求，水下市場則進一步考驗聲學設備、耐壓密封、能源及回收能力。

圖三：台船-奮進魔鬼魚(Endeavor Manta USV)

資料來源：台船、國泰證期研究部整理

另一方面，台海海象則是無人載具商業化不可忽略的限制。交通部觀測數據顯示，臺中港測站冬季平均波高約 2.1 公尺，比夏季高約 1.3 公尺，足以說明不同季節的差異性。對小型無人艇，波浪會增加船體拍擊與姿態變化，影響感測器穩定、推進效率、航速及回收；水下航行雖可減少直接受表面波浪影響，出入水、母船收放及浮出通信仍受海況約束。對無人機而言，浪高主要透過甲板運動

與海面飛濺影響艦載起降，飛行本身則更直接受到陣風、降雨及鹽霧影響。因此，廠商需要驗證的不只是最高航速，還包括不同波高、週期、浪向及酬載下的任務完成率、穩定通信、自主避碰、故障處置與回收成功率。研究部認為，能累積跨季節海試資料、維修經驗及可重複交付能力的船廠將更具競爭優勢。

從政策走向可持續交付，軍工供應鏈蓄勢

國防預算擴張、無人載具產業政策及非紅供應鏈需求，有利支持業者持續投入研發與量產。無人系統的產業效益亦將由載具製造延伸至感測器、通信、任務整合與反制設備，並隨部署規模增加，帶動備件、維修、軟體更新及訓練等後續需求，為供應鏈建立持續性收入來源。

無人機方面，長榮航太(2645)則以軍用商規艦載監偵無人機切入市場，具備航太製造與品質管理能力的延伸優勢。漢翔(2634)透過與 Shield AI 合作的 V-BAT 垂直起降無人機取得海巡案，後續成長機會包括在地整合、維保及新增採購；反無人機方面，創泓科技(7714)布局 C2 無人機防禦系統與 Sentrycs 反無人機方案，可望拓展偵測、辨識、反制及指揮控制整合業務。

表二：無人機規格與任務定位比較表

類型	代表機型 / 製造商	機體規模	續航/滯空時間	主要用途與產業重點
FPV 無人機	General Cherry/烏克蘭	多種尺寸機架	17~30 分鐘	前線任務；低成本、量產與快速補充
小型多旋翼監偵機	Mavic 3 Pro/DJI	展開尺寸約 35cm	43 分鐘	近距離監偵；便攜、影像品質與快速部署
艦載垂直起降監偵機	V-BAT/Shield AI	翼展 3.8m	12 小時以上	艦載監偵；垂直起降、酬載整合與維保
中型固定翼海洋監偵機	銳鳶二型/中科院	翼展約 12m	16 小時以上	海域監視與預警；資料鏈、感測及指管整合
大型中高度長航時無人機	MQ-9B SkyGuardian/ General Atomics	翼展約 24m	40 小時以上	長航時廣域監偵；地面站、通信與後勤支援

資料來源：國泰證期研究部整理

無人艇方面，台船(2208)的「奮進魔鬼魚」採三體船設計與模組化酬載，具備拓展多元海上任務的潛力；龍德造船(6753)的「黑潮」則以高速操縱與匿蹤設計為特色，結合既有軍用小型船建造經驗，爭取量產與後勤需求。研究部認為，後續隨產品驗證與採購逐步落實，具備系統整合、穩定交付及長期維保能力的業者，將有機會將政策需求轉化為實質營運成長。

國泰期貨 國泰證券子公司

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國泰期貨 國泰證券子公司

投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%

註:潛在報酬率=(6個月內目標價/本次報告評等最近一天收盤價－1) 100

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國泰期貨 國泰證券子公司

聯友金屬 7610 TT (綠能環保)

供應跟不上需求擴大

投資與評價

1. 3Q26 受惠國際鎢供應持續吃緊、鎢價維持高檔，加上屏東二廠新增 30%產能陸續開出，自售比重持續提升，預估營收季增 47.5%，毛利率 66.5%，EPS 27.32 元創單季新高。
2. 2027 年成長動能由鎢價轉向擴產與產品高值化，屏南三廠、屏科廠陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售比重有機會挑戰 60%，預估 2026/27 EPS 83.77/109.27 元。
3. 目前股價評價位於近兩年本益比 10～24 倍區間中段，考量非中鎢供應鏈稀缺性、自售比重提升及 2027 年新產能接棒，給予區間中間偏上的 22 倍 FY27E PER，給予強力買進評等，目標價 2,400 元(22x FY27E EPS)。

重點更新

3Q26 鎢價與自售業務推升獲利續增

預估 3Q26 營收 28.43 億元 (+47.5% QoQ)，主要受惠國際鎢價維持高檔、鎢酸鈉自售業務成長，以及屏東二廠新增 30%產能陸續開出；毛利 18.90 億元 (+44.8% QoQ)，毛利率 66.5% (-1.3ppts QoQ)，營收與毛利仍維持高速成長。營業費用 1.88 億元(+55.6% QoQ)，稅後淨利 13.27 億元(+40.3% QoQ)，EPS 27.32 元創單季新高。預估 4Q26 隨新產能進一步量產、自售比重過半，營收 29.59 億元 (+4.1% QoQ)，惟費用投入增加，EPS 降至 26.17 元。

2027 年產能與高值化產品接棒

預估 2027 年營收 132.79 億元 (+52.1% YoY)，成長動能由鎢價逐步轉向產能擴張及產品高值化，屏南三廠與屏科廠新產能陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售比重有機會挑戰 60%；毛利 87.22 億元 (+48.9% YoY)，毛利率 65.7% (-1.4ppts YoY)。惟新廠與新工序投入增加，營業費用 20.62 億元(+202.8% YoY)，使營業利益 66.60 億元(+28.7% YoY)、營益率 50.2%(-9.1ppts YoY)；稅後淨利 53.09 億元(+30.4% YoY)，EPS 109.27 元再創年度新高。後續營運重點將轉向新產能利用率，以及高值化產品的提升速率。

強力買進 – 初次評等

目標價：2,400 元 +34%
收盤價：1,790 元

報告日期	評等	目標價
2026/10/07	買進	2,400 元

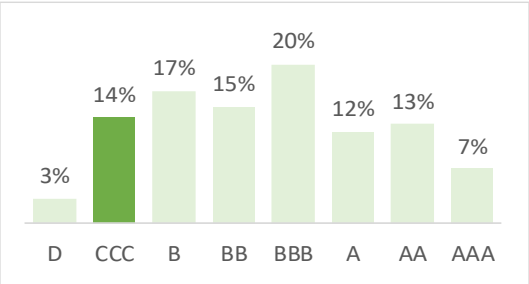
	國泰	市場	差異
目標價	2,400	--	--
2026 EPS	83.77	77.60	+7.9%
2027 EPS	109.27	107.51	+1.6%
現在 P/E	16.38	--	--

公司資訊

市值(百萬元)	105,430
自由流通股數比例(free float ; %)	72.93
20 日平均成交量(百萬元)	1,770
每股淨值(元,2027E)	219.48
股價淨值比(倍)	7.68
現金股利率(%)	0.56
股東權益報酬率(%,2026E)	50.98
外資持股比率(%)	18.99

聯友金屬 ESG 評等

ESG 評等	社會(S)	經濟(E)	環境(E)	揭露(D)
CCC	40%	39%	16%	33%



呂理舜/ 綠色能源、電動車
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項目(百萬元)	2Q'26	3Q'26(E)	4Q'26(E)	1Q'27(E)	2025	2026(E)	2027(E)
營業收入	1,928	2,843	2,959	3,010	1,406	8,731	13,279
稅前盈餘	1,188	1,702	1,620	1,599	303	5,176	6,655
稅後盈餘(母公司業主)	946	1,327	1,271	1,263	241	4,070	5,309
營收成長率(%)(QoQ/YoY)	92.42	47.50	4.07	1.72	27.90	521.20	52.08
毛利率(%)	67.73	66.47	64.94	64.10	30.78	67.08	65.68
營益率(%)	61.47	59.87	54.81	53.22	22.00	59.28	50.15
稅後盈餘成長率(%)(QoQ/YoY)	80.22	40.25	-4.2	-0.64	311	1587.07	30.44
每股盈餘 EPS(元)	19.48	27.32	26.17	26.00	4.97	83.77	109.27

EPS 以調整後股數 48.59 百萬股計算

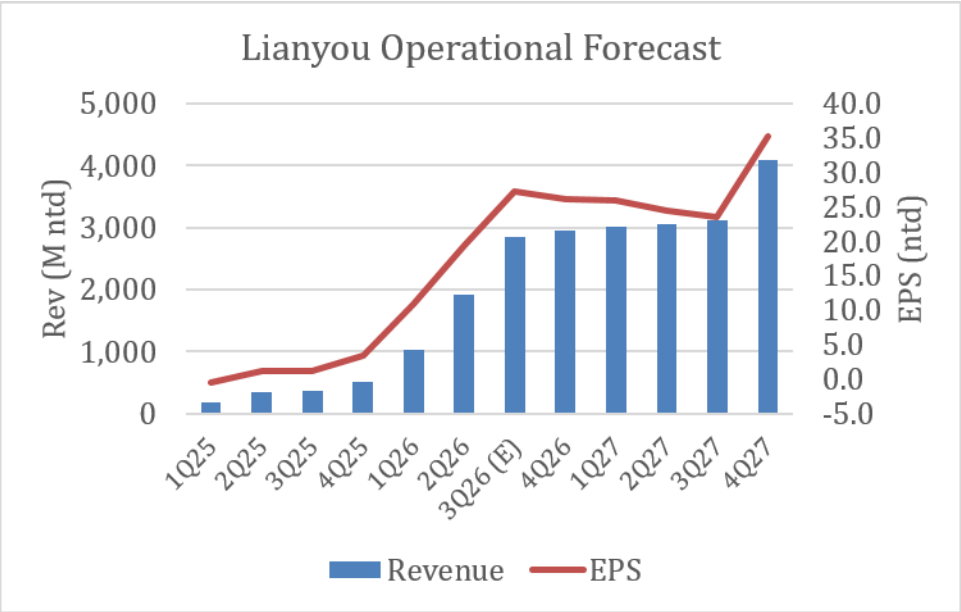
聯友金屬 7610 TT

訪談內容

3Q26 鎢價與自售業務推升獲利續增

預估 3Q26 營收 28.43 億元（+47.5% QoQ），主要受惠國際鎢供應持續吃緊、鎢價維持高檔，加上鎢酸鈉自售業務成長，以及屏東二廠增加的 30%新產能部分陸續開出；國際市場中國持續限制鎢相關產品出口，尤其對美國與日本出口限制趨嚴，使國際客戶加速建立非中供應鏈，聯友以二次硬質合金回收生產鎢產品，在競爭者有限的情況下，可望維持穩定的需求與產品售價。預估毛利 18.90 億元（+44.8% QoQ），毛利率 66.5%（-1.3ppts QoQ），雖自售產品比例提高，但毛利率較 2Q26 高基期略降；營業費用 1.88 億元（+55.6% QoQ），營業利益 17.02 億元（+43.7% QoQ），營益率 59.9%（-1.6ppts QoQ）。預估稅後淨利 13.27 億元（+40.3% QoQ），EPS 27.32 元創歷史新高。除鎢產品外，公司亦持續推進循環再製鈷粉客戶認證與出貨，作為後續產品高值化方向。

圖 1：聯友金屬獲利預估



資料來源：國泰證期研究部整理

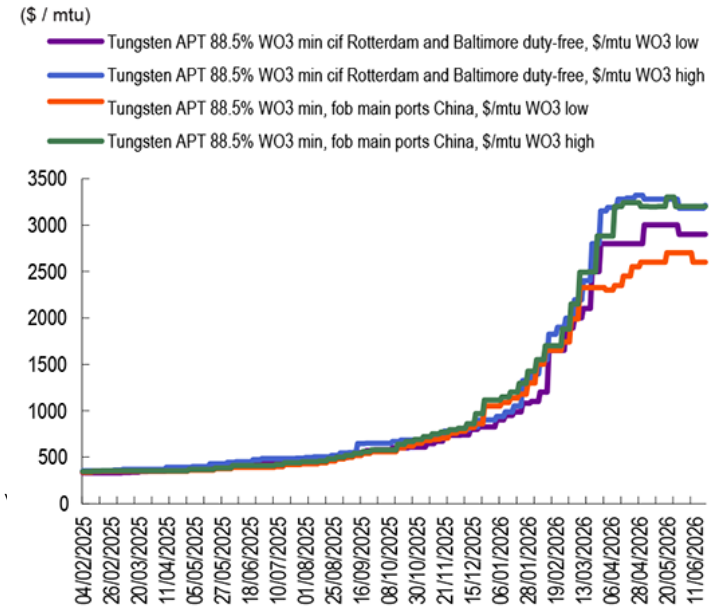
4Q26 營收續創高峰

預估 4Q26 營收 29.59 億元（+4.1% QoQ），從上季大幅成長後仍維持季增，主要受惠鎢產品自售業務持續成長，以及預期國際鎢供需仍偏緊，且公司新產能進一步進入量產階段，使單季自售比重過半。中國掌握全球主要原生鎢供應，出口限制使海外供應鏈短期難以快速補足缺口，軍工、半導體及 AI 相關精密加工需求則持續支撐國際鎢價。預估毛利 19.21 億元（+1.7% QoQ），毛利率 64.9%（-1.5ppts QoQ）；惟營業費用增至 3.00 億元（+59.6% QoQ），在於新的工序流程投資持續投入，使營業利益 16.22 億元（-4.7% QoQ），營益率 54.8%（-5.1ppts QoQ）。預估稅後淨利 12.71 億元（-4.2% QoQ），EPS 26.17 元；整體而言 4Q26 營收與毛利仍向上，但費用增加使獲利較 3Q26 高峰略為回落。

2026 年鎢價大漲與自售比例增加

預估 2026 年營收 87.31 億元(+521.2% YoY)，受惠國際鎢價大幅上漲、鎢酸鈉自售業務放量，以及非中鎢供應鏈需求快速提升。中國鎢出口限制使國際市場供給趨緊，聯友具備廢硬質合金回收、濕法冶金及既有海外客戶基礎，使鎢價上漲透過自售產品，直接轉化為獲利成長。預估毛利 58.57 億元(+1,254.0% YoY)，毛利率由 2025 年的 30.8%大幅提升至 67.1%(+36.3ppts YoY)，主因仍在產品國際報價及自售業務成長明顯；營業費用 6.81 億元 (+452.3% YoY)，營業利益 51.76 億元 (+1,573.7% YoY)，營益率 59.3% (+37.3ppts YoY)。預估稅後淨利 40.70 億元 (+1,587.1% YoY)，EPS 83.77 元創歷史新高，2026 年為公司材料處理量由加工大幅轉向自售產品後，獲利結構明顯轉變。

圖 2：鎢金屬價格走勢



資料來源：Fastmarkets, 國泰證期研究部整理

2027 年產能與高值化產品接棒

預估 2027 年營收 132.79 億元 (+52.1% YoY)，在 2026 年鎢價與自售業務快速成長後，2027 年成長動能逐步由價格轉向產能擴張及產品高值化。公司規劃由既有鎢酸鈉進一步向 APT、氧化鎢、鎢粉及碳化鎢延伸，屏南三廠與屏科廠相關新產能預計陸續開出，鎢相關加工產能有機會擴增倍翻，若產品認證與量產順利，可望提高公司在鎢產業鏈的垂直整合程度。預估毛利 87.22 億元 (+48.9% YoY)，毛利率 65.7% (-1.4ppts YoY)，在營收快速擴張下仍維持六成以上高毛利率；惟配合新廠、新產品及營運規模擴張，營業費用增至 20.62 億元 (+202.8% YoY)，使營業利益 66.60 億元 (+28.7% YoY)，營益率降至 50.2% (-9.1ppts YoY)。預估稅後淨利 53.09 億元 (+30.4% YoY)，EPS 109.27 元再創年度新高；2027 年營收與毛利仍維持高速成長，但費用投入增加使獲利增速低於營收，後續營運重點將轉向新產能利用率，及 APT、氧化鎢、鎢粉、碳化鎢等高值化產品的提升速率。

非中鎢供應鏈稀缺性提升

中國鎢出口限制使國際供應持續偏緊，聯友藉二次硬質合金回收與濕法冶金技術，受惠國際客戶加速建立非中供應鏈，2026 年鎢價上漲、自售比重提升及屏東二廠新增產能開出，帶動獲利明顯成長；2027 年成長動能則由格進一步轉向產能擴張與產品高值化，隨屏南三廠及屏科廠新產能陸續開出，產品由鎢酸鈉延伸至 APT、氧化鎢、鎢粉及碳化鎢，自售部分有機會在 2027 年挑戰 60%。預估 2026/27 EPS 為 83.77/109.27 元。3. 目前股價評價位於近兩年本益比 10～24 倍區間中段，考量非中鎢供應鏈稀缺性、自售比重提升及 2027 年新產能接棒，給予區間中間偏上的 22 倍 FY27E PER，給予強力買進評等，目標價 2,400 元(22x FY27E EPS)。

產業概況

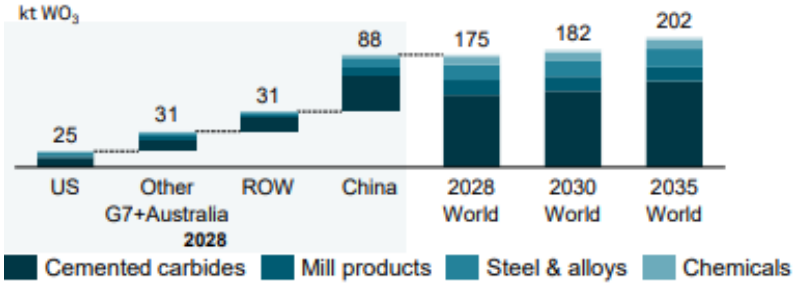
鎢金屬需求持續由工業與國防驅動

2026 年中國鎢總供給約 8.51 萬噸 W、YoY +12.4%，其中原生礦約 4.36 萬噸、幾乎零成長，回收鎢則由高價刺激增加約 50%至約 3.0 萬噸，使回收料占供給由 2025 年的 25–26%升至約 35%；換言之 2026 幾乎所有可快速增加的供給彈性都來自回收處理，使得非新礦山供應的競爭也明顯升高。顯示高鎢價已開始抑制價格敏感度較高的傳統用途，但切削工具、航太與國防等剛性需求仍維持成長。全球鎢需求主要仍來自硬質合金，應用於切削工具、礦山鑽頭、汽車與精密加工；其中礦業與營建約占 2025 年全球 WC 市場 38.5%。

國防需求的重要性在 2026 年進一步提高，美國與以色列對伊朗軍事行動、俄烏戰爭持續消耗飛彈與穿甲彈用鎢，這類用途與硬質合金不同，使用後無法回收，使其對原生鎢形成永久性消耗。因全球每年鎢需求約 10～11 萬噸，其中原生礦約占 65%、回收料約 35%，現有需求須依賴二次資源補足。短期新礦與冶煉產能建置時間長，因此即使高價刺激新增供給，供需緊張仍不容易快速逆轉。

各應用來看硬質合金(Cemented carbide)佔比最大，主要用於切削刀具、採礦鑽頭、耐磨零件與國防彈藥等。加工材/材加工品 (Mill products): 如鎢絲、鎢棒、鎢板等。鋼鐵與合金 (Steel & alloys) 用於高速鋼、耐熱合金鋼等。化學品 (Chemicals): 催化劑、顏料及化學試劑。其中 PCB 鑽針需求在近年也快速增加，據估計因 AI server PCB 變得更厚、更多層、更難鑽，高速低損耗材料使 WC 鑽針磨耗大幅提高。AI server PCB 已朝 28–46 層甚至更高層數發展，而一般鑽針原本約可加 工 3,000 孔，在高階材料下可能降到約 600 孔，在相同鑽孔量可能需要接近 4–5 倍鑽針消耗。半導體需求在 4Q26 仍然偏強，其中第一類是半導體製程本身使用的高純度鎢/WF_6 等材料，另一類是晶圓、封裝、設備與零組件加工所使用的 WC 超硬工具。整體而言隨著 AI GPU/ASIC、HBM 與先進封裝擴產，使高純度 W 與精密 WC 工具需求維持穩定成長。

圖 3：鎢金屬主要應用



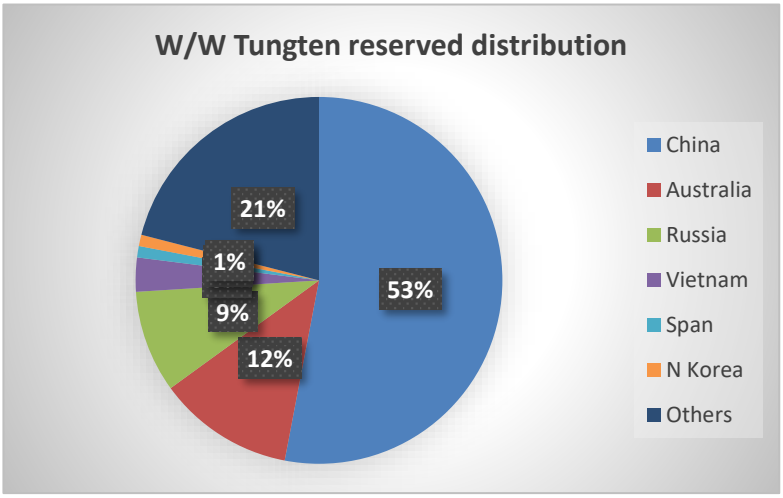
資料來源：S&P Global, 國泰證期研究部整理

中國七成產能限制鎢出口影響大

全球鎢供應高度集中，在 2025 年全球鎢精礦開採產量約 8 萬多噸，中國占約 75%，且中國同時掌握從礦山、APT、鎢粉到碳化鎢的完整加工鏈，鎢的供應風險除了礦源集中，更包含中游精煉能力集中。2H25 開始中國開始收緊鎢礦開採配額並實施出口管制後，非中國市場供給明顯轉緊，價格也顯著提升。

全球鎢市場主要依賴中國供應，因此中國價格大致主導全球定價；但出口限制與供應鏈安全政策出現後，國際買家開始建立中國以外的礦源、冶煉及回收體系，使得國際報價持續走高，鎢回收目前約占全球供給三成，雖然產地來源風險較低，但是既有供應鏈的產能瓶頸，使得回收流程取得鎢的價格持續走高。雖陸續有些新礦場宣告將投入開發，但仍需兩年以上時間準備，舊有礦場的重啟也面臨到回收率與經濟性的挑戰，且對於環境衝擊高未必適合歐美國家，目前最為環保與的作法仍是從回收鎢開始加工。透過廢鎢回收陸續轉化為鎢酸鈉、APT、氧化鎢、鎢粉、碳化鎢，將有利於賣方市場的精鍊供應商，邁向更高值化的末端產品發展。

圖 4：鎢礦蘊藏分布



資料來源：USGS, 國泰證期研究部整理

美國積極多元化鎢礦來源

3Q26 美國政府承諾向 Elmet Group 投入 US$450mn，並搭配 DLA 最高 US$2bn 鎢戰略庫存採購，政策目標已從單純取得礦源轉向礦山+APT/精煉+零組件的垂直整合。US$450mn 投資將用於擴張美國鎢製造能力、取得長期原料、投資美國及盟國礦山與建立新的 refining/conversion 能力；其中初期資金約 US$200mn。另一方面 DLA 與 Elmet 簽訂最高 US$2bn IDIQ 合約，最低保證採購 US$150mn，品項直接涵蓋鎢礦、鎢精礦及鎢酸鈉；Elmet 待新增礦山、offtake 與 processing capacity 形成後才向戰略庫存交貨，以避免政府囤貨擠壓既有民間供應。EQ Resources 規劃向美國供應 4,000 噸鎢精礦；但 Nevada APT processing facility 全面恢復營運預計要到 2H28，因此 4Q26 政策首先增加的是戰略採購與長約需求，而非立即增加 APT 現貨供給。這代表短期甚至可能強化非中國的原料競爭，但預計要數年後後才有機會逐步改善美國中游加工缺口。

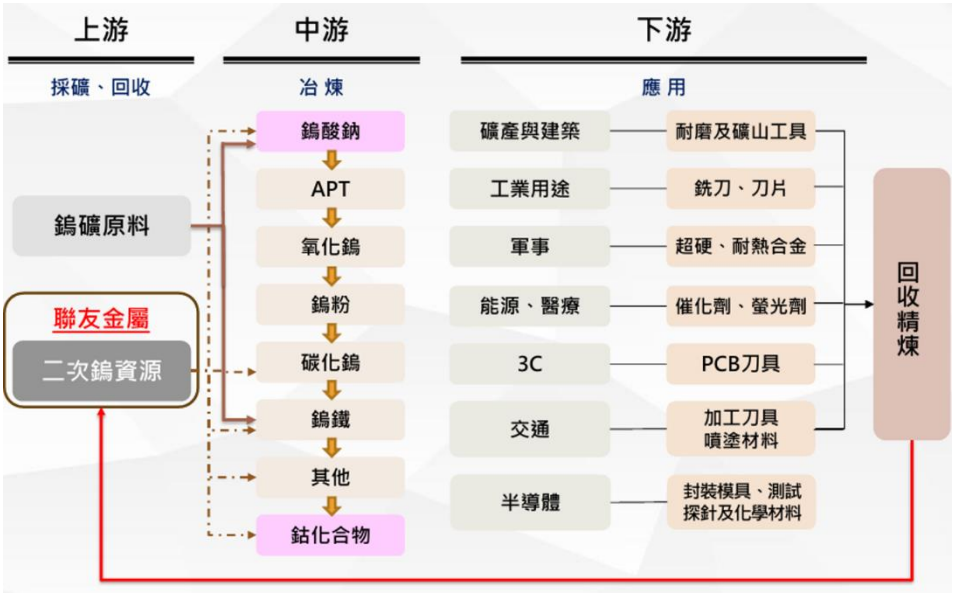
鈷金屬供給仍有短缺風險

2025 年全球鈷礦產量約 31 萬噸，其中剛果民主共和國約 23 萬噸、占七成比重，全球約 50%儲量亦集中於剛果，而中國又掌握全球七成以上鈷精煉能力，因此上游礦源與中游精煉均高度集中。需求方面 2025 年全球鈷消費動力電池占比超過 65%，EV 仍是最大需求來源；雖電池高鎳化、低鈷化以及 LFP 滲透率提升，會限制單位電池鈷用量成長，但 IT 電池的使用仍以氧化鈷為主要材料。而長期更值得注意的是航空發動機、燃氣渦輪、國防設備所使用的高溫合金，以及硬質合金與高性能磁鐵需求。高溫合金目前約占鈷需求 15～20%，工業工具、硬質合金與催化劑約一成，使鈷逐漸從單一新能源金屬轉變成兼具新能源與戰略材料屬性的關鍵金屬；在供應集中及可追溯要求提高下，循環再生鈷、高純度鈷粉與非中國精煉能力的戰略價值亦將持續提高。

公司簡介

聯友金屬不依賴原生鎢礦，而是以廢硬質合金、鎢鋼及研磨屑等二次料作為鎢來源，經濕式冶金回收後製成鎢酸鈉（Sodium Tungstate），同時回收鈷，形成 WC/鎢鋼廢料、鎢酸鈉、APT、氧化鎢的城市礦山供應鏈。公司目前鎢酸鈉年產能約 6,000 噸，2025 年鎢酸鈉出貨約 4,728 噸、YoY +8%，其中代工約占 52%、自售約 48%，相較 2024 年代工 42%、自售 58%，可看出自有設備利用率已處相對高檔；公司規劃透過新廠將鎢酸鈉產能進一步提高至約 12,000 噸/年、較現有產能增加 100%，若以鎢酸鈉 $Na_2WO_4{\cdot}2H_2O$ 含鎢重量約 55.7%換算，滿產相當於一年可處理約 6,700 噸 W contained。公司並規劃進一步向下游 APT 與氧化鎢延伸，使產品由目前以鎢酸鈉為主，逐步建立回收料、鎢酸鈉、APT、WO_3的垂直整合能力。

圖 5：聯友金屬產業地位



資料來源：聯友金屬，國泰證期研究部整理

ESG 政策

公司在社會面評分上屬中上，然而需在經濟與環境面與揭露面強化:
經濟面: 股東權益方面，公司優點為董事長親自出席股東常會；股東會結束後公開揭露逐項議案之詳細表決資訊。公司可加強處為當年度未召開 2 次以上法人說明會。
法規遵循方面，公司於題項評估期間未有不合理或受主管機關處分之情事。獨立董事方面，公司優點為至少 2 名獨立董事之連續任期不超過 9 年。公司可加強處為獨立董事席次未達董事席次二分之一以上。董事會永續治理方面，公司優點為有設置永續發展相關委員會；有制訂環境、社會及治理之中長期發展策略；當年度永續報告書經董事會通過。公司可加強處為負責永續發展之最終負責人非董事長或總經理。董事會績效與薪酬方面，公司優點為將董事績效評估與薪資報酬連結。公司可加強處為企業永續績效尚未與高階主管及董事薪酬進行連結。
董事會多元性方面，公司優點為揭露董事會多元化政策並揭露落實情形；董事會成員具備不同專業背景。公司可加強處為任一性別董事席次未達 1/3。永續創新方面，公司優點為取得綠色認證。公司可加強處為未揭露綠色相關部門設置情形與職責；未揭露綠色/永續採購之政策或金額。
此外，產業創新方面，公司優點為對其專利之智慧財產權進行管理與保護，並揭露量化績效。
法規遵循方面，公司近 3 年並未受公平交易委員會懲罰。

環境面: 環境系統與治理方面，公司優點為積極將綠色概念融入營運過程，打造綠色價值鏈；榮獲 ISO 14001 環境管理認證；將環境管理作為與聯合國永續發展目標(SDGs)連結。空氣管理方面，公司優點為制定空氣管理政策、揭露空氣汙染排放數據；公司可加強處為未揭露量化空氣汙染減量目標。法規遵循方面，評估期間未違反空氣汙染相關法規或已針對違反情形提出具體改善措施。氣候變遷管理方面，公司可加強處為未說明董事會或管理階層對氣候變遷議題之監督頻率或事項；未揭露使用具公信力的 1.5 °C 以下及 3 °C 以上溫度情境評估氣候風險；未揭露與供應鏈或合作夥伴共同推動因應氣候變遷相關策略；未揭露氣候變遷對公司所產生的機會項目。
溫室氣體管理方面，公司優點為揭露至少三年的溫室氣體範疇一與二排放量，並揭露涵蓋範圍或佔營收比率。公司可加強處為未揭露溫室氣體管理專責單位及相關管理政策；未揭露範疇三排放量；未揭露量化溫室氣體減量目標；未揭露是否通過碳盤查認證。節能管理方面，公司優點為揭露用電或節能數據。公司可加強處為未揭露能源管理專責單位及相關管理政策；未揭露節能量化目標；未揭露能源風險對營運的影響評估結果。再生能源管理方面，公司可加強處為未揭露再生能源管理專責單位及相關推動政策；未揭露再生能源使用量或發電量；未揭露量化再生能源發展目標。廢水管理方面，公司優點為制定廢水管理政策、揭露廢污水排放量；公司可加強處為未揭露量化廢汙水減量目標。
揭路面: ESG 揭露方面，公司優點為依據 GRI 準則編寫永續報告書；於永續報告書清楚說明揭露範疇；建置英文網站，並包含財務、業務及公司治理相關資訊；於永續報告書揭露重大性分析細節。公司可加強處為永續報告書未依據 SASB 進行對應揭露；未上傳英文版永續報告書於公開資訊觀測站；永續報告書尚未經過第三方機構確信或保證；未於官網揭露利害關係人問卷，供利害關係人回饋；未說明公司營運對應 SDGs 目標之具體作為；未揭露使用雙重重大性分析重大性議題。

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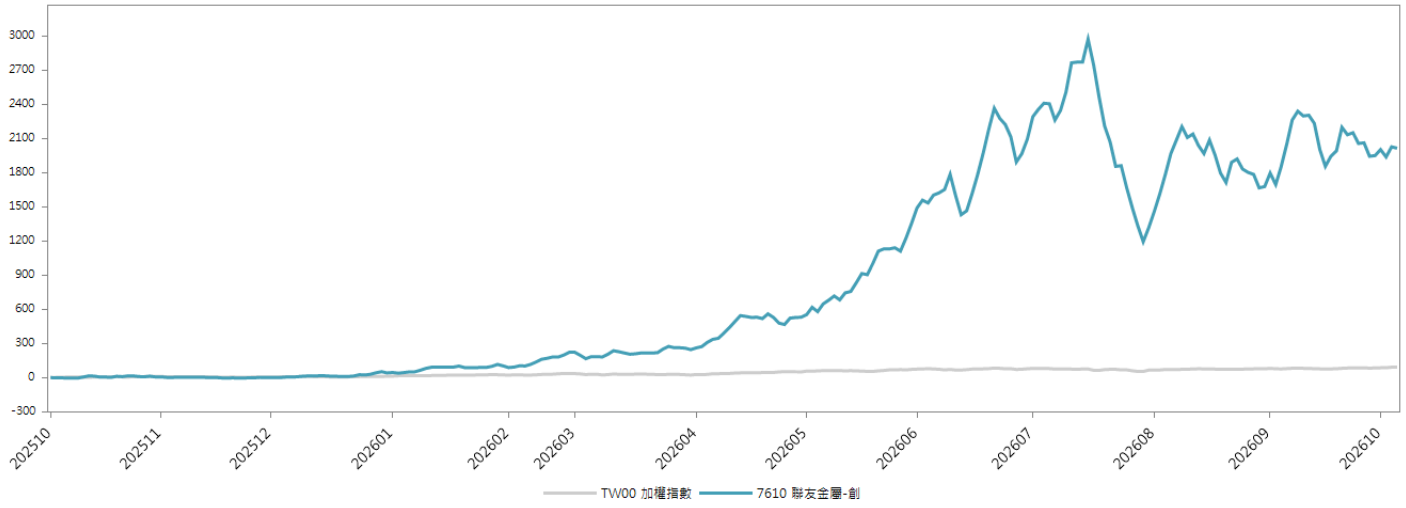
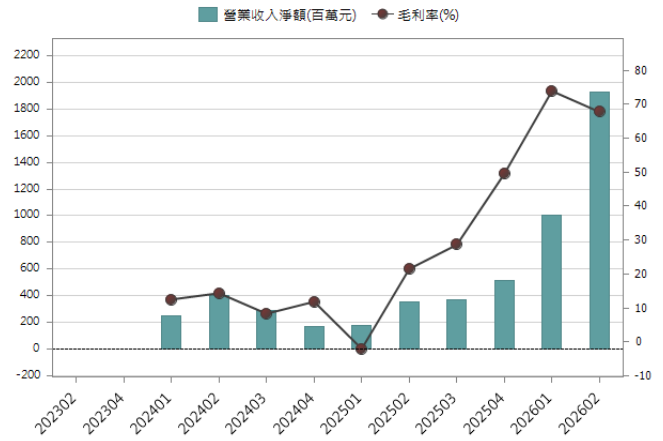
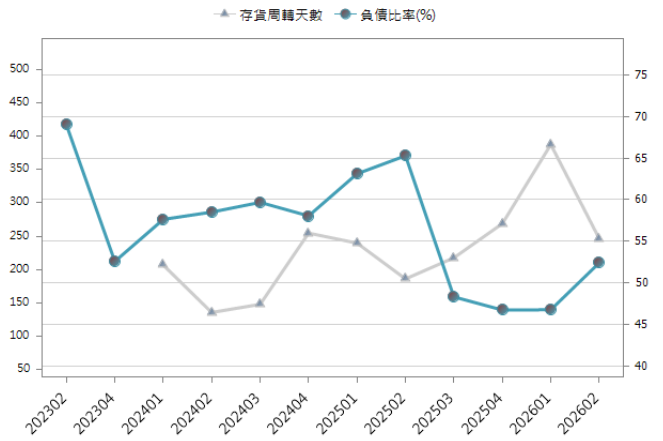
股價相對大盤走勢圖



近三年單季營收與毛利率走勢圖

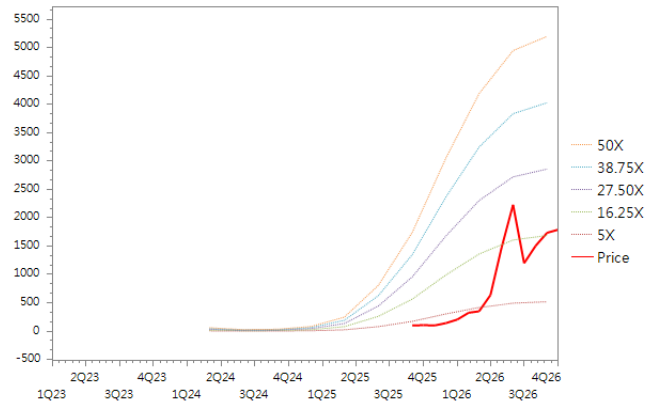


近三年負債比率與存貨週轉天數



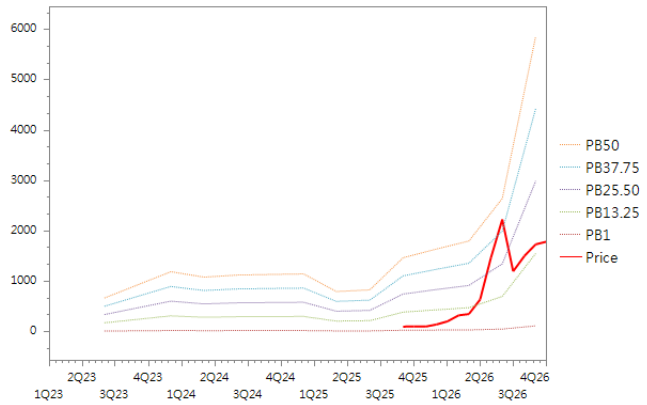
未來 12 個月 P/E(本益比)區間圖

PE band



未來 12 個月 P/B(股價淨值比)區間圖

PB band



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損益表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業收入	916	1,099	1,406	8,731	13,279
營業成本	835	967	973	2,874	4,557
營業毛利	82	132	433	5,857	8,722
營業費用	66	69	123	681	2,062
營業利益	16	63	309	5,176	6,660
營業外收入與支出	-14	3	-6	0	-5
其他收入	2	2	2	5	4
其他利益與損失	0	11	13	25	23
財務成本	17	16	28	40	41
權益法認列之投資損益	0	0	0	0	0
其他項目	1	6	7	10	9
稅前純益	1	66	303	5,176	6,655
所得稅費用	-3	7	62	1,105	1,345
稅後純益	4	59	241	4,071	5,310
母公司業主- 稅後純益	4	59	241	4,070	5,309
非控制權益- 稅後純益	0	0	0	1	1
稅前息前折舊攤銷前淨利	61	114	387	5,272	6,660
每股盈餘 (元)	0.08	1.21	4.97	83.77	109.27

資料來源:公司資料及國泰預估

資產負債表

12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
資產總額	1,554	1,894	3,025	11,001	21,003
流動資產	747	761	1,731	8,904	18,017
現金及約當現金	48	85	261	4,279	12,526
短期投資	14	95	49	66	66
存貨	435	431	923	2,758	3,302
應收帳款及票據	141	42	167	845	1,167
其他流動資產	108	109	331	957	957
非流動資產	807	1,133	1,294	2,097	2,985
長期投資	0	0	3	11	4
固定資產	460	980	1,228	1,965	2,861
其他非流動資產	347	153	63	121	121
負債總額	817	1,100	1,413	5,287	10,323
流動負債	709	599	881	3,473	8,011
應付帳款及票據	188	132	207	667	793
短期借款	199	438	591	2,352	6,764
其他流動負債	322	29	82	454	454
長期負債	108	500	533	1,813	2,313
長期借款	105	496	511	1,735	2,235
其他非流動負債	3	4	22	78	78
股東權益總額	737	794	1,611	5,714	10,679
股本	309	347	486	486	486
資本公積	399	370	843	916	916
保留盈餘	31	81	268	4,201	9,165
非控制權益	0	0	15	15	15
其他權益	-3	-3	-1	97	97

資料來源:公司資料及國泰預估

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現金流量表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業活動之現金流量	-157	152	-361	1,821	1,881
稅前純益	1	66	303	5,176	6,655
折舊及攤提	45	51	78	96	188
本期營運資金變動	-144	34	-548	-1,916	-2,101
其他營業活動現金流量	-60	1	-194	-1,536	-2,861
投資活動之現金流量	-148	-742	-186	-863	-1,496
資本支出淨額	-128	-236	-220	-829	-1,642
本期長期投資變動	2	0	-3	6	14
其他投資活動現金流量	-21	-506	37	-40	132
籌資活動之現金流量	341	627	723	3,059	7,426
現金增資	289	0	553	0	0
本期借款變動	65	631	168	4,086	8,824
發放現金股利	-7	-3	-7	-24	-486
其他籌資活動現金流量	-6	-1	9	-1,018	-1,038
匯率調整數	0	0	0	0	0
自由現金流量	-285	-84	-581	992	239
本期產生現金流量	36	37	176	4,017	7,810

資料來源:公司資料及國泰預估

資產負債表主要財務報表分析

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
年成長率(%)					
營業收入	53.0	19.9	27.9	521.2	52.1
營業利益	-85.72	299.64	391.8	1573.66	28.67
稅前息前折舊攤銷前淨利	-57.17	86.58	240.41	1261.88	26.32
稅後純益	-94.89	1364.86	311	1587.07	30.44
每股盈餘	-94.89	1364.86	311	1587.07	30.44
獲利能力分析(%)					
營業毛利率	8.9	12.0	30.8	67.1	65.7
營業利益率	1.7	5.7	22.0	59.3	50.2
稅前息前折舊攤銷前淨利率	6.7	10.3	27.5	60.4	50.2
稅後純益率	0.4	5.3	17.2	46.6	40.0
資產報酬率	0.3	3.4	9.8	58.0	25.3
股東權益報酬率	0.7	7.7	20.2	111.1	49.7
穩定 / 償債能力分析					
負債/資產（%）	52.6	58.1	46.7	48.1	49.2
淨負債比率（%）	32.7	95.0	49.6	-4.5	-33.6
利息保障倍數（X）	1.1	5.1	11.8	129.3	161.9
利息及短期債保障倍數	0.1	0.2	0.6	2.2	1.0
現金流量 / 利息保障倍數(X)	-9.0	9.4	-12.9	45.1	45.5
現金流量 / 利息,短債保障倍數(X)	-0.9	0.4	-0.7	0.8	0.3
流動比率(%)	105.4	127.0	196.6	256.3	224.9
速動比率(%)	32.0	40.3	57.7	149.4	171.8
淨負債(百萬元)	241	755	792	-257	-3,593
每股資料分析(元)					
每股盈餘	0.1	1.2	5.0	83.8	109.3
每股營業現金流量	-3.2	3.1	-7.4	37.5	38.7
每股淨值	15.2	16.4	32.8	117.3	219.5
每股營收	18.9	22.6	28.9	179.7	273.3
每股稅前息前折舊攤銷前淨利	1.3	2.3	8.0	108.5	137.1
每股現金股利	0.1	0.2	0.5	10.0	13.0

資料來源:公司資料及國泰預估

國泰期貨 國泰證券子公司

聯友金屬 7610 TT

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投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%

註:潛在報酬率=(6 個月內目標價/本次報告評等最近一天收盤價 – 1) * 100

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本報告係根據本公司所取得資訊加以彙集及研究分析， 本公司雖盡力使用可靠且廣泛的資訊， 但本公司並不保證各項資訊之完整性及正確性。本報告中所提出之意見係為本報告出版當時的意見，邇後相關資訊或意見若有變更， 本公司將不會另行通知。本公司亦無義務持續更新本報告之內容或追蹤研究本報告所涵蓋之主題。本報告中提及的標的物價格、價值及收益隨時可能因各種本公司無法控制之政治、經濟、市場等因素而產生變化。本報告中之各項預測， 如營收及目標價格， 均係基於對目前所得資訊作合理假設下所完成， 所以並不必然實現。從事特定交易， 如涉及期貨、選擇權及衍生性金融商品， 因具有高度風險並不適合每一個投資人。本報告不得視為買賣有價證券或其他金融商品的要約或要約之引誘。

國泰金融控股公司所屬子公司暨關係企業(以下稱 “ 國泰金融集團”) 從事廣泛金融業務，包括但不限於銀行、保險、證券經紀、自有資金投資與交易、資產管理及證券投資信託等。國泰金融集團對於本報告所涵蓋之標的公司可能有投資或其他的業務往來關係。國泰金融集團及其所屬員工， 可能會投資本報告所涵蓋之標的公司， 或以該等公司為標的的衍生性金融商品， 而其交易的方向與本報告中所提及的交易方向可能不一致。除此之外， 國泰金融集團於法令許可的範圍內，亦有可能以本報告所涵蓋之標的公司作為發行衍生性

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2026 年 10 月 6 日

研究員：王美珍 iris.wang@capital.com.tw

前日收盤價	2,005.00 元
目標價	
3 個月	2,680.00 元
12 個月	2,680.00 元

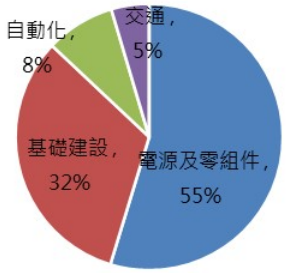
近期報告日期、評等及前日股價

07/31/2026	Buy	1,530 元
06/26/2026	Trading Buy	1,980 元
05/04/2026	Buy	2,165 元
03/02/2026	Buy	1,430 元
01/07/2026	Buy	1,050 元

公司基本資訊

目前股本(百萬元)	25,975
市值(億元)	52,080
目前每股淨值(元)	113.07
外資持股比(%)	61.71
投信持股比(%)	4.42
董監持股比(%)	7.32
融資餘額(張)	9,854
現金股息配發率(%)	50.13

產品組合



股價相對大盤走勢

台達電(2308 TT) Buy

產品組合較佳，獲利持續提升。

投資建議：受惠於 AI、Data Center 領域需求成長，營收與獲利維持成長趨勢，投資評等為 Buy，目標價為 2,680 元。

預估 3Q26 營收 QoQ+7.28%：受缺料影響，預估 3Q26 營收為 1,965.94 億元、QoQ+7.28%；在無較大業外收支下，稅後淨利為 257.42 億元、QoQ+2.41%，稅後 EPS 為 9.91 元。受惠於新一代 AI 伺服器放量與缺料緩解，預估 4Q26 營收為 2,218.52 億元、QoQ+12.85%；稅後淨利為 309.71 億元、QoQ+20.32%，稅後 EPS 為 11.92 元。

產品規格提升，電源類產品展望佳：台達電為 AI 伺服器電源解決方案主要供應商，受惠於 AI 需求提升，帶動 AC/DC 電源大幅成長外，CSP 客戶在新一代 ASIC 伺服器平台採用 DC/DC 電源模組，亦成為另一成長動能。800 VDC 與±400VDC 預計於 4Q26 出貨，2026 年營收比重約 1%，預期 2H27 營收貢獻較顯著。

水冷散熱 2026 年營收維持成長趨勢：水冷散熱產品有水冷板、Manifold、CDU、Sidecar。目前出貨以系統類的 Sidecar 為主，歸類於基礎建設；CDU 則歸類於電源類與零組件。水冷散熱 2025 年營收比重為 9%，台達電 Sidecar 產品 2025 年市佔率極高，雖有競爭者加入，但整體市場需求持續增長，2026 年 Sidecar 營收仍將維持成長趨勢，預估 2026 年水冷散熱營收比重為 12%。

固態變壓器為長期成長動能：SST(固態變壓器)目前已少量出貨，客戶為中系 CSP、800V 高壓直流電動車充電站，亦出貨給美系客戶測試。SST 導入需從資料中心設計初期即納入規劃，預期 2~3 年後會產生顯著的營收貢獻，仍持續改良產品規格中。

(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	761,055	1,032,458	1,312,055	150,318	161,613	159,353	183,256	196,594	221,852	218,056	252,669	281,284
營業毛利淨額	274,229	364,987	466,513	52,416	55,903	58,961	65,308	70,186	79,775	76,350	89,480	99,787
營業利益	132,954	182,531	298,210	24,809	26,418	28,417	30,570	33,684	40,282	39,550	51,103	33,684
稅後純益	102,404	160,037	230,476	18,606	17,324	20,556	25,136	25,742	30,971	30,878	38,884	45,411
稅後 EPS(元)	39.42	61.61	88.73	7.16	6.67	7.91	9.68	9.91	11.92	11.89	14.97	17.48
毛利率(%)	36.03%	35.35%	35.56%	34.87%	34.59%	37.00%	35.64%	35.70%	35.96%	35.01%	35.41%	35.48%
營業利益率(%)	17.47%	17.68%	22.73%	16.50%	16.35%	17.83%	16.68%	17.13%	18.16%	18.14%	20.23%	11.98%
稅後純益率(%)	13.46%	15.50%	17.57%	12.38%	10.72%	12.90%	13.72%	13.09%	13.96%	14.16%	15.39%	16.14%
營業收入 YoY/QoQ(%)	37.16%	35.66%	27.08%	21.19%	7.51%	-1.40%	15.00%	7.28%	12.85%	-1.71%	15.87%	11.33%
稅後純益 YoY/QoQ(%)	70.37%	56.28%	44.01%	33.39%	-6.89%	18.66%	22.28%	2.41%	20.32%	-0.30%	25.93%	16.78%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 259.75 億元計算。

ESG：

台達電超過一半的董事會成員實際參與台達永續委員會、節能管理、節水減廢管理，及再生電力的重要會議；永續長每季向董事會報告氣候變遷的管理進展，並且由董事會針對重要議題做決策。在進行重要併購或是新建廠房等決策時，亦會考量氣候變遷的風險與機會。

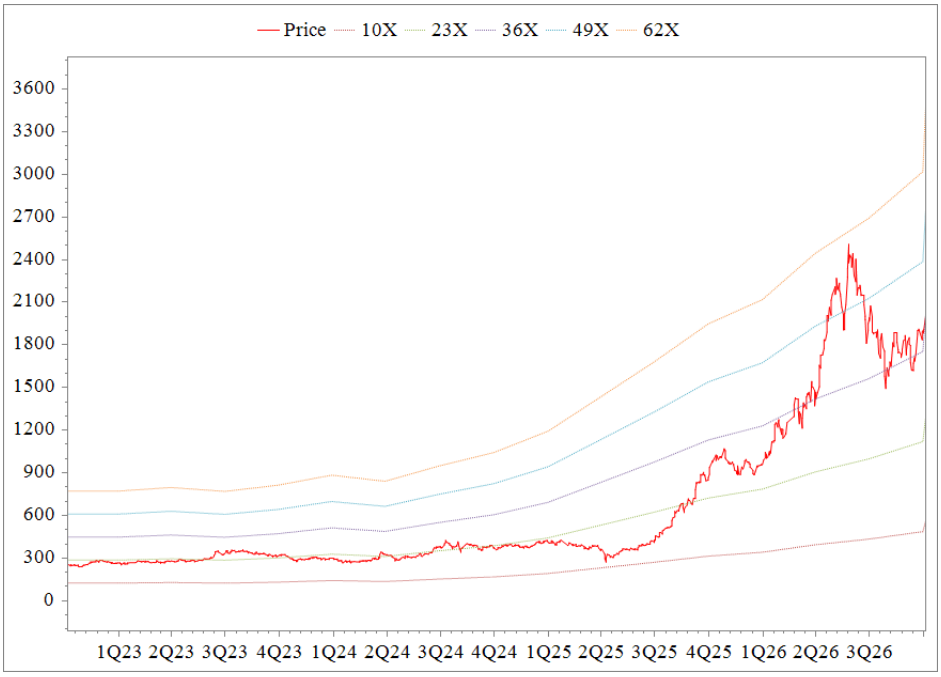
ESG 委員會為內部最高層級氣候相關推動組織。管理職責回歸包含企業永續發展部、節能服務部、財務、採購、人資、事業群等單位策略。依企業策略、氣候風險大數據、各類研究報告，以及外部評比指標鑑別風險與機會項目。

在治理層面，本報告新增「產品安全」與「品質管理」專章，藉由制定符合國際標準的相關政策與指引，強化控管機制，確保產品皆經嚴謹把關。此外，台達確立風險組織架構，由風管推動委員會辨別集團短期至長期（新興）風險，並完善風險管理流程。在供應鏈管理上，台達則首次舉辦「供應商 ESG 溝通大會」，明確傳達 ESG 與低碳供應鏈管理策略。

為推動價值鏈減碳，台達成立兩項子委員會，分別針對 C1(採購的產品與服務)與 C11(售出產品的使用)開展減碳行動。透過跨部門合作，盤點原物料採購與產品使用階段之碳排數據，並推動包括產品能效提升、低碳材料設計、以及與供應商與客戶議合，致力於實現範疇三減碳目標。

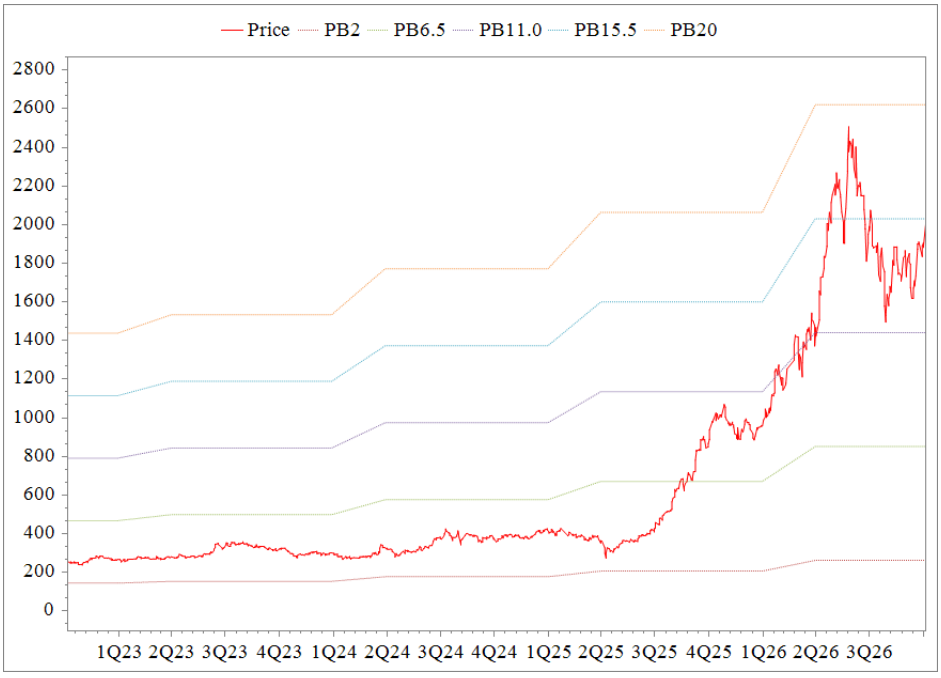
在自然資本保育方面，台達目標於 2050 年實現淨值正面效益(NPI)。2024 年成為 TNFD Early Adopters，並首次發表《氣候與自然生物多樣性報告書》，依循 TCFD、TNFD 與 IFRS S2 等國際準則，提升揭露品質與透明度。同年，參與聯合國生物多樣性 COP16 分享自身實踐，並連署 Business for Nature 發起之「Renewed policy ambition on nature」企業聲明，倡議將自然納入政策決策。

Forward PE Band



資料來源：CMoney，群益預估彙整

Forward PB Band



資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	531,898	639,619	781,606	1,024,618	1,372,413
流動資產	310,925	395,781	577,622	772,878	946,158
現金及約當現金	117,459	151,172	219,564	294,364	360,057
應收帳款與票據	89,921	121,717	172,363	235,415	280,420
存貨	83,856	101,478	146,309	179,292	258,536
採權益法之投資	234	380	224	397	1,042
不動產、廠房設備	115,710	142,040	143,865	145,462	147,082
負債總計	**252,332**	**314,230**	**373,776**	**487,043**	**651,229**
流動負債	152,283	209,665	287,089	381,198	466,547
應付帳款及票據	69,223	94,452	117,644	156,149	211,488
非流動負債	100,049	104,565	63,494	44,148	67,646
權益總計	279,565	325,389	407,830	537,575	721,184
普通股股本	**25,975**	**25,975**	**25,975**	**25,975**	**25,975**
保留盈餘	141,468	183,526	265,967	395,713	579,321
母公司業主權益	230,087	268,058	340,330	458,806	632,137
負債及權益總計	**531,898**	**639,619**	**781,606**	**1,024,618**	**1,372,413**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	72,895	98,474	139,167	220,752	314,346
稅前純益	51,316	87,866	144,118	215,680	304,946
折舊及攤銷	25,016	27,839	31,075	31,420	31,637
營運資金變動	-2,486	-24,190	-72,285	-57,530	-68,911
其他營運現金	-951	6,959	36,259	31,182	46,674
投資活動現金	-40,360	-53,214	-8,476	-90,987	-217,249
資本支出淨額	-33,380	-45,580	-50,000	-45,000	-45,000
長期投資變動	-547	-2,349	0	0	0
其他投資現金	-6,433	-5,285	41,524	-45,987	-172,249
籌資活動現金	-11,118	-9,401	-62,299	-54,967	-31,404
長借/公司債變動	8,552	11,283	-41,070	-19,346	23,497
現金增資	0	0	0	0	0
發放現金股利	-16,702	-18,183	-30,132	-41,561	-57,146
其他籌資現金	-2,967	-2,501	8,903	5,941	2,245
淨現金流量	30,003	33,713	68,392	74,799	65,693
期初現金	87,456	117,459	151,172	219,564	294,364
期末現金	117,459	151,172	219,564	294,364	360,057

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	421,148	554,885	761,055	1,032,458	1,312,055
營業成本	284,567	364,729	486,826	626,157	845,541
營業毛利淨額	136,580	190,157	274,229	364,987	466,513
營業費用	88,928	106,224	141,276	155,495	168,304
營業利益	47,652	83,932	132,954	182,531	298,210
EBITDA	74,472	114,204	154,768	203,660	210,360
業外收入及支出	3,664	3,934	11,164	11,154	7,872
稅前純益	51,316	87,866	144,118	215,680	304,946
所得稅	10,925	19,930	31,546	44,374	64,191
稅後純益	35,229	60,108	102,404	160,037	230,476
稅後 EPS(元)	13.56	23.14	39.42	61.61	88.73
完全稀釋 EPS**	13.56	23.14	39.42	61.61	88.73

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 259.75【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 259.75 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	4.97%	31.76%	37.16%	35.66%	27.08%
營業毛利淨額	16.52%	39.23%	44.21%	33.10%	27.82%
營業利益	16.37%	76.14%	58.41%	37.29%	63.38%
稅後純益	5.50%	70.62%	70.37%	56.28%	44.01%
獲利能力分析(%)					
毛利率	32.43%	34.27%	36.03%	35.35%	35.56%
EBITDA(%)	17.68%	20.58%	20.34%	19.73%	16.03%
營益率	11.31%	15.13%	17.47%	17.68%	22.73%
稅後純益率	8.36%	10.83%	13.46%	15.50%	17.57%
總資產報酬率	6.62%	9.40%	13.10%	15.62%	16.79%
股東權益報酬率	12.60%	18.47%	25.11%	29.77%	31.96%
償債能力檢視					
負債比率(%)	47.44%	49.13%	47.82%	47.53%	47.45%
負債/淨值比(%)	90.26%	96.57%	91.65%	90.60%	90.30%
流動比率(%)	204.18%	188.77%	201.20%	202.75%	202.80%
其他比率分析					
存貨天數	102.67	92.74	92.89	94.90	94.50
應收帳款天數	73.36	69.61	70.52	72.08	71.75

資料來源：CMoney、群益

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	124,035	150,318	161,613	159,353	183,256	196,594	221,852	218,056	252,669	281,284	280,449	276,346
營業成本	79,986	97,902	105,710	100,392	117,948	126,409	142,077	100,392	163,188	181,498	181,078	178,964
營業毛利淨額	44,049	52,416	55,903	58,961	65,308	70,186	79,775	76,350	89,480	99,787	99,371	97,382
營業費用	25,380	27,607	29,485	30,544	34,738	36,502	39,493	36,800	38,377	40,825	39,493	39,300
營業利益	18,669	24,809	26,418	28,417	30,570	33,684	40,282	39,550	51,103	33,684	58,194	58,082
業外收入及支出	902	2,174	-770	2,676	4,430	2,080	1,980	2,676	4,507	2,035	1,935	1,915
稅前純益	19,571	26,983	25,649	31,093	35,000	35,764	42,262	41,466	53,088	60,997	60,129	59,997
所得稅	4,237	6,062	6,011	7,258	7,831	7,582	8,875	7,258	11,679	12,809	12,627	11,999
稅後純益	13,948	18,606	17,324	20,556	25,136	25,742	30,971	30,878	38,884	45,411	44,864	45,492
最新股本	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975	25,975
稅後EPS(元)	5.37	7.16	6.67	7.91	9.68	9.91	11.92	11.89	14.97	17.48	17.27	17.51
獲利能力(%)												
毛利率(%)	35.51%	34.87%	34.59%	37.00%	35.64%	35.70%	35.96%	35.01%	35.41%	35.48%	35.43%	35.24%
營業利益率(%)	15.05%	16.50%	16.35%	17.83%	16.68%	17.13%	18.16%	18.14%	20.23%	11.98%	20.75%	21.02%
稅後純益率(%)	11.25%	12.38%	10.72%	12.90%	13.72%	13.09%	13.96%	14.16%	15.39%	16.14%	16.00%	16.46%
QoQ(%)												
營業收入淨額	4.30%	21.19%	7.51%	-1.40%	15.00%	7.28%	12.85%	-1.71%	15.87%	11.33%	-0.30%	-1.46%
營業利益	33.01%	32.89%	6.49%	7.57%	7.58%	10.19%	19.59%	-1.82%	29.21%	-34.09%	72.76%	-0.19%
稅前純益	24.95%	37.87%	-4.95%	21.23%	12.56%	2.18%	18.17%	-1.88%	28.03%	14.90%	-1.42%	-0.22%
稅後純益	36.33%	33.39%	-6.89%	18.66%	22.28%	2.41%	20.32%	-0.30%	25.93%	16.78%	-1.20%	1.40%
YoY(%)												
營業收入淨額	19.91%	33.97%	41.51%	34.00%	47.75%	30.79%	37.27%	36.84%	37.88%	43.08%	26.41%	26.73%
營業利益	42.16%	51.06%	146.87%	102.46%	63.75%	35.77%	52.48%	39.18%	67.17%	0.00%	44.47%	46.86%
稅前純益	30.16%	52.67%	160.67%	98.51%	78.83%	32.54%	64.77%	33.36%	51.68%	70.56%	42.28%	44.69%
稅後純益	40.21%	50.79%	141.36%	100.92%	80.21%	38.35%	78.78%	50.22%	54.70%	76.41%	44.86%	47.33%

註1：稅後EPS以股本259.75億元計算

註2：稅後純益係指本期淨利歸屬於母公司業主。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

個股報告

2026 年 10 月 6 日

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前日收盤價	817.00 元
目標價	
3 個月	1,080.00 元
12 個月	1,080.00 元

近期報告日期、評等及前日股價

禾伸堂(3026 TT) Buy

NP0 MLCC 受惠 AI SERVER 電源架構升級

投資建議：受惠 AI SERVER 電源架構升級，禾伸堂在 NP0 具領先同業優勢，營收及毛利率將同步攀升，投資建議為 Buy，目標價 1,080 元。

高階 MLCC 占比持續攀升，2Q26 財報表現優異：

禾伸堂 2Q26 營收 40.97 億元(QOQ+13.2%/YOY+25.2%)，優於市場預期，主因受惠於高階 MLCC 產出與銷售增加，帶動毛利率達 26.18% (QOQ+1.94ppts/YOY+6.37ppts)，費用率 10.89%與 2Q26 相當，本業淨利 6.27 億元(QOQ+29.6%/YOY+111.2%)，稅後純益 6.42 億元(QOQ+35.2%/YOY+179.1%)，較去年同期倍增，2Q26 稅後 EPS 3.87 元，獲利呈現大幅增長，優於市場預期。

展望未來，NP0 MLCC 市佔率有望持續攀升，獲利將同步上揚：

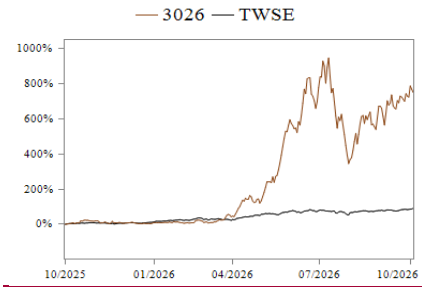
以 NP0 與 X7R/X8R 競爭態勢來看，目前電源類產品的驗證需要經過直接客戶、終端客戶以及安規認證，整體認證時間至少需要 2~3 年，以 VR 機種來看，台廠目前通過認證家數不到 3 家，當前 NP0 以禾伸堂為主要份額。日韓競爭方面，因高階小尺寸 MLCC 產能持續緊缺，日韓等主要大廠傾向優先轉向生產 0201、0402、0603 等小尺寸規格，且擴產意願不高，故客戶有望轉向要求禾伸堂增加產能與供給，2027 年 NP0 市佔率有望進一步提升。禾伸堂 2Q26 AI 應用占比約為 32%，考量 Vera Ribin 將於 4Q26 開始小量生產，於 2027 年正式放量，加上 TPU 與 ASIC 平台需求快速成長，皆有望帶動高階 MLCC 用量需求持續提升，群益預估禾伸堂 2026 年底 AI 應用占比有望達 45%，2027 年底將提升至 60%。綜上所述，預估 2026 年營收為 171.22 億元(YOY+27.5%)、2027 年營收為 245.70 億元(YOY+43.5%)，毛利率受惠 AI 比重持續拉升分別為 26.8%及 34%，2026 年、2027 年稅後 EPS 為 16.43 元及 31.59 元，獲利呈現大幅增長。

公司基本資訊

目前股本(百萬元)	1,659
市值(億元)	1,355
目前每股淨值(元)	62.28
外資持股比(%)	11.02
投信持股比(%)	16.10
董監持股比(%)	9.55
融資餘額(張)	13,652
現金股息配發率(%)	88.15

產品組合

股價相對大盤走勢



(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	17,122	24,570	31,986	3,433	3,320	3,618	4,097	4,494	4,912	5,296	5,887	6,687
營業毛利淨額	4,584	8,365	12,818	625	585	877	1,073	1,231	1,404	1,625	1,930	2,332
營業利益	2,715	5,674	9,309	268	193	483	627	753	853	1,040	1,284	1,610
稅後純益	2,725	5,240	8,304	280	303	475	642	746	863	979	1,197	1,479
稅後 EPS(元)	16.43	31.59	50.06	1.69	1.83	2.86	3.87	4.49	5.20	5.90	7.21	8.91
毛利率(%)	26.77%	34.05%	40.07%	18.19%	17.62%	24.24%	26.18%	27.38%	28.58%	30.68%	32.78%	34.88%
營業利益率(%)	15.86%	23.09%	29.10%	7.81%	5.81%	13.36%	15.29%	16.74%	17.36%	19.63%	21.81%	24.08%
稅後純益率(%)	15.91%	21.33%	25.96%	8.15%	9.14%	13.12%	15.66%	16.59%	17.57%	18.49%	20.33%	22.11%
營業收入 YoY/QoQ(%)	27.52%	43.49%	30.18%	4.92%	-3.30%	9.00%	13.23%	9.69%	9.29%	7.82%	11.15%	13.59%
稅後純益 YoY/QoQ(%)	149.60%	92.31%	58.47%	21.71%	8.37%	56.45%	35.24%	16.17%	15.73%	13.48%	22.21%	23.56%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 16.59 億元計算。

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ESG(Environment, Social, Governance)：

禾伸堂（3026 TT）致力於永續發展，在不斷精進技術研發及調整產品組合之下，同時將保護環境及節能減廢納入重要管理項目。在溫室氣體管理的部分，2025 年臺灣地區溫室氣體排放已進行盤查，並於 2026 年完成第三方查驗，2025 年也依據 ISO 14067:2018 標準完成積層陶瓷電容產品碳足跡查證，透過了解產品碳排放量，以制定有效的減碳計畫；此外，為了有效減少溫室氣體排放，透過自發自用太陽能光電 2025 年已取得國家再生能源憑證中心發放之綠電憑證 T-REC 628 張（626,064 度），推估減少溫室氣體排放約 309.28 公噸。在能源與資源管理方面，落實廢棄物分類管理，並進行製程原物料與廢溶劑回收再利用的「轉廢為能」計畫，與供應商合作將廢塑料作為固體再生燃料（SRF），2025 年共計回收 117.05 公噸，大幅減少廢棄物直接焚化的處理量；另外，2025 年回收溶劑量約 194.70 公噸，有效減少溶劑（原物料）採購及廢棄物處理費用，並降低對環境的衝擊。

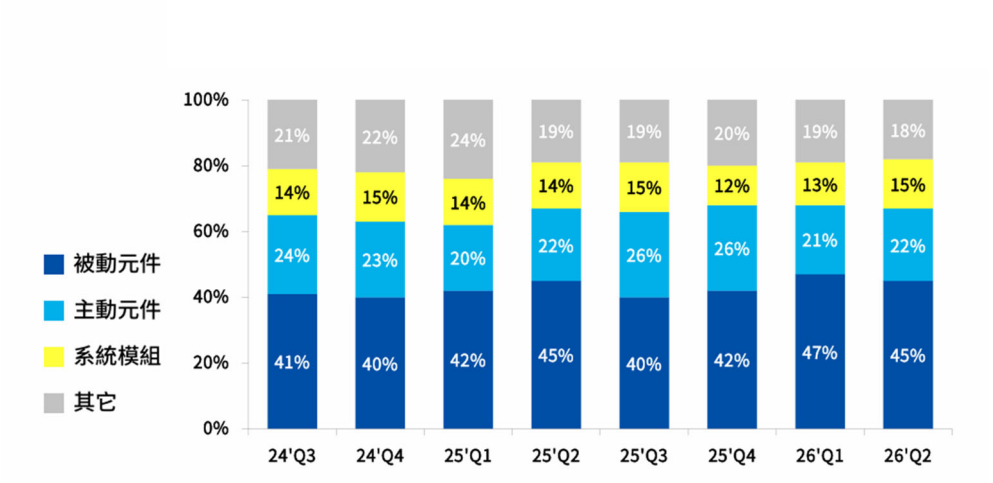
公司簡介：

禾伸堂(3026 TT)成立於 1981 年。公司總部位於台灣台北，生產基地分布於桃園龍潭、宜蘭利澤以及日本北海道。早期以被動元件代理商起家，後期切入利基型積層陶瓷電容器(MLCC)及固態電容研發製造。主要產品包含：積層陶瓷電容(MLCC)、主動元件、系統模組與其他代理產品。公司代理產品以銷售通訊、車用及消費性電子市場為主。2Q26 產品營收比重:被動元件 45%、主動元件 22%、系統模組 15%、其他 18%。

圖一、禾伸堂 MLCC 產品圖:

資料來源：禾伸堂、群益投顧彙整

圖二、禾伸堂產品營收比重:



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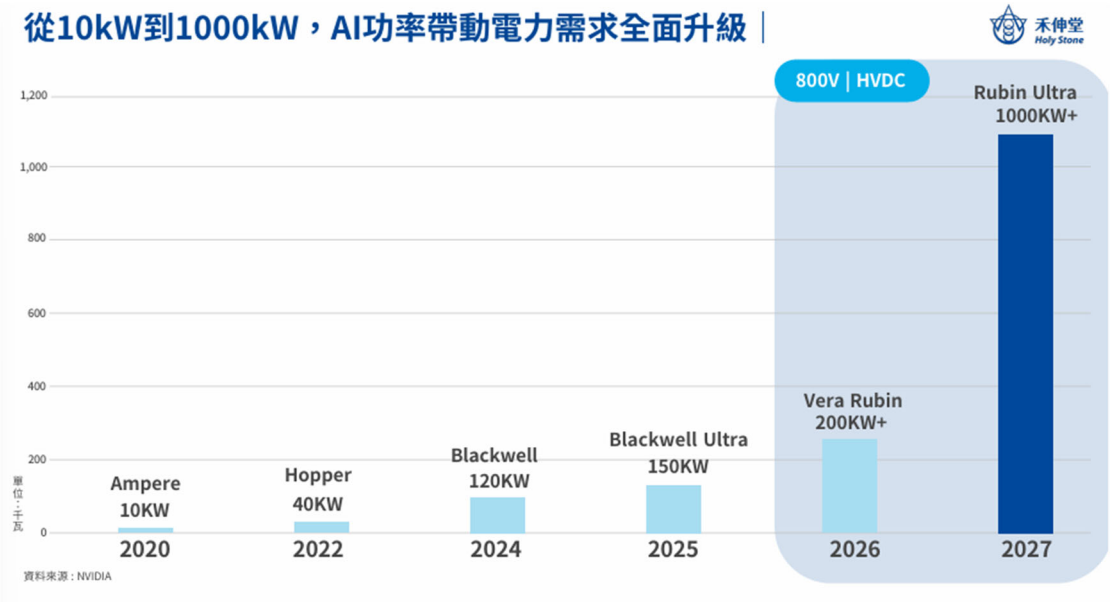
因 AI 伺服器電源架構升級，帶動 NP0 MLCC 用量顯著大增：

禾伸堂目前產品主要應用於電源類，尤其是 AI 伺服器隨著電源供應器瓦數增加，電源從 IT Rack 移出獨立成 Power Rack(Sidecar Rack)，新一代 Rubin 平台架構也將朝向更高功率密度的獨立電源櫃與多元電力備援系統發展。這項架構轉型大幅推升了高電壓、高效能與高穩定性被動元件的需求，其中 NP0 材質之積層陶瓷電容（MLCC）因具備極佳的溫度穩定性、低介質損耗與高可靠度，成為高功率電源供應器與電壓調節模組（VRM）中不可或缺的關鍵零組件，因過去多採用單相 PSU(如 3.3kW、4.2kW、4.3kW 等)，而下一世代為了在高瓦數下增加效率，Power Rack 已轉成三相 AC/DC PSU，此外，內部仍需要進行 DC/DC 轉換(400V 降至 48V/50V，或 800V 降至 48V)，群益預估單一機櫃所需的 NP0 MLCC 數量可提升 3~5 倍。

圖三、AI 功率持續提升，電力需求全面升級:



資料來源：禾伸堂、群益投顧彙整

圖四、AI Server Power、BBU 使用客製化、低損耗、高壓、高容 NP0:

年份	NVIDIA GPU 型號	伺服器電源供應器瓦數	機櫃電壓	客製化低損耗 (NP0) 高壓、高容需求規格	單一機櫃使用數量	AI Server Power 供應商	全球主要CSP
2020	A100	1600W~2200W	208~240 VAC	1206 NP0 10nF 630V	800~1000	DELTA LITEON flex 欧陆通 HONOTO	NVIDIA aws Microsoft Google Meta
2022	H100	3000W~3200W	240~415 VAC	1210 NP0 22nF 1000V	1800~2000		
2024	H200	3300W~3600W	415~480 VAC	1210 NP0 33nF 630V	1800~2000		
2025	GB200	4200W~5500W	480 VAC	1210 NP0 33nF 1000V	3000~5000		
2026	GB300	5500W~18KW	480 VAC	1210 NP0 33nF 1000V	3000~5000		
	Vera Rubin	5500W~18KW	480 VAC / 800V	1210 NP0 33nF 1000V	3000~5000		
2027	Rubin Ultra	>30KW	800V | HVDC	1210 NP0 47nF 630V	N/A		
				1210 NP0 33nF 1250V	N/A		

資料來源：禾伸堂、群益投顧彙整

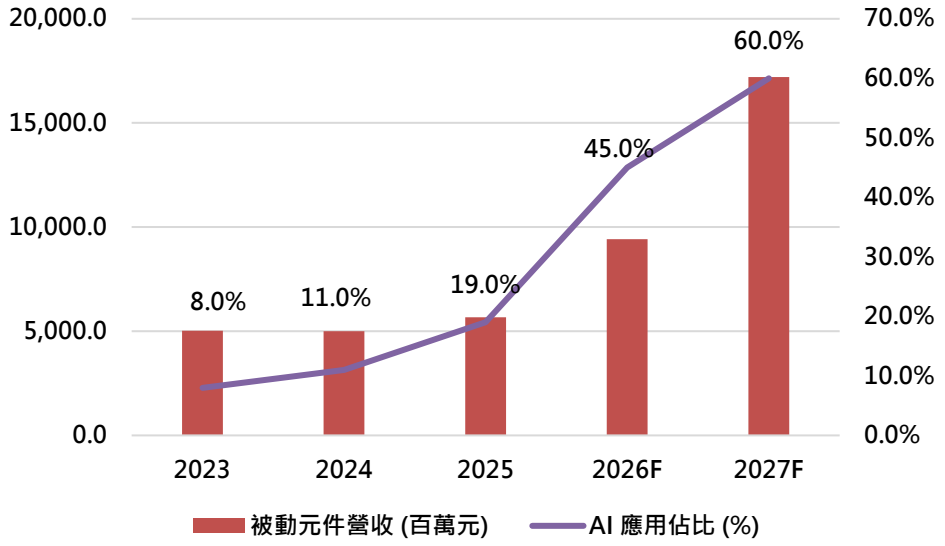
<u>展望未來，NP0 MLCC 市佔率有望持續攀升，獲利將同步上揚</u>：

以 NP0 與 X7R/X8R 競爭態勢來看，目前電源類產品的驗證需要經過直接客戶、終端客戶以及安規認證，整體認證時間至少需要 2~3 年，以 VR 機種來看，台廠目前通過認證家數不到 3 家，當前 NP0 以禾伸堂為主要份額。日韓競爭方面，因高階小尺寸 MLCC 產能持續緊缺，日韓等主要大廠傾向優先轉向生產 0201、0402、0603 等小尺寸規格，且擴產意願不高，故客戶有

望轉向要求禾伸堂增加產能與供給，2027 年 NP0 市佔率有望進一步提升。

禾伸堂 2Q26 AI 應用占比約為 32%，考量 Vera Ribin 將於 4Q26 開始小量生產，於 2027 年正式放量，加上 TPU 與 ASIC 平台需求快速成長，皆有望帶動高階 MLCC 用量需求持續提升，群益預估禾伸堂 2026 年底 AI 應用占比有望達 45%，2027 年底將提升至 60%。綜上所述，預估 2026 年營收為 171.22 億元(YOY+27.5%)、2027 年營收為 245.70 億元(YOY+43.5%)，毛利率受惠 AI 比重持續拉升分別為 26.8%及 34%，2026 年、2027 年稅後 EPS 為 16.43 元及 31.59 元，獲利呈現大幅增長。

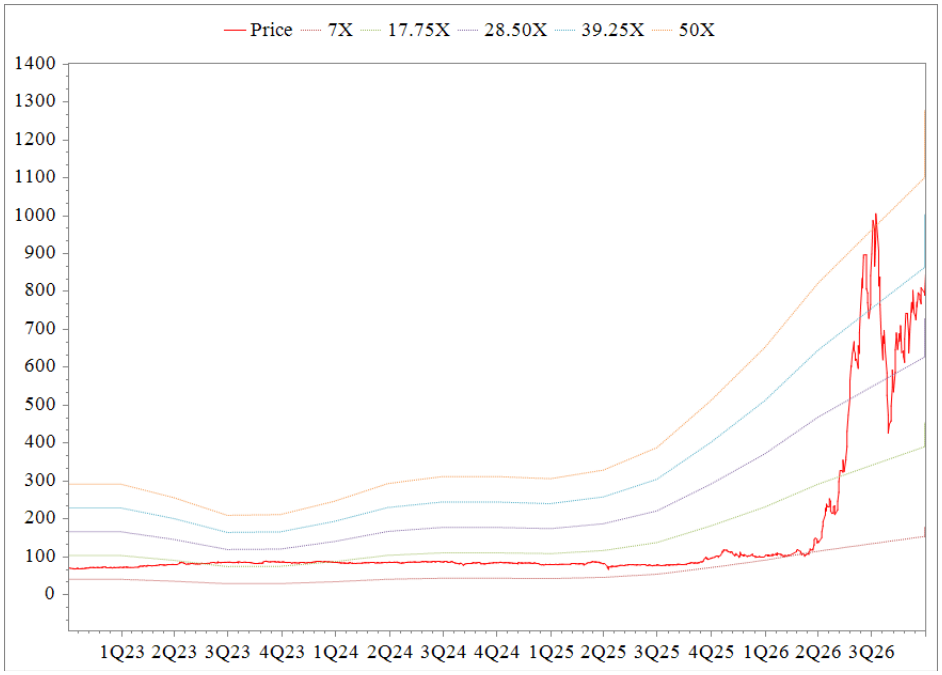
圖五、禾伸堂 AI 占比 2027 年預估達 60%:



資料來源：群益投顧預估

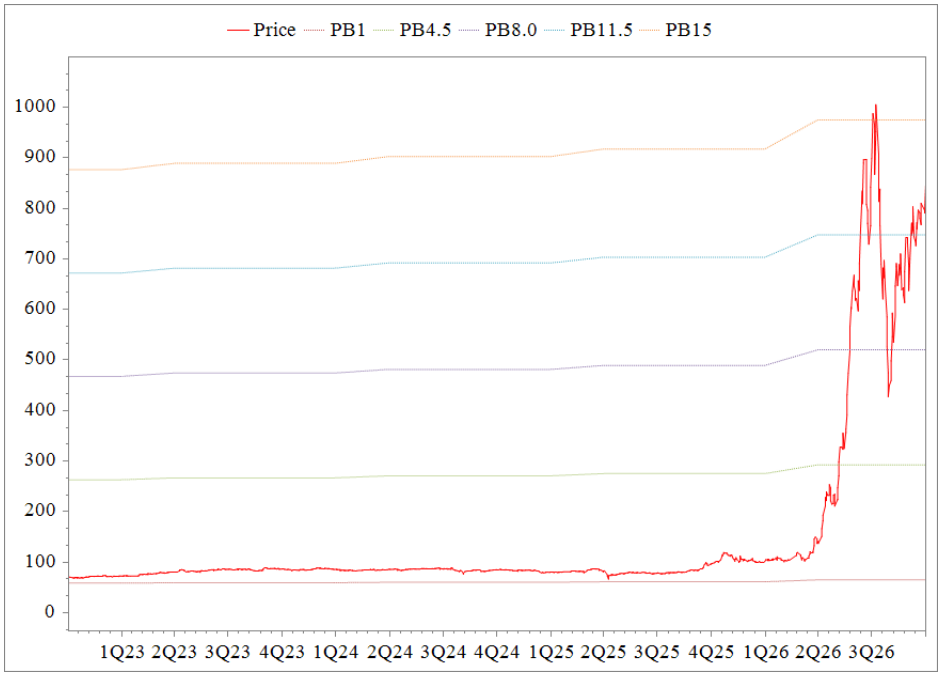
投資建議：受惠 AI SERVER 電源架構升級，禾伸堂在 NP0 具領先同業優勢，營收及毛利率將同步攀升，投資建議為 Buy，目標價 1,080 元。

Forward PE Band



資料來源：CMoney，群益預估彙整

Forward PB Band



資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	15,826	16,188	18,568	24,253	32,784
流動資產	10,912	11,724	13,996	19,777	28,405
現金及約當現金	4,502	4,619	6,135	11,838	20,464
應收帳款與票據	3,087	3,784	3,937	4,380	3,941
存貨	2,575	2,156	2,255	2,646	2,277
採權益法之投資	443	444	581	581	581
不動產、廠房設備	4,231	3,764	3,603	3,365	3,117
負債總計	**5,446**	**5,789**	**7,327**	**10,411**	**14,784**
流動負債	3,791	4,932	5,804	6,733	5,415
應付帳款及票據	989	946	1,024	1,047	1,049
非流動負債	1,655	857	1,523	3,678	9,369
權益總計	10,379	10,399	11,038	13,639	17,797
普通股股本	**1,659**	**1,659**	**1,659**	**1,659**	**1,659**
保留盈餘	5,259	5,442	6,401	9,021	13,173
母公司業主權益	9,983	10,148	10,786	13,387	17,545
負債及權益總計	**15,826**	**16,188**	**18,365**	**24,050**	**32,581**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	1,611	1,080	2,156	4,696	7,065
稅前純益	1,103	1,191	3,149	5,545	8,607
折舊及攤銷	724	734	698	687	687
營運資金變動	393	-322	-810	208	172
其他營運現金	-610	-522	-881	-1,743	-2,401
投資活動現金	146	-253	-677	-584	-587
資本支出淨額	138	-266	-486	-448	-438
長期投資變動	1	0	0	0	0
其他投資現金	6	13	-190	-137	-150
籌資活動現金	-379	-702	4	1,077	1,154
長借/公司債變動	160	-680	608	1,077	1,153
現金增資	0	0	0	0	0
發放現金股利	-829	-912	-962	-2,402	-4,193
其他籌資現金	290	890	1,567	1,402	3,183
淨現金流量	1,422	117	1,516	5,220	7,658
期初現金	3,080	4,502	4,619	6,135	11,355
期末現金	4,502	4,619	6,135	11,355	19,012

資料來源：CMoney、群益

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	12,786	13,427	17,122	24,570	31,986
營業成本	10,697	10,907	12,538	16,205	19,168
營業毛利淨額	2,089	2,520	4,584	8,365	12,818
營業費用	1,468	1,442	1,869	2,691	3,509
營業利益	621	1,078	2,715	5,674	9,309
EBITDA	1,855	1,985	3,414	6,360	9,996
業外收入及支出	482	113	434	447	449
稅前純益	1,103	1,191	3,149	6,121	9,758
所得稅	177	165	524	984	1,557
稅後純益	973	1,092	2,725	5,240	8,304
稅後 EPS(元)	5.87	6.58	16.43	31.59	50.06
完全稀釋 EPS**	5.87	6.58	16.43	31.59	50.06

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 16.59【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 16.59 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	-3.43%	5.01%	27.52%	43.49%	30.18%
營業毛利淨額	-8.84%	20.65%	81.90%	82.46%	53.23%
營業利益	-12.76%	73.70%	151.83%	108.95%	64.07%
稅後純益	14.44%	12.17%	149.60%	92.31%	58.47%
獲利能力分析(%)					
毛利率	16.34%	18.77%	26.77%	34.05%	40.07%
EBITDA(%)	14.50%	14.78%	19.94%	25.89%	31.25%
營益率	4.85%	8.03%	15.86%	23.09%	29.10%
稅後純益率	7.61%	8.13%	15.91%	21.33%	25.96%
總資產報酬率	6.15%	6.74%	14.68%	21.61%	25.33%
股東權益報酬率	9.38%	10.50%	24.69%	38.42%	46.66%
償債能力檢視					
負債比率(%)	34.41%	35.76%	39.46%	42.93%	45.10%
負債/淨值比(%)	52.47%	55.67%	66.38%	76.33%	83.07%
流動比率(%)	287.84%	237.73%	241.14%	293.74%	524.58%
其他比率分析					
存貨天數	97.29	79.15	64.20	55.19	46.87
應收帳款天數	83.61	93.40	82.30	61.78	47.48

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	3,272	3,433	3,320	3,618	4,097	4,494	4,912	5,296	5,887	6,687	6,700	7,219
營業成本	2,624	2,808	2,735	2,741	3,025	3,264	3,508	3,671	3,957	4,354	4,222	4,463
營業毛利淨額	648	625	585	877	1,073	1,231	1,404	1,625	1,930	2,332	2,478	2,756
營業費用	352	356	392	394	446	478	551	585	646	722	738	796
營業利益	297	268	193	483	627	753	853	1,040	1,284	1,610	1,740	1,960
業外收入及支出	-1	68	74	85	136	98	115	111	116	111	109	111
稅前純益	295	336	267	569	762	850	968	1,150	1,400	1,721	1,849	2,072
所得稅	58	71	0	113	150	129	132	197	229	268	290	335
稅後純益	230	280	303	475	642	746	863	979	1,197	1,479	1,585	1,762
最新股本	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659	1,659
稅後EPS(元)	1.39	1.69	1.83	2.86	3.87	4.49	5.20	5.90	7.21	8.91	9.56	10.62
獲利能力(%)												
毛利率(%)	19.81%	18.19%	17.62%	24.24%	26.18%	27.38%	28.58%	30.68%	32.78%	34.88%	36.98%	38.18%
營業利益率(%)	9.06%	7.81%	5.81%	13.36%	15.29%	16.74%	17.36%	19.63%	21.81%	24.08%	25.97%	27.15%
稅後純益率(%)	7.03%	8.15%	9.14%	13.12%	15.66%	16.59%	17.57%	18.49%	20.33%	22.11%	23.67%	24.41%
QoQ(%)												
營業收入淨額	-3.82%	4.92%	-3.30%	9.00%	13.23%	9.69%	9.29%	7.82%	11.15%	13.59%	0.19%	7.76%
營業利益	-7.46%	-9.60%	-28.01%	150.44%	29.61%	20.12%	13.32%	21.92%	23.49%	25.39%	8.06%	12.66%
稅前純益	1.04%	13.87%	-20.44%	112.66%	34.08%	11.50%	13.88%	18.82%	21.71%	22.91%	7.47%	12.02%
稅後純益	-17.42%	21.71%	8.37%	56.45%	35.24%	16.17%	15.73%	13.48%	22.21%	23.56%	7.22%	11.15%
YoY(%)												
營業收入淨額	1.01%	0.64%	3.88%	6.36%	25.22%	30.92%	47.97%	46.37%	43.68%	48.78%	36.39%	36.30%
營業利益	95.35%	37.77%	70.37%	50.81%	111.23%	180.69%	341.88%	115.11%	104.95%	113.94%	104.01%	88.51%
稅前純益	19.12%	-17.26%	46.71%	94.65%	158.31%	152.94%	262.08%	102.30%	83.63%	102.41%	91.01%	80.07%
稅後純益	0.41%	-6.23%	33.03%	70.41%	179.09%	166.38%	184.47%	106.34%	86.47%	98.33%	83.73%	79.96%

註1：稅後EPS以股本16.59億元計算

註2：稅後純益係指本期淨利歸屬於母公司業主。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

2026/10/07

紡織

聚陽 (1477 TT)

維持評等

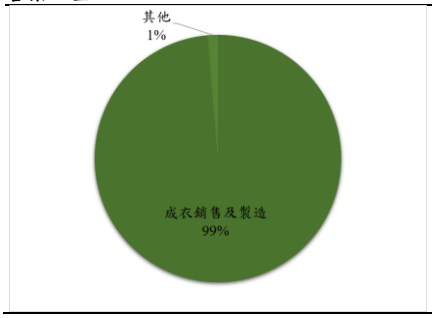
增加持股(Overweight)

潛在漲幅	18.9%
前日收盤價	203.50 元
目標價	242.00 元
前次目標價	242.00 元
前次評等	增加持股

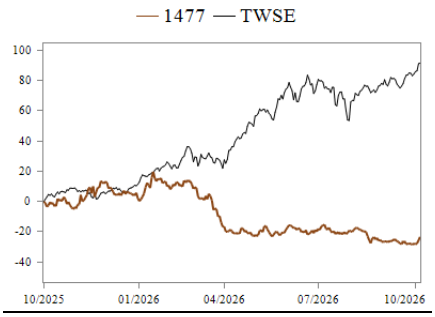
公司基本資訊

目前股本(百萬元)	2,467
市值(億元)	502
目前每股淨值(元)	55.89
外資持股比(%)	15.32
董監持股比(%)	9.39

營業比重



個股與大盤走勢(%)



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4Q26 營運展望正向；訂單篩選策略啟動

- **3Q26 成長動能延續；管理層審慎樂觀展望 4Q26**
- **下單模式轉變；客戶與訂單篩選支撐獲利能力**
- **2027 年復甦幅度仍取決於客戶補庫存時點**

營運維持溫和復甦，4Q26 展望轉趨正向：聚陽 3Q26 營收 94.8 億元、YoY+6.7%；前 9 月累計營收由負轉正為年增 0.2%，整體出貨大致符合公司預期。9 月營收 29.3 億元、YoY+24.1%，因去年同期低基期而有較高成長，但適逢聖誕節與感恩節出貨旺季導致船班擁擠，約 8,000 萬元營收遞延至次月。3Q26 出貨量估維持個位數年增，ASP 大致持平，預期毛利率仍可維持 24%以上水準。展望 4Q26，管理層看法轉趨正向，並啟動訂單篩選，包含挑選客戶、產品類別及提高短單比重，優先承接獲利能力較佳的業務，預期將有助於提升獲利表現。

短單占比提升，客戶與訂單篩選支撐獲利能力：受中東戰爭、油價及終端需求不確定性影響，客戶下單模式轉變，初期訂單通常未一次下滿，後續再以追加單補足，此趨勢預計延續至 4Q26。客戶下單模式轉向短單，使訂單能見度維持在約 4 個月，但聚陽具備快速反應與急短單生產優勢，而短單報價相對較佳，反而有利毛利率表現。公司將篩選價格及獲利條件較佳的訂單，減少關稅與原物料成本壓力較大的品項；4Q26 成長動能主要來自百貨公司、美系品牌及通路商回補庫存需求，在訂單結構及產品組合優化帶動下，獲利能力有望持續改善。

東南亞持續擴產，2027 年復甦幅度仍取決於客戶補庫存時點：產能佈局方面，由於菲律賓廠關閉抵銷部分新增產能，2026 年整體產能增幅有限，未來將持續以印尼與孟加拉作為主要擴產據點。展望 2027 年，客戶庫存已處低檔，管理層認為待美國期中選舉結束、政策與外部環境明朗後，客戶下單週期有望逐步恢復正常，將對具規模的大型供應商相對有利。不過，目前終端需求仍屬溫和，回補庫存啟動時點及力道尚未明確，因此短期營運可望延續緩和改善趨勢。公司表示內部營運體質已完成調整，後續較明顯的成長仍須等待外部需求回溫。

4Q26 營運正向，2027 年也有望持續復甦，惟時程與幅度仍須觀察。因此維持審慎的 14 倍 PER，以未來四季預估 EPS 計算，目標價 242 元。**風險因子：**外部環境變化、終端需求、客戶回補庫存速度。

(百萬元)	2022	2023	2024	2025	2026F	2027F
營業收入淨額	32,083	32,459	35,524	34,428	35,190	37,741
營業毛利淨額	8,256	8,366	8,991	8,564	8,899	9,626
營業利益	4,432	4,840	5,300	4,681	5,053	5,531
稅後純益	3,515	3,991	4,115	3,614	3,901	4,282
公告基本 EPS(元)	14.53	16.18	16.68	14.65	--	--
調整後 EPS(元)	14.53	16.50	16.68	14.65	15.81	17.36
營業收入 YoY(%)	10.90%	1.17%	9.44%	-3.08%	2.21%	7.25%
稅後純益 YoY(%)	32.38%	13.56%	3.10%	-12.17%	7.94%	9.77%
本益比	12.07	17.58	17.16	18.51	12.87	11.72
本淨比	2.76	4.35	4.82	4.76	3.40	3.35
股息(元)	14.50	16.40	17.10	15.00	16.00	17.20
殖利率(%)	6.16%	4.13%	4.98%	5.60%	7.86%	8.45%
ROE(%)	24.18%	25.60%	27.02%	25.20%	27.08%	28.77%

圖表一：季度財報預估差異

損益表 (單位：百萬元)	2Q26					3Q26(F)				
	實際數	中信預估	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	8,166	8,166	0.0%	8,413	-2.9%	9,484	9,303	1.9%	9,501	-0.2%
營業毛利	1,990	2,012	-1.1%	2,057	-3.3%	2,329	2,316	0.6%	2,326	0.1%
營業利益	1,086	1,191	-8.8%	1,179	-7.9%	1,354	1,442	-6.1%	1,338	1.3%
稅前純益	1,072	1,177	-8.9%	1,178	-9.0%	1,331	1,387	-4.1%	1,316	1.1%
稅後淨利	814	904	-9.9%	904	-9.9%	1,037	1,081	-4.1%	1,029	0.7%
EPS(元)	3.30	3.66	-10.0%	3.61	-8.5%	4.20	4.38	-4.1%	4.15	1.3%
財務比率(%/百分點)										
毛利率	24.4%	24.6%	-0.3 ppts	24.5%	-0.1 ppts	24.6%	24.9%	-0.3 ppts	24.5%	0.1 ppts
營益率	13.3%	14.6%	-1.3 ppts	14.0%	-0.7 ppts	14.3%	15.5%	-1.2 ppts	14.1%	0.2 ppts
稅後淨利率	10.0%	11.1%	-1.1 ppts	10.7%	-0.8 ppts	10.9%	11.6%	-0.7 ppts	10.8%	0.1 ppts

資料來源：Bloomberg、中信投顧預估

圖表二：年度財報預估差異

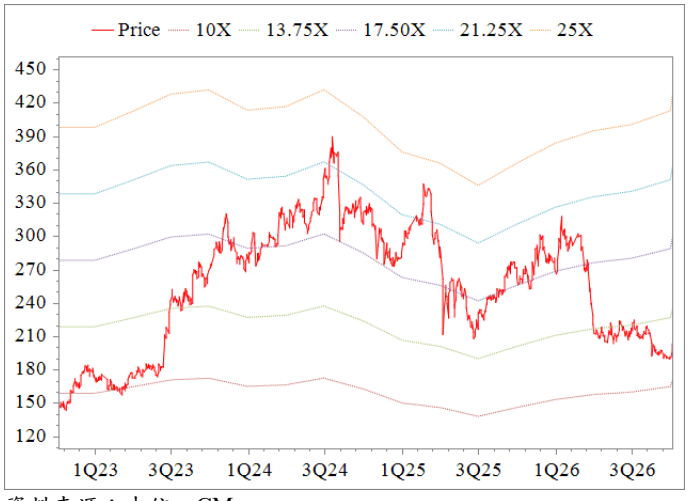
損益表 (單位：百萬元)	2026(F)					2027(F)				
	調整後	調整前	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	35,190	34,759	1.2%	35,276	-0.2%	37,741	37,584	0.4%	37,758	0.0%
營業毛利	8,899	8,807	1.1%	8,888	0.1%	9,626	9,632	-0.1%	9,627	0.0%
營業利益	5,053	5,173	-2.3%	4,973	1.6%	5,531	5,724	-3.4%	5,451	1.5%
稅前純益	4,981	5,068	-1.7%	4,891	1.8%	5,439	5,602	-2.9%	5,382	1.1%
稅後淨利	3,901	3,976	-1.9%	3,810	2.4%	4,282	4,455	-3.9%	4,189	2.2%
EPS(元)	15.81	16.12	-1.9%	15.51	1.9%	17.36	18.06	-3.9%	17.07	1.7%
財務比率(%/百分點)										
毛利率	25.3%	25.3%	0.0 ppts	25.2%	0.1 ppts	25.5%	25.6%	-0.1 ppts	25.5%	0.0 ppts
營益率	14.4%	14.9%	-0.5 ppts	14.1%	0.3 ppts	14.7%	15.2%	-0.6 ppts	14.4%	0.2 ppts
稅後淨利率	11.1%	11.4%	-0.4 ppts	10.8%	0.3 ppts	11.3%	11.9%	-0.5 ppts	11.1%	0.3 ppts

資料來源：Bloomberg、中信投顧預估

2026/10/07

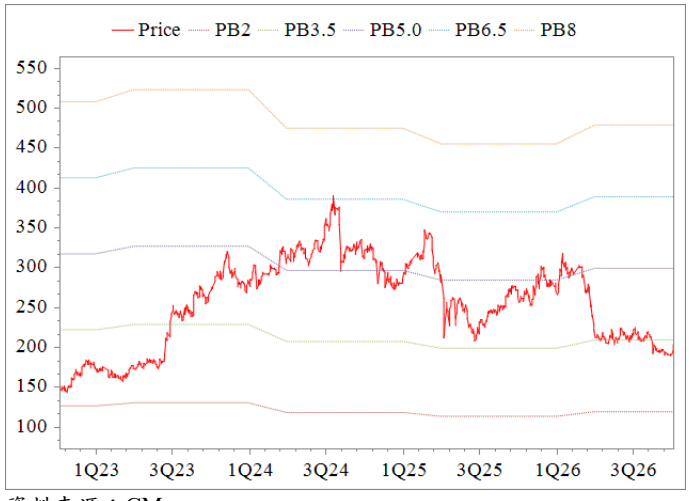
(百萬元)	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F
營業收入	8,897	8,166	9,484	8,643	9,451	8,706	10,334	9,250
營業成本	6,571	6,176	7,155	6,389	6,968	6,549	7,748	6,851
營業毛利	2,326	1,990	2,329	2,255	2,484	2,157	2,586	2,399
營業費用	949	903	975	1,020	1,034	906	1,059	1,096
營業利益	1,377	1,086	1,354	1,235	1,449	1,251	1,527	1,303
營業外收支淨額	-20	-15	-24	-14	-15	-35	-25	-16
稅前純益	1,357	1,072	1,331	1,221	1,434	1,216	1,502	1,287
所得稅	279	236	283	234	298	261	301	247
稅後純益	1,073	814	1,037	977	1,130	934	1,188	1,030
調整後 EPS(元)	4.35	3.30	4.20	3.96	4.58	3.79	4.82	4.17
毛利率(%)	26.14%	24.37%	24.56%	26.08%	26.28%	24.78%	25.03%	25.94%
營業利益率(%)	15.48%	13.30%	14.28%	14.29%	15.33%	14.37%	14.78%	14.09%
稅後純益率(%)	12.06%	9.97%	10.93%	11.30%	11.96%	10.73%	11.50%	11.13%
營業收入								
QoQ(%)	12.25%	-8.22%	16.14%	-8.86%	9.35%	-7.88%	18.70%	-10.49%
YoY(%)	-10.35%	6.20%	6.69%	9.05%	6.23%	6.62%	8.97%	7.02%
營業利益								
QoQ(%)	29.11%	-21.13%	24.68%	-8.82%	17.34%	-13.65%	22.07%	-14.69%
YoY(%)	-12.84%	15.37%	24.04%	15.76%	5.21%	15.18%	12.77%	5.51%
稅後純益								
QoQ(%)	23.45%	-24.11%	27.32%	-5.81%	15.72%	-17.34%	27.22%	-13.36%
YoY(%)	-15.54%	32.19%	20.85%	12.34%	5.31%	14.71%	14.62%	5.43%

PER Band



資料來源：中信、CMoney

PBR Band



資料來源：CMoney

2026/10/07

資產負債表

(百萬元)	2022	2023	2024	2025	2Q26
資產總計	24,429	24,876	23,198	23,904	23,967
流動資產	17,963	18,419	16,093	15,905	16,021
現金及約當現金	2,760	3,014	2,900	2,624	2,255
應收帳款與票據	1,199	1,527	829	899	930
存貨	5,241	4,843	5,702	5,034	5,362
採權益法之投資	289	364	315	346	323
不動產、廠房設備	5,227	5,264	5,815	5,978	5,960
負債總計	8,984	8,949	8,429	9,730	10,077
流動負債	8,511	8,597	8,019	8,565	8,894
應付帳款及票據	1,741	2,052	2,115	2,070	2,474
非流動負債	474	352	411	1,166	1,183
權益總計	15,445	15,927	14,769	14,174	13,890
普通股股本	2,419	2,419	2,467	2,467	2,467
保留盈餘	6,725	7,209	5,706	5,350	5,017
母公司業主權益	15,357	15,819	14,644	14,045	13,789
負債及權益總計	24,429	24,876	23,198	23,904	23,967

損益表

(百萬元)	2023	2024	2025	2026F	2027F
營業收入淨額	32,459	35,524	34,428	35,190	37,741
營業成本	24,092	26,532	25,864	26,291	28,115
營業毛利淨額	8,366	8,991	8,564	8,899	9,626
營業費用	3,527	3,691	3,883	3,846	4,096
營業利益	4,840	5,300	4,681	5,053	5,531
EBITDA	5,764	6,080	5,442	5,880	6,391
業外收入及支出	188	-57	-89	-73	-92
稅前純益	5,027	5,243	4,592	4,981	5,439
所得稅	992	1,071	922	1,033	1,107
稅後純益	3,991	4,115	3,614	3,901	4,282
公告基本 EPS(元)	16.18	16.68	14.65	- -	- -
調整後 EPS(元)	16.50	16.68	14.65	15.81	17.36

註：稅後純益係指本期淨利歸屬於母公司業主。

現金流量表

(百萬元)	2022	2023	2024	2025	2Q26
營業活動現金	3,715	5,785	4,731	4,057	957
稅前純益	4,589	5,027	5,243	4,592	1,072
折舊及攤銷	562	548	565	608	167
營運資金變動	-223	397	-72	486	-625
其他營運現金	-1,213	-188	-1,004	-1,629	342
投資活動現金	-1,066	-1,478	1,177	-150	568
資本支出淨額	-390	-437	-791	-692	-109
長期投資變動	-17	28	28	20	-7
其他投資現金	-659	-1,069	1,941	522	684
籌資活動現金	-2,640	-4,037	-6,169	-4,084	-2,283
長借/公司債變動	-81	-164	0	0	0
現金增資	0	0	0	0	0
發放現金股利	-2,298	-3,507	-5,659	-3,972	0
其他籌資現金	-261	-366	-510	-113	-2,283
淨現金流量	118	254	-115	-276	-764
期初現金	2,643	2,760	3,014	2,900	3,019
期末現金	2,760	3,014	2,900	2,624	2,255

資料來源：中信、CMoney

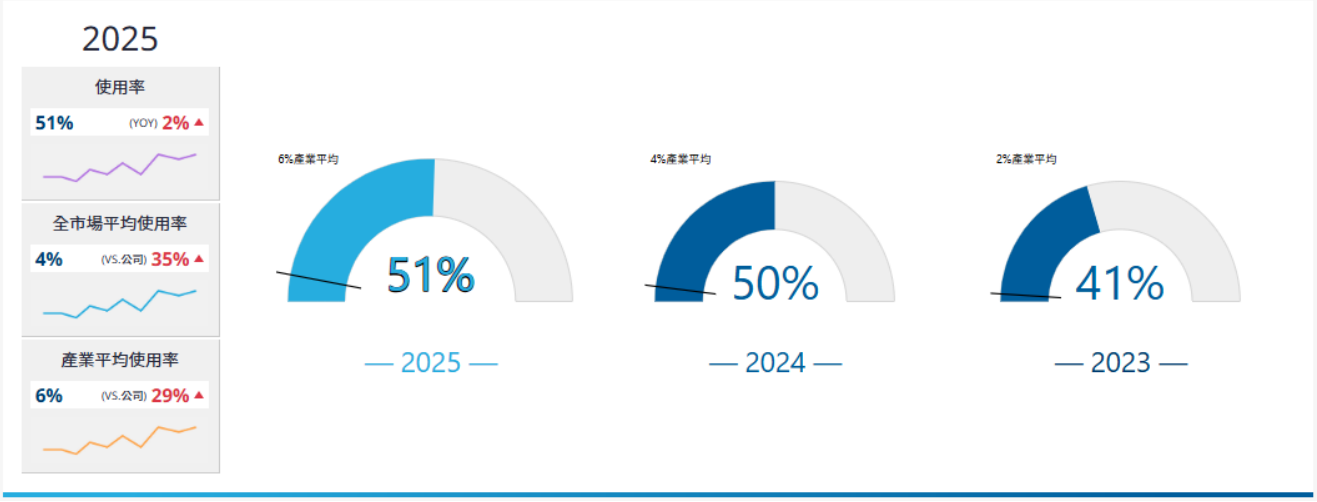
比率分析

(百萬元)	2023	2024	2025	2026F	2027F
成長力分析(%)					
營業收入淨額	1.17%	9.44%	-3.08%	2.21%	7.25%
營業毛利淨額	1.34%	7.47%	-4.75%	3.92%	8.17%
營業利益	9.20%	9.50%	-11.68%	7.96%	9.45%
稅後純益	13.98%	3.39%	-12.04%	7.94%	9.77%
獲利能力分析(%)					
毛利率	25.78%	25.31%	24.87%	25.29%	25.51%
EBITDA(%)	17.76%	17.12%	15.81%	16.71%	16.93%
營益率	14.91%	14.92%	13.60%	14.36%	14.65%
稅後純益率	12.43%	11.74%	10.66%	11.09%	11.35%
ROE(%)	25.60%	27.02%	25.20%	27.08%	28.77%

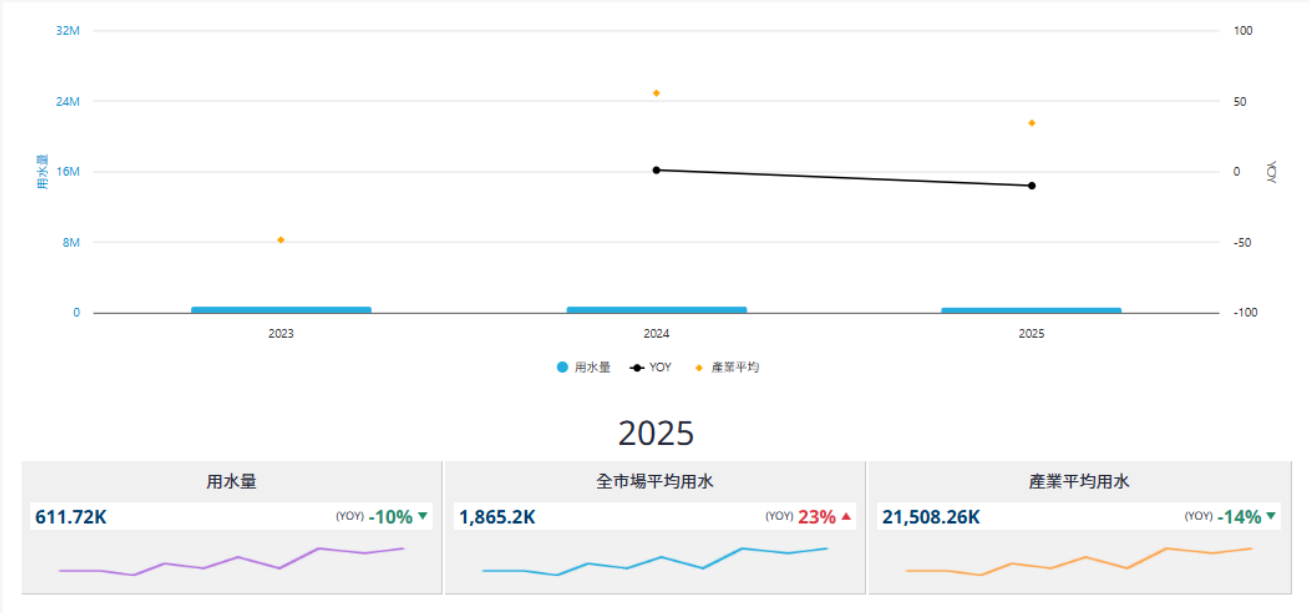
ESG 資訊-環境保護(Environmental)

溫室氣體(範疇一/二)排放量

再生能源使用率

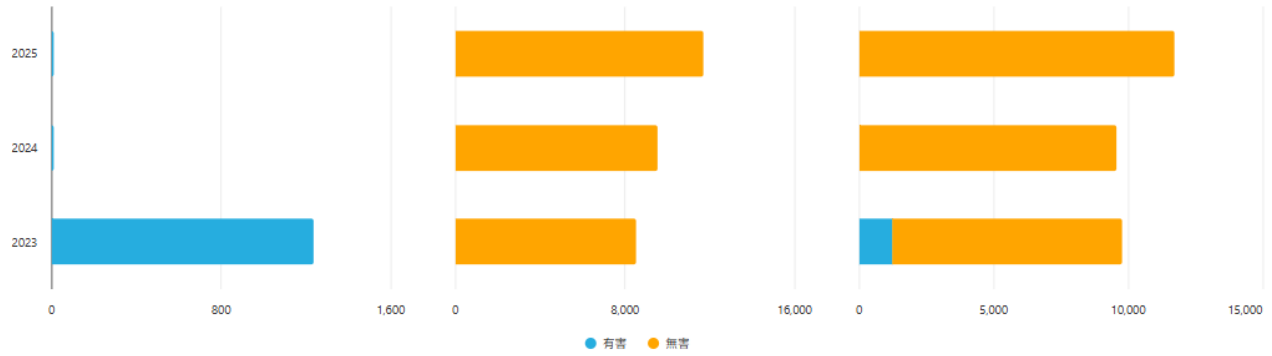


用水量



2026/10/07

廢棄物量



2025

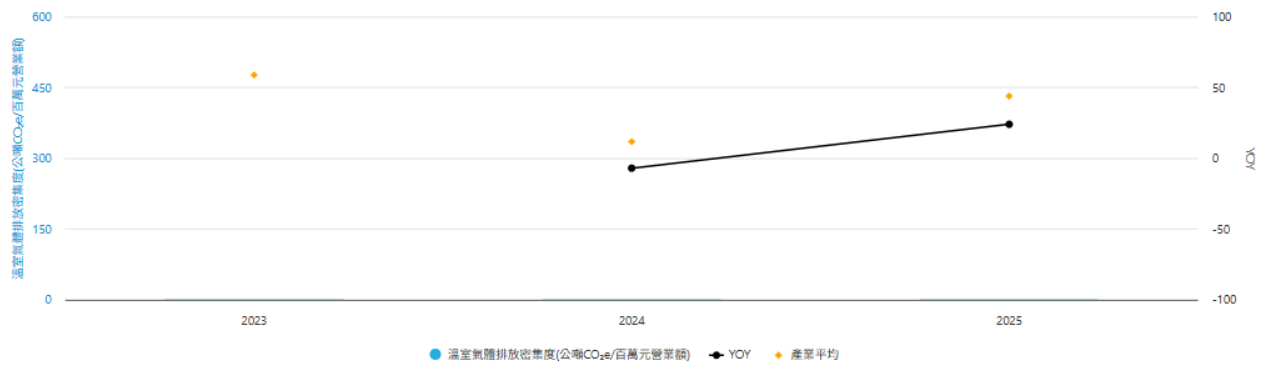
有害	無害	總量
0.01K (YOY) -13% ▼	11.7K (YOY) 23% ▲	11.71K (YOY) 23% ▲

密集度

2025

溫室氣體排放密集度	用水密集度	廢棄物密集度
1.081	17.768	0.34

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)

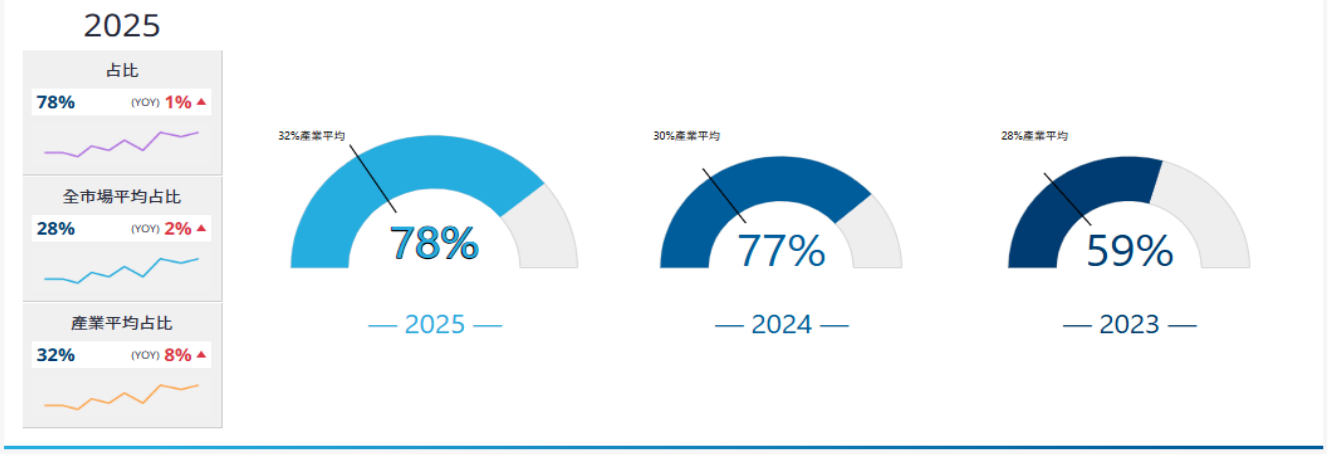


2025

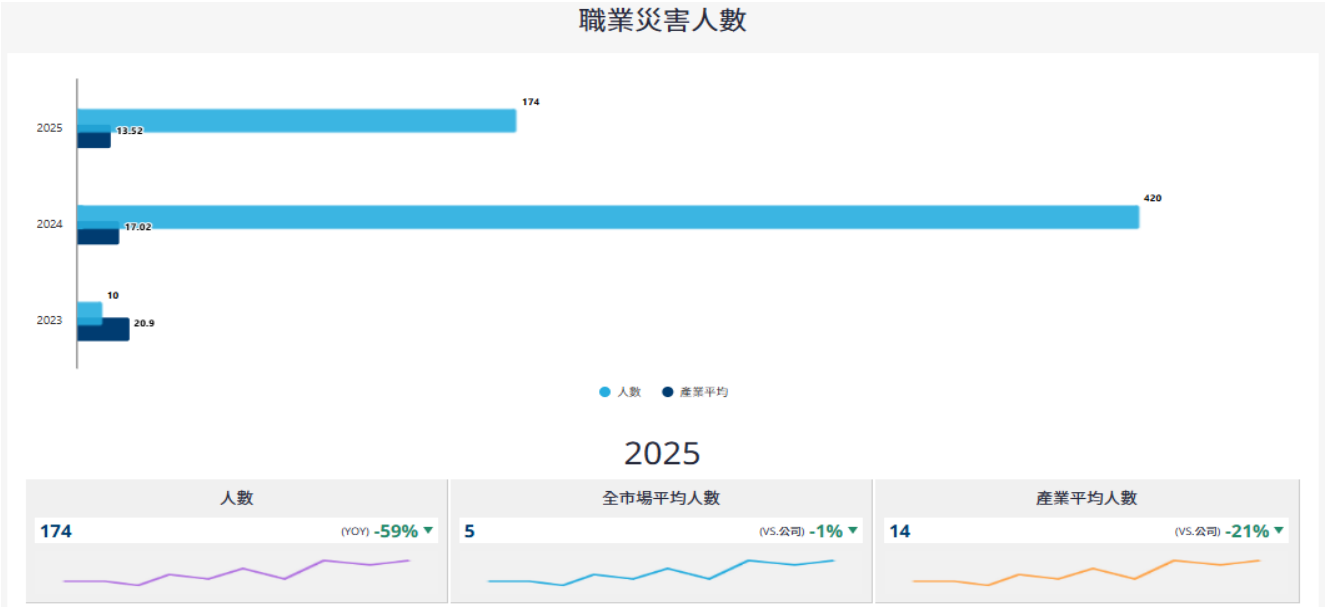
溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	全市場平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	產業平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)
0K (YOY) 24% ▲	0.03K (YOY) -24% ▼	0.43K (YOY) 29% ▲

ESG 資訊-社會責任(Social)

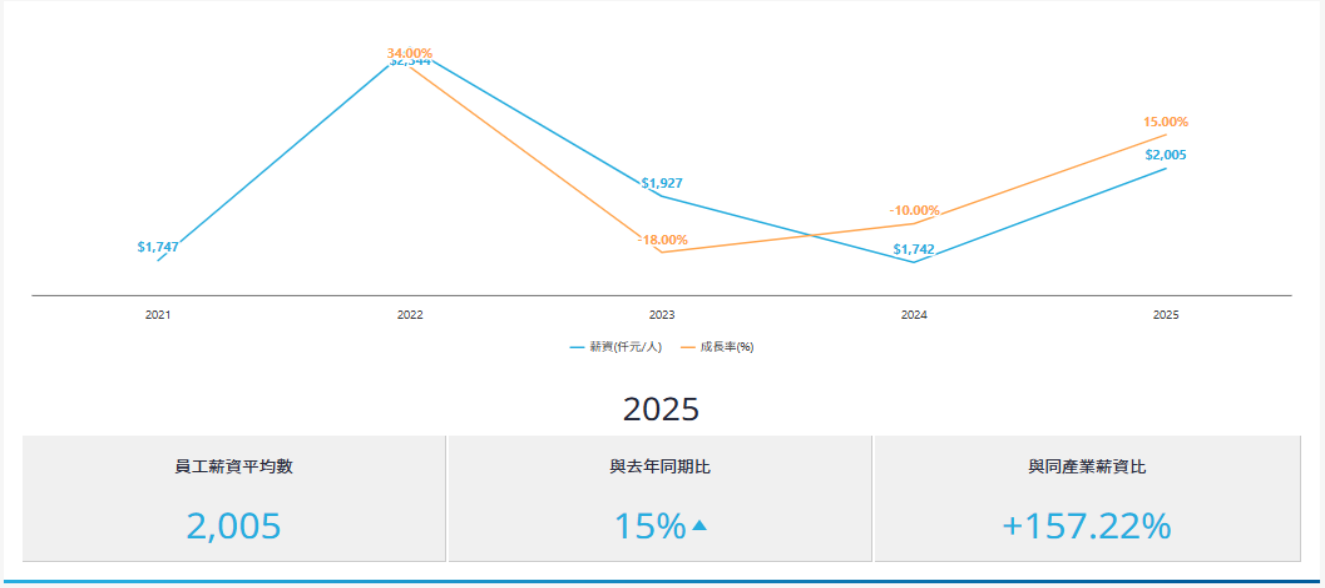
管理職女性主管占比



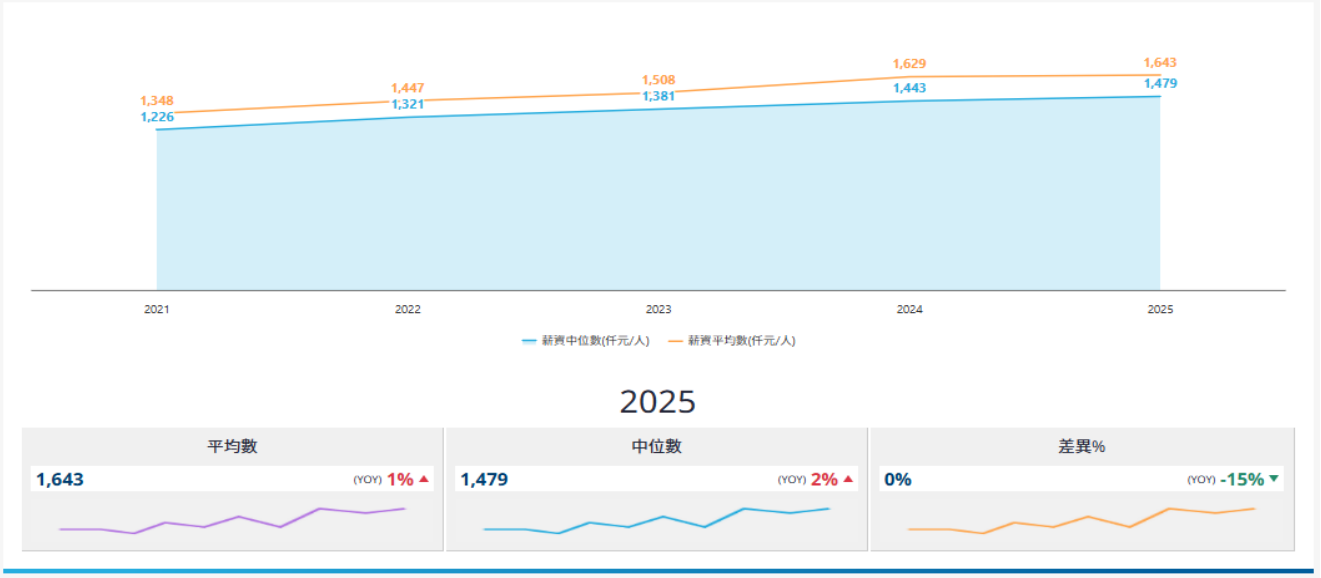
職業災害人數



員工薪資年度趨勢



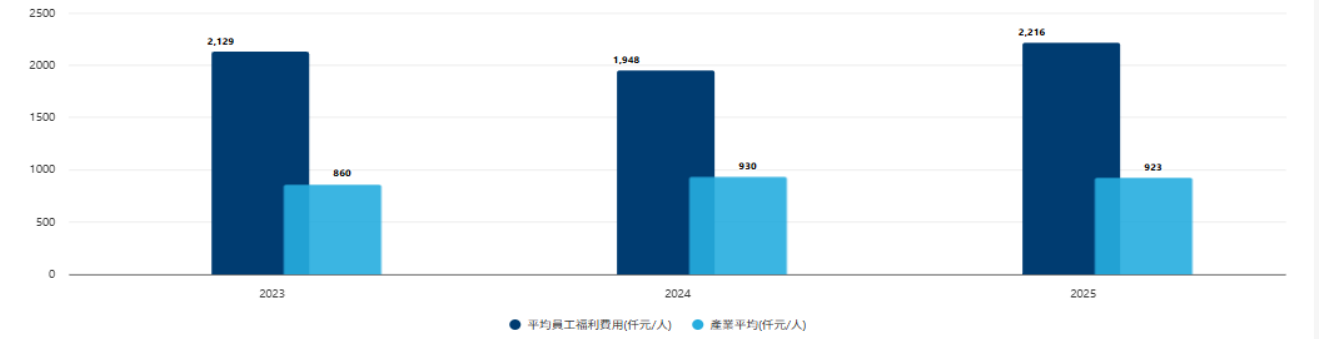
非擔任主管職務之全時員工薪資



2025

平均數	中位數	差異%
1,643 (YOY) 1% ▲	1,479 (YOY) 2% ▲	0% (YOY) -15% ▼

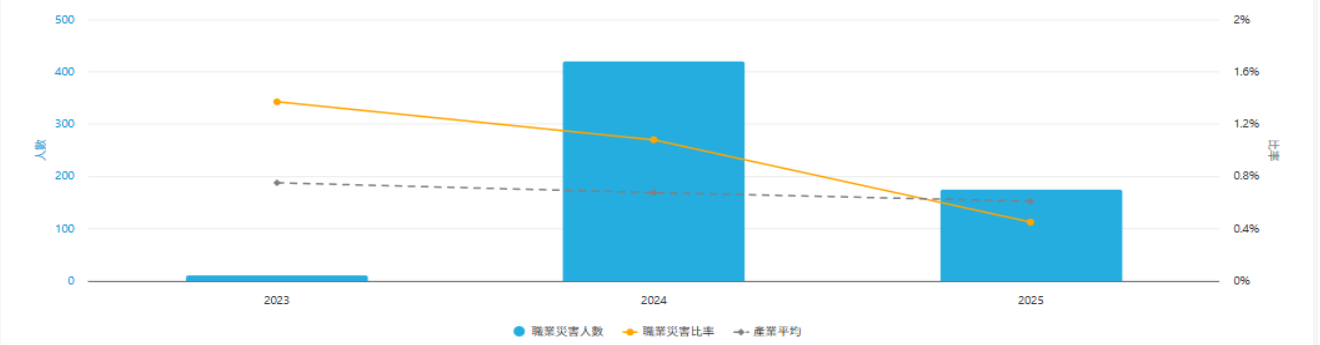
平均員工福利費用



2025

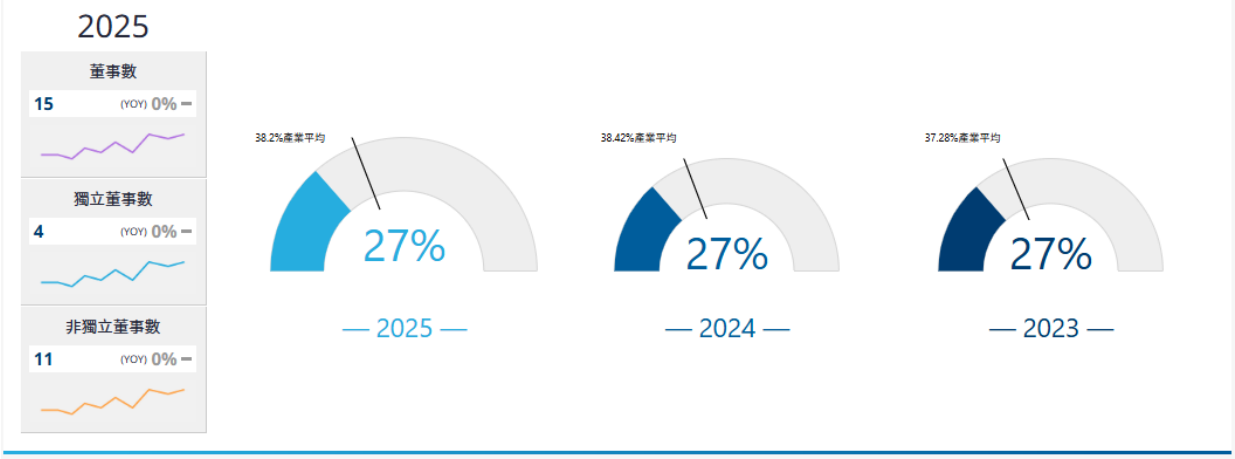
平均福利	全市場平均福利	產業平均福利
2,216 (YOY) 14% ▲	1,350 (YOY) 3% ▲	923 (YOY) -1% ▼

職業災害人數與比率

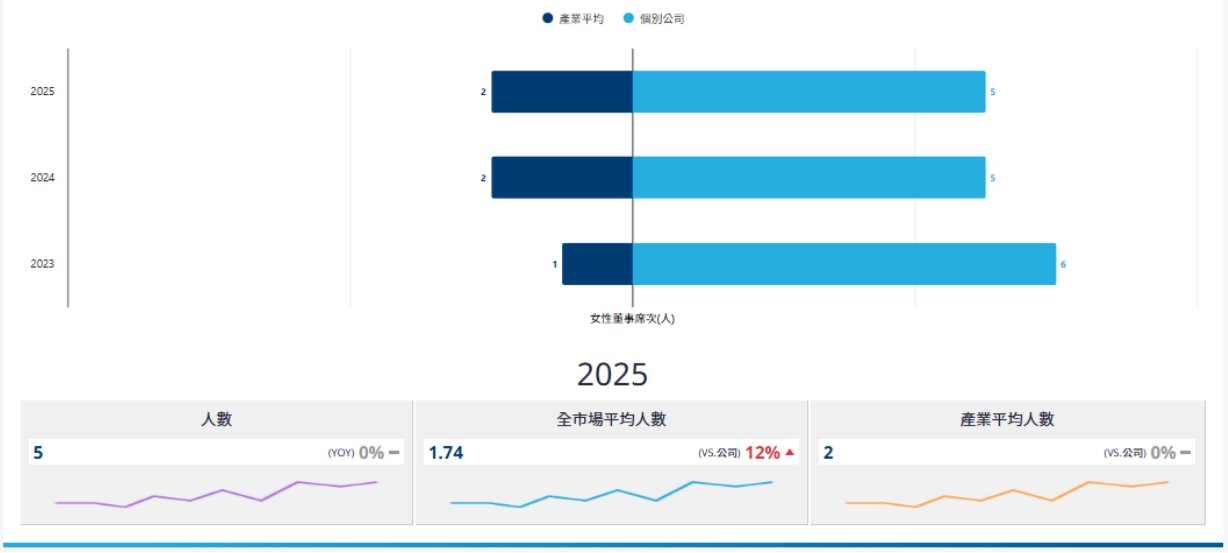


ESG 資訊-公司治理(Governance)

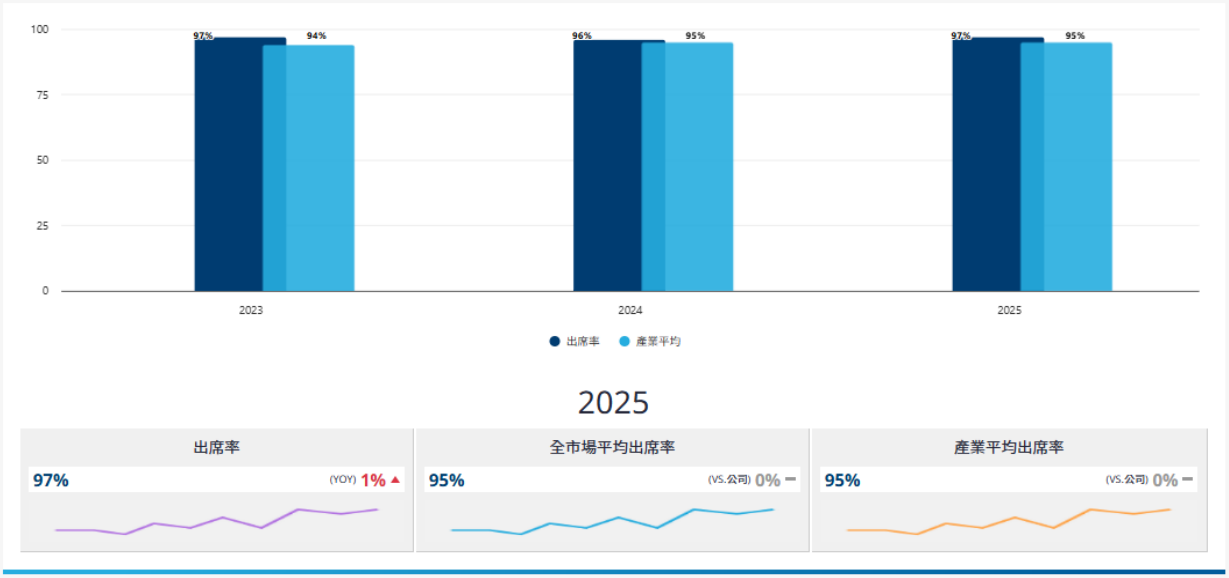
獨立董事vs.董事席次占比



女性董事席次



董事之董事會出席率



2026/10/07

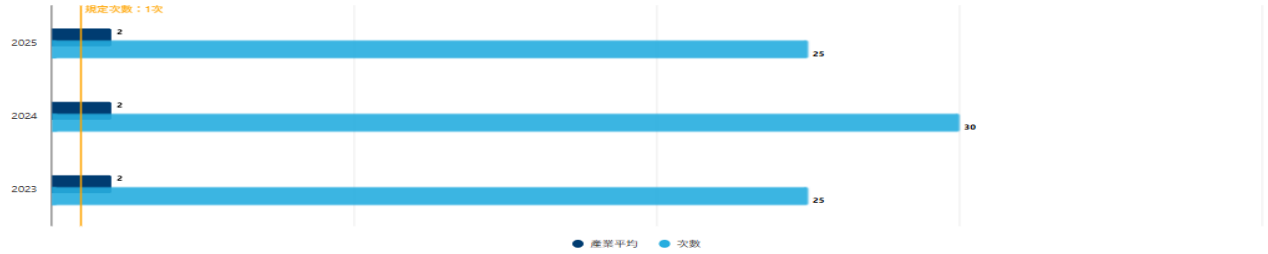
董事進修符合進修要點比率

內圈:個別公司　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0%	94% (VS.公司) 1%	93% (VS.公司) 3%

法說會次數



2025

次數	全市場平均次數	產業平均次數
25 (YOY) -17%	2 (VS.公司) 0%	2 (VS.公司) 0%

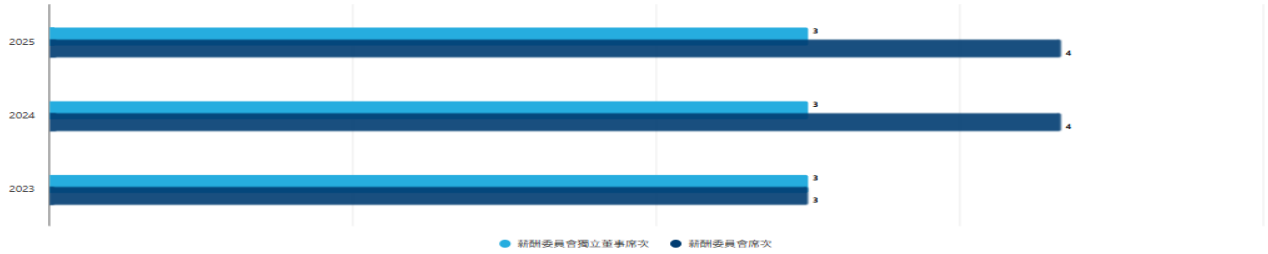
薪酬委員會出席率

內圈:個別公司出席率　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
92% (YOY) 0%	97% (VS.公司) 1%	97% (VS.公司) 3%

薪酬委員會席次



資料來源：證交所

2026/10/07

投資評等說明

個股評等	Rating	定義
買進	Buy (B)	相較於市場指數之回報潛力極具吸引力
增加持股	Overweight (OW)	相較於市場指數之回報潛力略有吸引力
中立	Neutral (N)	相較於市場指數之回報潛力相似
降低持股	Underweight (UW)	相較於市場指數之回報潛力稍低
賣出	Sell (S)	相較於市場指數之回報潛力較低
未評等	Not Rated (NR)	

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台北市南港區經貿二路 188 號 5 樓及 18 樓 / (02) 5569-1988
營業執照字號：114 金管投顧新字第 014 號

06 Oct 2026 05:09:42 ET | 11 pages

Chroma ATE (2360.TW)

3Q26 Another Strong Beat; September Revenue 3x YoY

CITI'S TAKE

Chroma ATE reported Sep revenue that tripled YoY to NT$6,860mn (+22% MoM), concluding 3Q26 with revenue of NT$17,926mn (+180% YoY/+32% QoQ), another quarterly record and beating Street/Citi estimates by 23%/25%, respectively. We believe the robust revenue was mainly driven by datacenter power testers (HDVC, China's ESS build) and photonics (CPO ramp-up), and the 3Q revenue beat suggests upside to Street estimates for FY26 and FY27. Looking ahead, we expect robust AI-driven revenue growth to continue for Chroma across segments, including power tester (HVDC), SLT (GPU & ASIC), metrology (CoWoS, panel), and photonics. CPO shipment remains at an early stage and could expand several times next year. New products in metrology, FT handler, burn-in system could see more meaningful contribution starting 2027. We maintain our positive view and Buy rating for Chroma.

Buy	
Price (06 Oct 26 13:30)	NT$2,350.00
Target price	NT$2,500.00
Expected share price return	6.4%
Expected dividend yield	0.8%
Expected total return	7.2%
Market Cap	NT$999,327M
	US$31,497M

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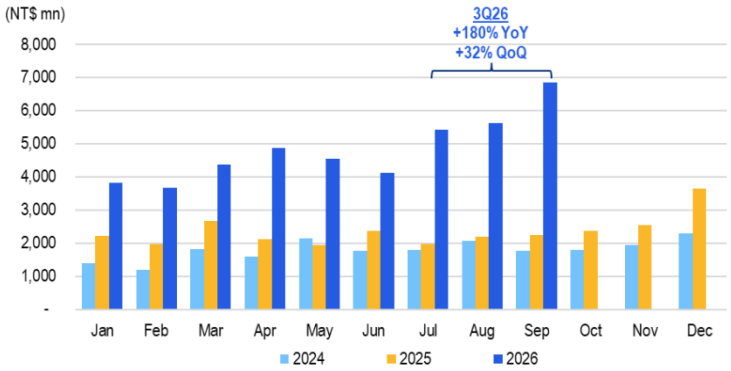
Figure 1. Chroma ATE – Monthly Revenue Trend



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Source: Company Reports, Citi Research

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations

Figure 2. Chroma ATE – 3Q26 Preliminary Results vs. Estimates

(NT$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26 Prelim	3Q26 BBG	3Q26 Diff	3Q26 Citi est	3Q26 Diff
Revenue	**6,865**	**6,455**	**6,410**	**8,580**	**11,859**	**13,529**	**17,926**	**14,618**	**23%**	**14,314**	**25%**
YoY %	*55%*	*17%*	*14%*	*42%*	*73%*	*110%*	*180%*	*128%*		*123%*	
Gross Profit	4,148	4,224	3,835	5,217	7,424	8,185		9,032		8,755	
GPM	**60.4%**	**65.4%**	**59.8%**	**60.8%**	**62.6%**	**60.5%**		**61.8%**		**61.2%**	
Opex	(1,984)	(2,027)	(1,987)	(2,229)	(2,627)	(2,943)		(3,036)		(2,816)	
Opex %	28.9%	31.4%	31.0%	26.0%	22.2%	21.8%		20.8%		19.7%	
EBIT	**2,164**	**2,197**	**1,848**	**2,988**	**4,797**	**5,242**		**5,997**		**5,938**	
EBIT margin	31.5%	34.0%	28.8%	34.8%	40.5%	38.7%		41.0%		41.5%	
Pre-Tax Profit- Adj.	2,597	2,438	2,288	3,408	5,094	6,514		6,268		6,183	
Net Profit - Adj.	**2,123**	**1,953**	**1,880**	**2,549**	**3,864**	**5,123**		**5,091**		**4,923**	
YoY %	*122%*	*39%*	*32%*	*73%*	*82%*	*162%*		*171%*		*162%*	
EPS (NT$) - Adj.	**4.98**	**4.60**	**4.42**	**5.99**	**9.07**	**12.10**		**11.85**		**11.63**	

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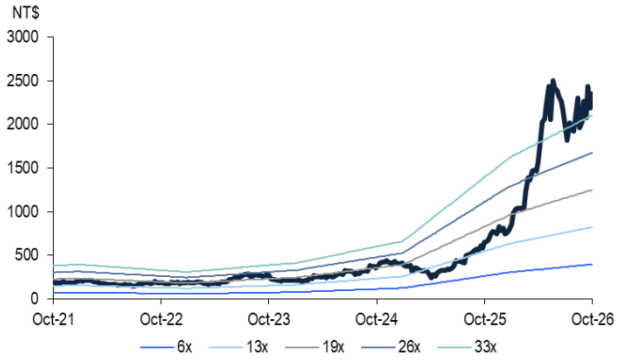
Source: Company Reports, Bloomberg, Citi Research Estimates

Figure 3. Chroma ATE – 12M Fwd. P/E Band



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Source: dataCentral, Citi Research

Figure 4. Chroma ATE – 12M Fwd. P/E Chart

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Source: dataCentral, Citi Research

RELATED:

Chroma ATE (2360.TW): August Revenue +156% YoY, Another Record High

Chroma ATE (2360.TW): Chroma Plans 13.6mn Share GDR Issuance; Dilution Manageable with Long-Term Upside Potential

Chroma ATE (2360.TW): 2Q Beat on Non-Op, Positive Outlook Intact

Chroma ATE

Valuation

Our target price of NT$2,500 is based on 40x 2027E P/E, at close to 1x PEG. Chroma shares have re-rated above its historical range as its growth outlook is now strongly tied to AI infrastructure build out. We estimate 60%+ of its revenue is now AI-related, with strong growth across product segments including power testers, SLT, metrology, and photonics. Chroma has a commanding market share in most of these segments and should be well positioned to capitalize on the global AI expansion.

Risks

Downside risks to achieving our target price include: 1) softer AI/CPO growth expectation due to changes in capex outlook or production bottleneck; 2) fewer SLT needed as a result of shortened cycle time as GPU yield rate improves; 3) disappointing metrology shipment due to changes at ramp-up schedule; 4) testing process change leading to loss in market share; 5) FX risks (50-60% of Chroma's sales are in US$, 10-20% in Rmb).

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Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Chroma ATE (2360.TW)

Ratings and Target Price History
Fundamental Research

Analyst: Kevin Chen

	Date	Rating	Target Price	Closing Price
1	11-Oct-23 14:29:57	1	*330.00	253.00
2	21-Feb-24 11:50:58	1	*280.00	227.50
3	30-Apr-24 13:48:08	1	*320.00	266.50
4	31-Jul-24 15:19:03	1	*360.00	301.00
5	04-Nov-24 17:53:01	1	*500.00	437.00
6	30-Apr-25 14:29:54	1	*425.00	292.50
7	29-Jun-25 19:17:54	1	*500.00	429.50
8	06-Aug-25 09:55:28	1	*550.00	494.50
9	30-Oct-25 18:40:38	1	*900.00	804.00
10	22-Feb-26 19:26:56	1	*1,200.00	1,050.00
11	11-Mar-26 02:42:34	1	*1,650.00	1,495.00
12	03-May-26 18:43:42	1	*2,500.00	2,120.00

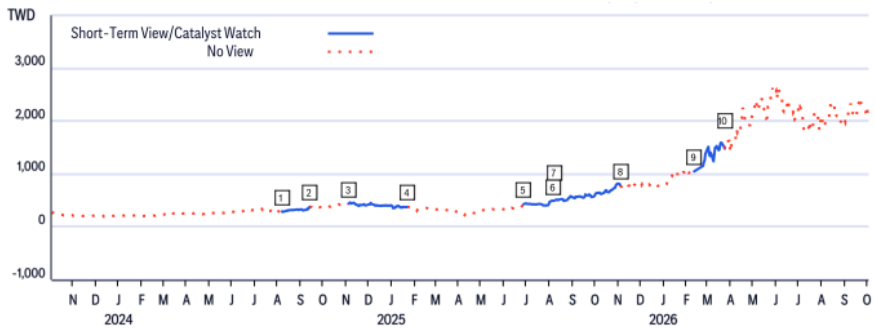
*Indicates Change

Rating/target price changes above reflect Eastern Time

Chroma ATE (2360.TW)

Short-Term View/Catalyst Watch Research

Analyst: Kevin Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	07-Aug-24 18:28:11	Add CW	Upside	90 Days	292.00
2	13-Sep-24 00:34:47	Remove CW	Upside	90 Days	380.00
3	04-Nov-24 12:53:01	Add STV	Upside	90 Days	437.00
4	02-Feb-25 19:44:53	Remove STV	Upside	90 Days	379.50
5	29-Jun-25 15:17:54	Add STV	Upside	90 Days	429.50
6	06-Aug-25 05:55:28	Remove STV	Upside	90 Days	494.50
7	06-Aug-25 05:55:28	Add CW	Upside	90 Days	494.50
8	04-Nov-25 19:35:17	Remove CW	Upside	90 Days	776.00
9	22-Feb-26 14:26:56	Add CW	Upside	30 Days	1,050.00
10	24-Mar-26 21:33:21	Remove CW	Upside	30 Days	1,480.00

CW - Catalyst Watch , STV - Short-Term View

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	12 Month Rating			Catalyst Watch		
Data current as of 01 Oct 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
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% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

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06 Oct 2026 07:28:09 ET | 11 pages

WinWay Technology (6515.TW)

Sep-26 Sales Dip on Probe Card, Mix Shift to Support Margins

CITI'S TAKE

WinWay reported Sep-26 revenues of NT$1.25bn (-27% MoM/+88% YoY). QTD 3Q26 revenues of NT$4.56bn (+29% QoQ/+153% YoY) beat Citi/cons forecasts by 3%/7%, on robust AI/HPC test interface demand. The MoM decline was mainly driven by lower probe card revenues as major project shipments have temporarily wrapped up, and we expect probe card contribution to stay light in 4Q26E. While this may weigh on the top line near term, we see margins could recover gradually as product mix improves alongside the continued AI GPU and ASIC socket ramp. In the long term, we continue to favor WinWay as a core beneficiary of rising testing complexity for advanced chips across diverse platforms. Reiterate Buy/High Risk.

Buy / High Risk	
Price (06 Oct 26 13:30)	NT$6,130.00
Target price	NT$11,500.00
Expected share price return	87.6%
Expected dividend yield	1.1%
Expected total return	88.7%
Market Cap	NT$223,407M
	US$7,041M

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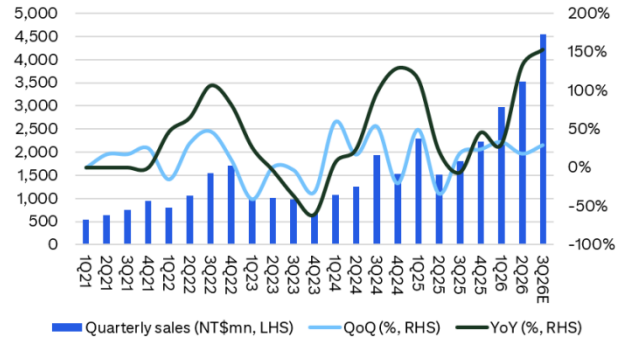
Figure 1. WinWay - Quarterly sales trend



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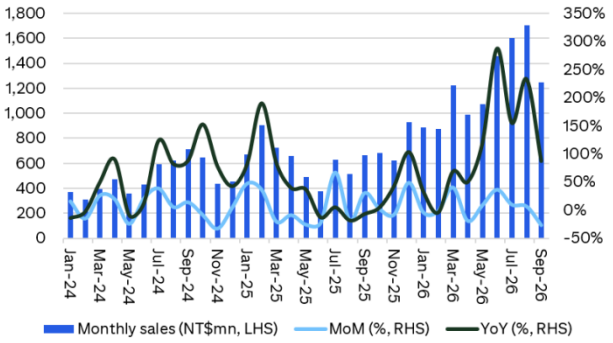
Source: Company Reports, Citi Research

Figure 2. WinWay - Monthly sales trend



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WinWay Technology

Valuation

We value WinWay using a PE-based approach that factors in earnings momentum and likely expansion of its addressable market. Our target price of NT$11,500 is derived from applying c.41x to 2H27E-1H28E EPS, higher than the upper end of its long-term PE range, which we believe is justified given the company's entrenched position in the high-end AI GPU testing supply chain, rising contribution from coaxial socket and SLT solutions, and the increasing value of test content per module. With these structural drivers in place, and as we believe the market is starting to value it as an AI/HPC play (rather than a consumer electronics stock), we expect the multiple to hold at a premium relative to its historical averages.

Risks

Our quant rating system rates the WinWay stock High Risk based on its historical volatility.

Key downside risks that could impede the stock from reaching our target price and our earnings forecasts include: 1) client and sector concentration – specifically any delays in US AI GPU product roadmap could impact the business outlook for WinWay; 2) intensifying competition in coaxial and hypersockets from global peers; 3) faster-than-expected testing algorithm optimization, which could shorten testing durations and reduce the anticipated hardware demand multiplier; and 4) delay of MEMS Phase 2 mass production.

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Appendix A-1

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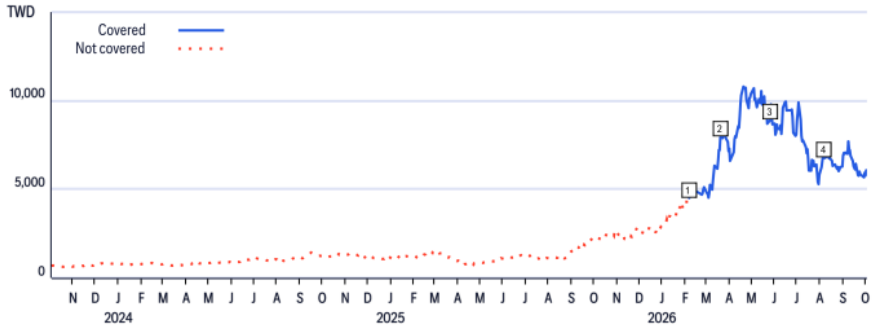
WinWay Technology (6515.TW)
Ratings and Target Price History
Fundamental Research

Analyst: Michael Hung



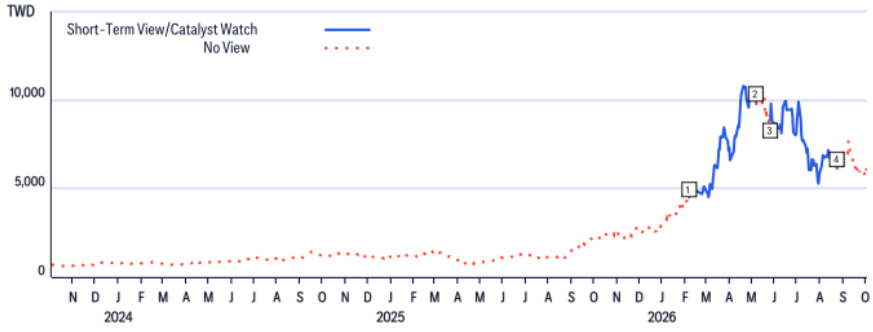
	Date	Rating	Target Price	Closing Price
1	06-Feb-26 05:59:37	*1H	*6,000.00	4,510.00
2	20-Mar-26 02:14:37	1H	*10,000.00	7,930.00
3	26-May-26 19:30:25	1H	*13,000.00	8,905.00
4	06-Aug-26 12:43:24	1H	*11,500.00	6,775.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

WinWay Technology (6515.TW)
Short-Term View/Catalyst Watch Research

Analyst: Michael Hung



	Date	Action	Expected Direction	Duration	Closing Price
1	06-Feb-26 00:59:37	Add CW	Upside	90 Days	4,510.00
2	07-May-26 21:43:12	Remove CW	Upside	90 Days	9,860.00
3	26-May-26 15:30:25	Add CW	Upside	90 Days	8,905.00
4	24-Aug-26 21:35:40	Remove CW	Upside	90 Days	6,145.00

CW - Catalyst Watch , STV - Short-Term View

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	12 Month Rating			Catalyst Watch		
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% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

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Kinsus Interconnect Technology (3189 TT)

Target price: **TWD1,580.00** (from TWD1,068.00)

Share price (6 Oct): **TWD1,070.00** | Up/downside: **+47.7%**

5 4 3 2 **1**

Buy (unchanged)

3Q26 results preview: several favourable trends ahead

- Rising high-end ABF substrate likely to drive blended ASP increase
- Multicore and optics likely to be the next growth drivers
- Reaffirming our Buy (1) call; raising 12M TP to TWD1,580

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What's new: We believe an improving product mix, driven by rising Nvidia's CPU contribution, to be a key near-term share price catalyst.

What's the impact: **Near-term update.** After Kinsus released its August preliminary results on 23 September, we note that we had underestimated the 3Q26E gross margin expansion on product optimisation and price hike scale. The results imply a GM of 33% in August (vs. 26% in 2Q26). We therefore lift our 3Q-4Q26E gross margin to 33.2-35.1% (from 28.9-30.3%) to reflect an improving product mix, including greater CPU substrate contribution. We believe the gross margin of high-end chipset substrate may be c.50%, supporting margin expansion. We also expect a c.10% QoQ price hike for both BT and ABF substrate in 3Q26, above Kinsus' guidance of a high-single-digit% QoQ, making these substrates the other growth driver. We see further gross margin expansion in 2027-28 on the ongoing price hike trend and increasing high-end ABF substrate product mix.

Business outlook. We believe several favourable trends lie ahead. First, we see a rising contribution from high-end GPU and CPU ABF substrates. We believe Nvidia's CPU currently accounts for 5-10% of total revenue, with Kinsus holding a 70-80% market share. We also expect Kinsus to acquire a teens% market share in AMD's Venice, which should gradually ramp up from 4Q26. These will likely become key growth drivers of the ABF substrate segment in 2027, in our view. We believe the ramp-up progress for GPU substrate for these two clients is slightly slower than that for CPU. Second, we believe Kinsus will benefit from the rising BT substrate demand for optical transceivers. In our view, migration to on-board optics (OBO) and NPO/CPO will require ABF substrate beneath the optics engine (OE) in OBO; the OE and IC will need a bigger ABF substrate as we move to the NPO/CPO generation. As such, we expect the total ABF substrate demand to increase with rising adoption of optics in AI servers. Lastly, we expect the adoption of multiple-core ABF from 2028 to improve the rigidity and transmission performance before the official launch of ceramic and glass-core substrate, which we expect to increase the ABF TAM as well.

What we recommend: We raise our 2026-28E earnings by 19-29% to reflect the better-than-expected gross margin in August. We reaffirm our Buy (1) call and raise our 12M TP to TWD1,580, based on an unchanged target PER of 36x (high end of 3-year-trading range of 10-36x), applied to the 1-year-forward EPS forecast. Key downside risks: weaker-than-expected demand for memory, PCs, gaming GPUs or AI servers.

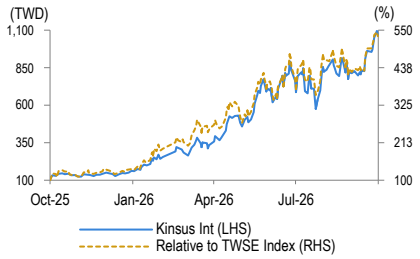
How we differ: Our 2026-28E EPS are 27-57% above the Bloomberg consensus, likely on our more bullish view on its gross margin expansion.

Forecast revisions (%)

Year to 31 Dec	26E	27E	28E
Revenue change	1.3	-	2.2
Net profit change	29.1	28.7	18.8
Core EPS (FD) change	18.6	18.3	9.2

Source: Daiwa forecasts

Share price performance



12-month range	111.18-1,095.00
Market cap (USDbn)	17.18
3m avg daily turnover (USDm)	428.04
Shares outstanding (m)	510
Major shareholder	Asus Investment Co Ltd (13.2%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	54,062	88,448	125,676
Operating profit (m)	9,142	24,773	43,198
Net profit (m)	7,558	22,325	39,673
Core EPS (fully-diluted)	14.824	43.787	77.810
EPS change (%)	322.9	195.4	77.7
Daiwa vs Cons. EPS (%)	29.1	57.1	46.7
PER (x)	72.2	24.4	13.8
Dividend yield (%)	0.3	0.4	0.4
DPS	3.5	4.5	4.5
PBR (x)	11.6	8.1	5.3
EV/EBITDA (x)	30.7	15.1	8.8
ROE (%)	15.2	32.4	40.0

Source: FactSet, Daiwa forecasts

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Flash

Kinsus: revenue and earnings forecast revisions

	2026E			2027E			2028E		
(TWDm)	Previous	New	Consensus	Previous	New	Consensus	Previous	New	Consensus
Revenue	53,385	54,062	53,912	88,484	88,448	80,794	122,920	125,676	120,243
Diff (%)		1.3%	0.3%		0.0%	9.5%		2.2%	4.5%
Gross Margin (%)	27.0%	29.65%	27.3%	35.6%	40.5%	33.7%	43.2%	46.37%	38.5%
Operating profit	7,578	9,142	7,907	20,449	24,773	18,170	38,330	43,198	32,345
Op Margin (%)	14.2%	16.9%	14.7%	23.1%	28.0%	22.5%	31.2%	34.4%	26.9%
Net profit	5,855	7,558	5,857	17,344	22,325	14,212	33,395	39,673	27,051
EPS	12.50	14.82	11.49	37.02	43.79	27.87	71.29	77.81	53.06
Diff (%)		18.6%	29.1%		18.3%	57.1%		9.2%	46.7%

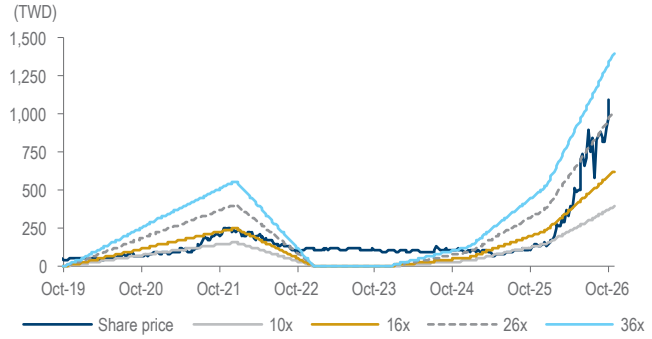
Source: Bloomberg, Daiwa forecasts

Kinsus: quarterly and annual P&L statement

	2026E				2027E				2025	2026E	2027E	2028E
(TWDm)	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE				
Net revenue	**11,105**	**12,496**	**14,539**	**15,922**	**18,046**	**21,099**	**23,941**	**25,361**	**39,351**	**54,062**	**88,448**	**125,676**
COGS	(8,755)	(9,235)	(9,716)	(10,326)	(11,304)	(12,750)	(14,067)	(14,480)	(31,037)	(38,032)	(52,601)	(67,397)
Gross profit	**2,349**	**3,261**	**4,824**	**5,596**	**6,743**	**8,349**	**9,874**	**10,881**	**8,314**	**16,030**	**35,847**	**58,279**
Operating expenses	1,459	1,582	1,841	2,006	2,274	2,637	2,993	3,170	5,645	6,888	11,074	15,081
Operating profit	**891**	**1,679**	**2,983**	**3,590**	**4,469**	**5,712**	**6,881**	**7,711**	**2,670**	**9,142**	**24,773**	**43,198**
Non-operating profit	74	106	88	90	89	84	90	92	417	358	354	359
Pre-tax profit	**964**	**1,784**	**3,071**	**3,680**	**4,558**	**5,796**	**6,971**	**7,803**	**3,087**	**9,499**	**25,127**	**43,557**
Income taxes	(89)	(101)	(174)	(208)	(258)	(328)	(394)	(441)	(370)	(571)	(1,420)	(2,462)
Net profit	**549**	**1,311**	**2,586**	**3,112**	**3,986**	**5,132**	**6,230**	**6,978**	**1,596**	**7,558**	**22,325**	**39,673**
Net EPS (TWD)	**1.17**	**2.57**	**5.07**	**6.10**	**7.82**	**10.06**	**12.22**	**13.69**	**3.5**	**14.8**	**43.8**	**77.8**
Operating Ratios												
Gross margin	21.2%	26.1%	33.2%	35.1%	37.4%	39.6%	41.2%	42.9%	21.1%	29.7%	40.5%	46.4%
Operating margin	8.0%	13.4%	20.5%	22.5%	24.8%	27.1%	28.7%	30.4%	6.8%	16.9%	28.0%	34.4%
Pre-tax margin	8.7%	14.3%	21.1%	23.1%	25.3%	27.5%	29.1%	30.8%	7.8%	17.6%	28.4%	34.7%
Net margin	4.9%	10.5%	17.8%	19.5%	22.1%	24.3%	26.0%	27.5%	4.1%	14.0%	25.2%	31.6%
YoY (%)												
Net revenue	29%	31%	40%	47%	63%	69%	65%	59%	29%	37%	64%	42%
Gross profit	21%	62%	146%	134%	187%	156%	105%	94%	-4%	93%	124%	63%
Operating profit	78%	171%	375%	290%	402%	240%	131%	115%	144%	242%	171%	74%
Pre-tax profit	38%	188%	331%	249%	373%	225%	127%	112%	93%	208%	165%	73%
Net profit	99%	288%	662%	384%	626%	292%	141%	124%	3164%	374%	195%	78%
QoQ (%)												
Net revenue	3%	13%	16%	10%	13%	17%	13%	6%				
Gross profit	-2%	39%	48%	16%	20%	24%	18%	10%				
Operating profit	-3%	88%	78%	20%	24%	28%	20%	12%				
Pre-tax profit	-9%	85%	72%	20%	24%	27%	20%	12%				
Net profit	-14%	139%	97%	20%	28%	29%	21%	12%				

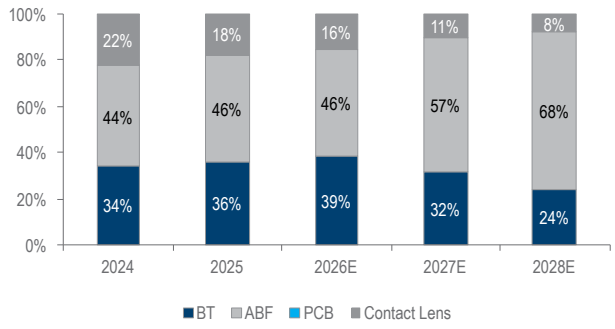
Source: Company, Daiwa forecasts

Kinsus: 1-year forward PER bands



Source: TEJ, Daiwa forecasts

Kinsus: revenue breakdown by product



Source: Company, Daiwa forecasts

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Global PC shipment (mn)	361	301	260	263	287	244	244	244
Regular server shipmnet (mn)	14	15	12	14	16	18	22	26
Global smartphone shipment (mn)	1,655	1,437	1,380	1,437	1,429	1,249	1,255	1,300

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABF	10,191	16,526	11,978	13,319	17,998	24,558	48,820	82,562
BT	17,565	18,796	8,065	10,398	14,314	21,234	30,530	33,105
Other Revenue	5,580	6,336	6,790	6,817	7,039	8,271	9,098	10,008
Total Revenue	**33,336**	**41,658**	**26,832**	**30,535**	**39,351**	**54,062**	**88,448**	**125,676**
Other income	0	0	0	0	0	0	0	0
COGS	(22,583)	(25,771)	(20,075)	(21,867)	(31,037)	(38,032)	(52,601)	(67,397)
SG&A	(2,879)	(3,532)	(3,301)	(5,010)	(3,433)	(4,241)	(6,960)	(9,312)
Other op.expenses	(2,413)	(2,484)	(2,397)	(2,612)	(2,212)	(2,647)	(4,114)	(5,769)
Operating profit	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
Net-interest inc./(exp.)	(68)	(172)	(332)	(394)	(391)	(336)	(356)	(350)
Assoc/forex/extraord./others	253	607	716	902	809	694	710	709
Pre-tax profit	**5,641**	**10,287**	**1,426**	**1,603**	**3,087**	**9,499**	**25,127**	**43,557**
Tax	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Min. int./pref. div./others	(633)	(957)	(1,123)	(1,282)	(1,121)	(1,370)	(1,381)	(1,422)
Net profit (reported)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
Net profit (adjusted)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
EPS (reported)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted fully-diluted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
DPS (TWD)	**4.500**	**6.476**	**1.000**	**1.000**	**1.518**	**3.500**	**4.500**	**4.500**
EBIT	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
EBITDA	**9,833**	**14,958**	**6,394**	**7,228**	**9,914**	**17,313**	**33,761**	**53,085**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	5,641	10,287	1,426	1,603	3,087	9,499	25,127	43,557
Depreciation and amortisation	4,377	5,107	5,351	6,133	7,244	8,171	8,988	9,887
Tax paid	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Change in working capital	(1,347)	(176)	1,528	(906)	(1,959)	(3,452)	(2,982)	(3,707)
Other operational CF items	2,575	2,901	(1,793)	1,147	91	(634)	0	0
Cash flow from operations	**10,574**	**15,961**	**6,256**	**7,705**	**8,093**	**13,013**	**29,713**	**47,274**
Capex	(13,384)	(17,094)	(10,136)	(10,408)	(6,117)	(7,832)	(5,483)	(3,838)
Net (acquisitions)/disposals	0	0	0	0	0	0	0	0
Other investing CF items	489	559	(2,351)	1,301	(1,752)	(2,374)	(44)	(85)
Cash flow from investing	**(12,895)**	**(16,535)**	**(12,486)**	**(9,107)**	**(7,869)**	**(10,206)**	**(5,526)**	**(3,922)**
Change in debt	6,544	1,903	6,043	1,624	(1,684)	(8,891)	0	(45)
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(451)	(2,443)	(3,438)	(454)	(457)	(774)	(1,785)	(2,294)
Other financing CF items	(94)	2,411	2,730	(173)	(1,130)	8,625	0	0
Cash flow from financing	**6,000**	**1,870**	**5,335**	**996**	**(3,270)**	**(1,041)**	**(1,785)**	**(2,340)**
Forex effect/others	(12)	56	(88)	71	(39)	51	39	35
Change in cash	**3,667**	**1,352**	**(983)**	**(335)**	**(3,084)**	**1,817**	**22,441**	**41,047**
Free cash flow	(2,809)	(1,133)	(3,880)	(2,703)	1,976	5,181	24,230	43,436

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	15,968	17,923	21,013	16,676	16,336	21,188	43,628	84,676
Inventory	3,466	3,481	2,612	2,981	4,828	6,237	8,747	10,924
Accounts receivable	5,679	5,050	4,299	5,284	6,804	10,300	16,405	22,824
Other current assets	1,312	1,647	1,445	4,735	970	1,080	1,080	1,080
Total current assets	**26,424**	**28,100**	**29,369**	**29,677**	**28,938**	**38,804**	**69,860**	**119,503**
Fixed assets	31,586	44,101	48,209	49,658	50,897	51,643	48,138	42,089
Goodwill & intangibles	0	0	0	0	0	0	0	0
Other non-current assets	376	432	489	100	138	434	478	562
Total assets	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
Short-term debt	1,100	377	1,409	2,452	3,053	915	915	869
Accounts payable	10,150	10,798	8,457	7,521	10,030	14,004	19,638	24,526
Other current liabilities	2,733	4,199	4,393	7,018	7,023	5,571	5,571	5,571
Total current liabilities	**13,983**	**15,374**	**14,259**	**16,990**	**20,106**	**20,490**	**26,124**	**30,966**
Long-term debt	9,387	10,770	15,280	13,779	11,197	6,219	6,219	6,219
Other non-current liabilities	2,027	7,674	9,369	8,788	7,297	6,318	6,318	6,318
Total liabilities	**25,397**	**33,818**	**38,908**	**39,557**	**38,600**	**33,027**	**38,661**	**43,503**
Share capital	4,508	4,528	4,544	4,566	4,568	5,269	5,269	5,269
Reserves/R.E./others	28,481	34,288	34,615	35,312	36,805	52,585	74,546	113,382
Shareholders' equity	**32,989**	**38,816**	**39,159**	**39,878**	**41,373**	**57,854**	**79,815**	**118,651**
Minority interests	0	0	0	0	0	0	0	0
Total equity & liabilities	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
EV	540,075	538,779	541,231	545,110	543,470	531,501	509,061	467,968
Net debt/(cash)	**(5,481)**	**(6,776)**	**(4,325)**	**(445)**	**(2,085)**	**(14,054)**	**(36,495)**	**(77,588)**
BVPS (TWD)	**64.478**	**75.804**	**69.748**	**69.314**	**71.656**	**92.357**	**131.414**	**202.420**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	23.0	25.0	(35.6)	13.8	28.9	37.4	63.6	42.1
EBITDA (YoY)	70.8	52.1	(57.3)	13.1	37.1	74.6	95.0	57.2
Operating profit (YoY)	307.0	80.6	(89.4)	5.1	143.7	242.4	171.0	74.4
Net profit (YoY)	700.2	65.4	(99.3)	2.9	3,164.4	373.6	195.4	77.7
Core EPS (fully-diluted) (YoY)	698.1	65.3	(99.3)	2.5	3,153.4	322.9	195.4	77.7
Gross-profit margin	32.3	38.1	25.2	28.4	21.1	29.7	40.5	46.4
EBITDA margin	29.5	35.9	23.8	23.7	25.2	32.0	38.2	42.2
Operating-profit margin	16.4	23.6	3.9	3.6	6.8	16.9	28.0	34.4
Net profit margin	13.0	17.2	0.2	0.2	4.1	14.0	25.2	31.6
ROAE	13.9	20.0	0.1	0.1	3.9	15.2	32.4	40.0
ROAA	8.6	10.9	0.1	0.1	2.0	8.8	21.3	28.3
ROCE	14.0	21.1	2.0	2.0	4.8	15.2	32.6	40.6
ROIC	19.9	26.1	2.6	2.4	6.0	20.7	53.7	96.6
Net debt to equity	net cash	net cash	2.5	4.6	4.8	net cash	net cash	net cash
Effective tax rate	11.9	21.0	17.9	17.0	12.0	6.0	5.7	5.7
Accounts receivable (days)	55.2	47.0	63.6	57.3	56.1	57.7	55.1	57.0
Current ratio (x)	1.9	1.8	2.1	1.7	1.4	1.9	2.7	3.9
Net interest cover (x)	84.0	61.0	5.3	5.1	8.9	29.3	71.6	125.5
Net dividend payout	52.6	41.9	909.2	909.2	43.2	23.5	10.3	5.8
Free cash flow yield	n.a.	n.a.	n.a.	n.a.	0.4	0.9	4.4	8.0

Source: FactSet, Daiwa forecasts

Company profile

Kinsus was founded in 2000 as a subsidiary of Asustek. The company mainly focus on niche ball grid array (BGA) substrates to avoid price competition among mainstream products like graphics card and chipsets, and it has become top-3 flip chip manufacturer in global. Besides, its manufacturing sites are based in Taiwan and mainland China.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	2	Kinsus's leadership team comprises experienced professionals dedicated to steering the company towards sustainable growth. This diverse board composition ensures a comprehensive range of expertise and perspectives, contributing to Kinsus's strategic decision-making and corporate governamce.
	Capital management	2	Kinsus maintains a robust financial framework, ensuring the availability of resources necessary for operational efficiency and strategic investments. The company's financial health is regularly disclosed through annual reports, providing transparency to stakeholders.
	Related party & transaction	1	The company adheres to stringent corporate governance policies to manage related party transactions, ensuring transparency and compliance with regulatory standards. The company discloses such transactions in its financial reports, maintaining accountability and stakeholder trust. For instance, the company's annual general meeting documents provide details on related party transactions, ensuring transparency and accountability.
S	Supply chain management	1	Kinsus is committed to sustainable supply chain practices, emphasizing environmental, social, and governance considerations. The company has established comprehensive ESG policies, including commitments to labor standards, environmental health and safety, ethics, and conflict-free minerals. These policies guide the company's interactions with suppliers and partners, promoting sustainability throughout the supply chain.
E	Data security	3	The company recognizes the importance of information security in its operations. As part of its corporate governance framework, Kinsus is likely to implement standard data protection practices to safeguard sensitive information and ensure compliance with relevant regulations. Stakeholders are encouraged to consult the company's official communications or contact the company directly for detailed information on data security policies.

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 31 Jan 2026

Source: Daiwa, Company

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Disclosure of investment ratings

Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

Notes: data is for single-branded Daiwa research in Asia (ex Japan) and correct as of 30 September 2026.
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* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

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